

Stylus Common UI Database Object Information

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Types and Definitions

The following is the list of allowed data types and their descriptions:

Type	Description	Accepted Input (from TCL)	Output Format (to TCL)
bool	A boolean value.	true: true/1/on/yes false: false/0/off/no	true or false
int	Signed 32-bit integer corresponding to Tcl integer.	Any integer value.	26
double	Double-precision (64-bit) signed float.	Any numeric value.	26.4 A .0 is added to the output of integer values to force it to be a double (e.g. 1.0 rather than 1).
string	A string. Unlike other types, the empty string is a valid value.	Any Tcl string value, like "abc" or {abc}.	abc
enum	One of a fixed set of string values (provided in the type qualification).	If "true" or "false" are enum values, these are interpreted as booleans. For example: enum (true false auto) accepts - "0", "no" and "off" as synonyms for "false" - "1", "yes" and "on" as synonyms for "true" - "auto" for "auto"	auto

obj	A database object pointer. It is followed by the list of object types allowed for this attribute. It is optionally followed by * object list of 0 or more + object list of 1 or more ? object link may be empty For example, obj(inst hinst)+ is a list of 1 or more inst or hinst objects. A mixture of both inst and hinst objects is possible.	A dual-ported Tcl object or object list. The string form for netlist objects is <obj_type>:<design_name>/<object_name>, for example: inst:my_design/i1/i2 For technology or library objects it is just <obj_type>:<name>, for example: layer:metal1 For objects with no useful name (like a wire), it is just the hex memory pointer, for example: wire:0x123300012 Type "man get_db" for more details on dual-ported Tcl objects.	layer:metal1 inst:my_design/i1/i2 wire:0x123300012
in_file	A file name that will be read in.	A Unix path name (relative or absolute). The file must exist and be readable.	my_dir/myfile.tcl
out_file	A file that will be written to.	A Unix path name (relative or absolute). The file must be writable.	my_dir/my_file.tcl
in_dir	A directory to read from.	A Unix path name (relative or absolute). The directory must exist and be readable.	my_dir
out_dir	A directory to write into.	A Unix path name (relative or absolute). The directory must be writable.	my_dir

Stylus Common UI Database Object Information
Types and Definitions

coord	A single, linear distance in microns that type double..	Any numerical value in units of microns.	26.4 A .0 is added to the output of integer values to force it to be a double (e.g. 1.0 rather than 1).
point	A point in 2D space, expressed as a two-element Tcl list of coord types: {<x> <y>}. Has sub types: .x - the first coordinate .y - the second coordinate	Any list format that can be translated into two coordinates. For example, the following are allowed: {1.0 2.2} {{1} {2.2}}	{1.0 2.2}
line	A line in 2D space, expressed as a two-element Tcl list of points: {{<x1> <y1>} {<x2> <y2>}}. Has sub types: .begin - first point .end - second point .length - length of the line .dx - difference in x coordinates .dy - difference in y coordinates	Any list format that can be translated into two points. For example, the following are allowed as input. {{1.0 2} {3 4.0}} {1 2 3 4} {{1} {2} {3} {4.0}}	{{1.0 2.2} {3.0 4.0}}

rect	A rectangle in 2D space, expressed as an ordered list of coordinates {<left_x> <lower_y> <right_x> <upper_y>}. Has sub types: .ll - lower left point .ur - upper right point .dx - difference in x coordinates .dy - difference in y coordinates .area - area of the rectangle .width - shorter of dx & dy .length - longer of dx and dy .perimeter - perimeter of the rect	Any list format that can be translated into two points. For example, the following are equivalent: {1 2} {3 4} {1 2 3 4} The two X coordinates and two Y coordinates can be input in either order. They will be ordered as lower-left for the first point, and upper-right as the second point for output.	A list of four coords: {1.0 2.0 3.0 4.0}
polygon	A polygon in 2D space, expressed as a list of points {{<x1> <y1>} {<x2> <y2>} {<x3> <y3>}...}. Has sub types: .bbox - bounding box (rect type) .area - area of the rectangle .perimeter - perimeter of the rectangle	Any list format that can be translated into multiple points. For example, the following are equivalent: {1 2} {3 4} {5 6} {1 2 3 4 5 6}	A list of points: {{1.0 2.0} {3.0 4.0} {5.0 6.0}}.

area	An area in units of micron squared that is type double.	Any numerical value in units of microns squared. 26.4	Value in micron squared without units. Example: 26.4 A .0 is added to the output of integer values to force it to be a double (e.g. 1.0 rather than 1).
delay	A time duration that is type double.	Any numerical value in user units. The units can be queried with "get_db timing_time_unit".	26.4 A .0 is added to the output of integer values to force it to be a double (e.g. 1.0 rather than 1).
resistance	A resistance that is type double.	Any numerical value in user units. The units can be queried with "get_resistance_unit".	26.4 A .0 is added to the output of integer values to force it to be a double (e.g. 1.0 rather than 1).
capacitance	A capacitance that is type double.	Any numerical value in user units. The units can be queried with "get_db timing_cap_unit".	26.4 A .0 is added to the output of integer values to force it to be a double (e.g. 1.0 rather than 1).
voltage	A voltage that is type double.	Any numerical value in units of volts.	1.4 A .0 is added to the output of integer values to force it to be a double (e.g. 1.0 rather than 1).

Database Objects

analysis_view

Parent Objects

design, root

Definition

An analysis view binds together a constraint mode with a delay corner, providing all the information needed to control a single MMMC analysis. Use the create_analysis_view and update_analysis_view commands to create and modify analysis_views.

Attribute	Description
constraint_mode	
	The constraint_mode object associated with the analysis_view. Type: obj(constraint_mode) Default: "" Edit: No
delay_corner	
	The delay_corner object associated with the analysis_view. Type: obj(delay_corner) Default: "" Edit: No
is_active	
	Indicates that the analysis_view is part of any leakage/dynamic/setup/hold designation Type: bool Default: "" Edit: No
is_dormant	
	Indicates that the analysis_view is not being timing analyzed, but a minimum set of data is loaded to keep the view in synch with potential design changes. Type: bool Default: "" Edit: No
is_drv	
	Indicates that the analysis_view is used as drv view. Type: bool Default: "" Edit: No
is_dynamic	
	Indicates that the analysis_view is dynamic power view. Type: bool Default: "" Edit: No
is_em	

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Database Objects--analysis_view

	Indicates that the analysis_view is used for electromigration analysis. Type: <code>bool</code> Default: "" Edit: No
is_hold	
	Indicates that the analysis_view is active for hold analysis Type: <code>bool</code> Default: "" Edit: No
is_hold_default	
	Indicates that the analysis_view is the default view for hold analysis Type: <code>bool</code> Default: "" Edit: No
is_ir	
	Indicates that the analysis_view is IR view. Type: <code>bool</code> Default: "" Edit: No
is_leakage	
	Indicates that the analysis_view is leakage power view. Type: <code>bool</code> Default: "" Edit: No
is_setup	
	Indicates that the analysis_view is active for setup analysis Type: <code>bool</code> Default: "" Edit: No
is_setup_default	
	Indicates that the analysis_view is the default view for setup analysis Type: <code>bool</code> Default: "" Edit: No
latency_file	
	Specifies an optional file containing view-specific set_clock_latency constraints used for balancing post_CTS IO timing. This file is normally generated automatically as part of the clock tree implementation flow. Type: <code>string</code> Default: "" Edit: No
name	
	Provides the name of this analysis_view object as specified by the create_analysis_view command. Type: <code>string</code> Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: <code>enum</code> (analysis_view) Default: "" Edit: No
power_modes	

Specifies an optional list of power_mode objects defined by the power intent specification.
Type: [string](#)
Default: ""
Edit: No

antenna_data

Parent Objects

[base_pin](#), [port](#)

Definition

Antenna information for terminals

Attribute	Description
area	
	Area value for non *Car type cases, 0 value used for *Car type cases as area is not applicable in those cases. Type: area Default: "" Edit: No
layer	
	The layer of antenna data. If layer is null(0x0), data applies to all layers. Type: obj(layer) Default: "" Edit: No
model	
	Oxide model, none is used for cases where oxide model does not apply. Type: enum Enum Values: none oxide1 oxide2 oxide3 oxide4 Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (antenna_data) Default: "" Edit: No
ratio	
	Ratio value for *Car type cases. 0 value used for non *Car enums as ratio is not applicable in those cases. Type: double Default: "" Edit: No
type	
	Type of antenna data. Equivalent to LEF MACRO PIN ANTENNA* constructs. Type: enum Enum Values: lib_gate_area lib_diff_area lib_partial_metal_area lib_partial_metal_side_area lib_max_area_car lib_max_side_area_car lib_partial_cut_area lib_max_cut_car top_partial_metal_area top_partial_metal_side_area top_gate_area top_diff_area top_max_area_car top_max_side_area_car top_partial_cut_area top_max_cut_car Default: "" Edit: No

antenna_model

Parent Objects

layer

Definition

Antenna model information for one layer & oxide

Attribute	Description
area_factor	
	Specifies the multiply factor for the antenna metal area calculation (default value 1.0), LEF(ANTENNAAREAFACTOR) Type: double Default: "" Edit: No
area_factor_diff_use_only	
	Specifies that the current antenna area factor should only be used when the corresponding layer is connected to the diffusion, LEF(DIFFUSEONLY) Type: bool Default: "" Edit: No
area_ratio	
	Specifies the maximum legal antenna ratio, using the area of the metal wire that is not connected to the diffusion diode (0 indicates that the attribute does not apply), LEF(ANTENNAAREARATIO) Type: double Default: "" Edit: No
cum_area_ratio	
	Specifies the cumulative antenna ratio, using the area of the wire that is not connected to the diffusion diode (0 indicates that the attribute does not apply), LEF(ANTENNACUMAREARATIO) Type: double Default: "" Edit: No
cum_routing_plus_cut	
	Indicates that the cumulative ratio rules (ANTENNACUMAREARATIO and ANTENNACUMDIFFAREARATIO) accumulate with the previous cut layer instead of the previous metal layer, LEF(ANTENNACUMROUTINGPLUSCUT) Type: bool Default: "" Edit: No
cum_side_area_ratio	
	Specifies the cumulative antenna ratio, using the side wall area of the metal wire that is not connected to the diffusion diode (0 indicates that the attribute does not apply), LEF(ANTENNACUMSIDEAREARATIO) Type: double Default: "" Edit: No
gate_minus_diff	
	Indicates that the antenna ratio metal area should subtract the diffusion area connected to it (0 indicates that the attribute does not apply), LEF(ANTENNAAREAMINUSDIFF) Type: double Default: "" Edit: No
obj_type	

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Database Objects--arc

	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (antenna_model) Default: "" Edit: No
side_area_factor	
	Specifies the multiply factor for the antenna metal side wall area calculation (default value 1.0), LEF(ANTENNASIDEAREAFACOR) Type: double Default: "" Edit: No
side_area_factor_diff_use_only	
	Specifies that the current antenna side area factor should only be used when the corresponding layer is connected to the diffusion, LEF(DIFFUSEONLY) Type: bool Default: "" Edit: No
side_area_ratio	
	Specifies the antenna ratio, using the side wall area of the metal wire that is not connected to the diffusion diode (0 indicates that the attribute does not apply), LEF(ANTENNASIDEAREARATIO) Type: double Default: "" Edit: No

arc

Parent Objects

[inst](#), [pin](#), [net](#)

Definition

cte timing arc

Attribute	Description
aocv_derate_capture_clock_early_fall	
	Returns graph based AOCV derate factors in AOCV mode for a given timing arc that is a part of early capture clock paths with fall sink pin transitions. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view Default: "" Edit: No
aocv_derate_capture_clock_early_rise	
	Returns graph based AOCV derate factors in AOCV mode for a given timing arc that is a part of early capture clock paths with rise sink pin transitions. You can use -index to return the value for a specific view Type: double Allowed -index values: analysis_view Default: "" Edit: No
aocv_derate_capture_clock_late_fall	
	Returns graph based AOCV derate factors in AOCV mode for a given timing arc that is a part of late capture clock paths with fall sink pin transitions. You can use -index to return the value for a specific view Type: double Allowed -index values: analysis_view Default: "" Edit: No

Stylus Common UI Database Object Information

Database Objects--arc

aocv_derate_capture_clock_late_rise	
	Returns graph based AOCV derate factors in AOCV mode for a given timing arc that is a part of late capture clock paths with rise sink pin transitions. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view Default: "" Edit: No
aocv_derate_data_early_fall	
	Returns AOCV derate values for a timing arc on an early data path with fall sink pin transitions. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view Default: "" Edit: No
aocv_derate_data_early_rise	
	Returns AOCV derate values for a timing arc on an early data path with rise sink pin transitions. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view Default: "" Edit: No
aocv_derate_data_late_fall	
	Returns AOCV derate values for a timing arc on a late data path with fall sink pin transitions. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view Default: "" Edit: No
aocv_derate_data_late_rise	
	Returns AOCV derate values for a timing arc on a late data path with rise sink pin transitions. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view Default: "" Edit: No
aocv_derate_launch_clock_early_fall	
	Returns graph based AOCV derate factors in AOCV mode for a given timing arc that is a part of early launch clock paths with fall sink pin transitions. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view Default: "" Edit: No
aocv_derate_launch_clock_early_rise	
	Returns graph based AOCV derate factors in AOCV mode for a given timing arc that is a part of early launch clock paths with rise sink pin transitions. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view Default: "" Edit: No
aocv_derate_launch_clock_late_fall	
	Returns graph based AOCV derate factors in AOCV mode for a given timing arc that is a part of late launch clock paths with fall sink pin transitions. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view Default: "" Edit: No

Stylus Common UI Database Object Information

Database Objects--arc

aocv_derate_launch_clock_late_rise	
	Returns graph based AOCV derate factors in AOCV mode for a given timing arc that is a part of late launch clock paths with rise sink pin transitions. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view Default: "" Edit: No
aocv_stage_count_capture_clock_early	
	Returns graph based AOCV stage count values in AOCV mode for a given timing arc that is a part of early capture clock paths. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view Default: "" Edit: No
aocv_stage_count_capture_clock_late	
	Returns graph based AOCV stage count values in AOCV mode for a given timing arc that is a part of late capture clock paths. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view Default: "" Edit: No
aocv_stage_count_data_early	
	Returns AOCV stage count values for a timing arc on an early data path. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view Default: "" Edit: No
aocv_stage_count_data_late	
	Returns AOCV stage count values for a timing arc on a late data path. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view Default: "" Edit: No
aocv_stage_count_launch_clock_early	
	Returns AOCV stage count values in AOCV mode for a given timing arc that is a part of early launch clock paths. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view Default: "" Edit: No
aocv_stage_count_launch_clock_late	
	Returns AOCV stage count values in AOCV mode for a given timing arc that is a part of late launch clock paths. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view Default: "" Edit: No
aocv_weight	
	Returns the AOCV stage weight for this arc. By default, all cells and arcs have default stage weight of 1.0. The aocv_weight property is specified as a user-defined library attribute in the Liberty timing library explicitly - or, by asserting it via command. This attribute is inherited from the associated library arc. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view Default: "" Edit: No

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Database Objects--arc

delay_max_fall	Returns the largest falling delay through the arc across all concurrent MMMC views. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view Default: "" Edit: No
delay_max_rise	Returns the largest rising delay through the arc across all concurrent MMMC views. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view Default: "" Edit: No
delay_mean_max_fall	In SOCV analysis mode, this returns the mean component of the largest falling delay through the arc across all concurrent MMMC views. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view Default: "" Edit: No
delay_mean_max_rise	In SOCV analysis mode, this returns the mean component of the largest rising delay through the arc across all concurrent MMMC views. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view Default: "" Edit: No
delay_mean_min_fall	In SOCV analysis mode, this returns the mean component of the smallest falling delay through the arc across all concurrent MMMC views. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view Default: "" Edit: No
delay_mean_min_rise	In SOCV analysis mode, this returns the mean component of the smallest rising delay through the arc across all concurrent MMMC views. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view Default: "" Edit: No
delay_min_fall	Returns the smallest falling delay through the arc across all concurrent MMMC views. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view Default: "" Edit: No
delay_min_rise	Returns the smallest rising delay through the arc across all concurrent MMMC views can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view Default: "" Edit: No
delay_sigma_max_fall	

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Database Objects--arc

	In SOCV analysis mode, this returns the variation component of the largest falling delay through the arc across all concurrent MMMC views. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view Default: "" Edit: No
delay_sigma_max_rise	
	In SOCV analysis mode, this returns the variation component of the largest rising delay through the arc across all concurrent MMMC views. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view Default: "" Edit: No
delay_sigma_min_fall	
	In SOCV analysis mode, this returns the variation component of the smallest falling delay through the arc across all concurrent MMMC views. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view Default: "" Edit: No
delay_sigma_min_rise	
	In SOCV analysis mode, this returns the variation component of the smallest rising delay through the arc across all concurrent MMMC views. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view Default: "" Edit: No
delta_delay_max_fall	
	Returns the delta/SI delay component of the largest falling delay through this arc across all concurrent MMMC views. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view Default: "" Edit: No
delta_delay_max_rise	
	Returns the delta/SI delay component of the largest rising delay through this arc across all concurrent MMMC views. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view Default: "" Edit: No
delta_delay_min_fall	
	Returns the delta/SI delay component of the smallest falling delay through this arc across all concurrent MMMC views. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view Default: "" Edit: No
delta_delay_min_rise	
	Returns the delta/SI delay component of the smallest rising delay through this arc across all concurrent MMMC views. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view Default: "" Edit: No
dynamic_delta_delay_max_fall	

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Database Objects--arc

	Returns the dynamic delay component of the largest falling delay through this arc across all concurrent MMMC views. This is computed if dynamic voltages have been specified. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view Default: "" Edit: No
dynamic_delta_delay_max_rise	
	Returns the dynamic delay component of the largest rising delay through this arc across all concurrent MMMC views. This is computed if dynamic voltages have been specified. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view Default: "" Edit: No
dynamic_delta_delay_min_fall	
	Returns the dynamic delay component of the smallest falling delay through this arc across all concurrent MMMC views. This is computed if dynamic voltages have been specified. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view Default: "" Edit: No
dynamic_delta_delay_min_rise	
	Returns the dynamic delay component of the smallest rising delay through this arc across all concurrent MMMC views. This is computed if dynamic voltages have been specified. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view Default: "" Edit: No
from_pin	
	Returns a pin object for the beginning pin of this timing arc Type: obj(pin)* obj(hpin)* obj(hport)* Allowed -index values: analysis_view Default: "" Edit: No
has_mis_fall	
	Return library attribute has_mis for fall output transition. Type: bool Allowed -index values: analysis_view Default: "" Edit: No
has_mis_rise	
	Return library attribute has_mis for rise output transition. Type: bool Allowed -index values: analysis_view Default: "" Edit: No
is_cell_arc	
	Returns true if the arc is a cell arc. Type: bool Allowed -index values: analysis_view Default: "" Edit: No
is_disabled	

Stylus Common UI Database Object Information

Database Objects--arc

	Returns a value of true if this library timing arc has been explicitly disabled by the user via a set_disable_timing constraint Type: bool Allowed -index values: analysis_view Default: "" Edit: No
lib_arcs	
	Returns a list of the associated lib_arc objects. You can use -index to return the value for a specific view. Type: obj(lib_arc)* Allowed -index values: analysis_view Default: "" Edit: No
mode	
	If the associated lib_arc is defined as part of a Liberty mode group, this attribute will return the name of the library group Type: string Allowed -index values: analysis_view Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (arc) Default: "" Edit: No
sdf_cond	
	Returns the value of the Liberty sdf_cond attribute if specified for the associated lib_arc Type: string Allowed -index values: analysis_view Default: "" Edit: No
sdf_cond_end	
	Returns the value of the Liberty sdf_cond_end attribute if specified for the associated lib_arc Type: string Allowed -index values: analysis_view Default: "" Edit: No
sdf_cond_start	
	Returns the value of the Liberty sdf_cond_start attribute if specified for the associated lib_arc Type: string Allowed -index values: analysis_view Default: "" Edit: No
sense	
	Returns the Liberty timing_sense value associated with this arc. This value is one of: positive_unate, negative_unate, or non_unate. This is inherited from the lib_arc. Type: string Allowed -index values: analysis_view Default: "" Edit: No
ssi_derate_fall	
	Returns the fall ssi_derate for the arc. Type: double Allowed -index values: analysis_view Default: "" Edit: No
ssi_derate_rise	

Stylus Common UI Database Object Information

Database Objects--attribute

	Returns the rise ssi_derate for the arc. Type: double Allowed -index values: analysis_view Default: "" Edit: No
timing_type	
	Returns the Liberty timing_type associated with this arc. You can consult the Liberty documentation for the list of possible values for this attribute. This is inherited from the lib_arc. Type: string Allowed -index values: analysis_view Default: "" Edit: No
to_pin	
	Returns a pin object for the terminating pin of this timing arc Type: obj(pin)* obj(hpin)* obj(hport)* Allowed -index values: analysis_view Default: "" Edit: No
when	
	Returns the value of the Liberty 'when' attribute if specified for the related lib_arc Type: string Allowed -index values: analysis_view Default: "" Edit: No
when_end	
	Returns the value of the Liberty 'when_end' attribute if specified for the related lib_arc Type: string Allowed -index values: analysis_view Default: "" Edit: No
when_library_string	
	Returns the exact value of the Liberty 'when' attribute if specified for the related lib_arc Type: string Allowed -index values: analysis_view Default: "" Edit: No
when_start	
	Returns the value of the Liberty 'when_start' attribute if specified for the related lib_arc Type: string Allowed -index values: analysis_view Default: "" Edit: No

attribute

Parent Objects

[root](#), [obj_type](#)

Definition

Attribute	Description
additional_help	

Stylus Common UI Database Object Information

Database Objects--attribute

	Range info or other additional help. Type: string Default: "" Edit: No
base_name	
	The name without the obj_type (e.g. place_status rather than inst/place_status). Type: string Default: "" Edit: No
category	
	Defines the category of the attribute. Categories group attributes that perform similar functions whereas object types describe where in the design an attribute is valid. You can specify any category name: both new and existing category names are valid. (examples: physical, timing, etc). Note: the value also be an empty string "". Type: string Default: "" Edit: No
check_function	
	Specifies a previously defined Tcl procedure's name in order to ensure that the newly defined attribute is valid. Procedure argument list: object value Type: string Default: "" Edit: No
compute_function	
	Specifies a previously defined Tcl procedure's name in order to get the newly defined attribute's value later command. Procedure argument list: object Type: string Default: "" Edit: No
data_type	
	data type attribute Type: string Default: "" Edit: No
default_value	
	Specifies a default value for the attribute, most attributes do not have default values and will use "". Type: string Default: "" Edit: No
help	
	Specifies the help text for the attribute Type: string Default: "" Edit: No
indices	
	List of object type names that represent indices for an attribute value. Type: enum Enum Values: analysis_view clock_delay_corner clock_tree skew_group power_domain layer Default: "" Edit: No
is_computed	

Stylus Common UI Database Object Information

Database Objects--base_cell

	Specifies whether the defined attribute will be computed. For .? use, the value will not be displayed if it needs to be recomputed. For direct access of the attribute the value will be returned or computed as needed (0 = static, 1 = computed) Type: bool Default: "" Edit: No
is_settable	
	whether the defined attribute is a read-only or settable attribute (0 = read-only, 1 = settable) Type: bool Default: "" Edit: No
is_user_defined	
	Specifies whether the defined attribute was defined by the user via Tcl (define_attribute) or is a system defined attribute (0 = system defined, 1 = user defined) Type: bool Default: "" Edit: No
name	
	name of attribute Type: string Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (attribute) Default: "" Edit: No
parent	
	The attribute's parent object Type: obj(obj_type) Default: "" Edit: No
possible_values	
	enum possible values attribute Type: string Default: "" Edit: No
set_function	
	Specifies a previously defined Tcl procedure's name. This option allows you to override user-defined values provided it conforms to the parameters in the Tcl procedure you created. Procedure argument list: object new_value current_value Type: string Default: "" Edit: No
skip_in_db	
	Prevents write_db command to write out the defined Type: bool Default: "" Edit: No

base_cell

Parent Objects

[lib_cell](#), [inst](#), [pin_group](#), [pg_base_pin](#), [tech](#), [bump](#), [pin_guide](#), [power_domain](#), [root](#), [base_pin](#)

Stylus Common UI Database Object Information

Database Objects--base_cell

Definition

base_cell

Attribute	Description
antenna_lef_file_name	
	Specifies the LEF file name of the base_cell terminals' antenna data Type: string Default: "" Edit: No
aocv_weight	
	Returns the AOCV stage weight specified for the cell either explicitly in the library or as a user-defined library attribute Type: double Default: "" Edit: No
area	
	The area of the cell Type: area Default: "" Edit: No
base_class	
	The base class is the prefix of the .class enum value. It is useful to separate the major cell categories. See the .class description for the definition of each enum value Type: enum Enum Values: none cover block pad core corner Default: "" Edit: No
base_name	
	Base name of the cell. Type: string Default: "" Edit: No
base_pins	
	The base_pins for this cell that are part of the logical netlist. This does not include pg_base_pins, or must_join_base_pins, or internal_base_pins. Type: obj(base_pin)* Default: "" Edit: No
bbox	
	The bounding box of the overlap rects that define the placement area used by this cell. Type: rect Default: "" Edit: No
bottom_edge_type	
	Name of cell edge type for the bottom edge of the cell (R0/N orientation), used to indicate which cells need extra spacing to other cells. Type: string Default: "" Edit: No
bottom_padding	
	The placer will leave this much extra space to the bottom side of the cell (in r0 orientation). It is only valid for standard cells. It is in units of site width. Type: int Default: 0 Edit: Yes

Stylus Common UI Database Object Information

Database Objects--base_cell

class	All the LEF CLASS and PROPERTY LEF58_CLASS values (and equivalent OpenAccess values). Refer to the LEF documentation for the complete list and descriptions. They are separated into 5 classes with a unique prefix based on their usage as described here: CLASS COVER types all start with cover. CLASS RING or BLOCK start with block. CLASS PAD start with pad. CLASS CORE start with core. CLASS ENDCAP that are corner cells (TOPLEFT, TOPRIGHT, BOTTOMLEFT, BOTTOMRIGHT) start with corner. CLASS ENDCAP that are not corner cells all start with core because they are all placed in the core rows like CLASS CORE cells. No CLASS means the value is none. Modifications are valid only for the current session. Type: enum Enum Values: none cover cover_bump cover_fill block_ring block_blackbox block_soft pad pad_input pad_output pad_inout pad_power pad_spacer pad_area_io core core_feedthru core_tie_high core_tie_low core_spacer core_antenna core_welltap core_endcap_pre core_endcap_post corner_top_left corner_top_right corner_bottom_left corner_bottom_right core_endcap_top_edge core_endcap_bottom_edge core_endcap_left_edge core_endcap_right_edge core_endcap_left_top_edge core_endcap_right_top_edge core_endcap_left_bottom_edge core_endcap_right_bottom_edge core_endcap_left_top_corner core_endcap_right_top_corner core_endcap_left_bottom_corner core_endcap_right_bottom_corner core_endcap_left_even_site_edge core_endcap_left_odd_site_edge core_endcap_right_even_site_edge core_endcap_right_odd_site_edge core_endcap_left_top_even_site_edge core_endcap_left_top_odd_site_edge core_endcap_right_top_even_site_edge core_endcap_right_top_odd_site_edge core_endcap_left_bottom_even_site_edge core_endcap_right_bottom_even_site_edge core_endcap_left_top_even_site_corner core_endcap_left_top_odd_site_corner core_endcap_right_top_even_site_corner core_endcap_right_top_odd_site_corner core_endcap_left_bottom_even_site_corner core_endcap_left_bottom_odd_site_corner core_endcap_right_bottom_even_site_corner core_endcap_right_bottom_odd_site_corner core_endcap_left_edge_bottom_border core_endcap_left_edge_top_border core_endcap_right_edge_bottom_border core_endcap_right_edge_top_border core_endcap_left_bottom_edge_neighbor core_endcap_left_top_edge_neighbor core_endcap_right_bottom_edge_neighbor core_endcap_right_top_edge_neighbor core_endcap_left_bottom_corner_neighbor core_endcap_left_top_corner_neighbor core_endcap_right_bottom_corner_neighbor core_endcap_right_top_corner_neighbor Default: "" Edit: Yes
cts_cell_halo_x	Specifies the clock halo distance in the x direction. The default value of this attribute is auto. The following attributes can be used to assign x direction clock halos within CCOpt: cts_cell_halo_x cts_cell_density cts_cell_halo_sites Only one of these attributes is used to determine the clock halo in the x direction. The following rules determine which: . If cts_cell_halo_x is set to a non-auto value, then this defines the x direction clock halo. The attributes cts_cell_density and cts_cell_halo_sites have no effect. . If cts_cell_halo_x is set to auto and cts_cell_density is set to a non-auto value then cts_cell_density defines the clock halo in the x direction. The attribute cts_cell_halo_sites has no effect. . If both cts_cell_halo_x and cts_cell_density are set to auto then cts_cell_halo_sites defines the clock halo in the x direction. See also: . cts_cell_density . cts_cell_halo_sites Type: string Allowed -index values: clock_tree power_domain Default: auto Edit: Yes
cts_cell_halo_y	

Stylus Common UI Database Object Information

Database Objects--base_cell

	Specifies the clock halo distance in the y direction. The default value of this attribute is auto. The following attributes can be used to assign y direction clock halos within CCOpt: cts_cell_halo_y cts_adjacent_rows_legal cts_cell_halo_rows Only one of these attributes is used to determine the clock halo in the y direction. The following rules determine which: . If cts_cell_halo_y is set to a non-auto value, then this defines the y direction clock halo. The attributes cts_adjacent_rows_legal and cts_cell_halo_rows have no effect. . If cts_cell_halo_y is set to auto and cts_adjacent_rows_legal is set to a non-auto value then cts_adjacent_rows_legal defines the clock halo in the y direction. The attribute cts_cell_halo_rows has no effect. . If both cts_cell_halo_y and cts_adjacent_rows_legal are set to auto then cts_cell_halo_rows defines the clock halo in the y direction. See also: . cts_adjacent_rows_legal . cts_cell_halo_rows Type: string Allowed -index values: clock_tree power_domain Default: auto Edit: Yes
dont_touch	
	This attribute says any inst of this base_cell cannot be modified during optimization. This is the effective dont_touch value for all lib_cells. It is set to the worst case of the lib_cells during init_design and can only be updated by set_db / set_dont_touch after that (subsequent library reads will not affect). This attribute will get restored back to the state during write_db regardless if the library files have been altered. Type: bool Default: false Edit: Yes
dont_use	
	This attribute says do not use this base_cell during optimization. This is the effective dont_use value for all lib_cells. It is set to the worst case of the lib_cells during init_design and can only be updated by set_db / set_dont_use after that (subsequent library reads will not affect). This attribute is saved by write_db, and restored by read_db regardless if the library files have been altered. Type: bool Default: false Edit: Yes
drive_strength	
	This attribute allows the user to assign a drive strength to each cell (X1, X2, etc.) to be used for metric capture. Type: string Default: "" Edit: Yes
eeq_cells	
	Electrically equivalent base_cells. Same as LEF MACRO EEQ statement. Type: obj(base_cell)* Default: "" Edit: No
eeq_variant	
	The LEF EEQ cell variant number from the LEF58_EEQ property 'EEQ macroName VARIANT num' Type: int Default: no_value Edit: Yes
escaped_name	
	Escaped name of the cell. Type: string Default: "" Edit: No
foreign_cells	

Stylus Common UI Database Object Information

Database Objects--base_cell

	List of foreign references. Equivalent to LEF MACRO FOREIGN. Type: <code>obj(foreign_cell)*</code> Default: "" Edit: No
has_pin_access_violation	
	Returns True if the cell has pin access violation. Note the attribute is valid after check_library is executed. Type: <code>bool</code> Default: no_value Edit: No
internal_base_pins	
	Internal pins from .lib that do not exist in the logical netlist. They do not appear in Verilog or DEF files, but may appear in timing reports. Type: <code>obj(base_pin)*</code> Default: "" Edit: No
is_always_on	
	Specifies the cell is an always-on cell. An always-on cell normally has two power pins. One is primary which aligns with the normal cell power-rail, and the other is the secondary which actually powers the cell, even when the primary power is off. This attribute can be set by liberty files, or by CPF commands. Type: <code>bool</code> Default: false Edit: No
is_black_box	
	Returns a value of true if the cell is a black box cell. Type: <code>bool</code> Default: "" Edit: No
is_buffer	
	This cell is a buffer. Type: <code>bool</code> Default: "" Edit: No
is_combinational	
	Returns a value of true if the cell is a combinational cell (not a sequential cell). Type: <code>bool</code> Default: "" Edit: No
is_fall_edge_triggered	
	Returns a value of true if the cell is triggered by the falling edge of the clock. Type: <code>bool</code> Default: "" Edit: No
is_fixed_mask	
	Indicates the cell has FIXEDMASK keyword in LEF. Type: <code>bool</code> Default: "" Edit: No
is_flop	
	Returns true if the cell is recognized as flip-flop/register type cell Type: <code>bool</code> Default: "" Edit: No
is_generic	

Stylus Common UI Database Object Information

Database Objects--base_cell

	This cell is Synthesis Generic cell. Type: <code>bool</code> Default: <code>""</code> Edit: No
is_integrated_clock_gating	
	Returns true if the cell has the Liberty clock_gating_integrated_cell set to true Type: <code>bool</code> Default: <code>""</code> Edit: No
is_interface_timing	
	Returns a value of true if a cell has the Liberty interface_timing attribute set to true Type: <code>bool</code> Default: <code>""</code> Edit: No
is_inverter	
	This cell is an inverter. Type: <code>bool</code> Default: <code>""</code> Edit: No
is_iso_nor	
	Specifies the cell is an ISONOR cell. An ISONOR cell is a kind of isolation cell, which has only one primary power pin and one primary ground pin. An ISONOR cell is defined by library files. In cell library, it has permit_power_down true for primary power pin, alive_during_power_up true for input signal pin, and alive_during_partial_power_down true for enable pin and output signal pin. The attribute should be queried after read and commit power intent. Type: <code>bool</code> Default: <code>""</code> Edit: No
is_isolation_cell	
	Specifies the cell is an isolation cell. An isolation cell is used to clamp the signal to high or low when its input is shutoff(unknown). This attribute can be set by liberty files, or by CPF commands. Type: <code>bool</code> Default: <code>false</code> Edit: No
is_latch	
	Returns true if the cell is recognized as latch type cell Type: <code>bool</code> Default: <code>""</code> Edit: No
is_level_shifter	
	Specifies the cell is a level shifter cell. A level-shifter cell is used to shift the signal voltage from low(high) to high(low). This attribute can be set by liberty files, or by CPF commands. Type: <code>bool</code> Default: <code>false</code> Edit: No
is_macro	
	Returns a value of true if the Liberty attribute is_macro_cell is set true Type: <code>bool</code> Default: <code>""</code> Edit: No
is_master_slave_flop	

Stylus Common UI Database Object Information

Database Objects--base_cell

	Returns true if the cell is recognized as a master/slave flip-flop type cell Type: <code>bool</code> Default: "" Edit: No
is_master_slave_lssd_flop	
	Returns true if this cell has been recognized as a master/slave LSSD type cell Type: <code>bool</code> Default: "" Edit: No
is_memory	
	Returns true or false depending on whether the associated library cell is recognized as a Liberty memory cell. Cells which include a Liberty memory group definition are recognized as memory cells. Type: <code>bool</code> Default: "" Edit: No
is_negative_level_sensitive	
	Returns a value of true if the cell is identified as a negative level-sensitive latch Type: <code>bool</code> Default: "" Edit: No
is_pad	
	Returns a value of true if Liberty attribute is_pad is set to true Type: <code>bool</code> Default: "" Edit: No
is_physical_defined	
	This cell has a physical abstract loaded from LEF or OA so the various physical attributes have been set Type: <code>bool</code> Default: "" Edit: No
is_pll	
	Returns a value of true if the Liberty library is_pll_cell attribute is set to true for this cell Type: <code>bool</code> Default: "" Edit: No
is_positive_level_sensitive	
	Returns a value of true if the cell is identified as a positive level-sensitive latch Type: <code>bool</code> Default: "" Edit: No
is_power_on_bottom	
	Indicates this standard cell has a power pin along the bottom of the cell. This is derived from the power and ground pin information in the cell. It is used by the placer to align multi-height cells properly to the rows. It is not meaningful for non standard cells. Modifications are not saved and are only valid for the current session. Type: <code>bool</code> Default: false Edit: Yes
is_power_switch	
	Specifies the cell is a power switch cell. The power switch cell is used to switch off the power/ground during shutoff. This attribute can be set by liberty files, or by CPF commands. Type: <code>bool</code> Default: false Edit: No

Stylus Common UI Database Object Information

Database Objects--base_cell

is_quick_abstract	
	Indicates that the base_cell was created from reading an OpenAccess layout view and inferred abstract (LEF MACRO equivalent) information from that view. The oa_abstract_lib and oa_abstract_view attributes indicate which layout view was read. For most uses, the inferred abstracts are intended for floorplanning only and are not recommended for final implementation. Type: bool Default: "" Edit: No
is_retention_cell	
	Specifies the cell is a state-retention cell. A state-retention cell is used to retain its state during shutoff. It has a secondary power pin which powers the cell and retains its state, even when the primary power is off. This attribute can be set by liberty files, or by CPF commands. Type: bool Default: false Edit: No
is_rise_edge_triggered	
	Returns a value of true if the cell is triggered by the rising edge of the clock Type: bool Default: "" Edit: No
is_scan_cell	
	Indicates the base_cell is a scan cell. A scan cell has test_cell defined in liberty files. Type: bool Default: "" Edit: No
is_sequential	
	This cell is sequential. Derived from .lib information Type: bool Default: "" Edit: No
is_timing_defined	
	This cell has a .lib file definition loaded, so the various timing attributes have been set Type: bool Default: "" Edit: No
is_tristate	
	Returns a value of true if the cell definition includes the Liberty attribute three_state set to true . Type: bool Default: "" Edit: No
is_vdd_on_bottom	
	Indicates this standard cell has a power pin along the bottom of the cell. This is derived from the power and ground pin information in the cell. It is used by the placer to align multi-height cells properly to the rows. It is not meaningful for non standard cells. Modifications are not saved and are only valid for the current session. Type: bool Default: false Edit: Yes
lde_data_left	
	The COPDE depth to the left side of the cell. Type: string Default: "" Edit: No
lde_data_right	

Stylus Common UI Database Object Information

Database Objects--base_cell

	The COPDE depth to the right side of the cell. Type: string Default: "" Edit: No
lef_file_name	
	Specifies the LEF file name of the base_cell Type: string Default: "" Edit: No
left_edge_type	
	Name of cell edge type for the left edge of the cell (R0/N orientation), used to indicate which cells need extra spacing to other cells. Type: string Default: "" Edit: No
left_padding	
	The placer will leave this much extra space to the left side of the cell (in r0 orientation). It is only valid for standard cells. It is in units of site width. Type: int Default: 0 Edit: Yes
legal_avg_pass_ratio	
	Returns the cell's legal average pass ratio. Range: [0,1]. Note the attribute is valid after check_library is executed. Type: double Default: no_value Edit: No
lib_cells	
	Returns a list of lib_cell objects which are associated with this base_cell Type: obj(lib_cell)* Default: "" Edit: No
logic_function	
	This attribute allows the user to assign a function to each cell (AND, OR, etc.) to be used for metric capture. Type: string Default: "" Edit: Yes
must_join_base_pins	
	Must join pins are physical-only pins that do not exist in the logical netlist. They do not appear in Verilog or DEF files. The router will connect the must-join base_pin to its corresponding logical pin. See the base_pin .must_join_pins attribute, and the LEF MACRO PIN MUSTJOIN statement for more details Type: obj(base_pin)* Default: "" Edit: No
name	
	Name of the cell. Type: string Default: "" Edit: No
num_base_pins	
	Number of signal base_pins for this cell. It does not include pg_base_pins or must_join_base_pins or internal_base_pins. Type: int Default: "" Edit: No
num_drivers	

Stylus Common UI Database Object Information

Database Objects--base_cell

	The number of driver base_pins for this base_cell. Signal base_pins that have direction = out or inout are considered drivers. Type: int Default: "" Edit: No
num_loads	
	The number of load base_pins for this base_cell. Signal base_pins that have direction = in or inout are considered loads. Type: int Default: "" Edit: No
oa_abstract_lib	
	OpenAccess library name of the physical abstract for the cell if read from OA. Type: string Default: "" Edit: No
oa_abstract_view	
	OpenAccess view name for the physical abstract for this cell if read from OA (equivalent of LEF MACRO data). The OA cell name is the same as this cell's 'name' attribute Type: string Default: "" Edit: No
oa_layout_lib	
	OpenAccess library name of physical layout for the cell if read from OA. Type: string Default: "" Edit: No
oa_layout_view	
	OpenAccess view name for the physical layout for this cell if read from OA (equivalent of GDSII data). The layout view data can optionally be displayed instead of the abstract view data, but otherwise this layout data is not used in Innovus. The OA cell name is the same as this cell's 'name' attribute Type: string Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (base_cell) Default: "" Edit: No
obs_layer_shapes	
	A list of cell obstruction layer_shapes (LEF OBS or OA abstract blockages that are not vias). Note, use obs_shape_vias for obstructions that are from vias (LEF OBS VIA). Type: obj(layer_shape)* Default: "" Edit: No
obs_shape_vias	
	A list of cell obstruction shape_vias (LEF OBS VIA or OA abstract via blockages). Use obs_layer_shapes for obstructions on a single layer. Type: obj(shape_via)* Default: "" Edit: No
pg_base_pins	
	The power/ground base_pins for this cell. Type: obj(pg_base_pin)* Default: "" Edit: No

Stylus Common UI Database Object Information

Database Objects--base_cell

right_edge_type	
	Name of cell edge type for the right edge of the cell (R0/N orientation), used to indicate which cells need extra spacing to other cells. Type: string Default: "" Edit: No
right_padding	
	The placer will leave this much extra space to the right side of the cell (in r0 orientation). It is only valid for standard cells. It is in units of site width. Type: int Default: 0 Edit: Yes
site	
	The site for this cell. Modifications are valid only for the current session. Type: obj(site) Default: "" Edit: Yes
symmetry	
	The allowed orientations for the placer to try. none means r0 only, x means ok to flip about X axis (r0, mx), y means ok to flip about Y axis (r0, my), xy means ok to flip about X or Y axis (r0, mx, my, r180), any means all orientations are allowed. Equivalent to LEF MACRO SYMMETRY or OA oaSymmetry. LEF values of R90, X R90, Y R90, and X Y R90 are all equivalent to the value = any. Modifications are not saved and are only valid for the current session. Type: enum Enum Values: none x y xy any Default: "" Edit: Yes
tap_type	
	Specifies the name of a well tap type for this cell. Various rules for well taps are grouped together for each tap_type. See the LEF documentation on the TAPTYPE keyword for more details. Type: string Default: "" Edit: Yes
tap_wall	
	Specifies a special well tap cell (LEF CLASS CORE WELLTAP) or a special end_cap (LEF CLASS ENDCAP ...) cell that can be used for a tap wall purpose, which is used to break OD diffusion and aligned vertically to form a tap wall. See the LEF docs about the keyword TAPWALL for more details. Type: bool Default: false Edit: Yes
tech	
	The tech for this base_cell. Type: obj(tech) Default: "" Edit: No
timing_model_type	
	Returns the Liberty model type for a given cell or instance. The supported values are abstracted , extracted , and qtm . Type: string Default: "" Edit: No
top_edge_type	
	Name of cell edge type for the top edge of the cell (R0/N orientation), used to indicate which cells need extra spacing to other cells. Type: string Default: "" Edit: No

Stylus Common UI Database Object Information

Database Objects--base_pin

top_padding	
	The placer will leave this much extra space to the top side of the cell (in r0 orientation). It is only valid for standard cells. It is in units of site width. Type: int Default: 0 Edit: Yes
voltage_threshold_group	
	This attribute allows the user to assign a base_cell to a voltage threshold group for metric capture. The threshold group name can be any valid string. Type: string Default: "" Edit: Yes

base_pin

Parent Objects

[partition](#), [bump_pin](#), [pin_blockage](#), [base_cell](#), [pin_group](#), [hpin](#), [pin](#), [tech](#), [root](#), [lib_pin](#),

Definition

base pin

Attribute	Description
antenna_data	
	The various process antenna_data values for this pin, including area of gate, diffusion, metal, cut, and cumulative area ratios for metal, cut attached to this pin. Type: obj(antenna_data)* Default: "" Edit: No
base_cell	
	The base_cell of base pin. Type: obj(base_cell) Default: "" Edit: No
base_name	
	The name without the base_cell name (e.g. out rather than and2/out). Type: string Default: "" Edit: No
cts_max_fanout	
	The maximum fanout at any point in the clock tree. Valid values: integer ranged between 2 and 1000 inclusive Type: int Default: 100 Edit: Yes
cts_max_source_to_sink_net_length	

Stylus Common UI Database Object Information
Database Objects--base_pin

	The maximum routing length in microns between driving source pin and driven sink pin on each net that clock tree synthesis should observe. This constraint can be applied to either a pin, a clock tree, or a net type. By default (if this attribute is not set) no explicit clock tree net length constraint is enforced. However, other clock tree constraints such as maximum slew (transition) and maximum capacitance will indirectly limit the maximum net length. Valid values: double Type: string Allowed -index values: clock_tree Default: top auto trunk auto leaf auto Edit: Yes
cts_max_source_to_sink_net_length_leaf	
	The maximum routing length in microns between driving source pin and driven sink pin on each net that clock tree synthesis should observe. This constraint can be applied to either a pin, a clock tree, or a net type. By default (if this attribute is not set) no explicit clock tree net length constraint is enforced. However, other clock tree constraints such as maximum slew (transition) and maximum capacitance will indirectly limit the maximum net length. Valid values: double Type: string Allowed -index values: clock_tree Default: auto Edit: Yes
cts_max_source_to_sink_net_length_top	
	The maximum routing length in microns between driving source pin and driven sink pin on each net that clock tree synthesis should observe. This constraint can be applied to either a pin, a clock tree, or a net type. By default (if this attribute is not set) no explicit clock tree net length constraint is enforced. However, other clock tree constraints such as maximum slew (transition) and maximum capacitance will indirectly limit the maximum net length. Valid values: double Type: string Allowed -index values: clock_tree Default: auto Edit: Yes
cts_max_source_to_sink_net_length_trunk	
	The maximum routing length in microns between driving source pin and driven sink pin on each net that clock tree synthesis should observe. This constraint can be applied to either a pin, a clock tree, or a net type. By default (if this attribute is not set) no explicit clock tree net length constraint is enforced. However, other clock tree constraints such as maximum slew (transition) and maximum capacitance will indirectly limit the maximum net length. Valid values: double Type: string Allowed -index values: clock_tree Default: auto Edit: Yes
cts_spec_config_base_pin_trace_through_to	

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Database Objects--base_pin

	Clock tree definition will, by default, not continue through certain types of cell arc (for instance, the clock to Q arc in a DFF). This attribute allows you to override this default behavior, permitting the clock tree to trace through all instances of such a cell. This attribute serves the same function as <code>trace_through_to</code>, except that here the clock path is specified at the level of the library cell. The attribute should be configured on the input library pin at which the clock will arrive. The value of the attribute specifies the output library pin to which the clock should propagate. The specified output pin must be another pin on the same library cell. The output pin may be specified either by its fully qualified name (i.e. inclusive of the cell name), or else simply by its local (cell-relative) name. There must be a pre-existing (library-defined) chain of one or more delay arcs that connect the input and output pins together. It is not possible to use <code>library_trace_through_to</code> to synthesize delay arcs. If multiple input pins are annotated on a given library cell, the value of <code>library_trace_through_to</code> at each of those pins must select the same output pin: i.e. the configuration must identify a single clock output for the cell. If multiple clock outputs are necessary then <code>library_trace_through_to</code> should not be used: instead for each instance of the library cell, define a generated clock tree at each of the clock-carrying outputs. If the configuration of <code>library_trace_through_to</code> settings for a given library cell does not meet these requirements, a warning will be issued and the settings for that cell will be ignored. All instances of the library cell will be affected by this setting. If both <code>trace_through_to</code> and <code>library_trace_through_to</code> are applicable at a given instance pin, the <code>trace_through_to</code> value will take precedence. Valid values: <code>base_pin</code> Type: <code>string</code> Default: <code>""</code> Edit: Yes
cts_stack_via_rule	
	The preferred stack via rule for terminal connections. This attribute helps guide the choice of stack via rule (via pillar) used for connecting routes to netlist terminals. If the specified value names a valid candidate for terminal in question (it is a member of list the candidate rules associated with the terminal's cell pin), then it will be used as the preferred stack via rule for connecting to that terminal. Valid values: 'auto' or stack via rule name Type: <code>string</code> Allowed -index values: <code>clock_tree</code> Default: <code>""</code> Edit: Yes
cts_stack_via_rule_leaf	
	The preferred stack via rule for terminal connections. This attribute helps guide the choice of stack via rule (via pillar) used for connecting routes to netlist terminals. If the specified value names a valid candidate for terminal in question (it is a member of list the candidate rules associated with the terminal's cell pin), then it will be used as the preferred stack via rule for connecting to that terminal. Valid values: 'auto' or stack via rule name Type: <code>string</code> Allowed -index values: <code>clock_tree</code> Default: <code>""</code> Edit: Yes
cts_stack_via_rule_required	
	The pin-specific required field for stack via rule connections. Type: <code>bool</code> Allowed -index values: <code>clock_tree</code> Default: <code>false</code> Edit: Yes

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Database Objects--base_pin

cts_stack_via_rule_required_leaf	
	The pin-specific required field for stack via rule connections. Type: bool Allowed -index values: clock_tree Default: false Edit: Yes
cts_stack_via_rule_required_top	
	The pin-specific required field for stack via rule connections. Type: bool Allowed -index values: clock_tree Default: false Edit: Yes
cts_stack_via_rule_required_trunk	
	The pin-specific required field for stack via rule connections. Type: bool Allowed -index values: clock_tree Default: false Edit: Yes
cts_stack_via_rule_top	
	The preferred stack via rule for terminal connections. This attribute helps guide the choice of stack via rule (via pillar) used for connecting routes to netlist terminals. If the specified value names a valid candidate for terminal in question (it is a member of list the candidate rules associated with the terminal's cell pin), then it will used as the preferred stack via rule for connecting to that terminal. Valid values: 'auto' or stack via rule name Type: string Allowed -index values: clock_tree Default: "" Edit: Yes
cts_stack_via_rule_trunk	
	The preferred stack via rule for terminal connections. This attribute helps guide the choice of stack via rule (via pillar) used for connecting routes to netlist terminals. If the specified value names a valid candidate for terminal in question (it is a member of list the candidate rules associated with the terminal's cell pin), then it will used as the preferred stack via rule for connecting to that terminal. Valid values: 'auto' or stack via rule name Type: string Allowed -index values: clock_tree Default: "" Edit: Yes
cts_target_max_capacitance	
	The target maximum capacitive load to allow during clock tree synthesis. This attribute specifies a maximum (combined pin and wire) capacitance that the clock tree synthesis algorithm will allow any given base_pin to drive in a given clock tree when driving a given net_type. It is specified in library units. It currently only constrains the primary delay corner capacitance values - other delay corners can be specified but will not be constrained. This attribute is applied in addition to the max_capacitance constraints read from the liberty library data - the tightest (lowest) of the constraint specified by this attribute and the constraint present in the liberty data will be used. It also does not apply at the root pins of clock trees - to constrain those nets the cts_clock_tree_source_max_capacitance CCOpt attribute should be used instead. Valid values: auto double Type: string Allowed -index values: delay_corner clock_tree Default: top auto trunk auto leaf auto Edit: Yes

Stylus Common UI Database Object Information

Database Objects--base_pin

cts_target_max_capacitance_leaf

The target maximum capacitive load to allow during clock tree synthesis. This attribute specifies a maximum (combined pin and wire) capacitance that the clock tree synthesis algorithm will allow any given base_pin to drive in a given clock tree when driving a given net_type. It is specified in library units. It currently only constrains the primary delay corner capacitance values - other delay corners can be specified but will not be constrained. This attribute is applied in addition to the max_capacitance constraints read from the liberty library data - the tightest (lowest) of the constraint specified by this attribute and the constraint present in the liberty data will be used. It also does not apply at the root pins of clock trees - to constrain those nets the cts_clock_tree_source_max_capacitance CCOpt attribute should be used instead. Valid values: auto | double

Type: string

Allowed -index values: delay_corner clock_tree

Default: auto

Edit: Yes

cts_target_max_capacitance_top

The target maximum capacitive load to allow during clock tree synthesis. This attribute specifies a maximum (combined pin and wire) capacitance that the clock tree synthesis algorithm will allow any given base_pin to drive in a given clock tree when driving a given net_type. It is specified in library units. It currently only constrains the primary delay corner capacitance values - other delay corners can be specified but will not be constrained. This attribute is applied in addition to the max_capacitance constraints read from the liberty library data - the tightest (lowest) of the constraint specified by this attribute and the constraint present in the liberty data will be used. It also does not apply at the root pins of clock trees - to constrain those nets the cts_clock_tree_source_max_capacitance CCOpt attribute should be used instead. Valid values: auto | double

Type: string

Allowed -index values: delay_corner clock_tree

Default: auto

Edit: Yes

cts_target_max_capacitance_trunk

The target maximum capacitive load to allow during clock tree synthesis. This attribute specifies a maximum (combined pin and wire) capacitance that the clock tree synthesis algorithm will allow any given base_pin to drive in a given clock tree when driving a given net_type. It is specified in library units. It currently only constrains the primary delay corner capacitance values - other delay corners can be specified but will not be constrained. This attribute is applied in addition to the max_capacitance constraints read from the liberty library data - the tightest (lowest) of the constraint specified by this attribute and the constraint present in the liberty data will be used. It also does not apply at the root pins of clock trees - to constrain those nets the cts_clock_tree_source_max_capacitance CCOpt attribute should be used instead. Valid values: auto | double

Type: string

Allowed -index values: delay_corner clock_tree

Default: auto

Edit: Yes

direction

Base_pins's direction from .lib if available, otherwise from LEF/OA. It can be in, out, inout or internal. Internal means it is an internal pin from a .lib file for the timing model, and is not part of the netlist. If there is no .lib for this cell, then the direction comes from the LEF PIN DIRECTION or equivalent OA oaTermType. The LEF DIRECTION values (and equivalent oaTermType values) are mapped this way: INPUT = in, OUTPUT = out, OUTPUT TRISTATE = out, INOUT = inout, FEEDTHRU = inout, and the oaTermType unknown = inout.

Type: enum

Enum Values: in out inout internal

Default: ""

Edit: No

Stylus Common UI Database Object Information

Database Objects--base_pin

effective_stack_via_rule	
	The stack_via_rule that is expected, but not required, to be used by the router for connecting to the instance pins instantiated from this base_pin. The actual stack via rule used (if any) may be effected by other pin and base_pin attributes, or by choices made by the software (optimization, clock tree synthesis, the router etc.) Type: obj(stack_via_rule) Default: "" Edit: No
is_always_on	
	The always on pin of an always on cell. Type: bool Default: false Edit: No
is_analog	
	Specifies the pin is an analog signal. This attribute can be set by liberty files. Type: bool Default: false Edit: No
is_iq_function	
	Output is an IQ pin. Type: bool Default: "" Edit: No
is_iqn_function	
	Output is an IQN pin. Type: bool Default: "" Edit: No
is_isolated	
	Specifies the pin is isolated internally in the cell. It is used for cells where some pins are internally isolated and some are not. This attribute can be set by liberty files. Type: bool Default: false Edit: No
is_isolation_cell_enable	
	Identifies the pin is an isolation enable pin. This pin is used to control when to clamp the output and isolate it from the input. This attribute can be set by liberty files, or by CPF commands. Type: bool Default: false Edit: No
is_level_shifter_enable	
	Identifies the pin is a level shifter enable pin. This pin is used to control when to clamp the output and isolate it from the input. This attribute can be set by liberty files, or by CPF commands. Type: bool Default: false Edit: No
is_must_join_all_ports	
	Indicates that the port has the LEF MUSTJOINALLPORTS attribute set. Type: bool Default: "" Edit: No
is_power_switch_enable	

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Database Objects--base_pin

	Identifies the pin is a power switch enable pin. This pin is used to control when to turn on/off the power switch. This attribute can be set by liberty files, or by CPF commands. Type: <code>bool</code> Default: <code>false</code> Edit: No
is_retention_cell_enable	
	Identifies the pin as a retention cell enable pin. This pin is used to control when to retain the state and ignore other inputs. This attribute can be set by liberty files, or by CPF commands. Type: <code>bool</code> Default: <code>false</code> Edit: No
is_unconnected	
	Specifies the pin is floating internally. This is used for cells where some of the inputs or outputs are unused by the cell. This attribute can be set by liberty files. Type: <code>bool</code> Default: <code>false</code> Edit: No
is_via_in_pin_only	
	Indicates that the pin has a LEF VIAINPINONLY property. It means that vias must be dropped inside the original pin shapes to connect to the pin. In some advanced nodes, the pin shapes can be extended for metal alignment purposes. However, via insertion is not allowed in that extended portion if this attribute is true. Type: <code>bool</code> Default: <code>""</code> Edit: No
layer	
	Layer of the base_pin. For base_pins with more than one shape, it is the layer of the first shape (which is the same shape used for the .location value). Type: <code>obj(layer)</code> Default: <code>""</code> Edit: No
must_join_pins	
	The must join base_pin Type: <code>obj(base_pin)</code> Default: <code>""</code> Edit: No
name	
	Terminal name Type: <code>string</code> Default: <code>""</code> Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: <code>enum (base_pin)</code> Default: <code>""</code> Edit: No
pg_type	
	Specifies the type of the power or ground pin from liberty data. Signal pins and PG pins with no liberty entry will have "invalid". Type: <code>enum</code> Enum Values: <code>primary_power primary_ground backup_power backup_ground internal_power internal_ground pwell nwell deepnpwell</code> Default: <code>invalid</code> Edit: No
physical_direction	

Stylus Common UI Database Object Information

Database Objects--base_pin

	Direction of this pin. Type: <code>enum</code> Enum Values: input output inout unknown feedthrough tristate Default: "" Edit: No
physical_pins	
	Physical pins for the base_pin. One logical base_pin can have multiple physical_pins. Each physical_pin had a list of strongly connected shapes (equivalent to LEF or DEF PORT or OA pin). The internal connection between physical_pins is weak (e.g. through poly or other high resistance paths) Type: <code>obj(physical_pin)*</code> Default: "" Edit: No
pin_edge	
	If this base_pin is for a partition, or inst that is a black-box, and the base_pin is assigned, the edge value indicates along which edge of the boundary polygon the base_pin is assigned. The edge number starts from the lowest Y, then left-most X vertex, starting with 0, and then counting clock-wise. See the set_pin_constraint command document for a figure showing the edge numbering. If the base_pin is not assigned, or not for a partition, or black-box inst, the value of -1 is returned. Type: <code>int</code> Default: "" Edit: No
related_ground_pin	
	Specifies which ground pin drives this signal pin. It must be one of the ground pins defined for this cell. It can be set by CPF, Liberty, or LEF/OA, with CPF having highest precedence, then Liberty, then LEF/OA. It is often only set when there is more than one ground pin for the cell. Type: <code>obj(base_pin)</code> Default: "" Edit: No
related_power_pin	
	Specifies which power pin drives this signal pin. It must be one of the power pins defined for this cell. It can be set by CPF, Liberty, or LEF/OA, with CPF having highest precedence, then Liberty, then LEF/OA. It is often only set when there is more than one power pin for the cell. Type: <code>obj(base_pin)</code> Default: "" Edit: No
stack_via_required	
	Sets whether stack vias are required for this term. Type: <code>bool</code> Default: false Edit: Yes
stack_via_rule_list	
	List of stack_via_rule that are valid choices for connecting to the instance pins instantiated from this base_pin. If the list is empty, no stack via is allowed. Type: <code>obj(stack_via_rule)*</code> Default: "" Edit: Yes
stack_via_rule_required	
	Specifies whether a stack via is required when connecting to the instance pins instantiated from this base_pin. If set to 0, a stack via is optional. If set to 1, the min one of the stack_via_rule from stack_via_rules must be used to generate a stack via even if a design rule violation occurs. If set to 2, the min one the stack_via_rule from stack_via_rules is used to generate a stack via if a design rule violation does not occur. If set to 3, tool enforces multiple tries from large to min one of the stack_via_rule from stack_via_rules to generate a stack via even if a design rule violation occurs. If set to 4, tool performs multiple tries from large to min one the stack_via_rule from stack_via_rules to generate a stack via if a design rule violation does not occur. Type: <code>enum</code> Enum Values: 0 1 2 3 4 Default: "" Edit: Yes
taper_rule	

Stylus Common UI Database Object Information

Database Objects--boundary

	The taper route_rule for the pin. By default, if tapered routing is needed to access the pin, the default route_rule will be used unless this attribute is set (see LEF MACRO PIN TAPERRULE). Type: obj(route_rule) Default: "" Edit: No
tech	
	The tech for this base_pin. Type: obj(tech) Default: "" Edit: No
tied_to	
	Specifies the PG pin name or 'empty' which the PG pin tied to. Type: string Default: "" Edit: No
use	
	Indicates how this pin is used from the LEF USE value or OA equivalent. The legal values are: signal analog power ground clock. Note that timing analysis does not use these values, it uses the .lib data instead (e.g. is_clock, is_analog, etc.). So the signal, clock or analog values are not normally used by applications. The power/ground values are used by many applications. Type: enum Enum Values: signal analog power ground clock Default: "" Edit: No

boundary

Parent Objects

[hinst](#), [design](#), [root](#)

Definition

A boundary object can only be attached to an hinst, normally when the hinst is a partition. It is used to constrain the placement of all the insts of the hinst. See the group object for placement constraint information attached to the group object (e.g. a list of insts, hinsts, or groups).

Attribute	Description
area	
	Sum of the area of the .rects that define the boundary Type: area Default: "" Edit: No
bbox	
	Bounding box of the boundary rects. Type: rect Default: "" Edit: No
hinst	
	The parent hinst. This boundary affects all the insts inside the parent hinst. Type: obj(hinst) Default: "" Edit: Yes
is_floating	

Stylus Common UI Database Object Information

Database Objects--bump

	Only affects boundary with .type = fence or region. If true, the global placer can move the fence or region. The .bbox and .rects value must also be set. The global placer will not change the size of the rect, but may move it. Type: bool Default: false Edit: Yes
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (boundary) Default: "" Edit: No
rects	
	List of non-overlapping rectangles that defines the shape of the boundary Type: rect* Default: "" Edit: No
type	
	fence: all the insts are inside the boundary, and only these insts can be inside. region: all the insts are inside, but other insts can also be inside. guide: lower cost for insts to be inside boundary, but not required. cluster: keep insts near each other (the .rects field is empty). none: no affect on placement. Type: enum Enum Values: guide region fence cluster none Default: "" Edit: Yes

bump

Parent Objects

[marker](#), [bump_pin](#), [design](#), [io_constraint](#), [root](#), [port](#)

Definition

Bump instance

Attribute	Description
base_cell	
	The base_cell master of the bump. Type: obj (base_cell) Default: "" Edit: No
base_name	
	The base_name of the bump Type: string Default: "" Edit: No
bbox	
	Bounding box of the bump shape for a package connection, which is the widest shape found on the top routing layer in the base_cell. If there is more than one wide shape of the same size, the bbox of all the wide shapes is returned. Type: rect Default: "" Edit: No
bump_connect_target	

Stylus Common UI Database Object Information

Database Objects--bump

	The string property added to the bump by create_bump_connect_target_constraint. The value takes the format like inst[:pin [:port]]. Type: string Default: "" Edit: No
bump_pins	
	The pins of the bump. A bump with ALLPINSCONNECTED LEF syntax can have multiple bump_pins which are internally connected. Type: obj(bump_pin)* Default: "" Edit: No
center	
	The center of the bump shape for a package connection, which is the center of the .bbox attribute. This is not the same as the .location of the bump which is normally at the lower-left corner of the cell. Type: point Default: "" Edit: No
escaped_name	
	The escaped name of the bump Type: string Default: "" Edit: No
foreign_excluded_center	
	The center of the bump. This is not the same as bump shape center which is the center point of its bounding box. This attribute is calculated from .point which is the lower left point of the bump by considering the FOREIGN offsets. Type: point Default: "" Edit: No
is_fixed_assignment	
	Indicates that the bump is connected to one port Type: bool Default: "" Edit: No
location	
	Location of the bump. Type: point Default: "" Edit: Yes
name	
	Name of the bump Type: string Default: "" Edit: No
net	
	The net connected to the bump. Type: obj(net) Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (bump) Default: "" Edit: No
orient	

Stylus Common UI Database Object Information

Database Objects--bump_grid

	Bump orientation. Type: enum Enum Values: r0 r90 r180 r270 mx mx90 my my90 Default: "" Edit: Yes
place_status	
	Placement status of bump. Type: enum Enum Values: unplaced placed fixed cover soft_fixed Default: "" Edit: Yes
ports	
	The ports the bump is assigned to. Type: obj(port)* Default: "" Edit: No

bump_grid

Parent Objects

[design](#), [root](#)

Definition

The grid to guide bump placement.

Attribute	Description
layers	
	List of layers. User assign relative bump pin layer to bump grid. Type: obj(layer)* Default: "" Edit: No
name	
	Bump grid name. Type: string Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (bump_grid) Default: "" Edit: No
origin	
	The coordinates of the first bump grid. Type: point Default: "" Edit: No
pitch_x	
	The spacing from the center of the bump to the center of the neighbor bump on the x-axis. Type: coord Default: "" Edit: No

Stylus Common UI Database Object Information

Database Objects--bump_pin

pitch_y	
	The spacing from the center of the bump to the center of the neighbor bump on the y-axis. Type: coord Default: "" Edit: No

bump_pin

Parent Objects

[bump](#), [port](#)

Definition

PIN in bump LEF

Attribute	Description
base_name	
	The bump_pin name without bump path. Type: string Default: "" Edit: No
base_pin	
	The equivalent base_pin. Type: obj(base_pin) Default: "" Edit: No
bump	
	The bump that the bump_pin belongs to. Type: obj(bump) Default: "" Edit: No
direction	
	bump_pin's direction from the corresponding base_pin. Type: enum Enum Values: in out inout Default: "" Edit: No
escaped_name	
	The bump_pin name including bump path. Type: string Default: "" Edit: No
layer	
	The layer of the bump_pin. For bump_pin with more than one shape, it is the layer of the first shape (which is the same shape used for the .location value). Type: obj(layer) Default: "" Edit: No
location	

Stylus Common UI Database Object Information

Database Objects--bus

	The location of the bump_pin. For bump_pin with more than one shape, it is the location of the first shape (which is the same shape used for the .layer value). Type: point Default: "" Edit: No
name	
	The bump_pin name including bump path. Type: string Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (bump_pin) Default: "" Edit: No
port	
	The port that the bump_pin connects to. Type: obj(port) Default: "" Edit: No

bus

Parent Objects

[design](#), [net](#), [root](#)

Definition

The Verilog bus definitions in the top Verilog module.

Attribute	Description
bits	
	The nets of the bus. All the bits of one bus will have the same obj_type. Type: obj(net)* Default: "" Edit: No
lsb	
	Bottom bit index of bus. Type: int Default: "" Edit: No
msb	
	Top bit index of bus. Type: int Default: "" Edit: No
name	
	Non-vector portion of the bus name (eg. A for bus A[7:0]) Type: string Default: "" Edit: No

obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (bus) Default: "" Edit: No

bus_guide

Parent Objects

[design](#)

Definition

Bus guide

Attribute	Description
area	
	Area of the bus_guide as defined by the LEF MACRO SIZE or OVERLAP information Type: area Default: "" Edit: No
bottom_layer	
	The bottom layer of allowed layer range Type: obj(layer) Default: "" Edit: Yes
net_group	
	The net group with list of nets to be routed within the bus_guide Type: obj(net_group)* Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (bus_guide) Default: "" Edit: No
rect	
	Rectangel that defines the bus_guide shape. Type: rect Default: "" Edit: No
top_layer	
	The top layer of allowed layer range Type: obj(layer) Default: "" Edit: Yes
type	

	The type of bus_guide. hard: Specifies the bus_guide as a hard constraint for routing. eGR, NR and NRHF should obey the path of bus guide. soft: Specifies the bus_guide as a soft constraint for routing. eGR, NR and NRHF should be guided by the route path. But tool can route the net out of the bus guide. Type: enum Enum Values: soft hard Default: "" Edit: Yes
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bus_sink_group

Parent Objects

[design](#), [root](#)

Definition

A group of sinks (loads) that some floorplan and routing commands use to control adding buffers and routing for a bus. See 'help *bus_sink*' for a list of commands related to this object.

Attribute	Description
name	
	The name of the bus_sink_group. Type: string Default: {} Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (bus_sink_group) Default: "" Edit: No
sinks	
	A list of sinks (pins of insts, or ports of the design) for this bus. Type: obj(pin)* obj(port)* Default: {} Edit: No

clock

Parent Objects

[context_data](#), [pin](#), [timing_path](#), [root](#), [port](#)

Definition

cte clock

Attribute	Description
base_name	
	The base_name of the clock. Type: string Default: "" Edit: No
child_clocks	

Stylus Common UI Database Object Information

Database Objects--bus_sink_group

	Returns a list of clock object which are the immediate generated clocks derived from this master clock. Type: <code>obj(clock)*</code> Default: "" Edit: No
clock_hold_uncertainty	
	clock_hold_uncertainty Type: <code>double</code> Default: "" Edit: No
clock_network_pins	
	Returns a list of pin and hpin objects in the extended fanout of the clock source. The extended fanout includes the fanout of generated clocks which the current clock is the master for. Type: <code>obj(port)* obj(pin)*</code> Default: "" Edit: No
clock_setup_uncertainty	
	clock_setup_uncertainty Type: <code>double</code> Default: "" Edit: No
comment	
	Returns the -comment option of a clock. Type: <code>string</code> Default: "" Edit: No
delay_max_fall	
	Returns the maximum falling delay value for the clock. Type: <code>delay</code> Default: "" Edit: No
delay_max_rise	
	Returns the maximum rising delay value for the clock. Type: <code>delay</code> Default: "" Edit: No
delay_min_fall	
	Returns the minimum falling delay value for the clock. Type: <code>delay</code> Default: "" Edit: No
delay_min_rise	
	Returns the minimum rising delay value for the clock. Type: <code>delay</code> Default: "" Edit: No
divide_by	
	Returns the create_generated_clock -divide_by option of a generated clock. Type: <code>int</code> Default: "" Edit: No
driving_network_pins	

Stylus Common UI Database Object Information

Database Objects--bus_sink_group

	Returns a list of pin object which are present between target pins of generated clock and master clock. Type: <code>obj(port)* obj(pin)*</code> Default: "" Edit: No
duty_cycle	
	Returns the create_generated_clock -duty_cycle option of a generated clock. Type: <code>double</code> Default: "" Edit: No
edge_shift	
	Returns the create_generated_clock -edge_shift option of a generated clock. Type: <code>string</code> Default: "" Edit: No
edges	
	Returns the create_generated_clock -edges option of a generated clock. Type: <code>string</code> Default: "" Edit: No
exceptions	
	Reports the path exceptions Type: <code>string</code> Allowed -index values: analysis_view clock Default: "" Edit: No
generated_clocks	
	Returns a list of clock object which are the generated clocks derived from this master clock. Type: <code>obj(clock)*</code> Default: "" Edit: No
hierarchical_name	
	hierarchical_name Type: <code>string</code> Default: "" Edit: No
ideal_transition_max_fall	
	Returns the set_clock_transition -max -fall asserted on the clock and used during ideal mode analysis. Type: <code>delay</code> Default: "" Edit: No
ideal_transition_max_rise	
	Returns the set_clock_transition -max -rise asserted on the clock and used during ideal mode analysis. Type: <code>delay</code> Default: "" Edit: No
ideal_transition_min_fall	
	Returns the set_clock_transition -min -fall asserted on the clock and used during ideal mode analysis. Type: <code>delay</code> Default: "" Edit: No
ideal_transition_min_rise	

Stylus Common UI Database Object Information

Database Objects--bus_sink_group

	Returns the set_clock_transition -min -rise asserted on the clock and used during ideal mode analysis. Type: delay Default: "" Edit: No
is_active	
	Returns true if the clock has been set active by the set_active_clocks constraint. Type: bool Default: "" Edit: No
is_combinational_source_path	
	Returns true if the create_generated_clock -combinational option was specified for the creation of this generated clock. Type: bool Default: "" Edit: No
is_context_mapped	
	Returns true if the clock is mapped in the block level SDC in the timing context reading session Type: bool Default: "" Edit: No
is_generated	
	Returns true if this clock object represents a generated clock Type: bool Default: "" Edit: No
is_inverted	
	Returns true if the create_generated_clock -invert option was specified for the creation of this generated clock. Type: bool Default: "" Edit: No
is_library_created	
	Returns true if this generated_clock was specified in a Liberty generated_clock group and not created by a create_generated_clock constraint Type: bool Default: "" Edit: No
is_propagated	
	Returns true if set_propagated_clock was asserted explicitly on this clock object. Type: bool Default: "" Edit: No
is_source_inverted	
	Returns true if the create_generated_clock -source_invert option was specified for the creation of this generated clock. Type: bool Default: "" Edit: No
is_virtual	
	Returns true if this clock is virtual Type: bool Default: "" Edit: No
master_clock	

Stylus Common UI Database Object Information

Database Objects--bus_sink_group

	Returns the master clock of this generated clock as specified by the create_generated_clock -master_clock option when this generated clock was created. Type: <code>obj(clock)*</code> Default: "" Edit: No
master_source	
	Returns the master clock source pin of this generated clock as specified by the create_generated_clock -source option when this generated clock was created. Type: <code>obj(pin)* obj(hpin)* obj(hport)*</code> Default: "" Edit: No
max_capacitance_clock_path_fall	
	Returns the value asserted by set_max_capacitance -clock_path -fall on this clock. Type: <code>delay</code> Default: "" Edit: No
max_capacitance_clock_path_rise	
	Returns the value asserted by set_max_capacitance -clock_path -rise on this clock. Type: <code>delay</code> Default: "" Edit: No
max_capacitance_data_path_fall	
	Returns the value asserted by set_max_capacitance -data_path -fall on this clock. Type: <code>delay</code> Default: "" Edit: No
max_capacitance_data_path_rise	
	Returns the value asserted by set_max_capacitance -data_path -rise on this clock. Type: <code>delay</code> Default: "" Edit: No
max_transition_clock_path_fall	
	Returns the value asserted by set_max_transition -clock_path -fall on this clock. Type: <code>delay</code> Default: "" Edit: No
max_transition_clock_path_rise	
	Returns the value asserted by set_max_transition -clock_path -rise on this clock. Type: <code>delay</code> Default: "" Edit: No
max_transition_data_path_fall	
	Returns the value asserted by set_max_transition -data_path -fall on this clock. Type: <code>delay</code> Default: "" Edit: No
max_transition_data_path_rise	
	Returns the value asserted by set_max_transition -data_path -rise on this clock. Type: <code>delay</code> Default: "" Edit: No
min_pulse_width_high	

Stylus Common UI Database Object Information

Database Objects--bus_sink_group

	Returns the minimum pulse width high requirement set by set_min_pulse_width . Type: double Default: "" Edit: No
min_pulse_width_low	
	Returns the minimum pulse width low requirement set by set_min_pulse_width. Type: double Default: "" Edit: No
multiply_by	
	Returns the create_generated_clock -multiply_by option of a generated clock. Type: int Default: "" Edit: No
name	
	The name of the clock. Type: string Default: "" Edit: No
network_latency_fall_max	
	Returns the maximum fall insertion delay specified by an explicit set_clock_latency on the clock. Type: delay Default: "" Edit: No
network_latency_fall_min	
	Returns the minimum fall insertion delay specified by an explicit set_clock_latency on the clock. Type: delay Default: "" Edit: No
network_latency_rise_max	
	Returns the maximum rise insertion delay specified by an explicit set_clock_latency on the clock. Type: delay Default: "" Edit: No
network_latency_rise_min	
	Returns the minimum rise insertion delay specified by an explicit set_clock_latency on the clock. Type: delay Default: "" Edit: No
num_exceptions	
	Reports the number of path exceptions Type: int Allowed -index values: analysis_view clock Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (clock) Default: "" Edit: No
period	

Stylus Common UI Database Object Information

Database Objects--bus_sink_group

	Returns the period of the clock. The period is either specified directly with create_clock or derived from a generated clock. Type: delay Default: "" Edit: No
source_jitter_early_fall_max	
	Returns the amount of the maximum early fall source insertion delay that is due to cycle-to-cycle variation. This is specified by using the set_clock_latency -jitter option. Type: double Default: "" Edit: No
source_jitter_early_fall_min	
	Returns the amount of the minimum early fall source insertion delay that is due to cycle-to-cycle variation. This is specified by using the set_clock_latency -jitter option. Type: double Default: "" Edit: No
source_jitter_early_rise_max	
	Returns the amount of the maximum early rise source insertion delay that is due to cycle-to-cycle variation. This is specified by using the set_clock_latency -jitter option. Type: double Default: "" Edit: No
source_jitter_early_rise_min	
	Returns the amount of the minimum early rise source insertion delay that is due to cycle-to-cycle variation. This is specified by using the set_clock_latency -jitter option. Type: double Default: "" Edit: No
source_jitter_late_fall_max	
	Returns the amount of the maximum late fall source insertion delay that is due to cycle-to-cycle variation. This is specified by using the set_clock_latency -jitter option. Type: double Default: "" Edit: No
source_jitter_late_fall_min	
	Returns the amount of the minimum late fall source insertion delay that is due to cycle-to-cycle variation. This is specified by using the set_clock_latency -jitter option. Type: double Default: "" Edit: No
source_jitter_late_rise_max	
	Returns the amount of the maximum late rise source insertion delay that is due to cycle-to-cycle variation. This is specified by using the set_clock_latency -jitter option. Type: double Default: "" Edit: No
source_jitter_late_rise_min	
	Returns the amount of the minimum late rise source insertion delay that is due to cycle-to-cycle variation. This is specified by using the set_clock_latency -jitter option. Type: double Default: "" Edit: No
source_latency_early_fall_max	

Stylus Common UI Database Object Information

Database Objects--bus_sink_group

	Returns the maximum early fall source insertion delay specified by an explicit set_clock_latency on the clock. Type: delay Default: "" Edit: No
source_latency_early_fall_min	
	Returns the minimum early fall source insertion delay specified by an explicit set_clock_latency on the clock. Type: delay Default: "" Edit: No
source_latency_early_rise_max	
	Returns the maximum early rise source insertion delay specified by an explicit set_clock_latency on the clock. Type: delay Default: "" Edit: No
source_latency_early_rise_min	
	Returns the minimum early rise source insertion delay specified by an explicit set_clock_latency on the clock. Type: delay Default: "" Edit: No
source_latency_late_fall_max	
	Returns the maximum late fall source insertion delay specified by an explicit set_clock_latency on the clock. Type: delay Default: "" Edit: No
source_latency_late_fall_min	
	Returns the minimum late fall source insertion delay specified by an explicit set_clock_latency on the clock. Type: delay Default: "" Edit: No
source_latency_late_rise_max	
	Returns the maximum late rise source insertion delay specified by an explicit set_clock_latency on the clock. Type: delay Default: "" Edit: No
source_latency_late_rise_min	
	Returns the minimum late rise source insertion delay specified by an explicit set_clock_latency on the clock. Type: delay Default: "" Edit: No
source_latency_pins	
	Returns a list of pin/port object(s) which are present between target pins of generated clock and master clock. Type: obj(port)* obj(pin)* Default: "" Edit: No
sources	
	Returns the pin or port objects where the clock is attached to the design - as specified by create_clock or create_generated_clock Type: obj(port)* obj(pin)* obj(hport)* Default: "" Edit: No
view_name	

Stylus Common UI Database Object Information

Database Objects--clock_spine

	The name of the analysis view for the clock. Type: string Default: "" Edit: No
waveform	
	Returns a list of the sequence of rising and falling edge times of a single period of the clock. The first list entry is always the first rising edge time, and the second entry is always the subsequent falling edge time. Type: string Default: "" Edit: No

clock_spine

Parent Objects

Definition

clock tree spine

Attribute	Description
name	
	name of clock_spine Type: string Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (clock_spine) Default: "" Edit: No

clock_tree

Parent Objects

[clock_tree_source_group](#), [pin](#), [root](#), [port](#)

Definition

clock tree in ccopt

Attribute	Description
clock_tree_source_group	
	the clock_tree_source_group this clock_trees is a member of Type: obj (clock_tree_source_group)* Default: "" Edit: No
cts_buffer_cells	

Stylus Common UI Database Object Information

Database Objects--clock_spine

	Specifies the buffer cells for CTS. If none are specified CCOpt will choose buffers from the libraries. Cell names may be specified as a Tcl list of names, or as a Tcl list of patterns to be expanded to match names. If set explicitly, CCOpt will ignore any dont_use settings for the cells specified. Different buffer cells may be specified for any combination of clock tree and power domain. To use different buffers for each net type set the <code>cts_buffer_cells_top</code> and <code>cts_buffer_cells_leaf</code> attributes . Some examples follow: To specify buffer cells for all clock trees and all power domains: <code>set_db cts_buffer_cells {bufAX* bufBX*}</code> To specify buffer cells for a particular clock tree and all power domains: <code>set_db clock_tree:<clk> .cts_buffer_cells {bufX20 bufX18}</code> To specify buffer cells for a particular clock tree and power domain: <code>set_db clock_tree:<clk> .cts_buffer_cells -index {power_domain <pd>} {bufX12 bufX8}</code> Valid values: a list of library cell names, or a list of patterns to expand to library cell names Type: string Allowed -index values: <code>power_domain</code> Default: "" Edit: Yes
cts_buffer_cells_leaf	
	Specifies the buffer cells available for CTS to use on leaf nets. If none are specified CCOpt will use the same buffers as on trunk nets (as specified in the <code>cts_buffer_cells</code> attribute). Cell names may be specified as a Tcl list of names, or as a Tcl list of patterns to be expanded to match names. If set explicitly, CCOpt will ignore any dont_use settings for the cells specified. Different leaf buffer cells may be specified for any combination of clock tree and power domain. Valid values: a list of library cell names, or a list of patterns to expand to library cell names Type: string Allowed -index values: <code>power_domain</code> Default: "" Edit: Yes
cts_buffer_cells_top	
	Specifies the buffers cells available for CTS to use on top nets. If none are specified CCOpt will use the same buffers as on trunk nets (as specified in the <code>cts_buffer_cells</code> attribute). Cell names may be specified as a Tcl list of names, or as a Tcl list of patterns to be expanded to match names. If set explicitly, CCOpt will ignore any dont_use settings for the cells specified. Different top buffer cells may be specified for any combination of clock tree and power domain. Valid values: a list of library cell names, or a list of patterns to expand to library cell names Type: string Allowed -index values: <code>power_domain</code> Default: "" Edit: Yes
cts_cell_density	

Stylus Common UI Database Object Information
Database Objects--clock_spine

	Specifies the clock halo in the x direction. Any x direction clock halo defined by this attribute is proportional to the cell width. The constant of proportionality is defined by the attribute value. For example: . If <code>cts_cell_density</code> = 0.25 then the x direction clock halo equals 3 * cell width. . If <code>cts_cell_density</code> = 0.5 then the x direction clock halo equals cell width. . If <code>cts_cell_density</code> = 0.75 then the x direction clock halo equals cell width / 3. . If <code>cts_cell_density</code> = 1 then the x direction clock halo equals zero. This attribute can specify the x direction clock halo for all clock trees via: <code>set_db cts_cell_density 0.8</code> This attribute can specify the x direction clock halo per-clock tree via: <code>set_db clock_tree:<clk> .cts_cell_density 0.9</code> The following attributes can be used to assign x direction clock halos within CCOpt: <code>cts_cell_halo_x</code> <code>cts_cell_density</code> <code>cts_cell_halo_sites</code> Only one of these attributes is used to determine the clock halo in the x direction. The following rules determine which: . If <code>cts_cell_halo_x</code> is set to a non-auto value, then this defines the x direction clock halo. The attributes <code>cts_cell_density</code> and <code>cts_cell_halo_sites</code> have no effect. . If <code>cts_cell_halo_x</code> is set to auto and <code>cts_cell_density</code> is set to a non-auto value then <code>cts_cell_density</code> defines the clock halo in the x direction. The attribute <code>cts_cell_halo_sites</code> has no effect. . If both <code>cts_cell_halo_x</code> and <code>cts_cell_density</code> are set to auto then <code>cts_cell_halo_sites</code> defines the clock halo in the x direction. Valid values: 0.01 to 1 See also: . <code>cts_cell_halo_x</code> . <code>cts_cell_halo_sites</code> Type: string Default: 0.75 Edit: Yes
<code>cts_cell_halo_x</code>	
	Specifies the clock halo distance in the x direction. The default value of this attribute is auto. The following attributes can be used to assign x direction clock halos within CCOpt: <code>cts_cell_halo_x</code> <code>cts_cell_density</code> <code>cts_cell_halo_sites</code> Only one of these attributes is used to determine the clock halo in the x direction. The following rules determine which: . If <code>cts_cell_halo_x</code> is set to a non-auto value, then this defines the x direction clock halo. The attributes <code>cts_cell_density</code> and <code>cts_cell_halo_sites</code> have no effect. . If <code>cts_cell_halo_x</code> is set to auto and <code>cts_cell_density</code> is set to a non-auto value then <code>cts_cell_density</code> defines the clock halo in the x direction. The attribute <code>cts_cell_halo_sites</code> has no effect. . If both <code>cts_cell_halo_x</code> and <code>cts_cell_density</code> are set to auto then <code>cts_cell_halo_sites</code> defines the clock halo in the x direction. See also: . <code>cts_cell_density</code> . <code>cts_cell_halo_sites</code> Type: string Allowed -index values: <code>power_domain</code> Default: auto Edit: Yes
<code>cts_cell_halo_y</code>	

Stylus Common UI Database Object Information

Database Objects--clock_spine

	Specifies the clock halo distance in the y direction. The default value of this attribute is auto. The following attributes can be used to assign y direction clock halos within CCOpt: cts_cell_halo_y cts_adjacent_rows_legal cts_cell_halo_rows Only one of these attributes is used to determine the clock halo in the y direction. The following rules determine which: . If cts_cell_halo_y is set to a non-auto value, then this defines the y direction clock halo. The attributes cts_adjacent_rows_legal and cts_cell_halo_rows have no effect. . If cts_cell_halo_y is set to auto and cts_adjacent_rows_legal is set to a non-auto value then cts_adjacent_rows_legal defines the clock halo in the y direction. The attribute cts_cell_halo_rows has no effect. . If both cts_cell_halo_y and cts_adjacent_rows_legal are set to auto then cts_cell_halo_rows defines the clock halo in the y direction. See also: . cts_adjacent_rows_legal . cts_cell_halo_rows Type: string Allowed -index values: power_domain Default: auto Edit: Yes
cts_clock_gating_cells	
	Specifies the clock gates for CTS. If none are specified CCOpt will choose clock gates from the libraries. Cell names may be specified as a Tcl list of names, or as a Tcl list of patterns to be expanded to match names. If set explicitly, CCOpt will ignore any dont_use settings for the cells specified. A list of lists can be used to specify separate clock gate sub-lists. All CCOpt transforms will resize clock gates within the clock gate sub-list they are initially part of. If an incoming clock gate is not part of any clock gate sub-list, it will be resized to the first clock gate sub-list. Different clock gates may be specified for any combination of clock tree and power domain. Some examples follow: To specify clock gates for all clock trees and all power domains: <pre>set_db cts_clock_gating_cells {cgAX* cgBX*}</pre> To specify clock gates for a particular clock tree and all power domains: <pre>set_db clock_tree:<clk> .cts_clock_gating_cells {cgX20 cgX18}</pre> To specify clock gates for a particular clock tree and power domain: <pre>set_db clock_tree:<clk> .cts_clock_gating_cells -index {power_domain <pd>} {cgX12 cgX8}</pre> To specify separate clock gate sub-lists: <pre>set_db .cts_clock_gating_cells {{cgX12A1 cgX8A1} {cgX12A2 cgX8A2}}</pre> Valid values: a list of library cell names, or a list of patterns to expand to library cell names, a list of lists to specify separate clock gate sub-lists Type: string Allowed -index values: power_domain Default: "" Edit: Yes
cts_clock_source_cells	
	Specifies the cells available for CTS to size clock sources if the cts_size_clock_sources attribute is set to true. If none are specified the tool will choose cells from the libraries. Cell names may be specified as a Tcl list of names, or as a Tcl list of patterns to be expanded to match names. If set explicitly, the tool will ignore any dont_use settings for the cells specified. Different cells may be specified for clock trees or power domains. Only clock sources that are buffers, inverters, logic and clock gating cells with a single output can be resized. Valid values: a list of library cell names, or a list of patterns to expand to library cell names Type: string Allowed -index values: power_domain Default: "" Edit: Yes
cts_clock_tree_parents	
	The list of parent clock trees from which this clock tree is generated, if any. Valid values: list cts_clock_tree Type: string Default: "" Edit: No
cts_clock_tree_source_driver	

Stylus Common UI Database Object Information
Database Objects--clock_spine

	Specifies the base_pin which is assumed to drive this clock tree. It is either a single base_pin (in which case all arcs to that base_pin shall be used when timing the clock tree root) or a pair of base_pins (in which case only arcs from the first specified pin to the second will be considered). By default this is generated from clock tree extraction Valid values: base_pin {base_pin base_pin} Type: string Default: "" Edit: Yes
cts_clock_tree_source_group	
	Specifies the clock tree source group to which this clock tree belongs. Valid values: list cts_clock_tree_source_group Type: string Default: "" Edit: No
cts_clock_tree_source_group_clock_trees	
	A list of the clock trees relevant to this source group. Valid values: list cts_clock_trees Type: string Default: "" Edit: Yes
cts_clock_tree_source_input_max_transition_time	
	The slew which will be assumed at the input of the root driver. Valid values: double Type: double Allowed -index values: delay_corner Default: 0 Edit: Yes
cts_clock_tree_source_max_capacitance	
	The maximum capacitive load which this clock tree is permitted to drive. Valid values: double auto Auto: from clock tree extraction Type: string Default: auto Edit: Yes
cts_clock_tree_source_output_max_transition_time	
	If non-zero, the slew which will be assumed at the output of the root driver. This overrides the value from SDC. Valid values: double Type: double Allowed -index values: delay_corner Default: 0 Edit: Yes
cts_delay_cells	

Stylus Common UI Database Object Information

Database Objects--clock_spine

	Specifies the delay cells available for CTS. If none are specified CCOpt will not use delay cells. Setting this attribute to the string 'auto' means that CCOpt will choose delay cells from the libraries to use. Cell names may be specified as a Tcl list of names, or as a Tcl list of patterns to be expanded to match names. If set explicitly, CCOpt will ignore any dont_use settings for the cells specified. Different delay cells may be specified for any combination of clock tree and power domain, or by omitting those arguments a global setting can be applied. Some examples follow: To specify delay cells for all clock trees and power domains: <code>set_db cts_delay_cells {delayAX* delayBX*}</code> To specify delay cells for a particular clock tree and all power domains: <code>set_db clock_tree:<clk> .cts_delay_cells {delayX1 delayX2}</code> To specify delay cells for a particular clock tree and power domain: <code>set_db clock_tree:<clk> .cts_delay_cells -index {power_domain <pd>} {delayX2 delayX3}</code> Valid values: a list of library cell names, or a list of patterns to expand to library cell names, or the string 'auto' Type: string Allowed -index values: power_domain Default: "" Edit: Yes
cts_inverter_cells	
	Specifies the inverter cells available for CTS. If none are specified CCOpt will choose inverters from the libraries. Cell names may be specified as a Tcl list of names, or as a Tcl list of patterns to be expanded to match names. If set explicitly, CCOpt will ignore any dont_use settings for the cells specified. Different inverter cells may be specified for any combination of clock tree and power domain. To use different inverters for each net type set the cts_inverter_cells_top and cts_inverter_cells_leaf attributes. Some examples follow: To specify inverter cells for all clock trees and all power domains: <code>set_db cts_inverter_cells {invAX* invBX*}</code> To specify inverter cells for a particular clock tree and all power domains: <code>set_db clock_tree:<clk> .cts_inverter_cells {invX20 invX18}</code> To specify inverter cells for a particular clock tree and power domain: <code>set_db clock_tree:<clk> .cts_inverter_cells -index {power_domain <pd>} {invX12 invX8}</code> Valid values: a list of library cell names, or a list of patterns to expand to library cell names Type: string Allowed -index values: power_domain Default: "" Edit: Yes
cts_inverter_cells_leaf	
	Specifies the inverter cells available for CTS to use on leaf nets. If none are specified CCOpt will use the same inverters as on trunk nets (as specified in the cts_inverter_cells attribute). Cell names may be specified as a Tcl list of names, or as a Tcl list of patterns to be expanded to match names. If set explicitly, CCOpt will ignore any dont_use settings for the cells specified. Different leaf inverter cells may be specified for any combination of clock tree and power domain. Valid values: a list of library cell names, or a list of patterns to expand to library cell names Type: string Allowed -index values: power_domain Default: "" Edit: Yes
cts_inverter_cells_top	
	Specifies the inverter cells available for CTS to use on top nets. If none are specified CCOpt will use the same inverters as on trunk nets (as specified in the cts_inverter_cells attribute). Cell names may be specified as a Tcl list of names, or as a Tcl list of patterns to be expanded to match names. If set explicitly, CCOpt will ignore any dont_use settings for the cells specified. Different top inverter cells may be specified for any combination of clock tree and power domain. Valid values: a list of library cell names, or a list of patterns to expand to library cell names Type: string Allowed -index values: power_domain Default: "" Edit: Yes
cts_load_capacitance_cells	

Stylus Common UI Database Object Information

Database Objects--clock_spine

	Specifies the load capacitance cells available for CTS. CTS will use cells from this collection for load capacitance optimizations. Cell names may be specified as a Tcl list of names, or as a Tcl list of patterns to be expanded to match names. Type: string Allowed -index values: power_domain Default: "" Edit: Yes
cts_logic_cells	
	Specifies the clock logics for CTS. If none are specified CCOpt will choose clock logics from the libraries. Cell names may be specified as a Tcl list of names, or as a Tcl list of patterns to be expanded to match names. If set explicitly, CCOpt will ignore any dont_use settings for the cells specified. Different logic cells may be specified for any combination of clock tree and power domain. Some examples follow: To specify logic cells for all clock trees and all power domains: set_db cts_logic_cells {and* mux*} To specify logic cells for a particular clock tree and all power domains: set_db clock_tree:<clk> .cts_logic_cells {andX20 andX18} To specify logic cells for a particular clock tree and power domain: set_db clock_tree:<clk> .cts_logic_cells -index {power_domain <pd>} {andX12 andX8} Valid values: a list of library cell names, or a list of patterns to expand to library cell names Type: string Allowed -index values: power_domain Default: "" Edit: Yes
cts_route_type_leaf	
	Specifies the route type. Setting this attribute binds an existing user-defined route_type to one or more types of clock tree nets. Binding a route_type to a type of clock tree nets means that all nets of that type (including the nets created by CTS) will be routed according to the specification of that route_type. In the most common usage, the route_type is bound to one of the three types of clock tree nets (top, trunk, or leaf) with the optional -net_type argument. Omitting the -net_type argument causes the route_type to be bound to all three types of clock tree nets. The optional -clock_tree <pattern> argument limits the binding to the clock trees whose name matches <pattern>. Omitting the -clock_tree argument causes the binding to apply to all clock trees. For a route_type to be used in CTS, it must be bound to at least one net type. If net type is not bound to any route_type, a default route_type will be created for that net type at the start of CTS. Valid values: names of route_types created with create_route_type Type: string Default: default Edit: Yes
cts_route_type_top	
	Specifies the route type. Setting this attribute binds an existing user-defined route_type to one or more types of clock tree nets. Binding a route_type to a type of clock tree nets means that all nets of that type (including the nets created by CTS) will be routed according to the specification of that route_type. In the most common usage, the route_type is bound to one of the three types of clock tree nets (top, trunk, or leaf) with the optional -net_type argument. Omitting the -net_type argument causes the route_type to be bound to all three types of clock tree nets. The optional -clock_tree <pattern> argument limits the binding to the clock trees whose name matches <pattern>. Omitting the -clock_tree argument causes the binding to apply to all clock trees. For a route_type to be used in CTS, it must be bound to at least one net type. If net type is not bound to any route_type, a default route_type will be created for that net type at the start of CTS. Valid values: names of route_types created with create_route_type Type: string Default: default Edit: Yes
cts_route_type_trunk	

Stylus Common UI Database Object Information

Database Objects--clock_spine

	Specifies the route type. Setting this attribute binds an existing user-defined route_type to one or more types of clock tree nets. Binding a route_type to a type of clock tree nets means that all nets of that type (including the nets created by CTS) will be routed according to the specification of that route_type. In the most common usage, the route_type is bound to one of the three types of clock tree nets (top, trunk, or leaf) with the optional -net_type argument. Omitting the -net_type argument causes the route_type to be bound to all three types of clock tree nets. The optional -clock_tree <pattern> argument limits the binding to the clock trees whose name matches <pattern>. Omitting the -clock_tree argument causes the binding to apply to all clock trees. For a route_type to be used in CTS, it must be bound to at least one net type. If net type is not bound to any route_type, a default route_type will be created for that net type at the start of CTS. Valid values: names of route_types created with create_route_type Type: string Default: default Edit: Yes
cts_source_latency	
	Specifies a delay value between the global clock source and this clock tree. This additional delay will be included in all timing analysis involving skew groups for which this clock tree is a source. The default is 0. Valid values: double Type: double Allowed -index values: delay_corner Default: 0 Edit: Yes
cts_target_max_transition_time	
	The target slew used for clock tree synthesis. This attribute specifies a maximum slew time that the clock tree synthesis algorithm will allow in this clock tree, in library units. 'default' means 'auto' in primary half corner and 'ignore' in other half corners. If set to 'auto', CTS picks an appropriate value based on the collection of allowed buffer sizes and library parameters, although this may not give optimal quality of results. If set to 'ignore', CTS does not constrain the corner. Valid values: default auto ignore double Type: string Allowed -index values: delay_corner power_domain Default: top default trunk default leaf default Edit: Yes
cts_target_max_transition_time_leaf	
	The target slew used for clock tree synthesis. This attribute specifies a maximum slew time that the clock tree synthesis algorithm will allow in this clock tree, in library units. 'default' means 'auto' in primary half corner and 'ignore' in other half corners. If set to 'auto', CTS picks an appropriate value based on the collection of allowed buffer sizes and library parameters, although this may not give optimal quality of results. If set to 'ignore', CTS does not constrain the corner. Valid values: default auto ignore double Type: string Allowed -index values: delay_corner power_domain Default: default Edit: Yes
cts_target_max_transition_time_sdc	
	If non-zero, the target slew used for clock tree synthesis, overriding the SDC. This attribute specifies a maximum slew time that the clock tree synthesis algorithm will allow, in library units obtained from SDC. Valid values: double Type: double Allowed -index values: delay_corner Default: 0 Edit: Yes
cts_target_max_transition_time_top	

Stylus Common UI Database Object Information

Database Objects--clock_spine

	The target slew used for clock tree synthesis. This attribute specifies a maximum slew time that the clock tree synthesis algorithm will allow in this clock tree, in library units. 'default' means 'auto' in primary half corner and 'ignore' in other half corners. If set to 'auto', CTS picks an appropriate value based on the collection of allowed buffer sizes and library parameters, although this may not give optimal quality of results. If set to 'ignore', CTS does not constrain the corner. Valid values: default auto ignore double Type: string Allowed -index values: delay_corner power_domain Default: default Edit: Yes
cts_target_max_transition_time_trunk	
	The target slew used for clock tree synthesis. This attribute specifies a maximum slew time that the clock tree synthesis algorithm will allow in this clock tree, in library units. 'default' means 'auto' in primary half corner and 'ignore' in other half corners. If set to 'auto', CTS picks an appropriate value based on the collection of allowed buffer sizes and library parameters, although this may not give optimal quality of results. If set to 'ignore', CTS does not constrain the corner. Valid values: default auto ignore double Type: string Allowed -index values: delay_corner power_domain Default: default Edit: Yes
cts_top_fanout_threshold	
	Minimum number of transitive fanout in the clock tree for a net to be routed as a top net. Nets with at least this many sinks in their transitive fanout in the clock tree will have the special routing rules applied to them. Valid values: integer Type: string Default: unset Edit: Yes
insts	
	instances in the clock tree Type: obj(inst)* Default: "" Edit: No
name	
	name of clock_tree Type: string Default: "" Edit: No
nets	
	nets in the clock tree Type: obj(net)* Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (clock_tree) Default: "" Edit: No
sinks	

	sinks (pins or ports) of the clock tree Type: obj(pin) * obj(port) * Default: "" Edit: No
source	
	The source pin for this cts_clock_tree. Valid values: pin Type: obj(pin) obj(port) Default: "" Edit: No

clock_tree_source_group

Parent Objects

[clock_tree](#), [root](#)

Definition

clock tree source group

Attribute	Description
clock_trees	
	clock_tree_source_group member clock_trees Type: obj(clock_tree) * Default: "" Edit: Yes
is_active	
	Determines whether the clock tree source group is active or not. Only active source groups will take part in sink allocation. Valid values: true false Type: bool Allowed -index values: clock_tree Default: true Edit: Yes
name	
	name of clock_tree_source_group Type: string Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (clock_tree_source_group) Default: "" Edit: No

constraint_mode

Parent Objects

[analysis_view](#), [design](#), [root](#)

Stylus Common UI Database Object Information

Database Objects--clock_tree_source_group

Definition

Associates a list of SDC constraint files with a specified constraint mode name, for multi-mode multi-corner analysis. This constraint mode name can be referred to later when creating analysis views. Use the `create_constraint_mode` and `update_constraint_mode` commands to create and modify constraint_modes.

Attribute	Description
ilm_sdc_files	
	Specifies an optional list of constraint files to use when using ILM mode Type: string Default: "" Edit: No
is_active	
	Indicates that the constraint_mode is associated with an analysis_view that is used in the active analysis_view. Type: bool Default: "" Edit: No
is_dynamic	
	Indicates that the constraint_mode is associated with an analysis_view that is used in the active dynamic analysis_view. Type: bool Default: "" Edit: No
is_hold	
	Indicates that the constraint_mode is associated with an analysis_view that is active for hold analysis. Type: bool Default: "" Edit: No
is_leakage	
	Indicates that the constraint_mode is associated with an analysis_view that is used in the active leakage analysis_view. Type: bool Default: "" Edit: No
is_setup	
	Indicates that the constraint_mode is associated with an analysis_view that is active for setup analysis. Type: bool Default: "" Edit: No
name	
	Provides the name of this constraint_mode object as specified by <code>create_constraint_mode</code> . Type: string Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (constraint_mode) Default: "" Edit: No
sdc_files	
	Specifies an optional list of constraint files to associate with the mode Type: string Default: "" Edit: No
tcl_vars	

Specifies an optional list of user-specified Tcl variable name and value pairs - e.g: {{my_var1 true} {my_var2 0.1}} - which are set before the constraint files are read, and can be used inside the constraint files.

Type: [string](#)

Default: ""

Edit: No

context_data

Parent Objects

Definition

context data objects

Attribute	Description
all_block_clocks	
	Reports all the mapped and unmapped block clocks of property context Type: obj(clock)* Default: "" Edit: No
all_top_clocks	
	Reports all the mapped and unmapped top clocks of property context Type: obj(context_top_clock)* Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (context_data) Default: "" Edit: No
top_instance	
	Reports the top view of proeprty context_data Type: string Allowed -index values: clock Default: "" Edit: No
top_view	
	Reports the top view of proeprty context_data Type: string Allowed -index values: clock Default: "" Edit: No
unmapped_block_clocks	
	Reports the unmapped block clocks of property context Type: obj(clock)* Default: "" Edit: No
unmapped_top_clocks	
	Reports the mapped top clocks of property context Type: obj(context_top_clock)* Default: "" Edit: No

context_top_clock

Parent Objects

[context_data](#)

Definition

context top clocks

Attribute	Description
hierarchical_name	
	Reports the name of top clock Type: string Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (context_top_clock) Default: "" Edit: No

def_scan_chain

Parent Objects

[design](#), [root](#)

Definition

A single scan-chain equivalent to one DEF SCANCHAIN. It is used by the placer to reorder scan-chain elements to reduce routing and congestion.

Attribute	Description
end	
	Specifies the last input_pin pin or an output port of the scan chain. Type: obj(pin) Default: "" Edit: No
floating_insts	
	Specifies a list of floating insts that can be re-ordered by scan-chain ordering commands. The format is a list of insts, where each inst is a sub-list of the scan-in pin, scan-out pin, and bit_length. Type: string Default: "" Edit: No
max_bits	
	Specify a value that is at least as large as the size of the current chain. When specified with chains that include a partition_name, sets the maximum bit length (flip-flop bit count) that the chain can grow to in the partition. Type: int Default: "" Edit: No
name	

Stylus Common UI Database Object Information

Database Objects--delay_corner

	The name of the def_scan_chain. Type: string Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (def_scan_chain) Default: "" Edit: No
ordered_insts	
	: Specifies one or more ordered lists of insts that should not be re-ordered by scan-chain ordering commands. The format is a list of <ordered lists> where each ordered list has an optional "name <string>", and then an "ordered <list>" where <list> has each inst as a sub-list of scan-in pin, scan-out pin, and bit length. Type: string Default: "" Edit: No
scan_partition	
	Specifies a partition name (same as DEF SCANCHAIN PARTITION <name>). Type: string Default: "" Edit: No
scan_pins	
	The list of all the scan-in and scan-out pins or ports in scan-order that make up this def_scan_chain. The first object can be a scan-in pin or external input port, and the last object can be a scan-out pin or external output port. All the intermediate objects must be scan-in/scan-out pins. Type: obj(pin) * obj(port) * Default: "" Edit: No
start	
	Specifies the first output_pin pin or an input port of the scan chain. Type: obj(pin) Default: "" Edit: No

delay_corner

Parent Objects

[analysis_view](#), [design](#), [root](#)

Definition

The delay_corner object provides references to rc_corner objects which define the interconnect corner and timing_condition-to-domain bindings which define device operating corners - providing the majority of the configuration information needed to drive delay calculation. An analysis_view object will reference the delay_corner and combine it with a constraint_mode to complete the configuration. Use the create_delay_corner and update_delay_corner commands to create and modify delay_corner objects.

Attribute	Description
default_early_timing_condition	
	Specifies the timing condition to be used for early path analysis if one has not be explicitly specified for a given power domain. Type: obj(timing_condition) Default: "" Edit: No
default_late_timing_condition	

Stylus Common UI Database Object Information

Database Objects--delay_corner

	Specifies the timing condition to be used for late path analysis if one has not be explicitly specified for a given power domain. Type: obj(timing_condition) Default: "" Edit: No
early_irdrop_files	
	Specifies a file with instance-specific voltage data for early path analysis. Instance voltage data can be either exported from power rail analysis tools such as Voltus, or specified with set_instance_voltage constraints Type: string Default: "" Edit: No
early_pg_net_voltages	
	Specifies the power/ground net voltages to use for early analysis Type: string Default: "" Edit: No
early_rc_corner	
	Specifies the associated rc_corner object for early path analysis Type: obj(rc_corner) Default: "" Edit: No
early_temperature_files	
	Specifies a file with instance-specific temperature data for early path analysis Type: string Default: "" Edit: No
early_timing_condition	
	Provides the early timing_condition object associated with the specified power_domain index. If no power_domain index is given, the default early timing_condition will be returned. Type: obj(timing_condition) Allowed -index values: power_domain Default: "" Edit: No
early_timing_condition_string	
	Specifies the list of power_domain to timing_condition bindings for early path analysis - using <power_domain>@<timing_condition> list syntax, for example 'defaultTC PD1@TC1 PD2@TC2'. Type: string Default: "" Edit: No
is_active	
	Indicates that the delay_corner is associated with an analysis_view that is used in the active analysis_view Type: bool Default: "" Edit: No
is_dynamic	
	Indicates that the delay_corner is associated with an analysis_view that is used in the active dynamic analysis_view Type: bool Default: "" Edit: No
is_hold	
	Indicates that the delay_corner is associated with an analysis_view that is active for hold analysis Type: bool Default: "" Edit: No

Stylus Common UI Database Object Information

Database Objects--delay_corner

is_leakage	
	Indicates that the delay_corner is associated with an analysis_view that is used in the active leakage analysis_view Type: bool Default: "" Edit: No
is_setup	
	Indicates that the delay_corner is associated with an analysis_view that is active for setup analysis Type: bool Default: "" Edit: No
is_si_enabled	
	Indicates that analysis_views associated with this delay_corner should have SI analysis performed on them Type: bool Default: "" Edit: No
late_irdrop_files	
	Specifies a file with instance-specific voltage data for late path analysis. Instance voltage data can be either exported from power rail analysis tools such as Voltus, or specified with set_instance_voltage constraints Type: string Default: "" Edit: No
late_pg_net_voltages	
	Specifies the power/ground net voltages to use for late analysis Type: string Default: "" Edit: No
late_rc_corner	
	Specifies the associated rc_corner object for late path analysis Type: obj(rc_corner) Default: "" Edit: No
late_temperature_files	
	Specifies a file with instance-specific temperature data for late path analysis Type: string Default: "" Edit: No
late_timing_condition	
	Provides the late timing_condition object associated with the specified power_domain index. If no power_domain index is given, the default late timing_condition will be returned. Type: obj(timing_condition) Allowed -index values: power_domain Default: "" Edit: No
late_timing_condition_string	
	Specifies the list of power_domain to timing_condition bindings for late path analysis - using <power_domain>@<timing_condition> list syntax, for example 'defaultTC PD1@TC1 PD2@TC2'. Type: string Default: "" Edit: No
name	

Stylus Common UI Database Object Information

Database Objects--delay_noise_obj

	Provides the name of this delay_corner object as specified by create_delay_corner. This name can referenced by other commands to refer to this object Type: string Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (delay_corner) Default: "" Edit: No
pg_net_voltages	
	Specifies the power/ground net voltages to use for both early/late analysis Type: string Default: "" Edit: No

delay_noise_obj

Parent Objects

[timing_point](#)

Definition

description

Attribute	Description
annotated_glitch	
	annotated_glitch Type: double Default: "" Edit: No
attackers	
	Returns the attackers property for a delay_noise_obj Type: obj(delay_noise_obj)* Default: "" Edit: No
capacitance	
	capacitance Type: double Default: "" Edit: No
coupling_capacitance	
	coupling_capacitance Type: double Default: "" Edit: No
glitch_area	
	glitch_area Type: double Default: "" Edit: No

Stylus Common UI Database Object Information
Database Objects--delay_noise_obj

glitch_width	
	glitch_width Type: double Default: "" Edit: No
hierarchical_name	
	Returns the hierarchical_name property for delay_noise_obj Type: string Default: "" Edit: No
input_failure_threshold	
	input_failure_threshold Type: double Default: "" Edit: No
input_peak	
	input_peak Type: double Default: "" Edit: No
input_peak_slack	
	input_peak_slack Type: double Default: "" Edit: No
level	
	Returns the level property for delay_noise_obj Type: string Default: "" Edit: No
num_active_attackers	
	num_active_attackers Type: double Default: "" Edit: No
num_attackers	
	num_attackers Type: double Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (delay_noise_obj) Default: "" Edit: No
propagated_glitch	
	Reports the propagated glitch seen at the victim driver output Type: double Default: "" Edit: No
receiver_peak	

Stylus Common UI Database Object Information

Database Objects--density_shape

	receiver_peak Type: double Default: "" Edit: No
receiver_peak_slack	
	Returns the receiver_peak_slack property for delay_noise_obj Type: string Default: "" Edit: No
receiver_peak_threshold	
	receiver_peak_threshold Type: double Default: "" Edit: No
resistance	
	Returns the resistance property for delay_noise_obj Type: string Default: "" Edit: No
transition	
	Returns the transition property for delay_noise_obj Type: string Default: "" Edit: No
view_type	
	Returns the view_type property for delay_noise_obj Type: string Default: "" Edit: No

density_shape

Parent Objects

[place_blockage](#)

Definition

Rectangle with metal/cut or placement density information

Attribute	Description
density	
	Density value (range: 0.0-1.0 Type: double Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (density_shape) Default: "" Edit: No
rect	

	Rectangle. For cellDensity, the coordinates are local to cell, not relative to the design. Type: rect Default: "" Edit: No
--	---

design

Parent Objects

[port_bus](#), [hinst](#), [inst](#), [module](#), [pin_group](#), [pin_guide](#), [root](#), [port](#)

Definition

top cell

Attribute	Description
actual_scan_chains	
	List of all actual_scan_chains Type: obj(actual_scan_chain)* Default: "" Edit: No
actual_scan_segments	
	List of all actual_scan_segments Type: obj(actual_scan_segment)* Default: "" Edit: No
analysis_type	
	Returns analysis type for the session as single, bestcase-worsecase or on-chip-variation. Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
analysis_views	
	Returns the information about the analysis views in the design. Type: obj(analysis_view)* Default: "" Edit: No
arch_filename	
	Source file that contains the specified module (Verilog) or architecture (VHDL). Type: string Default: "" Edit: Yes
arch_name	
	The name of the specified module (Verilog) or architecture (VHDL). Type: string Default: "" Edit: Yes
area	
	The area of all the insts contained inside this design and below it. This does not include phys_insts. Use 'get_db \$design .boundary.area' if you want the area of the physical design boundary. Type: area Default: "" Edit: No

Stylus Common UI Database Object Information

Database Objects--design

base_name	
	Name of design Type: string Default: "" Edit: No
bbox	
	Bounding box of the design boundary, Type: rect Default: "" Edit: No
boundaries	
	All the boundary objects (fence, region, etc.) in the design. Type: obj(boundary)* Default: "" Edit: No
boundary	
	Polygon that define the shape of the floorplan boundary Type: polygon* Default: "" Edit: No
boundary_model_analysis	
	Returns true if the design is boundary model Type: bool Allowed -index values: analysis_view Default: "" Edit: No
boundary_type	
	Specifies the IEEE boundary-scan architecture {IEEE_11491 IEEE_11496}. Type: string Default: "" Edit: Yes
bump_grids	
	List of bump_grid object. Type: obj(bump_grid)* Default: "" Edit: No
bumps	
	List of bumps in the cell Type: obj(bump)* Default: "" Edit: No
bus_guides	
	List of bus guide objects. Type: obj(bus_guide)* Default: "" Edit: No
bus_sink_groups	
	A group of sinks (loads) that some floorplan and routing commands use to control adding buffers and routing for a bus. See 'help *bus_sink*' for a list of commands related to this object. Type: obj(bus_sink_group)* Default: {} Edit: No

Stylus Common UI Database Object Information

Database Objects--design

busses	
	The Verilog bus definitions for nets in the top Verilog module. Type: <code>obj(bus)*</code> Default: "" Edit: No
capacitance_unit_in_farad	
	Returns session units for capacitance in farad Type: <code>double</code> Allowed -index values: analysis_view clock Default: "" Edit: No
chip_size	
	Returns the chip size that will be used for bounding box computation Type: <code>double</code> Allowed -index values: analysis_view clock Default: "" Edit: No
clock_gating_add_obs_port	
	Set clock-gating observation point. Type: <code>bool</code> Default: false Edit: Yes
clock_gating_auto_cost_group_initial_target	
	The initial target slack for automatic clock gate cost groups. Type: <code>string</code> Default: "" Edit: Yes
clock_gating_auto_cost_grouping	
	Automatically creates a cost group (per clock) for clock gate enable paths. Type: <code>bool</code> Default: true Edit: Yes
clock_gating_auto_path_adjust	
	Controls automatic timing adjustment on clock gate enable paths. Type: <code>enum</code> Enum Values: none fixed variable Default: none Edit: Yes
clock_gating_auto_path_adjust_fixed_delay	
	Fixes the path adjustment automatically added to clock gate enable pins. Type: <code>string</code> Default: 0.0 Edit: Yes
clock_gating_auto_path_adjust_modes	
	The timing modes in which clock gate enable path adjusts will be applied. Type: <code>string</code> Default: "" Edit: Yes
clock_gating_auto_path_adjust_multiplier	
	Scales the automatic load-dependent path adjustment added to clock gate enable pins. Type: <code>double</code> Default: 1.0 Edit: Yes

Stylus Common UI Database Object Information

Database Objects--design

clock_gating_control_point	
	Set clock-gating control point. Type: <code>enum</code> Enum Values: none precontrol postcontrol Default: precontrol Edit: Yes
clock_gating_exclude	
	Exclude this design from clock-gating. Type: <code>bool</code> Default: false Edit: Yes
clock_gating_extract_common_enable	
	Inserts clock-gating by extracting common logic from synchronous enable logic of flops. Type: <code>bool</code> Default: true Edit: Yes
clock_gating_hierarchical	
	Create hierarchical clock gates in this design. Type: <code>bool</code> Default: false Edit: Yes
clock_gating_max_flops	
	Set fanout limit to clock-gating cell. Type: <code>int</code> Default: no_value Edit: Yes
clock_gating_min_flops	
	Set minimum number of flops needed to add a clock-gating cell. Type: <code>int</code> Default: 3 Edit: Yes
clock_gating_module	
	Module used for clock gating circuitry. Type: <code>string</code> Default: "" Edit: Yes
clock_gating_style	
	Set clock-gating style. Type: <code>enum</code> Enum Values: none latch ff Default: latch Edit: Yes
clock_gating_test_signal	
	Set specified test signal to connect up test logic in clock-gating circuitry. Type: <code>string</code> Default: "" Edit: Yes
constraint_modes	
	Returns the information about the constraint modes in the design. Type: <code>obj(constraint_mode)*</code> Default: "" Edit: No

Stylus Common UI Database Object Information

Database Objects--design

control_logic_optimization	
	Controls the effort level for optimization of control logic on this module (basic/advanced/none/inherited). Type: enum Enum Values: basic advanced none inherited Default: inherited Edit: Yes
core_bbox	
	Rectangle that defines the core row area Type: rect Default: "" Edit: No
core_boundary	
	List of polygons that define the shape of the core boundary. Type: polygon* Default: "" Edit: No
core_site	
	The site object to use during floorplan creation Type: obj(site) Default: "" Edit: No
core_to_bottom	
	Distance between core edge and its die/io box Type: coord Default: "" Edit: No
core_to_left	
	Distance between core edge and its die/io box Type: coord Default: "" Edit: No
core_to_right	
	Distance between core edge and its die/io box Type: coord Default: "" Edit: No
core_to_top	
	Distance between core edge and its die/io box Type: coord Default: "" Edit: No
cpf_macro_inherit_parent_power_domain	
	Specify if the macro instances in CPF will inherit their power domain from parent instance. Type: bool Default: false Edit: Yes
cpi_inverter_name_prefix	
	Prefix for inverters added by 'commit_power_intent'. Type: string Default: cpi_inv Edit: Yes
cpi_output_net_name_prefix	

Stylus Common UI Database Object Information

Database Objects--design

	The prefix for net name on output side of an instance inserted by 'commit_power_intent'. Type: string Default: "" Edit: Yes
current_unit_in_amp	
	Returns session units for current in ampere. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
def_scan_chains	
	List of all def_scan_chains in the design Type: obj(def_scan_chain)* Default: "" Edit: No
delay_corners	
	Returns the information about the delay corners in the design. Type: obj(delay_corner)* Default: "" Edit: No
delete_unloaded_seqs	
	Control deletion of unloaded sequential instances. Type: enum Enum Values: 1 0 true false inherited Default: inherited Edit: Yes
dft_boundary_scan_exists	
	Attribute that indicates if boundary scan has been inserted into the design. Type: bool Default: false Edit: Yes
dft_clock_edge_for_head_of_scan_chains	
	Capture clock edge of scan chain (leading). Type: string Default: "" Edit: Yes
dft_clock_edge_for_tail_of_scan_chains	
	Launch clock edge of scan chain (leading trailing). Type: string Default: "" Edit: Yes
dft_configuration_modes	
	List of all dft_configuration_modes Type: obj(dft_configuration_mode)* Default: "" Edit: No
dft_connect_scan_data_pins_during_mapping	
	DFT scan connection mode {loopback floating ground high}. Type: enum Enum Values: loopback floating ground high Default: loopback Edit: Yes
dft_connect_shift_enable_during_mapping	

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Database Objects--design

	DFT shift enable connection mode {tie_off floating}. Type: enum Enum Values: tie_off floating Default: tie_off Edit: Yes
dft_core_max_pipe_depth	
	Maximum pipeline depth of core. Type: int Default: 0 Edit: Yes
dft_dont_scan	
	Exclude this design from scan-synthesis {inherited true false}. Type: enum Enum Values: false true inherited Default: inherited Edit: Yes
dft_lockup_element_type	
	Type of lockup element in scan chains {level_sensitive edge_sensitive preferred_level_sensitive preferred_edge_sensitive}. Type: enum Enum Values: level_sensitive edge_sensitive preferred_level_sensitive preferred_edge_sensitive Default: preferred_level_sensitive Edit: Yes
dft_lockup_element_type_for_tail_of_scan_chains	
	Launch element type for scan chains (level_sensitive edge_sensitive). Type: string Default: level_sensitive Edit: Yes
dft_max_length_of_scan_chains	
	Default max length for any scan chain (unless overridden by chain-specific max length). Type: int Default: no_value Edit: Yes
dft_min_number_of_scan_chains	
	Minimum number of total scan chains to be created. Type: int Default: no_value Edit: Yes
dft_mix_clock_edges_in_scan_chains	
	Allow mixing in a same scan chain flip-flops triggered by either edge of a same clock. Type: bool Default: false Edit: Yes
dft_scan_map_mode	
	DFT scan conversion mode {tdrc_pass preserve force_all}. Type: enum Enum Values: tdrc_pass preserve force_all Default: tdrc_pass Edit: Yes
dft_scan_output_preference	

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Database Objects--design

	DFT scan output preference {auto non_inverted inverted}. Type: enum Enum Values: auto non_inverted inverted Default: auto Edit: Yes
dft_tap_tck_period	
	Tap TCK period. Type: int Default: 50000 Edit: Yes
dont_touch	
	This attributes defines the user preservation status of the design during optimization. This setting will apply to all insts within the design unless overridden at a lower level hinst or on the inst object itself. The dont_touch_effective attribute on each child inst and hinst will return the resolved value. Use 'help inst dont_touch' to see the enum value definitions. Type: enum Enum Values: none false true delete_ok const_prop_size_delete_ok const_prop_delete_ok size_delete_ok size_ok size_same_height_ok size_same_footprint_ok Default: none Edit: Yes
dont_touch_effective	
	This attribute defines the effective preservation status of the design during optimization from the dont_touch_sources values. If the read_only source is 'true' then the effective value for the hinst is 'true'. Otherwise, the user value has precedence. See help on inst dont_touch attribute for details on possible values. Type: enum Enum Values: none false true delete_ok const_prop_size_delete_ok const_prop_delete_ok size_delete_ok size_ok size_same_height_ok size_same_footprint_ok Default: false Edit: No
dont_touch_file	
	Specifies the sourced file to set dont_touch. Type: string Default: "" Edit: Yes
dont_touch_sources	
	Dictionary of {source <value>} pairs contributing to the dont_touch_effective attribute for this object : {user <value>} {read_only <value>} user # This hinst dont_touch value read_only # The design is read_only (set_module_view -top_read_only true) Type: string Default: "" Edit: No
dont_use_cells	
	List of cell names (wildcards supported) to disallow for this design during optimization. This can be overridden at a lower hinst level. Overrides any library dont_use values. Type: string Default: "" Edit: Yes
dont_use_cells_effective	
	The resolved list of all cell names to disallow during optimization for the design based on the library dont_use and the dont_use_cells and use_cells attributes of this design. The precedence is: use_cells of this hinst), then dont_use_cells of this design. Type: string Default: "" Edit: No
dp_csa	

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Database Objects--design

	Controls CSA operation on design/subdesign in 'synthesize -to_generic' (basic/none/inherited). Type: enum Enum Values: basic none inherited Default: inherited Edit: Yes
dp_csa_factorize	
	Controls datapath csa_factorize on a design in 'syn_generic'. Type: enum Enum Values: basic none inherited Default: inherited Edit: Yes
dp_opt_power_opt	
	Controls datapath power optimization flow on a given subdesign. Type: enum Enum Values: 1 0 true false inherited Default: inherited Edit: Yes
dp_rename	
	Attribute to control the datapath rewriting optimization level on a subdesign (basic/advanced/none/inherited). Type: enum Enum Values: basic advanced none inherited Default: inherited Edit: Yes
dp_sharing	
	To control the datapath sharing on a design/subdesign in 'synthesize -to_generic' (advanced/none/inherited). Type: enum Enum Values: advanced none inherited Default: inherited Edit: Yes
dp_speculation	
	Controls datapath speculation on a design in 'syn_generic'. Type: enum Enum Values: basic none inherited Default: inherited Edit: Yes
early_clk_cell_derate_factor	
	Returns the derating factor for early clock paths specified through the set_timing_derate command with the -early parameter. Type: double Allowed -index values: analysis_view Default: "" Edit: No
early_data_cell_derate_factor	
	Returns the derating factorfor early paths specified through the set_timing_derate command with the -early parameter. Type: double Allowed -index values: analysis_view Default: "" Edit: No
early_fall_cell_check_derate_factor	
	Returns the early cell check derating factor specified through the set_timing_derate command with the -fall parameter. Type: double Allowed -index values: analysis_view Default: "" Edit: No
early_fall_clk_cell_derate_factor	

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Database Objects--design

	Returns the early clock path derating factor specified through the set_timing_derate command with the -fall parameter. Type: double Allowed -index values: analysis_view Default: "" Edit: No
early_fall_data_cell_derate_factor	
	Returns the early data cell check derating factor specified through the set_timing_derate command with the -fall parameter. Type: double Allowed -index values: analysis_view Default: "" Edit: No
early_rise_cell_check_derate_factor	
	Returns the early cell check derating factor specified through the set_timing_derate command with the -rise and -cell_check parameter. Type: double Allowed -index values: analysis_view Default: "" Edit: No
early_rise_clk_cell_derate_factor	
	Returns the early clock cell check derating factor specified through the set_timing_derate command with the -rise parameter. Type: double Allowed -index values: analysis_view Default: "" Edit: No
early_rise_data_cell_derate_factor	
	Returns the early data cell check derating factor specified through the set_timing_derate command with the -rise parameter. Type: double Allowed -index values: analysis_view Default: "" Edit: No
embedded_script	
	Stores the SDC command script embedded in the source file that contains the module (Verilog) or entity (VHDL) specification for this design. Type: string Default: "" Edit: Yes
entity_filename	
	Source file that contains the specified module (Verilog) or entity (VHDL). Type: string Default: "" Edit: Yes
entity_name	
	Name of the specified module (Verilog) or entity (VHDL). Type: string Default: "" Edit: Yes
exceptions	
	Reports the path exceptions Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
formal_verification_constraints	

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Database Objects--design

	List of all formal_verification_constraints Type: <code>obj(formal_verification_constraint)*</code> Default: "" Edit: No
gcell_grids	
	All the gcell_grid objects in the design. Type: <code>obj(gcell_grid)*</code> Default: "" Edit: No
gcells	
	The gcells for the current_design. The gcells are only created after global route has occurred. Type: <code>obj(gcell)*</code> Default: "" Edit: No
groups	
	List of groups Type: <code>obj(group)</code> Default: "" Edit: No
gui_lines	
	All the gui_line objects in the design. Type: <code>obj(gui_line)*</code> Default: "" Edit: No
gui_polygons	
	All the gui_polygon objects in the design. Type: <code>obj(gui_polygon)*</code> Default: "" Edit: No
gui_rects	
	All the gui_rect objects in the design. Type: <code>obj(gui_rect)*</code> Default: "" Edit: No
gui_shapes	
	All the gui_shape objects in the design. Type: <code>obj(gui_rect)* obj(gui_line)* obj(gui_polygon)*</code> Default: "" Edit: No
gui_texts	
	All the gui_text objects in the design. Type: <code>obj(gui_text)*</code> Default: "" Edit: No
has_dp_rewrite_gint	
	This design/subdesign has rewritten instance. Type: <code>bool</code> Default: false Edit: No
has_dp_shared_gint	

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Database Objects--design

	This design/subdesign has shared instance. Type: <code>bool</code> Default: <code>false</code> Edit: No
hdl_all_filelist	
	Attribute to get all the files along with the included hdl files and package files for a design or subdesign. Type: <code>string</code> Default: <code>""</code> Edit: Yes
hdl_config_name	
	Name of the configuration used to build this module. Type: <code>string</code> Default: <code>""</code> Edit: Yes
hdl_cw_list	
	List of CW components used by a design. Type: <code>string</code> Default: <code>""</code> Edit: Yes
hdl_elab_command_params	
	List of {parameter value} pair that are passed to 'elaborate -parameters' command. Type: <code>string</code> Default: <code>""</code> Edit: Yes
hdl_filelist	
	List of HDL files read prior to the elaboration of a design. Type: <code>string</code> Default: <code>""</code> Edit: Yes
hdl_parameters	
	List of {parameter value cmdset is _default} used to 'elaborate' a design. 'cmdset' is '1' only if the parameter is set using the 'elaborate -parameters' command. Type: <code>string</code> Default: <code>""</code> Edit: Yes
hdl_pipeline_comp	
	Specifies whether the design contains a \$pipeline component. Type: <code>bool</code> Default: <code>false</code> Edit: Yes
hdl_user_name	
	Name of the module (Verilog) or entity (VHDL) for this design. Type: <code>string</code> Default: <code>""</code> Edit: Yes
hdl_vdp_list	
	Information about datapath function primitives used in a module. Type: <code>string</code> Default: <code>""</code> Edit: Yes
hierarchical_name	

Stylus Common UI Database Object Information

Database Objects--design

	Returns the hierarchical name of the design Type: <code>string</code> Allowed -index values: analysis_view clock Default: "" Edit: No
hinsts	
	all hinsts inside this design Type: <code>obj(hinst)*</code> Default: "" Edit: No
hnets	
	all hnets inside this design Type: <code>obj(hnet)*</code> Default: "" Edit: No
hpin_busses	
	Returns all the hpin busses of the design. Type: <code>obj(hpin_bus)*</code> Default: "" Edit: No
hpins	
	all hpins inside this design Type: <code>obj(hpin)*</code> Default: "" Edit: No
hport_busses	
	Returns all the hport busses of the design. Type: <code>obj(hport_bus)*</code> Default: "" Edit: No
ignore_library_drc	
	Do not use design rule constraints from the library. Type: <code>bool</code> Default: false Edit: Yes
ignore_library_max_fanout	
	Do not use max_fanout rule constraints from the library. Type: <code>bool</code> Default: false Edit: Yes
insert_pmbist_without_liberty_files	
	Ignore liberty file information. Type: <code>bool</code> Default: false Edit: Yes
insts	
	all the logical insts inside this design. Does not include physical-insts. Type: <code>obj(inst)*</code> Default: "" Edit: No
io_bbox	

Stylus Common UI Database Object Information

Database Objects--design

	Rectangle that defines the IO area Type: <code>rect</code> Default: "" Edit: No
io_constraints	
	List of io_constrains. Type: <code>obj(io_constraint)*</code> Default: "" Edit: No
is_clock_synthesized	
	Design status: Clock synthesized Type: <code>bool</code> Default: false Edit: Yes
is_core_to_io	
	Indicates whether core2* attributes are measured between core edge (core_bbox) and design boundary (box) or between core edge and io box edge (io_bbox) Type: <code>bool</code> Default: false Edit: Yes
is_detail_routed	
	Design status: detail routed, true if route_design was run. Type: <code>bool</code> Default: false Edit: Yes
is_dont_touch	
	Returns true if design is marked dont touch Type: <code>bool</code> Allowed -index values: analysis_view clock Default: "" Edit: No
is_generic	
	Determine whether the design is generic. The attribute is set true after elaborate_design. Type: <code>bool</code> Default: false Edit: No
is_io_placed	
	Design status: Is IO placed Type: <code>bool</code> Default: "" Edit: No
is_mapped	
	Determine whether the design is mapped. The attribute is set true after synthesize_design. Type: <code>bool</code> Default: false Edit: No
is_max_transition_asserted	
	Returns true if max transition constraint has been applied on design Type: <code>bool</code> Allowed -index values: analysis_view clock Default: "" Edit: No
is_min_transition_asserted	

Stylus Common UI Database Object Information

Database Objects--design

	Returns true if min transition constraint has been applied on design Type: <code>bool</code> Allowed -index values: analysis_view clock Default: "" Edit: No
is_od_info_saved	
	Indicates if OD info was saved in the DB. Type: <code>bool</code> Default: false Edit: No
is_placed	
	Design status: Placed Type: <code>bool</code> Default: false Edit: Yes
is_proto_model_committed	
	Indicates that the design has committed FlexModels Type: <code>bool</code> Default: "" Edit: No
is_proto_model_specified	
	Indicates that the design has specified FlexModels Type: <code>bool</code> Default: "" Edit: No
is_rc_extracted	
	Design status: Parasitic extracted Type: <code>bool</code> Default: false Edit: Yes
is_retimed	
	Indicates that the design has been retimed. Type: <code>bool</code> Default: false Edit: Yes
is_routed	
	Design status: routed. true if design has global routes from place_design/route_early_global or detail routes from route_design. Type: <code>bool</code> Default: false Edit: Yes
is_scan_optimized	
	Design status: Scan optimized Type: <code>bool</code> Default: false Edit: Yes
isolate_zero_pin_retention	
	Insert isolation cell on zero pin SRPG cells. Type: <code>bool</code> Default: true Edit: Yes
language	

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Database Objects--design

	Name of the language in which module (Verilog) or entity (VHDL) is written. Type: string Default: "" Edit: Yes
late_cell_check_derate_factor	
	Returns the derating factor for late paths specified through the set_timing_derate command with the -late parameter. Type: double Allowed -index values: analysis_view Default: "" Edit: No
late_clk_cell_derate_factor	
	Returns the derating factor for late clock paths specified through the set_timing_derate command with the -late parameter. Type: double Allowed -index values: analysis_view Default: "" Edit: No
late_data_cell_derate_factor	
	Returns the derating factor for late data paths specified through the set_timing_derate command with the -late parameter. Type: double Allowed -index values: analysis_view Default: "" Edit: No
late_fall_cell_check_derate_factor	
	Returns the late cell check derating factor specified through the set_timing_derate command with the -fall parameter. Type: double Allowed -index values: analysis_view Default: "" Edit: No
late_fall_clk_cell_derate_factor	
	Returns the late clock cell check derating factor specified through the set_timing_derate command with the -fall parameter. Type: double Allowed -index values: analysis_view Default: "" Edit: No
late_fall_data_cell_derate_factor	
	Returns the late data cell check derating factor specified through the set_timing_derate command with the -fall parameter. Type: double Allowed -index values: analysis_view Default: "" Edit: No
late_rise_cell_check_derate_factor	
	Returns the late cell check derating factor specified through the set_timing_derate command with the -rise parameter. Type: double Allowed -index values: analysis_view Default: "" Edit: No
late_rise_clk_cell_derate_factor	
	Returns the late clock cell check derating factor specified through the set_timing_derate command with the -rise parameter. Type: double Allowed -index values: analysis_view Default: "" Edit: No
late_rise_data_cell_derate_factor	

Stylus Common UI Database Object Information

Database Objects--design

	Returns the late data cell check derating factor specified through the set_timing_derate command with the -rise parameter. Type: <code>double</code> Allowed -index values: analysis_view Default: "" Edit: No
library_name	
	Name of the (logical) library in which module (Verilog) or entity (VHDL) reside. Type: <code>string</code> Default: "" Edit: Yes
local_hinsts	
	List of hinsts in the current level of this design (e.g. from inside the design looking down at the hinsts at this level) Type: <code>obj(hinst)*</code> Default: "" Edit: No
local_hnets	
	List of hnets in the current level of this design (e.g. from inside the design looking down at the hnets at this level) Type: <code>obj(hnet)*</code> Default: "" Edit: No
local_hpins	
	List of hpins in the current level of this design (e.g. from inside the design looking down at the hpins of hinsts at this level) Type: <code>obj(hpin)*</code> Default: "" Edit: No
local_insts	
	List of insts in the current level of this design (e.g. from inside the design looking down at the insts at this level) Type: <code>obj(inst)*</code> Default: "" Edit: No
local_pins	
	List of pins in the current level of this design (e.g. from inside the design looking down at the pins of the insts at this level) Type: <code>obj(pin)*</code> Default: "" Edit: No
lp_clock_tree_buffers	
	Specify the buffers or inverters to be used in clock tree synthesis. Type: <code>string</code> Default: "" Edit: Yes
lp_clock_tree_leaf_max_fanout	
	Specify the number of flops driven by leaf clock buffer. Type: <code>int</code> Default: 0 Edit: Yes
lp_pso_aware_tcf	
	Specify that asserted TCF is generated by simulating the design while taking into account when power domains are shutoff. Type: <code>bool</code> Default: false Edit: Yes
markers	

Stylus Common UI Database Object Information

Database Objects--design

	All the marker objects in the design. Type: <code>obj(marker)*</code> Default: "" Edit: No
max_capacitance	
	Returns max capacitance value applied on design Type: <code>double</code> Allowed -index values: analysis_view clock Default: "" Edit: No
max_fanout	
	Returns max fanout value applied on design Type: <code>double</code> Allowed -index values: analysis_view clock Default: "" Edit: No
max_transition	
	Returns max transition value applied on design Type: <code>double</code> Allowed -index values: analysis_view clock Default: "" Edit: No
mbist_enable_shared_library_domain_set	
	Library domains across which memories can be shared. Type: <code>string</code> Default: "" Edit: Yes
min_capacitance	
	Returns min capacitance value applied on design Type: <code>double</code> Allowed -index values: analysis_view clock Default: "" Edit: No
min_fanout	
	Returns min fanout value applied on design Type: <code>double</code> Allowed -index values: analysis_view clock Default: "" Edit: No
min_transition	
	Returns min transition value applied on design Type: <code>double</code> Allowed -index values: analysis_view clock Default: "" Edit: No
modules	
	all modules inside this design Type: <code>obj(module)*</code> Default: "" Edit: No
name	

Stylus Common UI Database Object Information

Database Objects--design

	Name of design Type: <code>string</code> Default: "" Edit: No
net_groups	
	List of net groups Type: <code>obj(net_group)*</code> Default: "" Edit: No
nets	
	All the nets inside this design, including logical Verilog nets, physical-only nets, Verilog supply0/supply1, and Verilog 1'b0/1'b1 nets. Type: <code>obj(net)*</code> Default: "" Edit: No
num_core_rows	
	Number of rows within the core area Type: <code>int</code> Default: "" Edit: No
num_exceptions	
	Reports the number of path exceptions Type: <code>int</code> Allowed -index values: analysis_view clock Default: "" Edit: No
num_insts	
	Number of instances in the cell Type: <code>int</code> Default: "" Edit: No
num_nets	
	Number of canonical (flat) nets in the cell Type: <code>int</code> Default: "" Edit: No
num_pg_nets	
	Number of physical nets in the cell Type: <code>int</code> Default: "" Edit: No
num_phys_insts	
	Number of physical instances in the cell Type: <code>int</code> Default: "" Edit: No
oa_design_lib	
	Library name for the design in OpenAccess cellview Type: <code>string</code> Default: "" Edit: No
oa_design_view	

Stylus Common UI Database Object Information

Database Objects--design

	View name for the design in OpenAccess cellview Type: string Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (design) Default: "" Edit: No
operating_condition_max	
	Returns max operating condition of design Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
operating_condition_min	
	Returns min operating condition of design Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
opt_retime_clocks	
	Perform retiming on these clocks. Type: object* Default: "" Edit: Yes
opt_retime_move_mux_loop_with_register	
	Move the feedback mux loop along with the register during retiming. Module level control. Type: bool Default: true Edit: Yes
package_components	
	List of other chips in the package Type: obj(package_component)* Default: "" Edit: No
package_objects	
	List of objects in the package Type: obj(package_object)* Default: "" Edit: No
partitions	
	all partitions inside this design Type: obj(partition)* Default: "" Edit: No
pg_nets	
	List of power ground nets in the design. This include physical-only PG nets, and Verilog supply0/supply1 nets. Note that Verilog nets assigned to 1'b0/1'b1 are returned by the nets attribute, not the pg_nets attribute. Type: obj(net)* Default: "" Edit: No
pg_pins	

Stylus Common UI Database Object Information

Database Objects--design

	All PG pins inside this design. Type: obj(pg_pin)* Default: "" Edit: No
pg_ports	
	List of power/ground terminals in the cell Type: obj(port)* Default: "" Edit: No
phys_insts	
	List of physical instances in the cell Type: obj(inst)* Default: "" Edit: No
phys_ports	
	List of physical (unused) signal terminals in the cell Type: obj(port)* Default: "" Edit: No
pi_relax_map_iso_cell_checks	
	Relax sanity checks on map_isolation_cell list. Type: bool Default: false Edit: Yes
pi_relax_map_ls_cell_checks	
	Relax sanity checks on map_level_shifter_cell list. Type: bool Default: false Edit: Yes
pin_blockages	
	List of partition pin blockage objects Type: obj(pin_blockage)* Default: "" Edit: No
pin_groups	
	List of all the pin groups Type: obj(pin_group)* Default: "" Edit: No
pin_guides	
	List of all the pin guide object Type: obj(pin_guide)* Default: "" Edit: No
pin_to_corner_tracks	
	List of distance constraints (in tracks) of pins from design/partition corners where the lower left corner is listed first and the remaining corners are listed in clockwise order. Type: string* Default: "" Edit: No
pins	

Stylus Common UI Database Object Information

Database Objects--design

	Short-cut for [get_db current_design .pins] Type: obj(pin)* Default: "" Edit: No
place_blockages	
	List of placement blockages. Type: obj(place_blockage)* Default: "" Edit: No
pmbist_hri_async_reset	
	Asynchronous reset port/pin along with the active state for repair used by programmable MBIST specified as {<pin/port name> active_[high low]}. Type: string Default: "" Edit: Yes
port_busses	
	The Verilog bus definitions for ports in the top Verilog module. Type: obj(port_bus)* Default: "" Edit: No
port_shapes	
	These are all the individual physical_pin shapes of the top-level ports including signal and PG ports. Type: obj(port_shape)* Default: "" Edit: No
ports	
	ports of design. Does not include pg_ports, unless PG port is explicitly in the Verilog netlist. Type: obj(port)* Default: "" Edit: No
power_domains	
	List of power domains (pd) in the design Type: obj(power_domain)* Default: "" Edit: No
power_modes	
	List of the power mode names in the design. Type: string Default: "" Edit: No
preserve_power_domain_boundary	
	Preserve value for power domain boundaries. Type: enum Enum Values: false true const_prop_delete_ok delete_ok Default: false Edit: Yes
process_max	
	Returns max process value set on design Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
process_min	

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Database Objects--design

	Returns min process value set on design Type: <code>double</code> Allowed -index values: analysis_view clock Default: "" Edit: No
read_db_design_name	
	The design name of the restored design. Type: <code>string</code> Default: "" Edit: No
read_db_tool_name	
	The tool name save the restored design in current session. Type: <code>string</code> Default: "" Edit: No
read_db_tool_version	
	The tool version number save the restored design in current session. Type: <code>string</code> Default: "" Edit: No
read_only	
	This attribute is set by set_module_view to identify if the top_level is read only or not. It means the partition cannot be optimized, and cells inside will not be moved. Setting this attribute will set the dont_touch_effective attribute on all insts and hinsts within the top level partition unless overridden at a lower level partition. It cannot be overridden by other hinst or inst values. Supported values: false: The top level is allowed to be optimized. true: The top level is read_only. none: No constraint. Type: <code>enum</code> Enum Values: none false true Default: none Edit: No
resistance_unit_in_ohm	
	Returns session units for resistance in ohm Type: <code>double</code> Allowed -index values: analysis_view clock Default: "" Edit: No
resize_blockages	
	List of resize_blockage objects Type: <code>obj(resize_blockage)*</code> Default: "" Edit: No
retime	
	Retime this design during synthesis. Type: <code>bool</code> Default: false Edit: Yes
retime_period_percentage	
	Adjust target clock period in retiming. Type: <code>double</code> Default: 1.0 Edit: Yes
retime_ssw_sync_enable	

Stylus Common UI Database Object Information

Database Objects--design

	Directs retiming to use SAT-sweep to find equivalence classes for synchronous enable signals. Module level control. Type: <code>bool</code> Default: <code>false</code> Edit: Yes
route_blockages	
	List of routing blockages Type: <code>obj(route_blockage)*</code> Default: <code>""</code> Edit: No
route_halo_bottom_layer	
	The bottom layer for which routing halo will be created Type: <code>obj(layer)</code> Default: <code>""</code> Edit: No
route_halo_to_boundary	
	Specifies routing halo inside the design boundary (honored by the signal router). Only positive values are used and indicate the halo is inside of the design boundary. Type: <code>coord</code> Default: <code>no_value</code> Edit: Yes
route_halo_top_layer	
	The top layer for which routing halo will be created Type: <code>obj(layer)</code> Default: <code>""</code> Edit: No
row_flip	
	Specification of floorplan row creation, none = no flipping; first = first row is flipped, other rows alternate; second = first row is not flipped, other rows alternate Type: <code>enum</code> Enum Values: none second first Default: <code>""</code> Edit: No
row_spacing	
	Specification of the floorplan row spacing Type: <code>coord</code> Default: <code>no_value</code> Edit: Yes
row_spacing_type	
	Indicates whether the rowSpacing is applied between each row (1) or between each pair of rows (2) Type: <code>enum</code> Enum Values: 0 1 2 Default: <code>""</code> Edit: Yes
rows	
	List of rows Type: <code>obj(row)*</code> Default: <code>""</code> Edit: No
scan_chains	
	List of all scan_chains Type: <code>obj(scan_chain)*</code> Default: <code>""</code> Edit: No

Stylus Common UI Database Object Information

Database Objects--design

scan_in_pipeline_clock_edge	
	Edge of scan in pipeline flop {rise fall}. Type: enum Enum Values: rise fall Default: rise Edit: Yes
scan_out_pipeline_clock_edge	
	Edge of scan out pipeline flop {rise fall}. Type: enum Enum Values: rise fall Default: rise Edit: Yes
scan_segments	
	List of all scan_segments Type: obj(scan_segment)* Default: "" Edit: No
sdp_files	
	List of SDP files that have been read in. Type: string Default: no_value Edit: Yes
sdp_type	
	Specify the SDP file type read in for the design to create SDP Constraints. Valid values are {datapath ff_column both no_value}. Type: string Default: no_value Edit: Yes
seq_reason_deleted	
	Deleted sequential instances for a design with reason. Type: string Default: {} Edit: Yes
setup_views	
	Returns the information about the setup analysis views in the design. Type: obj(analysis_view)* Default: "" Edit: No
slew_threshold_percent_fall_high	
	Returns fall percentage value of high slew threshold Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slew_threshold_percent_fall_low	
	Returns fall percentage value of low slew threshold Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slew_threshold_percent_rise_high	

Stylus Common UI Database Object Information

Database Objects--design

	Returns rise percentage value of high slew threshold Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slew_threshold_percent_rise_low	
	Returns rise percentage value of low slew threshold Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
symmetry	
	The allowed orientations for this design. This value only affects the SYMMETRY value written out by write_lef_abstract. none means r0 only (no SYMMETRY statement), x means ok to flip about X axis (SYMMETRY X), y means ok to flip about Y axis (SYMMETRY Y), xy means ok to flip about X or Y axis (SYMMETRY X Y), any means all orientations are allowed (SYMMETRY X Y R90). Type: enum Enum Values: none x y xy any Default: any Edit: Yes
tech	
	The tech for this design. Type: obj(tech) Default: "" Edit: No
temperature_max	
	Returns max temperature value set on design Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
temperature_min	
	Returns min temperature value set on design Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
test_clock_domains	
	List of all test_clock_domains Type: obj(test_clock_domain)* Default: "" Edit: No
test_clocks	
	List of all test_clocks Type: obj(test_clock)* Default: "" Edit: No
test_signals	
	List of all test_signals Type: obj(test_signal)* Default: "" Edit: No
texts	

Stylus Common UI Database Object Information

Database Objects--design

	All the text objects in the design. Type: <code>obj(text)*</code> Default: "" Edit: No
time_unit_in_second	
	Returns session units for time in second. Type: <code>double</code> Allowed -index values: analysis_view clock Default: "" Edit: No
timing_context_analysis	
	Returns true if the design is context analysis Type: <code>bool</code> Allowed -index values: analysis_view Default: "" Edit: No
top_sdps	
	List of all top level structured data path (sdp) objects. Type: <code>obj(sdp)*</code> Default: "" Edit: No
total_coverage_mode_depth	
	This property is obsoleted Type: <code>double</code> Allowed -index values: analysis_view clock Default: "" Edit: No
track_patterns	
	All the track_pattern objects in the design. Type: <code>obj(track_pattern)*</code> Default: "" Edit: No
tree_type_max	
	Returns max tree type set on design through operating condition Type: <code>string</code> Allowed -index values: analysis_view clock Default: "" Edit: No
tree_type_min	
	Returns min tree type set on design through operating condition Type: <code>string</code> Allowed -index values: analysis_view clock Default: "" Edit: No
trim_grids	
	All the trim_grid objects in the design from the LEF TRIMMETALTRACK statement. Type: <code>obj(trim_grid)*</code> Default: "" Edit: No
use_cells	

Stylus Common UI Database Object Information

Database Objects--design

	List of cell names to allow for this design during optimization. This can be overridden at a lower hinst level. All lib_cells of each base_cell will be allowed. Overrides cells in the dont_use_cells list and any library dont_use values. Type: string Default: "" Edit: Yes
use_only_on_power_critical_nets	
	Use library cells from this base cell set only on power critical nets. Type: string Default: "" Edit: Yes
verification_directory	
	Directory in which verification files are written. The directory is created using the 'verification_directory_naming_style' attribute if it did not already exist. Type: string Default: "" Edit: Yes
violations	
	List of all violations Type: obj(violation)* Default: "" Edit: No
voltage_max	
	Return max voltage set on design Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
voltage_min	
	Returns min voltage set on design Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
voltage_unit_in_volt	
	Returns session units for voltage in volt. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
wcdc_synchronizer_type	
	Specify the type of synchronizer circuit used in design for CDC checks. Valid values are dff or mux or <module name>. Type: string Default: dff mux Edit: Yes
write_lec_dft_constraints	
	This attribute is a TCL dict and holds the LEC dft pin constraints from the last LEC run by Genus. It is passed forward through write_design. The write_do_lec command will automatically include this constraints in the do file. Type: string Default: "" Edit: Yes
write_lec_directory_naming_style	

Stylus Common UI Database Object Information

Database Objects--flexible_htree

The directory name where 'write_do_lec' will write verification files when the 'write_lec_files' attribute is 'true'. The directory will be created if it does not already exist, and will overwrite an existing directory of the same name. A %s in the string is replaced with the design name (e.g. it will overwrite the directory for the same design but not for a different design). A %d in the string is replaced with a unique integer to avoid overwriting any existing directory.

Type: [string](#)
Default: fv/invs/%s
Edit: Yes

flexible_htree

Parent Objects

[pin](#), [root](#)

Definition

flexible htree

Attribute	Description
cts_flexible_htree_adjust_sink_grid_for_aspect_ratio	
	If true adjust the sink grid for the aspect ratio of the sink grid box. Valid values: boolean Type: bool Default: true Edit: Yes
cts_flexible_htree_auto_sinks	
	The number of sinks based on the density of clock sinks under the root of the flexible H-tree. It can also be specified per partition hinst and top level. This attribute is not supported for designs containing master/clone partitions. Type: string Default: "" Edit: Yes
cts_flexible_htree_auto_sinks_rects	
	The automatically computed sink rectangles of the flexible H-tree when auto sinks are specified. This attribute is not supported for designs containing master/clone partitions. Type: string Default: "" Edit: No
cts_flexible_htree_auto_sinks_sink_area	
	The initial width and height of automatically computed sink rectangles of the flexible H-tree, which can be increased due to floorplan restrictions. This attribute is not supported for designs containing master/clone partitions. The attribute is mandatory if auto_sinks attribute is specified. Type: string Default: "" Edit: Yes
cts_flexible_htree_final_cell	
	The library cell to use for the H-tree sinks. Valid values: string Type: string Default: "" Edit: Yes
cts_flexible_htree_hv_balance	

Stylus Common UI Database Object Information

Database Objects--flexible_htree

	Specifies whether horizontal and vertical wires can only be balanced against other wires of the same orientation (true), or whether any wire can be balanced against any other wire (false). Valid values: true false Type: bool Default: true Edit: Yes
cts_flexible_htree_image_directory	
	Name of the directory to which images generated by the H-tree synthesis algorithm are written. Valid values: string Color coding of images: - White Unobstructed edges of the synthesis grid - Red Grid points that are blocked for trunk cell placement in all modules - Orange Grid points that are blocked for final cell placement in all modules - Red orange Grid points that are blocked for trunk and final cell placement in all modules - Yellow circle The grid point of the source - Yellow points Candidate grid points of H-tree sinks - Yellow rectangle The sink area containing target grid point candidates of H-sinks, adjusted to the synthesis grid - Brown If specified, the sink grid box adjusted to the synthesis grid - Green/blue The edges of the synthesized H-tree - Purple H-tree repeaters - Cyan Partition boundaries in hierarchical designs Type: string Default: "" Edit: Yes
cts_flexible_htree_inverting	
	Specifies whether the flexible H-tree will invert its input. Valid values: boolean Type: bool Default: false Edit: Yes
cts_flexible_htree_layer_density	
	The assumed layer density used to compute the parasitics for timing estimates of H-tree nets during H-tree synthesis based on the non default rule for top nets. Valid values: Any float in the range 0 to 1. Type: double Default: 1 Edit: Yes
cts_flexible_htree_max_driver_distance	
	If specified, ignore DRVs and use the given value as the maximum length of the nets connecting H-tree drivers. Valid values: float Type: string Default: auto Edit: Yes
cts_flexible_htree_max_root_distance	
	If specified, ignore DRVs and use the given value as the maximum length of the net connecting the root and the first driver of the H-tree. This value overrides the attribute cts_max_driver_distance of this net and can only be specified if -max_driver_distance is also specified. Valid values: float Type: string Default: auto Edit: Yes
cts_flexible_htree_mode	

Stylus Common UI Database Object Information

Database Objects--flexible_htree

	The driver insertion mode for the H-tree. If set to 'drv', the algorithm inserts drivers to avoid DRV's. It also minimizes the insertion delay of the H-tree if the power weight is less than 1.0. If set to 'distance', the attributes 'cts_max_driver_distance' and 'cts_max_root_distance' determine the maximum net lengths allowed. Transitions and delays are not computed in this case. Valid values: drv distance Type: string Default: drv Edit: Yes
cts_flexible_htree_omit_symmetry	
	Controls the omission of symmetry features to balance a flexible H-tree. Symmetry drivers are drivers that are added to balance pin capacitances at branch points. For example, if buffer pairs are inserted at branch points, one of these buffers may not drive any fanout and is inserted to match other buffer pairs at the same level of the flexible H-tree. Omitting symmetry drivers can reduce the power of the H-tree but increase its skew. Drivers that are added to terminate symmetry branches (see below) do not count as symmetry drivers and are never omitted. Similarly, symmetry branches are branches that are needed to balance the wire load at branch points. They are added to match other branchpoints at the same level of the flexible H-tree. Symmetry branches are always terminated with drivers, which are not counted as symmetry drivers. By default, symmetry branches and drivers are added. Possible values for this attribute: false Add symmetry branches and drivers true Omit both symmetry drivers and branches drivers Omit symmetry drivers branches Omit symmetry branches {drivers branches} Omit both symmetry drivers and branches { } Add symmetry branches and drivers Type: string Default: auto Edit: Yes
cts_flexible_htree_partition_boundary_polarity	
	Specifies polarity of the clock signal with regard to the source of the flexible H-tree when entering partitions. Valid values: non_inverting inverting ignore Type: string Default: non_inverting Edit: Yes
cts_flexible_htree_partition_groups	
	The groups in which partition are clustered in channelless designs. Nested lists imply allowed crossings between groups. Each group has zero or one input port and each partition must only be specified once. Optionally, a maximum pre-route net length from the boundary of a partition group can be specified. This argument must be specified as the second parameter of a nested partition group. Optionally, the next argument specifies the clock phase when entering the partition group in relation to the root pin of the H-tree. Allowed values are 'inverting' and 'non-inverting'. If no value is specified, the clock phase is unconstrained. The clock phase is unconstrained when crossing the boundaries of partitions within the same group. All specified clock phases must be either inverting or non-inverting. Example: <pre>{{A} {{C D} 50 non_inverting {{E F} {{G}}}}</pre> - The H-tree starts in partition A and descends into group C/D. - From partition group C/D, the tree descends into groups E/F and G. - Any clustering of sinks inside the C/D and E/F groups are allowed, potentially crossing internal partition boundaries several times. - Partition A has no clock input port and one clock output port - Partition group C/D has one clock input port and two clock output ports - Partition group E/F and G have one clock input port and no clock output port - The maximum net length from the entry point into partition group C/D is 50um (pre-route) - The clock phase is non-inverting when entering partition group C/D. The clock phase is unconstrained when entering other partition groups. Valid values: string Type: string Default: "" Edit: Yes
cts_flexible_htree_power_weight	

Stylus Common UI Database Object Information

Database Objects--flexible_htree

	The power versus insertion delay trade-off. The value specifies the weight that is put on the optimization of power related attributes , in particular the number of repeaters in the H-tree, during the synthesis of a flexible H-tree. Valid values: Any float in the range 0 to 1. Type: string Default: auto Edit: Yes
cts_flexible_htree_root	
	The pin under which the H-tree is created. This pin must be part of a clock tree at the time of synthesis. Valid values: pin Type: string Default: "" Edit: Yes
cts_flexible_htree_sink_grid	
	Specifies the columns and rows of a grid of H-tree sinks, where the columns and rows may be swapped if the cts_flexible_htree_adjust_sink_grid_for_aspect_ratio attribute is true. Valid values: {columns rows} Type: string Default: "" Edit: Yes
cts_flexible_htree_sink_grid_box	
	The box describing the area that the grid of H-tree sinks should cover. This attribute only has an effect if the cts_flexible_htree_sink_grid attribute of the flexible H-tree is set. Valid values: {xmin ymin xmax ymax} Type: string Default: "" Edit: Yes
cts_flexible_htree_sink_grid_exclusion_areas	
	Boxes describing zones that should not be covered by the grid of H-tree sinks. This attribute only has an effect if the cts_flexible_htree_sink_grid attribute of the flexible H-tree is set. Valid values: list {xmin ymin xmax ymax} Type: string Default: "" Edit: Yes
cts_flexible_htree_sink_grid_sink_area	
	The approximate size of the rectangle describing valid locations for final cells (given by -final_cell) per H-tree sink in the grid. This attribute only has an effect if the cts_flexible_htree_sink_grid attribute of the flexible H-tree is set. Valid values: {width height} Type: string Default: "" Edit: Yes
cts_flexible_htree_sink_instance_prefix	
	Prefix used for instance names of final cells (given by -final_cell). The name of the cell will be <prefix>_<htree_name>_<id>, where id is a running index. Valid values: string Type: string Default: "" Edit: Yes
cts_flexible_htree_sinks	
	Specifies H-tree sinks as approximate rectangular areas for locations of final cells (given by -final_cell) or pins to wire to. Valid values: list {pin {xmin ymin xmax ymax}} Type: string Default: "" Edit: Yes
cts_flexible_htree_stop_at_sdc_clock_roots	

Stylus Common UI Database Object Information

Database Objects--flow

	If specified, stop searching for parts of the clock tree through SDC clock roots when defining generated clock trees for H-tree sinks. Type: <code>bool</code> Default: <code>false</code> Edit: Yes
cts_flexible_htree_trunk_cell	
	The library cell to use inside the H-tree. Valid values: string Type: <code>string</code> Default: <code>""</code> Edit: Yes
cts_route_type_flexible_htree	
	Route type to use for flexible h-trees. Type: <code>string</code> Default: <code>default</code> Edit: Yes
cts_route_type_flexible_htree_backside	
	Route type to use for backside flexible h-trees. Type: <code>string</code> Default: <code>default</code> Edit: Yes
insts	
	instances in the clock tree Type: <code>obj(inst)*</code> Default: <code>""</code> Edit: No
name	
	name of flexible_htree Type: <code>string</code> Default: <code>""</code> Edit: No
nets	
	nets in the clock tree Type: <code>obj(net)*</code> Default: <code>""</code> Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: <code>enum (flexible_htree)</code> Default: <code>""</code> Edit: No

flow

Parent Objects

[root](#)

Definition

flow

Attribute	Description
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Stylus Common UI Database Object Information
Database Objects--flow

end_steps	
	List of steps for the flow Type: string Default: "" Edit: Yes
feature_values	
	Feature values for instances of this flow Type: string Default: "" Edit: Yes
features	
	Features defined for this flow Type: string Default: "" Edit: Yes
name	
	The name of the flow Type: string Default: "" Edit: Yes
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (flow) Default: "" Edit: No
owner	
	Owner of this flow Type: string Default: "" Edit: Yes
run_count	
	An integer indicating how many times the flow has been run. The value is initially zero, so it may be used as a Boolean indicating that the flow has been run at all. Type: int Default: no_value Edit: Yes
skip_metric	
	Do not create a metric snapshot for the flow Type: bool Default: false Edit: Yes
start_steps	
	List of steps for the flow Type: string Default: "" Edit: Yes
steps	
	List of steps for the flow Type: string Default: "" Edit: Yes
tool	

Stylus Common UI Database Object Information

Database Objects--flow_step

	The tool to run this flow Type: string Default: "" Edit: Yes
tool_options	
	The tool options for the tool for this flow Type: string Default: "" Edit: Yes

flow_step

Parent Objects

[root](#)

Definition

flow step

Attribute	Description
begin_tcl	
	The Tcl body to run at the start of the step Type: string Default: "" Edit: Yes
body_tcl	
	The Tcl body of the step. The initial value for this attribute is provided by the create_flow_step command. Type: string Default: "" Edit: Yes
categories	
	Metric categories to calculate for this flow step. Type: string Default: "" Edit: Yes
check_tcl	
	A block of Tcl to run to perform step-specific static checks. Type: string Default: "" Edit: Yes
description	
	Purpose for this flow step. Type: string Default: "" Edit: Yes
end_steps	
	A list of flows or flow_steps that will be run at the end of this step. Type: string Default: "" Edit: Yes
end_tcl	

Stylus Common UI Database Object Information
Database Objects--flow_step

	The Tcl body to run at the end of the step. Type: string Default: "" Edit: Yes
exclude_time_metric	
	Do not include cpu and wall time of this step in any parent steps Type: bool Default: false Edit: Yes
feature_values	
	Feature values for instances for this flow step Type: string Default: "" Edit: Yes
features	
	Features defined for this flow step Type: string Default: "" Edit: Yes
name	
	The name of the step Type: string Default: "" Edit: Yes
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (flow_step) Default: "" Edit: No
owner	
	Owner of this step. Type: string Default: "" Edit: Yes
run_count	
	An integer indicating how many times the step has been run. The value is initially zero, so it may be used as a Boolean indicating that the step has been run at all. Type: int Default: no_value Edit: Yes
skip_db	
	Do not auto-save a db at the end of this step Type: bool Default: true Edit: Yes
skip_metric	
	Do not auto-save a db at the end of this step Type: bool Default: false Edit: Yes
start_steps	

	A list of flows or flow_steps that will be run at the start of this step. Type: string Default: "" Edit: Yes
status	
	Flow step status Type: string Default: "" Edit: Yes

foreign_cell

Parent Objects

[base_cell](#), [root](#)

Definition

LEF MACRO FOREIGN information

Attribute	Description
name	
	Name of the foreign cell (can be the same as the master if only a shift/offset is required) Type: string Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (foreign_cell) Default: "" Edit: No
orient	
	Orientation of the foreign cell (non-R0 values are only allowed if the cell name is different from the master cell) Type: enum Enum Values: r0 r90 r180 r270 mx mx90 my my90 Default: "" Edit: No
point	
	Location (offset) of the foreign cell Type: point Default: "" Edit: No

gcell

Parent Objects

[design](#), [root](#)

Definition

A global-routing cell. It only exists after global routing has been run, and includes the number of routing tracks available (supply), and the tracks used (demand) for each gcell.

Stylus Common UI Database Object Information
Database Objects--foreign_cell

Attribute	Description
base_name	
	Object name derived from index value Type: string Default: "" Edit: No
demand	
	Number of routing tracks used in the preferred routing direction for the layer. If no layer index is given, the value is not useful, so -1 is returned. Type: int Allowed -index values: layer Default: 0 Edit: No
horizontal_demand	
	Number of routing tracks used in horizontal direction for all routing layers. Type: int Default: 0 Edit: No
horizontal_remaining	
	Number of routing tracks remaining in horizontal direction for all routing layers. Type: int Default: 0 Edit: No
horizontal_supply	
	Number of routing tracks available in horizontal direction for all routing layers. Type: int Default: 0 Edit: No
index	
	The X and Y index values into the gcell array. The bottom, left gcell has index = {0 0} Type: int* Default: "" Edit: No
name	
	Object name derived from index value Type: string Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (gcell) Default: "" Edit: No
rect	
	Physical dimensions {{llx lly} {urx ury}} Type: rect Default: "" Edit: No
remaining	

Stylus Common UI Database Object Information

Database Objects--gcell_grid

	Number of routing tracks remaining in the preferred routing direction for the layer. A negative value indicates overflow. If no layer index is given, the value is not useful, so -1 is returned. Type: int Allowed -index values: layer Default: 0 Edit: No
supply	
	Number of routing tracks available in the preferred routing direction for the layer. If no layer index is given, the value is not useful, so -1 is returned. Type: int Allowed -index values: layer Default: 0 Edit: No
vertical_demand	
	Number of routing tracks used in vertical direction for all routing layers. Type: int Default: 0 Edit: No
vertical_remaining	
	Number of routing tracks remaining in vertical direction for all routing layers. Type: int Default: 0 Edit: No
vertical_supply	
	Number of routing tracks available in vertical direction for all routing layers. Type: int Default: 0 Edit: No

gcell_grid

Parent Objects

[design](#), [root](#)

Definition

Equivalent to DEF GCELLGRID statements

Attribute	Description
direction	
	Specifies the location and direction of the first grid defined. x indicates vertical lines; y indicates horizontal lines. Type: enum Enum Values: y x Default: "" Edit: No
num_grids	
	Specifies the number of grid lines to create (number of rows or columns is numGrids-1) Type: int Default: "" Edit: No
obj_type	

Stylus Common UI Database Object Information

Database Objects--group

	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (gcell_grid) Default: "" Edit: No
start	
	Specifies the coordinate of the first line Type: coord Default: "" Edit: No
step	
	Specifies the spacing between the grids Type: coord Default: "" Edit: No

group

Parent Objects

[hinst](#), [inst](#), [design](#), [power_domain](#), [root](#),

Definition

group of hinsts, insts, or groups

Attribute	Description
area	
	Area of the group as defined by the LEF MACRO SIZE or OVERLAP information Type: area Default: "" Edit: No
base_name	
	Name of group Type: string Default: "" Edit: No
constraint_type	
	Constraint type for the group Type: enum Enum Values: guide region fence cluster none Default: "" Edit: Yes
density	
	Group density (legal range: 0.0-1.0) Type: double Default: no_value Edit: Yes
exclusive_group_gap	
	This is the gap should be maintained between exclusive_groups (per safety island groups). The value is measured in microns. It can be set by command create_exclusive_groups -gap. It is only valid when the group's constraint_type is region or fence. Type: coord Default: 5 Edit: No

Stylus Common UI Database Object Information

Database Objects--group

exclusive_group_type	
	This attribute specifies the logical type of this group. It describes the logical relationship of the group relative to other groups which can be used to implement safety islands in automotive application designs. Type: enum Enum Values: none master checker mutually_exclusive Default: none Edit: No
exclusive_groups	
	This attribute specifies a list of groups that are exclusive of this group. It can be used to implement safety islands in automotive application designs. Type: string Default: "" Edit: No
is_floating	
	Only affects groups with .constraint_type = fence or region. If true, the global placer can move the fence or region. The .rects value must also be set, and is currently restricted to a single rect. The global placer will not change the size of the rect, but may move it. Type: bool Default: false Edit: Yes
members	
	List of group's members Type: obj(hinst)* obj(inst)* obj(group)* Default: "" Edit: No
name	
	Name of group Type: string Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (group) Default: "" Edit: No
parent	
	The parent group if is a sub-group Type: obj(group) Default: "" Edit: No
power_domain	
	The Power Domain (if group is a power domain) Type: obj(power_domain) Default: "" Edit: No
rects	
	List of non-overlapping rectangles that defines the shape of the group Type: rect* Default: "" Edit: Yes
safety_dcls_isolate_clock	

Stylus Common UI Database Object Information

Database Objects--group

	The type of isolation required for the clock signals driving this dcls (dual core lock step) group. Supported values: Type: enum Enum Values: none split_net split_network Default: none Edit: No
safety_dcls_isolate_inputs	
	The type of isolation required for all the input signals driving this dcls (dual core lock step) group. Supported values: Type: enum Enum Values: none split_net Default: none Edit: No
safety_dcls_isolate_is_halo	
	When true, this group is an isolation halo for a dcls (dual core lock step) group. The isolation cells on the input/output nets for the dcls-group will be placed inside this halo around the dcls-group boundary. The halo-group will be the parent-group of the dcls-group. Type: bool Default: false Edit: No
safety_dcls_isolate_outputs	
	The type of isolation required for all the output signals driven by this dcls (dual core lock step) group. Supported values: Type: enum Enum Values: none split_net Default: none Edit: No
safety_dcls_isolate_reset	
	The type of isolation required for the reset signals driving this dcls (dual core lock step) group. Supported values: Type: enum Enum Values: none split_network split_net Default: none Edit: No
safety_dcls_isolate_scan_enable	
	The type of isolation required for the scan_enable signals driving this dcls (dual core lock step) group. Supported values: Type: enum Enum Values: none split_network split_net Default: none Edit: No
safety_dcls_isolate_use_halo	
	When true, the isolation cells on the input/output nets for this dcls (dual core lock step) group will be placed inside a halo around the dcls-group boundary. The halo-group will be the parent-group of the dcls-group. Type: bool Default: false Edit: No
safety_dcls_route_use_halo	
	When true, the internal nets for this dcls (dual core lock step) group will be routing inside a halo around the dcls-group boundary. Type: bool Default: false Edit: No
safety_failure_mode	

Stylus Common UI Database Object Information

Database Objects--gui_line

	The name of the failure-mode to which this group belongs, as defined in the USF (Unified Safety Format) file. Type: string Default: "" Edit: No
safety_mechanism	
	The name of the safety-mechanism applied to this group, as defined in the USF (Unified Safety Format) file. Type: string Default: "" Edit: No
safety_mechanism_type	
	The type of safety-mechanism applied to this group, as defined in the USF (Unified Safety Format) file. Type: string Default: "" Edit: No

gui_line

Parent Objects

[design](#), [root](#)

Definition

A line that can be displayed on the GUI, and is not output to DEF.

Attribute	Description
arrow	
	Draw arrow in the middle of the line with direction from start point to end point. Type: bool Default: "" Edit: No
gui_layer_name	
	Normally a GUI-only layer name that is not a tech-file layer. If the name is the same as a tech-file layer, it will be drawn on the GUI with other shapes on that layer, but it will not be output to DEF. Type: string Default: "" Edit: No
name	
	GUI line name. Type: string Default: "" Edit: Yes
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (gui_line) Default: "" Edit: No
pixel_width	
	Number of pixels in the border of the object. Type: int Default: "" Edit: No

points	
	2 points for line Type: point* Default: "" Edit: No

gui_polygon

Parent Objects

[design](#), [root](#)

Definition

The polygon shape that can be displayed on the GUI, and is not output to DEF.

Attribute	Description
gui_layer_name	
	Normally a GUI-only layer name that is not a tech-file layer. If the name is the same as a tech-file layer, it will be drawn on the GUI with other shapes on that layer, but it will not be output to DEF. Type: string Default: "" Edit: No
name	
	GUI polygon name. Type: string Default: "" Edit: Yes
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (gui_polygon) Default: "" Edit: No
pixel_width	
	Number of pixels in the border of the object. Type: int Default: "" Edit: No
points	
	point list for poly Type: point* Default: "" Edit: No

gui_rect

Parent Objects

[design](#), [root](#)

Stylus Common UI Database Object Information

Database Objects--gui_text

Definition

A rect that can be displayed on the GUI, and is not output to DEF.

Attribute	Description
gui_layer_name	
	Normally a GUI-only layer name that is not a tech-file layer. If the name is the same as a tech-file layer, it will be drawn on the GUI with other shapes on that layer, but it will not be output to DEF. Type: string Default: "" Edit: No
name	
	GUI rect name. Type: string Default: "" Edit: Yes
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (gui_rect) Default: "" Edit: No
pixel_width	
	Number of pixels in the border of the object. Type: int Default: "" Edit: No
rect	
	Rectangle that defines the gui_rect shape. Type: rect Default: "" Edit: No

gui_text

Parent Objects

[design](#), [root](#)

Definition

A text that can be displayed on the GUI, and is not output to DEF.

Attribute	Description
gui_layer_name	
	GUI layer name Type: string Default: "" Edit: No
height	
	Text height Type: coord Default: "" Edit: No

Stylus Common UI Database Object Information

Database Objects--hinst

label	
	Text string Type: string Default: "" Edit: No
location	
	Text location (lower left) Type: point Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (gui_text) Default: "" Edit: No
orient	
	Text Orientation Type: enum Enum Values: r0 r90 r180 r270 mx mx90 my my90 Default: "" Edit: No

hinst

Parent Objects

[violation](#), [group](#), [hport_bus](#), [hpin](#), [hnet](#), [actual_scan_segment](#), [partition](#), [design](#), [root](#), [boundary](#), [hport](#), [module](#), [inst](#), [scan_segment](#), [pin](#), [hpin_bus](#)

Definition

Hierarchical instance

Attribute	Description
apply_booth_encoding	
	Apply booth encoding by using user specified method (for a specified instance). Type: enum Enum Values: nonbooth auto_bitwidth auto_togglerate manual inherited Default: inherited Edit: Yes
area	
	The area of all the insts contained inside this hinst and below it. Type: area Default: "" Edit: No
base_name	
	The name at the base of a hierarchical name. So the base_name of i1/i2/i3 is i3. Type: string Default: "" Edit: No
bbox	

Stylus Common UI Database Object Information

Database Objects--hinst

	Bounding box of all the insts inside this hinst. Type: <code>rect</code> Default: "" Edit: No
boundary	
	The placement boundary (e.g. fence, region, group, cluster) for this hinst if it exists. Type: <code>obj(boundary)</code> Default: "" Edit: No
clock_gating_test_signal	
	Set specified test signal to connect up test logic on the clock-gating instance. Type: <code>string</code> Default: "" Edit: Yes
dft_abstract_dont_scan	
	Exclude this instance from scan-synthesis as an abstract segment has been defined on it. Type: <code>bool</code> Default: false Edit: No
dft_dont_scan	
	Exclude this design from scan-synthesis (inherited true false). Type: <code>enum</code> Enum Values: false true inherited Default: inherited Edit: Yes
dft_exclude_flop_from_los_pipeline	
	Avoid insertion of launch-off-shift pipelined logic to flop's shift enable. Type: <code>bool</code> Default: false Edit: Yes
dft_force_blackbox_for_atpg	
	Determines is the instance is blackbox for ATPG. Type: <code>enum</code> Enum Values: none true false Default: none Edit: Yes
dft_is_blackbox_for_atpg	
	Blackbox for ATPG. Type: <code>bool</code> Default: false Edit: No
dft_is_los_pipeline_flop	
	Determines if this flop is launch-off-shift pipeline flop. Type: <code>bool</code> Default: false Edit: Yes
dft_part_of_segment	
	The type of segment this instance belongs to {abstract fixed floating preserved shift_register}. Type: <code>enum</code> Enum Values: fixed floating preserve abstract shift_register Default: "" Edit: No
dft_scan_chain	

Stylus Common UI Database Object Information

Database Objects--hinst

	The actual scan chain this instance belongs to. Type: <code>obj(scan_chain) obj(scan_segment) obj(actual_scan_segment) obj(actual_scan_chain)</code> Default: "" Edit: No
dft_scan_chain_in_multi_mode	
	The actual scan chains this instance belongs to in multi mode stitching. Type: <code>obj(scan_chain)* obj(scan_segment)* obj(actual_scan_segment)* obj(actual_scan_chain)*</code> Default: "" Edit: No
dont_retime	
	Do not retime this instance. Type: <code>bool</code> Default: false Edit: Yes
dont_touch	
	This attributes defines the user preservation status of the hinst during optimization. Setting this attribute will set the dont_touch attribute on the parent module of this hinst and all hinsts of the same module. This setting will apply to all insts within the hinst unless overridden at a lower level hinst or on the inst object itself. The dont_touch_effective attribute on each child inst and hinst will return the resolved value. Use 'help inst dont_touch' to see the enum value definitions but note that map_size_ok is only allowed at the instance level. Type: <code>enum</code> Enum Values: none false true delete_ok const_prop_size_delete_ok const_prop_delete_ok size_delete_ok size_ok size_same_height_ok size_same_footprint_ok Default: none Edit: Yes
dont_touch_effective	
	This attribute defines the effective preservation status of this hinst during optimization from the dont_touch_sources values. If the partition source is 'true' or the ilm source is 'true', then the effective value for the hinst is 'true'. Otherwise, the user value has precedence. If the user value is false, then the parent value is returned. See help on inst dont_touch attribute for details on possible values. Type: <code>enum</code> Enum Values: none false true delete_ok const_prop_size_delete_ok const_prop_delete_ok size_delete_ok size_ok size_same_height_ok size_same_footprint_ok Default: false Edit: No
dont_touch_file	
	Specifies the sourced file to set dont_touch. Type: <code>string</code> Default: "" Edit: Yes
dont_touch_hports	
	This attribute defines the user preservation status for the hports of this hinst during optimization. Supported values: - none: Unconstrained - false: Can add/remove ports - true: Cannot add/remove ports - delete_ok: Can delete ports (if they have no fanout) - add_ok: We cannot delete, or change the polarity or any hport but can add or duplicate hports - invert_ok: We cannot delete, duplicate, or add but can change the polarity of any hport - add_invert_ok: We cannot delete any hport but can add, duplicate, and change the polarity Type: <code>enum</code> Enum Values: none false true delete_ok add_ok invert_ok add_invert_ok Default: none Edit: Yes
dont_touch_sources	

Stylus Common UI Database Object Information

Database Objects--hinst

	Dictionary of {source <value>} pairs contributing to the dont_touch_effective attribute for this object : {user <value>} {parent <value>} {read_only_effective <value>} user # This hinst dont_touch value parent # The effective dont_touch value from an hinst above (the closest hinst above that is not false) read_only_effective # True when this hinst is read_only because it is an ilm or set_module_view set it or a parent partition to read_only. Local read_only value overrides parent read_only value Type: string Default: "" Edit: No
dont_use_cells	
	List of cell names (wildcards supported) to disallow for this hinst during optimization. Setting on an hinst sets the attribute on the module of the hinst (setting on the hinst is just for convenience). If a cell is added to this list that is already in the .use_cells list, it will be removed from the .use_cells list so that the lists are non-overlapping. Type: string Default: "" Edit: Yes
dont_use_cells_effective	
	The resolved list of all cell names to disallow during optimization for this hinst, based on the library dont_use and the dont_use_cells and use_cells attributes of this hinst or the closest parent hinst with a non-empty list. The precedence is: union of the use_cells of this hinst (or closest parent if empty), then dont_use_cells of this hinst (or closest parent if empty), then the library dont_use setting. When there are multiple hinsts that share the same module, the dont_touch_effective is calculated for the master hinst and the other (clone) hinsts inherit. A cell C is in the don't use cell effective list for a hinst H if and only if. (1) Walk from the hinst H up in the parent hierarchy (including H) to find the closest parent hinst where C is referred to in use cell list or don't use cell list. (2) If such a hinst is found and if it occurs as use cell in hinst then it is not in the effective don't use list. If it occurs in the don't_use cell list of hinst then it is in the effective don't use list. (3) If such a hinst is not found the library don't use determines if the cell is in the effective don't use list or not. Type: string Default: "" Edit: No
dp_sharing	
	To control the datapath sharing on a datapath instance in 'synthesize -to_generic' (advanced/none/inherited). Type: enum Enum Values: advanced none inherited Default: inherited Edit: Yes
dpopt_instance_constant_input_based_speculation	
	Controls constant input based datapath speculation of datapath module in 'syn_generic'. Type: bool Default: false Edit: Yes
escaped_name	
	The full hierarchical name including escaped chars (if any). It follows DEF escaping syntax, so bus-bit chars [], or a hierarchy char \ that is not a bus-bit or hierarchy char has a \ in front of it. So i1/i2 is a two-level hierarchical name while i1\i2 is single level name, and a[0] is a bus-bit, while a\0\ is a scalar. Type: string Default: "" Edit: No
exceptions	
	Reports the path exceptions Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
fcu_instruction_set	

Stylus Common UI Database Object Information

Database Objects--hinst

	Attribute to define different repair specific instructions for the different instances of the same block on which fuse control unit has been inserted. Type: <code>string</code> Default: "" Edit: Yes
group	
	The parent group Type: <code>obj(group)</code> Default: "" Edit: No
hdl_proc_name	
	Name of the process (VHDL) or always block (Verilog) from which the instance is inferred. Type: <code>string</code> Default: "" Edit: Yes
hdl_vname	
	The name for this instance as specified in the input HDL file (Verilog or VHDL). Type: <code>string</code> Default: "" Edit: No
hinsts	
	List of all the hinsts at the current level and below this hinst Type: <code>obj(hinst)*</code> Default: "" Edit: No
hnets	
	List of all the hnets at the current level and below this hinst Type: <code>obj(hnet)*</code> Default: "" Edit: No
hpin_busses	
	List of all the hpin_bus at the current level and below this hinst Type: <code>obj(hpin_bus)*</code> Default: "" Edit: No
hpins	
	List of all the hpins at the current level and below this hinst Type: <code>obj(hpin)*</code> Default: "" Edit: No
hport_busses	
	List of all the hport_bus of this hinst. Type: <code>obj(hport_bus)*</code> Default: "" Edit: No
hports	
	List of all the hports of this hinst (e.g. from inside the hinst looking up at the hinst boundary). Type: <code>obj(hport)*</code> Default: "" Edit: No
ilm_inst	

Stylus Common UI Database Object Information

Database Objects--hinst

	Specifies the inst object of this ilm when it is under unflatten view. This attribute is only valid when the ilm is under flatten view. Type: obj(inst) Default: "" Edit: No
instance_internal_power	
	The instance's internal power. Type: double Default: no_value Edit: Yes
instance_leakage_power	
	The instance's leakage power. Type: double Default: no_value Edit: Yes
insts	
	List of all the insts at the current level and below this hinst Type: obj(inst)* Default: "" Edit: No
is_ilm	
	This attribute is true if the hinst is an ILM. It will affect the read_only_effective and dont_touch_effective of this hinst and all hinsts within it, as well as the dont_touch_effective of all insts within it. It cannot be overridden by other hinst or inst values. Type: bool Default: false Edit: No
is_partition_defined	
	Describes if there is any partition definitions on the hinst. Type: bool Default: "" Edit: No
local_hinsts	
	List of hinsts in the current level of this hinst (e.g. from inside the hinst looking down at the hinsts at this level) Type: obj(hinst)* Default: "" Edit: No
local_hnets	
	List of hnets in the current level of this hinst (e.g. from inside the hinst looking down at the hnets at this level) Type: obj(hnet)* Default: "" Edit: No
local_hpins	
	List of hpins in the current level of this hinst (e.g. from inside the hinst looking down at the hpins of hinsts at this level) Type: obj(hpin)* Default: "" Edit: No
local_insts	
	List of insts in the current level of this hinst (e.g. from inside the hinst looking down at the insts at this level) Type: obj(inst)* Default: "" Edit: No
local_pins	

Stylus Common UI Database Object Information

Database Objects--hinst

	List of pins in the current level of this hinst (e.g. from inside the hinst looking down at the pins of the insts at this level) Type: obj(pin)* Default: "" Edit: No
lp_dynamic_analysis_scope	
	Set VCD analysis scope attribute on an instance in the design hierarchy. Valid values for this attribute are true or false. Type: bool Default: false Edit: Yes
merge_combinational_hier_instance	
	Allow the combinational hierarchical instance to be merged. Type: enum Enum Values: 1 0 true false inherited Default: inherited Edit: Yes
module	
	The module cell that corresponds to the hInst Type: obj(module) Default: "" Edit: No
name	
	The same as .escaped_name without any \ escape chars. This name is commonly used to avoid problems with \ escape chars in Tcl scripts unless you carefully use list operators. Like SDC commands, 'get_db insts i1/i2' will first try to match i1/i2, and if not found then match i1\i2 so that flattening hierarchical names (e.g. with ungroup) does not require changing the names in Tcl scripts. Type: string Default: "" Edit: No
num_exceptions	
	Reports the number of path exceptions Type: int Allowed -index values: analysis_view clock Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (hinst) Default: "" Edit: No
orient	
	Orientation of the partition. Type: enum Enum Values: r0 r90 r180 r270 mx mx90 my my90 Default: "" Edit: Yes
original_name	
	Original rtl names of single-bit flops. Type: string Default: "" Edit: Yes
original_names_for_split	

Stylus Common UI Database Object Information

Database Objects--hinst

	Returns the original name of single-bit flops merged. Type: string Default: "" Edit: Yes
parent	
	The parent of the hinst, which is either another hinst, or the design object. Type: obj(hinst) obj(design) Default: "" Edit: No
part_power_intent_file	
	The path of file which will contain final power intent for this block. Type: string Default: "" Edit: Yes
partition	
	If the hinst is a partition/blackbox, this partition will contain the pin constraints and related data for partition/physical blackbox. Otherwise it is empty. Type: obj(partition) Default: "" Edit: No
pmbist_instruction_set	
	Attribute to define different programmable MBIST instructions for the different instances of the same block on which programmable MBIST has been inserted. Type: string Default: "" Edit: Yes
power_domain	
	Power domain of the hierarchical inst if in a group Type: obj(power_domain) Default: "" Edit: No
power_dynamic	
	The sum of the power_dynamic value for all the insts inside this hinst. A value of no_value is treated as 0. Type: double Default: "" Edit: No
power_internal	
	The sum of the power_internal value for all the insts inside this hinst. A value of no_value is treated as 0. Type: double Default: "" Edit: No
power_leakage	
	The sum of the power_leakage value for all the insts inside this hinst. A value of no_value is treated as 0. Type: double Default: "" Edit: No
power_switching	
	The sum of the power_switching value for all the insts inside this hinst. A value of no_value is treated as 0. Type: double Default: "" Edit: No
power_toggle_rate	

Stylus Common UI Database Object Information

Database Objects--hinst

	The average of the power_toggle_rate value for all the insts inside this <hinst/design>. A value of no_value is treated as 0 Type: double Default: "" Edit: No
power_total	
	The sum of the power_total value for all the insts inside this hinst. A value of no_value is treated as 0. Type: double Default: "" Edit: No
propagate_constant_from_timing_model	
	Allow constant propagation from specific timing model instance. Type: enum Enum Values: 1 0 true false inherited Default: inherited Edit: Yes
read_only	
	This attribute is true if the hinst is a read_only. This is set by the set_module_view command. When true, this hinst cannot be optimized and cells inside will not be moved. This attribute will affect the dont_touch_effective and place_status_effective attributes on all insts and hinsts within this hinst. It cannot be overridden by other hinst or inst values. Supported values: false: This hinst is allowed to be optimized true: The hinst read_only due to partition or ILM status none: No constraint. Type: enum Enum Values: none false true Default: none Edit: No
read_only_effective	
	This attribute defines the read_only status of this hinst. This can be true when the local .read_only attribute is true or if a parent .read_only attribute is true. The hinst cannot be optimized and cells inside will not be moved. Supported values: false: This hinst is allowed to be optimized true: The hinst read_only due to local or parent read_only partition Type: bool Default: false Edit: No
ref_name	
	ref_name Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
retime_original_registers	
	Original registers before retiming. Type: string Default: "" Edit: Yes
single_bit_original_name	
	Returns the hdl name of the original single bits. Type: string Default: "" Edit: Yes
trace_retime	

Stylus Common UI Database Object Information

Database Objects--hnet

	Trace this instance through retiming, use the attribute 'retime_original_registers' to get the name of the original instance. Type: <code>bool</code> Default: <code>false</code> Edit: Yes
ungroup_allowed	
	Controls whether this hierarchical instance should be (manually or automatically) ungrouped. true: allows ungrouping on all modules/hinsts. false: disallows ungrouping on all RTL modules/hinsts but still allows ungrouping of tool-generated modules/hinsts (identified by <code>hdl_instantiated=false</code>). <code>strict_no</code> : disallows ungrouping on all modules/hinsts. Type: <code>enum</code> Enum Values: <code>false</code> <code>true</code> <code>strict_no</code> Default: <code>true</code> Edit: Yes
unresolved	
	Returns true if the module is not from input netlist. Type: <code>bool</code> Default: <code>""</code> Edit: No
use_cells	
	List of <code>base_cell</code> names to allow for this hinst during optimization. Setting on an hinst sets the attribute on the module of the hinst (setting on the hinst is just for convenience). All <code>lib_cells</code> of each <code>base_cell</code> will be allowed. If a cell is added to this list that is already in the <code>.dont_use_cells</code> list, it will be removed from the <code>.dont_use_cells</code> list so that the lists are non-overlapping. Type: <code>string</code> Default: <code>""</code> Edit: Yes

hnet

Parent Objects

[hinst](#), [design](#), [hpin](#), [pin](#), [net](#), [root](#), [hport](#), [pg_pin](#), [port](#)

Definition

Hierarchical net

Attribute	Description
base_name	
	The name at the base of a hierarchical name. So the <code>base_name</code> of <code>i1/i2/i3</code> is <code>i3</code> . Type: <code>string</code> Default: <code>""</code> Edit: No
connections	
	Returns a list of connections of the hnet (hpins and hports). Type: <code>obj(pin)* obj(hpin)* obj(port)* obj(hport)*</code> Default: <code>""</code> Edit: No
constant	
	Returns if this hnet has a constant logic value in the Verilog (<code>supply0/supply1</code> or assigned <code>1'b0/1'b1</code>). If it is <code>no_constant</code> , and it is connected to a net, you must check the net <code>.constant</code> value to see if the net is constant for other reasons. A <code>supply0/1</code> hnet will cause its net to have <code>.use = power</code> or <code>ground</code> . Type: <code>enum</code> Enum Values: <code>0</code> <code>1</code> <code>no_constant</code> Default: <code>no_constant</code> Edit: No

Stylus Common UI Database Object Information

Database Objects--hnet

dft_constant_value	
	Constant net value in test mode {logic_0 logic_1 logic_z no_value}. Type: <code>string</code> Default: "" Edit: No
dont_touch	
	This attribute defines the preservation status of an hnet during optimization. Setting this will preserve all connections on the hnet at the level of hierarchy where the hnet exists (i.e. will stop at the hpins and hports connected to this hnet). Supported values: false: Unconstrained true: Cannot touch delete_ok: Can delete (if net has no sinks) Type: <code>enum</code> Enum Values: false true delete_ok const_prop_delete_ok Default: false Edit: Yes
dont_touch_file	
	Specifies the sourced file to set dont_touch. Type: <code>string</code> Default: "" Edit: Yes
drivers	
	local to this hinst: pins, hpins that are output or bider, and hports. Ports that are input or bidir Type: <code>obj(pin)* obj(hpin)* obj(port)* obj(hport)*</code> Default: "" Edit: No
escaped_name	
	The full hierarchical name including escaped chars (if any). It follows DEF escaping syntax, so bus-bit chars [], or a hierarchy char \ that is not a bus-bit or hierarchy char has a \ in front of it. So i1/i2 is a two-level hierarchical name while i1\i2 is single level name, and a[0] is a bus-bit, while a\0\ is a scalar. Type: <code>string</code> Default: "" Edit: No
hinst	
	The parent hinst for this hnet Type: <code>obj(hinst)</code> Default: "" Edit: No
loads	
	local to this hinst: pins, hpins that are input or bider, and hports. Ports that are output or bidir Type: <code>obj(pin)* obj(hpin)* obj(port)* obj(hport)*</code> Default: "" Edit: No
lp_asserted_probability	
	Probability of signal to be high on net as specified by user. Type: <code>double</code> Default: no_value Edit: Yes
lp_asserted_toggle_rate	
	Signal toggle rate assigned to net by user. Type: <code>double</code> Default: no_value Edit: Yes

Stylus Common UI Database Object Information

Database Objects--hpin

name	
	The same as .escaped_name without any \ escape chars. This name is commonly used to avoid problems with \ escape chars in Tcl scripts unless you carefully use list operators. Like SDC commands, 'get_db insts i1/i2' will first try to match i1/i2, and if not found then match i1\i2 so that flattening hierarchical names (e.g. with ungroup) does not require changing the names in Tcl scripts. Type: string Default: "" Edit: No
net	
	canonical (flat) net associated with the hierarchical net Type: obj(net) Default: "" Edit: No
num_drivers	
	number of local drivers Type: int Default: "" Edit: No
num_loads	
	number of local loads Type: int Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (hnet) Default: "" Edit: No
timing_constant	
	Reports constant value on net Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
timing_constant_source	
	Reports source of constant value on net Type: string Allowed -index values: analysis_view clock Default: "" Edit: No

hpin

Parent Objects

[test_clock](#), [clock](#), [port](#), [violation](#), [arc](#), [hinst](#), [scan_chain](#), [hnet](#), [actual_scan_segment](#), [dft_configuration_mode](#), [design](#), [partition](#), [root](#), [hport](#), [test_signal](#), [inst](#), [scan_segment](#), [actual_scan_chain](#), [hpin_bus](#)

Definition

Hierarchical instance pin

Attribute	Description
base_name	

Stylus Common UI Database Object Information

Database Objects--hpin

	The name at the base of a hierarchical name. So the base_name of i1/i2/i3 is i3. Type: <code>string</code> Default: "" Edit: No
boundary_feedthru	
	Captures feedthrough created during boundary optimization and used during mapping file generation. Type: <code>string</code> Default: "" Edit: Yes
boundary_optimize_constant_hpins	
	Allow constant propagation through hierarchical boundary pins. Type: <code>enum</code> Enum Values: 1 0 true false inherited Default: inherited Edit: Yes
boundary_optimize_equal_opposite_hpins	
	Allow connection of hierarchical boundary pins through equal/opposite boundary pins. Type: <code>enum</code> Enum Values: 1 0 true false inherited Default: inherited Edit: Yes
boundary_optimize_feedthrough_hpins	
	Allow routing of feedthrough boundary pins around a hierarchical instance. Type: <code>enum</code> Enum Values: 1 0 true false inherited Default: inherited Edit: Yes
boundary_optimize_invert_hpins	
	Allow inversion of hierarchical boundary pins. Type: <code>enum</code> Enum Values: 1 0 true false inherited Default: inherited Edit: Yes
bus	
	The bus for this hpin. Type: <code>obj(hpin_bus)</code> Default: "" Edit: No
clock_gating_exclude_signal	
	Exclude this pin from clock gating enable functions. Type: <code>bool</code> Default: false Edit: Yes
constant	
	Returns if this hpin has a constant logic value due to a 1'b0 or 1'b1 asserted directly on the hpin in the Verilog. If it is no_constant, and it is connected to a net, you must check the net.constant value to see if the net is constant for other reasons. Type: <code>enum</code> Enum Values: 0 1 no_constant Default: no_constant Edit: No
cts_isolated	

Stylus Common UI Database Object Information

Database Objects--hpin

	Is this hpin isolated Type: <code>string</code> Default: "" Edit: Yes
cts_isolated_effective	
	Is this hpin isolated Type: <code>bool</code> Default: "" Edit: No
depth	
	Depth constraint of the hpin in microns. This is only valid for partition hpins (see <code>.partition_pin</code>) affected by pin assignment commands, and returns "" if never set. The <code>edit_pin</code> command can be used to set it. Type: <code>coord</code> Default: "" Edit: No
dft_constant_value	
	Constant pin value in test mode { <code>logic_0</code> <code>logic_1</code> <code>logic_z</code> <code>no_value</code> }. Type: <code>string</code> Default: "" Edit: No
dft_controllable	
	In test mode the output pin can take either the <code>non_inverting</code> or <code>inverting</code> value of the input pin. Type: <code>string</code> Default: "" Edit: Yes
dft_driven_by_clock	
	If pin is driven by clock. Type: <code>enum</code> Enum Values: <code>false</code> <code>true</code> Default: <code>false</code> Edit: Yes
direction	
	Direction of pin. Type: <code>enum</code> Enum Values: <code>internal</code> <code>in</code> <code>out</code> <code>inout</code> Default: "" Edit: No
dont_touch	
	The preservation status of an hpin during optimization. A preserved hpin means the logical function of the hpin must be preserved to maintain a simulation or test-point hpin in the netlist. However, the name does not need to be preserved. Supported values: <code>false</code> : Unconstrained <code>true</code> : Cannot add/remove ports <code>delete_ok</code> : Can delete ports (if they have no fanout) <code>add_ok</code> : We cannot delete, or change the polarity of any hport but can add or duplicate hports <code>invert_ok</code> : We cannot delete, duplicate, or add but can change the polarity of any hport <code>add_invert_ok</code> : We cannot delete any hport but can add, duplicate, and change the polarity <code>none</code> : No user setting; will inherit from the module/hinst Type: <code>enum</code> Enum Values: <code>none</code> <code>false</code> <code>true</code> <code>delete_ok</code> <code>add_ok</code> <code>invert_ok</code> <code>add_invert_ok</code> <code>const_prop_delete_ok</code> Default: <code>none</code> Edit: Yes
dont_touch_effective	

Stylus Common UI Database Object Information

Database Objects--hpin

	This attribute defines the effective preservation status of an hpin during optimization based on the .dont_touch_sources. If the local hpin .dont_touch is not "none", it will be used. Otherwise, the hinst .dont_touch_effective is used if not "none" or "false". If not, then the hinst .dont_touch_hports is used. Type: enum Enum Values: none false true delete_ok add_ok invert_ok add_invert_ok const_prop_delete_ok Default: false Edit: No
dont_touch_file	
	Specifies the sourced file to set dont_touch. Type: string Default: "" Edit: Yes
dont_touch_sources	
	Dictionary of {source value>} pairs contributing to the dont_touch_effective attribute for this object : {user <value>} {power_intent <value>} {hinst_dont_touch_hports <value>} {hinst_dont_touch_effective}. user # This pins dont_touch value power_intent # Value set by committing UPF/CPF dft # This <obj> is part of design-for-test logic constraints hinst_dont_touch_effective # The hinst .dont_touch_effective value hinst_dont_touch_hports # The hinst .dont_touch_hports value Type: string Default: {user none} {power_intent false} Edit: No
edge	
	If this hpin is for a partition, or inst that is a black-box, and the hpin is assigned, the edge value indicates along which edge of the boundary polygon the hpin is assigned. The edge number starts from the lowest Y, then left-most X vertex, starting with 0, and then counting clock-wise. See the set_pin_constraint command document for a figure showing the edge numbering. If the hpin is not assigned, or not for a partition or black-box inst, the value of -1 is returned. Type: int Default: "" Edit: No
escaped_name	
	The full hierarchical name including escaped chars (if any). It follows DEF escaping syntax, so bus-bit chars [], or a hierarchy char \ that is not a bus-bit or hierarchy char has a \ in front of it. So i1/i2 is a two-level hierarchical name while i1\i2 is single level name, and a[0] is a bus-bit, while a[0] is a scalar. Type: string Default: "" Edit: No
exceptions	
	Reports the path exceptions Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
hdl_vname	
	The name for this hpin as specified in the input HDL file (Verilog or VHDL). Type: string Default: "" Edit: No
hinst	
	The hinst for this hpin. Type: obj(hinst) Default: "" Edit: No
hnet	

Stylus Common UI Database Object Information

Database Objects--hpin

	The hnet connected to this hpin (above the hinst for this hpin). Type: <code>obj(hnet)</code> Default: "" Edit: No
hport	
	The hport is the internal view of this hpin from inside the hinst. Type: <code>obj(hport)</code> Default: "" Edit: No
is_booth_encoded	
	Booth encoded setting assigned to input port by user. Type: <code>bool</code> Default: false Edit: Yes
is_boundary_duplicate	
	Captures duplicated ports after boundary optimizations and attributes is used during mapping file generation. Type: <code>string</code> Default: "" Edit: Yes
is_boundary_feedback	
	Captures feedback ports after boundary optimizations and attributes is used during mapping file generation. Type: <code>string</code> Default: "" Edit: Yes
is_boundary_feedthru	
	Captures feedthrough created during boundary optimization and used during mapping file generation. Type: <code>bool</code> Default: false Edit: Yes
is_boundary_phase_inverted	
	Captures phase inversion during boundary optimization and attribute is used during mapping file generation. Type: <code>bool</code> Default: false Edit: Yes
is_initial_phase_inverted	
	This attribute is set during init_design and holds the initial value of the .is_phase_inverted attribute for the pin Type: <code>bool</code> Default: "" Edit: No
is_phase_inverted	
	Identifies the hpin for which phase is inverted during optimizations. Type: <code>bool</code> Default: false Edit: Yes
layer	
	Returns the layer if this hpin is also for a partition (see .partition_pin), otherwise it returns "". The edit_pin command can be used to change a partition hpin .layer. Type: <code>obj(layer)</code> Default: "" Edit: Yes
location	

Stylus Common UI Database Object Information

Database Objects--hpin

	Returns the location if this hpin is also for a partition (see .partition_pin) or physical hierarchical hinst, otherwise it returns "". The edit_pin command can be used to change a partition hpin .location. Type: point Default: "" Edit: No
lp_asserted_probability	
	Probability of signal to be high on port/subport/pin as specified by user. Type: double Default: no_value Edit: Yes
lp_asserted_toggle_rate	
	Signal toggle rate assigned to pin by user. Type: double Default: no_value Edit: Yes
lp_probability_type	
	Signal probability type of a pin. Type: enum Enum Values: invalid default computed clock constant asserted Default: invalid Edit: Yes
lp_system_asserted_probability	
	System/Internal asserted probability. Type: double Default: no_value Edit: Yes
lp_system_asserted_toggle_rate	
	System/Internal asserted toggle rate. Type: double Default: no_value Edit: Yes
lp_toggle_rate_type	
	Signal toggle rate type of a pin. Type: enum Enum Values: invalid default computed clock constant asserted Default: invalid Edit: Yes
lssd_master_clock	
	Defines the master clock port related to this slave clock port. Type: string Default: "" Edit: Yes
name	
	The same as .escaped_name without any \ escape chars. This name is commonly used to avoid problems with \ escape chars in Tcl scripts unless you carefully use list operators. Like SDC commands, 'get_db insts i1/i2' will first try to match i1/i2, and if not found then match i1\i2 so that flattening hierarchical names (e.g. with ungroup) does not require changing the names in Tcl scripts. Type: string Default: "" Edit: No
net	

Stylus Common UI Database Object Information

Database Objects--hpin

	The canonical (flat) net attached to this hpin. Type: obj(net) Default: "" Edit: No
num_exceptions	
	Reports the number of path exceptions Type: int Allowed -index values: analysis_view clock Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (hpin) Default: "" Edit: No
original_name	
	Stores hdl_name on output pin Type: string Default: "" Edit: Yes
partition_pin	
	If this hpin is also for an uncommitted partition (the hinst for this hpin has a .partition object) then it returns the corresponding base_pin of the partition object, otherwise it returns "". The hinst (and its .hpins and .hports) for an uncommitted partition carry the logical netlist connectivity, while the partition object carries the physical pin data (.base_pins) and pin constraints for pin assignment commands. Some of partition_pin physical and constraint data are also available directly from the hpin (see .layer, .location, .width, .depth, .side, .place_status). Type: obj(base_pin) Default: "" Edit: No
place_status	
	Returns the place_status if this hpin is also for a partition (see .partition_pin), otherwise it returns "". The edit_pin command can be used to set it. Type: enum Enum Values: unplaced placed fixed cover Default: unplaced Edit: Yes
prune_unused_logic	
	Allow pruning of logic driving a hierarchical pin. Type: bool Default: true Edit: Yes
safety_dcls_isolate_pin	
	The type of isolation required for the signal driving this pin inside a dcls (dual core lock step) group. Supported values: Type: enum Enum Values: none split_network split_net Default: none Edit: No
safety_dmr_error_signal_endpoint	
	This object is the endpoint for a safety-mechanism's error-signal. Type: string Default: "" Edit: No
side	

Stylus Common UI Database Object Information

Database Objects--hport

	Side constraint of the hpin. This is only valid for partition hpins (see .partition_pin) affected by pin assignment commands, and returns "" if never set. The edit_pin command can be used to set it. Type: enum Enum Values: north west south east up none Default: "" Edit: No
timing_constant	
	Reports constant value on pin Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
timing_constant_source	
	Reports source of constant value on pin Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
width	
	Width constraint of the hpin in microns. This is only valid for partition hpins (see .partition_pin) affected by pin assignment commands, and returns "" if never set. The edit_pin command can be used to set it. Type: coord Default: "" Edit: No
wrapper_control	
	Specifies as wrapper control signal. Type: bool Default: false Edit: Yes
wrapper_segment	
	Specifies the wrapper segments associated with the pin. Type: obj(scan_chain)* obj(scan_segment)* obj(actual_scan_segment)* obj(actual_scan_chain)* Default: "" Edit: Yes

hport

Parent Objects

[test_clock](#), [clock](#), [port](#), [violation](#), [arc](#), [hinst](#), [hport_bus](#), [hpin](#), [hnet](#), [scan_chain](#), [actual_scan_segment](#), [dft_configuration_mode](#), [root](#), [test_signal](#), [inst](#), [scan_segment](#), [actual_scan_chain](#)

Definition

hierarchical terminal

Attribute	Description
base_name	
	The name without the leading hinst hierarchy (e.g. p3 for h1/h2/p3). Type: string Default: "" Edit: No
bus	

Stylus Common UI Database Object Information

Database Objects--hport

	The bus for this hport. Type: <code>obj(hport_bus)</code> Default: "" Edit: No
clock_gating_exclude_signal	
	Exclude this pin from clock gating enable functions. Type: <code>bool</code> Default: false Edit: Yes
dft_driven_by_clock	
	If pin is driven by clock. Type: <code>enum</code> Enum Values: false true Default: false Edit: Yes
direction	
	Direction of pin. Type: <code>enum</code> Enum Values: internal in out inout Default: "" Edit: No
escaped_name	
	The escaped name of the hport. Type: <code>string</code> Default: "" Edit: No
hdl_vname	
	The name for this port as specified in the input HDL file (Verilog or VHDL). Type: <code>string</code> Default: "" Edit: No
hinst	
	The hinst for this hport. Type: <code>obj(hinst)</code> Default: "" Edit: No
hnet	
	The hnet connected to this hport (inside the hinst of this hport). Type: <code>obj(hnet)</code> Default: "" Edit: No
hpin	
	The hpin is the external view from outside the hinst for this hport. Type: <code>obj(hpin)</code> Default: "" Edit: No
is_booth_encoded	
	Booth encoded setting assigned to input port by user. Type: <code>bool</code> Default: false Edit: Yes
lp_asserted_probability	

Stylus Common UI Database Object Information

Database Objects--hport

	Probability of signal to be high on port/subport/pin as specified by user. Type: double Default: no_value Edit: Yes
lp_asserted_toggle_rate	
	Signal toggle rate assigned to pin by user. Type: double Default: no_value Edit: Yes
lp_probability_type	
	Signal probability type of a pin. Type: enum Enum Values: invalid default computed clock constant asserted Default: invalid Edit: Yes
lp_system_asserted_probability	
	System/Internal asserted probability. Type: double Default: no_value Edit: Yes
lp_system_asserted_toggle_rate	
	System/Internal asserted toggle rate. Type: double Default: no_value Edit: Yes
lp_toggle_rate_type	
	Signal toggle rate type of a pin. Type: enum Enum Values: invalid default computed clock constant asserted Default: invalid Edit: Yes
lssd_master_clock	
	Defines the master clock port related to this slave clock port. Type: string Default: "" Edit: Yes
name	
	The fully qualified path name of the hport (e.g. h1/h2/p3). Type: string Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (hport) Default: "" Edit: No
original_name	
	Stores port_vname on port bits. Type: string Default: "" Edit: Yes
wrapper_control	

Stylus Common UI Database Object Information

Database Objects--inst

	Specifies as wrapper control signal. Type: <code>bool</code> Default: <code>false</code> Edit: Yes
wrapper_segment	
	Specifies the wrapper segments associated with the pin. Type: <code>obj(scan_chain)* obj(scan_segment)* obj(actual_scan_segment)* obj(actual_scan_chain)*</code> Default: <code>""</code> Edit: Yes

inst

Parent Objects

[flexible_htree](#), [marker](#), [io_constraint](#), [place_blockage](#), [violation](#), [group](#), [hinst](#), [sdp](#), [actual_scan_segment](#), [partition](#), [design](#), [clock_tree](#), [root](#), [pg_pin](#), [route_blockage](#), [scan_segment](#), [pin](#), [inst_obs_shape](#)

Definition

instance

Attribute	Description
allow_seq_in_out_phase_opto	
	Allow phase inversion on sequential instance. Type: <code>bool</code> Default: <code>true</code> Edit: Yes
aocv_bootstrap_fall_derate_factor	
	Returns aocv coefficient aocv_bootstrap_fall_derate_factor in bootstrap mode on type cell Type: <code>double</code> Allowed -index values: <code>analysis_view</code> Default: <code>""</code> Edit: No
aocv_bootstrap_rise_derate_factor	
	Returns aocv coefficient aocv_bootstrap_rise_derate_factor in bootstrap mode on type cell Type: <code>double</code> Allowed -index values: <code>analysis_view</code> Default: <code>""</code> Edit: No
apply_booth_encoding	
	Apply booth encoding by using user specified method (for a specified instance). Type: <code>enum</code> Enum Values: <code>nonbooth auto_bitwidth auto_togglerate manual inherited</code> Default: <code>inherited</code> Edit: Yes
arcs	
	Returns a list of arc objects that are associated with this inst Type: <code>obj(arc)*</code> Allowed -index values: <code>analysis_view</code> Default: <code>""</code> Edit: No
area	

Stylus Common UI Database Object Information

Database Objects--inst

	area of inst Type: <code>area</code> Default: "" Edit: No
base_cell	
	cell of inst Type: <code>obj(base_cell)</code> Default: "" Edit: No
base_name	
	The name at the base of a hierarchical name. So the base_name of i1/i2/i3 is i3. Type: <code>string</code> Default: "" Edit: No
bbox	
	Bounding box of the overlap rects that define the placement area used by this inst. Type: <code>rect</code> Default: "" Edit: No
bottom_padding	
	Value of bottom padding for the instance. Type: <code>int</code> Default: 0 Edit: Yes
clock_gating_exclude	
	Exclude this instance from clock-gating. Type: <code>bool</code> Default: false Edit: Yes
clock_gating_integrated_cell	
	clock_gating_integrated_cell Type: <code>string</code> Allowed -index values: analysis_view Default: "" Edit: No
clock_gating_method	
	The clock gating method used. Type: <code>enum</code> Enum Values: sena_feedback sena_and_feedback common_enable_extraction combine other ungated Default: ungated Edit: Yes
clock_gating_test_signal	
	Set specified test signal to connect up test logic on the clock-gating instance. Type: <code>string</code> Default: "" Edit: Yes
cts_cannot_clone_reasons	
	A list of reasons to explain why the inst could not be cloned. This attribute will not be set if attributes cts_clone_clock_gates/cts_clone_clock_logic are false. The attribute returns a list of reasons why the inst cannot be cloned. The reasons why a inst cannot be cloned can be divided into three categories: 1. The inst is marked as dont_touch 2. Reasons specific to the clock node 3. General reasons

The possible dont_touch reasons are:

dont_touch.add_port_driver Set if inst was added above an output port or below an input port via attribute cts_add_port_driver

dont_touch.add_driver_cell Set if inst was added below the root via attribute cts_add_driver_cell

dont_touch.clock_root Set during clock tree extraction if identified as having the root pin

dont_touch.clock_sink Set on inst if any pin is a clock sink in a clock tree

dont_touch.clock_tree_generator_path Set if nodes and wires are in clock tree generator paths

dont_touch.clock_wire Set on inst if clock input wire is user dont_touch

dont_touch.clockgate_no_power_domain Set if clock gate / clock logic is a clocknode in a no clock gate / clock logic power domain

dont_touch.drives_multi_driver_net Set if inst drives wires with other drivers

dont_touch.external_skew_group_pin Set if a pin on the inst is a skew group sink, source or ignore pin for a skew group created by the user

dont_touch.flexible_htree Set if inst was added to a flexible H-tree

dont_touch.internal_skew_group_pin Set if a pin on the inst is a skew group sink, source or ignore pin for a skew group created by CCOpt

dont_touch.neg_edge_clock_gate Set if a clock gate gates the falling edge

dont_touch.non_flop_clock_gating Set during CTS on insts where cloning them could cause a non-equivalence failure in LEC, including clock gates whose transitive fanout contains flops triggered by the opposite clock edge, macros, or latches

dont_touch.non_standard_inputs_clock_gate Set if inst is a clock gate with 'non-standard' inputs, to avoid disconnecting them

dont_touch.observability_clock_gate Set if inst is a clock gate with an observability output pin

dont_touch.output_wire Set if inst drives a net that is dont_touch

dont_touch.placer.lock Set if inst is user locked or locked by DEF

dont_touch.power_management Set if inst is a power management inst

dont_touch.prevent_assign Set if inst is necessary to prevent an assign

dont_touch.sdc Set if constrained by SDC timing

dont_touch.sdc_path_group Set if there is an SDC path group start / endpoint on one of the pins of this inst

dont_touch.sub_block Set if inst is in dont_touch module

dont_touch.unmergeable_composite_clock_gate Set if some of the clock gate is dont_touch

dont_touch.user Set by user

Contact Cadence Support if any of the following reasons are listed:

dont_touch.cannot_understand_clock_gate Set if clock gate not recognized

dont_touch.composite_clock_gate_enable_test_or_gate

The possible reasons specific to the clock node are:

PowerManagement Set if inst is a power management inst

PowerManagementInconsistency Set if a disconnection of the inst from the netlist would cause a power management inconsistency

PreservedUMPB Set if clock node is a User Module Port Bit which is preserved

Contact Cadence Support if any of the following reasons are listed:

ClockGatesAlways Set if inst is clock gate, considered always dont_touch

ClockLogicAlways Set if inst is clock logic, considered always dont_touch

NodesRoot Set if the clock node is the clock root

NoRoot Set if the clock node has no root

RootIgnored Set if the clock root is ignored

SpineCell Set if cell is a clock spine inst and is marked as dont_touch

The possible general reasons are:

DrivesAcrossPowerDomains The inst drives across power domain boundaries

NodeHasSGConstraints The inst has user mode skew group constraints in default mode

ClockDriverCannotCloneInverter

ClockDriverCreatedByBuffLongNets Set if the inst was created by 'Buffering long nets'.

ClockDriverInverterCloningDisabled The inst is an inverter and cloning inverters is disabled.

UnclonableClockSource Set if the inst carries the -source of an SDC clock

UnclonableIODelayReference Set if the inst is associated with an I/O timing constraint

UnclonablePLLReference Set if the inst is associated with a PLL constraint

UnclonablePLLFeedback Set if the inst is analyzed as a PLL feedback

UnclonablePLLOutput Set if the inst is analyzed as a PLL output

UnsupportedNodeType Set if the inst is not a supported type. Supported types are: clock gate, clock logic and clock driver.

SingleOrZeroFanoutNode Set if the inst has fewer than two fanout.

Contact Cadence Support if any of the following reasons are listed:

NoCellInst

ClockLogicMultiOutputCell

IgnoredTree

IsUnclonable

DiscreteClockGate

NotTreeViewNode

OutputPinCellPinNull

ClockDriverNeedToKeep

Valid values: list string

Type: string

Default: ""

Edit: Yes

Stylus Common UI Database Object Information

Database Objects--inst

cts_cannot_merge_reasons

A list of reasons to explain why the inst could not be merged.

The attribute returns a list of reasons why the inst cannot be merged. These attributes are set on clock gates and clock logic instances.

For clock gates, the merge is driven by the enable function. If two clock gates share the same enable function, potentially they may be merged together. The reasons explains why potentially mergeable clock gates could not in fact be merged.

For clock logic, the merge is driven by the logical expression, or logic function, at the clock output pin. If two clock logic express the same logic function, at least potentially they may be merged. The reasons lists why the potential merge could not be realized.

User configuration disabled merging:

ClockGateMergingDisabledOnTree Merging disabled: attribute ccopt/cts_merge_clock_gates is false

ClockLogicMergingDisabledOnTree Merging disabled: attribute ccopt/cts_merge_clock_logic is false

IsDontTouch The instance cannot be merged as it is marked dont_touch

Attributes of a clock node rendered it unmergeable:

GloballyUnique The clock node has no potential candidate with which it might merge

UniqueUnderParent There are potential merge candidates, but not on the same clock net

HasSkewGroupConstraints The instance carries a skew group source, sink or ignore pin

UndrivenEnablePins The clock gate has a floating enable input

UndrivenRetentionPins The clock gate has a floating retention input

UndrivenTestPins The clock gate has a floating test enable input

DifferentSplitOnInputs The multi-input clock logic has different skew groups at its clock inputs

SiblingsNotMergeable Could not merge because all other candidates were unmergeable

Otherwise mergeable clock nodes may have incompatibilities that prevent merging:

DifferentSkewGroupForInputs Two clock nodes have different skew groups at their clock inputs

DifferentSkewGroupForOutputs Two clock nodes have different skew groups at their clock output

DifferentCellFamilies Two clock nodes belong to incompatible cell families

DifferentNumberOfInputPins Two clock nodes have different numbers of input pins

DifferentNonStandardInputs Two clock gates have non-standard* inputs which are non-equivalent

MismatchingNonClockInputs Two clock nodes have non-clock inputs which receive non-equivalent signals

MismatchingClockInputs Two clock nodes have clock inputs which receive non-equivalent signals

MismatchingPowerDomains Two clock nodes are in incompatible power domains

NotLogicalClones Two clock nodes have outputs which are not logically equivalent

DifferentNumberOfParents Two clock nodes have different numbers of clock-carrying inputs

DifferentOutputPowerContext Two clock nodes have outputs with different power domain/voltage

DifferentInputPowerContext Two clock nodes have inputs with different power domain/voltage

IncompatibleRestrictedRegions Two clock nodes have incompatible restricted regions

DifferentAnnotations Two clock nodes have different annotations on its pins

(*) Non-standard inputs on a clock gate are inputs besides the clock, enable, test enable and retention inputs.

Valid values: list string

Type: string

Default: ""

Edit: Yes

cts_effective_clock_halo_x

This read only attribute shows the x component of the clock halo The units are um.

The following attributes can be used to assign x direction clock halos within CCOpt:

cts_cell_halo_x

cts_cell_density

cts_cell_halo_sites

See also:

. cts_effective_clock_halo_x_source

. cts_effective_clock_halo_y_source

. cts_effective_clock_halo_y

Type: string

Default: ""

Edit: No

cts_effective_clock_halo_x_source

Stylus Common UI Database Object Information

Database Objects--inst

	This read only attribute shows which attribute defines the x component of the clock halo. The following attributes can be used to assign x direction clock halos within CCOpt: cts_cell_halo_x cts_cell_density cts_cell_halo_sites See also: . cts_effective_clock_halo_x . cts_effective_clock_halo_y . cts_effective_clock_halo_y_source Type: string Default: "" Edit: No
cts_effective_clock_halo_y	
	This read only attribute shows the x component of the clock halo. The units are um. The following attributes can be used to assign y direction clock halos within CCOpt: cts_cell_halo_y cts_adjacent_rows_legal cts_cell_halo_rows See also: . cts_effective_clock_halo_x_source . cts_effective_clock_halo_y_source . cts_effective_clock_halo_x Type: string Default: "" Edit: No
cts_effective_clock_halo_y_source	
	This read only attribute shows which attribute defines the y component of the clock halo. The following attributes can be used to assign y direction clock halos within CCOpt: cts_cell_halo_y cts_adjacent_rows_legal cts_cell_halo_rows See also: . cts_effective_clock_halo_x . cts_effective_clock_halo_y . cts_effective_clock_halo_x_source Type: string Default: "" Edit: No
cts_node_type	
	node type of instance within cts graph Type: string Default: "" Edit: No
cts_original_names	
	Specifies for a clockgate or clocklogic which has been merged or is a clone a list of names from the original netlist which are equivalent to the clockgate/clocklogic. For example: If A and B are merged to form C then original_names for C is { A B }. If D_clone is a clone of D then original_names for D_clone is { D }. If E is a clone of C then original_names for E is { A B } (remembering C was a merger of A and B). Valid values: list string Type: string Default: "" Edit: Yes
cts_preserve_from_deletion	

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	Do not allow this instance to be deleted. Default false. Valid values are true or false. Type: <code>bool</code> Default: false Edit: Yes
delta_temperature	
	temperature of the inst Type: <code>string</code> Default: "" Edit: No
dft_abstract_dont_scan	
	Exclude this instance from scan-synthesis as an abstract segment has been defined on it. Type: <code>bool</code> Default: false Edit: No
dft_custom_se	
	Shift Enable for flop. Type: <code>obj(test_signal)</code> Default: "" Edit: Yes
dft_dont_scan	
	Exclude this design from scan-synthesis (inherited true false). Type: <code>enum</code> Enum Values: false true inherited Default: inherited Edit: Yes
dft_exclude_flop_from_los_pipeline	
	Avoid insertion of launch-off-shift pipelined logic to flop's shift enable. Type: <code>bool</code> Default: false Edit: Yes
dft_exclude_from_shift_register	
	Determines if flop should be excluded from shift register. Type: <code>bool</code> Default: false Edit: Yes
dft_exempt_from_system_clock_check	
	Exempt flop from system clock check. Type: <code>bool</code> Default: false Edit: Yes
dft_force_blackbox_for_atpg	
	Determines is the instance is blackbox for ATPG. Type: <code>enum</code> Enum Values: none true false Default: none Edit: Yes
dft_is_blackbox_for_atpg	
	Blackbox for ATPG. Type: <code>bool</code> Default: false Edit: No
dft_is_los_pipeline_flop	

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	Determines if this flop is launch-off-shift pipeline flop. Type: <code>bool</code> Default: <code>false</code> Edit: Yes
dft_is_testpoint	
	Attribute for flop which indicates if it is a testpoint flop or not. Type: <code>bool</code> Default: <code>false</code> Edit: Yes
dft_mapped	
	This flop is mapped to scan for DFT purposes. Type: <code>bool</code> Default: <code>false</code> Edit: No
dft_part_of_segment	
	The type of segment this instance belongs to {abstract fixed floating preserved shift_register}. Type: <code>enum</code> Enum Values: fixed floating preserve abstract shift_register Default: "" Edit: No
dft_scan_chain	
	The actual scan chain this instance belongs to. Type: <code>obj(scan_chain)</code> <code>obj(scan_segment)</code> <code>obj(actual_scan_segment)</code> <code>obj(actual_scan_chain)</code> Default: "" Edit: No
dft_scan_chain_in_multi_mode	
	The actual scan chains this instance belongs to in multi mode stitching. Type: <code>obj(scan_chain)*</code> <code>obj(scan_segment)*</code> <code>obj(actual_scan_segment)*</code> <code>obj(actual_scan_chain)*</code> Default: "" Edit: No
dft_status	
	DFT-rule-check status as analyzed by 'check_dft_rules' {Passes DFT rules Fails DFT rules Abstract Segment Dont scan Dont scan Misc. non scan}. Type: <code>string</code> Default: "" Edit: No
dft_test_clock	
	Test clock as analyzed by 'check_dft_rules'. Type: <code>obj(test_clock)</code> Default: "" Edit: No
dft_test_clock_edge	
	Test clock edge as analyzed by 'check_dft_rules'. Type: <code>string</code> Default: "" Edit: No
dft_test_clock_source	
	Test clock source as analyzed by 'check_dft_rules'. Type: <code>obj(pin)</code> <code>obj(hpin)</code> <code>obj(port)</code> <code>obj(hport)</code> Default: "" Edit: No
dft_test_clock_waveform_edge	

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	Test clock waveform leading or trailing edge as analyzed by 'check_dft_rules'. Type: string Default: "" Edit: No
dft_violation	
	DFT rule violation as analyzed by 'check_dft_rules'. Type: string Default: "" Edit: No
dont_merge_multibit	
	This attribute is denotes whether the instance can be merged during multibit optimization. This is enabled only when the root attribute 'use_multibit_cells' is set to true. Supported values: true: Cannot merge false: Can be merged Type: bool Default: false Edit: Yes
dont_retime	
	Do not retime this instance. Type: bool Default: false Edit: Yes
dont_split_multibit	
	This attribute is denotes whether the instance can be split (unmerged) during multibit optimization. This is enabled only when the root attribute 'use_multibit_cells' is set to true. Supported values: true: Cannot split false: Can be split Type: bool Default: false Edit: Yes
dont_touch	
	This attribute defines the user preservation status of an instance during optimization. Supported values: none (default): Unconstrained false: Can be mapped, sized, deleted, and constants can be propagated through it const_prop_size_delete_ok: Can be resized or deleted and constants can be propagated through it const_prop_delete_ok: Can be deleted and constants can be propagated through it size_delete_ok: Can be resized or deleted if no fanout delete_ok: Can be deleted if it has no fanout, but cannot be resized size_ok: Can only be resized size_same_height_ok: Can only be resized to a cell of the same height size_same_footprint_ok: Can only be resized to a cell of the same footprint that has exactly the same pin shapes map_size_ok: Can be mapped or sized (but not deleted). Applies only to sequential instances so cannot be applied on the module or hinst object. true: Cannot be touched Type: enum Enum Values: none false true delete_ok const_prop_size_delete_ok const_prop_delete_ok size_delete_ok size_ok size_same_height_ok size_same_footprint_ok map_size_ok Default: none Edit: Yes
dont_touch_effective	

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	This attribute defines the effective (most pessimistic) preservation status of an instance during optimization based on the 'sources'. Type: enum Enum Values: none false true delete_ok const_prop_size_delete_ok const_prop_delete_ok size_delete_ok size_ok size_same_height_ok size_same_footprint_ok map_size_ok Default: "" Edit: No
dont_touch_file	
	Specifies the sourced file to set dont_touch. Type: string Default: "" Edit: Yes
dont_touch_pins	
	Type: enum Enum Values: none false true delete_ok const_prop_size_delete_ok const_prop_delete_ok size_delete_ok size_ok size_same_height_ok size_same_footprint_ok map_size_ok Default: none Edit: Yes
dont_touch_sources	
	Dictionary of {source <value>} pairs contributing to the dont_touch_effective attribute for this object: {user <value>} {lib <value>} {parent <value>} {scan <value>} user # This inst dont_touch value lib # The base_cell dont_touch value parent # The effective dont_touch value from an hinst above (the closest hinst above that is not false) scan # Is this inst part of a scan-chain (value is either size_ok or none). There is an additional inst attribute that holds this information. dft # This <obj> is part of design-for-test logic constraints Type: string Default: "" Edit: No
dont_use_qbar_pin	
	Do not use inverted output of sequential instance. Type: bool Default: false Edit: Yes
dpopt_instance_constant_input_based_speculation	
	Controls constant input based datapath speculation of datapath module in 'syn_generic'. Type: bool Default: false Edit: Yes
early_cell_check_derate_factor	
	Returns the derating factor on timing check values for early paths (e.g Hold checks) Type: double Allowed -index values: analysis_view Default: "" Edit: No
early_cell_check_mean_derate_factor	
	Reports early mean derate factor for timing check arc of the specified cell. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
early_cell_check_sigma_derate_factor	

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	Reports early sigma derate factor for timing check arc of the specified cell. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
early_clk_cell_derate_factor	
	Returns the derating factor for instances on early clock paths Type: double Allowed -index values: analysis_view Default: "" Edit: No
early_clk_cell_mean_derate_factor	
	Reports early mean timing derate factor for clock line Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
early_clk_cell_sigma_derate_factor	
	Reports early sigma timing derate factor for clock line Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
early_clock_check_incremental_derate_factor	
	Reports incremental derate factor specified with -early -clock for check arcs Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
early_data_cell_derate_factor	
	Returns the derating factor for instances on early data paths Type: double Allowed -index values: analysis_view Default: "" Edit: No
early_data_cell_mean_derate_factor	
	Reports mean derate factor specified with -early -data Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
early_data_cell_sigma_derate_factor	
	Reports sigma derate factor specified with -early -data Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
early_data_check_incremental_derate_factor	
	Reports incremental derate factor specified with -early -data for check arcs Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
early_fall_cell_check_derate_factor	

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	Returns the early cell check derating factor for falling arrivals Type: double Allowed -index values: analysis_view Default: "" Edit: No
early_fall_cell_check_mean_derate_factor	
	Reports mean derate factor specified with -early -fall for check arcs Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
early_fall_cell_check_sigma_derate_factor	
	Reports sigma derate factor specified with -early -fall for check arcs Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
early_fall_clk_cell_derate_factor	
	Returns the early derating factor for falling delays through clock tree instances Type: double Allowed -index values: analysis_view Default: "" Edit: No
early_fall_clk_cell_mean_derate_factor	
	Reports mean derate factor specified with -early -fall for clock logic Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
early_fall_clk_cell_sigma_derate_factor	
	Reports sigma derate factor specified with -early -fall for clock logic Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
early_fall_clock_incremental_derate_factor	
	Reports incremental derate factor specified with -early -fall -clock Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
early_fall_data_cell_derate_factor	
	Returns the early derating factor for falling delays through data path instances Type: double Allowed -index values: analysis_view Default: "" Edit: No
early_fall_data_cell_mean_derate_factor	
	Reports mean derate factor specified with -early -fall -data Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
early_fall_data_cell_sigma_derate_factor	

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Database Objects--inst

	Reports sigma derate factor specified with -early -rise -data for check arcs Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
early_fall_data_incremental_derate_factor	
	Reports incremental derate factor specified with -early --fall -data Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
early_rise_cell_check_derate_factor	
	Returns the early cell check derating factor for rising arrivals Type: double Allowed -index values: analysis_view Default: "" Edit: No
early_rise_cell_check_mean_derate_factor	
	Reports mean derate factor specified with -early -rise for check arcs Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
early_rise_cell_check_sigma_derate_factor	
	Reports sigma derate factor specified with -early -rise for check arcs Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
early_rise_clk_cell_derate_factor	
	Returns the early derating factor for rise delays through clock tree instances Type: double Allowed -index values: analysis_view Default: "" Edit: No
early_rise_clk_cell_mean_derate_factor	
	Reports mean derate factor specified with -early -rise -clock Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
early_rise_clk_cell_sigma_derate_factor	
	Reports sigma derate factor specified with -early -rise -clock Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
early_rise_clock_incremental_derate_factor	
	Reports incremental derate factor specified with -early -rise -clock Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
early_rise_data_cell_derate_factor	

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Database Objects--inst

	Returns the early derating factor for rising delays through data path instances Type: double Allowed -index values: analysis_view Default: "" Edit: No
early_rise_data_cell_mean_derate_factor	
	Reports mean derate factor specified with -early -rise -data Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
early_rise_data_cell_sigma_derate_factor	
	Reports sigma derate factor specified with -early -rise -data Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
early_rise_data_incremental_derate_factor	
	Reports incremental derate factor specified with -early -rise -data Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
eco_opt_use_cells	
	Specifies the cells to be used by eco opt engine for replacing the inst. Type: string Default: "" Edit: Yes
escaped_name	
	The full hierarchical name including escaped chars (if any). It follows DEF escaping syntax, so bus-bit chars [], or a hierarchy char \ that is not a bus-bit or hierarchy char has a \ in front of it. So i1/i2 is a two-level hierarchical name while i1/i2 is single level name, and a[0] is a bus-bit, while a[0\] is a scalar. Type: string Default: "" Edit: No
exceptions	
	Reports the path exceptions Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
fcu_instruction_set	
	Attribute to define different repair specific instructions for the different instances of the same block on which fuse control unit has been inserted. Type: string Default: "" Edit: Yes
group	
	The floorplan group that contains this inst. The instances in a group will be placed close together. Type: obj(group) Default: "" Edit: No
has_aging	

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	Returns true if the associated library cell is recognized has aging information Type: <code>bool</code> Allowed -index values: analysis_view Default: "" Edit: No
has_bti_aging	
	Returns true if the associated library cell is recognized has BTI aging information Type: <code>bool</code> Allowed -index values: analysis_view Default: "" Edit: No
has_hci_aging	
	Returns true if the associated library cell is recognized has HCI aging information Type: <code>bool</code> Allowed -index values: analysis_view Default: "" Edit: No
has_lvf	
	Returns true if library arc(s) of libcell has LVF information Type: <code>bool</code> Allowed -index values: analysis_view Default: "" Edit: No
has_sdc_assertions	
	Reports true if SDC assertions are asserted Type: <code>bool</code> Allowed -index values: analysis_view clock Default: "" Edit: No
hdl_name	
	Same name attribute on inst from genus pass on the value to innovus attribute . to be used to get hdl name of Hard Macro/BBOX during 1 PAss LEC flow . Type: <code>string</code> Default: "" Edit: Yes
hdl_proc_name	
	Name of the process (VHDL) or always block (Verilog) from which the instance is inferred. Type: <code>string</code> Default: "" Edit: Yes
hdl_vname	
	The name for this instance as specified in the input HDL file (Verilog or VHDL). For flip-flop or latch instances, hdl_vname is the name of the variable for which the flip-flop or latch instance was inferred. Type: <code>string</code> Default: "" Edit: No
hierarchical_level	
	Returns hierarchical level for an Instance Type: <code>int</code> Allowed -index values: analysis_view clock Default: "" Edit: No
hierarchical_name	

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	hierarchical_name Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
instance_internal_power	
	The instance's internal power. Type: double Default: no_value Edit: Yes
instance_leakage_power	
	The instance's leakage power. Type: double Default: no_value Edit: Yes
instance_model	
	Reports the noise instance model Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
inverted_phase	
	The sequential instance was implemented using a inverted phase. Type: bool Default: false Edit: Yes
is_always_on	
	Returns true if the associated library cell is recognized as always on type cell Type: bool Allowed -index values: analysis_view Default: "" Edit: No
is_black_box	
	Returns a value of true if the cell is a black box cell. Type: bool Allowed -index values: analysis_view Default: "" Edit: No
is_buffer	
	This inst is a buffer. Type: bool Default: "" Edit: No
is_clock_gating_check	
	Returns true if a clock gating check is performed by this instance. The clock gating check may be inferred from logic in netlist, asserted by SDC constraints, or may be due to this instance being an integrated clock gating cell Type: bool Allowed -index values: analysis_view Default: "" Edit: No
is_clock_network_inst	

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	Reports true if specified inst is part of clock network Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
is_combinational	
	Returns a value of true if the instance is a combinational cell (not a sequential cell). Type: bool Allowed -index values: analysis_view Default: "" Edit: No
is_disable_timing	
	Returns true if set_disable_timing has been set on this cell instance Type: bool Allowed -index values: analysis_view Default: "" Edit: No
is_fall_edge_triggered	
	Returns a value of true if the instance is triggered by the falling edge of the clock. Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
is_fixed_mask	
	Indicates the base_cell of the inst has FIXEDMASK keyword in LEF, so mask-shifting is not allowed, except for the layers with LAYERMASKSHIFT keyword. Type: bool Default: "" Edit: No
is_flop	
	Returns true if the library cell is recognized as flip-flop/register type cell Type: bool Default: "" Edit: No
is_genus_clock_gate	
	Set by the tool in the iSpatial flow to indicate that this integrated clock gating instance was added by Genus clock gating Type: bool Default: false Edit: Yes
is_hierarchical	
	is_hierarchical Type: bool Allowed -index values: analysis_view Default: "" Edit: No
is_inside_ilm	
	This attribute denotes whether the inst is the child of a parent hinst the has an ILM specified for it Type: bool Default: "" Edit: No
is_integrated_clock_gating	

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	Returns true if this instance's library cell has the Liberty clock_gating_integrated_cell set to true Type: <code>bool</code> Allowed -index values: analysis_view Default: "" Edit: No
is_interface_timing	
	Returns a value of true if this instance's library cell has the Liberty interface_timing attribute set to true Type: <code>bool</code> Allowed -index values: analysis_view Default: "" Edit: No
is_inverter	
	This inst is an inverter. Type: <code>bool</code> Default: "" Edit: No
is_isolation	
	This inst is an isolation. Type: <code>bool</code> Default: "" Edit: No
is_isolation_cell	
	Returns true if the associated library cell is recognized as isolation type cell Type: <code>bool</code> Allowed -index values: analysis_view Default: "" Edit: No
is_jtag	
	The inst is a JTAG element. Type: <code>bool</code> Default: false Edit: Yes
is_latch	
	Returns true if the associated library cell is recognized as latch type cell Type: <code>bool</code> Default: "" Edit: No
is_legal	
	Specifies the legal status that was seen by the last place_detail or check_place. Type: <code>bool</code> Default: "" Edit: No
is_level_shifter	
	Returns true if the associated library cell is recognized as level shifter type cell Type: <code>bool</code> Allowed -index values: analysis_view Default: "" Edit: No
is_macro	

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	Returns a value of true if the associated library cell has the Liberty attribute is_macro_cell is set true Type: bool Allowed -index values: analysis_view Default: "" Edit: No
is_master_slave_flop	
	Returns true if the Liberty clock_on_also attribute is specified for the associated library cell Type: bool Default: "" Edit: No
is_master_slave_lssd_flop	
	Returns true if this cell has been recognized as a master/slave LSSD device Type: bool Default: "" Edit: No
is_memory	
	Returns true or false depending on whether the associated library cell of the inst is recognized as a Liberty memory cell. Cells which include a Liberty memory group definition are recognized as memory cells. Type: bool Allowed -index values: analysis_view Default: "" Edit: No
is_msv	
	Returns true if object is a multiple supply voltage cell Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
is_mux	
	Returns true if object is a mux cell Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
is_negative_level_sensitive	
	Returns a value of true if the associated library cell is identified as a negative level-sensitive latch. Type: bool Allowed -index values: analysis_view Default: "" Edit: No
is_pad	
	Returns a value of true if the associated library cell's Liberty is_pad attribute is set to true Type: bool Allowed -index values: analysis_view Default: "" Edit: No
is_physical	
	is physical only of inst Type: bool Default: "" Edit: No
is_place_halo	

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	Indicates that the instance has a placement halo. Type: <code>bool</code> Default: <code>false</code> Edit: Yes
is_positive_level_sensitive	
	Returns a value of true if the associated library cell is identified as a positive level-sensitive latch. Type: <code>bool</code> Allowed -index values: <code>analysis_view</code> Default: <code>""</code> Edit: No
is_power_switch	
	Returns true if the associated library cell is recognized as power switch type cell Type: <code>bool</code> Allowed -index values: <code>analysis_view</code> Default: <code>""</code> Edit: No
is_retention	
	Returns true if the associated library cell is recognized as retention type cell Type: <code>bool</code> Allowed -index values: <code>analysis_view</code> Default: <code>""</code> Edit: No
is_rise_edge_triggered	
	Returns a value of true if the instance is triggered by the rising edge of the clock. Type: <code>bool</code> Allowed -index values: <code>analysis_view clock</code> Default: <code>""</code> Edit: No
is_sequential	
	Returns a value of true if the library cell is a latch or flip-flop, or if the cell has sequential timing checks. Type: <code>bool</code> Allowed -index values: <code>analysis_view</code> Default: <code>""</code> Edit: No
is_spare	
	The inst is a spare instance. These are used by post-mask ECO flows. Type: <code>bool</code> Default: <code>false</code> Edit: Yes
is_synthesis_clock_gate	
	Attribute stored on clock gate instances to indicate if the clock gate was added by the tool or not. Type: <code>bool</code> Default: <code>false</code> Edit: Yes
is_tristate	
	Returns a value of true if the associated library cell definition includes the Liberty attribute <code>three_state</code> set to true . Type: <code>bool</code> Allowed -index values: <code>analysis_view</code> Default: <code>""</code> Edit: No
is_unmapped	

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	Returns true if object is not mapped in netlist Type: <code>bool</code> Allowed -index values: analysis_view clock Default: "" Edit: No
late_cell_check_derate_factor	
	Returns the derating factor on timing check values for late paths (e.g Setup checks) Type: <code>double</code> Allowed -index values: analysis_view Default: "" Edit: No
late_cell_check_mean_derate_factor	
	Returns mean derate factor for -late for check arcs Type: <code>double</code> Allowed -index values: analysis_view clock Default: "" Edit: No
late_cell_check_sigma_derate_factor	
	Returns sigma derate factor for -late for check arcs Type: <code>double</code> Allowed -index values: analysis_view clock Default: "" Edit: No
late_clk_cell_derate_factor	
	Returns the derating factor for instances on late clock paths Type: <code>double</code> Allowed -index values: analysis_view Default: "" Edit: No
late_clk_cell_mean_derate_factor	
	Returns mean derate factor for -late -clock Type: <code>double</code> Allowed -index values: analysis_view clock Default: "" Edit: No
late_clk_cell_sigma_derate_factor	
	Returns sigma derate factor for -late -clock Type: <code>double</code> Allowed -index values: analysis_view clock Default: "" Edit: No
late_clock_check_incremental_derate_factor	
	Returns incremental derate factor for -late -clock Type: <code>double</code> Allowed -index values: analysis_view clock Default: "" Edit: No
late_data_cell_derate_factor	
	Returns the derating factor for instances on late data paths Type: <code>double</code> Allowed -index values: analysis_view Default: "" Edit: No
late_data_cell_mean_derate_factor	

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	Returns mean derate factor for -late -data Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
late_data_cell_sigma_derate_factor	
	Returns sigma derate factor for -late -data Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
late_data_check_incremental_derate_factor	
	Returns incremental derate factor for -late -data Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
late_fall_cell_check_derate_factor	
	Returns the late cell check derating factor for falling arrivals Type: double Allowed -index values: analysis_view Default: "" Edit: No
late_fall_cell_check_mean_derate_factor	
	Returns mean derate factor for -late -fall for check arcs Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
late_fall_cell_check_sigma_derate_factor	
	Returns sigma derate factor for -late -fall for check arcs Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
late_fall_clk_cell_derate_factor	
	Returns the late derating factor for falling delays through clock tree instances Type: double Allowed -index values: analysis_view Default: "" Edit: No
late_fall_clk_cell_mean_derate_factor	
	Returns mean derate factor for -late -fall -clock Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
late_fall_clk_cell_sigma_derate_factor	
	Returns sigma derate factor for -late -fall -clock Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
late_fall_clock_incremental_derate_factor	

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	Returns incremental derate factor for -late -fall -clock Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
late_fall_data_cell_derate_factor	
	Returns the late derating factor for falling delays through data path instances Type: double Allowed -index values: analysis_view Default: "" Edit: No
late_fall_data_cell_mean_derate_factor	
	Returns mean derate factor for -late -fall -data Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
late_fall_data_cell_sigma_derate_factor	
	Returns sigma derate factor for -late -fall -data Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
late_fall_data_incremental_derate_factor	
	Returns incremental derate factor for -late -fall -data Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
late_rise_cell_check_derate_factor	
	Returns the late cell check derating factor for rising arrivals Type: double Allowed -index values: analysis_view Default: "" Edit: No
late_rise_cell_check_mean_derate_factor	
	Returns mean derate factor for -late -rise for check arcs Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
late_rise_cell_check_sigma_derate_factor	
	Returns sigma derate factor for -late -rise for check arcs Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
late_rise_clk_cell_derate_factor	
	Returns the late derating factor for rise delays through clock tree instances Type: double Allowed -index values: analysis_view Default: "" Edit: No
late_rise_clk_cell_mean_derate_factor	

Stylus Common UI Database Object Information
Database Objects--inst

	Returns mean derate factor for -late -rise -clock Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
late_rise_clk_cell_sigma_derate_factor	
	Returns sigma derate factor for -late -rise -clock Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
late_rise_clock_incremental_derate_factor	
	Returns incremental derate factor for -late -rise -clock Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
late_rise_data_cell_derate_factor	
	Returns the late derating factor for rising delays through data path instances Type: double Allowed -index values: analysis_view Default: "" Edit: No
late_rise_data_cell_mean_derate_factor	
	Returns mean derate factor for -late -rise -data Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
late_rise_data_cell_sigma_derate_factor	
	Returns sigma derate factor for -late -rise -data Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
late_rise_data_incremental_derate_factor	
	Returns incremental derate factor for -late -rise -data Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
Ide_distance_left	
	The COPDE spacing to the left side of the inst. Type: string Default: "" Edit: No
Ide_distance_right	
	The COPDE spacing to the right side of the inst. Type: string Default: "" Edit: No
leakage_power	

Stylus Common UI Database Object Information

Database Objects--inst

	Returns the leakage power of the instance from library cell Type: <code>double</code> Allowed -index values: <code>analysis_view</code> Default: "" Edit: No
left_padding	
	Value of left padding for the instance. Type: <code>int</code> Default: 0 Edit: Yes
level_shifter_type	
	Returns the Level Shifter type for associated library cell. The supported values are LH, HL and HL_LH. Type: <code>string</code> Allowed -index values: <code>analysis_view</code> Default: "" Edit: No
lib_binding_info_max	
	Returns max voltage library binding information Type: <code>string</code> Allowed -index values: <code>analysis_view clock</code> Default: "" Edit: No
lib_binding_info_min	
	Returns min voltage library binding information Type: <code>string</code> Allowed -index values: <code>analysis_view clock</code> Default: "" Edit: No
lib_cell	
	Returns the bind library cell Type: <code>obj(lib_cell)*</code> Allowed -index values: <code>analysis_view</code> Default: "" Edit: No
lib_cells	
	Returns a list of library cell objects associated with this instance. You can use -index to refine this list to lib_cells associated with a specific view Type: <code>obj(lib_cell)*</code> Allowed -index values: <code>analysis_view</code> Default: "" Edit: No
litho_halo	
	A litho_halo on an inst forces parallelrouting away from the block boundary to meet lithography DRC rules on a few routing layers but allows perpendicular access to pins. It is only allowed if LEF LITHOMACROHALO values exists for some routing layers. See the LEF manual for figures and more details of this DRC rule. It cannot be added to standard cells (cells with a CLASS CORE SITE). Type: <code>bool</code> Default: false Edit: Yes
location	
	The location of the inst. It is always the lower-left corner of the inst bounding-box, independent of the inst orientation. Type: <code>point</code> Default: "" Edit: Yes
lp_dynamic_analysis_scope	

Stylus Common UI Database Object Information

Database Objects--inst

	Set VCD analysis scope attribute on an instance in the design hierarchy. Valid values for this attribute are true or false. Type: bool Default: false Edit: Yes
map_to_multibit_bank_label	
	Defines a bank label for a sequential instance. Instances with the same label can be considered for multibit merging. Type: string Default: default_bank Edit: Yes
map_to_multibit_register	
	Enables predefined multibit cell merging for the sequential instance and limits the multibit mapping of this instance to the specified multibit library cells. Since this attribute enables predefined multibit cell inferencing, it implies a forced type of mapping and is therefore not QoR-driven. Type: string Default: "" Edit: Yes
mask_shift	
	Digit encoded value indicates the mask shifting for the instance contents (0 = unshifted, for other shift cases refer to the DEF COMP + MASKSHIFT documentation). Type: string Default: "" Edit: Yes
merge_combinational_hier_instance	
	Allow the combinational hierarchical instance to be merged. Type: enum Enum Values: 1 0 true false inherited Default: inherited Edit: Yes
merge_multibit	
	Infer multibit cell for this instance. Type: bool Default: false Edit: Yes
multibit_allow_async_phase_map	
	Allow interchanging async pins for multibit mapping. Type: enum Enum Values: 1 0 true false inherited Default: inherited Edit: Yes
multibit_flop_rejection_reason	
	Reports the reason why the instance could not be merged into a multibit cell. Type: string Default: "" Edit: Yes
multibit_rejection_reason	
	Reports rejection reason for not creating multibit. Type: string Default: "" Edit: Yes
multibit_rejection_reason_id	
	Reports the genus rejection reason category why the instance could not be merged into a multibit cell in map. Type: string Default: "" Edit: Yes

Stylus Common UI Database Object Information

Database Objects--inst

name	
	The same as .escaped_name without any \ escape chars. This name is commonly used to avoid problems with \ escape chars in Tcl scripts unless you carefully use list operators. Like SDC commands, 'get_db insts i1/i2' will first try to match i1/i2, and if not found then match i1\i2 so that flattening hierarchical names (e.g. with ungroup) does not require changing the names in Tcl scripts. Type: string Default: "" Edit: No
num_exceptions	
	Reports the number of path exceptions Type: int Allowed -index values: analysis_view clock Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (inst) Default: "" Edit: No
optimize_constant_0_seq	
	Allow constant 0 propagation through sequential instances. Type: enum Enum Values: 1 0 true false inherited Default: inherited Edit: Yes
optimize_constant_1_seq	
	Allow constant 1 propagation through sequential instances. Type: enum Enum Values: 1 0 true false inherited Default: inherited Edit: Yes
optimize_constant_feedback_seq	
	Allow constant propagation through feedback sequential instances. Type: bool Default: true Edit: Yes
optimize_merge_seq	
	Allow merging of equivalent sequential instances. Type: enum Enum Values: 1 0 true false inherited Default: inherited Edit: Yes
orient	
	Instance placement orientation. Type: enum Enum Values: r0 r90 r180 r270 mx mx90 my my90 Default: r0 Edit: Yes
original_name	
	Original rtl names of single-bit flops. Type: string Default: "" Edit: Yes
original_names_for_split	

Stylus Common UI Database Object Information

Database Objects--inst

	Original names of single-bit flops merged. Type: string Default: "" Edit: Yes
overlap_rects	
	List of rectangles that define the shape of instance Type: rect Default: "" Edit: No
parent	
	The parent of the inst, which is either a hinst for an instance within hierarchy or the design object for an instance at the top level. Type: obj(design) obj(hinst) Default: "" Edit: No
parent_cell	
	Returns the enclosing parent hinst Type: obj(hinst) * Allowed -index values: analysis_view clock Default: "" Edit: No
partition	
	If this inst is a physical black_box, this partition will carry the pin constraints and related data for the blackbox. Otherwise it is empty. Type: obj(partition) Default: "" Edit: No
peak_current	
	Peak current value of the inst Type: string Default: "" Edit: No
pg_pins	
	List of pg_pins for this inst. These are declared as power or ground pins in the .lib or LEF/OA library, and separated from the signal pins of the inst. Type: obj(pg_pin) * Default: "" Edit: No
pin_count	
	pin_count Type: int Allowed -index values: analysis_view clock Default: "" Edit: No
pins	
	The pins of this inst. Type: obj(pin) * Default: "" Edit: No
place_halo_bbox	
	Bounding box of the inst placement halo. For a rectilinear block this might include area that is not covered by the place halo. Type: rect Default: "" Edit: No
place_halo_bottom	

Stylus Common UI Database Object Information

Database Objects--inst

	Specifies an extra halo of space along the bottom edges of the inst that should not be used during placement. This area can still be used later during optimization and clock tree creation to add repeaters. It can only be used on blocks and cannot be added to standard-cells. See the padding attributes on the base_cell to add extra space to standard-cells. Type: coord Default: "" Edit: No
place_halo_left	
	Specifies an extra halo of space along the left edges of the inst that should not be used during placement. This area can still be used later during optimization and clock tree creation to add repeaters. It can only be used on blocks and cannot be added to standard-cells. See the padding attributes on the base_cell to add extra space to standard-cells. Type: coord Default: "" Edit: No
place_halo_polygon	
	Specifies the polygon shape of the inst placement halo. When the inst is unplaced, it pretends the inst is placed at coordinate origin and use it's base_cell to calculate polygon. For Example, if the base_cell's size is X by Y, then the polygon of the unplaced inst is {0 0 . . . }. Based on that, the place_halo_polygon is calculated. Type: point Default: "" Edit: No
place_halo_right	
	Specifies an extra halo of space along the right edges of the inst that should not be used during placement. This area can still be used later during optimization and clock tree creation to add repeaters. It can only be used on blocks and cannot be added to standard-cells. See the padding attributes on the base_cell to add extra space to standard-cells. Type: coord Default: "" Edit: No
place_halo_top	
	Specifies an extra halo of space along the top edges of the inst that should not be used during placement. This area can still be used later during optimization and clock tree creation to add repeaters. It can only be used on blocks and cannot be added to standard-cells. See the padding attributes on the base_cell to add extra space to standard-cells. Type: coord Default: "" Edit: No
place_status	
	This attribute is the placement status of an instance during placement and optimization. The placer will look at both place_status and place_status_cts and use the more restrictive value. When this attribute is explicitly set by the user, it will reset the place_status_cts attribute so the user intent has precedence. Supported values: unplaced: Unconstrained placed: Is placed but can be moved soft_fixed: Is fixed, but can move a short distance to legalize it fixed: Is fixed and can only be moved by the user, will not be moved by placement or optimization commands. cover: Is fixed and cannot be moved by any commands, you must change the place_status to move it. Type: enum Enum Values: unplaced placed fixed cover soft_fixed Default: unplaced Edit: Yes
place_status_cts	

Stylus Common UI Database Object Information

Database Objects--inst

	This attribute is the CTS placement status of an instance during placement and optimization. The placer will look at both place_status and place_status_cts and use the more restrictive value. Supported values: unset: Unconstrained soft_fixed: For future use fixed: Cannot be moved Type: enum Enum Values: unset fixed soft_fixed Default: unset Edit: Yes
place_status_effective	
	This attribute is the effective placement status for the instance. It is the worst case of the instance place_status, instance place_status_cts, and hinst parent read only attribute (closest hinst above that is not set to none). See place_status for the description of possible values. Type: enum Enum Values: unplaced placed fixed cover soft_fixed Default: "" Edit: No
pmbist_instruction_set	
	Attribute to define different programmable MBIST instructions for the different instances of the same block on which programmable MBIST has been inserted. Type: string Default: "" Edit: Yes
power_domain	
	The power_domain for the inst. Type: obj(power_domain) Default: "" Edit: No
power_dynamic	
	The switching + internal power of this inst computed by report_power. Type: double Default: "" Edit: No
power_frequency_domain	
	signifies Frequency of the inst with which it is constrained, sdc/twf file has clock frequency related information for the instances Type: string Default: "" Edit: No
power_internal	
	The internal power of this inst computed by report_power. Type: double Default: "" Edit: No
power_internal_density	
	Internal power density of the inst which is inst power internal / inst area Type: string Default: "" Edit: No
power_leakage	
	The leakage power of this inst computed by report_power. Type: double Default: "" Edit: No
power_leakage_density	

Stylus Common UI Database Object Information

Database Objects--inst

	Leakage power density of the inst which is inst power leakage / inst area Type: string Default: "" Edit: No
power_loading_capacitance	
	Loading capacitance of the inst, it signifies the output net cap + pin cap of the next instance it is connecting to. Type: string Default: "" Edit: No
power_ref_clock	
	The reference clock for toggle and activity calculations. By default this is the fastest clock for multi clock domains but can be modified by some power commands. Type: string Default: "" Edit: No
power_switch_type	
	Returns the switch cell type associated with library cell. The supported values are coarse_grain, fine_grain. Type: string Allowed -index values: analysis_view Default: "" Edit: No
power_switching	
	The switching power of the nets driven by this inst computed by report_power. Type: double Default: "" Edit: No
power_switching_density	
	Switching power density of the inst which is power switching / inst area Type: string Default: "" Edit: No
power_toggle_rate	
	The average number of toggles read from VCD,TCF,SAIF, etc. or from propagation that occur in 1 second on all the pins of this inst. no_value is returned if it cannot be computed. Type: double Default: "" Edit: No
power_total	
	The switching + internal + leakage power of this inst computed by report_power. Type: double Default: "" Edit: No
power_total_density	
	Total power density of the inst which is inst power total / inst area Type: string Default: "" Edit: No
propagate_constant_from_timing_model	
	Allow constant propagation from specific timing model instance. Type: enum Enum Values: 1 0 true false inherited Default: inherited Edit: Yes

Stylus Common UI Database Object Information

Database Objects--inst

rail_domain_voltage_drop	
	Worst Effective drop (VDD-VSS) of the inst in elapse or timing/switching window. Values are loaded in GUI after user loads the voltus state directory Type: string Default: "" Edit: No
rail_ground_voltage_drop	
	Ground bounce for ground rail of the inst, Values are loaded in GUI after user loads the voltus state directory Type: string Default: "" Edit: No
rail_power_voltage_drop	
	Voltage drop for power rail of the inst, Values are loaded in GUI after user loads the voltus state directory Type: string Default: "" Edit: No
rail_reff	
	signifies equivalent resistance for the instance pin from the voltage source, values are loaded in GUI after user load the effective resistance state directory. Type: string Default: "" Edit: No
rail_rlrp	
	signifies least resistance path value of the inst from a voltage source to a pin of the instance, values are loaded in GUI after user load the voltus state directory. It is enabled when enable_rlrp_analysis option of set_rail_analysis_config is turned on Type: string Default: "" Edit: No
ref_lib_cell_name	
	ref_lib_cell_name Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
ref_name	
	ref_name Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
retention_cell	
	Returns the retention_cell derived from Liberty Type: string Allowed -index values: analysis_view Default: "" Edit: No
retime_original_registers	
	Original registers before retiming. Type: string Default: "" Edit: Yes
right_padding	

Stylus Common UI Database Object Information

Database Objects--inst

	Value of right padding for the instance. Type: int Default: 0 Edit: Yes
route_halo_bottom	
	Specifies the size of a routing halo around the inst on the bottom side. A value of 0 means there is no route_halo. It is used to keep routes away from the block edges to reduce cross-coupling to wires inside the block. A route_halo can only be added to blocks and is not allowed on standard-cells. The router will only route through the halo to reach pins on the boundary of the block. The value applies to all sides of the block and only positive values are allowed. Type: coord Default: no_value Edit: Yes
route_halo_bottom_layer	
	The bottom layer for the routing halo. Type: obj(layer) Default: "" Edit: Yes
route_halo_left	
	Specifies the size of a routing halo around the inst on the left side. A value of 0 means there is no route_halo. It is used to keep routes away from the block edges to reduce cross-coupling to wires inside the block. A route_halo can only be added to blocks and is not allowed on standard-cells. The router will only route through the halo to reach pins on the boundary of the block. The value applies to all sides of the block and only positive values are allowed. Type: coord Default: no_value Edit: Yes
route_halo_polygon	
	Specifies the polygon shape of the inst routing halo. Type: point Default: "" Edit: No
route_halo_right	
	Specifies the size of a routing halo around the inst on the right side. A value of 0 means there is no route_halo. It is used to keep routes away from the block edges to reduce cross-coupling to wires inside the block. A route_halo can only be added to blocks and is not allowed on standard-cells. The router will only route through the halo to reach pins on the boundary of the block. The value applies to all sides of the block and only positive values are allowed. Type: coord Default: no_value Edit: Yes
route_halo_top	
	Specifies the size of a routing halo around the inst on the top side. A value of 0 means there is no route_halo. It is used to keep routes away from the block edges to reduce cross-coupling to wires inside the block. A route_halo can only be added to blocks and is not allowed on standard-cells. The router will only route through the halo to reach pins on the boundary of the block. The value applies to all sides of the block and only positive values are allowed. Type: coord Default: no_value Edit: Yes
route_halo_top_layer	
	The top layer of the routing halo. Type: obj(layer) Default: "" Edit: Yes
safety_dcls_cell_type	

Stylus Common UI Database Object Information

Database Objects--inst

	The type of cell added to support the implementation of a dcls (dual core lock step) safety mechanism. Supported values: Type: <code>enum</code> Enum Values: none isolate_split_net_group_input isolate_split_net_group_output isolate_split_network_start isolate_split_network_branch isolate_split_network_group_input common_cause_failure isolate_split_network_last_combo isolate_split_network_stop Default: none Edit: No
safety_dcls_isolate_groups	
	The dcls (dual core lock step) groups isolated by this inst. Type: <code>obj(group)*</code> Default: "" Edit: No
safety_dcls_isolate_is_group_input_cell	
	If true, then this inst is an isolation cell on an input net for a dcls (dual core lock step) group. Type: <code>bool</code> Default: false Edit: No
safety_dcls_isolate_is_group_output_cell	
	If true, then this inst is an isolation cell on an output net for a dcls (dual core lock step) group. Type: <code>bool</code> Default: false Edit: No
safety_dcls_isolate_signal_type	
	The type of signal isolated by this inst for a dcls (dual core lock step) group. Supported values: Type: <code>enum</code> Enum Values: none clock reset scan_enable pin input output Default: none Edit: No
safety_dmr_cell_type	
	The type of cell added to support the implementation of a dmr (dual modular redundancy) safety mechanism. Supported values: Type: <code>enum</code> Enum Values: none isolate_split_net isolate_split_network_start isolate_split_network_branch isolate_split_network_last_combo Default: none Edit: No
safety_dmr_parent	
	The dmr (dual modular redundancy) parent-flop of this object. Type: <code>obj(pin)</code> Default: "" Edit: No
safety_failure_mode	
	The name of the failure-mode to which this inst belongs, as defined in the USF (Unified Safety Format) file. Type: <code>string</code> Default: "" Edit: No
safety_mechanism	
	The name of the safety-mechanism applied to this inst, as defined in the USF (Unified Safety Format) file. Type: <code>string</code> Default: "" Edit: No
safety_mechanism_type	

Stylus Common UI Database Object Information

Database Objects--inst

	The type of safety-mechanism applied to this inst, as defined in the USF (Unified Safety Format) file. Type: string Default: "" Edit: No
safety_tmr_cell_type	
	The type of cell added to support the implementation of a tmr (triple modular redundancy) safety mechanism. Supported values: Type: enum Enum Values: none isolate_split_net isolate_split_network_start isolate_split_network_branch isolate_split_network_last_combo Default: none Edit: No
safety_tmr_clones	
	The tmr (triple modular redundancy) clone flops of this parent flop. This attribute is only stored on tmr parent-flops. Type: obj(inst)* Default: "" Edit: No
safety_tmr_errors	
	The tmr (triple modular redundancy) error-detection-logic of this parent-flop-bit. Type: obj(inst)* Default: "" Edit: No
safety_tmr_parent	
	The tmr (triple modular redundancy) parent flop of this inst. The tmr safety-mechanism is initially applied to the parent flop, adding clone flops and voting logic. Type: obj(inst) obj(pin) Default: "" Edit: No
safety_tmr_voters	
	The tmr (triple modular redundancy) voting logic of this parent flop. This attribute is only stored on tmr parent-flops. Type: obj(inst)* Default: "" Edit: No
safety_tmr_well_tap_left_padding	
	The padding used to reserve space for well-tap cells to be placed next to the left edges of tmr (triple modular redundancy) flops Type: int Default: 0 Edit: No
safety_tmr_well_tap_right_padding	
	The padding used to reserve space for well-tap cells to be placed next to the right edges of tmr (triple modular redundancy) flops. Type: int Default: 0 Edit: No
sdp	
	The parent sdp group which the instance belongs to. Type: obj(sdp) Default: "" Edit: No
single_bit_original_name	
	Stores the hdl name of the original single bits. Type: string Default: "" Edit: Yes
std_cell_main_rail_name	

Stylus Common UI Database Object Information

Database Objects--inst

	Returns the rail name associated with pg pin for which std_cell_main_rail is enabled. Type: string Allowed -index values: analysis_view Default: "" Edit: No
timing_model_type	
	Returns the Liberty model type for a given cell or instance. The supported values are abstracted , extracted , and qtm . Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
top_padding	
	Value of top padding for the instance. Type: int Default: 0 Edit: Yes
trace_retime	
	Trace this instance through retiming, use the attribute 'retime_original_registers' to get the name of the original instance. Type: bool Default: false Edit: Yes
transition_density	
	Transition density of the inst Type: string Default: "" Edit: No
unused_bit_count_in_multibit	
	Tells the number of unused bits in multibit cell. Type: int Default: 0 Edit: Yes
use_cells_only	
	This attribute is the list of allowable cells that this instance can be resized to. Wildcards are supported. If set, any parent (hinst) .dont_use_cells_effective values are ignored. This attribute only applies to operations on mapped designs. The .dont_touch_effective values size_same_height_ok and size_same_footprint_ok will filter this list further based on height and footprint, respectively. Type: string Default: "" Edit: Yes
x_coordinate_max	
	x_coordinate_max Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
x_coordinate_min	
	x_coordinate_min Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
y_coordinate_max	

Stylus Common UI Database Object Information

Database Objects--inst_obs

	y_coordinate_max Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
y_coordinate_min	
	y_coordinate_min Type: double Allowed -index values: analysis_view clock Default: "" Edit: No

inst_obs

Parent Objects

Definition

obstruction shape on an instance

Attribute	Description
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (inst_obs) Default: "" Edit: No

inst_obs_shape

Parent Objects

Definition

This corresponds to one of the .base_cell.obs_layer_shapes.shapes (LEF OBS shapes) for a given inst. It carries a link to the 'inst' object, that a 'shape' object does not have. This allows GUI operations that need both the shape and the inst object together to select, highlight or query the shape. It can only be accessed by selecting the object with the GUI.

Attribute	Description
inst	
	The inst of this OBS shape Type: obj(inst) Default: "" Edit: No
is_ignore_pg_net	
	Indicates that Power/Ground routing is ignored when checking for DRC violations (including shorts) involving the current shape (equivalent to LEF MACRO OBS LAYER EXCEPTPGNET). Type: bool Default: "" Edit: No
layer	

Stylus Common UI Database Object Information

Database Objects--io_constraint

	The layer of the obstruction Type: obj(layer) Default: "" Edit: No
mask	
	Indicates mask number for multiple mask layer usage. Refer to layer's .num_mask attribute for legal range, 0 indicates uncolored. Type: int Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (inst_obs_shape) Default: "" Edit: No
polygon	
	Points of the polygon (if type = polygon) in design coordinates (e.g. base_cell values are transformed for this inst). Type: point* Default: "" Edit: No
rect	
	Box of the shape (if type = rect) in design coordinates (e.g. base_cell values are transformed for this inst). Type: rect Default: "" Edit: No
spacing	
	LEF OBS SPACING value, no_value if not specified in LEF. Type: coord Default: "" Edit: No
type	
	Type of shape (rect, polygon) Type: enum Enum Values: rect polygon Default: "" Edit: No

io_constraint

Parent Objects

[design](#), [root](#)

Definition

IO object for block (term) or chip design (inst) constraints

Attribute	Description
area	
	Area of the instance as defined by the LEF MACRO SIZE or OVERLAP information Type: area Default: "" Edit: No

Stylus Common UI Database Object Information
Database Objects--io_constraint

bbox	
	Bounding box of the inst overlap rects. This is only correct if type = inst. Type: rect Default: "" Edit: Yes
indent	
	Indent of instance from the boundary of the design Type: coord Default: no_value Edit: Yes
inst	
	The IO instance (null/0x0 for block design case) Type: obj(inst) obj(bump) Default: "" Edit: No
is_assigned	
	Indicates that the IO location has been set by the IO placer Type: bool Default: false Edit: Yes
is_corner	
	Indicates that the IO refers to a corner cell Type: bool Default: false Edit: Yes
is_gap_fixed	
	Indicates that the IO spacing constraint is applied Type: bool Default: false Edit: Yes
is_ground	
	Indicates that the IO is a Ground Type: bool Default: "" Edit: No
is_offset_fixed	
	Indicates that the IO offset constraint is applied Type: bool Default: false Edit: Yes
is_power	
	Indicates that the IO is a Power Type: bool Default: "" Edit: No
name	
	IO name Type: string Default: "" Edit: No
obj_type	

Stylus Common UI Database Object Information

Database Objects--layer

	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (io_constraint) Default: "" Edit: No
offset	
	IO relative location from left (North & South) or bottom (East & West) die edge Type: coord Default: no_value Edit: Yes
order	
	IO order per side. Order is left-to-right (North & South) or bottom-to-top (East & West) Type: int Default: "" Edit: Yes
row	
	IO row/ring number (0 is outermost IO row) Type: int Default: 0 Edit: Yes
side	
	Side constraint of IO Type: enum Enum Values: north west south east none Default: "" Edit: Yes
spacing	
	Spacing between IO and previous IO (left or below) Type: coord Default: no_value Edit: Yes
term	
	The IO terminal (only used when 'type = term') Type: obj(port) Default: "" Edit: Yes
type	
	Type of IO (endSpace indicates from last IO to corner, inst indicates real instance case, obs indicates obstruction between IOs, term indicates block design case) Type: string Default: "" Edit: No

layer

Parent Objects

[marker](#), [bump_pin](#), [layer_shape](#), [text](#), [port](#), [via_def](#), [pin_blockage](#), [special_wire](#), [pg_base_pin](#), [hpin](#), [what_if_wire](#), [what_if_via](#), [trim_grid](#), [base_pin](#), [partition](#), [track_pattern](#), [design](#), [tap_layer_distance](#), [tech](#), [pin_guide](#), [root](#), [layer_rule](#), [virtual_wire](#), [port_shape](#), [via_def_rule](#), [route_blockage](#), [patch_wire](#), [antenna_data](#), [inst](#), [wire](#), [bus_guide](#), [net](#), [pin](#), [interposer_route_rule](#), [route_type](#), [bump_grid](#), [inst_obs_shape](#)

Definition

Stylus Common UI Database Object Information

Database Objects--layer

A layer from the LEF or OA technology file.

Attribute	Description
antenna_models	
	The list of antenna model objects for the Oxides of the layer Type: <code>obj(antenna_model)*</code> Default: "" Edit: No
area	
	Layer minimum area from LEF/OpenAccess. If AREA rule is not specified in LEF, the value of -1 in dbu will be returned. Type: <code>area</code> Default: "" Edit: No
backside	
	Indicates that the layer is a backside (underside of the die) layer. Type: <code>bool</code> Default: "" Edit: No
backside_cut_index	
	An index for BACKSIDE cut layers. It is 0 for the TSV cut-layer, -1 for the BACKSIDE cut layer between the first (.backside_route_index = -1) and second (.backside_route_index = -2) backside routing layers below the substrate, -2 for the BACKSIDE cut layer between the second and third backside routing layers, etc. It is 1000 for all other layers. Type: <code>int</code> Default: "" Edit: No
backside_route_index	
	An index for BACKSIDE routing layers. It is -1 for the first backside routing layer closest (just below) the substrate (e.g. the last LEF TYPE ROUTING layer with BACKSIDE property in the technology file), -2 for the second backside routing layer below the substrate, etc. It is 1000 for any other layer. Type: <code>int</code> Default: "" Edit: No
base_name	
	Layer name from LEF/OpenAccess technology file definition. Type: <code>string</code> Default: "" Edit: No
cut_index	
	An index into the cut layers used by vias in the routing. It is 1 for the cut-layer above the first routing layer, 2 for the cut-layer above the second routing layer, etc. It is 0 for the cut-layer just below the first routing layer. The cut_index is -1 for any other layer, including LEF cut layers with TYPE MIMCAP, TSV or BACKSIDE. In OpenAccess, the cut layers are determined by the LEFDefaultRouteSpec validVias list. Type: <code>int</code> Default: "" Edit: No
density_step_x	
	Layer density window step from LEF/OpenAccess Type: <code>coord</code> Default: "" Edit: No
density_step_y	
	Layer density window step from LEF/OpenAccess Type: <code>coord</code> Default: "" Edit: No

Stylus Common UI Database Object Information

Database Objects--layer

density_window_x	
	Layer density window length from LEF/OpenAccess Type: coord Default: "" Edit: No
density_window_y	
	Layer density window width from LEF/OpenAccess Type: coord Default: "" Edit: No
direction	
	Layer pref. direction from LEF/OpenAccess Type: enum Enum Values: horizontal vertical unassigned diag45 diag135 Default: "" Edit: No
eol_track_rules	
	PITCH and TRACKDISTANCE are stored in LEF while offset is stored in floorplan PLE grid. Type: string* Default: "" Edit: No
fill_active_spacing	
	Layer fill minimum spacing from LEF/OpenAccess Type: coord Default: "" Edit: No
fill_gap_spacing	
	Layer fill to fill spacing from LEF/OpenAccess Type: coord Default: "" Edit: No
is_trim_virtual	
	Indicates that the layer is a trim metal virtual layer. Type: bool Default: "" Edit: No
max_density	
	Layer fill maximum density from LEF/OpenAccess Type: double Default: "" Edit: No
max_width	
	Layer maximum wire width from LEF/OpenAccess Type: coord Default: "" Edit: No
mfg_grid	
	Manufacturing grid Type: coord Default: "" Edit: No
min_density	

Stylus Common UI Database Object Information

Database Objects--layer

	Layer fill minimum density from LEF/OpenAccess Type: double Default: "" Edit: No
min_spacing	
	Layer minimum spacing from LEF/OpenAccess Type: coord Default: "" Edit: No
min_width	
	Layer minimum wire width from LEF/OpenAccess Type: coord Default: "" Edit: No
name	
	Layer name from LEF/OpenAccess technology file definition. Type: string Default: "" Edit: No
num_masks	
	Indicates how many masks will be used for the layer (1 = single mask, 2 = double-patterning, 3 = triple-patterning) Type: int Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (layer) Default: "" Edit: No
offset_x	
	Layer offset X from LEF/OpenAccess Type: coord Default: "" Edit: No
offset_y	
	Layer offset Y from LEF/OpenAccess Type: coord Default: "" Edit: No
pitch_x	
	Layer wire pitch X from LEF/OpenAccess Type: coord Default: "" Edit: No
pitch_y	
	Layer wire pitch Y from LEF/OpenAccess Type: coord Default: "" Edit: No
route_index	

Stylus Common UI Database Object Information

Database Objects--layer

	An index into the routing layers. It is 1 for the first LEF TYPE ROUTING layer in the technology file, 2 for the second routing layer, etc. It is 0 for the layer just below the first routing layer (e.g a poly layer or non-routing metal layer that may have pin shapes in some cells). The route_index is -1 for any non-routing layers, or LEF routing layers with LEF TYPE MIMCAP or TYPE BACKSIDE. In OpenAccess, the routing layers are determined by the LEFDefaultRouteSpec validLayers list. Type: <code>int</code> Default: "" Edit: No
spacing_tables	
	A list of all the LEF spacing-table rules for this layer in the LEF SPACINGTABLE syntax. All SPACINGTABLE and LEF58_SPACINGTABLE property values are returned. Type: <code>string</code> Default: "" Edit: No
tech	
	The tech for the layer. Type: <code>obj(tech)</code> Default: "" Edit: No
type	
	The layer type (routing, cut, etc.) from the LEF LAYER TYPE statements or OA tech for this layer. Type: <code>enum</code> Enum Values: invalid masterslice cut overlap routing implant tsv mimcap passivation poly_routing nwell pwell stacked_die cut_mimcap above_die_edge below_die_edge diffusion ignore trim_poly trim_metal region cut_region routing_region trim_metal_region middle_eol high_resistance_routing high_resistance_cut tsv_metal pad_metal stacked_mimcap well_distance well_distance_boundary_tap pad_metal_cut special_cut copde rc_blockage trim_diffusion abut_logic abut_filler cut_pg_rail probe substract Default: "" Edit: No
width	
	Layer wire width from LEF/OpenAccess Type: <code>coord</code> Default: "" Edit: No
wrong_way_min_width	
	min width in the non-preferred direction (from LEF LAYER MINWIDTH WRONGDIRECTION). A value of 0 indicates that there is no special value assigned for a wrong direction behavior. Type: <code>coord</code> Default: "" Edit: No
wrong_way_spacing	
	min spacing in the non-preferred direction (from LEF LAYER SPACING WRONGDIRECTION). A value of 0 indicates that there is no special value assigned for a wrong direction behavior. Type: <code>coord</code> Default: "" Edit: No
wrong_way_width	
	min width in the non-preferred direction (from LEF LAYER WIDTH WRONGDIRECTION). A value of 0 indicates that there is no special value assigned for a wrong direction behavior. Type: <code>coord</code> Default: "" Edit: No
wsp_oa_width_spacing_pattern	

Stylus Common UI Database Object Information

Database Objects--layer_rule

	A list of Tcl dict style parameters that match the OA widthSpacingPattern parameters like this: {name <string> is_from_lib <bool> offset <coord>}. The 'name' and 'offset' match the corresponding OA parameters, and 'is_from_lib <bool>' is true if it is from the OA tech graph and false if it is from the cellview. Type: string Default: "" Edit: No
wsp_oa_width_spacing_snap_pattern_def	
	A list of Tcl dict style parameters that match the OA widthSpacingSnapPatternDef parameters like this: {name <string> is_from_lib <bool> offset <coord> offset_reference <string> wire_type <name> purpose <name>}. The 'name', 'offset', 'wire_type', and 'purpose' match the corresponding OA parameters. 'offset_reference' indicates whether the 'offset' is from the lower-left corner of the boundary or the origin. 'is_from_lib <bool>' is true if it is from the OA tech graph and false if it is from the cellview. Type: string Default: "" Edit: No
wsp_offset	
	Offset from lower-left corner of the core box to the first track. For a horizontal routing layer track, this is a Y offset. Type: coord Default: "" Edit: No
wsp_pattern	
	This is set by add_tracks or by reading an OA design with widthSpacingPatterns defined. It is a list of {width pitch repeat} values that created the tracks. For example, {1.5 2.5 1} {1.0 2.0 3} means a track of 1.5 width, 2.5 track-to-track pitch repeated 1 time, and then a track with width 1.0, track-to-track pitch 2.0 repeated 3 times, then the full pattern repeats. If this is not an SADP layer, there is no width assigned to the track, and a width value of 0.0 is returned. For a horizontal routing layer, the width and pitch are Y values. Type: string* Default: "" Edit: No
wsp_pattern_masks	
	A list of mask values for each track in the wsp_pattern after the pattern repeat sections are expanded. Type: int* Default: "" Edit: No

layer_rule

Parent Objects

[route_rule](#)

Definition

Layer Rule

Attribute	Description
layer	
	The layer of laye rule Type: obj(layer) Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (layer_rule) Default: "" Edit: No

Stylus Common UI Database Object Information
Database Objects--layer_shape

spacing	
	min spacing (from NONDEFAULTRULE LAYER SPACING, LEF LAYER SPACING or SPACINGTABLE) Type: coord Default: "" Edit: No
width	
	wire width (from NONDEFAULTRULE LAYER WIDTH or LEF LAYER WIDTH) Type: coord Default: "" Edit: No

layer_shape

Parent Objects

[base_cell](#), [physical_pin](#)

Definition

layer shape

Attribute	Description
is_ignore_pg_net	
	Indicates that Power/Ground routing is ignored when checking for DRC violations (including shorts) involving the current shape (equivalent to LEF MACRO OBS LAYER EXCEPTPGNET) Type: bool Default: "" Edit: No
layer	
	The layer of blockage Type: obj(layer) Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (layer_shape) Default: "" Edit: No
obs_type	
	The type of this LEF OBS shape corresponding to the LEF MACRO OBS REAL, ABSTRACT, NOROUTE keywords. The type should always be real for any LEF PIN shapes. Type: enum Enum Values: abstract real noroute Default: "" Edit: No
shapes	
	List of shapes that define the blockage area Type: obj(shape)* Default: "" Edit: No
spacing	

	LEF OBS SPACING equivalent min_spacing value, no_value if not specified in LEF. Type: coord Default: "" Edit: No
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lef_set

Parent Objects

Definition

A lef set is a group of lef files used by design(s).

Attribute	Description
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (lef_set) Default: "" Edit: No

lib_arc

Parent Objects

[lib_cell](#), [arc](#), [root](#), [lib_pin](#)

Definition

cte library timing arc

Attribute	Description
aocv_weight	
	Returns the AOCV stage weight for this library arc. By default, all library cells and library arcs have default stage weight of 1.0. The aocv_weight property is specified as a user-defined library attribute in the Liberty timing library explicitly - or, by asserting it via command. You can use -index to return a value for a specific view. Type: double Allowed -index values: analysis_view Default: "" Edit: No
base_name	
	Returns the simple name for this library timing arc in the form: inputPin_outputPin_uniqueIntegerSuffix Type: string Allowed -index values: analysis_view Default: "" Edit: No
from_lib_pin	
	Returns a lib_pin object for beginning pin of this timing arc. You can use -index to return a value for a specific view. Type: obj (lib_pin)* Allowed -index values: analysis_view Default: "" Edit: No
full_name	

Stylus Common UI Database Object Information

Database Objects--lef_set

	Returns the unique name for this library arc in the form: librarySet/libraryName/libraryCell/inputPin_outputPin_uniqueIntegerSuffix. You can use -index to return a value for a specific view. Type: <code>string</code> Default: "" Edit: No
has_bti_aging	
	Returns true if library arc(s) has BTI aging information Type: <code>bool</code> Allowed -index values: analysis_view Default: "" Edit: No
has_ccs_noise	
	Returns true if cell has ccs noise data. Type: <code>bool</code> Allowed -index values: analysis_view clock Default: "" Edit: No
has_hci_aging	
	Returns true if library arc(s) has HCI aging information Type: <code>bool</code> Allowed -index values: analysis_view Default: "" Edit: No
has_mis_fall	
	Return library attribute has_mis for fall output transition. Type: <code>bool</code> Allowed -index values: analysis_view Default: "" Edit: No
has_mis_rise	
	Return library attribute has_mis for rise output transition. Type: <code>bool</code> Allowed -index values: analysis_view Default: "" Edit: No
has_socv	
	Returns true if library arc(s) has SOCV information Type: <code>bool</code> Allowed -index values: analysis_view Default: "" Edit: No
is_disabled	
	Returns a value of true if this library timing arc has been explicitly disabled by the user via a set_disable_timing constraint. You can use -index to return a value for a specific view. Type: <code>bool</code> Allowed -index values: analysis_view Default: "" Edit: No
lib_cell	
	Returns the parent library cell object for this library arc. You can use -index to return a value for a specific view. Type: <code>obj(lib_cell)*</code> Allowed -index values: analysis_view Default: "" Edit: No

Stylus Common UI Database Object Information

Database Objects--lef_set

mode	
	If this timing arc is defined as part of a Liberty mode group, this attribute will return the name of the group. You can use -index to return a value for a specific view. Type: string Allowed -index values: analysis_view Default: "" Edit: No
name	
	Returns an identifier for this library arc in the form: libraryName/libraryCell/inputPin_outputPin_uniqueIntegerSuffix Type: string Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (lib_arc) Default: "" Edit: No
output_signal_level_high	
	Returns the upper/higher limit of partial voltage swing value that is defined for a library cell output and inout pins. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
output_signal_level_low	
	Returns the lower limit of partial voltage swing value that is defined for a library cell output and inout pins. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
sdf_cond	
	Returns the value of the Liberty sdf_cond attribute if specified for this arc. You can use -index to return a value for a specific view. Type: string Allowed -index values: analysis_view Default: "" Edit: No
sense	
	Returns the Liberty timing_sense value associated with this arc. This value is one of: positive_unate, negative_unate, or non_unate. You can use -index to return a value for a specific view. Type: string Allowed -index values: analysis_view Default: "" Edit: No
timing_type	
	Returns the Liberty timing_type associated with this arc. You can consult the Liberty documentation for the list of possible values for this attribute. You can use -index to return a value for a specific view. Type: string Allowed -index values: analysis_view Default: "" Edit: No
to_lib_pin	

Stylus Common UI Database Object Information
Database Objects--lib_cell

	Returns a lib_pin object for terminating pin of this timing arc. You can use -index to return a value for a specific view. Type: <code>obj(lib_pin)*</code> Allowed -index values: analysis_view Default: "" Edit: No
when	
	Returns the value of the Liberty 'when' attribute if specified for this arc. You can use -index to return a value for a specific view. Type: <code>string</code> Allowed -index values: analysis_view Default: "" Edit: No
when_end	
	Returns the value of the Liberty 'when_end' attribute if specified for this arc. You can use -index to return a value for a specific view. Type: <code>string</code> Allowed -index values: analysis_view Default: "" Edit: No
when_library_string	
	Returns the exact value of the Liberty 'when' attribute if specified for this arc. You can use -index to return a value for a specific view. Type: <code>string</code> Allowed -index values: analysis_view Default: "" Edit: No
when_start	
	Returns the value of the Liberty 'when_start' attribute if specified for this arc. You can use -index to return a value for a specific view. Type: <code>string</code> Allowed -index values: analysis_view Default: "" Edit: No

lib_cell

Parent Objects

[seq_function](#), [inst](#), [base_cell](#), [lib_arc](#), [root](#), [library](#), [lib_pin](#)

Definition

cte lib cell

Attribute	Description
aocv_bootstrap_fall_derate_factor	
	Returns aocv coefficient aocv_bootstrap_fall_derate_factor in bootstrap mode on the lib_cell Type: <code>float</code> Allowed -index values: analysis_view Default: "" Edit: No
aocv_bootstrap_rise_derate_factor	
	Returns aocv coefficient aocv_bootstrap_rise_derate_factor in bootstrap mode on the lib_cell Type: <code>float</code> Allowed -index values: analysis_view Default: "" Edit: No
aocv_weight	

Stylus Common UI Database Object Information

Database Objects--lib_cell

	Returns the AOCV stage weight specified for the library cell either explicitly in the library or as a user-defined library attribute Type: double Allowed -index values: analysis_view Default: "" Edit: No
area	
	Returns the area of the library cell as specified by the Liberty timing library Type: double Allowed -index values: analysis_view Default: "" Edit: No
async_clear_pins	
	Returns a list of lib_pins that have both the is_async and is_clear attributes set to true Type: obj(lib_pin)* Default: "" Edit: No
async_preset_pins	
	Returns a list of lib_pins that have both the is_async and is_clear attributes set to true Type: obj(lib_pin)* Default: "" Edit: No
base_cell	
	Returns a pointer the base_cell object associated with this library cell. You can use chaining to reach the attributes of the base_cell Type: obj(base_cell)* Default: "" Edit: No
base_name	
	The base_name of the cell. Type: string Default: "" Edit: No
bit_width	
	Returns the bit width of the cell Type: int Default: "" Edit: No
cell_footprint	
	Returns the library cell footprint. Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
clock_gating_integrated_cell	
	clock_gating_integrated_cell Type: string Allowed -index values: analysis_view Default: "" Edit: No
clock_pins	
	Returns a list of lib_pins which have the attribute is_clock equal to true Type: obj(lib_pin)* Default: "" Edit: No

Stylus Common UI Database Object Information

Database Objects--lib_cell

data_pins	
	Returns a list of lib_pins which have the attribute is_data equal to true Type: obj(lib_pin)* Default: "" Edit: No
dft_mapper_usable	
	The lib_cell is mapper usable. Type: bool Default: false Edit: Yes
dft_timing_model	
	libcell has timing model Type: bool Default: false Edit: Yes
early_clock_check_incremental_derate_factor	
	Returns the early clock path library cell check incremental derating factor specified through the set_timing_derate -incremental_adjust parameter. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
early_data_check_incremental_derate_factor	
	Returns the early data path library cell check incremental derating factor specified through the set_timing_derate -incremental_adjust parameter. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
early_fall_clock_incremental_derate_factor	
	Returns the early clock path library cell check incremental derating factor specified through the set_timing_derate -incremental_adjust parameter - when the -fall parameter is specified. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
early_fall_data_incremental_derate_factor	
	Returns the early data path library cell check incremental derating factor specified through the set_timing_derate -incremental_adjust parameter - when the -fall parameter is specified. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
early_fall_mean_slew_derate_factor	
	Returns early fall mean transition derate factor on the lib cell Type: double Allowed -index values: analysis_view Default: "" Edit: No
early_fall_sigma_slew_derate_factor	
	Returns early fall sigma transition derate factor on the lib cell Type: double Allowed -index values: analysis_view Default: "" Edit: No

Stylus Common UI Database Object Information
Database Objects--lib_cell

early_fall_slew_derate_factor	
	Returns early fall transition derate factor on the lib cell Type: double Allowed -index values: analysis_view Default: "" Edit: No
early_rise_clock_incremental_derate_factor	
	Returns the early clock path library cell check incremental derating factor specified through the set_timing_derate -incremental_adjust parameter - when the -rise parameter is specified. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
early_rise_data_incremental_derate_factor	
	Returns the early data path library cell check incremental derating factor specified through the set_timing_derate -incremental_adjust parameter - when the -rise parameter is specified. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
early_rise_mean_slew_derate_factor	
	Returns early rise mean transition derate factor on the lib cell Type: double Allowed -index values: analysis_view Default: "" Edit: No
early_rise_sigma_slew_derate_factor	
	Returns early rise sigma transition derate factor on the lib cell Type: double Allowed -index values: analysis_view Default: "" Edit: No
early_rise_slew_derate_factor	
	Returns early rise transition derate factor on the lib cell Type: double Allowed -index values: analysis_view Default: "" Edit: No
fall_input_switching_derate_factor	
	Returns user or library defined value for controlling the arrival sensitivity window for simultaneous rising inputs to this cell type Type: double Allowed -index values: analysis_view Default: "" Edit: No
ground_pins	
	Returns a collection of the ground pins of a library cell. Type: obj(lib_pin)* Allowed -index values: analysis_view clock Default: "" Edit: No
has_aging	

Stylus Common UI Database Object Information
Database Objects--lib_cell

	Returns true if library cell has aging information Type: bool Allowed -index values: analysis_view Default: "" Edit: No
has_bti_aging	
	Returns true if library cell has BTI aging information Type: bool Allowed -index values: analysis_view Default: "" Edit: No
has_ccs_noise	
	Returns true if cell has ccs noise data. Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
has_hci_aging	
	Returns true if library cell has HCI aging information Type: bool Allowed -index values: analysis_view Default: "" Edit: No
has_lvf	
	Returns true if library arc(s) of libcell has LVF information Type: bool Allowed -index values: analysis_view Default: "" Edit: No
has_mis	
	Returns true if library cell has multi-instance switching data. Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
has_socv	
	Returns true if library cell has socv data. Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
has_spatial	
	Returns a value of true if a spatial derate exists for a library cell. Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
hierarchical_name	
	hierarchical_name Type: string Allowed -index values: analysis_view Default: "" Edit: No
is_always_on	

Stylus Common UI Database Object Information

Database Objects--lib_cell

	Returns a value of true if the lib_cell is identified as always on type Type: <code>bool</code> Allowed -index values: analysis_view Default: "" Edit: No
is_black_box	
	Returns a value of true if the cell is a black box cell. Type: <code>bool</code> Allowed -index values: analysis_view Default: "" Edit: No
is_buffer	
	Returns a value of true if the lib_cell is identified as a buffer type Type: <code>bool</code> Allowed -index values: analysis_view Default: "" Edit: No
is_combinational	
	Returns a value of true if the library cell is a combinational cell (not a sequential cell). Type: <code>bool</code> Allowed -index values: analysis_view Default: "" Edit: No
is_disable_timing	
	Returns true if set_disable_timing has been set on this library cell Type: <code>bool</code> Allowed -index values: analysis_view Default: "" Edit: No
is_dont_touch	
	Returns a value of true if a set_dont_touch constraint exists for the library cell. Type: <code>bool</code> Allowed -index values: analysis_view clock Default: "" Edit: No
is_exist	
	Returns true if library cell present in view. Type: <code>bool</code> Allowed -index values: analysis_view Default: "" Edit: No
is_fall_edge_triggered	
	Returns a value of true if the library cell is triggered by the falling edge of the clock. Type: <code>bool</code> Allowed -index values: analysis_view Default: "" Edit: No
is_flop	
	Returns true if the library cell is recognized as flip-flop/register type cell Type: <code>bool</code> Default: "" Edit: No
is_generic	

Stylus Common UI Database Object Information
Database Objects--lib_cell

	This cell is Synthesis Generic cell. Type: bool Default: "" Edit: No
is_instantiated	
	Returns a value of true if the library cell is instantiated. Type: bool Allowed -index values: analysis_view Default: "" Edit: No
is_integrated_clock_gating	
	Returns true if the library cell has the Liberty clock_gating_integrated_cell set to true Type: bool Allowed -index values: analysis_view Default: "" Edit: No
is_interface_timing	
	Returns a value of true if a library cell has the Liberty interface_timing attribute set to true Type: bool Allowed -index values: analysis_view Default: "" Edit: No
is_inverter	
	Returns a value of true if the lib_cell is identified as an inverter type Type: bool Allowed -index values: analysis_view Default: "" Edit: No
is_isolation_cell	
	Returns a value of true if the lib_cell is identified as an isolation cell type Type: bool Allowed -index values: analysis_view Default: "" Edit: No
is_latch	
	Returns true if the library cell is recognized as latch type cell Type: bool Default: "" Edit: No
is_level_shifter	
	Returns a value of true if the lib_cell is identified as a level shifter type Type: bool Allowed -index values: analysis_view Default: "" Edit: No
is_macro	
	Returns a value of true if the Liberty attribute is_macro_cell is set true Type: bool Allowed -index values: analysis_view Default: "" Edit: No
is_master_slave_flop	

Stylus Common UI Database Object Information

Database Objects--lib_cell

	Returns true if the lib_cell is recognized as a master/slave flip-flop Type: bool Default: "" Edit: No
is_master_slave_lssd_flop	
	Returns true if this cell has been recognized as a master/slave LSSD device Type: bool Default: "" Edit: No
is_memory	
	Returns true or false depending on whether the associated library cell is recognized as a Liberty memory cell. Cells which include a Liberty memory group definition are recognized as memory cells. Type: bool Allowed -index values: analysis_view Default: "" Edit: No
is_msv	
	Returns true if cell has multi-supply voltages. Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
is_negative_level_sensitive	
	Returns a value of true if the library cell is identified as a negative level-sensitive latch. Type: bool Allowed -index values: analysis_view Default: "" Edit: No
is_pad	
	Returns a value of true if Liberty attribute is_pad is set to true Type: bool Allowed -index values: analysis_view Default: "" Edit: No
is_pll	
	Returns a value of true if the Liberty library is_pll_cell attribute is set to true for this library cell. Type: bool Allowed -index values: analysis_view Default: "" Edit: No
is_positive_level_sensitive	
	Returns a value of true if the library cell is identified as a positive level-sensitive latch. Type: bool Allowed -index values: analysis_view Default: "" Edit: No
is_power_switch	
	Returns a value of true if the lib_cell is identified as power switch type Type: bool Allowed -index values: analysis_view Default: "" Edit: No
is_retention	

Stylus Common UI Database Object Information

Database Objects--lib_cell

	Returns a value of true if the lib_cell is identified as an retention cell type Type: <code>bool</code> Allowed -index values: analysis_view Default: "" Edit: No
is_rise_edge_triggered	
	Returns a value of true if the library cell is triggered by the rising edge of the clock. Type: <code>bool</code> Allowed -index values: analysis_view Default: "" Edit: No
is_sequential	
	Returns a value of true if the library cell is a latch or flip-flop, or if the cell has sequential timing checks. Type: <code>bool</code> Allowed -index values: analysis_view Default: "" Edit: No
is_tristate	
	Returns a value of true if the library cell definition includes the Liberty attribute three_state set to true. Type: <code>bool</code> Allowed -index values: analysis_view Default: "" Edit: No
is_unmapped	
	Returns true if library cell is unmapped. Type: <code>bool</code> Allowed -index values: analysis_view clock Default: "" Edit: No
is_usable	
	The lib_cell can be inferred during mapping/optimization. Type: <code>bool</code> Default: true Edit: Yes
latch_enable_pins	
	Returns a list of lib_pin objects which function as the enable/gate pin of a latch Type: <code>obj(lib_pin)*</code> Default: "" Edit: No
late_clock_check_incremental_derate_factor	
	Returns the late clock path library cell check incremental derating factor specified through the set_timing_derate -incremental_adjust parameter. Type: <code>double</code> Allowed -index values: analysis_view clock Default: "" Edit: No
late_data_check_incremental_derate_factor	
	Returns the late data path instances' incremental derating factor specified through the set_timing_derate -incremental_adjust parameter. Type: <code>double</code> Allowed -index values: analysis_view clock Default: "" Edit: No
late_fall_clock_incremental_derate_factor	

Stylus Common UI Database Object Information
Database Objects--lib_cell

	Returns the late clock path library cell check incremental derating factor specified through the set_timing_derate -incremental_adjust parameter - when the -fall parameter is specified. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
late_fall_clock_period_derate_factor	
	Returns clock period derate factor for fall transition on the lib cell Type: double Allowed -index values: analysis_view Default: "" Edit: No
late_fall_clock_period_mean_derate_factor	
	Returns mean clock period derate factor for fall transition on the lib cell Type: double Allowed -index values: analysis_view Default: "" Edit: No
late_fall_clock_period_sigma_derate_factor	
	Returns sigma clock period derate factor for fall transition on the lib cell Type: double Allowed -index values: analysis_view Default: "" Edit: No
late_fall_data_incremental_derate_factor	
	Returns the late data path library cell check incremental derating factor specified through the set_timing_derate -incremental_adjust parameter - when the -fall parameter is specified. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
late_fall_mean_slew_derate_factor	
	Returns late fall mean transition derate factor on the lib cell Type: double Allowed -index values: analysis_view Default: "" Edit: No
late_fall_pulse_width_derate_factor	
	Returns pulse width derate factor for fall transition on the lib cell Type: double Allowed -index values: analysis_view Default: "" Edit: No
late_fall_pulse_width_mean_derate_factor	
	Returns mean pulse width derate factor for fall transition on the lib cell Type: double Allowed -index values: analysis_view Default: "" Edit: No
late_fall_pulse_width_sigma_derate_factor	
	Returns sigma pulse width derate factor for fall transition on the lib cell Type: double Allowed -index values: analysis_view Default: "" Edit: No

Stylus Common UI Database Object Information
Database Objects--lib_cell

late_fall_sigma_slew_derate_factor	
	Returns late fall sigma transition derate factor on the lib cell Type: double Allowed -index values: analysis_view Default: "" Edit: No
late_fall_slew_derate_factor	
	Returns late fall transition derate factor on the lib cell Type: double Allowed -index values: analysis_view Default: "" Edit: No
late_rise_clock_incremental_derate_factor	
	Returns the late clock path library cell check incremental derating factor specified through the set_timing_derate -incremental_adjust parameter - when the -rise parameter is specified. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
late_rise_clock_period_derate_factor	
	Returns clock period derate factor for rise transition on the lib cell Type: double Allowed -index values: analysis_view Default: "" Edit: No
late_rise_clock_period_mean_derate_factor	
	Returns mean clock period derate factor for rise transition on the lib cell Type: double Allowed -index values: analysis_view Default: "" Edit: No
late_rise_clock_period_sigma_derate_factor	
	Returns sigma clock period derate factor for rise transition on the lib cell Type: double Allowed -index values: analysis_view Default: "" Edit: No
late_rise_data_incremental_derate_factor	
	Returns the late data path library cell check incremental derating factor specified through the set_timing_derate -incremental_adjust parameter - when the -rise parameter is specified. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
late_rise_mean_slew_derate_factor	
	Returns late rise mean transition derate factor on the lib cell Type: double Allowed -index values: analysis_view Default: "" Edit: No
late_rise_pulse_width_derate_factor	

Stylus Common UI Database Object Information

Database Objects--lib_cell

	Returns pulse width derate factor for rise transition on the lib cell Type: double Allowed -index values: analysis_view Default: "" Edit: No
late_rise_pulse_width_mean_derate_factor	
	Returns mean pulse width derate factor for rise transition on the lib cell Type: double Allowed -index values: analysis_view Default: "" Edit: No
late_rise_pulse_width_sigma_derate_factor	
	Returns sigma pulse width derate factor for rise transition on the lib cell Type: double Allowed -index values: analysis_view Default: "" Edit: No
late_rise_sigma_slew_derate_factor	
	Returns late rise sigma transition derate factor on the lib cell Type: double Allowed -index values: analysis_view Default: "" Edit: No
late_rise_slew_derate_factor	
	Returns late rise transition derate factor on the lib cell Type: double Allowed -index values: analysis_view Default: "" Edit: No
leakage	
	Returns a cell_leakage_power property for a lib cell Type: double Allowed -index values: analysis_view Default: "" Edit: No
leakage_power	
	Returns the leakage power of the library cell as specified by the Liberty timing library Type: double Allowed -index values: analysis_view Default: "" Edit: No
level_shifter_type	
	Returns the Level Shifter type for a given lib_cell. The supported values are LH, HL and HL_LH. Type: string Allowed -index values: analysis_view Default: "" Edit: No
lib_arcs	
	Returns a list of lib_arc objects contained with the lib_cell Type: obj(lib_aro)* Default: "" Edit: No
lib_pins	

Stylus Common UI Database Object Information
Database Objects--lib_cell

	Returns a list of lib_pin objects for the lib_cell Type: obj(lib_pin)* Default: "" Edit: No
library	
	Returns the parent library object for the lib_cell Type: obj(library)* Default: "" Edit: No
master_physical_variant_cell	
	Returns the master physical variant cell of the library cell. Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
name	
	Returns the unique name of the library cell in the form lib_cell:librarySetName/libraryName/libraryCellName Type: string Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (lib_cell) Default: "" Edit: No
pg_lib_pins	
	Returns a list of the power and ground lib_pin objects of the lib_cell Type: obj(lib_pin)* Default: "" Edit: No
physical_variant_cells	
	Returns the list of physical variant cells of the master cell. Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
pin_count	
	Returns the number of pins of the cell. Type: int Allowed -index values: analysis_view clock Default: "" Edit: No
power_pins	
	Returns the collection of power pins of the cell. Type: obj(lib_pin)* Allowed -index values: analysis_view clock Default: "" Edit: No
power_switch_type	

Stylus Common UI Database Object Information

Database Objects--lib_cell

	Returns the switch cell type associated with library cell. The supported values are coarse_grain, fine_grain. Type: string Allowed -index values: analysis_view Default: "" Edit: No
receiver_capacitance_type	
	Returns input cap group being used for delay calculations, C1C2 or C1CN Type: string Allowed -index values: analysis_view Default: "" Edit: No
retention_cell	
	Returns the retention_cell derived from Liberty Type: string Allowed -index values: analysis_view Default: "" Edit: No
rise_input_switching_derate_factor	
	Returns user or library defined value for controlling the arrival sensitivity window for simultaneous rising inputs to this cell type Type: double Allowed -index values: analysis_view Default: "" Edit: No
seq_function	
	Returns a list of seq_function associated with that library cell object. Type: obj(seq_function)* Default: "" Edit: No
signal_pins	
	Returns the collection of signal pins Type: obj(pin)* Allowed -index values: analysis_view clock Default: "" Edit: No
std_cell_main_rail_name	
	Returns the rail name associated with pg pin for which std_cell_main_rail is enabled. Type: string Allowed -index values: analysis_view Default: "" Edit: No
sync_clear	
	Returns the collection of sync clear pins Type: obj(lib_pin)* Allowed -index values: analysis_view clock Default: "" Edit: No
sync_enable	
	Returns the collection of sync enable pins Type: obj(lib_pin)* Allowed -index values: analysis_view clock Default: "" Edit: No
sync_preset	

Stylus Common UI Database Object Information
Database Objects--lib_cell

	Returns the collection of sync preset pins Type: obj(lib_pin)* Allowed -index values: analysis_view clock Default: "" Edit: No
test_scan_enable	
	Returns the collection of test_scan_enable pins Type: obj(lib_pin)* Allowed -index values: analysis_view clock Default: "" Edit: No
test_scan_enable_inverted	
	Returns the collection of test_scan_enable_inverted pins Type: obj(lib_pin)* Allowed -index values: analysis_view clock Default: "" Edit: No
test_scan_in	
	Returns the collection of test_scan_in pins Type: obj(lib_pin)* Allowed -index values: analysis_view clock Default: "" Edit: No
test_scan_in_inverted	
	Returns the collection of test_scan_in_inverted pins Type: obj(lib_pin)* Allowed -index values: analysis_view clock Default: "" Edit: No
test_scan_out	
	Returns the collection of test_scan_out pins Type: obj(lib_pin)* Allowed -index values: analysis_view clock Default: "" Edit: No
test_scan_out_inverted	
	Returns the collection of test_scan_out_inverted pins Type: obj(lib_pin)* Allowed -index values: analysis_view clock Default: "" Edit: No
threshold_voltage_group	
	Returns name of a group/category – that is specified based on the cell's threshold voltage (high or low) characteristics. Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
timing_model_type	
	Returns the Liberty model type for a given cell or instance. The supported values are abstracted , extracted , and qtm . Type: string Allowed -index values: analysis_view Default: "" Edit: No

lib_pin

Parent Objects

[lib_cell](#), [pin](#), [lib_arc](#), [root](#), [port](#)

Definition

cte lib pin

Attribute	Description
all_q_pin_of_d_pin	
	Returns the collection of all_q_pin_of_d_pin pins Type: obj(lib_pin) * Allowed -index values: analysis_view clock Default: "" Edit: No
base_name	
	Returns the simple name of this library pin. The base_name of BUFFERX1/A is 'A'. Type: string Allowed -index values: analysis_view Default: "" Edit: No
base_pin	
	Returns the related base_pin object of this library pin. Use chaining to access the base_pin attributes Type: obj(base_pin) * Default: "" Edit: No
capacitance	
	Returns the capacitance for the library pin. Type: double Allowed -index values: analysis_view Default: "" Edit: No
capacitance_max_fall	
	Returns the maximum value of the falling capacitance range. Type: double Allowed -index values: analysis_view Default: "" Edit: No
capacitance_max_rise	
	Returns the maximum value of the rising capacitance range. Type: double Allowed -index values: analysis_view Default: "" Edit: No
capacitance_min_fall	
	Returns the minimum value of the falling capacitance range. Type: double Allowed -index values: analysis_view Default: "" Edit: No
capacitance_min_rise	

Stylus Common UI Database Object Information

Database Objects--lib_pin

	Returns the minimum value of the rising capacitance range. Type: double Allowed -index values: analysis_view Default: "" Edit: No
cell_name	
	Returns the name of the lib_cell of the lib_pin mentioned. Type: string Allowed -index values: analysis_view Default: "" Edit: No
clock_polarity	
	Returns the clock polarity for the library pin. Type: string Allowed -index values: analysis_view Default: "" Edit: No
direction	
	Returns the direction of the library pin: in , out , inout or internal. Internal is for a lib_pin that is not visible in the netlist, but is used internally by the t appear in the GUI, or Verilog, but may appear in timing reports. Type: enum Enum Values: internal in out inout Allowed -index values: analysis_view Default: "" Edit: No
drive_resistance_fall	
	Specifies the driving resistance of the cell for falling transitions Type: double Allowed -index values: analysis_view Default: "" Edit: No
drive_resistance_rise	
	Specifies the driving resistance of the cell for falling transitions Type: double Allowed -index values: analysis_view Default: "" Edit: No
fanout_load	
	Returns the fanout load value of the library pin. Type: double Allowed -index values: analysis_view Default: "" Edit: No
from_lib_arcs	
	Returns the list of lib_arc objects that begin from this library pin Type: obj(lib_arc)* Default: "" Edit: No
full_name	
	Returns a unique reference for the library cell of the form: libset/lib_name/cell_name/pin_name Type: string Allowed -index values: analysis_view Default: "" Edit: No

Stylus Common UI Database Object Information

Database Objects--lib_pin

function	
	Returns the function property for a lib pin Type: string Allowed -index values: analysis_view Default: "" Edit: No
hierarchical_name	
	Returns the hierarchical name of the library pin. Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
input_signal_level	
	Returns the rail name associated with input library pin. Type: string Allowed -index values: analysis_view Default: "" Edit: No
input_signal_level_high	
	Returns partial high voltage swing of an input pin. Type: double Allowed -index values: analysis_view Default: "" Edit: No
input_signal_level_low	
	Returns partial low voltage swing of an input pin. Type: double Allowed -index values: analysis_view Default: "" Edit: No
input_signal_level_voltage	
	Returns rail voltage of signal which is driving the input pin. Type: double Allowed -index values: analysis_view Default: "" Edit: No
is_always_on	
	Returns a value of true if the Liberty library is_always_on attribute is set to true for this library pin. Type: bool Allowed -index values: analysis_view Default: "" Edit: No
is_analog	
	Returns true if lib pin has analog signal Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
is_async	
	Returns a value of true if the library pin is an asynchronous preset pin, or an asynchronous clear pin. Type: bool Allowed -index values: analysis_view Default: "" Edit: No

Stylus Common UI Database Object Information
Database Objects--lib_pin

is_clear	
	Returns a value of true if the pin is an asynchronous clear pin. Type: bool Allowed -index values: analysis_view Default: "" Edit: No
is_clock	
	Returns a value of true if the library pin definition includes clock:true . Type: bool Allowed -index values: analysis_view Default: "" Edit: No
is_clock_gate_clock	
	Returns a value of true if the pin has the Liberty attribute clock_gate_clock_pin set to true Type: bool Allowed -index values: analysis_view Default: "" Edit: No
is_clock_gate_enable	
	Returns a value of true if the pin has the Liberty attribute clock_gate_enable_pin set to true Type: bool Allowed -index values: analysis_view Default: "" Edit: No
is_clock_gating_clock	
	Returns a value true if the pin is defined as a clock pin of a clock gating cell. Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
is_clock_gating_enable	
	Returns a value of true if the pin corresponds to the enable pin of a clock gating cell. Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
is_clock_gating_test	
	Returns a value of true if the pin corresponds to the test pin of a clock gating cell. Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
is_clock_pin	
	Returns a value of true if the library pin definition includes timing_type:setup_rising/setup_falling/hold_rising/hold_falling/recovery_rising/recovery_falling/removal_rising/removal_falling/non_seq_setup_ris: Type: bool Allowed -index values: analysis_view Default: "" Edit: No
is_data	

Stylus Common UI Database Object Information

Database Objects--lib_pin

	Returns a value of true if the library pin is the data pin of a flip-flop. Type: <code>bool</code> Allowed -index values: analysis_view Default: "" Edit: No
is_dynamic_clock_gating_data	
	Returns true if pin is data pin of a dynamic clock gating cell Type: <code>bool</code> Allowed -index values: analysis_view clock Default: "" Edit: No
is_exist	
	Returns a value of true if the library pin exists in view. Type: <code>bool</code> Allowed -index values: analysis_view Default: "" Edit: No
is_fall_edge_triggered_clock	
	is_fall_edge_triggered_clock Type: <code>bool</code> Allowed -index values: analysis_view Default: "" Edit: No
is_fall_edge_triggered_data	
	Returns a value of true if the library pin is the data pin of a falling edge triggered device. Type: <code>bool</code> Allowed -index values: analysis_view Default: "" Edit: No
is_iq_function	
	Returns a value of true if the q pin has a function. Type: <code>bool</code> Allowed -index values: analysis_view clock Default: "" Edit: No
is_iqn_function	
	Returns a value of true if the q bar pin has a function. Type: <code>bool</code> Allowed -index values: analysis_view clock Default: "" Edit: No
is_isolated	
	Returns a value of true if the Liberty library is_isolated attribute is set to true for this library pin. Type: <code>bool</code> Allowed -index values: analysis_view Default: "" Edit: No
is_isolation_cell_clock	
	Returns true if a clock lib pin is part of an isolatoin cell, Type: <code>bool</code> Allowed -index values: analysis_view Default: "" Edit: No
is_isolation_cell_data	

Stylus Common UI Database Object Information

Database Objects--lib_pin

	Returns a value of true if the Liberty library is_isolation_cell_data attribute is set to true for this library pin. Type: <code>bool</code> Allowed -index values: analysis_view Default: "" Edit: No
is_isolation_cell_enable	
	Returns a value of true if the Liberty library is_isolation_cell_enable attribute is set to true for this library pin. Type: <code>bool</code> Allowed -index values: analysis_view Default: "" Edit: No
is_level_shifter_data	
	Returns a value of true if the Liberty library is_level_shifter_data attribute is set to true for this library pin. Type: <code>bool</code> Allowed -index values: analysis_view Default: "" Edit: No
is_level_shifter_enable	
	Returns a value of true if the Liberty library is_level_shifter_enable attribute is set to true for this library pin. Type: <code>bool</code> Allowed -index values: analysis_view Default: "" Edit: No
is_negative_level_sensitive_clock	
	Returns a value of true if the library pin is an enable pin of an active low level-sensitive device. Type: <code>bool</code> Allowed -index values: analysis_view Default: "" Edit: No
is_negative_level_sensitive_data	
	Returns a value of true if the pin is a data pin of an active low level-sensitive device. Type: <code>bool</code> Allowed -index values: analysis_view Default: "" Edit: No
is_pad	
	Returns the pad value of a pin Type: <code>bool</code> Allowed -index values: analysis_view Default: "" Edit: No
is_pll_feedback_pin	
	Returns a value of true if the Liberty library is_pll_feedback_pin attribute is set to true for this library pin. Type: <code>bool</code> Allowed -index values: analysis_view Default: "" Edit: No
is_pll_output_pin	
	Returns a value of true if the Liberty library is_pll_output_pin attribute is set to true for this library pin. Type: <code>bool</code> Allowed -index values: analysis_view Default: "" Edit: No
is_pll_reference_pin	

Stylus Common UI Database Object Information

Database Objects--lib_pin

	Returns a value of true if the Liberty library is_pll_reference_pin attribute is set to true for this library pin. Type: bool Allowed -index values: analysis_view Default: "" Edit: No
is_positive_level_sensitive_clock	
	Returns a value of true if the library pin is an enable pin of an active high level-sensitive device. Type: bool Allowed -index values: analysis_view Default: "" Edit: No
is_positive_level_sensitive_data	
	Returns a value of true if the pin is a data pin of an active high level-sensitive device. Type: bool Allowed -index values: analysis_view Default: "" Edit: No
is_power_switch_enable	
	Returns a value of true if the Liberty library is_power_switch_enable attribute is set to true for this library pin. Type: bool Allowed -index values: analysis_view Default: "" Edit: No
is_preset	
	Returns a value of true if the pin is a preset pin. Type: bool Allowed -index values: analysis_view Default: "" Edit: No
is_ref_clock_pin	
	Returns true if object is a reference clock pin Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
is_retention_cell_enable	
	Returns a value of true if the Liberty library is_retention_cell_enable attribute is set to true for this library pin. Type: bool Allowed -index values: analysis_view Default: "" Edit: No
is_rise_edge_triggered_clock	
	Returns a value of true if the library pin is the clock pin of a rising edge triggered device. Type: bool Allowed -index values: analysis_view Default: "" Edit: No
is_rise_edge_triggered_data	
	Returns a value of true if the library pin is the data pin of a rising edge triggered device. Type: bool Allowed -index values: analysis_view Default: "" Edit: No
is_tristate	

Stylus Common UI Database Object Information

Database Objects--lib_pin

	Returns a value of true if the library pin definition includes three_state:true . Type: <code>bool</code> Allowed -index values: analysis_view Default: "" Edit: No
is_tristate_enable	
	Returns a value of true if the library pin is part of a Liberty three_state_enable logical expression Type: <code>bool</code> Allowed -index values: analysis_view Default: "" Edit: No
is_tristate_output	
	Returns a value of true if the library pin definition includes three_state:true . Type: <code>bool</code> Allowed -index values: analysis_view Default: "" Edit: No
lib_cell	
	Returns the lib_cell parent object of this lib_pin Type: <code>obj(lib_cell)*</code> Default: "" Edit: No
max_capacitance	
	Returns the maximum capacitance limit for the library pin. Type: <code>double</code> Allowed -index values: analysis_view Default: "" Edit: No
max_fanout	
	Returns the maximum fanout value for the library pin. Type: <code>double</code> Allowed -index values: analysis_view Default: "" Edit: No
max_transition	
	Returns the maximum transition time limit specified for a given library pin. If the limit is not specified for a library pin, then the limit specified at the c Type: <code>double</code> Allowed -index values: analysis_view Default: "" Edit: No
min_capacitance	
	Returns the minimum capacitance limit for the library pin. Type: <code>double</code> Allowed -index values: analysis_view Default: "" Edit: No
min_fanout	
	Returns the minimum fanout value for the library pin. Type: <code>double</code> Allowed -index values: analysis_view Default: "" Edit: No
min_transition	

Stylus Common UI Database Object Information

Database Objects--lib_pin

	Returns the minimum transition time limit specified for a given library pin. If the limit is not specified for a library pin, then the limit specified at the cell level. Type: double Allowed -index values: analysis_view Default: "" Edit: No
name	
	Returns the name of the library pin in the form: <library_name>/<cell_name>/<pin_name> Type: string Allowed -index values: analysis_view Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (lib_pin) Default: "" Edit: No
output_signal_level	
	Returns the rail name associated with output library pin. Type: string Allowed -index values: analysis_view Default: "" Edit: No
output_signal_level_high	
	Returns partial high voltage swing of an output pin. Type: double Allowed -index values: analysis_view Default: "" Edit: No
output_signal_level_low	
	Returns partial low voltage swing of an output pin. Type: double Allowed -index values: analysis_view Default: "" Edit: No
pin_capacitance	
	Returns capacitance of library pin. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
pin_capacitance_max_fall	
	Returns max fall capacitance of lib pin. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
pin_capacitance_max_rise	
	Returns max rise capacitance of lib pin. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
pin_capacitance_min_fall	

Stylus Common UI Database Object Information
Database Objects--lib_pin

	Returns min fall capacitance of lib Pin. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
pin_capacitance_min_rise	
	Returns min rise capacitance of lib pin. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
related_ground_pin	
	Returns the related ground pin. Type: obj(pg_pin)* obj(pg_base_pin)* Allowed -index values: analysis_view clock Default: "" Edit: No
related_ground_pin_name	
	related_ground_pin_name Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
related_ground_pin_rail_voltage	
	Returns rail voltage of associated ground pin for the given pin. Type: double Allowed -index values: analysis_view Default: "" Edit: No
related_power_pin	
	Returns the related power pin. Type: obj(pg_pin)* obj(pg_base_pin)* Allowed -index values: analysis_view clock Default: "" Edit: No
related_power_pin_name	
	related_power_pin_name Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
related_power_pin_rail_voltage	
	Returns rail voltage of associated power pin for the given pin. Type: double Allowed -index values: analysis_view Default: "" Edit: No
slew_threshold_percent_fall_high	
	Specifies the upper threshold point used to model a falling transition on this pin. This value is typically inherited from the library-level specification Type: double Allowed -index values: analysis_view Default: "" Edit: No
slew_threshold_percent_fall_low	

Stylus Common UI Database Object Information

Database Objects--library

	Specifies the lower threshold point used to model a falling transition on this pin. This value is typically inherited from the library-level specification. Type: double Allowed -index values: analysis_view Default: "" Edit: No
slew_threshold_percent_rise_high	
	Specifies the upper threshold point used to model a rising transition on this pin. This value is typically inherited from the library-level specification. Type: double Allowed -index values: analysis_view Default: "" Edit: No
slew_threshold_percent_rise_low	
	Specifies the lower threshold point used to model a rising transition on this pin. This value is typically inherited from the library-level specification. Type: double Allowed -index values: analysis_view Default: "" Edit: No
to_lib_arcs	
	Returns the list of lib_arc objects that terminate at this pin Type: obj(lib_arc)* Default: "" Edit: No

library

Parent Objects

[timing_condition](#), [lib_cell](#), [library_set](#), [root](#)

Definition

cte lib

Attribute	Description
base_name	
	Returns the simple name of the library as defined by the Liberty library() group name Type: string Default: "" Edit: No
cap_scale_in_ff	
	Returns the capacitance scaling of the library in femtofarads Type: double Default: "" Edit: No
default_max_transition	
	Returns the default max transition of the library Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
default_power_rail	

Stylus Common UI Database Object Information

Database Objects--library

	Returns the power rail defined by the Liberty default_power_rail attribute Type: <code>string</code> Default: "" Edit: No
default_wireload	
	Returns the wire-load model defined by the Liberty default_wire_load attribute Type: <code>string</code> Default: "" Edit: No
files	
	Returns the list of library files associated with this library Type: <code>string</code> Allowed -index values: analysis_view Default: "" Edit: No
has_cells_having_power_ground_pins	
	Returns a value of true if the library cells in this library have power/ground pin modeling Type: <code>bool</code> Default: "" Edit: No
hierarchical_name	
	hierarchical_name Type: <code>string</code> Allowed -index values: analysis_view Default: "" Edit: No
input_threshold_pct_fall	
	Specifies the delay threshold for a falling input signal. This value is specified as a percentage. Type: <code>double</code> Allowed -index values: analysis_view Default: "" Edit: No
input_threshold_pct_rise	
	Specifies the delay threshold for a rising input signal. This value is specified as a percentage. Type: <code>double</code> Allowed -index values: analysis_view Default: "" Edit: No
is_exist	
	Returns a value of true if library is present in view. Type: <code>bool</code> Default: "" Edit: No
leakage_power_scale_in_nw	
	Returns the leakage power scaling used in this library in units of nanoWatts Type: <code>double</code> Default: "" Edit: No
lib_cells	
	Returns a list of library cell objects contained within this library Type: <code>obj(lib_cell)*</code> Default: "" Edit: No

Stylus Common UI Database Object Information

Database Objects--library

lib_file_path	
	Returns the library file path name. Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
lib_version	
	Returns the library version. Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
name	
	Returns an identifier for this library of the form: library_set/library_name Type: string Default: "" Edit: No
nominal_process	
	Returns the nominal process of the library as specified by the Liberty nom_process attribute Type: double Default: "" Edit: No
nominal_temperature	
	Returns the nominal temperature of the library as specified by the Liberty nom_temperature attribute Type: double Default: "" Edit: No
nominal_voltage	
	Returns the nominal voltage of the library as specified by the Liberty nom_voltage attribute Type: double Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (library) Default: "" Edit: No
operating_condition	
	Returns the default operating conditions from the library Type: string Allowed -index values: analysis_view Default: "" Edit: No
output_threshold_pct_fall	
	Specifies the delay threshold for a falling output signal. This value is specified as a percentage. Type: double Allowed -index values: analysis_view Default: "" Edit: No
output_threshold_pct_rise	

Stylus Common UI Database Object Information

Database Objects--library

	Specifies the delay threshold for a rising output signal. This value is specified as a percentage. Type: double Allowed -index values: analysis_view Default: "" Edit: No
power_rails	
	Returns the power rails defined for this library Type: string Default: "" Edit: No
slew_derate_from_library	
	Specifies the multiplier used to translate between the transition time range used during characterization and the transition time range used in the timing library tables. A characterization range of 30% and library range of 10% would result in a derate value of 0.5. Type: double Allowed -index values: analysis_view Default: "" Edit: No
slew_lower_threshold_pct_fall	
	Specifies the lower threshold point used to model a falling transition on this pin. This value is specified as a percentage. Type: double Allowed -index values: analysis_view Default: "" Edit: No
slew_lower_threshold_pct_rise	
	Specifies the lower threshold point used to model a rising transition on this pin. This value is specified as a percentage. Type: double Allowed -index values: analysis_view Default: "" Edit: No
slew_upper_threshold_pct_fall	
	Specifies the upper threshold point used to model a falling transition on this pin.. This value is specified as a percentage. Type: double Allowed -index values: analysis_view Default: "" Edit: No
slew_upper_threshold_pct_rise	
	Specifies the upper threshold point used to model a rising transition on this pin. This value is specified as a percentage. Type: double Allowed -index values: analysis_view Default: "" Edit: No
threshold_voltage_group	
	Returns name of a group/category – that is specified based on the timing library's threshold voltage (high or low) characteristics. Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
wireload_models	
	Returns a list of the wireload models defined in the library Type: string Default: "" Edit: No

library_set

Parent Objects

[timing_condition](#), [root](#)

Definition

Specifies a group of library files to be treated as a single entity so that higher-level descriptions (delay_corners) can simply refer to the library configuration by name. All non-physical libraries used by the timing must be part of a library_set - including Liberty, AOCV, SOCV, and signal integrity library formats. Use the create_library_set command to define new library_sets and the update_library_set command to update the attributes of existing library_sets.

Attribute	Description
aocv_files	
	Specifies the list of optional AOCV derating library files associated with the library_set Type: string Default: "" Edit: No
libraries	
	Specifies the list of timing library objects that result from the import of the library_files Type: obj(library)* Default: "" Edit: No
library_files	
	Specifies the list of Liberty timing library files associated with the library_set Type: string Default: "" Edit: No
name	
	Provides the name of this library_set object as specified by create_library_set. Type: string Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (library_set) Default: "" Edit: No
si_files	
	Specifies the list of optional signal integrity (.cdb) library files associated with the library_set Type: string Default: "" Edit: No
socv_files	
	Specifies the list of optional SOCV variation library files associated with the library_set Type: string Default: "" Edit: No

marker

Parent Objects

[design](#), [root](#)

Definition

A violation marker. All markers have a .bbox and .originator set, and optionally .polygon for polygon shapes. Other attributes depend on the marker. Markers created by internal check or report commands will have .originator != external and have a .layer, .type, .subtype, .message, and .message_id defined by those commands. Markers created with read_markers or create_marker will have .originator == external and will have .layer, .user_type, .user_subtype, .user_originator, and .message based on the external input.

Attribute	Description
bbox	
	Bounding box of the marker shape. Type: rect Default: "" Edit: No
is_false	
	Indicates the marker is marked as false violation by user. Type: bool Default: false Edit: Yes
is_hidden	
	Indicates the marker is hidden by user in violation browser. Type: bool Default: false Edit: Yes
layer	
	The layer of marker Type: obj(layer) Default: "" Edit: No
message	
	The message attached to the marker Type: string Default: "" Edit: No
message_id	
	The message ID used internally for this message Type: int Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (marker) Default: "" Edit: No
objects	

Stylus Common UI Database Object Information

Database Objects--message

	The objects which caused the DRC. The list may be empty, and is currently limited to at most 2 objects. Type: <code>obj(inst)* obj(net)* obj(pin)* obj(port)* obj(bump)*</code> Default: "" Edit: No
originator	
	The originator of the marker from an internal command, or 'external' if the marker is from create_marker or read_markers. Type: <code>enum</code> Enum Values: unknown check external check_place route_litho verify_power_domain check_filler nano_route euler route_design check_pin_assignment pg ccopt check_floorplan check_bus_guide check_safety_mechanism Default: "" Edit: No
polygon	
	Polygon boundary for the marker if it is not a rectangle. The first point is not repeated as the last point in the list. Type: <code>polygon</code> Default: "" Edit: No
subtype	
	The marker subtype if it was created be an internal check or report command (when .originator != external). Type: <code>string</code> Default: {} Edit: No
type	
	The marker type if it was created by an internal check or report command (when .originator != external). Type: <code>enum</code> Enum Values: none drc antenna connectivity floorplan overlap density ir_drop xtalk ac_limit mixed_signal electrical place route_litho busplanning msv ccopt safety scan_chain_probe Default: "" Edit: No
user_originator	
	The originator given by the user for external markers from create_marker or read_marker (when .originator == external). Type: <code>string</code> Default: {} Edit: No
user_subtype	
	The subtype given by the user for external markers from create_marker or read_marker (when .originator == external). Type: <code>string</code> Default: {} Edit: No
user_type	
	The type given by the user for external markers from create_marker or read_marker (when .originator == external). Type: <code>string</code> Default: {} Edit: No

message

Parent Objects

`root`

Definition

Formatted error, warning and info messages (e.g. ERROR)

Stylus Common UI Database Object Information

Database Objects--message

Attribute	Description
base_name	
	The message name (e.g. ABCDEF-123 for the message "ERROR: (ABCDEF-123):..."). The DPO name would be "message:ABCDEF-123". Type: string Default: "" Edit: No
count	
	The number of times message has been issued in current session. Type: int Default: "" Edit: No
long_help	
	Some messages have additional help information. It is also available with 'man ABCDEF-123' in the DESCRIPTION section. This is also the "DESCRIPTION" part of get_message -long. If the "DESCRIPTION" of a message is "There are no further details for this message", get_db doesn't need to return it, but return an empty string instead. Type: string Default: "" Edit: No
max_print	
	Specifies the maximum number of times the message will be printed to the log file and screen. -1 is unlimited. 0 is suppressed. Type: int Default: 20 Edit: Yes
message	
	A brief explanation of the message. It is also available with 'help ABCDEF-123'. It uses printf syntax with %s, %d, etc. for message specific arguments. This is the get_message -short string. Type: string Default: "" Edit: No
name	
	The message name (e.g. ABCDEF-123 for the message "ERROR: (ABCDEF-123):..."). The DPO name would be "message:ABCDEF-123". Type: string Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (message) Default: "" Edit: No
on_error	
	The error action for the specified message. This only applies if the message has severity = error, and is ignored for messages with severity of warning or info. Valid values are: msg_only: Write out message, with normal command behavior. exit: Fatal error. Exit and return to Linux. stop_script: If message occurs inside 'source <file>', then the source command stops and returns. Type: enum Enum Values: exit stop_script msg_only Default: msg_only Edit: Yes
severity	

	The severity of the message. The severity is "default" until some application writes it out and sets the severity, or the user sets it. Setting it to "default" will undo any previous setting, and then applications will set it if again the next time they write it out. Valid values are: default: A default severity is set to all messages until it is issued at least once. info: An information message. No user action needed. warning: A possible problem, but the command can continue. error: A problem causing wrong behavior. In some cases, the command cannot continue and stops, and sometimes it can continue to enable debugging or prototyping with errors. Type: enum Enum Values: default error warning info Default: default Edit: Yes
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module

Parent Objects

[hinst](#), [design](#), [root](#)

Definition

Cell module

Attribute	Description
allow_ilm_eco	
	The attribute is only valid when the module is an ILM. If true, optimizer can optimize the ILM boundary interface logic to improve timing Type: bool Default: false Edit: Yes
arch_filename	
	Source file that contains the specified module (Verilog) or architecture (VHDL). Type: string Default: "" Edit: Yes
arch_name	
	The name of the specified module (Verilog) or architecture (VHDL). Type: string Default: "" Edit: Yes
base_name	
	Name of cell Type: string Default: "" Edit: No
boundary_optimize_constant_hpins	
	Allow constant propagation through hierarchical boundary pins. Type: enum Enum Values: 1 0 true false inherited Default: inherited Edit: Yes
boundary_optimize_equal_opposite_hpins	
	Allow connection of hierarchical boundary pins through equal/opposite boundary pins. Type: enum Enum Values: 1 0 true false inherited Default: inherited Edit: Yes

Stylus Common UI Database Object Information

Database Objects--module

boundary_optimize_feedthrough_hpins	
	Allow routing of feedthrough boundary pins around a hierarchical instance. Type: enum Enum Values: 1 0 true false inherited Default: inherited Edit: Yes
boundary_optimize_invert_hpins	
	Allow inversion of hierarchical boundary pins. Type: enum Enum Values: 1 0 true false inherited Default: inherited Edit: Yes
boundary_opto	
	Control boundary-optimization on a subdesign. Type: enum Enum Values: true false strict_no Default: true Edit: Yes
clock_gating_auto_path_adjust	
	Controls automatic timing adjustment on clock gate enable paths. Type: enum Enum Values: inherited none fixed variable Default: inherited Edit: Yes
clock_gating_auto_path_adjust_fixed_delay	
	Fixes the path adjustment automatically added to clock gate enable pins. Type: string Default: "" Edit: Yes
clock_gating_auto_path_adjust_multiplier	
	Scales the automatic load-dependent path adjustment added to clock gate enable pins. Type: double Default: no_value Edit: Yes
clock_gating_exclude	
	Exclude this design from clock-gating. Type: bool Default: false Edit: Yes
clock_gating_hierarchical	
	Create hierarchical clock gates in this module. Type: enum Enum Values: auto false Default: auto Edit: Yes
clock_gating_max_flops	
	Set fanout limit to clock-gating cell. Type: int Default: no_value Edit: Yes
clock_gating_min_flops	

Stylus Common UI Database Object Information

Database Objects--module

	Set minimum number of flops needed to add a clock-gating cell. Type: <code>int</code> Default: <code>no_value</code> Edit: Yes
clock_gating_module	
	Module used for clock gating circuitry. Type: <code>string</code> Default: "" Edit: Yes
clock_gating_test_signal	
	Set specified test signal to connect up test logic in clock-gating circuitry. Type: <code>string</code> Default: "" Edit: Yes
control_logic_optimization	
	Controls the effort level for optimization of control logic on this module (basic/advanced/none/inherited). Type: <code>enum</code> Enum Values: basic advanced none inherited Default: inherited Edit: Yes
delete_unloaded_insts	
	Control deletion of unloaded hierarchical instances. Type: <code>enum</code> Enum Values: 1 0 true false inherited Default: inherited Edit: Yes
delete_unloaded_seqs	
	Control deletion of unloaded sequential instances. Type: <code>enum</code> Enum Values: 1 0 true false inherited Default: inherited Edit: Yes
design	
	The design that contains this module. Type: <code>obj(design)*</code> Default: "" Edit: No
dft_dont_scan	
	Exclude this design from scan-synthesis {inherited true false}. Type: <code>enum</code> Enum Values: false true inherited Default: inherited Edit: Yes
dont_touch	
	This attributes defines the user preservation status of the module during optimization. Setting this attribute will set the dont_touch attribute on all hinsts of the same module. This setting will apply to all insts within the hinst unless overridden at a lower level hinst or on the inst object itself. The dont_touch_effective attribute on each child inst and hinst will return the resolved value. Use 'help inst dont_touch' to see the enum value definitions. Type: <code>enum</code> Enum Values: none false true delete_ok const_prop_size_delete_ok const_prop_delete_ok size_delete_ok size_ok size_same_height_ok size_same_footprint_ok Default: none Edit: Yes
dont_touch_file	

Stylus Common UI Database Object Information

Database Objects--module

	Specifies the sourced file to set dont_touch. Type: string Default: "" Edit: Yes
dont_touch_hports	
	This attribute defines the user preservation status for the module object hports during optimization. Supported values: - none: Unconstrained - false: Can add/remove ports - true: Cannot add/remove ports - delete_ok: Can delete ports (if they have no fanout) - add_ok: We cannot delete, or change the polarity or any hport but can add or duplicate hports - invert_ok: We cannot delete, duplicate, or add but can change the polarity of any hport - add_invert_ok: We cannot delete any hport but can add, duplicate, and change the polarity Type: enum Enum Values: none false true delete_ok add_ok invert_ok add_invert_ok Default: none Edit: Yes
dont_use_cells	
	List of cell names (wildcards supported) to disallow for this module during optimization. Setting this applies to all hinsts sharing the module. Overrides any library dont_use values Type: string Default: "" Edit: Yes
dont_use_cells_effective	
	The resolved list of all cell names to disallow during optimization for hinsts of this module, based on the library dont_use and the dont_use_cells and use_cells attributes of this module or the closest parent hinst with a non-empty list. The precedence is: use_cells of this hinst (or closest parent if empty), then dont_use_cells of this hinst (or closest parent if empty), then the library dont_use setting. Type: string Default: "" Edit: No
dp_csa	
	Controls CSA operation on design/subdesign in 'synthesize -to_generic' (basic/none/inherited). Type: enum Enum Values: basic none inherited Default: inherited Edit: Yes
dp_csa_factorize	
	Controls datapath csa_factorize on a design in 'syn_generic'. Type: enum Enum Values: basic none inherited Default: inherited Edit: Yes
dp_opt_power_opt	
	Controls datapath power optimization flow on a given subdesign. Type: enum Enum Values: 1 0 true false inherited Default: inherited Edit: Yes
dp_rename	
	Attribute to control the datapath rewriting optimization level on a subdesign (basic/advanced/none/inherited). Type: enum Enum Values: basic advanced none inherited Default: inherited Edit: Yes

Stylus Common UI Database Object Information
Database Objects--module

dp_sharing	
	To control the datapath sharing on a design/subdesign in 'synthesize -to_generic' (advanced/none/inherited). Type: enum Enum Values: advanced none inherited Default: inherited Edit: Yes
dp_speculation	
	Controls datapath speculation on a design in 'syn_generic'. Type: enum Enum Values: basic none inherited Default: inherited Edit: Yes
embedded_script	
	Stores the SDC command script embedded in the source file that contains the module (Verilog) or entity (VHDL) specification for this design. Type: string Default: "" Edit: Yes
entity_filename	
	Source file that contains the specified module (Verilog) or entity (VHDL). Type: string Default: "" Edit: Yes
entity_name	
	Name of the specified module (Verilog) or entity (VHDL). Type: string Default: "" Edit: Yes
has_dp_rewrite_gint	
	This design/subdesign has rewritten instance. Type: bool Default: false Edit: No
has_dp_shared_gint	
	This design/subdesign has shared instance. Type: bool Default: false Edit: No
hdl_all_filelist	
	Attribute to get all the files along with the included hdl files and package files for a design or subdesign. Type: string Default: "" Edit: Yes
hdl_config_name	
	Name of the configuration used to build this module. Type: string Default: "" Edit: Yes
hdl_cw_list	
	List of CW components used by a design. Type: string Default: "" Edit: Yes

Stylus Common UI Database Object Information

Database Objects--module

hdl_elab_command_params	
	List of {parameter value} pair that are passed to 'elaborate -parameters' command. Type: string Default: "" Edit: Yes
hdl_filelist	
	List of HDL files read prior to the elaboration of a design. Type: string Default: "" Edit: Yes
hdl_parameters	
	List of {parameter value cmdset is_default} used to 'elaborate' a design. 'cmdset' is '1' only if the parameter is set using the 'elaborate -parameters' command. Type: string Default: "" Edit: Yes
hdl_pipeline_comp	
	Specifies whether the design contains a \$pipeline component. Type: bool Default: false Edit: Yes
hdl_user_name	
	Name of the module (Verilog) or entity (VHDL) for this design. Type: string Default: "" Edit: Yes
hdl_vdp_list	
	Information about datapath function primitives used in a module. Type: string Default: "" Edit: Yes
hinsts	
	The hinsts that are derived from (instantiate) this module. Type: obj(hinst)* Default: "" Edit: No
is_ilm	
	This attribute is true if the module is a ILM. This attribute will affect the read_only_effective and dont_touch_effective attribute on all insts and hinsts within the hinsts of this module. It cannot be overridden by other hinst or inst values. Supported values: false: This module is not an ILM true: This module is an ILM Type: bool Default: false Edit: No
is_retimed	
	Indicates that the design has been retimed. Type: bool Default: false Edit: Yes
language	

Stylus Common UI Database Object Information

Database Objects--module

	Name of the language in which module (Verilog) or entity (VHDL) is written. Type: string Default: "" Edit: Yes
library_name	
	Name of the (logical) library in which module (Verilog) or entity (VHDL) reside. Type: string Default: "" Edit: Yes
logical_hier	
	Indicates if the module is a logical hierarchy object, that is, it corresponds to a user-defined entity (in VHDL) or user-defined module (in Verilog). Type: bool Default: true Edit: Yes
minimize_uniquify	
	Try to avoid uniquification of specific subdesign. Type: enum Enum Values: 1 0 true false inherited Default: inherited Edit: Yes
multibit_allow_async_phase_map	
	Allow interchanging async pins for multibit mapping. Type: enum Enum Values: 1 0 true false inherited Default: inherited Edit: Yes
name	
	Name of cell Type: string Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (module) Default: "" Edit: No
opt_retime_move_mux_loop_with_register	
	Move the feedback mux loop along with the register during retiming. Module level control. Type: bool Default: true Edit: Yes
pbs_mim_keep_folded	
	Keep the module folded in MiM flow. Type: bool Default: false Edit: Yes
retime	
	Retime this design during synthesis. Type: bool Default: false Edit: Yes
retime_hard_region	

Stylus Common UI Database Object Information

Database Objects--module

	Do not move flops across the boundaries of this subdesigns. Type: <code>bool</code> Default: false Edit: Yes
retime_period_percentage	
	Adjust target clock period in retiming. Type: <code>double</code> Default: 1.0 Edit: Yes
retime_ssw_sync_enable	
	Directs retiming to use SAT-sweep to find equivalence classes for synchronous enable signals. Module level control. Type: <code>bool</code> Default: false Edit: Yes
speed_grade	
	Effective speed_grade for a subdesign. Type: <code>string</code> Default: "" Edit: No
sub_arch	
	Datapath sub architecture type. Type: <code>enum</code> Enum Values: no_value non_booth booth radix8 ao barrel Default: no_value Edit: Yes
ungroup_ok	
	To allow ungroup for this sub-design. Type: <code>enum</code> Enum Values: false true strict_no Default: true Edit: Yes
use_cells	
	List of cell names to allow for this module during optimization. Setting this applies to all hinsts sharing the module All lib_cells of each base_cell will be allowed. Overrides cells in the dont_use_cells list and any library dont_use values. Type: <code>string</code> Default: "" Edit: Yes
user_speed_grade	
	User defined speed grade. Type: <code>string</code> Default: "" Edit: Yes
user_sub_arch	
	User defined sub_architecture. Valid values 'booth', 'nonbooth', 'radix8', and 'radix8_fast' apply to multipliers. Valid values 'ao' and 'barrel' apply to shifters. Type: <code>enum</code> Enum Values: no_value non_booth booth radix8 ao barrel barrel_radix4 barrel_radix8 barrel_radix16 radix8_fast Default: "" Edit: Yes

net

Parent Objects

flexible_htree, marker, bus, resistor, port, special_wire, hpin, hnet, what_if_wire, via, what_if_via, timing_point, design, clock_tree, net_group, timing_path, bump, root, power_domain, virtual_wire, port_shape, pg_pin, patch_wire, wire, pin, interposer_route_rule, special_via, route_type

Definition

canonical net

Attribute	Description
annotated_capacitance_max	
	Returns maximum annotated capacitance of the net for late path analysis. You can use -index to return the derate factor for specific views. Type: double Allowed -index values: analysis_view Default: "" Edit: No
annotated_capacitance_min	
	Returns the minimum annotated capacitance of the net for early path analysis. You can use -index to return the derate factor for specific views. Type: double Allowed -index values: analysis_view Default: "" Edit: No
annotated_resistance_max	
	Returns the maximum annotated resistance of the net for late path analysis. You can use -index to return the derate factor for specific views. Type: double Allowed -index values: analysis_view Default: "" Edit: No
annotated_resistance_min	
	Returns the minimum annotated resistance of the net for early path analysis. You can use -index to return the derate factor for specific views. Type: double Allowed -index values: analysis_view Default: "" Edit: No
arcs	
	Returns a list of arc objects that are associated with this net. Type: obj(arc)* Default: "" Edit: No
attackers	
	Returns the RC value. Type: obj(net)* Allowed -index values: analysis_view clock Default: "" Edit: No
attackers_rc	
	Returns the RC value. Type: string Allowed -index values: analysis_view Default: "" Edit: No

Stylus Common UI Database Object Information
Database Objects--net

base_name	
	The name at the base of a hierarchical name. So the base_name of i1/i2/i3 is i3. Type: string Default: "" Edit: No
bbox	
	Bounding box of all the wires, vias, pins, and ports of the net. Does not include special wires or special vias. Type: rect Default: "" Edit: No
bottom_preferred_shield_layer	
	The bottom preferred layer on which to add shield on for the net. Type: obj(layer) Default: "" Edit: Yes
bus	
	The bus for this net. Type: obj(bus) Default: "" Edit: No
capacitance_max	
	Returns the total capacitance of the net used for late path analysis. You can use -index to return the capacitance for specific views. Type: double Allowed -index values: analysis_view Default: "" Edit: No
capacitance_min	
	Returns the total capacitance of the net for early path analysis. You can use -index to return the capacitance for specific views. Type: double Allowed -index values: analysis_view Default: "" Edit: No
connections	
	Returns a list of connections of the net (pins and ports). Type: obj(pin)* obj(port)* Default: "" Edit: No
constant	
	Returns if this net has a 1'b0 or 1'b1 on any hpin or hnet, or a supply0/supply1 for any hnet. It does not include timing set_case_analysis assertions, or propagation through cells, or the effects of a constant driver on the net. Type: enum Enum Values: 0 1 no_constant Default: no_constant Edit: No
coupling_capacitance_max	
	Returns the maximum value of coupling capacitance of a net over all the views. Type: double Allowed -index values: analysis_view Default: "" Edit: No
coupling_capacitance_min	

Stylus Common UI Database Object Information

Database Objects--net

	Returns the minimum value of coupling capacitance of a net over all the views. Type: <code>double</code> Allowed -index values: <code>analysis_view</code> Default: <code>""</code> Edit: No
cts_ideal_net	
	If set the clock tree timing engine will consider this net as ideal. Type: <code>bool</code> Default: <code>false</code> Edit: Yes
cts_net_type	
	For a net, return the CTS net type. Will be one of: . "top" - A clock net that has fanout above <code>cts_top_fanout_transitive_count</code>. . "trunk" - A CTS clock net that is not a "top" net nor a "leaf" net. . "leaf" - A CTS clock net that has fanout to only clock sinks. . "unknown" - A CTS clock net that has not been analyzed. . "" - Not a CTS clock net. See also the <code>get_clock_tree_nets</code> command. Type: <code>string</code> Default: <code>""</code> Edit: No
delta_delay_max	
	Return the delta delay of max corner Type: <code>double</code> Allowed -index values: <code>analysis_view clock</code> Default: <code>""</code> Edit: No
delta_delay_max_fall	
	Returns the delta fall delay of max corner. Type: <code>double</code> Allowed -index values: <code>analysis_view clock</code> Default: <code>""</code> Edit: No
delta_delay_max_rise	
	Returns the delta rise delay for max corner. Type: <code>double</code> Allowed -index values: <code>analysis_view clock</code> Default: <code>""</code> Edit: No
delta_delay_min	
	Retuurns the delta delay for min corner. Type: <code>double</code> Allowed -index values: <code>analysis_view clock</code> Default: <code>""</code> Edit: No
delta_delay_min_fall	
	Returns the delta fall delay for min corner. Type: <code>double</code> Allowed -index values: <code>analysis_view clock</code> Default: <code>""</code> Edit: No
delta_delay_min_rise	

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	Returns the delta rise delay for min corner. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
dont_touch	
	This attribute defines the preservation status of an net during optimization. Setting this attribute will preserve all connections on this net. When set, this overrides any setting on hnets of this net. Also, note that the .use attribute for the net can also cause the net to be preserved. Supported values: false: Unconstrained true: Cannot touch delete_ok: Can delete (if they have no sinks) Type: enum Enum Values: false true delete_ok const_prop_delete_ok Default: false Edit: Yes
driver_pins	
	These are the drivers of the net that are pins (no port drivers are returned). Pins that have direction = out or inout are considered drivers. Type: obj(pin)* Default: "" Edit: No
driver_ports	
	These are the drivers of the net that are ports (no pin drivers are returned). Ports that have direction = in or inout are considered drivers. Type: obj(port)* Default: "" Edit: No
drivers	
	Ports and pins that drive the net. These are pins that have direction = out or inout, and ports that have direction = in or inout. Type: obj(pin)* obj(port)* Default: "" Edit: No
duty_cycle	
	Returns the duty cycle. Type: string Allowed -index values: analysis_view Default: "" Edit: No
early_fall_clk_net_delta_derate_factor	
	Returns the early derating factor for falling SI/delta delays on data path nets. You can use -index to return the derate factor for specific views. Type: double Allowed -index values: analysis_view Default: "" Edit: No
early_fall_clk_net_derate_factor	
	Returns the early derating factor for falling static delays on clock path nets. You can use -index to return the derate factor for specific views. Type: double Allowed -index values: analysis_view Default: "" Edit: No
early_fall_data_net_delta_derate_factor	
	Returns the early derating factor for falling SI/delta delays on clock path nets. You can use -index to return the derate factor for specific views. Type: double Allowed -index values: analysis_view Default: "" Edit: No

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early_fall_data_net_derate_factor	
	Returns the early derating factor for falling static delays on data path nets. You can use -index to return the derate factor for specific views. Type: double Allowed -index values: analysis_view Default: "" Edit: No
early_rise_clk_net_delta_derate_factor	
	Returns the early derating factor for rising SI/delta delays on data path nets. You can use -index to return the derate factor for specific views. Type: double Allowed -index values: analysis_view Default: "" Edit: No
early_rise_clk_net_derate_factor	
	Returns the early derating factor for rising static delays on clock path nets. You can use -index to return the derate factor for specific views. Type: double Allowed -index values: analysis_view Default: "" Edit: No
early_rise_data_net_delta_derate_factor	
	Returns the early derating factor for rising SI/delta delays on clock path nets. You can use -index to return the derate factor for specific views. Type: double Allowed -index values: analysis_view Default: "" Edit: No
early_rise_data_net_derate_factor	
	Returns the early derating factor for rising static delays on data path nets. You can use -index to return the derate factor for specific views. Type: double Allowed -index values: analysis_view Default: "" Edit: No
eco_opt_add_cells	
	Specifies the cells to be used by eco opt engine for inserting buffer/inverter on the net. Type: string Default: "" Edit: Yes
escaped_name	
	The full hierarchical name including escaped chars (if any). It follows DEF escaping syntax, so bus-bit chars [], or a hierarchy char \ that is not a bus-bit or hierarchy char has a \ in front of it. So i1/i2 is a two-level hierarchical name while i1\i2 is single level name, and a[0] is a bus-bit, while a[0\] is a scalar. Type: string Default: "" Edit: No
frequency	
	Reports the frequency of the net. Type: double Default: "" Edit: No
has_detailed_parasitics	
	Returns a value of true if the net, or one of its parts, has detailed parasitics associated with it from either RC extraction or SPEF annotation. Type: bool Allowed -index values: analysis_view Default: "" Edit: No

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has_valid_parasitics	
	Returns true if the net has complete RC info from SPEF. Type: bool Allowed -index values: analysis_view Default: "" Edit: No
hierarchical_name	
	hierarchical_name Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
hnets	
	List of hNets which make up this net. Type: obj(hnet)* Default: "" Edit: No
is_clock	
	Indicates that net is a clock according to timing constraints and tracing. It is only valid if the timing-graph has been created. Type: bool Default: "" Edit: No
is_clock_source_latency	
	Returns true if object is part of source latency network Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
is_custom_route	
	Indicates that the net has Virtuoso/OA custom routing constraints that NR does not support, so NR should not route this net. Type: bool Default: "" Edit: No
is_dirty	
	Indicates that the net is dirty. Type: bool Default: false Edit: Yes
is_early_global_routed	
	Indicate that the net is from early global route Type: bool Default: false Edit: Yes
is_edit	
	Indicate that the wire/via on the net has been modified. This attribute only works when "set_db edit_wire_create_is_edit_flag 1". Please look up for more details by "help edit_wire_create_is_edit_flag". Type: bool Default: false Edit: Yes
is_external	

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	Indicates that net is connected to a to a top-level port. Type: <code>bool</code> Default: "" Edit: No
is_fixed_bump	
	Indicates that net is fixed to bump connection. Type: <code>bool</code> Default: "" Edit: No
is_ground	
	Indicate that the net is ground net Type: <code>bool</code> Default: "" Edit: No
is_hierarchical	
	Returns true if net is a part of hierarchichal module Type: <code>bool</code> Allowed -index values: analysis_view clock Default: "" Edit: No
is_ideal	
	Returns true if the net is part of an ideal network due to the assertion or propagation of the set_ideal_network constraint. You can use -index to return the is_ideal value for specific views. Type: <code>bool</code> Allowed -index values: analysis_view Default: "" Edit: No
is_ilm	
	Specify if a top level net is connecting to ILM (Interface Logic Model) modules and some or all terms (either drive or sink) of the net are inside ILM module. Type: <code>bool</code> Default: "" Edit: No
is_partial_parallel_route_net	
	Returns true if the net is a partial parallel routed net. Type: <code>bool</code> Default: false Edit: No
is_physical_only	
	Indicates that the net is a physical only net that does not get written to logical Verilog netlist. Type: <code>bool</code> Default: "" Edit: No
is_power	
	Indicate that the net is power net Type: <code>bool</code> Default: "" Edit: No
is_voltage_asserted	

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	Returns whether a pg net voltage is inferred by tool or user asserted. Type: <code>bool</code> Allowed -index values: analysis_view clock Default: "" Edit: No
late_fall_clk_net_delta_derate_factor	
	Returns the late derating factor for falling SI/delta delays on data path nets. You can use -index to return the derate factor for specific views. Type: <code>double</code> Allowed -index values: analysis_view Default: "" Edit: No
late_fall_clk_net_derate_factor	
	Returns the late derating factor for falling static delays on clock path nets. You can use -index to return the derate factor for specific views. Type: <code>double</code> Allowed -index values: analysis_view Default: "" Edit: No
late_fall_data_net_delta_derate_factor	
	Returns the late derating factor for falling SI/delta delays on clock path nets. You can use -index to return the derate factor for specific views. Type: <code>double</code> Allowed -index values: analysis_view Default: "" Edit: No
late_fall_data_net_derate_factor	
	Returns the late derating factor for falling static delays on data path nets. You can use -index to return the derate factor for specific views. Type: <code>double</code> Allowed -index values: analysis_view Default: "" Edit: No
late_rise_clk_net_delta_derate_factor	
	Returns the late derating factor for rising SI/delta delays on data path nets. You can use -index to return the derate factor for specific views. Type: <code>double</code> Allowed -index values: analysis_view Default: "" Edit: No
late_rise_clk_net_derate_factor	
	Returns the late derating factor for rising static delays on clock path nets. You can use -index to return the derate factor for specific views. Type: <code>double</code> Allowed -index values: analysis_view Default: "" Edit: No
late_rise_data_net_delta_derate_factor	
	Returns the late derating factor for rising SI/delta delays on clock path nets. You can use -index to return the derate factor for specific views. Type: <code>double</code> Allowed -index values: analysis_view Default: "" Edit: No
late_rise_data_net_derate_factor	
	Returns the late derating factor for rising static delays on data path nets. You can use -index to return the derate factor for specific views. Type: <code>double</code> Allowed -index values: analysis_view Default: "" Edit: No
load_pins	

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	These are the loads of the net that are pins (no port loads are returned). Pins that have direction = in or inout are considered loads. Type: obj(pin)* Default: "" Edit: No
load_ports	
	These are the loads of the net that are ports (no pin loads are returned). Ports that have direction = out or inout are considered loads. Type: obj(port)* Default: "" Edit: No
loads	
	Ports and pins that are loads for the net. These are pins that have direction = in or inout, and ports that have direction = out or inout. Type: obj(pin)* obj(port)* Default: "" Edit: No
lumped_capacitance_max	
	Returns the total lumped capacitance of the net used for late path analysis. You can use -index to return the capacitance for specific views. Type: double Allowed -index values: analysis_view Default: "" Edit: No
lumped_capacitance_min	
	Returns the total lumped capacitance of the net for early path analysis. You can use -index to return the capacitance for specific views. Type: double Allowed -index values: analysis_view Default: "" Edit: No
lumped_resistance_max	
	Returns the maximum resistance of the net for late path analysis. You can use -index to return the value for a specific view Type: double Allowed -index values: analysis_view Default: "" Edit: No
lumped_resistance_min	
	Returns the maximum resistance of the net for late path analysis. You can use -index to return the value for a specific view Type: double Allowed -index values: analysis_view Default: "" Edit: No
max_voltage	
	Max voltage for the net, determined by the max net voltage of all active hold views. Net voltage is determined using the following procedure: voltage from net's driver pin, associated power from CPF related_power_pins command or Liberty related_pg_pin attribute or LEF SUPPLYSENSITIVITY statement; power domain operating voltage; or default system voltage. Type: voltage Default: "" Edit: No
min_voltage	
	Min voltage for the net, determined by the min net voltage of all active setup views. Net voltage is determined using the following procedure: voltage from net's driver pin, associated power from CPF related_power_pins command or Liberty related_pg_pin attribute or LEF SUPPLYSENSITIVITY statement; power domain operating voltage; or default system voltage. Type: voltage Default: "" Edit: No
name	

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	The same as .escaped_name without any \ escape chars. This name is commonly used to avoid problems with \ escape chars in Tcl scripts unless you carefully use list operators. Like SDC commands, 'get_db insts i1/i2' will first try to match i1/i2, and if not found then match i1\i2 so that flattening hierarchical names (e.g. with ungroup) does not require changing the names in Tcl scripts. Type: string Default: "" Edit: No
num_connections	
	Number of connections to the net (number of pins + number of ports) Type: int Default: "" Edit: No
num_driver_pins	
	Returns the number of driver pins of the net. Type: int Allowed -index values: analysis_view clock Default: "" Edit: No
num_drivers	
	Number of drivers for the net. These are pins that have direction = out or inout, and ports that have direction = in or inout. Type: int Default: "" Edit: No
num_load_pins	
	Returns this number of load pins of the net. Type: int Allowed -index values: analysis_view clock Default: "" Edit: No
num_loads	
	Number of loads for the net. These are pins that have direction = in or inout, and ports that have direction = out or inout. Type: int Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (net) Default: "" Edit: No
patch_wires	
	List of patch_wire Type: obj(patch_wire)* Default: "" Edit: No
pin_capacitance_max	
	Returns the portion of the net's total capacitance which comes the library max pin capacitance values. This is used for late path delay calculation. You can use -index to return the derate factor for specific views. Type: double Allowed -index values: analysis_view Default: "" Edit: No
pin_capacitance_max_fall	

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	pin_capacitance_max_fall Type: double Allowed -index values: analysis_view Default: "" Edit: No
pin_capacitance_max_rise	
	pin_capacitance_max_rise Type: double Allowed -index values: analysis_view Default: "" Edit: No
pin_capacitance_min	
	Returns the portion of the net's total capacitance which comes the library min pin capacitance values. This is used for early path delay calculation. You can use -index to return the derate factor for specific views. Type: double Allowed -index values: analysis_view Default: "" Edit: No
pin_capacitance_min_fall	
	pin_capacitance_min_fall Type: double Allowed -index values: analysis_view Default: "" Edit: No
pin_capacitance_min_rise	
	pin_capacitance_min_rise Type: double Allowed -index values: analysis_view Default: "" Edit: No
pins	
	Returns all pins on the net. Type: obj(pin)* Default: "" Edit: No
power_duty_cycle	
	. Type: double Default: "" Edit: No
power_ref_clock	
	The static probability of the signal to stay high during one clock cycle. It is a value between 0.0 and 1.0. no_value is returned if it cannot be computed. Type: string Default: "" Edit: No
power_switching	
	The switching power of this net computed by report_power. Type: double Default: "" Edit: No
power_toggle_per_clock	

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	The number of toggles per clock cycle. It is computed from power_toggle_rate * <period of power_ref_clock>. It can be as high as 2.0 for a clock, and is between 0 and 1 for a signal net. If there is more than one output pin, then it is the average of all the output pin power_toggle_rates. no_value is returned if no value can be computed. Type: double Default: "" Edit: No
power_toggle_rate	
	The average number of toggles read from VCD, TCF, SAIF, etc. or from propagation that occur in 1 second on the net. Type: double Default: "" Edit: No
power_toggle_rate_max	
	The average number of toggles read from VCD, TCF, SAIF, etc. or from propagation that occur in 1 second on the net. Type: double Default: "" Edit: No
power_toggle_rate_source	
	The source of the power_toggle_rate value. The enum values mean: asserted: the value is user-specified by direct assertion or from a switching activity file. clock: the value is derived from the clock waveform. computed: the value is computed by propagating internal switching activity. constant: the value is driven by a constant value (e.g. from set_case_analysis or tie-offs). default: the value is not user-specified, but determined from the root power_default_toggle_rate attribute. Type: string Default: "" Edit: No
probe_marker	
	the probe marker port_shape of this net Type: obj(port_shape) Default: "" Edit: No
probe_pad	
	the probe pad port_shape of this net Type: obj(port_shape) Default: "" Edit: No
resistance_max	
	Returns the maximum resistance of the net for late path analysis. You can use -index to return the value for a specific view. Type: resistance Allowed -index values: analysis_view Default: "" Edit: No
resistance_min	
	Returns the minimum resistance of the net for early path analysis. You can use -index to return the value for a specific view. Type: resistance Allowed -index values: analysis_view Default: "" Edit: No
route_avoid_detour	
	Avoids detours of roughly more than a few gcell grids on the specified nets. This attribute affects global routing only. It's recommended that you use caution with this attribute, as it adds congestion to the design. Type: bool Default: false Edit: Yes
route_bottom_preferred_layer	

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	Specifies the preferred lowest routing layer. This attribute is a soft limit; that is, the router might use a layer below the specified layer if necessary to complete routing. You can set <code>route_preferred_layer_effort</code> to determine how strict the router needs to honor this limit. If the top routing layer is equal to or greater the 4th routing layer (layer with <code>.route_index = 4</code>), and no bottom preferred routing layer is specified for clock nets, the router automatically sets the <code>route_bottom_preferred_layer</code> for clock nets to the 3rd routing layer (layer with <code>.route_index = 3</code>), to prevent the clock nets from blocking other nets. Type: <code>obj(layer)</code> Default: "" Edit: Yes
route_coaxial_shield_interval	
	Specifies the interval of crossties for the shield wires by routing track. A single value applies to all layers, or you can use a list of <code><layer> <value></code> pairs for individual layer controls. The default of 0 means there are no gaps between cross-ties. For example, '2' means 2 empty tracks between cross-ties on all layers, while 'metal2 2 metal3 4' means metal2 has 2 empty track and, metal3 have 4 empty tracks, and the default of 0 for all the other layers. The <code><layer></code> args can be input as a DPO name 'layer:metal3' or a layer name 'metal3', or as a layer index '3'. The output is always in a DPO name format like {layer:metal2 2 layer:metal3 4}. Type: <code>string</code> Default: 0 Edit: Yes
route_coaxial_shield_layers	
	Specifies the bottom and top routing layer for coaxial shields added around a shielded net. For example {metal4 metal6} means no coaxial shield shapes are added below metal4 or above metal6. The <code><layer></code> names can be input as a DPO name "layer:metal3", or a layer name "metal3", or as a layer index "3". The output is always in a DPO name format like {layer:metal3 layer:metal6}. Type: <code>obj(layer)*</code> Default: {} Edit: Yes
route_coaxial_shield_layers_limit	
	Specifies the bottom and top routing layer limit for routing that should be shielded by coaxial shields. For example, {metal3 metal6} means the routing wires on layers metal3 to metal6 should have coaxial shields added around them. The <code><layer></code> names can be input as a DPO name 'layer:metal3' or a layer name 'metal3', or as a layer index '3'. The output is always in a DPO name format like {layer:metal3 layer:metal6}. Type: <code>obj(layer)*</code> Default: {} Edit: Yes
route_em_rule	
	Specifies a <code>route_rule</code> which is used for fixing EM issues. It is used to add a wider <code>route_rule</code> for the routing at output driver pins for a distance given by <code>route_em_rule_distance</code>. Type: <code>obj(route_rule)</code> Default: "" Edit: Yes
route_em_rule_distance	
	Specifies the routing distance in microns that routing at output driver pins should use the <code>route_em_rule</code> rather than the normal net <code>route_rule</code> to avoid EM problems. Type: <code>coord</code> Default: 0.0 Edit: Yes
route_em_width_rule	
	The layer width and distance of routing at output driver pins needed to fix EM issues. The format is {<code><layer> <width> <dist> ...</code>}. For example {metal2 0.4 5.0 metal3 0.5 6.0}. The router will choose a <code>route_rule</code> that meets the <code><width></code> required for each layer and use it for at least the <code><dist></code> from the output pin on each layer before changing back to the normal <code>route_rule</code> for the net. This attribute cannot be set at the same time as <code>route_em_rule</code> as only one parameter can be used to specify the <code>route_rule</code> to use. The <code><layer></code> names can be input as a DPO name 'layer:metal3' or a layer name 'metal3', or as a layer index '3'. The output is always in a DPO name format like {layer:metal2 0.4 5.0 layer:metal3 0.5 6.0}. Type: <code>string</code> Default: {} Edit: Yes
route_hard_spacing_length	

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	Specifies the parallel run length allowed on each layer to relax a route_rule with a hard spacing constraint (e.g. DEF NONDEFAULTRULE with SPACING constraint) into a soft constraint (e.g. only spacing needed by DRC rules are enforced rather than extra space from the route_rule). The soft constraint is allowed if the parallel run length between any wire on this net, and another wire is smaller than the specified value. For example, {metal3 2.0 metal4 3.0} would use soft-spacing for any metal3 wire on this net that does not have parallel lengths of more than 2.0 microns with other metal3 wires, and metal4 wires for 3.0 microns. The <layer> names can be input as a DPO name 'layer:metal3' or a layer name 'metal3', or as a layer index '3'. The output is always in a DPO name format like {layer:metal3 2.0 layer:metal4 3.0}. Type: string Default: {} Edit: Yes
route_initial_bottom_preferred_layer	
	The initial lowest layer set by set_route_attributes or set_db. Opt will use it as a reference to set bottom_preferred_layer to any value equal or greater than initial_bottom_preferred_layer and smaller than top_preferred_layer for better performance. Type: obj(layer) Default: "" Edit: Yes
route_interposer_interlayer_shielding_rules	
	Returns the interposer inter-layer shielding rules of the net. Type: obj(interposer_route_rule)* Default: "" Edit: No
route_interposer_properties	
	This attributes returns interposer routing related properties as a dict. Type: string Default: "" Edit: No
route_interposer_samelayer_shielding_rules	
	Returns the interposer same-layer shielding rules of the net. Type: obj(interposer_route_rule)* Default: "" Edit: No
route_interposer_shielding_mode	
	Returns the shielding mode of the net. The valid modes are GSG, GSSG, GxSG:n. Type: string Default: "" Edit: No
route_interposer_trunk_rules	
	Returns the interposer trunk rules of the net. Type: obj(interposer_route_rule)* Default: "" Edit: No
route_mask	
	Masks constraint setting for the net. Legal values are from 0 to the mask number for the layers. A value of 0 indicates that there is no mask constraint on the net. A non-zero value indicates that all wiring on the net should be created on the specified mask. Layers that do not support the specified value will be treated as unconstrained. Type: int Default: 0 Edit: Yes
route_mask_layer_range	
	Specifies the bottom and top layer range to which the mask constraint should be applied. For example, {metal3 metal5} means that the mask constraint should be applied on layers metal3 to metal5. The <layer> names can be input as a DPO name 'layer:metal3' or a layer name 'metal3', or as a layer index '3'. The output is always in a DPO name format like {layer:metal3 layer:metal5}. Type: obj(layer)* Default: {} Edit: Yes

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route_max_fanout	
	Specifies the maximum fanout value for each "steiner tree" for trunk routing (e.g. .route_pattern = trunk). If not set to 0, this value overrides the root attribute route_design_trunk_with_cluster_target_size value for this net. Type: int Default: 0 Edit: Yes
route_min_stack_via_layer	
	Specifies the lowest stack_via layer for output pins attached to this net. It indicates that stack_vias should be added up to the specified layer. If a pin already has a pre-defined custom stack_via, and the upper most layer of the via is still below the specified layer, additional stack_vias should be added up to the specified layer. Then regular routing would start from the specified layer. Type: obj(layer) Default: "" Edit: Yes
route_multi_cut_via_effort	
	Specifies how much effort the router should use to add multi-cut vias. Higher settings will result in more multi-cut vias at the expense of more congestion and higher total routing length. Type: enum Enum Values: low medium high Default: low Edit: Yes
route_one_side_spacing_layer_range	
	Specifies the bottom to top layer range that a one_side_spacing constraint should be applied on. This constraint means that extra spacing from the route_rule (e.g. LEF or DEF NONDEFAULTRULE SPACING values with or without HARDSPACING) is reserved on only one side. For example, {metal3 metal5} means only one side of the routing wires will get extra spacing for layers metal3 to metal5. Type: obj(layer)* Default: {} Edit: Yes
route_pattern	
	Specifies the routing pattern. - steiner: Routes all pins with a single Steiner tree. - trunk: Routes pins in clusters to the nearest special_route trunk on the net. Trunk routing will cluster pins into separate Steiner trees and connects each tree to the nearest special_route trunk. By default the max number of pins in any single Steiner tree is controlled by the root attribute route_design_trunk_with_cluster_target_size, unless this net has the route_max_fanout attribute set to override it. Type: enum Enum Values: steiner trunk fishbone Default: steiner Edit: Yes
route_preferred_extra_space_tracks	
	Specifies the preferred extra spacing tracks for this net. It adds extra spacing in units of "routing tracks" (which vary in width on different layers). Use this attribute to give critical nets extra space to reduce coupling. Setting it to 1 gives the routing 1 extra routing pitch of spacing (e.g. one empty routing track on both sides of each wire). This parameter creates a soft rule and, if the design is congested, might not give the extra space to the net. If you specify this attribute, Early Global Route also gives additional space to the specified nets. Type: int Default: 0 Edit: Yes
route_preferred_layer_effort	
	Determines how much effort the router uses to meet the preferred layer limits for this net. Use this attribute with route_top_preferred_layer and route_bottom_preferred_layer attributes. Higher values will make the router try to obey the preferred routing layer range more strongly at the expense of more congestion and longer total routing length. Type: enum Enum Values: low medium high Default: low Edit: Yes
route_rule	

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	The route_rule (e.g. LEF or DEF NONDEFAULTRULE) to use for this net. Use this attribute for critical nets, such as clock nets, that might need to have wider wires or wider spacing than other nets. Type: obj(route_rule) Default: "" Edit: Yes
route_rule_effort	
	Specifies the route_rule effort to use for this net. auto: Follows the route_rule .is_hard_spacing setting (e.g. LEF NONDEFAULTRULE HARDSPACING). hard: Makes the route_rule hard even if .is_hard_spacing == false for the route_rule. soft: Makes the route_rule soft even if .is_hard_spacing == true for the route_rule. Type: enum Enum Values: auto hard soft Default: auto Edit: Yes
route_shield_nets	
	Specifies one or two PG nets to use as shields for critical nets. You should route these PG nets before routing this net so the shields can be connected to the PG mesh properly when the signal router runs. Normally only one shield net is used for shielding both sides. You can also specify two route_shield_nets (one for each side), but you cannot control which side is used if two nets are given. The shields are added as special_wires on each side of the signal wires with .shield_net pointing at the net that is shielded. The signal wires should be shielded for the full length, except near pins. Type: obj(net)* Default: {} Edit: Yes
route_shield_side	
	By default both sides of a shielded signal wire have special_wire shields. If this value is set to 1, then only one side will be shielded, which will improve congestion at the expense of less effective shielding. Type: enum Enum Values: 1 2 Default: 2 Edit: Yes
route_si_post_repair	
	Repairs crosstalk violations on this net. This includes changing topology, layer assignments, spacing and location to avoid coupling to other nets. This attribute only has an effect if the root attribute route_design_with_si_driven is true. Type: bool Default: false Edit: Yes
route_skip	
	Stops the router from routing this net. When set true for a net, the router treats the net as if you had marked it + FIXED in the DEF file and it becomes an obstruction for other nets. If the net is partially routed, the router does not complete the routing. This attribute also applies to Early Global Route. Type: bool Default: false Edit: Yes
route_skip_antenna_repair	
	Prevents the router from repairing process antenna violations on this net by either layer-hopping to higher layers or by adding diodes. This attribute is useful on nets that will have antenna violations repaired at the next level of hierarchy or with some other method. Type: bool Default: false Edit: Yes
route_stack_via_distance	
	Specifies the maximum distance in microns that stack_vias can be added from any pin that requires a stack_via. By default, the value of 0 means stack_vias must be directly on top of the pin. Type: double Default: 0.0 Edit: Yes
route_stripe_layer_range	

Stylus Common UI Database Object Information

Database Objects--net

	This is used for nets that have pre-routed special_wires (e.g. clock trunks, or PG mesh for always-on PG nets, etc.) to limit the signal routing to connect to special_wires in the given layer range. The root attribute route_design_tie_net_to_shape controls which special_wire shapes (stripe, ring, etc.) can be used for connections. Type: obj(layer)* Default: {} Edit: Yes
route_top_preferred_layer	
	Specifies the preferred highest layer for routing this net. This attribute is a soft limit; that is, the router might use a layer above the specified layer if necessary to complete routing. You can set route_preferred_layer_effort to determine how strictly the router should honor this limit. Use this attribute to improve timing by restricting the routing layers available for critical nets, while simultaneously routing other nets on all routing layers. Type: obj(layer) Default: "" Edit: Yes
route_weight	
	Specifies a relative weight for routing nets. In each switch box, the router routes nets with the highest weight first, then the next highest weight, and so on. Specify a value higher than 2 to ensure a net is routed before other nets. A switch box is a collection of global routing cells used during detailed routing. When you specify a net weight and route a net, the router does not route the entire net according to the weight. It routes the nets in each switch box with the highest weight first, then the next highest weight, and so on. Net weight. Type: int Default: 2 Edit: Yes
shield_tap_instance_insertion_effort	
	different route type for clock shielding, including high frequency(high) and low frequency(standard) that main difference is what ground tie cell instance/via used to achieve shield segments electrical connection. None: default Standard: use existing standard cell vss m0 pins as much as possible, create a new instance of a ground tie cell under the shield route if cannot find an existing vss m0 pin within a user controlled search distance of a required connection, the new cell vss pins can have a "max fanout" of 2 shield nets. High: must create ground tie cell instances, and user can specify the physical distance of these new ground tie cells along the clock segments. new instance m0 vss pin to the clock shield segments will be a via stack of single vias on ever layer to the shield nets. It is acceptable for the router to create a n-1 metal shape to connect adjacent vss shield segments together and then route this single vss shape down to m0 ground tie pin. Shield segments may be electrically connected between adjacent route layers through vias. If this is done, there shall be no Manhattan distance between ground tie cells that is larger than the user provided value. Newly inserted ground tie cells cannot be shared between clocks. Type: enum Enum Values: none standard high Default: none Edit: Yes
special_vias	
	List of special vias. Type: obj(special_via)* Default: "" Edit: No
special_wires	
	List of special_wires (DEF SPECIALNETS equivalent) Type: obj(special_wire)* Default: "" Edit: No
timing_constant	
	Reports constant value on net Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
timing_constant_source	

Stylus Common UI Database Object Information

Database Objects--net

	Reports source of constant value on net Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
toggle_rate	
	Returns the toggle rate. Type: string Allowed -index values: analysis_view Default: "" Edit: No
top_preferred_shield_layer	
	The top preferred layer on which to add shield on for the net. Type: obj(layer) Default: "" Edit: Yes
total_capacitance_max_fall	
	Returns the total falling capacitance of the net used for late path analysis. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
total_capacitance_max_rise	
	Returns the total rising capacitance of the net used for late path analysis. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
total_capacitance_min_fall	
	Returns the total falling capacitance of the net used for early path analysis. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
total_capacitance_min_rise	
	Returns the total rising capacitance of the net used for early path analysis. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
total_lumped_capacitance_max_fall	
	Returns the total capacitance of the net used for late path analysis. You can use -index to return the capacitance for specific views. Type: double Allowed -index values: analysis_view Default: "" Edit: No
total_lumped_capacitance_max_rise	
	Returns the total capacitance of the net used for late path analysis. You can use -index to return the capacitance for specific views. Type: double Allowed -index values: analysis_view Default: "" Edit: No
total_lumped_capacitance_min_fall	

Stylus Common UI Database Object Information

Database Objects--net

	Returns the total capacitance of the net for early path analysis. You can use -index to return the capacitance for specific views. Type: double Allowed -index values: analysis_view Default: "" Edit: No
total_lumped_capacitance_min_rise	
	Returns the total capacitance of the net for early path analysis. You can use -index to return the capacitance for specific views. Type: double Allowed -index values: analysis_view Default: "" Edit: No
use	
	Indicates how this net is used. The enum values correspond to DEF and OpenAccess enum names. Note that the .use value is not used by any timing analysis commands because they only rely on SDC, .lib and Verilog data. You can change the .use value between signal, clock and analog, but you cannot change any power or ground values (you must delete and create a new net instead). signal: By default, nets have this value. clock: This value is set by clock-tree creation commands like ccopt_design. The 'real' clock status from SDC and .lib files is found in .is_clock after report_timing is run. So the .use 'clock' value will not match the .is_clock value unless you have run ccopt_design on this clock net. analog: This marks the net as an analog net. This can be used with some route_design controls to skip these nets or handle them in a special way. ground: A ground net. power: A power net. Type: enum Enum Values: signal analog clock ground power scan tieoff Default: "" Edit: Yes
vias	
	List of vialnsts. Type: obj(via)* Default: "" Edit: No
virtual_wires	
	List of virtual_wire Type: obj(virtual_wire)* Default: "" Edit: No
vl_vh_analysis_status	
	Reports the glitch analysis status for net Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
vlu_vho_analysis_status	
	Reports the glitch analysis status for net Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
what_if_vias	
	List of 'what if' vias. Type: obj(what_if_via)* Default: "" Edit: No
what_if_wires	

Stylus Common UI Database Object Information
Database Objects--net

	List of 'what if' wires Type: <code>obj(what_if_wire)*</code> Default: "" Edit: No
wire_capacitance_max	
	Returns the portion of the net's total capacitance which comes extracted or annotated wire capacitance values. This is used for late path delay calculation. You can use -index to return the derate factor for specific views. Type: <code>double</code> Allowed -index values: <code>analysis_view</code> Default: "" Edit: No
wire_capacitance_max_fall	
	Returns the wire capacitance of the net for the maximum value of the falling capacitance range. Type: <code>double</code> Allowed -index values: <code>analysis_view</code> Default: "" Edit: No
wire_capacitance_max_rise	
	Returns the wire capacitance of the net for the maximum value of the rising capacitance range. Type: <code>double</code> Allowed -index values: <code>analysis_view</code> Default: "" Edit: No
wire_capacitance_min	
	Returns the portion of the net's total capacitance which comes extracted or annotated wire capacitance values. This is used for early path delay calculation. You can use -index to return the derate factor for specific views. Type: <code>double</code> Allowed -index values: <code>analysis_view</code> Default: "" Edit: No
wire_capacitance_min_fall	
	Returns the wire capacitance of the net for the minimum value of the falling capacitance range. Type: <code>double</code> Allowed -index values: <code>analysis_view</code> Default: "" Edit: No
wire_capacitance_min_rise	
	Returns the wire capacitance of the net for the minimum value of the rising capacitance range. Type: <code>double</code> Allowed -index values: <code>analysis_view</code> Default: "" Edit: No
wire_lumped_capacitance_max	
	Returns the portion of the net's total lumped capacitance which comes extracted or annotated wire lumped capacitance values. Type: <code>double</code> Allowed -index values: <code>analysis_view</code> Default: "" Edit: No
wire_lumped_capacitance_max_fall	
	Returns the lumped wire capacitance of the net for the maximum value of the falling capacitance range. Type: <code>double</code> Allowed -index values: <code>analysis_view</code> Default: "" Edit: No

Stylus Common UI Database Object Information

Database Objects--net_group

wire_lumped_capacitance_max_rise	
	Returns the lumped wire capacitance of the net for the maximum value of the rising capacitance range. Type: double Allowed -index values: analysis_view Default: "" Edit: No
wire_lumped_capacitance_min	
	Returns the portion of the net's total lumped capacitance which comes extracted or annotated wire lumped capacitance values. Type: double Allowed -index values: analysis_view Default: "" Edit: No
wire_lumped_capacitance_min_fall	
	Returns the lumped wire capacitance of the net for the minimum value of the falling capacitance range. Type: double Allowed -index values: analysis_view Default: "" Edit: No
wire_lumped_capacitance_min_rise	
	Returns the lumped wire capacitance of the net for the minimum value of the rising capacitance range. Type: double Allowed -index values: analysis_view Default: "" Edit: No
wires	
	List of wires Type: obj(wire)* Default: "" Edit: No

net_group

Parent Objects

[design](#), [bus_guide](#), [pin_guide](#), [root](#)

Definition

Net Group

Attribute	Description
exclude_net	
	Indicates group net exclude type(all_layer, same_layer, inclusive, all_layer_in_guided_area) Type: enum Enum Values: all_layer_in_guided_area all_layer same_layer inclusive Default: "" Edit: Yes
is_compact	
	Indicate whether the nets in the group are assigned tightly together. By default, nets which are part of a group and associated to a guide can be spread inside the guide based on the area available and alignment to targets Type: bool Default: false Edit: Yes

Stylus Common UI Database Object Information

Database Objects--obj_type

is_guided	
	Indicates if net group is guided Type: bool Default: false Edit: Yes
is_mixed_signal	
	Returns if the net group is a mixed signal group Type: bool Default: "" Edit: No
is_optimize_order	
	Indicates whether net order will be optimized Type: bool Default: false Edit: Yes
is_spread	
	Indicates whether member nets distributed evenly Type: bool Default: false Edit: Yes
keep_out_space	
	Minimum spacing with pin of foreign nets (Unit: track) Type: int Default: 0 Edit: Yes
name	
	Group name Type: string Default: "" Edit: No
nets	
	List of member nets to this net group Type: obj(net) Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (net_group) Default: "" Edit: No
spacing	
	Specifies minimum spacing between the pins of the net group Type: int Default: "" Edit: No

obj_type

Parent Objects

[attribute](#), [root](#)

Definition

Attribute	Description
accept_user_defined_attributes	
	Specifies whether new attributes can be added by user. Type: bool Default: "" Edit: No
attributes	
	list of attribute for this obj_type Type: obj(attribute)* Default: "" Edit: No
help	
	Specifies the help text for the object Type: string Default: "" Edit: No
name	
	obj_type name like inst, lib_cell, etc. Type: string Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (obj_type) Default: "" Edit: No

opcond

Parent Objects

[timing_condition](#), [root](#)

Definition

An opcond represents a named operating condition which is defined by specific process, voltage, and temperature values. Opcond objects may preexist within a Liberty library or "virtual" opconds may be created using the create_opcond command. Virtual opconds can also be modified by using the update_opcond command.

Attribute	Description
base_name	
	The base name of the opcond without any leading library_set or library names (e.g. op1 for libset1/lib1/op1). Type: string Default: "" Edit: No
is_virtual	
	Indicates that this opcond object is a virtual opcond created by the create_opcond command vs. an opcond that is created by reading a Liberty timing library file. Type: bool Default: "" Edit: No

Stylus Common UI Database Object Information

Database Objects--package_component

name	
	The full name of the opcond. The opcond name from a Liberty file includes the library_set and library names (e.g. libset1/lib1/op1) while the full name from create_opcond is just the opcond name (e.g. op1). Type: string Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (opcond) Default: "" Edit: No
process	
	Specifies the Process value of the opcond Type: double Default: "" Edit: No
temperature	
	Specifies the Temperature value of the opcond in Celsius. Type: double Default: "" Edit: No
tree_type	
	The Liberty tree_type model. It is an enum with these choices: none means there is no tree_type value given; best_case_tree models each load pin is close to the driver, so wire capacitance is incurred, but wire resistance is ignored; balanced_tree models when all load pins are on equal branches, so each load pin has an equal portion of the total wire cap and resistance; worst_case_tree models when the load pin is at the extreme end of the wire, so each load pin incurs the full wire capacitance and resistance Type: enum Enum Values: none best_case_tree balanced_tree worst_case_tree Default: "" Edit: No
voltage	
	Specifies the Voltage value of the opcond in units of the library it came from. Type: double Default: "" Edit: No

package_component

Parent Objects

[design](#)

Definition

pkg component

Attribute	Description
cell_name	
	The name of other chip cell in the package Type: string Default: "" Edit: No
inst_name	

Stylus Common UI Database Object Information

Database Objects--package_object

	The name of other chips in the package Type: string Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (package_component) Default: "" Edit: No
pt	
	Location of the other chips in the package Type: point Default: "" Edit: No
ref_design	
	Another identifying name of the chip in the package Type: string Default: "" Edit: No
size	
	Size of the cell Type: point Default: "" Edit: No

package_object

Parent Objects

[design](#), [root](#)

Definition

pkg object

Attribute	Description
die_net_name	
	the name of die net which this package object connected to Type: string Default: "" Edit: No
layer_name	
	The package layer that this package object belongs to Type: string Default: "" Edit: No
name	
	The name of this package object Type: string Default: "" Edit: No
obj_type	

Stylus Common UI Database Object Information

Database Objects--partition

	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (package_object) Default: "" Edit: No
package_net_name	
	the name of package net which this package object connected to Type: string Default: "" Edit: No
pt	
	Location of the package object Type: point Default: "" Edit: No
size	
	Size of the package object Type: point Default: "" Edit: No
type	
	the type of this package object Type: enum Enum Values: bga_ball bond_finger bond_wire route flight_line component other Default: "" Edit: No

partition

Parent Objects

[hinst](#), [inst](#), [design](#), [pin_group](#), [pin_guide](#), [root](#)

Definition

Partition Object

Attribute	Description
base_pins	
	List of partition pins. It is valid only if partition is in non-committed state, i.e. partition module is instantiated as hinst. Otherwise it is set to NULL. Type: obj (base_pin)* Default: "" Edit: No
clones	
	List of clone inst/hinsts. The obj_type returned will be inst if the partition is committed or a blackbox, otherwise the obj_type will be hinst. Type: obj (inst)* obj (hinst)* Default: "" Edit: No
core_to_bottom	
	Spacing between the partition boundary and core design area of the partition module Type: coord Default: no_value Edit: Yes

Stylus Common UI Database Object Information

Database Objects--partition

core_to_left	
	Spacing between the partition boundary and core design area of the partition module Type: coord Default: no_value Edit: Yes
core_to_right	
	Spacing between the partition boundary and core design area of the partition module Type: coord Default: no_value Edit: Yes
core_to_top	
	Spacing between the partition boundary and core design area of the partition module Type: coord Default: no_value Edit: Yes
hpins	
	Hierarchical instance pin of the partition. It is only valid if the partition is uncommitted and still an hinst. Type: obj(hpin)* Default: "" Edit: No
is_black_box	
	Specifies if partition is a blackbox Type: bool Default: "" Edit: No
is_committed	
	Specifies if partition is committed Type: bool Default: "" Edit: No
master	
	The master inst/hInst. The obj_type returned will be inst if the partition is committed or a blackbox, otherwise the objType will be hInst. Type: obj(inst) obj(hinst) Default: "" Edit: No
min_pitch_bottom	
	Specifies the pin pitch (in tracks) Type: int Default: 0 Edit: Yes
min_pitch_left	
	Specifies the pin pitch (in tracks) Type: int Default: 0 Edit: Yes
min_pitch_right	
	Specifies the pin pitch (in tracks) Type: int Default: 0 Edit: Yes
min_pitch_top	

Stylus Common UI Database Object Information

Database Objects--partition

	Specifies the pin pitch (in tracks) Type: int Default: 0 Edit: Yes
name	
	name of the partition Type: string Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (partition) Default: "" Edit: No
pin_bottom_layers	
	List of bottom layers to use for the partition Type: obj(layer)* Default: "" Edit: Yes
pin_left_layers	
	List of left layers to use for the partition Type: obj(layer)* Default: "" Edit: Yes
pin_right_layers	
	List of right layers to use for the partition Type: obj(layer)* Default: "" Edit: Yes
pin_to_corner_tracks	
	Minimum number of routing tracks between pins and each corner. It is a list of int values, where the first value is for the lower-left corner, and the remaining corners are listed in clockwise order. Type: int* Default: "" Edit: No
pin_top_layers	
	List of top layers to use for the partition Type: obj(layer)* Default: "" Edit: Yes
place_halo_bottom	
	Specifies extra spacing around the partition that should not be used for placement Type: coord Default: no_value Edit: Yes
place_halo_left	
	Specifies extra spacing around the partition that should not be used for placement Type: coord Default: no_value Edit: Yes
place_halo_right	

Stylus Common UI Database Object Information

Database Objects--patch_wire

	Specifies extra spacing around the partition that should not be used for placement Type: coord Default: no_value Edit: Yes
place_halo_top	
	Specifies extra spacing around the partition that should not be used for placement Type: coord Default: no_value Edit: Yes
rail_width	
	Specifies the cell rail width Type: coord Default: "" Edit: No
read_only	
	This attributes is set by set_module_view to identify if a partition is read only or not. It means the partition cannot be optimized, and cells inside will not be moved. Setting this attribute will set the dont_touch_effective attribute on all insts and hinsts within the partition unless overridden at a lower level partition. It cannot be overridden by other hinst or inst values. Supported values: false: This partition is allowed to be optimized, even if a partition or design above has read_only = true. true: The partition is read_only. none: No constraint. This partition inherits any read_only from a partition or design above. Type: enum Enum Values: none false true Default: none Edit: No
reserved_layers	
	List of metal layers which are used for routing in the partition and generating partition pins. Any metal layers that are not specified, usually the top-most metal layers, are allowed to route over the partition Type: obj(layer)* Default: "" Edit: Yes
route_halo_bottom_layer	
	The bottom partition layer for which routing halo will be created Type: obj(layer) Default: "" Edit: Yes
route_halo_to_boundary	
	Specifies routing halo around the partition (honored by signal router). Positive values indicate the halo is outside the partition. Negative values indicate the halo is inside of the boundary of the partition and will be pushed into the partition when the partition is committed. Type: coord Default: no_value Edit: Yes
route_halo_top_layer	
	The top partition layer for which routing halo will be created Type: obj(layer) Default: "" Edit: Yes

patch_wire

Parent Objects

[net](#)

Stylus Common UI Database Object Information
Database Objects--patch_wire

Definition

DEF NETS RECT wire

Attribute	Description
has_trim_metal	
	Indicate the path wire has trim metal or not Type: <code>bool</code> Default: "" Edit: No
is_pin_probe	
	Returns true if the patch wire is pin probe patch. Type: <code>bool</code> Default: false Edit: No
layer	
	The layer of patch Type: <code>obj(layer)</code> Default: "" Edit: No
location	
	Reference point to symbolic location Type: <code>point</code> Default: "" Edit: No
mask	
	Indicates mask number for multiple mask layer usage. Refer to layer's .numMask attribute for legal range, 0 indicates uncolored. Type: <code>int</code> Default: 0 Edit: Yes
net	
	The net that the patch belongs to Type: <code>obj(net)</code> Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: <code>enum (patch_wire)</code> Default: "" Edit: No
rect	
	Rectangle that defines the patch_wire shape Type: <code>rect</code> Default: "" Edit: No
route_rule	
	The non-default rule corresponding to the wire, wires with the default routing rule will return NULL (0x0). Type: <code>obj(route_rule)</code> Default: "" Edit: No
status	

Stylus Common UI Database Object Information

Database Objects--path_group

	Wiring status (equivalent to DEF NETS regular wiring status) Type: enum Enum Values: unknown routed fixed cover noshield Default: "" Edit: Yes
trim_metal_color	
	Color of this patch wire trim metal Type: int Default: "" Edit: No
trim_metal_rect	
	The trim_metal rect if the patch_wire has a trim_metal shape attached. This only occurs for some advanced node layers that use self-aligned patterning. {0 0 0 0} is returned if there is no trim_metal attached. Type: rect Default: "" Edit: No

path_group

Parent Objects

[timing_path](#)

Definition

cte path group

Attribute	Description
name	
	The name of the path_group. Type: string Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (path_group) Default: "" Edit: No

pg_base_pin

Parent Objects

[base_cell](#), [pin](#), [tech](#), [root](#), [lib_pin](#), [pg_pin](#)

Definition

A power or ground pin for a base_cell.

Attribute	Description
base_cell	

Stylus Common UI Database Object Information

Database Objects--path_group

	The base_cell of this pg_base_pin. Type: obj(base_cell)* Default: "" Edit: No
base_name	
	The base name of this pg_base_pin without the base_cell name (e.g. vdd). Type: string Default: "" Edit: No
direction	
	The pin direction from .lib if available, otherwise from LEF/OA. It can be in, out, inout or internal. Internal means it is an internal pin from a .lib file for the timing model, and is not part of the netlist. If there is no .lib for this cell, then the direction comes from the LEF PIN DIRECTION or equivalent OA oaTermType. The LEF DIRECTION values (and equivalent oaTermType values) are mapped this way: INPUT = in, OUTPUT = out, OUTPUT TRISTATE = out, INOUT = inout, FEEDTHRU = inout, and the oaTermType unknown = inout. Type: enum Enum Values: in out inout internal Default: "" Edit: No
effective_stack_via_rule	
	The stack_via_rule that is expected, but not required, to be used by the router for connecting this pg_base_pin. The actual stack via rule used (if any) may be effected by other pg_pin and pg_base_pin attributes, or by choices made by the software (optimization, clock tree synthesis, the router etc.) Type: obj(stack_via_rule) Default: "" Edit: No
is_always_on	
	This pg_base_pin is an always on power pin. This attribute can be set by liberty files. Type: bool Default: "" Edit: No
name	
	The name of this pg_base_pin with the base_cell name (e.g. and2/vdd). Type: string Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (pg_base_pin) Default: "" Edit: No
pg_type	
	The type of the power or ground pin from liberty data. PG pins with no liberty entry will have "invalid". Type: enum Enum Values: primary_power primary_ground backup_power backup_ground internal_power internal_ground pwell nwell deep_pwell deep_nwell invalid Default: "" Edit: No
physical_direction	
	Direction of this pin. Type: enum Enum Values: input output inout unknown feedthrough tristate Default: "" Edit: No
physical_pins	

Stylus Common UI Database Object Information

Database Objects--path_group

	Physical pins for the pg_base_pin. One logical pg_base_pin can have multiple physical_pins (equivalent to separate LEF or DEF PORT statements for one LEF or DEF PIN). Each physical_pin normally has one shape, but may have multiple shapes that are all strongly connected to each other. By default, at least one shape of each physical_pin should be connected to the power-mesh. Type: obj(physical_pin)* Default: "" Edit: No
route_bottom_preferred_layer	
	Specifies the preferred lowest routing layer. This attribute is a soft limit; that is, the router might use a layer below the specified layer if necessary to complete routing. You can set route_preferred_layer_effort to determine how strict the router needs to honor this limit. This attribute is intended for usage on always-on power pins that are routed by the signal router rather than the power router (all instances of this pg_base_pin will have the same constraint). Type: obj(layer) Default: "" Edit: Yes
route_max_fanout	
	Specifies the maximum fanout value for each "steiner tree" for trunk routing (e.g. route_pattern = trunk). This attribute is intended for usage on always-on power pins that are routed by the signal router rather than the power router (all instances of this pg_base_pin will have the same constraint). Type: int Default: 0 Edit: Yes
route_pattern	
	Specifies the routing pattern. This attribute is intended for usage on always-on power pins that are routed by the signal router rather than the power router (all instances of this pg_base_pin will have the same constraint). steiner: Routes all pins with a single Steiner tree. trunk: Routes pins in clusters to the nearest special_route trunk on the net. Type: enum Enum Values: steiner trunk Default: steiner Edit: Yes
route_preferred_layer_effort	
	Determines how much effort the router uses to meet the preferred layer limits. Use this attribute with route_top_preferred_layer and route_bottom_preferred_layer attributes. Higher values will make the router try to obey the preferred routing layer range more strongly at the expense of more congestion and longer total routing length. This attribute is intended for usage on always-on power pins that are routed by the signal router rather than the power router (all instances of this pg_base_pin will have the same constraint). Type: enum Enum Values: low medium high Default: low Edit: Yes
route_rule	
	The route_rule (e.g. LEF or DEF NONDEFAULTRULE) to use. This attribute is intended for usage on always-on power pins that are routed by the signal router rather than the power router (all instances of this pg_base_pin will have the same constraint). Type: obj(route_rule) Default: "" Edit: Yes
route_stripe_layer_range	
	Limits the signal routing to connect to special_wires in the given layer range. Exactly two layers should be given for the bottom and top layers in the range. For example, {metal3 metal5} means the router will only connect to special_wires on layers metal3 to metal5. The <layer> names can be input as a DPO name 'layer:metal3' or a layer name 'metal3', or as a layer index '3'. The output is always in a DPO name format like {layer:metal3 layer:metal5}. This attribute is intended for usage on always-on power pins that are routed by the signal router rather than the power router (all instances of this pg_base_pin will have the same constraint). Type: obj(layer)* Default: {} Edit: Yes
route_top_preferred_layer	

Stylus Common UI Database Object Information

Database Objects--path_group

	Specifies the preferred highest layer for routing. This attribute is a soft limit; that is, the router might use a layer above the specified layer if necessary to complete routing. You can set <code>route_preferred_layer_effort</code> to determine how strictly the router should honor this limit. This attribute is intended for usage on always-on power pins that are routed by the signal router rather than the power router (all instances of this <code>pg_base_pin</code> will have the same constraint). Type: <code>obj(layer)</code> Default: "" Edit: Yes
stack_via_required	
	Sets whether stack vias are required for this <code>pg_base_pin</code>. Type: <code>bool</code> Default: false Edit: Yes
stack_via_rule_list	
	List of <code>stack_via_rule</code> that are valid choices for connecting this <code>pg_base_pin</code>. If the list is empty, no stack via is allowed. Type: <code>obj(stack_via_rule)*</code> Default: "" Edit: Yes
stack_via_rule_required	
	Specifies whether a stack via is required when connecting to this <code>pg_base_pin</code>. If set to 0, a stack via is optional. If set to 1, the min one of the <code>stack_via_rule</code> from <code>stack_via_rules</code> must be used to generate a stack via even if a design rule violation occurs. If set to 2, the min one the <code>stack_via_rule</code> from <code>stack_via_rules</code> is used to generate a stack via if a design rule violation does not occur. If set to 3, tool enforces multiple tries from large to min one of the <code>stack_via_rule</code> from <code>stack_via_rules</code> to generate a stack via even if a design rule violation occurs. If set to 4, tool performs multiple tries from large to min one the <code>stack_via_rule</code> from <code>stack_via_rules</code> to generate a stack via if a design rule violation does not occur. Type: <code>enum</code> Enum Values: 0 1 2 3 4 Default: "" Edit: Yes
taper_rule	
	The taper route_rule for the <code>pg_pin</code>. By default, if tapered routing is needed to access the pin, the default route_rule will be used unless this attribute is set (see LEF MACRO PIN TAPERRULE). Type: <code>obj(route_rule)</code> Default: "" Edit: No
tech	
	The tech for this <code>pg_base_pin</code>. Type: <code>obj(tech)</code> Default: "" Edit: No
tied_to	
	If this <code>pg_base_pin</code> is a substrate bias-pin (e.g. <code>pg_type</code> is <code>nwell</code> or <code>pwell</code>), and it is internally tied-to a "master" PG pin, this specifies the PG pin name (e.g. an <code>nwell</code> pin might have <code>tied_to</code> = <code>vdd</code> if it is internally connected to the <code>vdd</code> pin). Type: <code>string</code> Default: "" Edit: No
use	
	Indicates how this <code>pg_base_pin</code> is used from the LEF USE value or OA equivalent. Only power or ground enum values make sense in normal usage, but if there are conflicts between Liberty and LEF/OA definitions, you may have <code>pg_base_pins</code> that are marked in LEF/OA with the other values. Type: <code>enum</code> Enum Values: signal analog power ground clock Default: "" Edit: No

pg_pin

Parent Objects

[inst](#), [design](#), [pin](#), [root](#), [lib_pin](#)

Definition

Power or ground pin information

Attribute	Description
base_name	
	The base name for this pg_pin without the base_cell name (e.g. vdd). Type: string Default: "" Edit: No
effective_stack_via_rule	
	The stack_via_rule that is expected to be used by the router for connecting to this pg_pin. This value is derived from the other pg_pin and pg_base_pin attributes related to stack via. Type: obj(stack_via_rule) Default: "" Edit: No
escaped_name	
	The escaped name of this pg_pin. Type: string Default: "" Edit: No
hnet	
	The hierarchical net connected to this pg_pin. Type: obj(hnet) Default: "" Edit: No
inst	
	The inst containing this pg_pin. Type: obj(inst) Default: "" Edit: No
name	
	The pg_pin name including the inst path (e.g. i1/i2/vdd) Type: string Default: "" Edit: No
net	
	The net connected to this pg_pin. Type: obj(net) Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (pg_pin) Default: "" Edit: No

Stylus Common UI Database Object Information

Database Objects--physical_pin

pg_base_pin	
	The corresponding pg_base_pin on the base_cell for this pg_pin. Type: obj(pg_base_pin) Default: "" Edit: No
route_max_fanout	
	Specifies the maximum fanout value for each "steiner tree" for trunk routing (e.g. route_pattern = trunk). Normally, every pg_pin instantiated by the same pg_base_pin honors the same max fanout. So user only needs to set it on the pg_base_pin. For the few cases where pg_pin has different requirement than the pg_base_pin, NanoRoute will honor the one on pg_pin. This is intended for usage on always-on power pins routed by the signal router rather than the power router. It is most commonly used during late ECO routing to meet EM and IR limits by forcing an isolated pg_pin into its own "cluster" by setting the value to 1. Type: int Default: 0 Edit: Yes
stack_via_rule	
	This value serves as a pg_pin-specific override to use this stack_via_rule when connecting to this pg_pin. If set, it must match one of the elements from the stack_via_rule_list attribute of the corresponding pg_base_pin. The interpretation of this value depends on this pg_pin's stack_via_rule_required attribute. stack_via_rule == {} && stack_via_rule_required == false: stack_via_rule does not affect the choice. stack_via_rule == stackrule1 && stack_via_rule_required == false: Router will prefer stackrule1, but it may select another (or possibly none) if necessary to avoid design rule violations. stack_via_rule == {} && stack_via_rule_required == true: The required stack_via_rule is empty, so the router will not use any stack_via_rule for connecting to this pg_pin (even ignoring stack_via_rule_required on the pg_base_pin if necessary). stack_via_rule == stackrule1 && stack_via_rule_required == true: Router will use stackrule1, even if it leads to design rule violations. Type: obj(stack_via_rule) Default: "" Edit: Yes
stack_via_rule_required	
	Specifies whether the router must use this pg_pin's stack_via_rule value. If false, the stack_via_rule value will be preferred by the router. If true, the stack_via_rule value is required by the router. See the stack_via_rule attribute for more details. Type: bool Default: false Edit: Yes

physical_pin

Parent Objects

[pg_base_pin](#), [base_pin](#), [port](#)

Definition

physical pin

Attribute	Description
class	
	Physical pin class Type: enum Enum Values: undefined none core bump Default: "" Edit: Yes
layer_shapes	

Stylus Common UI Database Object Information

Database Objects--pin

	List of layer_shapes that define the terminal pin geometries. Type: obj(layer_shape)* Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (physical_pin) Default: "" Edit: No
port_number	
	The port number for geometries under the physical_pin. Type: int Default: "" Edit: No
shape_vias	
	List of shape_vias that define the terminal pin geometries. Type: obj(shape_via)* Default: "" Edit: No

pin

Parent Objects

[marker](#), [test_clock](#), [clock](#), [port](#), [def_scan_chain](#), [violation](#), [skew_group](#), [arc](#), [hinst](#), [scan_chain](#), [hnet](#), [actual_scan_segment](#), [timing_point](#), [dft_configuration_mode](#), [design](#), [bus_sink_group](#), [clock_tree](#), [timing_path](#), [root](#), [test_signal](#), [lib_cell](#), [inst](#), [scan_segment](#), [net](#), [actual_scan_chain](#)

Definition

Instance terminal

Attribute	Description
actual_latency_early_fall_max	
	Returns the computed early falling clock latency to this clock pin. If the analysis mode is set to best-case/worst-case, the value will be the Setup capture latency. When operating in OCV mode, the max qualification is ignored. If there are multiple arrival times from different clock phases - the most conservative latency value is reported. You can use the -index function to return latencies with respect to a specific clock. When operating in concurrent MMMC mode, the worst latency across all views is reported. You can use the -index function to filter latencies based on view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
actual_latency_early_fall_min	
	Returns the computed early falling clock latency to this clock pin. If the analysis mode is set to best-case/worst-case, the value will be the Hold launch latency. When operating in OCV mode, the min qualification is ignored. If there are multiple arrival times from different clock phases - the most conservative latency value is reported. You can use the -index function to return latencies with respect to a specific clock. When operating in concurrent MMMC mode, the worst latency across all views is reported. You can use the -index function to filter latencies based on view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
actual_latency_early_rise_max	

Stylus Common UI Database Object Information

Database Objects--pin

	Returns the computed early rising clock latency to this clock pin. If the analysis mode is set to best-case/worst-case, the value will be the Setup capture latency. When operating in OCV mode, the max qualification is ignored. If there are multiple arrival times from different clock phases - the most conservative latency value is reported. You can use the -index function to return latencies with respect to a specific clock. When operating in concurrent MMMC mode, the worst latency across all views is reported. You can use the -index function to filter latencies based on view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
actual_latency_early_rise_min	
	Returns the computed early rising clock latency to this clock pin. If the analysis mode is set to best-case/worst-case, the value will be the Hold launch latency. When operating in OCV mode, the min qualification is ignored. If there are multiple arrival times from different clock phases - the most conservative latency value is reported. You can use the -index function to return latencies with respect to a specific clock. When operating in concurrent MMMC mode, the worst latency across all views is reported. You can use the -index function to filter latencies based on view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
actual_latency_late_fall_max	
	Returns the computed late falling clock latency to this clock pin. If the analysis mode is set to best-case/worst-case, the value will be the Setup launch latency. When operating in OCV mode, the max qualification is ignored. If there are multiple arrival times from different clock phases - the most conservative latency value is reported. You can use the -index function to return latencies with respect to a specific clock. When operating in concurrent MMMC mode, the worst latency across all views is reported. You can use the -index function to filter latencies based on view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
actual_latency_late_fall_min	
	Returns the computed late falling clock latency to this clock pin. If the analysis mode is set to best-case/worst-case, the value will be the Hold capture latency. When operating in OCV mode, the min qualification is ignored. If there are multiple arrival times from different clock phases - the most conservative latency value is reported. You can use the -index function to return latencies with respect to a specific clock. When operating in concurrent MMMC mode, the worst latency across all views is reported. You can use the -index function to filter latencies based on view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
actual_latency_late_rise_max	
	Returns the computed late rising clock latency to this clock pin. If the analysis mode is set to best-case/worst-case, the value will be the Setup launch latency. When operating in OCV mode, the max qualification is ignored. If there are multiple arrival times from different clock phases - the most conservative latency value is reported. You can use the -index function to return latencies with respect to a specific clock. When operating in concurrent MMMC mode, the worst latency across all views is reported. You can use the -index function to filter latencies based on view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
actual_latency_late_rise_min	
	Returns the computed late rising clock latency to this clock pin. If the analysis mode is set to best-case/worst-case, the value will be the Hold capture latency. When operating in OCV mode, the min qualification is ignored. If there are multiple arrival times from different clock phases - the most conservative latency value is reported. You can use the -index function to return latencies with respect to a specific clock. When operating in concurrent MMMC mode, the worst latency across all views is reported. You can use the -index function to filter latencies based on view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
adaptive_ipba_clock_period_threshold	

Stylus Common UI Database Object Information

Database Objects--pin

	Reports percentage clock period threshold used for filtering violation during adaptive IPBA Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
arrival_clocks	
	Reports clocks that correspond to the arrival times Type: obj(clock)* Allowed -index values: analysis_view clock Default: "" Edit: No
arrival_level	
	Reports arrival level of the pin Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
arrival_max	
	Reports maximum arrival time Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
arrival_max_fall	
	Returns the latest falling arrival time to the specified pin across all concurrent MMMC views. You can use -index to determine the worst arrival for a specific view. You can also use -index to get the latest arrival time related to a specific clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
arrival_max_rise	
	Returns the latest rising arrival time to the specified pin across all concurrent MMMC views. You can use -index to determine the worst arrival for a specific view. You can also use -index to get the latest arrival time related to a specific clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
arrival_mean_max_fall	
	In SOCV analysis mode, this returns the mean value of the latest falling arrival time to the specified pin across all concurrent MMMC views. You can use -index to determine the latest arrival for a specific view. You can also use -index to get the worst arrival time related to a specific clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
arrival_mean_max_rise	
	In SOCV analysis mode, this returns the mean value of the latest rising arrival time to the specified pin across all concurrent MMMC views. You can use -index to determine the latest arrival for a specific view. You can also use -index to get the worst arrival time related to a specific clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
arrival_mean_min_fall	

Stylus Common UI Database Object Information

Database Objects--pin

	In SOCV analysis mode, this returns the mean value of the earliest falling arrival time to the specified pin across all concurrent MMMC views. You can use -index to determine the earliest arrival for a specific view. You can also use -index to get the earliest arrival time related to a specific clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
arrival_mean_min_rise	
	In SOCV analysis mode, this returns the mean value of the earliest rising arrival time to the specified pin across all concurrent MMMC views. You can use -index to determine the earliest arrival for a specific view. You can also use -index to get the earliest arrival time related to a specific clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
arrival_min	
	Reports minimum arrival time Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
arrival_min_fall	
	Returns the earliest falling arrival time to the specified pin across all concurrent MMMC views. You can use -index to determine the worst arrival for a specific view. You can also use -index to get the earliest arrival time related to a specific clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
arrival_min_rise	
	Returns the earliest falling arrival time to the specified pin across all concurrent MMMC views. You can use -index to determine the worst arrival for a specific view. You can also use -index to get the worst arrival time related to a specific clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
arrival_sigma_max_fall	
	In SOCV analysis mode, this returns the variation component of the latest falling arrival time to the specified pin across all concurrent MMMC views. You can use -index to determine the latest arrival for a specific view. You can also use -index to get the worst arrival time related to a specific clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
arrival_sigma_max_rise	
	In SOCV analysis mode, this returns the variation component of the latest rising arrival time to the specified pin across all concurrent MMMC views. You can use -index to determine the latest arrival for a specific view. You can also use -index to get the worst arrival time related to a specific clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
arrival_sigma_min_fall	

Stylus Common UI Database Object Information

Database Objects--pin

	In SOCV analysis mode, this returns the variation component of the earliest falling arrival time to the specified pin across all concurrent MMMC views. You can use -index to determine the earliest arrival for a specific view. You can also use -index to get the earliest arrival time related to a specific clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
arrival_sigma_min_rise	
	In SOCV analysis mode, this returns the mean value of the earliest rising arrival time to the specified pin across all concurrent MMMC views. You can use -index to determine the earliest arrival for a specific view. You can also use -index to get the earliest arrival time related to a specific clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
arrival_skewness_max_fall	
	In SOCV analysis mode, this returns the skewness value of the latest falling arrival time to the specified pin across all concurrent MMMC views. You can use -index to determine the latest arrival for a specific view. You can also use -index to get the worst arrival time related to a specific clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
arrival_skewness_max_rise	
	In SOCV analysis mode, this returns the skewness value of the latest rising arrival time to the specified pin across all concurrent MMMC views. You can use -index to determine the latest arrival for a specific view. You can also use -index to get the worst arrival time related to a specific clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
arrival_skewness_min_fall	
	In SOCV analysis mode, this returns the skewness value of the earliest falling arrival time to the specified pin across all concurrent MMMC views. You can use -index to determine the earliest arrival for a specific view. You can also use -index to get the earliest arrival time related to a specific clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
arrival_skewness_min_rise	
	In SOCV analysis mode, this returns the skewness value of the earliest rising arrival time to the specified pin across all concurrent MMMC views. You can use -index to determine the earliest arrival for a specific view. You can also use -index to get the earliest arrival time related to a specific clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
arrival_window	
	Returns a list of the earliest and latest, rising and falling arrival times per clock phase at the pin across all concurrent MMMC views. You can use -index to make the arrival times specific to a given view Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
attacker_fall_slew	

Stylus Common UI Database Object Information

Database Objects--pin

	Reports fall slew of attacker seen for net connected to this pin Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
attacker_rise_slew	
	Reports rise slew of attacker seen for net connected to this pin Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
base_name	
	The name at the base of a hierarchical name. So the base_name of i1/i2/i3 is i3. Type: string Default: "" Edit: No
base_pin	
	pointer to equivalent base pin Type: obj(base_pin) Default: "" Edit: No
capacitance_max	
	Reports maximum capacitance of pin Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
capacitance_max_fall	
	Returns the maximum value of the falling capacitance range of the corresponding library pin. You can use -index to return the value from a specific analysis view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
capacitance_max_rise	
	Returns the maximum value of the rising capacitance range of the corresponding library pin. You can use -index to return the value from a specific analysis view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
capacitance_min	
	Reports minimum capacitance of pin Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
capacitance_min_fall	
	Returns the minimum value of the falling capacitance range of the corresponding library pin. You can use -index to return the value from a specific analysis view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No

Stylus Common UI Database Object Information

Database Objects--pin

capacitance_min_rise	
	Returns the minimum value of the rising capacitance range of the corresponding library pin.You can use -index to return the value from a specific analysis view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
cell	
	Reports cell of pin Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
cell_name	
	Returns the name of the instance of the instance pin mentioned. Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
clock_gating_exclude_signal	
	Exclude this pin from clock gating enable functions. Type: bool Default: false Edit: Yes
clock_pulse_width_high	
	return pulse_width_high for arrival clocks Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
clock_pulse_width_low	
	return pulse_width_low for arrival clocks Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
clock_sources	
	Reports the collection of clock roots of clocks associated with pin Type: obj(pin)* Allowed -index values: analysis_view clock Default: "" Edit: No
clocks	
	Returns a list of all of the clock objects arriving at the pin. You can use -index to return the value from a specific analysis view. Type: obj(clock)* Allowed -index values: analysis_view clock Default: "" Edit: No
constant	

Stylus Common UI Database Object Information

Database Objects--pin

	Returns if this pin has a constant logic value of 1'b0 or 1'b1 directly on the pin in the Verilog. If it is no_constant, and it is connected to a net, you must also check the net.constant value to see if the net driving the pin is constant. Type: enum Enum Values: 0 1 no_constant Default: no_constant Edit: No
constant_value	
	Returns a constant value of 0 or 1 if logic state has been asserted or propagated to this pin. You can use -index to return the value from a specific analysis view. Type: int Allowed -index values: analysis_view clock Default: "" Edit: No
cts_add_pin_driver	
	Specifies a cell type or pair of cell types, so that a cell inst can to be added above an input pin or below an output pin. The pin is specified by the argument pin. No more than two cell types may be specified. A warning will be emitted if no AON cells are specified. Type: string Default: "" Edit: Yes
cts_add_port_driver	
	Specifies a cell type or pair of cell types, so that a cell inst can to be added above an output port or below an input port. The port is specified by the argument pin. If the pin specified is not a design IO pin or it is not in the clock network then CCOpt will emit a warning and will not add cell insts at that position. No more than two cell types may be specified. A warning will be emitted if no AON cells are specified. Type: string Default: "" Edit: Yes
cts_annotated_delay_to	
	Override any clock tree timing engine computed cell arc or net arc delays to this pin, in a similar manner to SDC set_annotated_delay. Type: string Allowed -index values: delay_corner Default: "" Edit: Yes
cts_annotated_transition	
	Override any clock tree timing engine computed transition at this pin, in a similar manner to SDC set_annotated_transition. Type: string Allowed -index values: delay_corner Default: auto Edit: Yes
cts_assign_clock_tree	
	Can be used to make sure a clock tree sink pin gets driven by a particular clock tree. The clock tree can alternatively be specified as the tree's root pin, in case extraction rennumbers the clock trees defined from a single SDC clock. By default CCOpt automatically selects appropriate pins. Valid values: cts_clock_tree list of pins Type: string Default: "" Edit: Yes
cts_case_analysis	

Stylus Common UI Database Object Information

Database Objects--pin

	Specifies a constant value to be used as pin signal when analyzing timing within CTS. Only applies to input pins of cells in the clock tree. If set to 'none' (the default), CTS will consider the input as non-constant. Valid values: 0 1 none Type: string Default: auto Edit: Yes
cts_clock_tree	
	The clock tree to which this object belongs. Flops do not belong to a clock tree, but their clock pins do. Valid values: cts_clock_tree Type: obj(clock_tree) Default: "" Edit: No
cts_clock_trees	
	A list of clock trees the pin is contained within. This includes parents of generated clock trees and all relevant parents when clock trees overlap. Valid values: list cts_clock_tree Type: obj(clock_tree) Default: "" Edit: No
cts_effective_mixed_buffer_budget	
	Allows the user to query the mixed buffer budget at a particular pin. This is the number of buffers allowed above the given pin, and is the smallest of the remaining budgets from any fanout below this point or the override set at this point if one has been set. Type: string Default: auto Edit: No
cts_effective_routing_preference	
	Returns the pin's effective routing preference; one of leaf, trunk, top, none. The value "none" can be returned for pins that have no preference such as antenna diodes or load cap inputs. The effective routing preference value for a pin does not solely determine the net_type of the driving net if the net has multiple fanout. The fanout routing preferences for all sinks are taken into account when determining the net_type for the net as a whole. Type: string Default: "" Edit: No
cts_flexible_htree	
	The flexible HTree, if any, associated with the given object. Type: obj(flexible_htree)* Default: "" Edit: No
cts_is_sdc_clock_root	
	Specifies whether the given pin is the root (source pin) of an SDC clock. The create_clock_tree_spec populates this attribute with the location of the SDC clock root (source) pins. This attribute controls the behavior of clock tree definition commands create_clock_tree and create_generated_clock_tree, when the -stop_at_sdc_clock_roots argument is specified. In such a case, pins and ports for which this attribute is true will be treated as being SDC clock root pins. Valid values: true false Type: bool Default: false Edit: Yes
cts_isolated_effective	

Stylus Common UI Database Object Information

Database Objects--pin

	This is the isolated value CCOpt will use for the given pin. Type: bool Default: false Edit: No
cts_net_unbufferable_reasons	
	This attribute contains a list of reasons why CCOpt was not able to buffer the clock net attached to the specified pin. Valid values: string Type: string Default: "" Edit: No
cts_node_type	
	node type of pin within cts graph Type: string Default: "" Edit: No
cts_pin_capacitance_sources	
	List of the places the capacitance could be retrieved from. Requesting this attribute for a pin not in the clock tree will result in an error. The possible places are: - library - CTS will retrieve capacitance values from the library for one or more timing_corner/event combinations. - cts_capacitance_override - CTS will retrieve capacitance values from the cts_capacitance_override attribute for one or more timing_corner/event combinations. - cts_blackbox_default_load_base_pin - CTS will retrieve capacitance values from the cts_blackbox_default_load_base_pin attribute for one or more timing_corner/event combinations. Type: string Default: "" Edit: No
cts_pin_insertion_delay	
	The amount of insertion delay under this pin. Clock tree synthesis will attempt to make the insertion delay to this pin less than that to other sinks in the same skew group by this amount if a positive value is set. A negative value should be used if you would like the insertion delay to this pin to be greater than that to other sinks. The value 'auto' means there is no insertion delay offset for the pin. This attribute exposes the 'total' pin insertion delay. For further discussion on this, consult the documentation for attribute cts_pin_insertion_delay_sources. Valid values: double ignore min auto Type: string Allowed -index values: delay_corner Default: auto Edit: Yes
cts_pin_insertion_delay_sources	

Stylus Common UI Database Object Information

Database Objects--pin

	The amount of insertion delay under this pin, broken down by the source of the pin insertion delay (PID). The value of this attribute is a Tcl dictionary of PID values, keyed on the source of that PID. This provides an 'exploded' view into the total pin insertion delay that is present, categorizing the various contributions by their origin. Valid Tcl dict keys: - user Asserted by user Tcl scripting or restored from an older DB. - sdv Extracted from SDC set_clock_latency by spec creation. - sink_type Asserted to reflect the impact of the cts_sink_type attribute. - ilm Records the mean clock latency of the ILM clock network that is rooted at this pin. - useful_skew A useful skew generated by pre-CTS optimization. - incremental An incremental offset, used to specify useful skews to incremental CTS. - total The formal sum of the above contributions. This value is the 'overall' pin insertion delay accessible via the cts_pin_insertion_delay attribute. Valid Tcl dict values: double ignore min auto Reading the attribute returns a dict value that exposes the current state of the PID, recording all the PID sources and the associated PID value of each one. Sources whose value are auto are omitted. If there are no non-auto sources, the returned dict is empty. For convenience, the total is always included in the dict if it is non-auto. Writing a dict value to the attribute will update each of the specified sources with the specified PID value. The total PID value will then be updated. Sources in the dict that are assigned a value of 'auto' will be reset. Sources omitted from the dict are left unchanged from their current value: assigning an empty dict is a no-op. Resetting the attribute will clear all the sources and the total back, resetting them to the default value of 'auto', thereby completely removing the PID from the pin. Valid values: A Tcl dictionary of PID values, keyed on PID source. Type: string Allowed -index values: delay_corner Default: "" Edit: Yes
cts_pin_route_type_propagation	
	Describes how cts_pin_route_type should be applied to the net driven by the specified pin. The possible values are: - none - it applies to net driven by the pin and only that net, irrespective of whether the net is buffered or not. - net - it applies to all nets generated by buffering the pre-CTS net driven by pin (default). Type: string Default: net Edit: Yes
cts_routing_trunk_override	
	Prefer trunk routing rules for this pin. Only applies to clock tree sinks. Valid values: true false Type: bool Default: false Edit: Yes
cts_sink_type	

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	The type of sink this pin represents. Valid values are as follows: auto The sink type will be automatically determined by CCOpt. through Through pin. Trace the clock tree through this pin. stop Stop pin. When defining clock trees, CCOpt stops searching for parts of the clock tree at stop pins. ignore Ignore pin. CCOpt stops searching for parts of the clock tree at ignore pins and it does not attempt to balance the insertion delay of ignore pins. exclude Exclude pin. Exclude this pin from the clock tree. Valid values: auto through stop ignore exclude Type: string Default: auto Edit: Yes
cts_sink_type_effective	
	Indicates how CCOpt will treat a given pin, taking into account both its cts_sink_type_implicit, and any cts_sink_type settings. Setting a non-default value for the cts_sink_type attribute will override the cts_sink_type_implicit attribute. Type: string Default: auto Edit: No
cts_sink_type_implicit	
	Indicates the type of sink that CCOpt classified this pin as. Note that the cts_sink_type attribute can override these internal classifications. Possible values are: exclude Indicates that this pin is a sink which represents a non-clock pin. ignore Indicates that CCOpt has determined not to search for more clock tree through this pin. Additionally, this pin will not be balanced. stop Indicates that CCOpt has determined not to search for more clock tree through this pin. An empty value for this attribute indicates either that this pin is either not a sink, or that it is a sink that is not implicitly exclude or ignore or stop. Type: string Default: exclude Edit: No
cts_sink_type_reasons	
	The reasons why this pin has the given cts_sink_type. This attribute is configured by the create_clock_tree_spec command in order to record the reasons that the cts_sink_type attribute has been configured for the given pin. This attribute is a list of values. Valid values are as follows: auto The cts_sink_type attribute is set to 'auto' implicit The pin is an implicit sink (flop/latch) user The user has set the cts_sink_type attribute design_io This pin is a design I/O multiple_outputs This pin is on an instance with multiple outputs, and the cts_spec_config_trace_through_to attribute has not been set set_disable_timing SDC set_disable_timing stops the clock at this pin set_case_analysis SDC set_case_analysis stops the clock at this pin generated_clock_tree This pin is the generator input to an SDC generated clock no_sdc_clock The SDC clock is stopped at this pin for other reasons ilm The create_clock_tree_spec command has detected an ILM below this pin Valid values: auto implicit user design_io multiple_outputs set_disable_timing set_case_analysis generated_clock_tree no_sdc_clock ilm Type: string Default: auto Edit: Yes
cts_skew_groups_active	

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	Returns the list of active skew groups for this pin. For sink pins, this attribute lists both skew groups that pass through this pin and skew groups for which this sink is an endpoint. For non-sink pins, shows skew groups that pass through this pin. Valid values: list skew_groups Type: obj(skew_group)* Default: "" Edit: No
cts_skew_groups_active_sink	
	Returns the list of skew groups for which this pin is an active sink. For sink pins, this attribute lists the skew groups for which this sink is an endpoint. Skew groups that pass through this pin are not included. For non-sink pins, this attribute always returns null. Valid values: list skew_groups Type: obj(skew_group)* Default: "" Edit: No
cts_skew_groups_constraining	
	The list of delay-constraining skew groups which are active at this pin. This attribute reports similar data to cts_skew_groups_active. The only difference is that the reporting-only skew groups are not included in this attribute's value. For sink pins, this attribute lists both delay-constraining skew groups that pass through this pin and delay-constraining skew groups for which this sink is an endpoint. For non-sink pins, shows delay-constraining skew groups that pass through this pin. Valid values: list skew_groups Type: obj(skew_group)* Default: "" Edit: No
cts_skew_groups_constraining_sink	
	The list of delay-constraining skew groups for which this pin is an active sink. This attribute lists the delay-constraining skew groups for which this pin is an active sink. This attribute reports similar data to cts_skew_groups_active_sink. The only difference is that the reporting-only (constrains none) skew groups are not included in this attribute's value. For sink pins, this attribute lists the delay-constraining skew groups for which this sink is an endpoint. Skew groups that pass through this pin are not included. For non-sink pins, this attribute always returns null. Valid values: list skew_groups Type: obj(skew_group)* Default: "" Edit: No
cts_skew_groups_ignore	
	The list of skew groups for which paths through this pin are ignored. Valid values: list skew_groups Type: obj(skew_group)* Default: "" Edit: No
cts_skew_groups_sink	
	The list of skew groups for which this pin is a sink. Valid values: list skew_groups Type: obj(skew_group)* Default: "" Edit: No

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cts_skew_groups_source_pin	
	The list of skew groups for which this clock tree or pin is specified as a source. Valid values: list skew_groups Type: obj(skew_group)* Default: "" Edit: No
cts_spec_config_trace_through_to	
	Clock tree definition will, by default, not continue through certain types of cell arc (for instance, the clock to Q arc in a DFF). This attribute allows you to override this default behavior, permitting the clock tree to trace through such a cell. The attribute should be configured on the input pin at which the clock arrives. The value of the attribute specifies the output pin to which the clock should propagate. The specified output pin must be another pin on the same instance. The output pin may be specified either by its fully qualified name (i.e. inclusive of the instance name), or else simply by its local (cell-relative) name. There must be a pre-existing (library-defined) chain of one or more delay arcs that connect the input and output pins together. It is not possible to use trace_through_to to synthesize delay arcs. If multiple input pins are annotated on a given instance, the value of trace_through_to at each of those pins must select the same output pin: i.e. the configuration must identify a single clock output for the instance. If multiple clock outputs are required then trace_through_to should not be used: instead define a generated clock tree at each of the clock-carrying outputs. If the configuration of trace_through_to settings for a given instance does not meet these requirements, a warning will be issued and the settings for that instance will be ignored. Note: if both trace_through_to and library_trace_through_to are applicable at a given netlist instance pin, the trace_through_to value takes precedence. Valid values: pin Type: string Default: "" Edit: Yes
cts_top_fanout_count_override	
	The number of clock sinks this sink counts for when applying the top routing rules. Note that this attribute is only valid for sink pins, and it returns auto for non-sink pins. For a sink pin, a non-auto value means that this sink is counted as though it were multiple sinks, for the purposes of determining which nets should have top routing. An auto value for a sink pin means that the sink counts as a single sink. Valid values: integer > 0 Type: string Default: auto Edit: Yes
cts_top_fanout_transitive_count	
	The number of clock sinks in the transitive fanout of the pin as counted for applying the top routing rules. This attribute is very similar to the cts_transitive_fanout attribute but counts sink fanout using the cts_top_fanout_count_override attribute instead of always counting sinks as a single item of fanout. Requesting this attribute for a pin not in the clock tree will result in an error. Valid values: integer Type: int Default: 0 Edit: No
cts_transitive_fanout	

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	The number of clock sinks in the transitive fanout of the pin, within the clock tree. Requesting this attribute for a pin not in the clock tree will result in an error. Valid values: int Type: int Default: 0 Edit: No
cts_virtual_delay_early_fall	
	The amount of virtual delay that has been applied under this pin. Valid values: double Type: double Allowed -index values: delay_corner Default: 0 Edit: Yes
cts_virtual_delay_early_rise	
	The amount of virtual delay that has been applied under this pin. Valid values: double Type: double Allowed -index values: delay_corner Default: 0 Edit: Yes
cts_virtual_delay_late_fall	
	The amount of virtual delay that has been applied under this pin. Valid values: double Type: double Allowed -index values: delay_corner Default: 0 Edit: Yes
cts_virtual_delay_late_rise	
	The amount of virtual delay that has been applied under this pin. Valid values: double Type: double Allowed -index values: delay_corner Default: 0 Edit: Yes
delay_max_fall	
	Returns the maximum falling delay. You can use -index to return the value from a specific analysis view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
delay_max_rise	
	Returns the maximum rising delay. You can use -index to return the value from a specific analysis view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
delay_min_fall	
	Returns the minimum falling delay. You can use -index to return the value from a specific analysis view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
delay_min_rise	

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Database Objects--pin

	Returns the minimum rising delay. You can use -index to return the value from a specific analysis view. Type: <code>double</code> Allowed -index values: analysis_view clock Default: "" Edit: No
dft_constant_value	
	Constant pin value in test mode {logic_0 logic_1 logic_z no_value}. Type: <code>string</code> Default: "" Edit: No
dft_controllable	
	In test mode the output pin can take either the non_inverting or inverting value of the input pin. Type: <code>string</code> Default: "" Edit: Yes
dft_driven_by_clock	
	If pin is driven by clock. Type: <code>enum</code> Enum Values: false true Default: false Edit: Yes
direction	
	Pin's direction from the corresponding base_pin. It can be in, out, inout or internal. Internal means it is an internal pin from a .lib file for the timing model, and is not part of the netlist. Type: <code>enum</code> Enum Values: internal in out inout Default: "" Edit: No
dont_invert_phase	
	Type: <code>bool</code> Default: false Edit: Yes
dont_touch	
	The preservation status of a pin during optimization. A preserved pin means the logical function of the pin must be preserved to maintain a simulation or test-point pin in the netlist. However, the name does not need to be preserved. Supported values: - false: Unconstrained - true: Cannot add/remove ports - delete_ok: Can delete ports (if they have no fanout) - invert_ok: We cannot delete, duplicate, or add but can change the polarity - none: No user setting; will inherit from the module/hinst Type: <code>enum</code> Enum Values: none false true delete_ok invert_ok const_prop_delete_ok Default: none Edit: Yes
dont_touch_file	
	Specifies the sourced file to set dont_touch. Type: <code>string</code> Default: "" Edit: Yes
driver_cell	

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	Reports driver cell of pin Type: obj(inst)* Allowed -index values: analysis_view clock Default: "" Edit: No
driver_cell_name	
	Reports driver cell name of pin Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
driver_ground_pin_voltage_max	
	Reports maximum ground pin voltage of driver cell Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
driver_ground_pin_voltage_min	
	Reports minimum ground pin voltage of driver cell Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
driver_power_pin_voltage_max	
	Reports maximum power pin voltage of driver cell Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
driver_power_pin_voltage_min	
	Reports minimum power pin voltage of driver cell Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
driver_ref_lib_cell_name	
	Reports library cell name of driver cell of the pin Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
edge	
	If this pin is for a partition, or inst that is a black-box, and the pin is assigned, the edge value indicates along which edge of the boundary polygon the pin is assigned. The edge number starts from the lowest Y, then left-most X vertex, starting with 0, and then counting clock-wise. See the set_pin_constraint command document for a figure showing the edge numbering. If the pin is not assigned, or not for a partition or black-box inst, the value of -1 is returned. Type: int Default: "" Edit: No
effective_stack_via_rule	
	The stack_via_rule that is expected to be used by the router for connecting to this pin. This value is derived from the other pin and base_pin attributes related to stack via. Type: obj(stack_via_rule) Default: "" Edit: No

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Database Objects--pin

escaped_name	
	The full hierarchical name including escaped chars (if any). It follows DEF escaping syntax, so bus-bit chars [], or a hierarchy char \ that is not a bus-bit or hierarchy char has a \ in front of it. So i1/i2 is a two-level hierarchical name while i1\i2 is single level name, and a[0] is a bus-bit, while a[0\] is a scalar. Type: string Default: "" Edit: No
exceptions	
	Reports the path exceptions Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
fanin	
	fanin Type: int Allowed -index values: analysis_view clock Default: "" Edit: No
fanout	
	fanout Type: int Allowed -index values: analysis_view clock Default: "" Edit: No
fanout_load	
	Returns the fanout load for the pin. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
fastest_clock_name	
	Reports the clock name with maximum frequency associated with the pin Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
fastest_clock_period	
	Reports clock period of maximum frequency clock associated with the pin Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
from_arcs	
	Returns a list of arc objects for which this pin is the starting pin of the timing arc. Type: obj(arc)* Default: "" Edit: No
hdl_name	

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	This is the original RTL name for this pin. It is used to map RTL simulation results with RTL names to the current netlist for switching activity analysis. It is only maintained properly on the output pins of sequential cells. Optimization will copy this name during any multi-bit merge or splitting transforms to the equivalent pin, but not on output pins of combinational cells. The is_phase_inverted attribute will be flipped if the phase is inverted. Type: string Default: "" Edit: Yes
hierarchical_level	
	Returns hierarchical level for an Instance Pin. Type: int Allowed -index values: analysis_view clock Default: "" Edit: No
hnet	
	pointer to the hnet connected to the pin Type: obj(hnet) Default: "" Edit: No
hold_uncertainty	
	Returns the most conservative uncertainty of all possible uncertainty assertions associated with the pin. You can use -index to return the value from a specific analysis view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
holding_resistance_high	
	Reports high holding resistance of pin Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
holding_resistance_low	
	Reports low holding resistance of pin Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
imm_parent_cell_id	
	Reports numeric identifier assigned to parent cell during inter module matrix analysis Type: int Allowed -index values: analysis_view clock Default: "" Edit: No
initial_name	
	This will be used for verification of initial input netlist (post-synthesis) to any other netlist generated during Innovus flow. Type: string Default: "" Edit: Yes
input_signal_level	
	Returns the rail name associated with input pin. Type: string Allowed -index values: analysis_view clock Default: "" Edit: No

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input_signal_level_voltage	
	Returns rail voltage of signal which is driving the input pin. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
inst	
	pointer to the inst containing the pin Type: obj(inst) Default: "" Edit: No
interpolation_info_max	
	Reports interpolation information related to max corner Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
interpolation_info_min	
	Reports interpolation information related to min corner Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
is_always_on	
	Returns a value of true if the pin is always on type pin. Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
is_async	
	Returns a value of true if the pin is an asynchronous clear or preset pin. Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
is_booth_encoded	
	Booth encoded setting assigned to input port by user. Type: bool Default: false Edit: Yes
is_clear	
	Returns a value of true if the pin is an asynchronous clear pin. Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
is_clock	
	Returns a value of true if the pin has the Liberty pin attribute: clock . Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
is_clock_gating	

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	Returns a value of true if the pin is defined as a pin of a clock gating cell. You can use -index to return the value from a specific analysis view. Type: <code>bool</code> Allowed -index values: analysis_view clock Default: "" Edit: No
is_clock_gating_clock	
	Returns a value of true if the pin is defined as a clock pin of a clock gating cell. Type: <code>bool</code> Allowed -index values: analysis_view clock Default: "" Edit: No
is_clock_gating_enable	
	Returns a value of true if the pin corresponds to the enable pin of a clock gating cell. Type: <code>bool</code> Allowed -index values: analysis_view clock Default: "" Edit: No
is_clock_network_pin	
	Reports true if specified pin is part of clock network Type: <code>string</code> Allowed -index values: analysis_view clock Default: "" Edit: No
is_clock_source_latency	
	Returns true if object is part of latency network Type: <code>bool</code> Allowed -index values: analysis_view clock Default: "" Edit: No
is_clock_source_pin	
	Returns true if object is a clock source pin Type: <code>bool</code> Allowed -index values: analysis_view clock Default: "" Edit: No
is_clock_used_as_clock	
	Returns a value of true if the pin lies in the clock network and at least one of the clocks arriving at the pin is used as a clock in the downstream network of the pin. You can use -index to return the value for a specific analysis view. Type: <code>bool</code> Allowed -index values: analysis_view clock Default: "" Edit: No
is_clock_used_as_data	
	Returns a value of true if the pin lies in the clock source path and at least one of the clocks arriving on the pin is used as data in the downstream network of the pin. You can use -index to return the value for a specific analysis view. Type: <code>bool</code> Allowed -index values: analysis_view clock Default: "" Edit: No
is_constant_in_all_views	
	Returns true if logic state has been asserted or propagated to this pin in all the views Type: <code>bool</code> Allowed -index values: analysis_view clock Default: "" Edit: No

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Database Objects--pin

is_constant_in_any_view	
	Returns true if logic state has been asserted or propagated to this pin in atleast one view. Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
is_constrained	
	Returns true if specified has a constraint applied Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
is_constrained_endpoint	
	Reports true if pin is a constrained endpoint Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
is_data	
	Returns a value of true if the pin is a data pin (that is, is not a clock pin). Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
is_disable_timing	
	Returns a value of true if the pin's timing has been disabled. You can use -index to return the value for a specific analysis view. Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
is_dynamic_clock_gating_data	
	Returns true if pin is data pin of a dynamic clock gating cell Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
is_endpoint	
	Returns true if specified pin is an endpoint Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
is_existing_cte_pin	
	Returns true if pin exists as part of timing graph during analysis Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
is_fall_edge_triggered_clock	
	Returns a value of true if the pin is a clock pin of a flop, and is triggered by the falling edge of a clock. Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No

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is_fall_edge_triggered_data	
	Returns a value of true if the pin corresponds to the data pin of a fall edge triggered device. Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
is_hier_crossing_needed	
	Returns true if needs hierarchical crossing arcs to be built Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
is_hierarchical	
	is_hierarchical Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
is_inside_partition	
	Returns whether pin is in partition or not. Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
is_inter_die_feedthru	
	Returns true if pin is a freethru inter-die pin Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
is_internal_disable	
	Returns the internal disabled assertion on endpoints. Such endpoints will have no timing computed on them. Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
is_isolation_cell_clock	
	Returns true if a clock pin is part of an isolatoin cell, Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
is_isolation_cell_data	
	Returns a value of true if the pin is isolation data pin type. Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
is_isolation_cell_enable	
	Returns a value of true if the pin is isolation enable pin type. Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No

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is_latency_network_pin	
	Returns a value of true if the pin has latency phase propagated to it. Type: <code>bool</code> Allowed -index values: analysis_view clock Default: "" Edit: No
is_level_shifter_data	
	Returns a value of true if the pin is level shifter data pin type. Type: <code>bool</code> Allowed -index values: analysis_view clock Default: "" Edit: No
is_level_shifter_enable	
	Returns a value of true if the pin is level shifter enable pin type. Type: <code>bool</code> Allowed -index values: analysis_view clock Default: "" Edit: No
is_marked	
	Returns true if pin is marked Type: <code>bool</code> Allowed -index values: analysis_view clock Default: "" Edit: No
is_multiple_clock_fanin_point	
	Returns a value of true if multiple clock phases converge at the pin. You can use -index to return the value for a specific analysis view. Type: <code>bool</code> Allowed -index values: analysis_view clock Default: "" Edit: No
is_mux_select_pin	
	Returns the is_mux_select_pin property for a pin. You can use -index to return the value for a specific view. Type: <code>bool</code> Allowed -index values: analysis_view clock Default: "" Edit: No
is_negative_level_sensitive_clock	
	Returns a value of true if the library pin is an enable pin of an active low level-sensitive device. Type: <code>bool</code> Allowed -index values: analysis_view clock Default: "" Edit: No
is_negative_level_sensitive_data	
	Returns a value of true if the pin is a data pin of an active low level-sensitive device. Type: <code>bool</code> Allowed -index values: analysis_view clock Default: "" Edit: No
is_owner	
	Returns true if client is the owner for pin in distributed STA. Type: <code>bool</code> Allowed -index values: analysis_view clock Default: "" Edit: No

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Database Objects--pin

is_pba_begin_point_candidate	
	Returns a value of true if the pin is PBA begin point candidate. Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
is_phase_inverted	
	Identifies the pin for which phase is inverted during optimizations. Type: bool Default: false Edit: Yes
is_pin	
	Returns true if object is a pin Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
is_port	
	Returns true if pin is a port Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
is_positive_level_sensitive_clock	
	Returns a value of true if the library pin is an enable pin of an active high level-sensitive device. Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
is_positive_level_sensitive_data	
	Returns a value of true if the pin is a data pin of an active high level-sensitive device. Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
is_power_mode_disabled	
	Returns true if power mode is disabled Type: bool Allowed -index values: analysis_view Default: "" Edit: No
is_preset	
	Returns a value of true if the pin is a preset pin. Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
is_propagated_clock	
	Returns a value of true if there is an explicit set_propagated_clock assertion at the pin. You can use -index to return the value for a specific analysis view. Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No

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is_ref_clock_pin	
	Returns true if object is a reference clock pin Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
is_rise_edge_triggered_clock	
	Returns a value of true if the pin is a clock pin of a flop, and is triggered by the rising edge of a clock. Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
is_rise_edge_triggered_data	
	Returns a value of true if the pin corresponds to the data pin of a rise edge-triggered device. Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
is_sequential_duplicate	
	Identifies the Pin which is cloned or duplicated. Type: string Default: "" Edit: Yes
is_sequential_merged	
	Keep track of Pins after merge or split during optimization. Type: string Default: "" Edit: Yes
is_sequential_retimed	
	is retimed of pin Type: bool Default: false Edit: Yes
is_shared	
	Returns true if pin is shared across clients in distributed STA Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
is_special	
	pin is belong DEF SPECIALNETS section Type: bool Default: false Edit: Yes
is_timing_ignored	
	Returns true if pin is ignored fur timing analysis Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
is_tristate	

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	Returns a value of true if the pin has the three_state attribute in the Liberty timing library. Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
is_tristate_enable	
	Returns a value of true if the pin is the source pin of timing arcs with either the three_state_enable or three_state_disable attribute in the Liberty timing library. Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
is_tristate_output	
	Returns a value of true if the pin corresponds to a three-state output pin. Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
is_valid_for_reports	
	Returns the pin is valid for reporting. Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
is_valid_pin_for_imm_report	
	Returns true if pin is classified for inter-module-matrix analysis Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
latch_time_given	
	In latch-based analysis, this returns the value by which the arrival time on this pin is adjusted to account for time borrowed at the previous stage. You can use -index to return the value for a specific analysis view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
layer	
	Layer of the pin. For pins with more than one shape, it is the layer of the first shape found in the DB. This is also the same shape used for the .location value. If the pin has no shapes or vias defined, then {} is returned. Type: obj(layer) Default: "" Edit: No
lib_binding_info_max	
	Reports the type of bindinf, i.e. Match/Interpolation/Extrapolation/UserSpecified Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
lib_binding_info_min	
	Reports the type of bindinf, i.e. Match/Interpolation/Extrapolation/UserSpecified Type: string Allowed -index values: analysis_view clock Default: "" Edit: No

Stylus Common UI Database Object Information

Database Objects--pin

lib_pins	
	Returns a list of library pin (lib_pin) objects which are associated with this pin. Type: obj(lib_pin)* Default: "" Edit: No
local_slack_max	
	Reports maximum local slack of pin Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
local_slack_max_fall	
	Reports maximum fall local slack of pin Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
local_slack_max_rise	
	Reports maximum rise local slack of pin Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
local_slack_min	
	Reports minimum local slack of pin Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
local_slack_min_fall	
	Reports minimum fall local slack of pin Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
local_slack_min_rise	
	Reports minimum rise local slack of pin Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
location	
	Location of the pin. The returned point will overlap some part of the pin shape. For pins with more than one shape, the point overlaps the first shape found in the DB. This is also the same shape used for the .layer value. If the pin has no shapes or vias defined, then the inst.location is returned. It is first initialized when the LEF is read in, but the location might be changed by the router to align with the routing tracks and default routing via when it first runs. Type: point Default: "" Edit: No
lp_asserted_probability	
	Probability of signal to be high on port/subport/pin as specified by user. Type: double Default: no_value Edit: Yes

Stylus Common UI Database Object Information

Database Objects--pin

lp_asserted_toggle_rate	
	Signal toggle rate assigned to pin by user. Type: double Default: no_value Edit: Yes
lp_system_asserted_probability	
	System/Internal asserted probability. Type: double Default: no_value Edit: Yes
lp_system_asserted_toggle_rate	
	System/Internal asserted toggle rate. Type: double Default: no_value Edit: Yes
lssd_master_clock	
	Defines the master clock port related to this slave clock port. Type: string Default: "" Edit: Yes
max_capacitance	
	Returns the maximum capacitance limit for the pin. You can use -index to return the value for a specific analysis view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
max_cpvr_early	
	Reports maximum early CPPR Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
max_cpvr_late	
	Reports maximum late CPPR Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
max_fanout	
	Returns the maximum fanout load that the pin can drive. This value is set using set_max_fanout or the default_max_fanout library attribute. You can use -index to return the value for a specific analysis view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
max_time_borrow	
	In latch-based analysis, returns the maximum time that a path arriving at this latch input can borrow from the next stage - as specified by set_max_time_borrow. You can use -index to return the value for a specific analysis view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
max_time_borrow_io	

Stylus Common UI Database Object Information

Database Objects--pin

	In latch-based analysis, returns the maximum time that a path arriving at this latch input can borrow from the next stage - as specified by set_max_time_borrow -interface. You can use -index to return the value for a specific analysis view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
max_transition	
	Returns the maximum transition time limit specified for the pin. You can use -index to return the value for a specific analysis view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
min_capacitance	
	Returns the minimum capacitance limit for the pin. You can use -index to return the value for a specific analysis view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
min_fanout	
	Returns the minimum fanout design rule limit of the corresponding library pin. You can use -index to return the value for a specific analysis view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
min_transition	
	Returns the minimum transition time limit specified for the pin. You can use -index to return the value for a specific analysis view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
multibit_aware_mapped	
	Set when the sequential instance is mapped for multibit. Type: bool Default: false Edit: Yes
name	
	The same as .escaped_name without any \ escape chars. This name is commonly used to avoid problems with \ escape chars in Tcl scripts unless you carefully use list operators. Like SDC commands, 'get_db insts i1/i2' will first try to match i1/i2, and if not found then match i1\i2 so that flattening hierarchical names (e.g. with ungroup) does not require changing the names in Tcl scripts. Type: string Default: "" Edit: No
net	
	pointer to the net connected to the pin Type: obj(net) Default: "" Edit: No
net_name	
	net_name Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
network_latency_fall_max	

Stylus Common UI Database Object Information

Database Objects--pin

	Returns the maximum fall insertion delay specified by an explicit set_clock_latency at the pin. You can use -index to return the value for a specific analysis view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
network_latency_fall_min	
	Returns the minimum fall insertion delay specified by an explicit set_clock_latency at the pin. You can use -index to return the value for a specific analysis view and/or clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
network_latency_rise_max	
	Returns the maximum rise insertion delay specified by an explicit set_clock_latency at the pin. You can use -index to return the value for a specific analysis view and/or clock Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
network_latency_rise_min	
	Returns the minimum rise insertion delay specified by an explicit set_clock_latency at the pin. You can use -index to return the value for a specific analysis view and/or clock Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
noise_victim	
	Reports associated victim noise object Type: obj(pin)* Allowed -index values: analysis_view clock Default: "" Edit: No
num_exceptions	
	Reports the number of path exceptions Type: int Allowed -index values: analysis_view clock Default: "" Edit: No
num_phases	
	Reports total number of phases on the pin Type: int Allowed -index values: analysis_view clock Default: "" Edit: No
num_violating_phase_count_max	
	Reports count of maximum violating phases on pin Type: int Allowed -index values: analysis_view clock Default: "" Edit: No
num_violating_phase_count_min	

Stylus Common UI Database Object Information

Database Objects--pin

	Reports count of minimum violating phases on pin Type: int Allowed -index values: analysis_view clock Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (pin) Default: "" Edit: No
opt_skew_max_allowed_delay	
	Specifies in nanoseconds the maximum permitted amount of insertion delay change that the software can make at this pin, when performing useful skew operations. When not specified at a given pin (value "no_value"), the tool will apply the globally configured skewing limits, for example as specified by attribute opt_skew_max_allowed_delay. This attribute only takes effect when configured on the load of a clock net. Type: double Default: no_value Edit: Yes
output_signal_level	
	Returns the rail name associated with output pin. Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
parent_cell	
	Reports the parent cell of the cell to which specified pin belongs Type: obj(hinst)* Allowed -index values: analysis_view clock Default: "" Edit: No
pin_capacitance_max	
	Reports maximum capacitance of pin Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
pin_capacitance_max_fall	
	Reports maximum fall capacitance of pin Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
pin_capacitance_max_rise	
	Reports maximum rise capacitance of pin Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
pin_capacitance_min	
	Reports minimum capacitance of pin Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
pin_capacitance_min_fall	

Stylus Common UI Database Object Information

Database Objects--pin

	Reports minimum fall capacitance of pin Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
pin_capacitance_min_rise	
	Reports minimum rise capacitance of pin Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
pin_direction	
	Reports direction of pin Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
power_domain	
	The parent power domain of the pin (equivalent to Design Browser effPD) Type: obj(power_domain) Default: "" Edit: No
power_rail_voltage_inout_input_max	
	Reports power rail voltage for input part of bidi pins/ports for maximum operating condition. You can use -index to return the value for a specific analysis view Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
power_rail_voltage_inout_input_min	
	Reports power rail voltage for input part of bidi pins/ports for minimum operating condition. You can use -index to return the value for a specific analysis view Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
power_rail_voltage_max	
	Reports power rail voltage for maximum operating condition for bidirectional, input and output pins. For bidirectional pins, the power rail voltage for output signal is reported. You can use -index to return the value for a specific analysis view Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
power_rail_voltage_min	
	Reports power rail voltage for minimum operating condition for bidirectional, input and output pins. For bidirectional pins, the power rail voltage for output signal is reported. You can use -index to return the value for a specific analysis view Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
receiver_ccc_type	

Stylus Common UI Database Object Information
Database Objects--pin

	Reports receiver ccc type of pin Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
related_ground_pin	
	Reports associated ground pin of specified pin Type: obj(pg_pin)* obj(pg_base_pin)* Allowed -index values: analysis_view clock Default: "" Edit: No
related_ground_pin_voltage_max	
	Reports maximum voltage of related ground pin of specified pin Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
related_ground_pin_voltage_min	
	Reports minimum voltage of related ground pin of specified pin Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
related_power_pin	
	Reports related power pin of specified pin Type: obj(pg_pin)* obj(pg_base_pin)* Allowed -index values: analysis_view clock Default: "" Edit: No
related_power_pin_voltage_max	
	Reports maximum voltage of related power pin of specified pin Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
related_power_pin_voltage_min	
	Reports minimum voltage of related power pin of specified pin Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
required_level	
	Reports required level of pin in timing graph Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
safety_dmr_clone	
	The dmr (dual modular redundancy) clone-flop of this parent-flop. This attribute is only stored on dmr parent-flops. Type: obj(pin) Default: "" Edit: No
safety_dmr_clone_inv	

Stylus Common UI Database Object Information

Database Objects--pin

	The dmr (dual modular redundancy) inverter driving the clone-flop of this parent-flop. This attribute is only stored on dmr parent-flops. Type: <code>obj(inst)</code> Default: "" Edit: No
safety_dmr_error_signal_endpoint	
	This object is the endpoint for a safety-mechanism's error-signal. Type: <code>string</code> Default: "" Edit: No
safety_dmr_errors	
	The dmr (dual modular redundancy) error-detection-logic of this parent-flop. This attribute is only stored on dmr parent-flops. Type: <code>obj(inst)*</code> Default: "" Edit: No
safety_dmr_parent	
	The dmr (dual modular redundancy) parent-flop of this object. Type: <code>obj(pin)</code> Default: "" Edit: No
safety_failure_mode	
	The name of the failure-mode to which this pin belongs, as defined in the USF (Unified Safety Format) file. Type: <code>string</code> Default: "" Edit: No
safety_mechanism	
	The name of the safety-mechanism applied to this pin, as defined in the USF (Unified Safety Format) file. Type: <code>string</code> Default: "" Edit: No
safety_mechanism_type	
	The type of safety-mechanism applied to this pin, as defined in the USF (Unified Safety Format) file. Type: <code>string</code> Default: "" Edit: No
safety_tmr_clones	
	The tmr (triple modular redundancy) clone-flop-bits of this parent-flop-bit. Type: <code>obj(pin)*</code> Default: "" Edit: No
safety_tmr_errors	
	The tmr (triple modular redundancy) error-detection-logic of this parent-flop-bit. Type: <code>obj(inst)*</code> Default: "" Edit: No
safety_tmr_parent	
	safety The tmr (triple modular redundancy) parent-flop-bit of this object. Type: <code>obj(pin)</code> Default: "" Edit: No
safety_tmr_voters	

Stylus Common UI Database Object Information

Database Objects--pin

	The tmr (triple modular redundancy) voting-logic of this parent-flop-bit. Type: <code>obj(inst)*</code> Default: "" Edit: No
safety_tmr_well_tap_left_padding	
	The padding used to reserve space for well-tap cells to be placed next to the left edges of tmr (triple modular redundancy) flops. Type: <code>int</code> Default: 0 Edit: No
safety_tmr_well_tap_right_padding	
	The padding used to reserve space for well-tap cells to be placed next to the right edges of tmr (triple modular redundancy) flops. Type: <code>int</code> Default: 0 Edit: No
setup_uncertainty	
	Returns the most conservative uncertainty of all possible uncertainty assertions associated with the pin. You can use -index to return the value for a specific analysis view Type: <code>double</code> Allowed -index values: analysis_view clock Default: "" Edit: No
si_slew_max	
	Reports the maximum slew considering incremental slew from signal integrity analysis Type: <code>double</code> Allowed -index values: analysis_view clock Default: "" Edit: No
si_slew_max_fall	
	Reports the maximum fall slew considering incremental slew from signal integrity analysis Type: <code>double</code> Allowed -index values: analysis_view clock Default: "" Edit: No
si_slew_max_rise	
	Reports the maximum rise slew considering incremental slew from signal integrity analysis Type: <code>double</code> Allowed -index values: analysis_view clock Default: "" Edit: No
si_slew_mean_max_fall	
	Reports the maximum fall mean slew considering incremental slew from signal integrity analysis Type: <code>double</code> Allowed -index values: analysis_view clock Default: "" Edit: No
si_slew_mean_max_rise	
	Reports the maximum rise mean slew considering incremental slew from signal integrity analysis Type: <code>double</code> Allowed -index values: analysis_view clock Default: "" Edit: No
si_slew_mean_min_fall	

Stylus Common UI Database Object Information

Database Objects--pin

	Reports the minimum fall mean slew considering incremental slew from signal integrity analysis Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
si_slew_mean_min_rise	
	Reports the minimum rise mean slew considering incremental slew from signal integrity analysis Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
si_slew_min	
	Reports the minimum slew considering incremental slew from signal integrity analysis Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
si_slew_min_fall	
	Reports the minimum fall slew considering incremental slew from signal integrity analysis Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
si_slew_min_rise	
	Reports the minimum rise slew considering incremental slew from signal integrity analysis Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
si_victim	
	Reports the victim object from signal integrity analysis Type: obj(pin)* Allowed -index values: analysis_view clock Default: "" Edit: No
slack_max	
	Returns the worst slack across all concurrent MMMC views for Setup-style late data path checks. You can use -index to return the value for a specific view and/or clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slack_max_edge	
	Returns the data edge of the path responsible for the slack_max value. You can use -index to return the value for a specific view and/or clock. Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
slack_max_fall	
	Returns the worst falling slack across all concurrent MMMC views for Setup-style late data path checks. You can use -index to return the value for a specific view and/or clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No

Stylus Common UI Database Object Information

Database Objects--pin

slack_max_rise	
	Returns the worst rising slack across all concurrent MMMC views for Setup-style late data path checks. You can use -index to return the value for a specific view and/or clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slack_mean_max	
	In SOCV analysis, this returns the mean component of the worst slack across all concurrent MMMC views for Setup-style late data path checks. You can use -index to return the value for a specific view and/or clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slack_mean_max_fall	
	In SOCV analysis, this returns the mean component of the worst falling slack across all concurrent MMMC views for Setup-style late data path checks. You can use -index to return the value for a specific view and/or clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slack_mean_max_rise	
	In SOCV analysis, this returns the mean component of the worst rising slack across all concurrent MMMC views for Setup-style late data path checks. You can use -index to return the value for a specific view and/or clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slack_mean_min	
	In SOCV analysis, this returns the mean component of the worst slack across all concurrent MMMC views for Hold-style early data path checks. You can use -index to return the value for a specific view and/or clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slack_mean_min_fall	
	In SOCV analysis, this returns the mean component of the worst falling slack across all concurrent MMMC views for Hold-style early data path checks. You can use -index to return the value for a specific view and/or clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slack_mean_min_rise	
	In SOCV analysis, this returns the mean component of the worst rising slack across all concurrent MMMC views for Hold-style early data path checks. You can use -index to return the value for a specific view and/or clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slack_min	
	Returns the worst slack across all concurrent MMMC views for Hold-style early data path checks. You can use -index to return the value for a specific view and/or clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No

Stylus Common UI Database Object Information

Database Objects--pin

slack_min_edge	
	Returns the data edge of the path responsible for the slack_min value. You can use -index to return the value for a specific view and/or clock. Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
slack_min_fall	
	Returns the worst falling slack across all concurrent MMMC views for Hold-style early data path checks. You can use -index to return the value for a specific view and/or clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slack_min_rise	
	Returns the worst rising slack across all concurrent MMMC views for Hold-style early data path checks. You can use -index to return the value for a specific view and/or clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slack_sigma_max	
	In SOCV analysis, this returns the variation component of the worst slack across all concurrent MMMC views for Setup-style late data path checks. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slack_sigma_max_fall	
	In SOCV analysis, this returns the variation component of the worst falling slack across all concurrent MMMC views for Setup-style late data path checks. You can use -index to return the value for a specific view and/or clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slack_sigma_max_rise	
	In SOCV analysis, this returns the variation component of the worst rising slack across all concurrent MMMC views for Setup-style late data path checks. You can use -index to return the value for a specific view and/or clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slack_sigma_min	
	In SOCV analysis, this returns the variation component of the worst slack across all concurrent MMMC views for Hold-style early data path checks. You can use -index to return the value for a specific view and/or clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slack_sigma_min_fall	
	In SOCV analysis, this returns the variation component of the worst falling slack across all concurrent MMMC views for Hold-style early data path checks. You can use -index to return the value for a specific view and/or clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No

Stylus Common UI Database Object Information

Database Objects--pin

slack_sigma_min_rise	
	In SOCV analysis, this returns the variation component of the worst rising slack across all concurrent MMMC views for Hold-style early data path checks. You can use -index to return the value for a specific view and/or clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slack_skewness_max_fall	
	In SOCV analysis, this returns the skewness component of the worst falling slack across all concurrent MMMC views for Setup-style late data path checks. You can use -index to return the value for a specific view and/or clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slack_skewness_max_rise	
	In SOCV analysis, this returns the skewness component of the worst rising slack across all concurrent MMMC views for Setup-style late data path checks. You can use -index to return the value for a specific view and/or clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slack_skewness_min_fall	
	In SOCV analysis, this returns the skewness component of the worst falling slack across all concurrent MMMC views for Hold-style early data path checks. You can use -index to return the value for a specific view and/or clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slack_skewness_min_rise	
	In SOCV analysis, this returns the skewness component of the worst rising slack across all concurrent MMMC views for Hold-style early data path checks. You can use -index to return the value for a specific view and/or clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slew_max	
	Reports the maximum base slew of pin Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slew_max_fall	
	Returns the maximum slew time for falling transitions across all concurrent MMMC views. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slew_max_rise	
	Returns the maximum slew time for rising transitions across all concurrent MMMC views. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No

Stylus Common UI Database Object Information

Database Objects--pin

slew_mean_max_fall	
	In SOCV mode, returns the mean component of the maximum slew time for falling transitions across all concurrent MMMC views. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slew_mean_max_rise	
	In SOCV mode, returns the mean component of the maximum slew time for rising transitions across all concurrent MMMC views. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slew_mean_min_fall	
	In SOCV mode, returns the mean component of the minimum slew time for falling transitions across all concurrent MMMC views. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slew_mean_min_rise	
	In SOCV mode, returns the mean component of the minimum slew time for rising transitions across all concurrent MMMC views. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slew_min	
	Reports minimum base slew of pin Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slew_min_fall	
	Returns the minimum slew time for falling transitions across all concurrent MMMC views. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slew_min_rise	
	Returns the minimum slew time for rising transitions across all concurrent MMMC views. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slew_sigma_max_fall	
	In SOCV mode, returns the variation component of the maximum slew time for falling transitions across all concurrent MMMC views. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No

Stylus Common UI Database Object Information

Database Objects--pin

slew_sigma_max_rise	
	In SOCV mode, returns the variation component of the maximum slew time for rising transitions across all concurrent MMMC views. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slew_sigma_min_fall	
	In SOCV mode, returns the variation component of the minimum slew time for falling transitions across all concurrent MMMC views. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slew_sigma_min_rise	
	In SOCV mode, returns the variation component of the minimum slew time for rising transitions across all concurrent MMMC views. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slew_skewness_max_fall	
	In SOCV mode, returns the skewness component of the maximum slew time for falling transitions across all concurrent MMMC views. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slew_skewness_max_rise	
	In SOCV mode, returns the skewness component of the maximum slew time for rising transitions across all concurrent MMMC views. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slew_skewness_min_fall	
	In SOCV mode, returns the skewness component of the minimum slew time for falling transitions across all concurrent MMMC views. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slew_skewness_min_rise	
	In SOCV mode, returns the skewness component of the minimum slew time for rising transitions across all concurrent MMMC views. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
source_latency_early_fall_max	
	Returns the maximum early fall source insertion delay specified by an explicit set_clock_latency at the pin. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No

Stylus Common UI Database Object Information

Database Objects--pin

source_latency_early_fall_min	
	Returns the minimum early fall source insertion delay specified by an explicit set_clock_latency at the pin. You can use -index to return the value for a specific view and/or clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
source_latency_early_rise_max	
	Returns the maximum early rise source insertion delay specified by an explicit set_clock_latency at the pin. You can use -index to return the value for a specific view and/or clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
source_latency_early_rise_min	
	Returns the minimum early rise source insertion delay specified by an explicit set_clock_latency at the pin. You can use -index to return the value for a specific view and/or clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
source_latency_late_fall_max	
	Returns the maximum late fall source insertion delay specified by an explicit set_clock_latency at the pin. You can use -index to return the value for a specific view and/or clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
source_latency_late_fall_min	
	Returns the minimum late fall source insertion delay specified by an explicit set_clock_latency at the pin. You can use -index to return the value for a specific view and/or clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
source_latency_late_rise_max	
	Returns the maximum late rise source insertion delay specified by an explicit set_clock_latency at the pin. You can use -index to return the value for a specific view and/or clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
source_latency_late_rise_min	
	Returns the minimum late rise source insertion delay specified by an explicit set_clock_latency at the pin. You can use -index to return the value for a specific view and/or clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
source_of_clocks	
	Reports source pins of clocks associated with this pin Type: obj(pin)* Allowed -index values: analysis_view clock Default: "" Edit: No

Stylus Common UI Database Object Information

Database Objects--pin

stack_via_rule	
	This value serves as a pin-specific override to use this stack_via_rule when connecting to this pin. If set, it must match one of the elements from the stack_via_rule_list attribute of the corresponding base_pin. The interpretation of this value depends on this pin's stack_via_rule_required attribute. stack_via_rule == {} && stack_via_rule_required == false: stack_via_rule does not affect the choice. stack_via_rule == stackrule1 && stack_via_rule_required == false: Router will prefer stackrule1, but it may select another (or possibly none) if necessary to avoid design rule violations. stack_via_rule == {} && stack_via_rule_required == true: The required stack_via_rule is empty, so the router will not use any stack_via_rule for connecting to this pin (even ignoring stack_via_required on the base_pin if necessary). stack_via_rule == stackrule1 && stack_via_rule_required == true: Router will use stackrule1, even if it leads to design rule violations. Type: obj(stack_via_rule) Default: "" Edit: Yes
stack_via_rule_required	
	Specifies whether the router must use this pin's stack_via_rule value. If false, the stack_via_rule value will be preferred by the router, if true the stack_via_rule value is required by the router. See the stack_via_rule attribute for more details. Type: bool Default: false Edit: Yes
stolen_slack	
	Reports stolen (borrowed) slack for latch objects Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
timing_case_computed_value	
	Returns the case computed value Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
timing_case_logic_value	
	Returns the case logic value Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
timing_constant	
	Reports constant value on pin Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
timing_constant_source	
	Reports source of constant value on pin Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
to_arcs	
	Returns a list of arc objects where the current pin is the termination point of the arc Type: obj(arc)* Default: "" Edit: No
unused_bit_in_multibit	

Stylus Common UI Database Object Information

Database Objects--pin

	Returns a value of true if the pin is unused in the multibit cell. Type: <code>bool</code> Default: false Edit: Yes
user_constant_value	
	Returns constant values from netlist or constraints. You can use -index to return the value for a specific view. Type: <code>int</code> Allowed -index values: analysis_view clock Default: "" Edit: No
waveform_max_fall	
	Reports maximum fall waveform on pin Type: <code>string</code> Allowed -index values: analysis_view clock Default: "" Edit: No
waveform_max_rise	
	Reports maximum rise waveform on pin Type: <code>string</code> Allowed -index values: analysis_view clock Default: "" Edit: No
waveform_min_fall	
	Reports minimum fall waveform on pin Type: <code>string</code> Allowed -index values: analysis_view clock Default: "" Edit: No
waveform_min_rise	
	Reports minimum rise waveform on pin Type: <code>string</code> Allowed -index values: analysis_view clock Default: "" Edit: No
wrapper_control	
	Specifies as wrapper control signal. Type: <code>bool</code> Default: false Edit: Yes
wrapper_segment	
	Specifies the wrapper segments associated with the pin. Type: <code>obj(scan_chain)* obj(scan_segment)* obj(actual_scan_segment)* obj(actual_scan_chain)*</code> Default: "" Edit: Yes
x_coordinate	
	Reports X co-ordinate of pin Type: <code>double</code> Allowed -index values: analysis_view clock Default: "" Edit: No
y_coordinate	

	Reports Y co-ordinate of pin Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
--	---

pin_blockage

Parent Objects

[design](#), [root](#)

Definition

Pin blockage objects

Attribute	Description
excluded_pins	
	Returns the excluded pins for pin_blockage. The pin list is set by create_pin_blockage -exclude_pin_patterns. If the .included_pins & .excluded_pins were both empty, the pin blockage would be applied to all pins on the cell. Type: obj(port)* obj(base_pin)* Default: "" Edit: No
included_pins	
	Returns the included pins for pin_blockage. The pin list is set by create_pin_blockage -include_pin_patterns. If the .included_pins & .excluded_pins were both empty, the pin blockage would be applied to all pins on the cell. Type: obj(port)* obj(base_pin)* Default: "" Edit: No
layer	
	The layer that is being blocked Type: obj(layer) Default: "" Edit: No
name	
	Partition pin blockage name Type: string Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (pin_blockage) Default: "" Edit: No
rect	
	Rectangle that define the pin_blockage shape. Type: rect Default: "" Edit: No

pin_group

Parent Objects

[design](#), [pin_guide](#), [root](#)

Definition

Pin group

Attribute	Description
base_pins	
	List of base pins or ports associated with the group. Type: obj(port)* obj(base_pin)* Default: "" Edit: No
exclude_pin	
	Indicates group pin exclude type(all_layer, same_layer, inclusive, all_layer_in_guided_area) Type: enum Enum Values: all_layer_in_guided_area all_layer same_layer inclusive Default: "" Edit: Yes
is_compact	
	Indicate whether the pins in the group are assigned tightly together. By default, pins which are part of a group and associated to a guide can be spread inside the guide based on the area available and alignment to targets Type: bool Default: false Edit: Yes
is_guided	
	Indicates if pin group is guided Type: bool Default: "" Edit: No
is_spread	
	Indicates whether member pins distributed evenly Type: bool Default: "" Edit: No
keep_out_space	
	Minimum spacing with foreign pins (Unit: track) Type: int Default: "" Edit: No
name	
	Name of pin group Type: string Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (pin_group) Default: "" Edit: No

Stylus Common UI Database Object Information

Database Objects--pin_guide

optimize_order	
	Name of pin group Type: <code>bool</code> Default: "" Edit: No
parent	
	The parent of the pin_group. It can be a design, a partition (if not committed yet), or base_cell for a black_box. Type: <code>obj(design)</code> <code>obj(partition)</code> <code>obj(base_cell)</code> Default: "" Edit: No
pin_spacing	
	Minimum spacing between adjacent pins (Unit: track) Type: <code>int</code> Default: "" Edit: No

pin_guide

Parent Objects

`design`, `root`

Definition

Pin guide

Attribute	Description
area	
	Area of the pin guide as defined by the LEF MACRO SIZE or OVERLAP information Type: <code>area</code> Default: "" Edit: No
layer_priority	
	layer priority Type: <code>bool</code> Default: "" Edit: No
layers	
	layers where pin_guide is present Type: <code>obj(layer)*</code> Default: "" Edit: No
name	
	Name of pin guide Type: <code>string</code> Default: "" Edit: No
net_group	
	net_group, if pin_guide is based on net_group otherwise NULL Type: <code>obj(net_group)</code> Default: "" Edit: No

Stylus Common UI Database Object Information
Database Objects--place_blockage

obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (pin_guide) Default: "" Edit: No
parent	
	The parent of the pin_guide. It can be a design, a partition (if not committed yet), or base_cell for a black_box. Type: obj (design) obj (partition) obj (base_cell) Default: "" Edit: No
pin_group	
	pin_group, if pin_guide is based on pin_group otherwise NULL Type: obj (pin_group) Default: "" Edit: No
rects	
	The rects that define the location of the pin_guide. Type: rect * Default: "" Edit: No

place_blockage

Parent Objects

[design](#), [root](#)

Definition

Placement blockage(hard, soft, partial).

Attribute	Description
base_name	
	Name of placement blockage Type: string Default: "" Edit: Yes
density	
	The max placement density percent allowed inside this place_blockage. It must be in the range of 5 to 100, in steps of 5. It is only valid if the type = partial or soft. For example, a partial placement percentage of 75 percent means that up to 75 percent of placement density is allowed in the area. If the type is not partial or soft, a value of 0 is returned. Type: double Default: "" Edit: Yes
inst	
	The instance that the placement blockage is associated with (equivalent to DEF BLOCKAGES + COMPONENT) Type: obj (inst) Default: "" Edit: No
is_no_flop	

Stylus Common UI Database Object Information

Database Objects--port

	Flip-flops and latches cannot be placed inside this place_blockage. Only has an effect if type = partial. Type: <code>bool</code> Default: false Edit: Yes
is_pushdown	
	This place_blockage has been pushed down from a higher level in the design hierarchy. Type: <code>bool</code> Default: false Edit: Yes
name	
	Name of placement blockage Type: <code>string</code> Default: "" Edit: Yes
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: <code>enum</code> (<code>place_blockage</code>) Default: "" Edit: No
rects	
	List of non-overlapping rectangles that defines the shape of place_blockage Type: <code>rect*</code> Default: "" Edit: Yes
shapes	
	List of shapes of this placement blockage Type: <code>obj(shape)* obj(density_shape)*</code> Default: "" Edit: No
type	
	The type of blockage. hard = no cells allowed, macro_only = standard-cells are allowed but blocks are not allowed, partial = allow cells until reach the .density limit, soft = most logic cells are not allowed, but 'repeater' cells like inverters, buffers, level-shifters, isolation cells, and clock-gating cells are allowed. See the set_selective_blockage_gate command for other methods to control what is allowed inside a soft place_blockage. Type: <code>enum</code> Enum Values: macro_only hard soft partial Default: "" Edit: Yes

port

Parent Objects

marker, bump_pin, test_clock, clock, io_constraint, violation, def_scan_chain, skew_group, pin_blockage, scan_chain, hnet, actual_scan_segment, timing_point, dft_configuration_mode, bus_sink_group, design, clock_tree, pin_group, timing_path, root, bump, test_signal, port_shape, inst, port_bus, scan_segment, net, actual_scan_chain

Definition

External logical ports of the design. See pg_ports for power/ground ports

Attribute	Description
actual_latency_early_fall_max	

Stylus Common UI Database Object Information

Database Objects--port

	Returns the computed early falling clock latency to this clock port. If the analysis mode is set to best-case/worst-case, the value will be the Setup capture latency. When operating in OCV mode, the max qualification is ignored. If there are multiple arrival times from different clock phases - the most conservative latency value is reported. You can use the -index function to return latencies with respect to a specific clock. When operating in concurrent MMMC mode, the worst latency across all views is reported. You can use the -index function to filter latencies based on view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
actual_latency_early_fall_min	
	Returns the computed early falling clock latency to this clock port. If the analysis mode is set to best-case/worst-case, the value will be the Hold launch latency. When operating in OCV mode, the min qualification is ignored. If there are multiple arrival times from different clock phases - the most conservative latency value is reported. You can use the -index function to return latencies with respect to a specific clock. When operating in concurrent MMMC mode, the worst latency across all views is reported. You can use the -index function to filter latencies based on view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
actual_latency_early_rise_max	
	Returns the computed early rising clock latency to this clock port. If the analysis mode is set to best-case/worst-case, the value will be the Setup capture latency. When operating in OCV mode, the max qualification is ignored. If there are multiple arrival times from different clock phases - the most conservative latency value is reported. You can use the -index function to return latencies with respect to a specific clock. When operating in concurrent MMMC mode, the worst latency across all views is reported. You can use the -index function to filter latencies based on view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
actual_latency_early_rise_min	
	Returns the computed early rising clock latency to this clock port. If the analysis mode is set to best-case/worst-case, the value will be the Hold launch latency. When operating in OCV mode, the min qualification is ignored. If there are multiple arrival times from different clock phases - the most conservative latency value is reported. You can use the -index function to return latencies with respect to a specific clock. When operating in concurrent MMMC mode, the worst latency across all views is reported. You can use the -index function to filter latencies based on view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
actual_latency_late_fall_max	
	Returns the computed late falling clock latency to this clock port. If the analysis mode is set to best-case/worst-case, the value will be the Setup launch latency. When operating in OCV mode, the max qualification is ignored. If there are multiple arrival times from different clock phases - the most conservative latency value is reported. You can use the -index function to return latencies with respect to a specific clock. When operating in concurrent MMMC mode, the worst latency across all views is reported. You can use the -index function to filter latencies based on view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
actual_latency_late_fall_min	
	Returns the computed late falling clock latency to this clock port. If the analysis mode is set to best-case/worst-case, the value will be the Hold capture latency. When operating in OCV mode, the min qualification is ignored. If there are multiple arrival times from different clock phases - the most conservative latency value is reported. You can use the -index function to return latencies with respect to a specific clock. When operating in concurrent MMMC mode, the worst latency across all views is reported. You can use the -index function to filter latencies based on view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
actual_latency_late_rise_max	

Stylus Common UI Database Object Information

Database Objects--port

	Returns the computed late rising clock latency to this clock port. If the analysis mode is set to best-case/worst-case, the value will be the Setup launch latency. When operating in OCV mode, the max qualification is ignored. If there are multiple arrival times from different clock phases - the most conservative latency value is reported. You can use the -index function to return latencies with respect to a specific clock. When operating in concurrent MMMC mode, the worst latency across all views is reported. You can use the -index function to filter latencies based on view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
actual_latency_late_rise_min	
	Returns the computed late rising clock latency to this clock port. If the analysis mode is set to best-case/worst-case, the value will be the Hold capture latency. When operating in OCV mode, the min qualification is ignored. If there are multiple arrival times from different clock phases - the most conservative latency value is reported. You can use the -index function to return latencies with respect to a specific clock. When operating in concurrent MMMC mode, the worst latency across all views is reported. You can use the -index function to filter latencies based on view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
antenna_data	
	List of antenna data for the terminal Type: obj(antenna_data)* Default: "" Edit: No
aocv_early_input_stage_weight	
	Returns the number of external path stages to consider in AOCV early analysis of paths from this port. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
aocv_early_output_stage_weight	
	Returns the number of external path stages to consider in AOCV early analysis of paths to this port. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
aocv_late_input_stage_weight	
	Returns the number of external path stages to consider in AOCV late analysis of paths from this port. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
aocv_late_output_stage_weight	
	Returns the number of external path stages to consider in AOCV late analysis of paths to this port. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
arrival_max_fall	
	Returns the latest falling arrival time to the specified port across all concurrent MMMC views. You can use -index to determine the worst arrival for a specific view. You can also use -index to get the latest arrival time related to a specific clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
arrival_max_rise	

Stylus Common UI Database Object Information

Database Objects--port

	Returns the latest rising arrival time to the specified port across all concurrent MMMC views. You can use -index to determine the worst arrival for a specific view. You can also use -index to get the latest arrival time related to a specific clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
arrival_mean_max_fall	
	In SOCV analysis mode, this returns the mean value of the latest falling arrival time to the specified port across all concurrent MMMC views. You can use -index to determine the latest arrival for a specific view. You can also use -index to get the worst arrival time related to a specific clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
arrival_mean_max_rise	
	In SOCV analysis mode, this returns the mean value of the latest rising arrival time to the specified port across all concurrent MMMC views. You can use -index to determine the latest arrival for a specific view. You can also use -index to get the worst arrival time related to a specific clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
arrival_mean_min_fall	
	In SOCV analysis mode, this returns the mean value of the earliest falling arrival time to the specified port across all concurrent MMMC views. You can use -index to determine the earliest arrival for a specific view. You can also use -index to get the earliest arrival time related to a specific clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
arrival_mean_min_rise	
	In SOCV analysis mode, this returns the mean value of the earliest rising arrival time to the specified port across all concurrent MMMC views. You can use -index to determine the earliest arrival for a specific view. You can also use -index to get the earliest arrival time related to a specific clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
arrival_min_fall	
	Returns the earliest falling arrival time to the specified port across all concurrent MMMC views. You can use -index to determine the worst arrival for a specific view. You can also use -index to get the earliest arrival time related to a specific clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
arrival_min_rise	
	Returns the earliest falling arrival time to the specified port across all concurrent MMMC views. You can use -index to determine the worst arrival for a specific view. You can also use -index to get the worst arrival time related to a specific clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
arrival_sigma_max_fall	

Stylus Common UI Database Object Information

Database Objects--port

	In SOCV analysis mode, this returns the variation component of the latest falling arrival time to the specified port across all concurrent MMMC views. You can use -index to determine the latest arrival for a specific view. You can also use -index to get the worst arrival time related to a specific clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
arrival_sigma_max_rise	
	In SOCV analysis mode, this returns the variation component of the latest rising arrival time to the specified port across all concurrent MMMC views. You can use -index to determine the latest arrival for a specific view. You can also use -index to get the worst arrival time related to a specific clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
arrival_sigma_min_fall	
	In SOCV analysis mode, this returns the variation component of the earliest falling arrival time to the specified port across all concurrent MMMC views. You can use -index to determine the earliest arrival for a specific view. You can also use -index to get the earliest arrival time related to a specific clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
arrival_sigma_min_rise	
	In SOCV analysis mode, this returns the mean value of the earliest rising arrival time to the specified port across all concurrent MMMC views. You can use -index to determine the earliest arrival for a specific view. You can also use -index to get the earliest arrival time related to a specific clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
arrival_window	
	Returns a list of the earliest and latest, rising and falling arrival times per clock phase at the port across all concurrent MMMC views. You can use -index to make the arrival times specific to a given view Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
base_name	
	The name at the base of a hierarchical name. So the base_name of i1/i2/i3 is i3. Type: string Default: "" Edit: No
bumps	
	Specifies the bumps and bump_pins assigned to this port. Type: obj(bump)* obj(bump_pin)* Default: "" Edit: No
bus	
	The bus of this port. Type: obj(port_bus) Default: "" Edit: No
clock_gating_exclude_signal	

Stylus Common UI Database Object Information

Database Objects--port

	Exclude this pin from clock gating enable functions. Type: <code>bool</code> Default: <code>false</code> Edit: Yes
clocks	
	Returns a list of all of the clock objects arriving at the port. You can use -index to return the value from a specific analysis view. Type: <code>obj(clock)*</code> Allowed -index values: <code>analysis_view clock</code> Default: <code>""</code> Edit: No
constant_value	
	Returns a constant value of 0 or 1 if logic state has been asserted or propagated to this pin. You can use -index to return the value from a specific analysis view. Type: <code>int</code> Allowed -index values: <code>analysis_view clock</code> Default: <code>""</code> Edit: No
context_constant_value	
	Returns a context_constant value of 0 or 1 if logic state has been asserted or propagated to this pin. You can use -index to return the value from a specific analysis view. Type: <code>int</code> Allowed -index values: <code>analysis_view clock</code> Default: <code>""</code> Edit: No
cts_add_pin_driver	
	Specifies a cell type or pair of cell types, so that a cell inst can to be added above an input pin or below an output pin. The pin is specified by the argument pin. No more than two cell types may be specified. A warning will be emitted if no AON cells are specified. Type: <code>string</code> Default: <code>""</code> Edit: Yes
cts_add_port_driver	
	Specifies a cell type or pair of cell types, so that a cell inst can to be added above an output port or below an input port. The port is specified by the argument pin. If the pin specified is not a design IO pin or it is not in the clock network then CCOpt will emit a warning and will not add cell insts at that position. No more than two cell types may be specified. A warning will be emitted if no AON cells are specified. Type: <code>string</code> Default: <code>""</code> Edit: Yes
cts_annotated_delay_to	
	Override any clock tree timing engine computed cell arc or net arc delays to this pin, in a similar manner to SDC set_annotated_delay. Type: <code>string</code> Allowed -index values: <code>delay_corner</code> Default: <code>""</code> Edit: Yes
cts_annotated_transition	
	Override any clock tree timing engine computed transition at this pin, in a similar manner to SDC set_annotated_transition. Type: <code>string</code> Allowed -index values: <code>delay_corner</code> Default: <code>auto</code> Edit: Yes
cts_assign_clock_tree	

Stylus Common UI Database Object Information

Database Objects--port

	Can be used to make sure a clock tree sink pin gets driven by a particular clock tree. The clock tree can alternatively be specified as the tree's root pin, in case extraction rennumbers the clock trees defined from a single SDC clock. By default CCOpt automatically selects appropriate pins. Valid values: cts_clock_tree list of pins Type: string Default: "" Edit: Yes
cts_case_analysis	
	Specifies a constant value to be used as pin signal when analyzing timing within CTS. Only applies to input pins of cells in the clock tree. If set to 'none' (the default), CTS will consider the input as non-constant. Valid values: 0 1 none Type: string Default: auto Edit: Yes
cts_clock_tree	
	The clock tree to which this object belongs. Flops do not belong to a clock tree, but their clock pins do. Valid values: cts_clock_tree Type: obj(clock_tree) Default: "" Edit: No
cts_clock_trees	
	A list of clock trees the pin is contained within. This includes parents of generated clock trees and all relevant parents when clock trees overlap. Valid values: list cts_clock_tree Type: obj(clock_tree) Default: "" Edit: No
cts_effective_mixed_buffer_budget	
	Allows the user to query the mixed buffer budget at a particular pin. This is the number of buffers allowed above the given pin, and is the smallest of the remaining budgets from any fanout below this point or the override set at this point if one has been set. Type: string Default: auto Edit: No
cts_effective_routing_preference	
	Returns the pin's effective routing preference; one of leaf, trunk, top, none. The value "none" can be returned for pins that have no preference such as antenna diodes or load cap inputs. The effective routing preference value for a pin does not solely determine the net_type of the driving net if the net has multiple fanout. The fanout routing preferences for all sinks are taken into account when determining the net_type for the net as a whole. Type: string Default: "" Edit: No
cts_is_sdc_clock_root	

Stylus Common UI Database Object Information

Database Objects--port

	Specifies whether the given pin is the root (source pin) of an SDC clock. The create_clock_tree_spec populates this attribute with the location of the SDC clock root (source) pins. This attribute controls the behavior of clock tree definition commands create_clock_tree and create_generated_clock_tree, when the -stop_at_sdc_clock_roots argument is specified. In such a case, pins and ports for which this attribute is true will be treated as being SDC clock root pins. Valid values: true false Type: bool Default: false Edit: Yes
cts_isolated_effective	
	This is the isolated value CCOpt will use for the given pin. Type: bool Default: false Edit: No
cts_net_unbufferable_reasons	
	This attribute contains a list of reasons why CCOpt was not able to buffer the clock net attached to the specified pin. Valid values: string Type: string Default: "" Edit: No
cts_node_type	
	node type of port within cts graph Type: string Default: "" Edit: No
cts_pin_capacitance_sources	
	List of the places the capacitance could be retrieved from. Requesting this attribute for a pin not in the clock tree will result in an error. The possible places are: - library - CTS will retrieve capacitance values from the library for one or more timing_corner/event combinations. - cts_capacitance_override - CTS will retrieve capacitance values from the cts_capacitance_override attribute for one or more timing_corner/event combinations. - cts_blackbox_default_load_base_pin - CTS will retrieve capacitance values from the cts_blackbox_default_load_base_pin attribute for one or more timing_corner/event combinations. Type: string Default: "" Edit: No
cts_pin_insertion_delay	
	The amount of insertion delay under this pin. Clock tree synthesis will attempt to make the insertion delay to this pin less than that to other sinks in the same skew group by this amount if a positive value is set. A negative value should be used if you would like the insertion delay to this pin to be greater than that to other sinks. The value 'auto' means there is no insertion delay offset for the pin. This attribute exposes the 'total' pin insertion delay. For further discussion on this, consult the documentation for attribute cts_pin_insertion_delay_sources. Valid values: double ignore min auto Type: string Allowed -index values: delay_corner Default: auto Edit: Yes
cts_pin_insertion_delay_sources	

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	The amount of insertion delay under this pin, broken down by the source of the pin insertion delay (PID). The value of this attribute is a Tcl dictionary of PID values, keyed on the source of that PID. This provides an 'exploded' view into the total pin insertion delay that is present, categorizing the various contributions by their origin. Valid Tcl dict keys: - user Asserted by user Tcl scripting or restored from an older DB. - sdcl Extracted from SDC set_clock_latency by spec creation. - sink_type Asserted to reflect the impact of the cts_sink_type attribute. - ilm Records the mean clock latency of the ILM clock network that is rooted at this pin. - useful_skew A useful skew generated by pre-CTS optimization. - incremental An incremental offset, used to specify useful skews to incremental CTS. - total The formal sum of the above contributions. This value is the 'overall' pin insertion delay accessible via the cts_pin_insertion_delay attribute. Valid Tcl dict values: double ignore min auto Reading the attribute returns a dict value that exposes the current state of the PID, recording all the PID sources and the associated PID value of each one. Sources whose value are auto are omitted. If there are no non-auto sources, the returned dict is empty. For convenience, the total is always included in the dict if it is non-auto. Writing a dict value to the attribute will update each of the specified sources with the specified PID value. The total PID value will then be updated. Sources in the dict that are assigned a value of 'auto' will be reset. Sources omitted from the dict are left unchanged from their current value: assigning an empty dict is a no-op. Resetting the attribute will clear all the sources and the total back, resetting them to the default value of 'auto', thereby completely removing the PID from the pin. Valid values: A Tcl dictionary of PID values, keyed on PID source. Type: string Allowed -index values: delay_corner Default: "" Edit: Yes
cts_pin_route_type_propagation	
	Describes how cts_pin_route_type should be applied to the net driven by the specified pin. The possible values are: - none - it applies to net driven by the pin and only that net, irrespective of whether the net is buffered or not. - net - it applies to all nets generated by buffering the pre-CTS net driven by pin (default). Type: string Default: net Edit: Yes
cts_routing_trunk_override	
	Prefer trunk routing rules for this pin. Only applies to clock tree sinks. Valid values: true false Type: bool Default: false Edit: Yes
cts_sink_type	

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	The type of sink this pin represents. Valid values are as follows: auto The sink type will be automatically determined by CCOpt. through Through pin. Trace the clock tree through this pin. stop Stop pin. When defining clock trees, CCOpt stops searching for parts of the clock tree at stop pins. ignore Ignore pin. CCOpt stops searching for parts of the clock tree at ignore pins and it does not attempt to balance the insertion delay of ignore pins. exclude Exclude pin. Exclude this pin from the clock tree. Valid values: auto through stop ignore exclude Type: string Default: auto Edit: Yes
cts_sink_type_effective	
	Indicates how CCOpt will treat a given pin, taking into account both its cts_sink_type_implicit, and any cts_sink_type settings. Setting a non-default value for the cts_sink_type attribute will override the cts_sink_type_implicit attribute. Type: string Default: auto Edit: No
cts_sink_type_implicit	
	Indicates the type of sink that CCOpt classified this pin as. Note that the cts_sink_type attribute can override these internal classifications. Possible values are: exclude Indicates that this pin is a sink which represents a non-clock pin. ignore Indicates that CCOpt has determined not to search for more clock tree through this pin. Additionally, this pin will not be balanced. stop Indicates that CCOpt has determined not to search for more clock tree through this pin. An empty value for this attribute indicates either that this pin is either not a sink, or that it is a sink that is not implicitly exclude or ignore or stop. Type: string Default: exclude Edit: No
cts_sink_type_reasons	
	The reasons why this pin has the given cts_sink_type. This attribute is configured by the create_clock_tree_spec command in order to record the reasons that the cts_sink_type attribute has been configured for the given pin. This attribute is a list of values. Valid values are as follows: auto The cts_sink_type attribute is set to 'auto' implicit The pin is an implicit sink (flop/latch) user The user has set the cts_sink_type attribute design_io This pin is a design I/O multiple_outputs This pin is on an instance with multiple outputs, and the cts_spec_config_trace_through_to attribute has not been set set_disable_timing SDC set_disable_timing stops the clock at this pin set_case_analysis SDC set_case_analysis stops the clock at this pin generated_clock_tree This pin is the generator input to an SDC generated clock no_sdc_clock The SDC clock is stopped at this pin for other reasons ilm The create_clock_tree_spec command has detected an ILM below this pin Valid values: auto implicit user design_io multiple_outputs set_disable_timing set_case_analysis generated_clock_tree no_sdc_clock ilm Type: string Default: auto Edit: Yes
cts_skew_groups_active	

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	Returns the list of active skew groups for this pin. For sink pins, this attribute lists both skew groups that pass through this pin and skew groups for which this sink is an endpoint. For non-sink pins, shows skew groups that pass through this pin. Valid values: list skew_groups Type: obj(skew_group)* Default: "" Edit: No
cts_skew_groups_active_sink	
	Returns the list of skew groups for which this pin is an active sink. For sink pins, this attribute lists the skew groups for which this sink is an endpoint. Skew groups that pass through this pin are not included. For non-sink pins, this attribute always returns null. Valid values: list skew_groups Type: obj(skew_group)* Default: "" Edit: No
cts_skew_groups_constraining	
	The list of delay-constraining skew groups which are active at this pin. This attribute reports similar data to cts_skew_groups_active. The only difference is that the reporting-only skew groups are not included in this attribute's value. For sink pins, this attribute lists both delay-constraining skew groups that pass through this pin and delay-constraining skew groups for which this sink is an endpoint. For non-sink pins, shows delay-constraining skew groups that pass through this pin. Valid values: list skew_groups Type: obj(skew_group)* Default: "" Edit: No
cts_skew_groups_constraining_sink	
	The list of delay-constraining skew groups for which this pin is an active sink. This attribute lists the delay-constraining skew groups for which this pin is an active sink. This attribute reports similar data to cts_skew_groups_active_sink. The only difference is that the reporting-only (constrains none) skew groups are not included in this attribute's value. For sink pins, this attribute lists the delay-constraining skew groups for which this sink is an endpoint. Skew groups that pass through this pin are not included. For non-sink pins, this attribute always returns null. Valid values: list skew_groups Type: obj(skew_group)* Default: "" Edit: No
cts_skew_groups_ignore	
	The list of skew groups for which paths through this pin are ignored. Valid values: list skew_groups Type: obj(skew_group)* Default: "" Edit: No
cts_skew_groups_sink	
	The list of skew groups for which this pin is a sink. Valid values: list skew_groups Type: obj(skew_group)* Default: "" Edit: No

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cts_skew_groups_source_pin	
	The list of skew groups for which this clock tree or pin is specified as a source. Valid values: list skew_groups Type: obj(skew_group)* Default: "" Edit: No
cts_spec_config_trace_through_to	
	Clock tree definition will, by default, not continue through certain types of cell arc (for instance, the clock to Q arc in a DFF). This attribute allows you to override this default behavior, permitting the clock tree to trace through such a cell. The attribute should be configured on the input pin at which the clock arrives. The value of the attribute specifies the output pin to which the clock should propagate. The specified output pin must be another pin on the same instance. The output pin may be specified either by its fully qualified name (i.e. inclusive of the instance name), or else simply by its local (cell-relative) name. There must be a pre-existing (library-defined) chain of one or more delay arcs that connect the input and output pins together. It is not possible to use trace_through_to to synthesize delay arcs. If multiple input pins are annotated on a given instance, the value of trace_through_to at each of those pins must select the same output pin: i.e. the configuration must identify a single clock output for the instance. If multiple clock outputs are required then trace_through_to should not be used: instead define a generated clock tree at each of the clock-carrying outputs. If the configuration of trace_through_to settings for a given instance does not meet these requirements, a warning will be issued and the settings for that instance will be ignored. Note: if both trace_through_to and library_trace_through_to are applicable at a given netlist instance pin, the trace_through_to value takes precedence. Valid values: pin Type: string Default: "" Edit: Yes
cts_top_fanout_count_override	
	The number of clock sinks this sink counts for when applying the top routing rules. Note that this attribute is only valid for sink pins, and it returns auto for non-sink pins. For a sink pin, a non-auto value means that this sink is counted as though it were multiple sinks, for the purposes of determining which nets should have top routing. An auto value for a sink pin means that the sink counts as a single sink. Valid values: integer > 0 Type: string Default: auto Edit: Yes
cts_top_fanout_transitive_count	
	The number of clock sinks in the transitive fanout of the pin as counted for applying the top routing rules. This attribute is very similar to the cts_transitive_fanout attribute but counts sink fanout using the cts_top_fanout_count_override attribute instead of always counting sinks as a single item of fanout. Requesting this attribute for a pin not in the clock tree will result in an error. Valid values: integer Type: int Default: 0 Edit: No
cts_transitive_fanout	

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	The number of clock sinks in the transitive fanout of the pin, within the clock tree. Requesting this attribute for a pin not in the clock tree will result in an error. Valid values: int Type: int Default: 0 Edit: No
cts_virtual_delay_early_fall	
	The amount of virtual delay that has been applied under this pin. Valid values: double Type: double Allowed -index values: delay_corner Default: 0 Edit: Yes
cts_virtual_delay_early_rise	
	The amount of virtual delay that has been applied under this pin. Valid values: double Type: double Allowed -index values: delay_corner Default: 0 Edit: Yes
cts_virtual_delay_late_fall	
	The amount of virtual delay that has been applied under this pin. Valid values: double Type: double Allowed -index values: delay_corner Default: 0 Edit: Yes
cts_virtual_delay_late_rise	
	The amount of virtual delay that has been applied under this pin. Valid values: double Type: double Allowed -index values: delay_corner Default: 0 Edit: Yes
delay_max_fall	
	Returns the maximum falling delay. You can use -index to return the value from a specific analysis view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
delay_max_rise	
	Returns the maximum rising delay. You can use -index to return the value from a specific analysis view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
delay_min_fall	
	Returns the minimum falling delay. You can use -index to return the value from a specific analysis view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
delay_min_rise	

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	Returns the minimum rising delay. You can use -index to return the value from a specific analysis view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
depth	
	Depth constraint of the port in microns. Pin assignment commands will create a pin shape for this port that extends 'depth' microns inside the design edge. The edit_pin command can be used to set it. Type: coord Default: "" Edit: No
design	
	The design of this port Type: obj(design) Default: "" Edit: No
dft_driven_by_clock	
	If pin is driven by clock. Type: enum Enum Values: false true Default: false Edit: Yes
dft_enable_hookup_pin	
	Hookup pin of shift enable port. Type: obj(pin) obj(hpin) obj(port) obj(hport) Default: "" Edit: Yes
dft_enable_hookup_polarity	
	Hookup polarity of shift enable port. Type: string Default: "" Edit: Yes
dft_sdi_output_hookup_pin	
	Input hookup pin of a bidirectional port. Type: obj(pin) obj(hpin) obj(port) obj(hport) Default: "" Edit: Yes
dft_sdo_input_hookup_pin	
	Output hookup pin of a bidirectional port. Type: obj(pin) obj(hpin) obj(port) obj(hport) Default: "" Edit: Yes
direction	
	Direction of the port from the Verilog netlist. Type: enum Enum Values: in out inout Default: "" Edit: No
drive_resistance_fall_max	

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	Returns the falling linear drive resistance at the port for late timing paths. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
drive_resistance_fall_min	
	Returns the falling linear drive resistance at the port for early timing paths. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
drive_resistance_rise_max	
	Returns the rising linear drive resistance at the port for late timing paths.. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
drive_resistance_rise_min	
	Returns the rising linear drive resistance at the port for early timing paths. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
driver_from_pin_fall_max	
	Returns the driving cell input pin specified for max falling delays at the port. You can use -index to return the value for a specific view. Type: obj(lib_pin)* Default: "" Edit: No
driver_from_pin_fall_min	
	Returns the driving cell input pin specified for min falling delays at the port. You can use -index to return the value for a specific view. Type: obj(lib_pin)* Default: "" Edit: No
driver_from_pin_rise_max	
	Returns the driving cell input pin specified for max rising delays at the port. You can use -index to return the value for a specific view. Type: obj(lib_pin)* Default: "" Edit: No
driver_from_pin_rise_min	
	Returns the driving cell input pin specified for min rising delays at the port. You can use -index to return the value for a specific view. Type: obj(lib_pin)* Default: "" Edit: No
driver_ignore_drc	
	driver_ignore_drc Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
driver_input_slew_fall_to_fall_max	

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	Returns the driver input pin slew used for the max fall-to-fall delay. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
driver_input_slew_fall_to_fall_min	
	Returns the driver input pin slew used for the min fall-to-fall delay. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
driver_input_slew_fall_to_rise_max	
	Returns the driver input pin slew used for the max fall-to-rise delay. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
driver_input_slew_fall_to_rise_min	
	Returns the driver input pin slew used for the min fall-to-rise delay. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
driver_input_slew_rise_to_fall_max	
	Returns the driver input pin slew used for the max rise-to-fall delay. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
driver_input_slew_rise_to_fall_min	
	Returns the driver input pin slew used for the min rise-to-fall delay. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
driver_input_slew_rise_to_rise_max	
	Returns the driver input pin slew used for the max rise-to-rise delay. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
driver_input_slew_rise_to_rise_min	
	Returns the driver input pin slew used for the min rise-to-rise delay. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
driver_pin_fall_max	
	Returns the driving cell output pin specified for max falling delays at the port. You can use -index to return the value for a specific view. Type: obj(lib_pin)* Allowed -index values: analysis_view Default: "" Edit: No
driver_pin_fall_min	

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	Returns the driving cell output pin specified for min falling delays at the port. You can use -index to return the value for a specific view. Type: <code>obj(lib_pin)*</code> Allowed -index values: analysis_view Default: "" Edit: No
driver_pin_rise_max	
	Returns the driving cell output pin specified for max rising delays at the port. You can use -index to return the value for a specific view. Type: <code>obj(lib_pin)*</code> Allowed -index values: analysis_view Default: "" Edit: No
driver_pin_rise_min	
	Returns the driving cell output pin specified for min rising delays at the port. You can use -index to return the value for a specific view. Type: <code>obj(lib_pin)*</code> Allowed -index values: analysis_view Default: "" Edit: No
driving_cell_fall_max	
	Returns the name of the library cell used to compute max falling delays at the port. You can use -index to return the value for a specific view. Type: <code>double</code> Allowed -index values: analysis_view clock Default: "" Edit: No
driving_cell_fall_min	
	Returns the name of the library cell used to compute min falling delays at the port. You can use -index to return the value for a specific view. Type: <code>double</code> Allowed -index values: analysis_view clock Default: "" Edit: No
driving_cell_from_pin_fall_max	
	Returns the driving cell input pin specified for max falling delays at the port. You can use -index to return the value for a specific view. Type: <code>double</code> Allowed -index values: analysis_view clock Default: "" Edit: No
driving_cell_from_pin_fall_min	
	Returns the driving cell input pin specified for min falling delays at the port. You can use -index to return the value for a specific view. Type: <code>double</code> Allowed -index values: analysis_view clock Default: "" Edit: No
driving_cell_from_pin_rise_max	
	Returns the driving cell input pin specified for max rising delays at the port. You can use -index to return the value for a specific view. Type: <code>double</code> Allowed -index values: analysis_view clock Default: "" Edit: No
driving_cell_from_pin_rise_min	
	Returns the driving cell input pin specified for min rising delays at the port. You can use -index to return the value for a specific view. Type: <code>double</code> Allowed -index values: analysis_view clock Default: "" Edit: No
driving_cell_library_fall_max	

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	Returns the driving cell library specified for max falling delays at the port. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
driving_cell_library_fall_min	
	Returns the driving cell library specified for min falling delays at the port. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
driving_cell_library_rise_max	
	Returns the driving cell library specified for max rising delays at the port. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
driving_cell_library_rise_min	
	Returns the driving cell library specified for min rising delays at the port. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
driving_cell_pin_fall_max	
	Returns the driving cell output pin specified for max falling delays at the port. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
driving_cell_pin_fall_min	
	Returns the driving cell output pin specified for min falling delays at the port. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
driving_cell_pin_rise_max	
	Returns the driving cell output pin specified for max rising delays at the port. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
driving_cell_pin_rise_min	
	Returns the driving cell output pin specified for min rising delays at the port. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
driving_cell_rise_max	
	Returns the name of the library cell used to compute max rising delays at the port. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
driving_cell_rise_min	

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	Returns the name of the library cell used to compute max rising delays at the port. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
escaped_name	
	The full hierarchical name including escaped chars (if any). It follows DEF escaping syntax, so bus-bit chars [], or a hierarchy char \ that is not a bus-bit or hierarchy char has a \ in front of it. So i1/i2 is a two-level hierarchical name while i1\i2 is single level name, and a[0] is a bus-bit, while a\0\ is a scalar. Type: string Default: "" Edit: No
exceptions	
	Reports the path exceptions Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
external_capacitance_max	
	Returns the total max external loading on the port from all sources - including set_load assertions or wireload models. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
external_capacitance_min	
	Returns the total min external loading on the port from all sources - including set_load assertions or wireload models. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
external_fanout_load	
	Returns the external fanout load specified by set_fanout_load. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
external_wireload_model	
	Returns the wireload model used to model external connections to the port. You can use -index to return the value for a specific view. Type: string Default: "" Edit: No
gpin_verilog_sim_type	
	To identify if the net is wand or wor type. Type: int Default: no_value Edit: Yes
hdl_name	

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	This is the original RTL name for this port. It is used to map RTL simulation results with RTL names to the current netlist for switching activity analysis. It is only maintained properly on the output ports of sequential cells. Optimization will copy this name during any multi-bit merge or splitting transforms to the equivalent port, but not on output ports of combinational cells. The is_phase_inverted attribute will be flipped if the phase is inverted. Type: string Default: "" Edit: Yes
hnet	
	hnet of this port Type: obj(hnet) Default: "" Edit: No
hold_uncertainty	
	Returns the most conservative uncertainty of all possible uncertainty Hold assertions associated with the port. You can use -index to return the value from a specific analysis view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
input_slew_max_fall	
	Returns the slowest falling transition on the port as specified by set_input_transition. You can use -index to return the value from a specific analysis view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
input_slew_max_rise	
	Returns the slowest rising transition on the port as specified by set_input_transition. You can use -index to return the value from a specific analysis view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
input_slew_min_fall	
	Returns the fastest falling transition on the port as specified by set_input_transition. You can use -index to return the value from a specific analysis view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
input_slew_min_rise	
	Returns the fastest rising transition on the port as specified by set_input_transition. You can use -index to return the value from a specific analysis view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
is_booth_encoded	
	Booth encoded setting assigned to input port by user. Type: bool Default: false Edit: Yes
is_clock_network_port	

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	Reports true if specified port is part of clock network Type: <code>string</code> Default: "" Edit: No
is_clock_source_latency	
	Returns true if object is part of latency network Type: <code>bool</code> Allowed -index values: analysis_view clock Default: "" Edit: No
is_clock_source_pin	
	Returns true if object is a clock source port Type: <code>bool</code> Allowed -index values: analysis_view clock Default: "" Edit: No
is_clock_used_as_clock	
	Returns a value of true if the port lies in the clock network and at least one of the clocks arriving at the pin is used as a clock in the downstream network of the pin. You can use -index to return the value for a specific analysis view. Type: <code>bool</code> Allowed -index values: analysis_view clock Default: "" Edit: No
is_clock_used_as_data	
	Returns a value of true if the port lies in the clock source path and at least one of the clocks arriving on the pin is used as data in the downstream network of the pin. You can use -index to return the value for a specific analysis view. Type: <code>bool</code> Allowed -index values: analysis_view clock Default: "" Edit: No
is_constant_in_all_views	
	Returns true if logic state has been asserted or propagated to this pin in all the views Type: <code>bool</code> Allowed -index values: analysis_view clock Default: "" Edit: No
is_constant_in_any_view	
	Returns true if logic state has been asserted or propagated to this pin in atleast one view. Type: <code>bool</code> Allowed -index values: analysis_view clock Default: "" Edit: No
is_disable_timing	
	Returns a value of true if the port's timing has been disabled. You can use -index to return the value for a specific analysis view. Type: <code>bool</code> Allowed -index values: analysis_view clock Default: "" Edit: No
is_ideal_driver	
	Returns true if the port is an ideal driver due to the set_ideal_latency or set_ideal_transition constraint. Type: <code>bool</code> Default: "" Edit: No
is_ideal_network	

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	Returns true if the port is part of an ideal network due to the set_ideal_network constraint. You can use -index to return the value for a specific analysis view. Type: bool Default: "" Edit: No
is_inside_partition	
	Returns whether port is inside partition or not. Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
is_internal_disable	
	Return is_internal_disable property for a pin. Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
is_owner	
	Returns true if client is the owner for pin in distributed STA. Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
is_power_mode_disabled	
	Returns true if power mode is disabled Type: bool Allowed -index values: analysis_view Default: "" Edit: No
is_propagated_clock	
	Returns a value of true if there is an explicit set_propagated_clock assertion at the port. You can use -index to return the value for a specific analysis view. Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
is_shared	
	Returns true if pin is shared across clients in distributed STA. Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
is_special	
	Indicates that the port is Special(not set for base_cell terms) Type: bool Default: false Edit: Yes
is_valid_for_reports	
	Return whether a pin is valid for reporting or not Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
layer	

Stylus Common UI Database Object Information

Database Objects--port

	Layer of the port. For ports with more than one shape, it is the layer of the first shape (which is the same shape used for the .location value). If there are no shapes for the port, {} is returned. Type: obj(layer) Default: "" Edit: No
location	
	Location of port. For port with only one shape, it is the center of the port shape edge abutting the design boundary. The edit_pin command can be used to set it. For ports that have more than one shape, the location will overlap the first shape of the port. Type: point Default: "" Edit: Yes
lp_asserted_probability	
	Probability of signal to be high on port/subport/pin as specified by user. Type: double Default: no_value Edit: Yes
lp_asserted_toggle_rate	
	Signal toggle rate assigned to pin by user. Type: double Default: no_value Edit: Yes
lp_system_asserted_probability	
	System/Internal asserted probability. Type: double Default: no_value Edit: Yes
lp_system_asserted_toggle_rate	
	System/Internal asserted toggle rate. Type: double Default: no_value Edit: Yes
lssd_master_clock	
	Defines the master clock port related to this slave clock port. Type: string Default: "" Edit: Yes
max_capacitance	
	max_capacitance Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
max_fanout	
	Returns the maximum fanout load that the port can drive. . You can use -index to return the value for a specific analysis view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
max_transition	

Stylus Common UI Database Object Information

Database Objects--port

	Returns the maximum transition time limit specified for the port. You can use -index to return the value for a specific analysis view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
min_capacitance	
	Returns the minimum capacitance limit for the port. You can use -index to return the value for a specific analysis view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
min_fanout	
	Returns the minimum fanout value for the port. Type: int Default: "" Edit: No
min_transition	
	Returns the minimum transition time limit specified for the port. Type: double Default: "" Edit: No
name	
	The same as .escaped_name without any \ escape chars. This name is commonly used to avoid problems with \ escape chars in Tcl scripts unless you carefully use list operators. Like SDC commands, 'get_db insts i1/i2' will first try to match i1/i2, and if not found then match i1\i2 so that flattening hierarchical names (e.g. with ungroup) does not require changing the names in Tcl scripts. Type: string Default: "" Edit: No
net	
	The canonical (flat) net connected to the terminal Type: obj(net) Default: "" Edit: No
network_latency_fall_max	
	Returns the maximum fall insertion delay specified by an explicit set_clock_latency at the port. You can use -index to return the value for a specific analysis view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
network_latency_fall_min	
	Returns the minimum fall insertion delay specified by an explicit set_clock_latency at the port. You can use -index to return the value for a specific analysis view and/or clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
network_latency_rise_max	
	Returns the maximum rise insertion delay specified by an explicit set_clock_latency at the port. You can use -index to return the value for a specific analysis view and/or clock Type: double Allowed -index values: analysis_view clock Default: "" Edit: No

Stylus Common UI Database Object Information

Database Objects--port

network_latency_rise_min	
	Returns the minimum rise insertion delay specified by an explicit set_clock_latency at the port. You can use -index to return the value for a specific analysis view and/or clock Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
num_exceptions	
	Reports the number of path exceptions Type: int Allowed -index values: analysis_view clock Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (port) Default: "" Edit: No
physical_pins	
	List of ports Type: obj (physical_pin)* Default: "" Edit: No
pin_edge	
	If this port is for a design, and the port is assigned (has place_status of placed/fixed/cover), the edge value indicates along which edge of the boundary polygon the port is assigned. The edge number starts from the lowest Y, then left-most X vertex, starting with 0, and then counting clock-wise. See the set_pin_constraint command document for a figure showing the edge numbering. If the port is not assigned, or not for a design, the value of -1 is returned. Type: int Default: "" Edit: No
place_status	
	Placement Status of terminal (not for base_cell terms) Type: enum Enum Values: unplaced placed fixed cover Default: unplaced Edit: Yes
power_domain	
	The power domain of the terminal (equivalent to Design Browser effPD) Type: obj (power_domain) Default: "" Edit: No
power_rail_voltage_max	
	power_rail_voltage_max Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
power_rail_voltage_min	
	power_rail_voltage_min Type: double Allowed -index values: analysis_view clock Default: "" Edit: No

Stylus Common UI Database Object Information

Database Objects--port

related_ground_pin	
	Specifies which ground pin drives this signal pin. It must be one of the ground pins defined for this cell. It can be set by CPF, Liberty, or LEF/OA, with CPF having highest precedence, then Liberty, then LEF/OA. It is often only set when there is more than one ground pin for the cell. Type: obj(port) Default: "" Edit: No
related_power_pin	
	Specifies which power pin drives this signal pin. It must be one of the power pins defined for this cell. It can be set by CPF, Liberty, or LEF/OA, with CPF having highest precedence, then Liberty, then LEF/OA. It is often only set when there is more than one power pin for the cell. Type: obj(port) Default: "" Edit: No
safety_dmr_error_signal_endpoint	
	This object is the endpoint for a safety-mechanism's error-signal. Type: string Default: "" Edit: No
setup_uncertainty	
	Returns the most conservative uncertainty of all possible uncertainty assertions associated with the port. You can use -index to return the value for a specific analysis view Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
shape	
	Terminal shape Type: enum Enum Values: none ring abutment feed_through Default: "" Edit: No
side	
	Side constraint of the port. The edit_pin command can be used to set this value. Type: enum Enum Values: north west south east up none Default: "" Edit: No
slack_max	
	Returns the worst slack across all concurrent MMMC views for Setup-style late data path checks. You can use -index to return the value for a specific view and/or clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slack_max_edge	
	Returns the edge (rise or fall) of the worst slack-causing path at the specified port in late mode. Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
slack_max_fall	

Stylus Common UI Database Object Information

Database Objects--port

	Returns the worst falling slack across all concurrent MMMC views for Setup-style late data path checks. You can use -index to return the value for a specific view and/or clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slack_max_rise	
	Returns the worst rising slack across all concurrent MMMC views for Setup-style late data path checks. You can use -index to return the value for a specific view and/or clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slack_mean_max	
	In SOCV analysis, this returns the mean component of the worst slack across all concurrent MMMC views for Setup-style late data path checks. You can use -index to return the value for a specific view and/or clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slack_mean_max_fall	
	In SOCV analysis, this returns the mean component of the worst falling slack across all concurrent MMMC views for Setup-style late data path checks. You can use -index to return the value for a specific view and/or clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slack_mean_max_rise	
	In SOCV analysis, this returns the mean component of the worst rising slack across all concurrent MMMC views for Setup-style late data path checks. You can use -index to return the value for a specific view and/or clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slack_mean_min	
	In SOCV analysis, this returns the mean component of the worst slack across all concurrent MMMC views for Hold-style early data path checks. You can use -index to return the value for a specific view and/or clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slack_mean_min_fall	
	In SOCV analysis, this returns the mean component of the worst falling slack across all concurrent MMMC views for Hold-style early data path checks. You can use -index to return the value for a specific view and/or clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slack_mean_min_rise	
	In SOCV analysis, this returns the mean component of the worst rising slack across all concurrent MMMC views for Hold-style early data path checks. You can use -index to return the value for a specific view and/or clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slack_min	

Stylus Common UI Database Object Information

Database Objects--port

	Returns the worst slack across all concurrent MMMC views for hold-style late data path checks Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slack_min_edge	
	Returns the edge (rise or fall) of the worst slack-causing path at the specified port in early mode. Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
slack_min_fall	
	Returns the worst hold slack for a falling signal at the port endpoint. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slack_min_rise	
	Returns the worst hold slack for a rising signal at the port endpoint. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slack_sigma_max	
	In SOCV analysis, this returns the variation component of the worst slack across all concurrent MMMC views for Setup-style late data path checks. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slack_sigma_max_fall	
	In SOCV analysis, this returns the variation component of the worst falling slack across all concurrent MMMC views for Setup-style late data path checks. You can use -index to return the value for a specific view and/or clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slack_sigma_max_rise	
	In SOCV analysis, this returns the variation component of the worst rising slack across all concurrent MMMC views for Setup-style late data path checks. You can use -index to return the value for a specific view and/or clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slack_sigma_min	
	In SOCV analysis, this returns the variation component of the worst slack across all concurrent MMMC views for Hold-style early data path checks. You can use -index to return the value for a specific view and/or clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slack_sigma_min_fall	

Stylus Common UI Database Object Information

Database Objects--port

	In SOCV analysis, this returns the variation component of the worst falling slack across all concurrent MMMC views for Hold-style early data path checks. You can use -index to return the value for a specific view and/or clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slack_sigma_min_rise	
	In SOCV analysis, this returns the variation component of the worst rising slack across all concurrent MMMC views for Hold-style early data path checks. You can use -index to return the value for a specific view and/or clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slew_max_fall	
	Returns the maximum slew time for falling transitions across all concurrent MMMC views. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slew_max_rise	
	Returns the maximum slew time for rising transitions across all concurrent MMMC views. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slew_mean_max_fall	
	In SOCV mode, returns the mean component of the maximum slew time for falling transitions across all concurrent MMMC views. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slew_mean_max_rise	
	In SOCV mode, returns the mean component of the maximum slew time for rising transitions across all concurrent MMMC views. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slew_mean_min_fall	
	In SOCV mode, returns the mean component of the minimum slew time for falling transitions across all concurrent MMMC views. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slew_mean_min_rise	
	In SOCV mode, returns the mean component of the minimum slew time for rising transitions across all concurrent MMMC views. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slew_min_fall	

Stylus Common UI Database Object Information

Database Objects--port

	Returns the minimum slew time for falling transitions across all concurrent MMMC views. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slew_min_rise	
	Returns the minimum slew time for rising transitions across all concurrent MMMC views. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slew_sigma_max_fall	
	In SOCV mode, returns the variation component of the maximum slew time for falling transitions across all concurrent MMMC views. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slew_sigma_max_rise	
	In SOCV mode, returns the variation component of the maximum slew time for rising transitions across all concurrent MMMC views. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slew_sigma_min_fall	
	In SOCV mode, returns the variation component of the minimum slew time for falling transitions across all concurrent MMMC views. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slew_sigma_min_rise	
	In SOCV mode, returns the variation component of the minimum slew time for rising transitions across all concurrent MMMC views. You can use -index to return the value for a specific view. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
source_latency_early_fall_max	
	Returns the maximum early fall source insertion delay specified by an explicit set_clock_latency at the port. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
source_latency_early_fall_min	
	Returns the minimum early fall source insertion delay specified by an explicit set_clock_latency at the port. You can use -index to return the value for a specific view and/or clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
source_latency_early_rise_max	

Stylus Common UI Database Object Information

Database Objects--port

	Returns the maximum early rise source insertion delay specified by an explicit set_clock_latency at the port. You can use -index to return the value for a specific view and/or clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
source_latency_early_rise_min	
	Returns the minimum early rise source insertion delay specified by an explicit set_clock_latency at the port. You can use -index to return the value for a specific view and/or clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
source_latency_late_fall_max	
	Returns the maximum late fall source insertion delay specified by an explicit set_clock_latency at the port. You can use -index to return the value for a specific view and/or clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
source_latency_late_fall_min	
	Returns the minimum late fall source insertion delay specified by an explicit set_clock_latency at the port. You can use -index to return the value for a specific view and/or clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
source_latency_late_rise_max	
	Returns the maximum late rise source insertion delay specified by an explicit set_clock_latency at the port. You can use -index to return the value for a specific view and/or clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
source_latency_late_rise_min	
	Returns the minimum late rise source insertion delay specified by an explicit set_clock_latency at the port. You can use -index to return the value for a specific view and/or clock. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
timing_case_computed_value	
	Returns the case computed value Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
timing_case_logic_value	
	Returns the case logic value Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
timing_constant	

Stylus Common UI Database Object Information

Database Objects--port

	Reports constant value on port Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
timing_constant_source	
	Reports source of constant value on port Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
use	
	Indicates how this port is used from the DEF PIN +USE value or OA equivalent. The legal values are: analog clock ground power signal. Note that timing analysis does not use these values, it uses the .lib data instead (e.g. is_clock, is_analog, etc.). So the signal, clock or analog values are not normally used by applications. The power/ground values are used by many applications. DEF PIN +USE RESET, SCAN or TIEOFF are not supported, so read_def will ignore them, and keep the existing DB use value. Type: enum Enum Values: signal clock power ground analog Default: "" Edit: Yes
width	
	Width constraint of the port in microns. The edit_pin command can be used to set it. Type: coord Default: "" Edit: No
wire_lumped_capacitance_max	
	Returns the portion of the net's total lumped capacitance which comes extracted or annotated wire lumped capacitance values. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
wire_lumped_capacitance_max_fall	
	Returns the portion of the net's total lumped capacitance which comes extracted or annotated wire lumped capacitance values. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
wire_lumped_capacitance_max_rise	
	Returns the portion of the net's total lumped capacitance which comes extracted or annotated wire lumped capacitance values. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
wire_lumped_capacitance_min	
	Returns the portion of the net's total lumped capacitance which comes extracted or annotated wire lumped capacitance values. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
wire_lumped_capacitance_min_fall	
	Returns the portion of the net's total lumped capacitance which comes extracted or annotated wire lumped capacitance values. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No

wire_lumped_capacitance_min_rise	
	Returns the portion of the net's total lumped capacitance which comes extracted or annotated wire lumped capacitance values. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
wrapper_control	
	Specifies as wrapper control signal. Type: bool Default: false Edit: Yes
wrapper_segment	
	Specifies the wrapper segments associated with the pin. Type: obj(scan_chain)* obj(scan_segment)* obj(actual_scan_segment)* obj(actual_scan_chain)* Default: "" Edit: Yes

port_shape

Parent Objects

[design](#), [net](#), [root](#)

Definition

This corresponds to one of a port's .physical_pins.layer_shapes.shapes, but is directly accessible from the design and root objects. It carries a link to the port object, that a layer_shape object does not have. This allows GUI operations that need both the layer_shape and the port object together as one object like delete_obj or wire-editing commands.

Attribute	Description
layer	
	Layer of this port shape Type: obj(layer) Default: "" Edit: Yes
mask	
	The mask number for this port_shape if this layer has multiple masks. 0 means it is uncolored. Refer to layer .num_masks for legal range. Type: int Default: no_value Edit: Yes
name	
	Name of the port for this port_shape. Note that all the port_shapes for one port will have the same name. Type: string Default: "" Edit: No
net	
	The net this port shape connects to. Type: obj(net) Default: "" Edit: No
obj_type	

Stylus Common UI Database Object Information

Database Objects--power_domain

	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (port_shape) Default: "" Edit: No
polygon	
	Points of the polygon of the port_shape for type rect or polygon. Type: point * Default: "" Edit: Yes
port	
	The port this port shape belongs to. Type: obj (port) Default: "" Edit: No
rect	
	Bounding box of the port_shape for type rect or polygon. Type: rect Default: "" Edit: Yes
type	
	Type of port_shape (rect polygon). Note if a polygon was used to create the shape, but the specified polygon actually is rectangle, the type will be rect. Type: enum Enum Values: rect polygon Default: "" Edit: No

power_domain

Parent Objects

[hinst](#), [inst](#), [design](#), [pin](#), [root](#), [group](#), [port](#)

Definition

Power domain

Attribute	Description
available_supply_nets	
	Specifies the power nets physically available for this power domain to use for secondary power pin connections. Type: obj (net)* Default: "" Edit: No
base_domains	
	Specifies the base power domains (always-on domains) that supply the power to this switchable power domain through power switch cells. Type: obj (power_domain)* Default: "" Edit: No
core_to_bottom	
	Distance between the power domain edge and its core box Type: coord Default: "" Edit: No

Stylus Common UI Database Object Information
Database Objects--power_domain

core_to_left	
	Distance between the power domain edge and its core box Type: coord Default: "" Edit: No
core_to_right	
	Distance between the power domain edge and its core box Type: coord Default: "" Edit: No
core_to_top	
	Distance between the power domain edge and its core box Type: coord Default: "" Edit: No
default_tech_site	
	default tech. site of the domain Type: string Default: "" Edit: No
disjoint_hinst_box_list	
	List of hierarchical instances and disjoint boxes defining power domain boundary. It is defined as an hinst and region pair. The listed hinst will be placed inside this domain Type: string Default: "" Edit: No
extend_power_bottom	
	Maximum search distance for power connections Type: coord Default: "" Edit: No
extend_power_edges	
	List of maximum search distances for power extension connections in clockwise order starting with the vertical edge at the lower-left corner (smallest Y, then smallest X) Type: coord* Default: "" Edit: No
extend_power_left	
	Maximum search distance for power connections Type: coord Default: "" Edit: No
extend_power_right	
	Maximum search distance for power connections Type: coord Default: "" Edit: No
extend_power_top	
	Maximum search distance for power connections Type: coord Default: "" Edit: No

Stylus Common UI Database Object Information
Database Objects--power_domain

first_row_site_index	
	Specifies the first row site index value. Type: <code>int</code> Default: <code>""</code> Edit: No
gap_bottom	
	Minimum spacing to other power domains or rows Type: <code>coord</code> Default: <code>""</code> Edit: No
gap_edges	
	List of minimum spacing values to other power domains or rows in clockwise order starting with the vertical edge at the lower-left corner (smallest Y, then smallest X) Type: <code>coord*</code> Default: <code>""</code> Edit: No
gap_left	
	Minimum spacing to other power domains or rows Type: <code>coord</code> Default: <code>""</code> Edit: No
gap_right	
	Minimum spacing to other power domains or rows Type: <code>coord</code> Default: <code>""</code> Edit: No
gap_top	
	Minimum spacing to other power domains or rows Type: <code>coord</code> Default: <code>""</code> Edit: No
group	
	The group of PD Type: <code>obj(group)</code> Default: <code>""</code> Edit: No
is_always_on	
	Indicates that the power domain is always on Type: <code>bool</code> Default: <code>""</code> Edit: No
is_default	
	The power domain is the default power domain or not. Type: <code>bool</code> Default: <code>""</code> Edit: No
is_internal	
	Indicates that the domain is created by tool for the supply set which is not the primary set of any domains defined in IEEE1801 file. It is created for supply set PVT specification and leaf instance pin's domain assignment. Type: <code>bool</code> Default: <code>false</code> Edit: No

Stylus Common UI Database Object Information
Database Objects--power_domain

is_logical	
	Indicates that the domain might have a physical constraint Type: bool Default: false Edit: No
is_macro_only	
	Indicates whether the domain member has only hardmacro members. Type: bool Default: "" Edit: No
is_physical	
	Indicates that the domain might have a physical constraint Type: bool Default: false Edit: No
is_virtual	
	Specifies the power domain is a virtual domain, meaning that this domain does not have any inst member. Type: bool Default: false Edit: No
isolation_rules	
	Get the domain's isolation_rules Type: obj(isolation_rule)* Default: "" Edit: No
last_row_site_index	
	Specifies the last row site index value. Type: int Default: "" Edit: No
level_shifter_rules	
	Get the domain's level_shifter_rules Type: obj(level_shifter_rule)* Default: "" Edit: No
logical_power_domains	
	Indicates the domain's associated logical domains. Type: string* Default: "" Edit: No
name	
	Name of the power domain Type: string Default: "" Edit: No
nwell_supply_net	
	Specifies the nwell bias net for this power_domain to use for nwell pin connections. Type: obj(net) Default: "" Edit: No
obj_type	

Stylus Common UI Database Object Information

Database Objects--power_domain

	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (power_domain) Default: "" Edit: No
physical_power_domain	
	Indicates the domain's physical domain. Type: string Default: "" Edit: No
power_switch_rule_name	
	Indicates the domain's power switch rule name. Type: string Default: "" Edit: No
power_switch_rules	
	Get the domain's power_switch_rules Type: obj(power_switch_rule)* Default: "" Edit: No
primary_ground_net	
	Specifies the main ground rail net of the power domain. Type: obj(net) Default: "" Edit: No
primary_power_net	
	Specifies the main power rail net of the power domain. Type: obj(net) Default: "" Edit: No
pwell_supply_net	
	Specifies the pwell bias net for this power_domain to use for pwell pin connections. Type: obj(net) Default: "" Edit: No
retention_rules	
	Get the domain's retention_rules Type: obj(retention_rule)* Default: "" Edit: No
row_flip	
	Flips the orientation of either the first or second row, then follows the flipped and abutted row pattern. Refer to "update_power_domain -row_flip" for more details. Type: enum Enum Values: noflip second first auto Default: auto Edit: Yes
row_pattern_site	
	Specifies the site name of the row pattern for the power doamin. For non-default power domain, the default value is "". For default power domain, the default value is the default row pattern site set by floorplan setting or "" if floorplan setting is not set. Type: obj(site) Default: "" Edit: Yes
row_space_type	

Stylus Common UI Database Object Information

Database Objects--preferred_cell_stripe

	Determines whether the row spacing value applies between no row (0), each row (1), or each pair of rows (2). Type: enum Enum Values: 0 1 2 Default: 2 Edit: Yes
row_spacing	
	Specifies the row spacing between each row (1) or each pair of rows (2) as specified in row_space_type. Type: double Default: 0 Edit: Yes
safety_ser_cells	
	The list of base_cells allowed for those insts inside this power_domain with an ser (soft error resilience) safety mechanism. Type: obj(base_cell)* Default: "" Edit: No
safety_tmr_well_tap_cells	
	The well-tap cells to be placed next to the left and right edges of tmr (triple modular redundancy) flops within this power_domain. Type: obj(base_cell)* Default: "" Edit: No
shutoff_condition	
	Indicates the power domain's shutoff condition. It is a logic expression with pin or hpin names. Type: string Default: "" Edit: No

preferred_cell_stripe

Parent Objects

Definition

preferred cell stripe

Attribute	Description
name	
	name of preferred_cell_stripe Type: string Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (preferred_cell_stripe) Default: "" Edit: No

rc_corner

Parent Objects

[delay_corner](#), [root](#)

Stylus Common UI Database Object Information

Database Objects--preferred_cell_stripe

Definition

The rc_corner object represents a specific interconnect parasitic corner. It contains configuration information for controlling the extraction of the parasitics and possible scaling. Parasitic files are annotated or exported with respect to an rc_corner object name. The rc_corner object is typically referenced from one or many higher-level delay_corner objects. Use the create_rc_corner command to define new rc_corners and the update_rc_corner command to update the attributes of existing rc_corners.

Attribute	Description
cap_table_file	
	Specifies the layer capacitance table file associated with the rc_corner. This is not recommended for designs below 32nm. See the write_cap_table command for more details on the file. Type: string Default: "" Edit: No
is_active	
	Indicates that the rc_corner is associated with an analysis_view that is used in the active analysis_view Type: bool Default: "" Edit: No
is_dynamic	
	Indicates that the rc_corner is associated with an analysis_view that is used in the active dynamic analysis_view Type: bool Default: "" Edit: No
is_hold	
	Indicates that the rc_corner is associated with an analysis_view that is used in the active hold analysis_view Type: bool Default: "" Edit: No
is_leakage	
	Indicates that the rc_corner is associated with an analysis_view that is used in the active leakage analysis_view Type: bool Default: "" Edit: No
is_setup	
	Indicates that the rc_corner is associated with an analysis_view that is used in the active setup analysis_view Type: bool Default: "" Edit: No
name	
	Provides the name of this rc_corner object as specified by create_rc_corner. Type: string Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (rc_corner) Default: "" Edit: No
post_route_cap	

Stylus Common UI Database Object Information

Database Objects--preferred_cell_stripe

	Specifies scaling factor(s) to be used for capacitances in post_route flow steps. 1 to 3 values may be specified to control each extraction effort level. Example: {lowEffortFactor}, {lowEffortFactor mediumEffortFactor}, {lowEffortFactor mediumEffortFactor highEffortFactor}. A scaling value of 1.0 is assumed for any effort level without an explicit setting. Type: double* Default: "" Edit: No
post_route_clock_cap	
	Specifies scaling factor(s) to be used for clock network capacitances in post_route flow steps. 1 to 3 values may be specified to control each extraction effort level. Example: {lowEffortFactor}, {lowEffortFactor mediumEffortFactor}, {lowEffortFactor mediumEffortFactor highEffortFactor}. A scaling value of 1.0 is assumed for any effort level without an explicit setting. Type: double* Default: "" Edit: No
post_route_clock_cross_cap	
	Specifies scaling factor(s) to be used for clock network coupling capacitances in post_route flow steps. 1 to 3 values may be specified to control each extraction effort level. Example: {lowEffortFactor}, {lowEffortFactor mediumEffortFactor}, {lowEffortFactor mediumEffortFactor highEffortFactor}. A scaling value of 1.0 is assumed for any effort level without an explicit setting. Type: double* Default: "" Edit: No
post_route_clock_res	
	Specifies scaling factor(s) to be used for clock network resistances in post_route flow steps. 1 to 3 values may be specified to control each extraction effort level. Example: {lowEffortFactor}, {lowEffortFactor mediumEffortFactor}, {lowEffortFactor mediumEffortFactor highEffortFactor}. A scaling value of 1.0 is assumed for any effort level without an explicit setting. Type: double* Default: "" Edit: No
post_route_cross_cap	
	Specifies scaling factor(s) to be used for coupling-capacitances in post_route flow steps. 1 to 3 values may be specified to control each extraction effort level. Example: {lowEffortFactor}, {lowEffortFactor mediumEffortFactor}, {lowEffortFactor mediumEffortFactor highEffortFactor}. A scaling value of 1.0 is assumed for any effort level without an explicit setting. Type: double* Default: "" Edit: No
post_route_res	
	Specifies scaling factor(s) to be used for resistances in post_route flow steps. 1 to 3 values may be specified to control each extraction effort level. Example: {lowEffortFactor}, {lowEffortFactor mediumEffortFactor}, {lowEffortFactor mediumEffortFactor highEffortFactor}. A scaling value of 1.0 is assumed for any effort level without an explicit setting. Type: double* Default: "" Edit: No
pre_route_cap	
	Specifies a scaling factor to be used for capacitances in pre_route flow steps. Defaults to 1.0 if not given. Type: double Default: "" Edit: No
pre_route_clock_cap	
	Specifies a scaling factor to be used for clock network capacitances in pre_route flow steps. Defaults to 1.0 if not given. Type: double Default: "" Edit: No
pre_route_clock_res	

Stylus Common UI Database Object Information

Database Objects--resistor

	Specifies a scaling factor to be used for clock network resistance in pre_route flow steps. Defaults to 1.0 if not given. Type: double Default: "" Edit: No
pre_route_res	
	Specifies a scaling factor to be used for resistances in pre_route flow steps. Defaults to 1.0 if not given. Type: double Default: "" Edit: No
qrc_tech_file	
	Specifies the QRC technology file associated with the rc_corner Type: string Default: "" Edit: No
temperature	
	Specifies the temperature, in units of Celsius, to use to derate resistance values for this rc_corner. Use this parameter when you want to override the default temperature specified in the capacitance table or the QRC technology file. By default, the RC extractor uses the temperature of 25 degrees Celsius, unless it is specified by the capacitance table or the QRC technology file. Note, the opcond temperature is not used for RC extraction, the temperature for RC extraction must be set with this attribute if you want to override the techfile or captable settings. Type: double Default: "" Edit: No
thru_substrate_cell_subckt_file	
	Specifies sub circuit RC for FTV cells. Type: string Default: "" Edit: No
via_variation_file	
	Specifies the via layer file so that all the VIA resistance in SPEF could be mapped to have own variation multiplier. Type: string Default: "" Edit: No

resistor

Parent Objects

Definition

After loading a Voltus IR-drop analysis result, you can select a resistor in the GUI. The pointer of the object is the same as its name, like resistor

Attribute	Description
capacitance	
	The capacitance for the two nodes before any effects of set_rail_what_if_capacitance in units of farads. Type: double* Default: "" Edit: No
current	
	The current of the resistor in units of A. For static rail_analysis it is the average value, for dynamic power analysis it is the rms value. Type: double Default: "" Edit: No

Stylus Common UI Database Object Information
Database Objects--resistor

current_direction	
	The direction of current. no_value is returned for non-orthogonal resistors or if it is not available. Type: enum Enum Values: no_value left right up down Default: "" Edit: No
em_jmax_dc_avg_ratio	
	The ratio of the current / em_jmax_dc_avg limit in the Quantus ICT file. This is computed during static rail analysis. The jmax name is used for legacy reasons, but the value is a current value ratio, not a current density ratio. Type: double Default: "" Edit: No
em_jmax_dc_rms_ratio	
	The ratio of the current / em_jmax_dc_rms limit in the Quantus ICT file. This is computed during dynamic rail analysis. The jmax name is used for legacy reasons, but the value is a current value ratio, not a current density ratio. Type: double Default: "" Edit: No
failure_in_time	
	The failure in time value of the resistor in units of 1e9 hours (e.g. 1.0 means one failure in 1e9 hours of operation). Type: double Default: "" Edit: No
layer	
	The layer of the resistor. A via resistor will have a via-layer, and a routing layer resistor will have a routing layer. Type: string Default: "" Edit: No
length	
	The length of the resistor in unit of um. Type: coord Default: "" Edit: No
location	
	The x, y location of the two nodes at each end of the resistor. Type: point* Default: "" Edit: No
name	
	This is an R followed by an integer to identify the resistor. Type: string Default: "" Edit: No
net	
	The net of the resistor. Type: obj(net) Default: "" Edit: No
node_names	
	This is an N followed by an integer to identify the two nodes at each end of the resistor. Type: string Default: "" Edit: No

Stylus Common UI Database Object Information

Database Objects--resize_blockage

node_reff	
	The values are in units of ohms and are valid whenever the node reff db is available in the loaded state directory. Type: double* Default: "" Edit: No
node_voltage_drop	
	The IR voltage drop of the two nodes in units of mV. Type: double* Default: "" Edit: No
node_voltages	
	The IR voltage drop of the two nodes in units of mV. Type: double* Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (resistor) Default: "" Edit: No
resistance	
	The resistance before any effects from set_rail_what_if_resistance in units of ohms. Type: double Default: "" Edit: No
switch_net	
	The switched net of the resistor. Type: obj (net) Default: "" Edit: No
what_if_capacitance	
	The capacitance after any effects of set_rail_what_if_capacitance. This is the capacitance actually used in the rail analysis in units of farads. Type: double* Default: "" Edit: No
what_if_resistance	
	The resistance after any effects of set_rail_what_if_resistance. This is the resistance actually used in the rail analysis in units of ohms. Type: double Default: "" Edit: No
width	
	The width of the resistor in unit of um. Type: coord Default: "" Edit: No

resize_blockage

Parent Objects

[design](#), [root](#)

Stylus Common UI Database Object Information

Database Objects--root

Definition

resize blkg

Attribute	Description
is_resizeable	
	Specifies that the size blockage can be resized, however alignment and the minimum space between the objects in the blockage area will be maintained during floorplan resize.(1 = resizable). Type: bool Default: false Edit: Yes
name	
	Name of the size blockage Type: string Default: "" Edit: Yes
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (resize_blockage) Default: "" Edit: No
rect	
	Rectangle that defines the resize_blockage shape. Type: rect Default: "" Edit: No

root

Parent Objects

Definition

root

Attribute	Description
-enable_sens_based_scaling	
	Type: bool Edit: Yes
-ptr_effort	
	Type: double Edit: Yes
actual_scan_chains	
	List of all actual_scan_chains Type: obj(actual_scan_chain)* Default: "" Edit: No
actual_scan_segments	

Stylus Common UI Database Object Information
Database Objects--root

	List of all actual_scan_segments Type: obj(actual_scan_segment)* Default: "" Edit: No
add_endcaps_antenna_tap_fill_wall_cells	
	N2 spr flow fillwall cells Type: string Default: "" Edit: Yes Reference: add_endcaps_antenna_tap_fill_wall_cells
add_endcaps_avoid_two_sites_cell_abut	
	avoid 2 sites boundary cell abut to incorner cell in N10 Type: bool Default: false Edit: Yes Reference: add_endcaps_avoid_two_sites_cell_abut
add_endcaps_barrier_border_cell_bottom_y	
	horizontal bottom border break cell Type: string Default: "" Edit: Yes Reference: add_endcaps_barrier_border_cell_bottom_y
add_endcaps_barrier_border_cell_top_y	
	horizontal top border break cell Type: string Default: "" Edit: Yes Reference: add_endcaps_barrier_border_cell_top_y
add_endcaps_barrier_cell_x	
	vertical break cell Type: string Default: "" Edit: Yes Reference: add_endcaps_barrier_cell_x
add_endcaps_barrier_cell_xy	
	horizontal vertical overlap break cell Type: string Default: "" Edit: Yes Reference: add_endcaps_barrier_cell_xy
add_endcaps_barrier_cell_y	
	horizontal break cell Type: string Default: "" Edit: Yes Reference: add_endcaps_barrier_cell_y
add_endcaps_barrier_keepout_from_boundary	
	break cell x and y Keepout distance from boundary value Type: string Default: "" Edit: Yes
add_endcaps_barrier_pitch	

Stylus Common UI Database Object Information
Database Objects--root

	break cell x and y pitch value Type: string Default: "" Edit: Yes Reference: add_endcaps_barrier_pitch
add_endcaps_barrier_termination_bottom_y	
	horizontal termination bottom break cell Type: string Default: "" Edit: Yes Reference: add_endcaps_barrier_termination_bottom_y
add_endcaps_barrier_termination_top_y	
	horizontal termination top break cell Type: string Default: "" Edit: Yes Reference: add_endcaps_barrier_termination_top_y
add_endcaps_barrier_termination_x	
	vertical termination break cell Type: string Default: "" Edit: Yes Reference: add_endcaps_barrier_termination_x
add_endcaps_barrier_y_mode	
	use mode 1 or 2 for barrier y cell insertion Type: enum Enum Values: 1 2 Default: 2 Edit: Yes Reference: add_endcaps_barrier_y_mode
add_endcaps_bottom_edge	
	EndCaps with n-well at the bottom edge Type: string Default: "" Edit: Yes Reference: add_endcaps_bottom_edge
add_endcaps_boundary_tap	
	New flow to add cap cell and tap cell together Type: bool Default: false Edit: Yes Reference: add_endcaps_boundary_tap
add_endcaps_boundary_tap_swap_flow	
	boundary tap swap flow Type: bool Default: false Edit: Yes Reference: add_endcaps_boundary_tap_swap_flow
add_endcaps_cells	
	Specify endCaps cell candidates for N10 that has defined in LEF Type: string Default: "" Edit: Yes Reference: add_endcaps_cells

Stylus Common UI Database Object Information
Database Objects--root

add_endcaps_create_rows	
	Create rows for endtap cell techSites which do not have rows in the floor plan Type: <code>bool</code> Default: <code>false</code> Edit: Yes Reference: add_endcaps_create_rows
add_endcaps_double_height_incorner_protrusion	
	allow double height in-corner cell to protrude into core area Type: <code>bool</code> Default: <code>false</code> Edit: Yes Reference: add_endcaps_double_height_incorner_protrusion
add_endcaps_enable_shrink_physical_cell_flow	
	Samsung shrink left/right edge cell flow Type: <code>bool</code> Default: <code>false</code> Edit: Yes Reference: add_endcaps_enable_shrink_physical_cell_flow
add_endcaps_flip_y	
	Flips the orientation of the endcap instances in Y direction if the site symmetry allows it. Type: <code>bool</code> Default: <code>false</code> Edit: Yes Reference: add_endcaps_flip_y
add_endcaps_incremental_left_edge	
	EndCap lists with n-well at the left edge Type: <code>string</code> Default: <code>""</code> Edit: Yes Reference: add_endcaps_incremental_left_edge
add_endcaps_incremental_right_edge	
	EndCap list with n-well at the right edge Type: <code>string</code> Default: <code>""</code> Edit: Yes Reference: add_endcaps_incremental_right_edge
add_endcaps_insert_nppp_wall	
	cells will be used for np wall Type: <code>string</code> Default: <code>""</code> Edit: Yes Reference: add_endcaps_insert_nppp_wall
add_endcaps_left_bottom_corner	
	EndCap with n-well at the left bottom corner Type: <code>string</code> Default: <code>""</code> Edit: Yes Reference: add_endcaps_left_bottom_corner
add_endcaps_left_bottom_corner_even	
	EndCap with n-well at the left bottom corner, even sites Type: <code>string</code> Default: <code>""</code> Edit: Yes Reference: add_endcaps_left_bottom_corner_even

Stylus Common UI Database Object Information
Database Objects--root

add_endcaps_left_bottom_corner_neighbor	
	The cell to be abutted with the left side of leftBottomCorner in R0 orientation Type: string Default: "" Edit: Yes Reference: add_endcaps_left_bottom_corner_neighbor
add_endcaps_left_bottom_corner_odd	
	EndCap with n-well at the left bottom corner, odd sites Type: string Default: "" Edit: Yes Reference: add_endcaps_left_bottom_corner_odd
add_endcaps_left_bottom_edge	
	EndCap with n-well at the left bottom edge Type: string Default: "" Edit: Yes Reference: add_endcaps_left_bottom_edge
add_endcaps_left_bottom_edge_even	
	EndCap with n-well at the left bottom edge, even sites Type: string Default: "" Edit: Yes Reference: add_endcaps_left_bottom_edge_even
add_endcaps_left_bottom_edge_neighbor	
	The cell to be abutted with the right edge of leftBottomEdge in R0 orientation Type: string Default: "" Edit: Yes Reference: add_endcaps_left_bottom_edge_neighbor
add_endcaps_left_bottom_edge_odd	
	EndCap with n-well at the left bottom edge, odd sites Type: string Default: "" Edit: Yes Reference: add_endcaps_left_bottom_edge_odd
add_endcaps_left_corner_bottom_border	
	inner corner's border cell Type: string Default: "" Edit: Yes Reference: add_endcaps_left_corner_bottom_border
add_endcaps_left_corner_top_border	
	inner corner's border cell Type: string Default: "" Edit: Yes Reference: add_endcaps_left_corner_top_border
add_endcaps_left_edge	
	EndCap with n-well at the left edge Type: string Default: "" Edit: Yes Reference: add_endcaps_left_edge

Stylus Common UI Database Object Information
Database Objects--root

add_endcaps_left_edge_bottom_border	
	EndCaps with n-well at the left edge, poly at the bottom boundary Type: string Default: "" Edit: Yes Reference: add_endcaps_left_edge_bottom_border
add_endcaps_left_edge_even	
	EndCap with n-well at the left edge, even sites Type: string Default: "" Edit: Yes Reference: add_endcaps_left_edge_even
add_endcaps_left_edge_odd	
	EndCap with n-well at the left edge, odd sites Type: string Default: "" Edit: Yes Reference: add_endcaps_left_edge_odd
add_endcaps_left_edge_top_border	
	EndCaps with n-well at the left edge, poly at the top boundary Type: string Default: "" Edit: Yes Reference: add_endcaps_left_edge_top_border
add_endcaps_left_top_corner	
	EndCap with n-well at the left top corner Type: string Default: "" Edit: Yes Reference: add_endcaps_left_top_corner
add_endcaps_left_top_corner_even	
	EndCap with n-well at the left top corner, even sites Type: string Default: "" Edit: Yes Reference: add_endcaps_left_top_corner_even
add_endcaps_left_top_corner_neighbor	
	The cell to be abutted with the left side of leftTopCorner in R0 orientation Type: string Default: "" Edit: Yes Reference: add_endcaps_left_top_corner_neighbor
add_endcaps_left_top_corner_odd	
	EndCap with n-well at the left top corner, odd sites Type: string Default: "" Edit: Yes Reference: add_endcaps_left_top_corner_odd
add_endcaps_left_top_edge	
	EndCap with n-well at the left top edge Type: string Default: "" Edit: Yes Reference: add_endcaps_left_top_edge

Stylus Common UI Database Object Information

Database Objects--root

add_endcaps_left_top_edge_even	
	EndCap with n-well at the left top edge, even sites Type: string Default: "" Edit: Yes Reference: add_endcaps_left_top_edge_even
add_endcaps_left_top_edge_neighbor	
	The cell to be abutted with the right edge of leftTopEdge in R0 orientation Type: string Default: "" Edit: Yes Reference: add_endcaps_left_top_edge_neighbor
add_endcaps_left_top_edge_odd	
	EndCap with n-well at the left top edge, odd sites Type: string Default: "" Edit: Yes Reference: add_endcaps_left_top_edge_odd
add_endcaps_min_horizontal_channel_width	
	Type: int Default: 0 Edit: Yes Reference: add_endcaps_min_horizontal_channel_width
add_endcaps_min_jog_height	
	Type: int Default: 0 Edit: Yes Reference: add_endcaps_min_jog_height
add_endcaps_min_jog_width	
	Type: int Default: 0 Edit: Yes Reference: add_endcaps_min_jog_width
add_endcaps_min_vertical_channel_width	
	Type: int Default: 0 Edit: Yes Reference: add_endcaps_min_vertical_channel_width
add_endcaps_nppp_wall_offset	
	Type: double Default: 0.0 Edit: Yes
add_endcaps_nppp_wall_pitch	
	Type: double Default: 0.0 Edit: Yes Reference: add_endcaps_nppp_wall_pitch

Stylus Common UI Database Object Information
Database Objects--root

add_endcaps_nppp_wall_shift_step	
	Type: double Default: 0.0 Edit: Yes
add_endcaps_nppp_wall_to_fillwall_spacing	
	Type: double Default: 0.0 Edit: Yes Reference: add_endcaps_nppp_wall_to_fillwall_spacing
add_endcaps_nppp_wall_to_incorner_spacing	
	Type: double Default: 0.0 Edit: Yes Reference: add_endcaps_nppp_wall_to_incorner_spacing
add_endcaps_prefix	
	Prefix of EndCap cells Type: string Default: "" Edit: Yes Reference: add_endcaps_prefix
add_endcaps_right_bottom_corner	
	EndCap with n-well at the right bottom corner Type: string Default: "" Edit: Yes Reference: add_endcaps_right_bottom_corner
add_endcaps_right_bottom_corner_even	
	EndCap with n-well at the right bottom corner, even sites Type: string Default: "" Edit: Yes Reference: add_endcaps_right_bottom_corner_even
add_endcaps_right_bottom_corner_neighbor	
	The cell to be abutted with the right side of rightBottomCorner in R0 orientation Type: string Default: "" Edit: Yes Reference: add_endcaps_right_bottom_corner_neighbor
add_endcaps_right_bottom_corner_odd	
	EndCap with n-well at the right bottom corner, odd sites Type: string Default: "" Edit: Yes Reference: add_endcaps_right_bottom_corner_odd
add_endcaps_right_bottom_edge	
	EndCap with n-well at the right bottom edge Type: string Default: "" Edit: Yes Reference: add_endcaps_right_bottom_edge

Stylus Common UI Database Object Information

Database Objects--root

add_endcaps_right_bottom_edge_even	
	EndCap with n-well at the right bottom edge, even sites Type: string Default: "" Edit: Yes Reference: add_endcaps_right_bottom_edge_even
add_endcaps_right_bottom_edge_neighbor	
	The cell to be abutted with the left edge of rightBottomEdge in R0 orientation Type: string Default: "" Edit: Yes Reference: add_endcaps_right_bottom_edge_neighbor
add_endcaps_right_bottom_edge_odd	
	EndCap with n-well at the right bottom edge, odd sites Type: string Default: "" Edit: Yes Reference: add_endcaps_right_bottom_edge_odd
add_endcaps_right_corner_bottom_border	
	inner corner's border cell Type: string Default: "" Edit: Yes Reference: add_endcaps_right_corner_bottom_border
add_endcaps_right_corner_top_border	
	inner corner's border cell Type: string Default: "" Edit: Yes Reference: add_endcaps_right_corner_top_border
add_endcaps_right_edge	
	EndCap with n-well at the right edge Type: string Default: "" Edit: Yes Reference: add_endcaps_right_edge
add_endcaps_right_edge_bottom_border	
	EndCaps with n-well at the right edge, poly at the bottom boundary Type: string Default: "" Edit: Yes Reference: add_endcaps_right_edge_bottom_border
add_endcaps_right_edge_even	
	EndCap with n-well at the right edge, even sites Type: string Default: "" Edit: Yes Reference: add_endcaps_right_edge_even
add_endcaps_right_edge_odd	
	EndCap with n-well at the right edge, odd sites Type: string Default: "" Edit: Yes Reference: add_endcaps_right_edge_odd

Stylus Common UI Database Object Information
Database Objects--root

add_endcaps_right_edge_top_border	
	EndCaps with n-well at the right edge, poly at the top boundary Type: string Default: "" Edit: Yes Reference: add_endcaps_right_edge_top_border
add_endcaps_right_top_corner	
	EndCap with n-well at the right top corner Type: string Default: "" Edit: Yes Reference: add_endcaps_right_top_corner
add_endcaps_right_top_corner_even	
	EndCap with n-well at the right top corner, even sites Type: string Default: "" Edit: Yes Reference: add_endcaps_right_top_corner_even
add_endcaps_right_top_corner_neighbor	
	The cell to be abutted with the right side of rightTopCorner in R0 orientation Type: string Default: "" Edit: Yes Reference: add_endcaps_right_top_corner_neighbor
add_endcaps_right_top_corner_odd	
	EndCap with n-well at the right top corner, odd sites Type: string Default: "" Edit: Yes Reference: add_endcaps_right_top_corner_odd
add_endcaps_right_top_edge	
	EndCap with n-well at the right top edge Type: string Default: "" Edit: Yes Reference: add_endcaps_right_top_edge
add_endcaps_right_top_edge_even	
	EndCap with n-well at the right top edge, even sites Type: string Default: "" Edit: Yes Reference: add_endcaps_right_top_edge_even
add_endcaps_right_top_edge_neighbor	
	The cell to be abutted with the left edge of rightTopEdge in R0 orientation Type: string Default: "" Edit: Yes Reference: add_endcaps_right_top_edge_neighbor
add_endcaps_right_top_edge_odd	
	EndCap with n-well at the right top edge, odd sites Type: string Default: "" Edit: Yes Reference: add_endcaps_right_top_edge_odd

Stylus Common UI Database Object Information
Database Objects--root

add_endcaps_top_bottom_edge	
	EndCaps with n-well at the both top and bottom edges Type: string Default: "" Edit: Yes Reference: add_endcaps_top_bottom_edge
add_endcaps_top_edge	
	EndCaps with n-well at the top edge Type: string Default: "" Edit: Yes Reference: add_endcaps_top_edge
add_endcaps_use_even_odd_sites	
	use even or odd poly sites Type: enum Enum Values: none even odd Default: none Edit: Yes Reference: add_endcaps_use_even_odd_sites
add_endcaps_wall_keepout_from_vertical_boundary	
	Type: double Default: 0.0 Edit: Yes Reference: add_endcaps_wall_keepout_from_vertical_boundary
add_endcaps_wall_offset	
	Type: double Default: 0.0 Edit: Yes Reference: add_endcaps_wall_offset
add_endcaps_wall_pitch	
	Type: double Default: 0.0 Edit: Yes Reference: add_endcaps_wall_pitch
add_endcaps_wall_shift_step	
	Type: double Default: 0.0 Edit: Yes Reference: add_endcaps_wall_shift_step
add_endcaps_wall_to_convex_corner_spacing	
	Type: double Default: 0.0 Edit: Yes Reference: add_endcaps_wall_to_convex_corner_spacing
add_fillers_avoid_abutment_patterns	

Stylus Common UI Database Object Information
Database Objects--root

	illegal abutment patterns Type: string Default: "" Edit: Yes Reference: add_fillers_avoid_abutment_patterns
add_fillers_cell_name_style	
	Add physical cells into hierarchical modules, or as top level cells (flat) Type: enum Enum Values: hier flat Default: hier Edit: Yes Reference: add_fillers_cell_name_style
add_fillers_cells	
	List of filler cells Type: string Default: "" Edit: Yes Reference: add_fillers_cells
add_fillers_check_different_cells	
	Check violations between different cell types Type: bool Default: false Edit: Yes Reference: add_fillers_check_different_cells
add_fillers_check_drc	
	DRC checking after filler insertion Type: bool Default: true Edit: Yes Reference: add_fillers_check_drc
add_fillers_check_trim_rule	
	check trim-2-trim rule between cell trim and routing trim. Type: bool Default: false Edit: Yes Reference: add_fillers_check_trim_rule
add_fillers_create_rows	
	Create rows for filler cell techSites which do not have rows in the floor plan Type: bool Default: true Edit: Yes Reference: add_fillers_create_rows
add_fillers_distribute_implant_evenly	
	distribute implant evenly for fillers Type: bool Default: false Edit: Yes Reference: add_fillers_distribute_implant_evenly
add_fillers_double_height_filler_insertion	
	enable double height filler during filler insertion. Type: bool Default: false Edit: Yes Reference: add_fillers_double_height_filler_insertion

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add_fillers_eco_mode	
	Fillers inserted in eco flow Type: bool Default: false Edit: Yes Reference: add_fillers_eco_mode
add_fillers_enable_restore_filler_flow	
	enable restore filler flow. Type: bool Default: true Edit: Yes
add_fillers_honor_preroute_as_obs	
	Filler command honors place_detail_preroute_as_obs value Type: bool Default: false Edit: Yes Reference: add_fillers_honor_preroute_as_obs
add_fillers_horizontal_exception_cell	
	List of exceptional cells for horizontal stack checking Type: string Default: "" Edit: Yes Reference: add_fillers_horizontal_exception_cell
add_fillers_horizontal_max_length	
	specify instance max length (in micron) aligned in horizontal direction Type: double Default: 0.0 Edit: Yes Reference: add_fillers_horizontal_max_length
add_fillers_horizontal_repair_cell	
	List of filler cells to fix horizontal max length violation Type: string Default: "" Edit: Yes Reference: add_fillers_horizontal_repair_cell
add_fillers_keep_fixed	
	'FIXED' fillers are kepted Type: bool Default: false Edit: Yes Reference: add_fillers_keep_fixed
add_fillers_no_single_site_gap	
	Filler command fit 1-site gap Type: bool Default: true Edit: Yes Reference: add_fillers_no_single_site_gap
add_fillers_prefix	
	Prefix of filler cells Type: string Default: "" Edit: Yes Reference: add_fillers_prefix

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add_fillers_preserve_user_order	
	Instruct tool to insert fillers by the user defined order Type: bool Default: false Edit: Yes Reference: add_fillers_preserve_user_order
add_fillers_scheme	
	Instruct tool to select locationFirst cellFirst scheme Type: enum Enum Values: locationFirst cellFirst Default: locationFirst Edit: Yes Reference: add_fillers_scheme
add_fillers_swap_cell	
	swap cell pairs Type: string Default: "" Edit: Yes Reference: add_fillers_swap_cell
add_fillers_vertical_stack_exception_cell	
	List of exceptional cells for vertical stack checking Type: string Default: "" Edit: Yes Reference: add_fillers_vertical_stack_exception_cell
add_fillers_vertical_stack_left_edge_exception_cell	
	List of cells only check right edge for vertical stack violation Type: string Default: "" Edit: Yes Reference: add_fillers_vertical_stack_left_edge_exception_cell
add_fillers_vertical_stack_max_length	
	specify instance max length (in micron) aligned in vertical direction Type: double Default: 0.0 Edit: Yes Reference: add_fillers_vertical_stack_max_length
add_fillers_vertical_stack_repair_cell	
	List of filler cells to fix vertical stack violation Type: string Default: "" Edit: Yes Reference: add_fillers_vertical_stack_repair_cell
add_fillers_vertical_stack_repair_edge	
	specify the edge to use of vertical repair cell Type: enum Enum Values: left right Default: "" Edit: Yes Reference: add_fillers_vertical_stack_repair_edge
add_fillers_vertical_stack_right_edge_exception_cell	

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	List of cells only check left edge for vertical stack violation Type: string Default: "" Edit: Yes Reference: add_fillers_vertical_stack_right_edge_exception_cell
add_fillers_with_drc	
	call second add_fillers without drc checking Type: bool Default: true Edit: Yes Reference: add_fillers_with_drc
add_fillers_y_flip_type	
	Defines first or last cell with certain cell size should be y-flipped Type: string Default: "" Edit: Yes Reference: add_fillers_y_flip_type
add_reinforce_pg_critical_path_slack	
	Specifies a slack threshold, in nanoseconds, for critical nets during timing-driven reinforce_pg. Type: double Default: 0.0 Edit: Yes Reference: add_reinforce_pg_critical_path_slack
add_reinforce_pg_irdrop_hierarchical_block	
	Specifies hierarchical blocks to fix irdrop. Type: string Default: "" Edit: Yes Reference: add_reinforce_pg_irdrop_hierarchical_block
add_reinforce_pg_irdrop_hierarchical_top_rail_analysis_directory	
	Specifies the directory of rail analysis information of full chip for hierarchical blocks to fix irdrop. Type: string Default: "" Edit: Yes Reference: add_reinforce_pg_irdrop_hierarchical_top_rail_analysis_directory
add_reinforce_pg_respect_defined_nets	
	Specifies reinforce_pg to respect user defined nets in file. Type: string Default: "" Edit: Yes Reference: add_reinforce_pg_respect_defined_nets
add_reinforce_pg_respect_routes	
	Specifies whether to honor pre-existing routes check during stripe generation Type: enum Enum Values: none fixed fixed_and_clock all Default: fixed_and_clock Edit: Yes Reference: add_reinforce_pg_respect_routes
add_reinforce_pg_respect_stdcell_geometry	
	Specifies the cell type to be respected in via generation. Type: enum Enum Values: none sequential all Default: all Edit: Yes Reference: add_reinforce_pg_respect_stdcell_geometry

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add_rings_avoid_short	
	Specifies whether to avoid short violation Type: bool Default: false Edit: Yes Reference: add_rings_avoid_short
add_rings_break_core_ring_io_list	
	Specifies io instance names for which core ring wires are broken Type: string Default: "" Edit: Yes Reference: add_rings_break_core_ring_io_list
add_rings_continue_on_no_selection	
	Continues w/o selection Type: bool Default: false Edit: Yes Reference: add_rings_continue_on_no_selection
add_rings_detailed_log	
	Outputs detailed log file addring.log Type: bool Default: false Edit: Yes Reference: add_rings_detailed_log
add_rings_extend_block_ring_search_distance	
	Specifies blocking extension search distance Type: double Default: 0.0 Edit: Yes Reference: add_rings_extend_block_ring_search_distance
add_rings_extend_core_ring_search_distance	
	Specifies corering extension search distance Type: double Default: 0.0 Edit: Yes Reference: add_rings_extend_core_ring_search_distance
add_rings_extend_merge_with_pre_wires	
	Specifies whether to merge with prerouted wires Type: bool Default: true Edit: Yes Reference: add_rings_extend_merge_with_pre_wires
add_rings_extend_over_row	
	Specifies whether to extend over row Type: bool Default: false Edit: Yes Reference: add_rings_extend_over_row
add_rings_extend_search_nets	
	Specifies search net names Type: string Default: "" Edit: Yes Reference: add_rings_extend_search_nets

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add_rings_extend_stripe_search_distance	
	Specifies stripe extension search distance Type: double Default: 0.0 Edit: Yes Reference: add_rings_extend_stripe_search_distance
add_rings_gap_width_without_io	
	Specifies the max missing IO Pads area width between IO Pads Type: double Default: 0.0 Edit: Yes Reference: add_rings_gap_width_without_io
add_rings_ignore_rows	
	Specifies whether to ignore all rows Type: bool Default: false Edit: Yes Reference: add_rings_ignore_rows
add_rings_max_via_size	
	the maximum size of a crossover via Type: string Default: "" Edit: Yes Reference: add_rings_max_via_size
add_rings_orthogonal_only	
	vias connect to orthogonal targets only Type: bool Default: true Edit: Yes Reference: add_rings_orthogonal_only
add_rings_skip_shared_inner_ring	
	Specifies whether to skip crossing trunks Type: enum Enum Values: horizontal vertical none Default: none Edit: Yes Reference: add_rings_skip_shared_inner_ring
add_rings_skip_via_on_pin	
	prevent vias from being generated on the specified type of pins Type: enum Enum Values: pad block cover standardcell physicalpin Default: standardcell Edit: Yes Reference: add_rings_skip_via_on_pin
add_rings_skip_via_on_wire_shape	
	prevent vias from being generated on the specified wire shapes Type: enum Enum Values: blockring stripe followpin corewire blockwire iowire padring ring fillwire noshape Default: noshape Edit: Yes Reference: add_rings_skip_via_on_wire_shape
add_rings_spacing_from_block	

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	Specifies the spacing from block Type: double Default: 0.0 Edit: Yes Reference: add_rings_spacing_from_block
add_rings_split_long_via	
	Split vias longer than <threshold> into smaller vias with specified <step> and bottom/left end <offset> and vertical/horizontal <length> Type: string Default: 0 0 -1 -1 Edit: Yes Reference: add_rings_split_long_via
add_rings_stacked_via_bottom_layer	
	lef layer name Type: string Default: bottomLayer Edit: Yes Reference: add_rings_stacked_via_bottom_layer
add_rings_stacked_via_top_layer	
	lef layer name Type: string Default: topLayer Edit: Yes Reference: add_rings_stacked_via_top_layer
add_rings_target	
	Specifies ring target Type: enum Enum Values: core_ring default first_ring pad_ring stripe Default: default Edit: Yes Reference: add_rings_target
add_rings_via_using_exact_crossover_size	
	generate partial vias of the exact crossover size Type: bool Default: true Edit: Yes Reference: add_rings_via_using_exact_crossover_size
add_rings_wire_center_offset	
	offset value is from wire center to domain/core boundary Type: bool Default: false Edit: Yes Reference: add_rings_wire_center_offset
add_route_vias_auto	
	enable auto via generation when design is loaded Type: bool Default: false Edit: Yes Reference: add_route_vias_auto
add_route_vias_ndr_only	
	generate vias for non default rules only Type: bool Default: false Edit: Yes Reference: add_route_vias_ndr_only

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add_stripes_allow_jog	
	Specifies the allowed jog type. Type: enum Enum Values: none padcore_ring block_ring Default: padcore_ring block_ring Edit: Yes Reference: add_stripes_allow_jog
add_stripes_allow_non_preferred_dir	
	Specifies whether to allow non-preferred direction to jog to target Type: enum Enum Values: none blocking corering padding stripe Default: padding Edit: Yes Reference: add_stripes_allow_non_preferred_dir
add_stripes_area	
	Specify the rectangular areas or rectilinear areas or a mix of them, the correct value should be {<lx ly ux uy> <x1 y1 x2 y2 x3 y3 ...>} Type: string Default: "" Edit: Yes Reference: add_stripes_area
add_stripes_blocks_without_same_net	
	Specifies the stripe break point. Type: enum Enum Values: none block_ring selected_block unassigned_bump overlap_ringpin outer_ring outside_ringmacro blocks_without_same_net Default: none Edit: Yes Reference: add_stripes_blocks_without_same_net
add_stripes_break_at	
	Specifies the stripe break point. Type: enum Enum Values: none block_ring selected_block unassigned_bump overlap_ringpin outer_ring outside_ringmacro blocks_without_same_net Default: none Edit: Yes Reference: add_stripes_break_at
add_stripes_continue_on_no_selection	
	Specifies whether to continue w/o selection Type: bool Default: false Edit: Yes Reference: add_stripes_continue_on_no_selection
add_stripes_detailed_log	
	Specifies whether to output detailed log file addStripe.log Type: bool Default: false Edit: Yes Reference: add_stripes_detailed_log
add_stripes_domain_offset_from_core	
	Specifies whether to use global offset for power domain stripes Type: bool Default: false Edit: Yes Reference: add_stripes_domain_offset_from_core
add_stripes_extend_to_closest_target	

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	Specifies the stripe to specified target Type: enum Enum Values: ring stripe same_dir_stripe area_boundary none Default: none Edit: Yes Reference: add_stripes_extend_to_closest_target
add_stripes_extend_to_first_ring	
	Specifies whether to extend stripe antennas to closest ring Type: bool Default: false Edit: Yes Reference: add_stripes_extend_to_first_ring
add_stripes_ignore_block_check	
	Specifies whether to ignore block check Type: bool Default: false Edit: Yes Reference: add_stripes_ignore_block_check
add_stripes_ignore_block_ring_during_break	
	Specifies whether to break stripe outside block Type: bool Default: false Edit: Yes Reference: add_stripes_ignore_block_ring_during_break
add_stripes_ignore_drc	
	Specifies whether to ignore DRC check during stripe generation, it is always used for prototype flow Type: bool Default: false Edit: Yes Reference: add_stripes_ignore_drc
add_stripes_ignore_non_default_domains	
	Specifies whether to break at non-default domain Type: bool Default: false Edit: Yes Reference: add_stripes_ignore_non_default_domains
add_stripes_in_cell_allow_shift	
	Specifies to allow the created pillar inside cell can be shifted when meeting DRC Type: bool Default: false Edit: Yes Reference: add_stripes_in_cell_allow_shift
add_stripes_in_cell_only	
	Specifies whether to generate stripes only inside selected or specified cell Type: bool Default: false Edit: Yes Reference: add_stripes_in_cell_only
add_stripes_keep_pitch_after_snap	
	Type: bool Default: false Edit: Yes Reference: add_stripes_keep_pitch_after_snap

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add_stripes_mask_color_balance	
	same_color would correspond to interlace with groups and alternate_color would correspond to interlace within group Type: enum Enum Values: same_color alternate_color none Default: none Edit: Yes Reference: add_stripes_mask_color_balance
add_stripes_max_extension_distance	
	Specifies maximum extension distance(default=max integer) when -extend_to_closest_target is specified. Type: double Default: 2.14748e+09 Edit: Yes Reference: add_stripes_max_extension_distance
add_stripes_max_via_size	
	the maximum size of a crossover via, the correct value should be {shape width% height% target_penetration%} Type: string Default: "" Edit: Yes Reference: add_stripes_max_via_size
add_stripes_merge_with_all_layers	
	Specifies whether to merge a stripe on all layers Type: bool Default: false Edit: Yes Reference: add_stripes_merge_with_all_layers
add_stripes_mesh_via	
	use mesh via to reduce memory Type: bool Default: false Edit: Yes Reference: add_stripes_mesh_via
add_stripes_offset_from_core	
	Specifies whether to offset first or last stripe from the core area Type: bool Default: true Edit: Yes Reference: add_stripes_offset_from_core
add_stripes_opt_stripe_for_routing_track	
	move stripe location to preserve routing resource Type: enum Enum Values: shift none Default: none Edit: Yes Reference: add_stripes_opt_stripe_for_routing_track
add_stripes_orthogonal_offset	
	Specifies the offsets for edges orthogonally, the correct value should be {{{edge1 orthogonal_offset1} {edge2 edge3 orthogonal_offset2} {edge4 edge5 ... orthogonal_offset3} ...} [all orthogonal_offset]} Type: string Default: {none} Edit: Yes Reference: add_stripes_orthogonal_offset
add_stripes_orthogonal_only	

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	vias connect to orthogonal targets only Type: <code>bool</code> Default: true Edit: Yes Reference: add_stripes_orthogonal_only
add_stripes_over_row_extension	
	Specifies whether to extend the stripe to cover followpin at the row end Type: <code>bool</code> Default: false Edit: Yes Reference: add_stripes_over_row_extension
add_stripes_partial_set_through_domain	
	Specifies whether to go over power domain if domain contains the specified net Type: <code>bool</code> Default: false Edit: Yes Reference: add_stripes_partial_set_through_domain
add_stripes_remove_floating_stapling	
	remove floating stapling stripe if there is no via connection Type: <code>bool</code> Default: true Edit: Yes Reference: add_stripes_remove_floating_stapling
add_stripes_remove_floating_stripe_over_block	
	Specifies whether to remove stripe fragments start and end inside same block or ring macro Type: <code>bool</code> Default: true Edit: Yes Reference: add_stripes_remove_floating_stripe_over_block
add_stripes_remove_stripe_under_ring	
	Specifies whether to remove those segments under rings. Type: <code>bool</code> Default: false Edit: Yes Reference: add_stripes_remove_stripe_under_ring
add_stripes_respect_routes	
	Specifies whether to honor pre-existing routes check during stripe generation Type: <code>enum</code> Enum Values: none fixed fixed_and_clock all Default: fixed_and_clock Edit: Yes Reference: add_stripes_respect_routes
add_stripes_route_over_rows_only	
	Specifies whether to route stripe over rows within the boundary Type: <code>bool</code> Default: false Edit: Yes Reference: add_stripes_route_over_rows_only
add_stripes_rows_without_stripes_only	
	Specifies whether to generate stripes over rows w/o stripes Type: <code>bool</code> Default: false Edit: Yes Reference: add_stripes_rows_without_stripes_only

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add_stripes_same_size_stack_vias	
	generate same-sized stacked vias Type: bool Default: false Edit: Yes Reference: add_stripes_same_size_stack_vias
add_stripes_skip_via_on_pin	
	prevent vias from being generated on the specified type of pins Type: enum Enum Values: pad block cover standardcell physicalpin Default: standardcell Edit: Yes Reference: add_stripes_skip_via_on_pin
add_stripes_skip_via_on_wire_shape	
	prevent vias from being generated on the specified wire shapes Type: enum Enum Values: blocking stripe followpin corewire blockwire iowire padding ring fillwire noshape Default: noshape Edit: Yes Reference: add_stripes_skip_via_on_wire_shape
add_stripes_spacing_from_block	
	Specifies the spacing from block Type: double Default: 0.0 Edit: Yes Reference: add_stripes_spacing_from_block
add_stripes_spacing_type	
	Specify the type of intra spacing for add_stripes -spacing Type: enum Enum Values: edge_to_edge center_to_center Default: edge_to_edge Edit: Yes Reference: add_stripes_spacing_type
add_stripes_split_long_via	
	Split vias longer than <threshold> into smaller vias with specified <step> and bottom/left end <offset> and vertical/horizontal <length>, the correct value should be {threshold step offset length} Type: string Default: 0 0 -1 -1 Edit: Yes Reference: add_stripes_split_long_via
add_stripes_split_vias	
	create multiple partial vias Type: bool Default: false Edit: Yes Reference: add_stripes_split_vias
add_stripes_split_wire_spacing	
	Specifies the spacing between each split wire Type: double Default: 0.0 Edit: Yes Reference: add_stripes_split_wire_spacing
add_stripes_split_wire_weight	

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	Deletes fewest IO wires to accommodate approximately n of 10 split stripe wires Type: double Default: 0.0 Edit: Yes Reference: add_stripes_split_wire_weight
add_stripes_split_wire_width	
	Specifies the width of each split wire Type: double Default: 0.0 Edit: Yes Reference: add_stripes_split_wire_width
add_stripes_stacked_via_bottom_layer	
	lef layer name Type: string Default: bottomLayer Edit: Yes Reference: add_stripes_stacked_via_bottom_layer
add_stripes_stacked_via_top_layer	
	lef layer name Type: string Default: topLayer Edit: Yes Reference: add_stripes_stacked_via_top_layer
add_stripes_stapling_extend_to_minimum_spacing	
	Specified to automatically extend the adjacent stapling as long as enough to satisfy the end-to-end minimum spacing with DRC free Type: bool Default: false Edit: Yes Reference: add_stripes_stapling_extend_to_minimum_spacing
add_stripes_stapling_nets_style	
	Specify the stapling style if multiple nets are specified Type: enum Enum Values: end_to_end side_to_side side_to_side_full_nets Default: end_to_end Edit: Yes Reference: add_stripes_stapling_nets_style
add_stripes_stapling_shift	
	shift stapling stripe to avoid the drc automatically Type: bool Default: true Edit: Yes Reference: add_stripes_stapling_shift
add_stripes_stop_at_closest_target	
	Specifies the stripe to specified target when over_pins is enabled. Type: enum Enum Values: none block_ring core_ring stripe Default: none Edit: Yes Reference: add_stripes_stop_at_closest_target
add_stripes_stop_at_last_wire_for_area	
	Specifies whether to trim back stripe antennas to the closest wire or pin of the same net Type: bool Default: false Edit: Yes Reference: add_stripes_stop_at_last_wire_for_area

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add_stripes_stripe_min_length	
	Specifies the minimum stripe length Type: string Default: stripe_width Edit: Yes Reference: add_stripes_stripe_min_length
add_stripes_stripe_min_width	
	Specifies the min width of remaining stripe after being trimmed Type: double Default: 0.0 Edit: Yes Reference: add_stripes_stripe_min_width
add_stripes_switch_cell_name	
	Specifies the switch cells to be treated as rings Type: string Default: none Edit: Yes Reference: add_stripes_switch_cell_name
add_stripes_switch_layer_overlap_length	
	Specifies the overlap distance used for the via between a stripe layer and the layer being switched to Type: double Default: 0.0 Edit: Yes Reference: add_stripes_switch_layer_overlap_length
add_stripes_trim_antenna_back_to_shape	
	Specifies the distance between the end of antenna and blocked target Type: enum Enum Values: none block_ring core_ring pad_ring stripe Default: none Edit: Yes Reference: add_stripes_trim_antenna_back_to_shape
add_stripes_trim_antenna_max_distance	
	Specifies the distance between the end of antenna and blocked target Type: double Default: 0.0 Edit: Yes Reference: add_stripes_trim_antenna_max_distance
add_stripes_trim_stripe	
	Specifies if stripe is trimmed by the specified shape. Type: enum Enum Values: none design_boundary core_boundary Default: none Edit: Yes Reference: add_stripes_trim_stripe
add_stripes_use_exact_spacing	
	Specifies whether to use exact spacing Type: bool Default: false Edit: Yes Reference: add_stripes_use_exact_spacing
add_stripes_use_point_to_point_route	

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	Specifies whether to use exact spacing Type: bool Default: false Edit: Yes Reference: add_stripes_use_point_to_point_route
add_stripes_use_stripe_width	
	Specifies whether to use stripe width Type: bool Default: true Edit: Yes Reference: add_stripes_use_stripe_width
add_stripes_via_using_exact_crossover_size	
	generate partial vias of the exact crossover size Type: bool Default: true Edit: Yes Reference: add_stripes_via_using_exact_crossover_size
add_stripes_width_file	
	Specify the values of recommended width in the file name Type: string Default: "" Edit: Yes Reference: add_stripes_width_file
add_target_pg_allow_weak_connect	
	Decrease specified pillar number to avoid drc Type: bool Default: false Edit: Yes Reference: add_target_pg_allow_weak_connect
add_target_pg_max_extension_distance	
	Specify the max distance that target PG pin can extend to target layer Type: double Default: 2.14748e+09 Edit: Yes Reference: add_target_pg_max_extension_distance
add_target_pg_pins	
	Specify the cell pins to be connected Type: string Default: "" Edit: Yes Reference: add_target_pg_pins
add_target_pg_pins_group_distance	
	Specify the side_to_side distance of pins to be treated as a group during connecting target PG pins Type: string Default: auto Edit: Yes Reference: add_target_pg_pins_group_distance
add_target_pg_respect_routes	
	Specifies whether to honor pre-existing routes check during stripe generation Type: enum Enum Values: none fixed fixed_and_clock all Default: fixed_and_clock Edit: Yes Reference: add_target_pg_respect_routes

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add_target_pg_share_resource	
	New added PG connect to the existing PG at middle layer Type: bool Default: true Edit: Yes Reference: add_target_pg_share_resource
add_tieoffs_cells	
	Set tie cell lists for set_db command Type: string Default: "" Edit: Yes Reference: add_tieoffs_cells
add_tieoffs_create_hports	
	Add tie-cells in different hierarchy than tie-off pins when possible Type: bool Default: false Edit: Yes Reference: add_tieoffs_create_hports
add_tieoffs_honor_dont_touch	
	honor dont_touch attribute on nets, insts, cells and modules Type: bool Default: false Edit: Yes Reference: add_tieoffs_honor_dont_touch
add_tieoffs_honor_dont_use	
	honor dont_use attribute on cells. Type: bool Default: false Edit: Yes Reference: add_tieoffs_honor_dont_use
add_tieoffs_max_distance	
	distance between tie-cell and tie-pins should be less than given value Type: double Default: 0.0 Edit: Yes Reference: add_tieoffs_max_distance
add_tieoffs_max_fanout	
	the number of tie-pins a tie-net can drive. A '0' means no-limit Type: int Default: 0 Edit: Yes Reference: add_tieoffs_max_fanout
add_tieoffs_module_prevention	
	add tie cell for module instantiation port Type: bool Default: false Edit: Yes Reference: add_tieoffs_module_prevention
add_tieoffs_prefix	
	Prefix of tie cell instances Type: string Default: "" Edit: Yes Reference: add_tieoffs_prefix

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add_tieoffs_report_hports	
	Report created ports when add_tieoffs_create_hports is set Type: bool Default: false Edit: Yes Reference: add_tieoffs_report_hports
add_well_taps_abut_boundary_tap_distance	
	Specifies boundary cells and their tap distance when they are vertically abutted Type: string Default: "" Edit: Yes
add_well_taps_antenna_tap_break_cell	
	Specifies cells can break antenna tap Type: string Default: "" Edit: Yes Reference: add_well_taps_antenna_tap_break_cell
add_well_taps_antenna_tap_cell	
	Specifies normal antenna tap cell Type: string Default: "" Edit: Yes Reference: add_well_taps_antenna_tap_cell
add_well_taps_antenna_tap_left_cell	
	Specifies antenna tap cell for left edge Type: string Default: "" Edit: Yes Reference: add_well_taps_antenna_tap_left_cell
add_well_taps_antenna_tap_pitch	
	Specifies the pitch of antenna tap cell Type: double Default: 0.0 Edit: Yes Reference: add_well_taps_antenna_tap_pitch
add_well_taps_antenna_tap_right_cell	
	Specifies antenna tap cell for right edge Type: string Default: "" Edit: Yes Reference: add_well_taps_antenna_tap_right_cell
add_well_taps_antenna_tap_right_top_edge_cell	
	Specifies antenna tap cell for right top edge Type: string Default: "" Edit: Yes Reference: add_well_taps_antenna_tap_right_top_edge_cell
add_well_taps_avoid_abutment	
	Specifies whether the inserted taps should avoid horizontal and vertical abutment Type: bool Default: false Edit: Yes Reference: add_well_taps_avoid_abutment

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add_well_taps_avoid_vertical_well_abutment	
	Specifies which type of well-tap cells to be forbidden from vertical abutment Type: enum Enum Values: none ntap ptap all_tap Default: none Edit: Yes Reference: add_well_taps_avoid_vertical_well_abutment
add_well_taps_block_boundary_only	
	Specifies whether inserting termination cells on the core boundary Type: bool Default: false Edit: Yes Reference: add_well_taps_block_boundary_only
add_well_taps_bottom_tap_cell	
	Cell to be used as bottom tap for add_endcaps boundary cell insertion flow Type: string Default: "" Edit: Yes Reference: add_well_taps_bottom_tap_cell
add_well_taps_bottom_termination_cell	
	Specifies the list of bottom termination cells to be used Type: string Default: "" Edit: Yes Reference: add_well_taps_bottom_termination_cell
add_well_taps_cell	
	Cell to be used as well_taps Type: string Default: "" Edit: Yes Reference: add_well_taps_cell
add_well_taps_channel_offset	
	Specifies the offset value between taps and blockage in channel Type: double Default: 0.0 Edit: Yes Reference: add_well_taps_channel_offset
add_well_taps_check_channel	
	Specifies whether checking blockage channel Type: bool Default: false Edit: Yes Reference: add_well_taps_check_channel
add_well_taps_column_cells	
	Specifies the list of cells which can be replaced by termination cell Type: string Default: "" Edit: Yes Reference: add_well_taps_column_cells
add_well_taps_create_rows	

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	Create rows for welltap cell techSites which do not have rows in the floor plan Type: <code>bool</code> Default: false Edit: Yes Reference: add_well_taps_create_rows
add_well_taps_disable_check_zone_at_boundary	
	disable check zone at bounadry Type: <code>enum</code> Enum Values: none vdd vss both Default: none Edit: Yes Reference: add_well_taps_disable_check_zone_at_boundary
add_well_taps_in_row_offset	
	Specifies the distance between the first wellTap in a row and the start of the row Type: <code>double</code> Default: 0.0 Edit: Yes Reference: add_well_taps_in_row_offset
add_well_taps_insert_cells	
	Insert well-taps based on cell, layer and inertval Type: <code>string</code> Default: "" Edit: Yes Reference: add_well_taps_insert_cells
add_well_taps_rule	
	Specifies the well latch-up rule Type: <code>double</code> Default: 0.0 Edit: Yes Reference: add_well_taps_rule
add_well_taps_site_offset	
	Specifies the offset value in number of sites that vertical taps need to be aware of Type: <code>int</code> Default: 1 Edit: Yes Reference: add_well_taps_site_offset
add_well_taps_tap_function_cells	
	Specifies the list of tap function cells to be used Type: <code>string</code> Default: "" Edit: Yes Reference: add_well_taps_tap_function_cells
add_well_taps_tap_termination_alignment	
	Specifies whether to align well-taps and termination cells Type: <code>bool</code> Default: false Edit: Yes Reference: add_well_taps_tap_termination_alignment
add_well_taps_termination_align	
	Specifies the termination cell align Type: <code>enum</code> Enum Values: left right center Default: left Edit: Yes Reference: add_well_taps_termination_align

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add_well_taps_termination_cells	
	Specifies the list of termination cells to be used Type: string Default: "" Edit: Yes Reference: add_well_taps_termination_cells
add_well_taps_top_tap_cell	
	Cell to be used as top tap for add_endcaps boundary cell insertion flow Type: string Default: "" Edit: Yes Reference: add_well_taps_top_tap_cell
add_well_taps_top_termination_cell	
	Specifies the list of top termination cells to be used Type: string Default: "" Edit: Yes Reference: add_well_taps_top_termination_cell
add_well_taps_vertical_boundary_spacing	
	Specifies the spacing between center of tap cell and vertical boundary Type: double Default: -1.0 Edit: Yes Reference: add_well_taps_vertical_boundary_spacing
add_well_taps_well_cut_cells	
	Specifies the list of cells with the well cuts Type: string Default: "" Edit: Yes Reference: add_well_taps_well_cut_cells
analysis_views	
	Returns the information about the analysis views in the design. Type: obj(analysis_view)* Default: "" Edit: No
aocv_library	
	To specify AOCV library. Type: string Default: "" Edit: Yes Reference: aocv_library
apply_booth_encoding	
	Apply booth encoding by using user specified method (for all instances). Type: enum Enum Values: nonbooth auto_bitwidth auto_togglerate manual inherited Default: auto_bitwidth Edit: Yes Reference: apply_booth_encoding
assemble_design_generic_hier	
	Read DEF for generic hierarchy. Type: bool Default: false Edit: Yes Reference: assemble_design_generic_hier

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assign_pins_advanced_node_rule_support	
	If you use this parameter, all the pin related command will use design object shapes for deciding pin location which will give accurate result but will have larger runtime. Type: bool Default: true Edit: Yes Reference: assign_pins_advanced_node_rule_support
assign_pins_allow_non_ndr_net_pins_on_ndr_tracks	
	A true value allows general pins to be placed on ndr tracks if available. Alternately, if the value is set to false, the ndr-track rule is followed strictly and the net's pins are placed on matching ndr tracks only. Type: bool Default: true Edit: Yes Reference: assign_pins_allow_non_ndr_net_pins_on_ndr_tracks
assign_pins_allow_unconnected_in_abutted_edge	
	If you use this parameter, unconnected partition pin can be placed in abutted edge Type: bool Default: true Edit: Yes Reference: assign_pins_allow_unconnected_in_abutted_edge
assign_pins_blocked_boundary_macro_distance	
	This option blocks automatic pin assignment, from putting pins on the part of partition boundary facing block on layers with OBS, in the places where channel width between partition boundary and block is less than the value specified. For channel width below specified value, block all pins on boundary facing this channel on layers with OBS. Type: double Default: 3.0 Edit: Yes Reference: assign_pins_blocked_boundary_macro_distance
assign_pins_bottom_layer	
	Specifies the minimum metal layer name for partition/block pin assignment Type: string Default: 2 Edit: Yes
assign_pins_edit_in_batch	
	To enable/disable pin-editing in batch mode. Type: bool Default: false Edit: Yes Reference: assign_pins_edit_in_batch
assign_pins_force_abutment_with_fixed	
	For fixed abutted pins's connected pin, ignore soft constraints checks Type: bool Default: true Edit: Yes Reference: assign_pins_force_abutment_with_fixed
assign_pins_inside_pin_search_distance	
	To control the search distance for legalizing the internal pins Type: int Default: 10 Edit: Yes Reference: assign_pins_inside_pin_search_distance
assign_pins_max_channel_width_as_abutted	

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	If you set this parameter value then all channel less than equal to specified value will be considered as abutted and abutment rules will be applied Type: double Default: 3.0 Edit: Yes Reference: assign_pins_max_channel_width_as_abutted
assign_pins_max_distance_pair	
	Ignores moving corresponding connected pins of 2-pin-connection nets where distance (in microns) between the pins is equal or greater than distance specified Type: double Default: 50.0 Edit: Yes Reference: assign_pins_max_distance_pair
assign_pins_off_stripe	
	Prevents the generation of pins on metal layers with respect to the power and ground stripe Type: enum Enum Values: below all none Default: none Edit: Yes Reference: assign_pins_off_stripe
assign_pins_pin_to_stripe_distance	
	Layer specific, keep distance (in microns) from pin to stripe Type: string Default: "" Edit: Yes Reference: assign_pins_pin_to_stripe_distance
assign_pins_pin_to_via_distance_non_preferred_direction	
	Layer specific, keep distance (in microns) from pin to via-stack Type: string Default: "" Edit: Yes Reference: assign_pins_pin_to_via_distance_non_preferred_direction
assign_pins_pin_to_via_distance_preferred_direction	
	Layer specific, keep distance (in microns) from pin to via-stack from the partition fence boundary Type: string Default: "" Edit: Yes Reference: assign_pins_pin_to_via_distance_preferred_direction
assign_pins_promoted_macro_bottom_layer	
	Specifies the minimum metal layer name for promoting macro pins Type: string Default: 31 Edit: Yes Reference: assign_pins_promoted_macro_bottom_layer
assign_pins_promoted_macro_top_layer	
	Specifies the maximum metal layer name for promoting macro pins Type: string Default: 31 Edit: Yes Reference: assign_pins_promoted_macro_top_layer
assign_pins_restricted_boundary_macro_distance	

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	This option blocks automatic pin assignment, from putting pins on the part of partition boundary facing block on layers with OBS, in the places where channel width between partition boundary and block is less than the value specified. For channel width below specified value: OBS area on the layer, channel length and channel width considered to allow suitable number of pins on boundary facing this channel. Type: double Default: 30.0 Edit: Yes Reference: assign_pins_restricted_boundary_macro_distance
assign_pins_strict_abutment	
	To relax abutment violations, for placing multi partition pin of a net and non neighbor pins of a net on abutted edges Type: bool Default: true Edit: Yes Reference: assign_pins_strict_abutment
assign_pins_top_layer	
	Specifies the maximum metal layer name for partition/block pin assignment Type: string Default: 31 Edit: Yes
assign_pins_use_common_color_engine_for_io	
	assignIoPins command will use coloring engine api to assign mask-color to pins. Type: bool Default: false Edit: Yes Reference: assign_pins_use_common_color_engine_for_io
attributes	
	Short-cut for [get_db obj_types .attributes] Type: obj(attribute)* Default: "" Edit: No
auto_file_dir	
	Represents the top-level directory to store files/sub-directories 'auto-generated' by the tool. Type: string Default: ./ Edit: Yes Reference: auto_file_dir
auto_file_prefix	
	Represents the prefix to be applied to all files/sub-directories 'auto-generated' by the tool. Type: string Default: "" Edit: Yes Reference: auto_file_prefix
auto_partition	
	Create temporary hierarchies based on connectivity to guide optimization. Type: bool Default: true Edit: Yes Reference: auto_partition
auto_ungroup	
	Automatically ungroup user hierarchies to improve optimization. Type: enum Enum Values: none area timing both Default: both Edit: Yes Reference: auto_ungroup

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auto_ungroup_max_threshold	
	To allow ungroup for the instance according to max threshold. Type: int Default: 0 Edit: Yes Reference: auto_ungroup_max_threshold
auto_voltage_scaling	
	To enable library scaling when libraries with different voltages mentioned in single list. Type: bool Default: true Edit: Yes
avoid_tied_inputs	
	Avoid creation of tied input pins of mapped gates in mapping if possible. Type: bool Default: false Edit: Yes Reference: avoid_tied_inputs
bank_based_multibit_inferencing	
	Removes map to multibit register constraint in predefined multibit. Type: bool Default: false Edit: Yes Reference: bank_based_multibit_inferencing
base_cells	
	All the base_cells from .lib, Verilog stubs, LEF or OA. Type: obj(base_cell)* Default: "" Edit: No
base_pins	
	All the base_pins in all the base_cells. Type: obj(base_pin)* Default: "" Edit: No
bit_blasted_port_style	
	the default bus bit blasted port name as defined by this tcl global Type: string Default: %s_%d Edit: Yes Reference: bit_blasted_port_style
boundaries	
	Short-cut to all the boundary objects (fence, region, etc.) in the design. Type: obj(boundary)* Default: "" Edit: No
boundary_optimize_constant_hpins	
	Allow constant propagation through hierarchical boundary pins. Type: bool Default: true Edit: Yes Reference: boundary_optimize_constant_hpins
boundary_optimize_equal_opposite_hpins	

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	Allow connection of hierarchical boundary pins through equal/opposite boundary pins. Type: bool Default: true Edit: Yes Reference: boundary_optimize_equal_opposite_hpins
boundary_optimize_feedthrough_hpins	
	Allow routing of feedthrough boundary pins around a hierarchical instance. Type: bool Default: true Edit: Yes Reference: boundary_optimize_feedthrough_hpins
boundary_optimize_invert_hpins	
	Allow inversion of hierarchical boundary pins. Type: bool Default: false Edit: Yes Reference: boundary_optimize_invert_hpins
boundary_optimize_invert_hpins_rename_nets	
	Rename nets driven by inverted hierarchical boundary output pins. Type: bool Default: true Edit: Yes Reference: boundary_optimize_invert_hpins_rename_nets
budget_abutted	
	Assumes all partitions are abutted Type: bool Default: false Edit: Yes Reference: budget_abutted
budget_accumulated	
	Write both interface and flat constraints for nested partitions under if {budget_use_accumulated != 1} {} else {} conditions. Type: bool Default: false Edit: Yes Reference: budget_accumulated
budget_boundary_condition_template_path	
	Directory path containing model files to be used with option useBoundaryCondition template Type: string Default: "" Edit: Yes Reference: budget_boundary_condition_template_path
budget_buffer_delay_lib_cell	
	user defined buffer which should be used for virtual buffer adjustment Type: string Default: "" Edit: Yes Reference: budget_buffer_delay_lib_cell
budget_buffer_delay_selection_effort	
	controls the tool effort for calculating the buffer delay adjustment internally Type: enum Enum Values: low high none Default: none Edit: Yes Reference: budget_buffer_delay_selection_effort

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budget_ccd	
	Generates scripts for verification of time budgeting top and block constraints against pre-budget chip constraints using the CCD Type: bool Default: false Edit: Yes Reference: budget_ccd
budget_constant_model	
	Specifies the type of timing model created Type: bool Default: true Edit: Yes Reference: budget_constant_model
budget_distribute_mmmc	
	Distribute multiple analysis views over multiple sessions Type: bool Default: false Edit: Yes Reference: budget_distribute_mmmc
budget_dont_write_wire_load	
	write set_load with -wire_load information Type: bool Default: false Edit: Yes Reference: budget_dont_write_wire_load
budget_drive_constraint	
	Driving boundary constraint to be written on block input port Type: enum Enum Values: drive drive_cell input_transition Default: drive_cell Edit: Yes Reference: budget_drive_constraint
budget_fix_top_level_paths	
	Slack value for which the top-level timing budget is fixed, and remaining budget is proportioned only for partitions Type: enum Enum Values: none all negative_only positive_only Default: none Edit: Yes Reference: budget_fix_top_level_paths
budget_handle_complex_sdc	
	handle sdc files with complex constructs and regular expressions in push down Type: bool Default: false Edit: Yes Reference: budget_handle_complex_sdc
budget_honor_port_boundary_condition	
	Push down chip IO constraints to block boundary pins which are directly connected to chip IO Type: bool Default: true Edit: Yes Reference: budget_honor_port_boundary_condition
budget_honor_port_delays	

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	Push down chip IO SID SOD values to block IO if the two are directly connected and abutting. Type: <code>bool</code> Default: true Edit: Yes Reference: budget_honor_port_delays
budget_honor_report_timing_format	
	honor set_global report_timing_format in justification reports Type: <code>bool</code> Default: true Edit: Yes Reference: budget_honor_report_timing_format
budget_ignore_dont_touch	
	Specifies the handling of don't touch objects Type: <code>bool</code> Default: false Edit: Yes Reference: budget_ignore_dont_touch
budget_include_latency	
	Specifies whether the clock latency is included in the set_input_delay and set_output_delay constraints Type: <code>bool</code> Default: true Edit: Yes Reference: budget_include_latency
budget_include_wire_loads_in_lib	
	Include the wire loads in library model pin capacitance Type: <code>bool</code> Default: false Edit: Yes Reference: budget_include_wire_loads_in_lib
budget_input_load	
	write set_load for input pins Type: <code>bool</code> Default: false Edit: Yes Reference: budget_input_load
budget_justify	
	controls the type of justification Type: <code>enum</code> Enum Values: delay exception lib_arc boundary_condition all Default: "" Edit: Yes Reference: budget_justify
budget_keep_pin_list_for_block_ports	
	Preserve all fan-in and fan-out lists Type: <code>bool</code> Default: true Edit: Yes Reference: budget_keep_pin_list_for_block_ports
budget_latency_on_clocks	
	write set_clock_latency on clocks Type: <code>bool</code> Default: true Edit: Yes Reference: budget_latency_on_clocks

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budget_local_latency	
	Write user specific set_clock_latency constraints, provided through set_partition_user_constraints_file command, in partition constraint files Type: bool Default: false Edit: Yes Reference: budget_local_latency
budget_local_uncertainty	
	Use local uncertainty specified through set_partition_user_constraints_file command for partitions Type: bool Default: false Edit: Yes Reference: budget_local_uncertainty
budget_make_negative_input_delay_zero	
	To be backward compatible where negative set input delays were made zero Type: bool Default: true Edit: Yes Reference: budget_make_negative_input_delay_zero
budget_make_reduced_timing_graph	
	Enable running ART before Budgeting. Type: bool Default: false Edit: Yes
budget_master_clone	
	controls the tool behaviour for master clones Type: enum Enum Values: master_only merged unique_view_per_hinst Default: master_only Edit: Yes Reference: budget_master_clone
budget_no_false_paths_for_unconstrained_ports	
	do not create false paths for unconstrained ports Type: bool Default: false Edit: Yes Reference: budget_no_false_paths_for_unconstrained_ports
budget_no_hold_view	
	Disable hold view budgeting Type: bool Default: false Edit: Yes Reference: budget_no_hold_view
budget_no_setup_view	
	Disable setup view budgeting Type: bool Default: false Edit: Yes Reference: budget_no_setup_view
budget_report_dir	
	Alternate directory name for justify reports when -justify is given with create_timing_budget Type: string Default: budget_justify Edit: Yes Reference: budget_report_dir

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budget_report_negative_slack_on_ports	
	save warnings for ports having slack less than the specified value Type: double Default: -1e+06 Edit: Yes Reference: budget_report_negative_slack_on_ports
budget_report_or_update_budget	
	Enable the collection of data to use the update_timing_budget command Type: bool Default: false Edit: Yes Reference: budget_report_or_update_budget
budget_sdc_contents	
	option to control SDC contents Type: enum Enum Values: full push_down_only boundary_constraints_only no_boundary_condition no_boundary_constraints Default: full Edit: Yes Reference: budget_sdc_contents
budget_stage_based_fanout_driving_factor	
	Fanout count of the pin, above which we will add buffer stages in Timing Graph Type: int Default: -1 Edit: Yes Reference: budget_stage_based_fanout_driving_factor
budget_stage_based_partition_multiplier	
	List of partition hinsts with its corresponding multiplication factor Type: string Default: "" Edit: Yes Reference: budget_stage_based_partition_multiplier
budget_stage_based_weight	
	List of weights given for {Sequential Combinational Buffer} cells Type: string Default: 1 1 1 Edit: Yes Reference: budget_stage_based_weight
budget_use_boundary_condition	
	Mechanism used for finding buffers to be used as driver/load Type: enum Enum Values: optimized template actual empty Default: template Edit: Yes Reference: budget_use_boundary_condition
budget_virtual_clock	
	Create additional virtual clocks for budgeting Type: bool Default: true Edit: Yes Reference: budget_virtual_clock
budget_virtual_opt_engine	

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	Specifies the virtual optimization engine to be used Type: enum Enum Values: none early_timing_engine proto Default: none Edit: Yes Reference: budget_virtual_opt_engine
budget_write_constraints_for_clock_output_ports	
	Write constraints for clock output ports Type: bool Default: false Edit: Yes Reference: budget_write_constraints_for_clock_output_ports
budget_write_false_path_for_hold	
	write false path -hold for all I/Os in setup budgeting Type: bool Default: false Edit: Yes Reference: budget_write_false_path_for_hold
budget_write_latency_per_clock	
	write latency per clock instead of per port Type: bool Default: false Edit: Yes Reference: budget_write_latency_per_clock
bump_grids	
	List of bump_grid object. Type: obj(bump_grid)* Default: "" Edit: No
bumps	
	Short-cut for [get_db current_design .bumps] Type: obj(bump)* Default: "" Edit: No
bus_sink_groups	
	A group of sinks (loads) that some floorplan and routing commands use to control adding buffers and routing for a bus. See 'help *bus_sink*' for a list of commands related to this object. Type: obj(bus_sink_group)* Default: {} Edit: No
busses	
	Short-cut for [get_db current_design .busses] Type: obj(bus)* Default: "" Edit: No
categories	
	available categories of root attributes Type: string Default: "" Edit: No
ccopt_auto_limit_insertion_delay_factor	

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	CCOpt attempts to keep the insertion delays of each clock tree a fixed multiple of the longest insertion delay that would result from a global skew approach. This multiple can be modified by design timing and the presence of other clock trees, but will start at a fixed fraction above the global skew insertion delay. This attribute specifies that fixed fraction. Valid values: real See also: .ccopt_auto_limit_insertion_delay_factor_skew_group Type: double Default: 1.5 Edit: Yes Reference: ccopt_auto_limit_insertion_delay_factor
ccopt_merge_clock_gates	
	If set to true, clock gate merging is enabled. If this is false, merging of all clock gates is disabled, including clock gates which may have been cloned by CTS. Note that this attribute has no impact on 'ccopt_design -cts'. See also attribute cts_merge_clock_gates. Valid values: true false Type: bool Default: true Edit: Yes Reference: ccopt_merge_clock_gates
ccopt_merge_clock_logic	
	If set to true, clock logic merging is enabled. If this is false, merging of all clock logics is disabled, including clock logics which may have been cloned by CTS. Note that this attribute has no impact on 'ccopt_design -cts'. See also attribute cts_merge_clock_logic. Valid values: true false Type: bool Default: true Edit: Yes Reference: ccopt_merge_clock_logic
change_cap_precision	
	Change cap precision to 0.001ff. Type: bool Default: false Edit: Yes Reference: change_cap_precision
check_ac_limit_additional_individual_violation_report	
	Report calculated duty ratio in addition with applied duty ratio in report file. Type: bool Default: false Edit: Yes Reference: check_ac_limit_additional_individual_violation_report
check_ac_limit_avg_recovery	
	override QRC tech em_recover factor for all layers used in lavg limits. Type: double Default: 1.0 Edit: Yes Reference: check_ac_limit_avg_recovery
check_ac_limit_check_thermal_aware_em	

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	check thermal aware EM Type: <code>bool</code> Default: false Edit: Yes
check_ac_limit_current_file	
	Specify current file to provide currents for cell/instance's pin. Type: <code>string</code> Default: "" Edit: Yes Reference: check_ac_limit_current_file
check_ac_limit_current_scale_factor	
	Scale Signal EM current, syntax: -current_scale_factor { [rms <rms_value>] [peak <peak_value>] [avg <avg_value>] } Type: <code>string</code> Default: "" Edit: Yes Reference: check_ac_limit_current_scale_factor
check_ac_limit_current_scale_table	
	Specifies the layer-based scale table for current Type: <code>string</code> Default: "" Edit: Yes Reference: check_ac_limit_current_scale_table
check_ac_limit_default_freq_for_unconstrained_nets	
	Specifies the frequency for EM calculation when a net has no defined frequency or a defined frequency of 0Hz in the design. Unit: Hz. Type: <code>double</code> Default: 1e+06 Edit: Yes Reference: check_ac_limit_default_freq_for_unconstrained_nets
check_ac_limit_delta_temperature	
	maximum change in temperature allowed in units of Celsius. Used in the QRC tech for RMS limits. Default=5. Type: <code>double</code> Default: 5.0 Edit: Yes Reference: check_ac_limit_delta_temperature
check_ac_limit_delta_temperature_layer_list	
	Specifies layer based delta temperature for EM RMS currnet limit analysis Type: <code>string</code> Default: "" Edit: Yes Reference: check_ac_limit_delta_temperature_layer_list
check_ac_limit_detailed	
	Generates a detailed report containing information for all signal nets, including those without violations Type: <code>bool</code> Default: false Edit: Yes Reference: check_ac_limit_detailed
check_ac_limit_effort_level	
	Enum_list legal values are: low, medium, high. Default: low. Type: <code>enum</code> Enum Values: low medium high Default: low Edit: Yes Reference: check_ac_limit_effort_level
check_ac_limit_em_cdf_percentage	

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	Specify em cumulative distribution factor (cdf) percentage. Type: double Default: 1e-09 Edit: Yes Reference: check_ac_limit_em_cdf_percentage
check_ac_limit_em_limit_scale_factor	
	Scale Signal EM limit, syntax: -em_limit_scale_factor { [rms <rms_value>] [peak <peak_value>] [avg <avg_value>] } Type: string Default: "" Edit: Yes Reference: check_ac_limit_em_limit_scale_factor
check_ac_limit_em_limit_scale_table	
	Specifies the layer-based scale table for em_limit Type: string Default: "" Edit: Yes Reference: check_ac_limit_em_limit_scale_table
check_ac_limit_em_res_width	
	Specifies width used to get EM limit. enum_list legal values are: drawn, silicon. Default: drawn. Type: enum Enum Values: drawn silicon Default: drawn Edit: Yes Reference: check_ac_limit_em_res_width
check_ac_limit_em_temperature	
	Specifies the temperature used to lookup the temperature scaling factor. By default, no scaling is done. Unit: Celsius. Type: double Default: -1.0 Edit: Yes Reference: check_ac_limit_em_temperature
check_ac_limit_em_temperature_layer_list	
	Specifies layer based EM temperature for EM AVG current limit analysis Type: string Default: "" Edit: Yes Reference: check_ac_limit_em_temperature_layer_list
check_ac_limit_em_threshold	
	The value is the ratio of signal current (avg/peak/rms) to the respective EM limit. Default: 1.0. Type: double Default: 1.0 Edit: Yes Reference: check_ac_limit_em_threshold
check_ac_limit_enable_seb	
	Enable SEB calculation Type: bool Default: false Edit: Yes Reference: check_ac_limit_enable_seb
check_ac_limit_env_temperature	
	environment temperature. Unit: Celsius. Type: double Default: -1.0 Edit: Yes Reference: check_ac_limit_env_temperature

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check_ac_limit_extraction_tech_file	
	Specify qrcTechFile name. Type: string Default: "" Edit: Yes Reference: check_ac_limit_extraction_tech_file
check_ac_limit_force_hold_view	
	Turn this on to force hold view. Type: bool Default: false Edit: Yes Reference: check_ac_limit_force_hold_view
check_ac_limit_handle_pin_obs_via	
	Turn on to handle pin obs via. Type: bool Default: false Edit: Yes Reference: check_ac_limit_handle_pin_obs_via
check_ac_limit_ict_em_models	
	Specify ict EM file to provide EM rule if EM rule not exists in qrcTechFile. Type: string Default: "" Edit: Yes Reference: check_ac_limit_ict_em_models
check_ac_limit_ircx_models	
	Specify ircx model file. Type: string Default: "" Edit: Yes Reference: check_ac_limit_ircx_models
check_ac_limit_lifetime	
	Specifies the hours of operation used to lookup the lifetime scaling factor. By default, no scaling is done. Unit: hour. Type: double Default: 87600.0 Edit: Yes Reference: check_ac_limit_lifetime
check_ac_limit_max_error	
	Specifies the maximam number of errors to report. Type: int Default: 10000 Edit: Yes Reference: check_ac_limit_max_error
check_ac_limit_method	
	Specifies the current waveform calculation method, check one or more of lrms, lpeak, and lavg limits. enum_list legal values are: rms, peak, avg. Default: rms. Type: string Default: "" Edit: Yes Reference: check_ac_limit_method
check_ac_limit_min_peak_duty_ratio	

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	Do not check lpeak for nets with duty ratio below <duty_ratio>. default = 0.05, range 0.0 to 1.0. Type: double Default: -1.0 Edit: Yes Reference: check_ac_limit_min_peak_duty_ratio
check_ac_limit_min_peak_freq	
	Do not check lpeak for nets with frequency below <freq> in units of hertz. default = 1e6. Unit: Hz. Type: double Default: 1e+06 Edit: Yes Reference: check_ac_limit_min_peak_freq
check_ac_limit_net_file	
	Specifies from file the net to check in signalEM analysis Type: string Default: "" Edit: Yes Reference: check_ac_limit_net_file
check_ac_limit_nets	
	Specifies whether to check named or selected nets Type: string Default: "" Edit: Yes Reference: check_ac_limit_nets
check_ac_limit_out_file	
	Specifies the report file for the violation data Type: string Default: "" Edit: Yes Reference: check_ac_limit_out_file
check_ac_limit_peak_td_method	
	Specifies Td calculation method for lpeak check. enum_list legal values are: effective_width_from_integration, effective_half_peak_width, max_equivalent_dc_peak, sum_half_peak_width. Default: effective_width_from_integration. Type: enum Enum Values: effective_width_from_integration effective_half_peak_width max_equivalent_dc_peak sum_half_peak_width Default: effective_width_from_integration Edit: Yes Reference: check_ac_limit_peak_td_method
check_ac_limit_read_detail_delta_temperature_file	
	Specify detail delta temperature file for thermal aware checking. Type: string Default: "" Edit: Yes Reference: check_ac_limit_read_detail_delta_temperature_file
check_ac_limit_report_db	
	Enable writing a db file to store data for GUI display. Type: bool Default: false Edit: Yes Reference: check_ac_limit_report_db
check_ac_limit_seb_lifetime	
	Specifies the hours of operation. By default, it's 5 years. Unit : hour. Type: double Default: 43800.0 Edit: Yes Reference: check_ac_limit_seb_lifetime

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check_ac_limit_seb_table	
	Specify seb table file. Type: string Default: "" Edit: Yes Reference: check_ac_limit_seb_table
check_ac_limit_seb_temperature	
	Specifies the temperature of operation. By default, it's 110c. Unit: Celsius. Type: double Default: -273.0 Edit: Yes Reference: check_ac_limit_seb_temperature
check_ac_limit_selected	
	Specifies whether to check named or selected nets Type: bool Default: false Edit: Yes Reference: check_ac_limit_selected
check_ac_limit_simulation_net	
	Specifies a list of nets to use simulation mode Type: string Default: "" Edit: Yes Reference: check_ac_limit_simulation_net
check_ac_limit_simulation_net_file	
	Specifies a file contained a list of nets to use simulation mode Type: string Default: "" Edit: Yes Reference: check_ac_limit_simulation_net_file
check_ac_limit_skip_category_mode	
	Specify the mode that under which condition a net should be skipped and which category a skipped net belongs to. Type: int Default: 1 Edit: Yes Reference: check_ac_limit_skip_category_mode
check_ac_limit_skip_net	
	Specifies the net to be skipped in signalEM analysis Type: string Default: "" Edit: Yes Reference: check_ac_limit_skip_net
check_ac_limit_skip_net_file	
	Specifies the net to be skipped in signalEM analysis Type: string Default: "" Edit: Yes Reference: check_ac_limit_skip_net_file
check_ac_limit_toggle	
	Specifies the toggle rate for signal nets Type: double Default: 1.0 Edit: Yes Reference: check_ac_limit_toggle

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check_ac_limit_use_db_freq	
	Uses the database frequency value as the effective frequency per net Type: bool Default: false Edit: Yes Reference: check_ac_limit_use_db_freq
check_ac_limit_use_qrc_tech	
	Turn this on to force lrms checks to use the QRC tech file rather than the LEF tech. If either lpeak or lavg is also checked, then all checks will use the QRC tech file, including lrms. Type: bool Default: true Edit: Yes Reference: check_ac_limit_use_qrc_tech
check_ac_limit_use_rms_delta_t	
	Turn on to use rms delta temperature. Type: bool Default: false Edit: Yes Reference: check_ac_limit_use_rms_delta_t
check_ac_limit_view	
	Specifies the name of the view Type: string Default: "" Edit: Yes Reference: check_ac_limit_view
check_design_across_hierarchy	
	Enables checking of unloaded, undriven, multi-driven and constant pins across design hierarchies Type: bool Default: true Edit: Yes Reference: check_design_across_hierarchy
check_design_report_async_pins	
	Includes async pins in netlist design checks. Type: bool Default: true Edit: Yes Reference: check_design_report_async_pins
check_design_report_scan_pins	
	Includes scan pins in netlist design checks. Type: bool Default: true Edit: Yes Reference: check_design_report_scan_pins
check_design_threshold_fanout	
	Upper threshold for netlist design checks. Type: int Default: 10000 Edit: Yes Reference: check_design_threshold_fanout
check_drc_area	

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	Specifies the coordinates of the area to verify Type: rect Default: 0.0 0.0 0.0 0.0 Edit: Yes Reference: check_drc_area
check_drc_check_illegal_trim_shapes	
	check illegal trim shapes. Type: bool Default: false Edit: Yes Reference: check_drc_check_illegal_trim_shapes
check_drc_check_only	
	Specifies what kind of shapes to be checked. Type: enum Enum Values: all regular special selected_net selected cell default Default: default Edit: Yes Reference: check_drc_check_only
check_drc_check_reverse	
	report CONVEXCORNER REVERSE violation. Type: bool Default: false Edit: Yes Reference: check_drc_check_reverse
check_drc_check_routing_halo	
	Check routing halo. Type: bool Default: false Edit: Yes Reference: check_drc_check_routing_halo
check_drc_check_routing_halo_corner	
	Check routing halo in the corner Type: bool Default: false Edit: Yes Reference: check_drc_check_routing_halo_corner
check_drc_disable_rules	
	Disable the rule that from the provided rule list. the rule list as follow: (jog2jog_spacing eol_spacing cut_spacing min_cut enclosure color min_step protrusion min_area out_of_die off_manufacturing_grid off_routing_track) Type: enum Enum Values: jog2jog_spacing eol_spacing cut_spacing min_cut enclosure color min_step protrusion min_area out_of_die off_manufacturing_grid off_routing_track Default: "" Edit: Yes Reference: check_drc_disable_rules
check_drc_enable_post_passive_fill	
	Check full fill trim. Type: bool Default: false Edit: Yes Reference: check_drc_enable_post_passive_fill
check_drc_exclude_pg_net	

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	Excludes the checking of power and ground nets. Type: bool Default: false Edit: Yes Reference: check_drc_exclude_pg_net
check_drc_ignore_cell_blockage	
	Ignore geometries of cell blockage. Type: bool Default: false Edit: Yes Reference: check_drc_ignore_cell_blockage
check_drc_ignore_fill_wire	
	Ignore metalfill wires when check drc. Type: bool Default: false Edit: Yes Reference: check_drc_ignore_fill_wire
check_drc_ignore_non_rectangle_shapes	
	ignore non-rectangle shapes. Type: bool Default: false Edit: Yes Reference: check_drc_ignore_non_rectangle_shapes
check_drc_ignore_trial_route	
	Ignore trial route when check drc. Type: bool Default: false Edit: Yes Reference: check_drc_ignore_trial_route
check_drc_inside_via_def	
	Check the cut spacing from the same via Type: bool Default: true Edit: Yes Reference: check_drc_inside_via_def
check_drc_layer_range	
	Checks between the range of metal layers including cut layers in between Type: string Default: "" Edit: Yes Reference: check_drc_layer_range
check_drc_limit	
	Specifies the maximum number of errors to report Type: int Default: 1000 Edit: Yes Reference: check_drc_limit
check_drc_max_wrong_way_halo	
	Specifies the maximum wrong way halo Type: double Default: -1.0 Edit: Yes Reference: check_drc_max_wrong_way_halo
check_drc_ndr_spacing	

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	Specifies what kind of non-default rule spacing to be checked Type: enum Enum Values: true false auto Default: auto Edit: Yes Reference: check_drc_ndr_spacing
check_drc_report	
	Specifies the report file that contains the violations information Type: string Default: "" Edit: Yes Reference: check_drc_report
check_drc_short_only	
	Exclude all DRC check except short. Type: bool Default: false Edit: Yes Reference: check_drc_short_only
check_drc_trim_length	
	Consider max length in the trim shape layer. Type: bool Default: false Edit: Yes Reference: check_drc_trim_length
check_drc_uncolored	
	report color change violation when the dpt layer shape is free color. Type: bool Default: false Edit: Yes Reference: check_drc_uncolored
check_drc_use_min_spacing_on_block_obs	
	Use min spacing to check for block OBS. Type: enum Enum Values: true false auto Default: auto Edit: Yes Reference: check_drc_use_min_spacing_on_block_obs
clock_gating_connect_test_clock_gating_types	
	Set the types of clock gates that will be considered for test connection. Type: enum Enum Values: all synthesis user Default: synthesis Edit: Yes Reference: clock_gating_connect_test_clock_gating_types
clock_gating_connect_test_consider_constant_enabled_as_connected	
	Clock gating test connection will consider a constant enabled clock gate as already connected. Type: bool Default: false Edit: Yes Reference: clock_gating_connect_test_consider_constant_enabled_as_connected
clock_gating_coverage_effort	

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	Effort level for clock gating enable discovery. Possible values are standard and high. Type: enum Enum Values: standard high Default: standard Edit: Yes Reference: clock_gating_coverage_effort
clock_gating_exceptions_aware	
	Check timing exceptions and copy onto flops during clock-gating. Type: bool Default: true Edit: Yes Reference: clock_gating_exceptions_aware
clock_gating_infer_enable	
	Use advanced algorithm to infer enable for clock-gating. Type: enum Enum Values: false true set_reset_flops_only Default: true Edit: Yes Reference: clock_gating_infer_enable
clock_gating_prefix	
	Set name prefix for clock-gating objects. Type: string Default: "" Edit: Yes Reference: clock_gating_prefix
clock_tree_source_groups	
	list of clock_tree_source_group Type: obj(clock_tree_source_group)* Default: "" Edit: No Reference: clock_tree_source_groups
clock_trees	
	list of clock_tree Type: obj(clock_tree)* Default: "" Edit: No Reference: clock_trees
clocks	
	clocks Type: obj(clock)* Allowed -index values: analysis_view Default: "" Edit: No
clp_consider_errors_as_warnings	
	Specify the CLP messages whose severity should be reduced to 'Warning'. Type: string Default: "" Edit: Yes
clp_enable_1801_hierarchical_bbox	
	To enable hierarchical bbox flow for 1801 based design. Type: bool Default: false Edit: Yes Reference: clp_enable_1801_hierarchical_bbox

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clp_ignore_level_shifter_high_to_low	
	Specify if high to low level shifter checks should be skipped. Type: bool Default: false Edit: Yes
cmd_file	
	The cmd file name of the program. It has each command run in the current session. It can be changed during the session to direct cmd file output to a different file. Type 'man log_file' for more details. Type: string Default: "" Edit: Yes Reference: cmd_file
combinational_sequential_merge_message_threshold	
	Merging message is printed when hier instance has cells more than this threshold value. Type: int Default: 10 Edit: Yes Reference: combinational_sequential_merge_message_threshold
constraint_modes	
	Returns the information about the constraint modes in the design. Type: obj(constraint_mode)* Default: "" Edit: No
control_logic_optimization	
	Controls the effort level for optimization of control logic (basic/advanced/none). Type: enum Enum Values: basic advanced none Default: basic Edit: Yes Reference: control_logic_optimization
cts_add_port_driver_max_distance_from_port	
	This attribute specifies the allowable distance, in um if no units suffix is specified, that a port driver is allowed to be placed from the port being buffered. If the driver cannot be added close enough to the port it will not be added at all. This attribute, if set, overrides any value set in the attribute cts_add_port_driver_max_distance_from_port_rows. Type: string Default: auto Edit: Yes Reference: cts_add_port_driver_max_distance_from_port
cts_add_port_driver_max_distance_from_port_rows	
	This attribute specifies the allowable distance, in standard cell row heights, that a port driver is allowed to be placed from the port being buffered. If the driver cannot be added close enough to the port it will not be added at all. The attribute cts_add_port_driver_max_distance_from_port, if set, overrides this value. Type: int Default: 50 Edit: Yes Reference: cts_add_port_driver_max_distance_from_port_rows
cts_adjacent_rows_legal	

Stylus Common UI Database Object Information

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	Defines the clock halo in the y direction. If set to true the y direction clock halo is zero and other clock instances are allowed in adjacent rows. The following attributes can be used to assign y direction clock halos within CCOpt: cts_cell_halo_y cts_adjacent_rows_legal cts_cell_halo_rows Only one of these attributes is used to determine the clock halo in the y direction. The following rules determine which: . If cts_cell_halo_y is set to a non-auto value, then this defines the y direction clock halo. The attributes cts_adjacent_rows_legal and cts_cell_halo_rows have no effect. . If cts_cell_halo_y is set to auto and cts_adjacent_rows_legal is set to a non-auto value then cts_adjacent_rows_legal defines the clock halo in the y direction. The attribute cts_cell_halo_rows has no effect. . If both cts_cell_halo_y and cts_adjacent_rows_legal are set to auto then cts_cell_halo_rows defines the clock halo in the y direction. Valid values: true false See also: . cts_cell_halo_y . cts_cell_halo_rows Type: string Default: false Edit: Yes Reference: cts_adjacent_rows_legal
cts_allow_non_std_clock_gate_inputs	
	If this attribute is set, CTS will allow the use of clock gates with non-standard pins. CCOpt considers the following pin types to be standard: clock pins, enable pins, test enable pins, retention pins and power gating pins. Before starting CTS CCOpt will emit a warning, indicating which pin(s) it considers non-standard. Type: bool Default: false Edit: Yes Reference: cts_allow_non_std_clock_gate_inputs
cts_allow_post_route_sizing_stack_via_rule_update	
	Whether to allow post-route resizing to make cell swaps which require stack via rules to change for a clock_tree pin. Enabling this may cause disruption to routed nets, i.e. ECO routing will have more to repair. Valid values: resize_allow_svr_change resize_dont_allow_svr_change Type: string Default: resize_dont_allow_svr_change Edit: Yes Reference: cts_allow_post_route_sizing_stack_via_rule_update
cts_balance_mode	
	Replace CCOpt mode setting cts_opt_type {full cluster trial}. If not full, causes CCOpt and CCOpt CTS to halt before final completion of the clock tree to facilitate clock tree inspection. The possible values for this attribute are as follows: - full - default value, a full CTS is performed. - cluster - a cluster-only CTS is performed. The clock tree has no balancing delay applied. - trial - The clock has only virtual (numeric annotation) balancing delays applied. Type: string Default: full Edit: Yes Reference: cts_balance_mode
cts_blackbox_default_driver_base_pin	
	Base pin that will be used for all timing and capacitance modeling for clock roots with a source pin at a blackbox output. Type: string Default: "" Edit: Yes Reference: cts_blackbox_default_driver_base_pin
cts_blackbox_default_load_base_pin	
	Base pin that will be used for all timing and capacitance modeling for clock sinks at blackbox inputs. Any cts_capacitance_override attribute will be used in preference to this. Type: string Default: "" Edit: Yes Reference: cts_blackbox_default_load_base_pin

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cts_buffer_cells

Specifies the buffer cells for CTS. If none are specified CCOpt will choose buffers from the libraries. Cell names may be specified as a Tcl list of names, or as a Tcl list of patterns to be expanded to match names. If set explicitly, CCOpt will ignore any dont_use settings for the cells specified. Different buffer cells may be specified for any combination of clock tree and power domain. To use different buffers for each net type set the cts_buffer_cells_top and cts_buffer_cells_leaf attributes . Some examples follow:
To specify buffer cells for all clock trees and all power domains:
set_db cts_buffer_cells {bufAX* bufBX*}
To specify buffer cells for a particular clock tree and all power domains:
set_db clock_tree:<clk> .cts_buffer_cells {bufX20 bufX18}
To specify buffer cells for a particular clock tree and power domain:
set_db clock_tree:<clk> .cts_buffer_cells -index {power_domain <pd>} {bufX12 bufX8}
Valid values: a list of library cell names, or a list of patterns to expand to library cell names
Type: [string](#)
Allowed -index values: power_domain
Default: ""
Edit: Yes
Reference: [cts_buffer_cells](#)

cts_buffer_cells_leaf

Specifies the buffer cells available for CTS to use on leaf nets. If none are specified CCOpt will use the same buffers as on trunk nets (as specified in the cts_buffer_cells attribute). Cell names may be specified as a Tcl list of names, or as a Tcl list of patterns to be expanded to match names. If set explicitly, CCOpt will ignore any dont_use settings for the cells specified. Different leaf buffer cells may be specified for any combination of clock tree and power domain. Valid values: a list of library cell names, or a list of patterns to expand to library cell names
Type: [string](#)
Allowed -index values: power_domain
Default: ""
Edit: Yes
Reference: [cts_buffer_cells_leaf](#)

cts_buffer_cells_top

Specifies the buffers cells available for CTS to use on top nets. If none are specified CCOpt will use the same buffers as on trunk nets (as specified in the cts_buffer_cells attribute). Cell names may be specified as a Tcl list of names, or as a Tcl list of patterns to be expanded to match names. If set explicitly, CCOpt will ignore any dont_use settings for the cells specified. Different top buffer cells may be specified for any combination of clock tree and power domain. Valid values: a list of library cell names, or a list of patterns to expand to library cell names
Type: [string](#)
Allowed -index values: power_domain
Default: ""
Edit: Yes
Reference: [cts_buffer_cells_top](#)

cts_cell_density

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	Specifies the clock halo in the x direction. Any x direction clock halo defined by this attribute is proportional to the cell width. The constant of proportionality is defined by the attribute value. For example: . If <code>cts_cell_density</code> = 0.25 then the x direction clock halo equals 3 * cell width. . If <code>cts_cell_density</code> = 0.5 then the x direction clock halo equals cell width. . If <code>cts_cell_density</code> = 0.75 then the x direction clock halo equals cell width / 3. . If <code>cts_cell_density</code> = 1 then the x direction clock halo equals zero. This attribute can specify the x direction clock halo for all clock trees via: <code>set_db cts_cell_density 0.8</code> This attribute can specify the x direction clock halo per-clock tree via: <code>set_db clock_tree:<clk> .cts_cell_density 0.9</code> The following attributes can be used to assign x direction clock halos within CCOpt: <code>cts_cell_halo_x</code> <code>cts_cell_density</code> <code>cts_cell_halo_sites</code> Only one of these attributes is used to determine the clock halo in the x direction. The following rules determine which: . If <code>cts_cell_halo_x</code> is set to a non-auto value, then this defines the x direction clock halo. The attributes <code>cts_cell_density</code> and <code>cts_cell_halo_sites</code> have no effect. . If <code>cts_cell_halo_x</code> is set to auto and <code>cts_cell_density</code> is set to a non-auto value then <code>cts_cell_density</code> defines the clock halo in the x direction. The attribute <code>cts_cell_halo_sites</code> has no effect. . If both <code>cts_cell_halo_x</code> and <code>cts_cell_density</code> are set to auto then <code>cts_cell_halo_sites</code> defines the clock halo in the x direction. Valid values: 0.01 to 1 See also: . <code>cts_cell_halo_x</code> . <code>cts_cell_halo_sites</code> Type: string Default: 0.75 Edit: Yes Reference: cts_cell_density
<code>cts_cell_halo_mode</code>	
	Specifies how clock halos are used to determine the minimum legal separation between a pair of clock instances. There are two possible modes 'max' and 'sum'. When set to 'max' the minimum legal separation is the larger of the two clock halos. When set to 'sum' the minimum legal separation is the sum of the two clock halos. Valid values: max sum See also: . <code>cts_cell_halo_x</code> . <code>cts_cell_halo_y</code> . <code>cts_cell_density</code> . <code>cts_adjacent_rows_legal</code> . <code>cts_cell_halo_sites</code> . <code>cts_cell_halo_rows</code> . <code>cts_effective_clock_halo_x</code> . <code>cts_effective_clock_halo_y</code> . <code>cts_effective_clock_halo_x_source</code> . <code>cts_effective_clock_halo_y_source</code> Type: string Default: max Edit: Yes Reference: cts_cell_halo_mode
<code>cts_cell_halo_rows</code>	

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	Specifies the clock halo in the y direction in rows for all clock cells. The following attributes can be used to assign y direction clock halos within CCOpt: cts_cell_halo_y cts_adjacent_rows_legal cts_cell_halo_rows Only one of these attributes is used to determine the clock halo in the y direction. The following rules determine which: . If cts_cell_halo_y is set to a non-auto value, then this defines the y direction clock halo. The attributes cts_adjacent_rows_legal and cts_cell_halo_rows have no effect. . If cts_cell_halo_y is set to auto and cts_adjacent_rows_legal is set to a non-auto value then cts_adjacent_rows_legal defines the clock halo in the y direction. The attribute cts_cell_halo_rows has no effect. . If both cts_cell_halo_y and cts_adjacent_rows_legal are set to auto then cts_cell_halo_rows defines the clock halo in the y direction. See also: . cts_cell_halo_y . cts_adjacent_rows_legal Type: int Default: 1 Edit: Yes Reference: cts_cell_halo_rows
cts_cell_halo_sites	
	Specifies the clock halo in the x direction in sites for all clock cells. The following attributes can be used to assign x direction clock halos within CCOpt: cts_cell_halo_x cts_cell_density cts_cell_halo_sites Only one of these attributes is used to determine the clock halo in the x direction. The following rules determine which: . If cts_cell_halo_x is set to a non-auto value, then this defines the x direction clock halo. The attributes cts_cell_density and cts_cell_halo_sites have no effect. . If cts_cell_halo_x is set to auto and cts_cell_density is set to a non-auto value then cts_cell_density defines the clock halo in the x direction. The attribute cts_cell_halo_sites has no effect. . If both cts_cell_halo_x and cts_cell_density are set to auto then cts_cell_halo_sites defines the clock halo in the x direction. See also: . cts_cell_halo_x . cts_cell_density Type: int Default: 4 Edit: Yes Reference: cts_cell_halo_sites
cts_cell_halo_x	
	Specifies the clock halo distance in the x direction. The default value of this attribute is auto. The following attributes can be used to assign x direction clock halos within CCOpt: cts_cell_halo_x cts_cell_density cts_cell_halo_sites Only one of these attributes is used to determine the clock halo in the x direction. The following rules determine which: . If cts_cell_halo_x is set to a non-auto value, then this defines the x direction clock halo. The attributes cts_cell_density and cts_cell_halo_sites have no effect. . If cts_cell_halo_x is set to auto and cts_cell_density is set to a non-auto value then cts_cell_density defines the clock halo in the x direction. The attribute cts_cell_halo_sites has no effect. . If both cts_cell_halo_x and cts_cell_density are set to auto then cts_cell_halo_sites defines the clock halo in the x direction. See also: . cts_cell_density . cts_cell_halo_sites Type: string Allowed -index values: power_domain Default: auto Edit: Yes Reference: cts_cell_halo_x
cts_cell_halo_y	

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Database Objects--root

	Specifies the clock halo distance in the y direction. The default value of this attribute is auto. The following attributes can be used to assign y direction clock halos within CCOpt: cts_cell_halo_y cts_adjacent_rows_legal cts_cell_halo_rows Only one of these attributes is used to determine the clock halo in the y direction. The following rules determine which: . If cts_cell_halo_y is set to a non-auto value, then this defines the y direction clock halo. The attributes cts_adjacent_rows_legal and cts_cell_halo_rows have no effect. . If cts_cell_halo_y is set to auto and cts_adjacent_rows_legal is set to a non-auto value then cts_adjacent_rows_legal defines the clock halo in the y direction. The attribute cts_cell_halo_rows has no effect. . If both cts_cell_halo_y and cts_adjacent_rows_legal are set to auto then cts_cell_halo_rows defines the clock halo in the y direction. See also: . cts_adjacent_rows_legal . cts_cell_halo_rows Type: string Allowed -index values: power_domain Default: auto Edit: Yes Reference: cts_cell_halo_y
cts_clock_gate_movement_limit	
	Each clock gate is restricted to a Manhattan ball centered on its original location with CTS flow. The radius of the ball is a multiple of the clock gate height. This controls the default value of that multiple. Type: string Default: 10 Edit: Yes Reference: cts_clock_gate_movement_limit
cts_clock_gating_cells	
	Specifies the clock gates for CTS. If none are specified CCOpt will choose clock gates from the libraries. Cell names may be specified as a Tcl list of names, or as a Tcl list of patterns to be expanded to match names. If set explicitly, CCOpt will ignore any dont_use settings for the cells specified. A list of lists can be used to specify separate clock gate sub-lists. All CCOpt transforms will resize clock gates within the clock gate sub-list they are initially part of. If an incoming clock gate is not part of any clock gate sub-list, it will be resized to the first clock gate sub-list. Different clock gates may be specified for any combination of clock tree and power domain. Some examples follow: To specify clock gates for all clock trees and all power domains: <pre>set_db cts_clock_gating_cells {cgAX* cgBX*}</pre> To specify clock gates for a particular clock tree and all power domains: <pre>set_db clock_tree:<clk> .cts_clock_gating_cells {cgX20 cgX18}</pre> To specify clock gates for a particular clock tree and power domain: <pre>set_db clock_tree:<clk> .cts_clock_gating_cells -index {power_domain <pd>} {cgX12 cgX8}</pre> To specify separate clock gate sub-lists: <pre>set_db .cts_clock_gating_cells {{cgX12A1 cgX8A1} {cgX12A2 cgX8A2}}</pre> Valid values: a list of library cell names, or a list of patterns to expand to library cell names, a list of lists to specify separate clock gate sub-lists Type: string Allowed -index values: power_domain Default: "" Edit: Yes Reference: cts_clock_gating_cells
cts_clock_source_cells	

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	Specifies the cells available for CTS to size clock sources if the <code>cts_size_clock_sources</code> attribute is set to true. If none are specified the tool will choose cells from the libraries. Cell names may be specified as a Tcl list of names, or as a Tcl list of patterns to be expanded to match names. If set explicitly, the tool will ignore any <code>dont_use</code> settings for the cells specified. Different cells may be specified for clock trees or power domains. Only clock sources that are buffers, inverters, logic and clock gating cells with a single output can be resized. Valid values: a list of library cell names, or a list of patterns to expand to library cell names Type: string Allowed -index values: <code>power_domain</code> Default: "" Edit: Yes Reference: cts_clock_source_cells
cts_clock_tree_source_group_clock_trees	
	A list of the clock trees relevant to this source group. Valid values: list <code>cts_clock_trees</code> Type: string Default: "" Edit: Yes Reference: cts_clock_tree_source_group_clock_trees
cts_clock_tree_source_input_max_transition_time	
	The slew which will be assumed at the input of the root driver. Valid values: double Type: double Allowed -index values: <code>delay_corner</code> Default: 0 Edit: Yes Reference: cts_clock_tree_source_input_max_transition_time
cts_cloning_effort	
	Specifies the amount of cloning during clustering. By default the cloning effort is high. Type: string Default: <code>high</code> Edit: Yes Reference: cts_cloning_effort
cts_cloning_inst_name_suffix	
	If set, the suffix will be used generally for all insts cloned by CTS, else the suffix will default to "clone". Type: string Default: <code>clone</code> Edit: Yes Reference: cts_cloning_inst_name_suffix
cts_cloning_inst_name_suffix_source_group_assignment	
	Controls the suffix used by multi-tap cloning. If set to a non-empty string, that string will be used when naming cloned instances created by multi-tap cloning. Otherwise, the suffix will be controlled by the attribute <code>cts_cloning_inst_name_suffix</code>. Type: string Default: "" Edit: Yes Reference: cts_cloning_inst_name_suffix_source_group_assignment
cts_delay_cells	

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	Specifies the delay cells available for CTS. If none are specified CCOpt will not use delay cells. Setting this attribute to the string 'auto' means that CCOpt will choose delay cells from the libraries to use. Cell names may be specified as a Tcl list of names, or as a Tcl list of patterns to be expanded to match names. If set explicitly, CCOpt will ignore any dont_use settings for the cells specified. Different delay cells may be specified for any combination of clock tree and power domain, or by omitting those arguments a global setting can be applied. Some examples follow: To specify delay cells for all clock trees and power domains: set_db cts_delay_cells {delayAX* delayBX*} To specify delay cells for a particular clock tree and all power domains: set_db clock_tree:<clk> .cts_delay_cells {delayX1 delayX2} To specify delay cells for a particular clock tree and power domain: set_db clock_tree:<clk> .cts_delay_cells -index {power_domain <pd>} {delayX2 delayX3} Valid values: a list of library cell names, or a list of patterns to expand to library cell names, or the string 'auto' Type: string Allowed -index values: power_domain Default: "" Edit: Yes Reference: cts_delay_cells
cts_detailed_cell_warnings	
	If set to true, CCOpt outputs detailed cell warning diagnostics when it encounters issues with library cell selection, power domains and/or signal levels. Valid values: true false Type: bool Default: false Edit: Yes Reference: cts_detailed_cell_warnings
cts_exit_if_no_placeable_area	
	Specifies that CTS should exit if the design has zero placeable area. Setting this attribute may be useful to temporarily work-around problems with row definition and/or blockages causing placeable area to be zero. Valid values: true false Type: bool Default: true Edit: Yes Reference: cts_exit_if_no_placeable_area
cts_exit_if_skew_target_over_constrained	
	Specifies that CCOpt should exit as current skew targets are likely to lead to poor results. By default, CCOpt computes a minimum skew target which should not lead to excessive buffering, and does not allow any skew target to be set lower than this minimum. This attribute overrides that check, and allows any skew target to be used. Valid values: true false Type: bool Default: true Edit: Yes Reference: cts_exit_if_skew_target_over_constrained
cts_exit_if_stage_delay_sigma_target_over_constrained	
	If set, allows any max stage delay sigma target to be set. If the target is too low, it is likely to lead to runtime problems. Type: bool Default: true Edit: Yes Reference: cts_exit_if_stage_delay_sigma_target_over_constrained
cts_fix_clock_sinks	

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	If set to true, we will DEF lock clock tree sinks after routing in addition to any clock node locking (fixed). If set to soft, we will DEF lock clock tree sinks after routing in addition to any clock node soft_locking (softFixed). Valid values: true false soft Type: string Default: false Edit: Yes Reference: cts_fix_clock_sinks
cts_flexible_htree_allow_mixed_sink_polarity	
	If set to true, the tool is allowed to create flexible H-trees with mixed sink polarities for sink pins and final cells, where the polarity is honored for the final cell (default false). Type: bool Default: false Edit: Yes
cts_flexible_htree_enable_metrics_reporting	
	A three valued enum comprising: false, post_route, true. Default is post_route. The values true/false enable/disable all reporting, while post_route enables only the report at the end of flexible H-Tree synthesis. Type: string Default: post_route Edit: Yes Reference: cts_flexible_htree_enable_metrics_reporting
cts_flexible_htree_placement_legalization_effort	
	The legalization effort for finding placement unblocked points on the synthesis grid for flexible H-trees. High placement legalization effort can avoid having to relax placement constraints when implementing H-trees but may lead to increased runtime. Valid values: low, high (default low) Type: string Default: low Edit: Yes Reference: cts_flexible_htree_placement_legalization_effort
cts_force_target_max_transition_time	
	Specifies that CCOpt should allow any transition target to be used, even if it is likely to lead to runtime problems. By default, CCOpt computes a minimum transition target which should not lead to runtime problems, and does not allow any transition target to be set lower than this minimum. This attribute overrides that check, and allows any transition target to be used. Valid values: true false Type: bool Default: false Edit: Yes Reference: cts_force_target_max_transition_time
cts_ignore_problematic_skew_as_result_of_dont_touch_nets	
	If set, sinks directly connected to nets that are causing unfixable skew problems will be ignored for skew balancing. Type: bool Default: false Edit: Yes Reference: cts_ignore_problematic_skew_as_result_of_dont_touch_nets
cts_inst_name_prefix	
	The name prefix of instances created by CTS. The default value is "CTS". The default names of instances are CTS_*. Valid values: string Type: string Default: CTS Edit: Yes Reference: cts_inst_name_prefix
cts_inverter_cells	

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	Specifies the inverter cells available for CTS. If none are specified CCOpt will choose inverters from the libraries. Cell names may be specified as a Tcl list of names, or as a Tcl list of patterns to be expanded to match names. If set explicitly, CCOpt will ignore any dont_use settings for the cells specified. Different inverter cells may be specified for any combination of clock tree and power domain. To use different inverters for each net type set the <code>cts_inverter_cells_top</code> and <code>cts_inverter_cells_leaf</code> attributes . Some examples follow: To specify inverter cells for all clock trees and all power domains: <code>set_db cts_inverter_cells {invAX* invBX*}</code> To specify inverter cells for a particular clock tree and all power domains: <code>set_db clock_tree:<clk> .cts_inverter_cells {invX20 invX18}</code> To specify inverter cells for a particular clock tree and power domain: <code>set_db clock_tree:<clk> .cts_inverter_cells -index {power_domain <pd>} {invX12 invX8}</code> Valid values: a list of library cell names, or a list of patterns to expand to library cell names Type: string Allowed -index values: <code>power_domain</code> Default: "" Edit: Yes Reference: cts_inverter_cells
<code>cts_inverter_cells_leaf</code>	
	Specifies the inverter cells available for CTS to use on leaf nets. If none are specified CCOpt will use the same inverters as on trunk nets (as specified in the <code>cts_inverter_cells</code> attribute). Cell names may be specified as a Tcl list of names, or as a Tcl list of patterns to be expanded to match names. If set explicitly, CCOpt will ignore any dont_use settings for the cells specified. Different leaf inverter cells may be specified for any combination of clock tree and power domain. Valid values: a list of library cell names, or a list of patterns to expand to library cell names Type: string Allowed -index values: <code>power_domain</code> Default: "" Edit: Yes Reference: cts_inverter_cells_leaf
<code>cts_inverter_cells_top</code>	
	Specifies the inverter cells available for CTS to use on top nets. If none are specified CCOpt will use the same inverters as on trunk nets (as specified in the <code>cts_inverter_cells</code> attribute). Cell names may be specified as a Tcl list of names, or as a Tcl list of patterns to be expanded to match names. If set explicitly, CCOpt will ignore any dont_use settings for the cells specified. Different top inverter cells may be specified for any combination of clock tree and power domain. Valid values: a list of library cell names, or a list of patterns to expand to library cell names Type: string Allowed -index values: <code>power_domain</code> Default: "" Edit: Yes Reference: cts_inverter_cells_top
<code>cts_load_capacitance_cells</code>	
	Specifies the load capacitance cells available for CTS. CTS will use cells from this collection for load capacitance optimizations. Cell names may be specified as a Tcl list of names, or as a Tcl list of patterns to be expanded to match names. Type: string Allowed -index values: <code>power_domain</code> Default: "" Edit: Yes Reference: cts_load_capacitance_cells
<code>cts_logic_cells</code>	

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	Specifies the clock logics for CTS. If none are specified CCOpt will choose clock logics from the libraries. Cell names may be specified as a Tcl list of names, or as a Tcl list of patterns to be expanded to match names. If set explicitly, CCOpt will ignore any dont_use settings for the cells specified. Different logic cells may be specified for any combination of clock tree and power domain. Some examples follow: To specify logic cells for all clock trees and all power domains: set_db cts_logic_cells {and* mux*} To specify logic cells for a particular clock tree and all power domains: set_db clock_tree:<clk> .cts_logic_cells {andX20 andX18} To specify logic cells for a particular clock tree and power domain: set_db clock_tree:<clk> .cts_logic_cells -index {power_domain <pd>} {andX12 andX8} Valid values: a list of library cell names, or a list of patterns to expand to library cell names Type: string Allowed -index values: power_domain Default: "" Edit: Yes Reference: cts_logic_cells
cts_manage_power_intent_violations	
	If this attribute is set, the CTS algorithm will work around power management illegalities in the clock tree, as opposed to failing with an error when it encounters them. This allows the clock tree to be synthesized, but any power management illegalities will remain in the exported design. Valid values: true false Type: bool Default: true Edit: Yes Reference: cts_manage_power_intent_violations
cts_max_fanout	
	The maximum fanout at any point in the clock tree. Valid values: integer ranged between 2 and 1000 inclusive Type: int Default: 100 Edit: Yes Reference: cts_max_fanout
cts_max_source_to_sink_net_length	
	The maximum routing length in microns between driving source pin and driven sink pin on each net that clock tree synthesis should observe. This constraint can be applied to either a pin, a clock tree, or a net type. By default (if this attribute is not set) no explicit clock tree net length constraint is enforced. However, other clock tree constraints such as maximum slew (transition) and maximum capacitance will indirectly limit the maximum net length. Valid values: double Type: string Default: top auto trunk auto leaf auto Edit: Yes Reference: cts_max_source_to_sink_net_length
cts_max_source_to_sink_net_length_leaf	
	The maximum routing length in microns between driving source pin and driven sink pin on each net that clock tree synthesis should observe. This constraint can be applied to either a pin, a clock tree, or a net type. By default (if this attribute is not set) no explicit clock tree net length constraint is enforced. However, other clock tree constraints such as maximum slew (transition) and maximum capacitance will indirectly limit the maximum net length. Valid values: double Type: string Default: auto Edit: Yes Reference: cts_max_source_to_sink_net_length_leaf

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cts_max_source_to_sink_net_length_top	
	The maximum routing length in microns between driving source pin and driven sink pin on each net that clock tree synthesis should observe. This constraint can be applied to either a pin, a clock tree, or a net type. By default (if this attribute is not set) no explicit clock tree net length constraint is enforced. However, other clock tree constraints such as maximum slew (transition) and maximum capacitance will indirectly limit the maximum net length. Valid values: double Type: string Default: auto Edit: Yes Reference: cts_max_source_to_sink_net_length_top
cts_max_source_to_sink_net_length_trunk	
	The maximum routing length in microns between driving source pin and driven sink pin on each net that clock tree synthesis should observe. This constraint can be applied to either a pin, a clock tree, or a net type. By default (if this attribute is not set) no explicit clock tree net length constraint is enforced. However, other clock tree constraints such as maximum slew (transition) and maximum capacitance will indirectly limit the maximum net length. Valid values: double Type: string Default: auto Edit: Yes Reference: cts_max_source_to_sink_net_length_trunk
cts_merge_clock_gates	
	If set to true, clock gate merging is enabled. If this is false, merging of all clock gates is disabled, including clock gates which may have been cloned by CTS. Note that this attribute only impacts 'ccopt_design -cts'. See also attribute ccopt_merge_clock_gates. Valid values: true false Type: bool Default: false Edit: Yes Reference: cts_merge_clock_gates
cts_merge_clock_logic	
	If set to true, clock logic merging is enabled. If this is false, merging of all clock logics is disabled, including clock logics which may have been cloned by CTS. Note that this attribute only impacts 'ccopt_design -cts'. See also attribute ccopt_merge_clock_logic. Valid values: true false Type: bool Default: false Edit: Yes Reference: cts_merge_clock_logic
cts_mixed_fanout_net_type	
	Controls how CCOpt considers nets that have fanout consisting partially but not entirely of sinks. By default, having any sinks (eg. DFFs) will make the net be considered leaf, but when set to trunk, a net has to drive only sinks to be considered leaf. For example, a net driving a clock gate and a DFF would no longer count as a leaf net. Valid values: leaf trunk Type: string Default: leaf Edit: Yes Reference: cts_mixed_fanout_net_type
cts_move_clock_gates	

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	If this attribute is set, the CTS algorithm will move clock gates that appear in the clock tree. Setting this attribute may cause the clock tree to have a lower insertion delay, but might break datapath timing. During optimization, this is not a problem, because the timing will be automatically recovered during the optimization process. If this attribute is false, CTS permits small movements of ICGs for legalization. Type: bool Default: true Edit: Yes Reference: cts_move_clock_gates
cts_move_logic	
	If this attribute is set, the CTS algorithm will move logic that appears in the clock tree. "Logic" does not include clock gates, buffers, and inverters in a clock tree, which are always moved unless they are locked, or clock generators that are above the root of the clock tree. Usually, this will affect multiplexers used for selecting one of a number of clocks, or for switching between a test clock and the main clock. Setting this attribute may cause the clock tree to have a lower insertion delay, but might break datapath timing. During optimization, this is not a problem, because the timing will be automatically recovered during the optimization process. Valid values: true false Type: bool Default: true Edit: Yes Reference: cts_move_logic
cts_net_name_prefix	
	The name prefix of nets created by CTS. The default value is "CTS". The default names of nets are CTS, CTS_1, CTS_2... ... Valid values: string Type: string Default: CTS Edit: Yes Reference: cts_net_name_prefix
cts_pin_insertion_delay_histogram_bin_size	
	This attribute specifies the preferred size of the histogram bins that should be used when reporting the distribution of pin insertion delays. The bin size is specified in time units and must be greater than zero. A value of auto (the default) leaves it to the tool to calculate a bin size. This attribute affects pin insertion delay histograms that are logged during various flow commands. It also determines the preferred histogram bin size that is used by the report_pin_insertion_delays command when an explicit -bin_size argument has not been specified. Valid values: auto, time value in library units Type: string Default: auto Edit: Yes Reference: cts_pin_insertion_delay_histogram_bin_size
cts_post_route_enable	
	Enable post-conditioning optimization after clock nets are routed. Valid values: true false Type: bool Default: true Edit: Yes Reference: cts_post_route_enable
cts_post_route_enable_routing_eco	
	If set to false, post-conditioning will skip its ECO-routing step. Valid values: true false Type: bool Default: false Edit: Yes Reference: cts_post_route_enable_routing_eco
cts_post_route_repair_drv	

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	If set to false, post-conditioning will skip its DRV-fixing step. Valid values: true false Type: bool Default: true Edit: Yes Reference: cts_post_route_repair_drv
cts_post_route_repair_skew_by_buffering	
	If set to true, post-conditioning will attempt skew-fixing using rebuffering. Valid values: true false Type: bool Default: false Edit: Yes Reference: cts_post_route_repair_skew_by_buffering
cts_primary_delay_corner	
	This specifies the delay corner in which clock tree balancing applies the slew and insertion delay targets. If more than one timing corner is defined, this must be set before running CCOpt. By default, this is set to the delay corner contained in the first member of the list of setup views provided to the set_analysis_view command. Valid values: corner name, or empty Type: string Default: "" Edit: Yes Reference: cts_primary_delay_corner
cts_primary_reporting_skew_groups	
	The primary skew groups used for reporting. By default, the value is specified as auto that automatically takes the skew group with maximum number of sinks as the primary reporting skew group. For invalid values, the default (auto) will be considered. Valid values: a list of existing skew group names, auto or none Type: string Default: auto Edit: Yes Reference: cts_primary_reporting_skew_groups
cts_primary_reporting_skew_groups_log_min_max_sinks	
	If set to on, the sinks with the shortest and longest paths in each primary reporting skew group will be logged. If set to logv, they will be logged only to the logv file. Valid values: on, off, logv Type: string Default: logv Edit: Yes Reference: cts_primary_reporting_skew_groups_log_min_max_sinks
cts_repair_congestion	
	If set, place_fix_congestion is called during CTS. This can reduce congestion and ease later routing issues. Valid values: true or false Type: bool Default: true Edit: Yes Reference: cts_repair_congestion
cts_repair_drv_by_buffering	
	If set, the standalone clock_post_route_repair command called after route_design and the clock_post_route_repair called inside opt_design - post_route will fix clock DRVs by adding buffers, otherwise the tool will not add buffers to fix DRVs. Valid values: true false Type: bool Default: false Edit: Yes Reference: cts_repair_drv_by_buffering
cts_report_skew_groups_only_with_targets	

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	The skew groups report (run using the <code>report_skew_groups</code> command) displays insertion delay, skew, and min/max path information for different combinations of skew group, timing corner, and early/late path. Set the <code>cts_report_skew_groups_only_with_targets</code> attribute to false (the default) to report on all skew group/timing corner/path combinations regardless of whether a skew target has been set. Set the <code>cts_report_skew_groups_only_with_targets</code> attribute to true to report only on skew group/timing corner/path combinations where a skew target has been set (either explicitly or using the 'auto' setting). Type: <code>bool</code> Default: false Edit: Yes Reference: cts_report_skew_groups_only_with_targets
cts_route_clock_tree_nets	
	Perform detailed routing, during the final implementation clock routing phase. Valid values: true or false Type: <code>bool</code> Default: true Edit: Yes Reference: cts_route_clock_tree_nets
cts_route_type_flexible_htree	
	Route type to use for flexible h-trees. Type: <code>string</code> Default: default Edit: Yes Reference: cts_route_type_flexible_htree
cts_route_type_flexible_htree_backside	
	Route type to use for backside flexible h-trees. Type: <code>string</code> Default: default Edit: Yes Reference: cts_route_type_flexible_htree_backside
cts_route_type_leaf	
	Specifies the route type. Setting this attribute binds an existing user-defined <code>route_type</code> to one or more types of clock tree nets. Binding a <code>route_type</code> to a type of clock tree nets means that all nets of that type (including the nets created by CTS) will be routed according to the specification of that <code>route_type</code>. In the most common usage, the <code>route_type</code> is bound to one of the three types of clock tree nets (top, trunk, or leaf) with the optional <code>-net_type</code> argument. Omitting the <code>-net_type</code> argument causes the <code>route_type</code> to be bound to all three types of clock tree nets. The optional <code>-clock_tree <pattern></code> argument limits the binding to the clock trees whose name matches <code><pattern></code>. Omitting the <code>-clock_tree</code> argument causes the binding to apply to all clock trees. For a <code>route_type</code> to be used in CTS, it must be bound to at least one net type. If net type is not bound to any <code>route_type</code>, a default <code>route_type</code> will be created for that net type at the start of CTS. Valid values: names of <code>route_types</code> created with <code>create_route_type</code> Type: <code>string</code> Default: default Edit: Yes Reference: cts_route_type_leaf
cts_route_type_top	

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	Specifies the route type. Setting this attribute binds an existing user-defined route_type to one or more types of clock tree nets. Binding a route_type to a type of clock tree nets means that all nets of that type (including the nets created by CTS) will be routed according to the specification of that route_type. In the most common usage, the route_type is bound to one of the three types of clock tree nets (top, trunk, or leaf) with the optional -net_type argument. Omitting the -net_type argument causes the route_type to be bound to all three types of clock tree nets. The optional -clock_tree <pattern> argument limits the binding to the clock trees whose name matches <pattern>. Omitting the -clock_tree argument causes the binding to apply to all clock trees. For a route_type to be used in CTS, it must be bound to at least one net type. If net type is not bound to any route_type, a default route_type will be created for that net type at the start of CTS. Valid values: names of route_types created with create_route_type Type: string Default: default Edit: Yes Reference: cts_route_type_top
cts_route_type_trunk	
	Specifies the route type. Setting this attribute binds an existing user-defined route_type to one or more types of clock tree nets. Binding a route_type to a type of clock tree nets means that all nets of that type (including the nets created by CTS) will be routed according to the specification of that route_type. In the most common usage, the route_type is bound to one of the three types of clock tree nets (top, trunk, or leaf) with the optional -net_type argument. Omitting the -net_type argument causes the route_type to be bound to all three types of clock tree nets. The optional -clock_tree <pattern> argument limits the binding to the clock trees whose name matches <pattern>. Omitting the -clock_tree argument causes the binding to apply to all clock trees. For a route_type to be used in CTS, it must be bound to at least one net type. If net type is not bound to any route_type, a default route_type will be created for that net type at the start of CTS. Valid values: names of route_types created with create_route_type Type: string Default: default Edit: Yes Reference: cts_route_type_trunk
cts_routing_preferred_layer_effort	
	Control preferred layer effort for clock routing. Type: string Default: standard Edit: Yes Reference: cts_routing_preferred_layer_effort
cts_safety_tmr_isolation_route_type	
	Sets a route type for nets driven by TMR buffers/inverters. These should be short single fanout nets and this route type is expected to be easily routable (eg. minimal shielding, spacing requirements) Type: string Default: default Edit: Yes Reference: cts_safety_tmr_isolation_route_type
cts_size_clock_gates	
	When set to true (the default), the CTS algorithm sizes clock gates that appear in the clock tree. Setting this attribute may cause the clock tree to have a lower insertion delay, but might change the cell types of logic gates in the clock tree, which in turn may require them to be moved slightly to find a legal location for the new cell. Type: bool Default: true Edit: Yes Reference: cts_size_clock_gates
cts_size_clock_sources	
	When set to true, CTS will try to size the clock source. Only clock sources that are buffers, inverters, logic and clock gating cells with a single output will be sized. The cells available for CTS to size clock sources can be set specified using the attribute 'cts_clock_source_cells'. Valid values: true false Type: bool Default: false Edit: Yes Reference: cts_size_clock_sources
cts_size_logic	

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	When set to true (the default), the CTS algorithm sizes logic that appears in the clock tree. "Logic" does not include clock gates, buffers, and inverters in a clock tree, which are always sized unless they are locked, or clock generators that are above the root of the clock tree. Usually, this affects multiplexers used for selecting one of a number of clocks, or for switching between a test clock and the main clock. Setting this attribute may cause the clock tree to have a lower insertion delay, but might change the cell types of logic gates in the clock tree, which in turn may require them to be moved slightly to find a legal location for the new cell. Valid values: true false Type: bool Default: true Edit: Yes Reference: cts_size_logic
cts_skew_group_report_columns	
	A Tcl list of columns to include in skew group reports produced by the skew group report. You can use this attribute to specify the columns you would like to include in the skew group report, and the order in which the columns should appear. Valid values: - capacitance - distance - event - fanout - increment - lib_cell - location - name - slew - status - time Type: string Default: name lib_cell event increment time slew capacitance location distance fanout status voltage Edit: Yes Reference: cts_skew_group_report_columns
cts_skew_group_report_histogram_bin_size	
	When set to a numeric value, that numeric value will be used as the histogram range size (in library units). For example, if the library time units are set to 1 nanosecond, a value of 0.010 for report_skew_groups_histogram_bin_size will result in histogram ranges of 10 picoseconds. When set to auto, the size of the histogram ranges are dependent on the skew targets that are set. If a skew target is set for a given half corner and skew group combination, then the histogram range size will be 10% of the skew target for that half corner and skew target combination. If no skew target is set for a half corner and skew group combination but a skew target is set for the primary half corner and skew group combination, then the histogram range size will be 10% of the skew target for the primary half corner and skew group combination. In the event that no skew targets are set and report_skew_groups_histogram_bin_size is set to auto, a default value of 10 picoseconds will be used for the histogram range size. Valid values: auto string Type: string Default: auto Edit: Yes Reference: cts_skew_group_report_histogram_bin_size
cts_spec_config_base_pin_trace_through_to	

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	Clock tree definition will, by default, not continue through certain types of cell arc (for instance, the clock to Q arc in a DFF). This attribute allows you to override this default behavior, permitting the clock tree to trace through all instances of such a cell. This attribute serves the same function as <code>trace_through_to</code>, except that here the clock path is specified at the level of the library cell. The attribute should be configured on the input library pin at which the clock will arrive. The value of the attribute specifies the output library pin to which the clock should propagate. The specified output pin must be another pin on the same library cell. The output pin may be specified either by its fully qualified name (i.e. inclusive of the cell name), or else simply by its local (cell-relative) name. There must be a pre-existing (library-defined) chain of one or more delay arcs that connect the input and output pins together. It is not possible to use <code>library_trace_through_to</code> to synthesize delay arcs. If multiple input pins are annotated on a given library cell, the value of <code>library_trace_through_to</code> at each of those pins must select the same output pin: i.e. the configuration must identify a single clock output for the cell. If multiple clock outputs are necessary then <code>library_trace_through_to</code> should not be used: instead for each instance of the library cell, define a generated clock tree at each of the clock-carrying outputs. If the configuration of <code>library_trace_through_to</code> settings for a given library cell does not meet these requirements, a warning will be issued and the settings for that cell will be ignored. All instances of the library cell will be affected by this setting. If both <code>trace_through_to</code> and <code>library_trace_through_to</code> are applicable at a given instance pin, the <code>trace_through_to</code> value will take precedence. Valid values: <code>base_pin</code> Type: <code>string</code> Default: <code>""</code> Edit: Yes Reference: <code>cts_spec_config_base_pin_trace_through_to</code>
<code>cts_spec_config_create_clock_tree_source_groups</code>	
	Causes <code>create_clock_tree_spec</code> to set up a clock tree source group for SDC clocks with multiple source pins. If this attribute is set to 'active', <code>create_clock_tree_spec</code> defines one clock tree for each source pin and then uses the <code>create_clock_tree_source_group</code> command to collect those clock trees together, so that CTS can distribute sinks between the clock trees. If this attribute is set to 'inactive' or 'true', the source groups are also created during <code>create_clock_tree_spec</code>, but they will be defined as inactive. Only active source groups will take part in sink allocation. If this attribute is set to 'false', no source groups are extracted from SDC multiple source pin clocks. Type: <code>string</code> Default: <code>false</code> Edit: Yes Reference: <code>cts_spec_config_create_clock_tree_source_groups</code>
<code>cts_spec_config_create_generator_skew_groups</code>	
	This attribute will cause the <code>create_clock_tree_spec</code> command to create skew groups for sequential generators and their adjacent registers. Such skew groups will be specified with the same highest rank so that they can be balanced from the other normal skew groups that share some sinks of them. The adjacent registers of a generator are registers that have a datapath timing path to talk with the generator directly. When this attribute is set to true, one skew group will be created per sequential generator instance, master clock and generated clock tree triple. The resulting skew groups will by default be named in the pattern: <code>_clock_gen_<master_clock_name>_<generator_local_name>_<optional_number>_<constraint_mode_name></code>. For example, for a pair of generators, with the same local name "reg_clkgen", CCOpt creates generated clock trees from the same master clock named "fclk" in a constraint mode named "func" the skew groups emitted into the clock tree specification file would be named: <code>_clock_gen_fclk_reg_clkgen_1/func</code> <code>_clock_gen_fclk_reg_clkgen_2/func</code> The prefix for the names of such skew groups is controlled by the <code>cts_spec_config_create_generator_skew_groups_name_prefix</code> CCOpt attribute and defaults to <code>"_clock_gen"</code> (the start underscore is used to group such skew groups at the end of any skew group listing ordered by name). Type: <code>bool</code> Default: <code>true</code> Edit: Yes Reference: <code>cts_spec_config_create_generator_skew_groups</code>

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cts_spec_config_create_generator_skew_groups_name_prefix	This attribute controls the skew group name prefix used for skew groups generated to balance generator flops with their adjacent flops. Default is "_clock_gen". The default has a start underscore at the beginning to cause listings of skew groups ordered by name to collect such skew groups together at the end of a list. Type: string Default: _clock_gen Edit: Yes Reference: cts_spec_config_create_generator_skew_groups_name_prefix
cts_spec_config_create_reporting_only_skew_groups	Controls whether spec creation will synthesize reporting-only skew groups. A reporting-only skew group is a skew group whose cts_skew_group_constrains attribute is set to 'none'. Such a skew group imposes no clock balancing constraint and will not be considered by CTS. The typical example is a generated clock that is synchronous to its master clock. The relationship between the two clocks implies that the sinks of both clocks should be balanced together. To model this, a clock spec created for such a design will have the skew group corresponding to the master clock span the domain of both the master clock and also the generated clock. The skew group of the generated clock is then a subset of the master clock skew group: as such it is a redundant constraint, and may be safely omitted. If is synthesized, such a skew group will be created as a reporting-only skew group. When this attribute is set to false, reporting-only skew groups are completely omitted from the generated spec. When set to true, reporting-only skew groups are included in the generated spec, but are marked with the cts_skew_group_constrains attribute set to 'none'. Either way, they impose no balancing constraint. Valid values: true false Type: bool Default: true Edit: Yes Reference: cts_spec_config_create_reporting_only_skew_groups
cts_spec_config_trace_multi_output_cells_with_single_clock_output	This attribute controls whether and how create_clock_tree_spec will extract through cells that have multiple outputs but where only one of those outputs has a clock emanating from it; call these types of instance multi-output single-clock instances. If the feature is disabled, spec creation will not trace through multi-output single-clock instances. The resulting clock spec will specify a generated clock tree on the clock-carrying outputs of every multi-output instance. Clock-carrying inputs that are connected to each clock output by way of timing arcs will be generator inputs to the corresponding generated clock tree. The presence of these generated clock trees may impact CTS operations on the instance such as sizing, moving, and cloning. If the feature is enabled, the resulting clock spec will include cts_spec_config_trace_through_to settings for multi-output single-clock instances. These settings will allow the instance to be treated as a normal clock instance, most commonly as a clock logic. In this case there will be no need for the generated clock tree on the clock-carrying output. The treatment of instances with multiple clock-carrying outputs is unchanged: on such an instance each clock-carrying output will take a generated clock tree, as before. The feature may be enabled conditionally, such that it is only applied to multi-output single-clock instances which are combinational. In this case multi-output single-clock instances which are sequential will continue to take a generated clock tree on the clock output. The possible values for this attribute are as follows: - false - The feature is completely disabled. - true - The feature is enabled for all cell types; both combinational and sequential. - combinational - The feature is enabled for combinational cells, but disabled for sequential cells. Valid values: false true combinational Type: string Default: combinational Edit: Yes Reference: cts_spec_config_trace_multi_output_cells_with_single_clock_output
cts_target_max_capacitance	

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	The target maximum capacitive load to allow during clock tree synthesis. This attribute specifies a maximum (combined pin and wire) capacitance that the clock tree synthesis algorithm will allow any given base_pin to drive in a given clock tree when driving a given net_type. It is specified in library units. It currently only constrains the primary delay corner capacitance values - other delay corners can be specified but will not be constrained. This attribute is applied in addition to the max_capacitance constraints read from the liberty library data - the tightest (lowest) of the constraint specified by this attribute and the constraint present in the liberty data will be used. It also does not apply at the root pins of clock trees - to constrain those nets the cts_clock_tree_source_max_capacitance CCOpt attribute should be used instead. Valid values: auto double Type: string Allowed -index values: delay_corner Default: top auto trunk auto leaf auto Edit: Yes Reference: cts_target_max_capacitance
cts_target_max_capacitance_leaf	
	The target maximum capacitive load to allow during clock tree synthesis. This attribute specifies a maximum (combined pin and wire) capacitance that the clock tree synthesis algorithm will allow any given base_pin to drive in a given clock tree when driving a given net_type. It is specified in library units. It currently only constrains the primary delay corner capacitance values - other delay corners can be specified but will not be constrained. This attribute is applied in addition to the max_capacitance constraints read from the liberty library data - the tightest (lowest) of the constraint specified by this attribute and the constraint present in the liberty data will be used. It also does not apply at the root pins of clock trees - to constrain those nets the cts_clock_tree_source_max_capacitance CCOpt attribute should be used instead. Valid values: auto double Type: string Allowed -index values: delay_corner Default: auto Edit: Yes Reference: cts_target_max_capacitance_leaf
cts_target_max_capacitance_top	
	The target maximum capacitive load to allow during clock tree synthesis. This attribute specifies a maximum (combined pin and wire) capacitance that the clock tree synthesis algorithm will allow any given base_pin to drive in a given clock tree when driving a given net_type. It is specified in library units. It currently only constrains the primary delay corner capacitance values - other delay corners can be specified but will not be constrained. This attribute is applied in addition to the max_capacitance constraints read from the liberty library data - the tightest (lowest) of the constraint specified by this attribute and the constraint present in the liberty data will be used. It also does not apply at the root pins of clock trees - to constrain those nets the cts_clock_tree_source_max_capacitance CCOpt attribute should be used instead. Valid values: auto double Type: string Allowed -index values: delay_corner Default: auto Edit: Yes Reference: cts_target_max_capacitance_top
cts_target_max_capacitance_trunk	

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	The target maximum capacitive load to allow during clock tree synthesis. This attribute specifies a maximum (combined pin and wire) capacitance that the clock tree synthesis algorithm will allow any given base_pin to drive in a given clock tree when driving a given net_type. It is specified in library units. It currently only constrains the primary delay corner capacitance values - other delay corners can be specified but will not be constrained. This attribute is applied in addition to the max_capacitance constraints read from the liberty library data - the tightest (lowest) of the constraint specified by this attribute and the constraint present in the liberty data will be used. It also does not apply at the root pins of clock trees - to constrain those nets the cts_clock_tree_source_max_capacitance CCOpt attribute should be used instead. Valid values: auto double Type: string Allowed -index values: delay_corner Default: auto Edit: Yes Reference: cts_target_max_capacitance_trunk
cts_target_max_stage_delay_sigma	
	The max per-stage SOCV delay sigma target to use for CTS. Type: string Allowed -index values: delay_corner Default: ignore Edit: Yes Reference: cts_target_max_stage_delay_sigma
cts_target_max_transition_time	
	The target slew used for clock tree synthesis. This attribute specifies a maximum slew time that the clock tree synthesis algorithm will allow in this clock tree, in library units. 'default' means 'auto' in primary half corner and 'ignore' in other half corners. If set to 'auto', CTS picks an appropriate value based on the collection of allowed buffer sizes and library parameters, although this may not give optimal quality of results. If set to 'ignore', CTS does not constrain the corner. Valid values: default auto ignore double Type: string Allowed -index values: delay_corner power_domain Default: top default trunk default leaf default Edit: Yes Reference: cts_target_max_transition_time
cts_target_max_transition_time_leaf	
	The target slew used for clock tree synthesis. This attribute specifies a maximum slew time that the clock tree synthesis algorithm will allow in this clock tree, in library units. 'default' means 'auto' in primary half corner and 'ignore' in other half corners. If set to 'auto', CTS picks an appropriate value based on the collection of allowed buffer sizes and library parameters, although this may not give optimal quality of results. If set to 'ignore', CTS does not constrain the corner. Valid values: default auto ignore double Type: string Allowed -index values: delay_corner power_domain Default: default Edit: Yes Reference: cts_target_max_transition_time_leaf
cts_target_max_transition_time_top	

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	The target slew used for clock tree synthesis. This attribute specifies a maximum slew time that the clock tree synthesis algorithm will allow in this clock tree, in library units. 'default' means 'auto' in primary half corner and 'ignore' in other half corners. If set to 'auto', CTS picks an appropriate value based on the collection of allowed buffer sizes and library parameters, although this may not give optimal quality of results. If set to 'ignore', CTS does not constrain the corner. Valid values: default auto ignore double Type: string Allowed -index values: delay_corner power_domain Default: default Edit: Yes Reference: cts_target_max_transition_time_top
cts_target_max_transition_time_trunk	
	The target slew used for clock tree synthesis. This attribute specifies a maximum slew time that the clock tree synthesis algorithm will allow in this clock tree, in library units. 'default' means 'auto' in primary half corner and 'ignore' in other half corners. If set to 'auto', CTS picks an appropriate value based on the collection of allowed buffer sizes and library parameters, although this may not give optimal quality of results. If set to 'ignore', CTS does not constrain the corner. Valid values: default auto ignore double Type: string Allowed -index values: delay_corner power_domain Default: default Edit: Yes Reference: cts_target_max_transition_time_trunk
cts_target_skew	
	This specifies the target skew for clock tree balancing. This may be set to a numeric value, or one of 'auto', 'ignore' or 'default'. If set to 'auto' this indicates that an appropriate skew target should be computed. If set to 'ignore' this indicates that skew should not be balanced for this corner/path combination. If unspecified then the value of this attribute is 'default'. If the value of the attribute is 'default' the target skew for late delays in the primary delay corner is interpreted as 'auto' and as 'ignore' otherwise. Valid values: default auto ignore double Type: string Allowed -index values: delay_corner Default: default Edit: Yes Reference: cts_target_skew
cts_timing_connectivity_based_skew_groups	

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	SDC false path assertions may render a generated SDC clock asynchronous to its master clock. Under such a condition, the sinks of the generated clock need not be balanced against the sinks of the master clock: there are no timing paths between the two clocks that can be affected by inter-clock skew. This balancing 'relaxation' is realized by adjusting the skew groups that are generated for the two clocks. This attribute specifies which SDC assertions are considered when deciding whether a generated clock is asynchronous to its master. Valid values for this attribute are as follows: off In this mode, every generated SDC clock is treated as being synchronous to its master SDC clock. clock_false_path Only Clock/clock set_false_path and set_clock_group assertions are considered. Valid values: off clock_false_path Type: string Default: off Edit: Yes Reference: cts_timing_connectivity_based_skew_groups
cts_timing_connectivity_based_skew_groups_balance_master_clocks	
	This attribute has no effect if cts_timing_connectivity_based_skew_groups is set to 'off'. With timing connectivity based skew groups enabled, this attribute controls how CCOpt will address the synchrony of disjoint (i.e. non-overlapping) master (i.e. non-generated) SDC clocks. When this attribute is set to false, all disjoint master clocks are assumed to be mutually asynchronous. The skew groups generated by create_clock_tree_spec will not constrain the sinks of two such clocks to be balanced together. When this attribute is set to true, the synchrony of disjoint master clocks is determined by consulting the SDC assertions in accordance with the value of attribute cts_timing_connectivity_based_skew_groups. Additional skew groups will be generated by create_clock_tree_spec, such that the sinks of each pair of synchronous disjoint master clocks will be balanced together by CTS. Valid values true false Type: bool Default: false Edit: Yes Reference: cts_timing_connectivity_based_skew_groups_balance_master_clocks
cts_timing_connectivity_info	

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	This attribute is populated by clock spec creation when timing connectivity based skew groups are enabled. Configuration of this attribute will be written to the generated spec. It documents the clock/clock balancing relationships that were determined for the production of timing connectivity based skew groups. This attribute is documenting only. If timing connectivity based skew groups are not enabled, this attribute is not populated by spec creation. The value of the attribute is a set of nested Tcl dictionaries, four levels in total. The top-level dictionary indicates the information category. Currently there is only one such category: 'clock_relationships'. The second level is keyed on constrained mode name; the lowest level dictionaries are both keyed on SDC clock name. These keys taken together identify two SDC clocks in a given constraint mode. The leaf value specifies the resolved balancing relationships between the two SDC clocks. It an enum with the following members: - direct - The two clocks are determined to belong to the same clock group and must be balanced together. - indirect - The two clocks do not belong to the same clock group; however by transitive closure they must balance together. Clocks in a direct or indirect balancing relationship will share one or more skew groups so as to ensure that their sinks are balanced together by CTS. Missing entries should be taken as 'need not balance'; i.e. the pair of SDC clocks have neither a direct nor indirect balancing relationship. The indirect balancing relationship can be explored in more detail using this same Tcl dictionary to probe out the transitive closure over the direct balancing relationships that link two indirect balance clocks together. For example if clock A directly balances with clock B only, and clock B directly balances with clock C only; then clock A indirectly balances with clock C. Below is an example dict value, corresponding to clock groups {clkA clkB} {clkB clkC} in constraint mode cm1, and clock groups {clkP clkQ} {clkR clkS} in constraint mode cm2. <pre>{clock_relationships {cm1 {clkA {clkB direct clkC indirect} clkB {clkA direct clkC direct} clkC {clkA indirect clkB direct}}} cm2 {clkP {clkQ direct} clkQ {clkP direct} clkR {clkS direct} clkS {clkR direct}}}}</pre> Notice that although they are in separate clock groups, clkA and clkC nevertheless indirectly balance. This is due to fact that clkB is present in both clock groups. Note also that we omit information about the relationship between clkP and clkS: they are 'need not balance'. Type: string Default: "" Edit: Yes Reference: cts_timing_connectivity_info
cts_top_fanout_threshold	
	Minimum number of transitive fanout in the clock tree for a net to be routed as a top net. Nets with at least this many sinks in their transitive fanout in the clock tree will have the special routing rules applied to them. Valid values: integer Type: string Default: unset Edit: Yes Reference: cts_top_fanout_threshold
cts_update_clock_latency	
	Determine whether to update IO latencies within ccopt_design. Valid values: true false Type: bool Default: true Edit: Yes Reference: cts_update_clock_latency
cts_use_inverters	

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	Specifies whether clock tree synthesis should prefer to use inverters rather than buffers when balancing the clock tree. If set to true, CTS will use inverters for the clock tree balancing process. If set to false, CTS will use the minimum number of levels of inverters required to maintain logical correctness. If set to auto (the default) CTS will build a tree that is built from buffers. A very small number of inverters may be inserted for surgical balancing optimizations. Valid values: auto true false Type: string Default: auto Edit: Yes Reference: cts_use_inverters
cts_use_receiver_model_capacitance_for_drv	
	If auto, CCOpt will use receiver model capacitance for drv if global variable timing_report_use_receiver_model_capacitance is true. If true, CCOpt will use receiver model capacitance for drv even if timing_report_use_receiver_model_capacitance is false. If false, CCOpt will not use receiver model capacitance for drv. Valid values: auto true false Type: string Default: false Edit: Yes Reference: cts_use_receiver_model_capacitance_for_drv
current_design	
	current design Type: obj(design) Default: "" Edit: No
current_tech	
	The tech for the current_design. This attribute is used to make it easier to type tech DPO names without the leading tech name. For example, if the current_tech = tech:tech1, then both "get_db layer:tech1/metal1" and "get_db layer:metal1" return the same layer object. current_tech cannot be directly modified—you must change current_design instead. Type: obj(tech) Default: "" Edit: No
cwd_setup_file	
	Developer setup file. Type: string Default: "" Edit: Yes Reference: cwd_setup_file
dataflow_hinsts	
	specify hinst to be placed Type: string Default: "" Edit: Yes
def_scan_chains	
	Short-cut for [get_db current_design .def_scan_chains]. Type: obj(def_scan_chain)* Default: "" Edit: No
delay_corners	
	Returns the information about the delay corners in the design. Type: obj(delay_corner)* Default: "" Edit: No
delaycal_accuracy_level	

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	set accuracy level for delay calculation Type: int Default: 0 Edit: Yes Reference: delaycal_accuracy_level
delaycal_advanced_accuracy_mode	
	advanced accuracy level Type: enum Enum Values: medium high ultra Default: medium Edit: Yes
delaycal_advanced_calculation_mode	
	Enable advanced calculation mode for base delay. Type: enum Enum Values: medium high Default: medium Edit: Yes
delaycal_advanced_node_pin_cap_settings	
	Enable advanced node pin cap settings. Type: bool Default: false Edit: Yes Reference: delaycal_advanced_node_pin_cap_settings
delaycal_advanced_pin_cap_mode	
	Enable advanced receiver pin cap mode for base delay. Type: enum Enum Values: 0 1 2 Default: 0 Edit: Yes Reference: delaycal_advanced_pin_cap_mode
delaycal_combine_mmmc	
	Specifies whether delay calculation runs are to be combined for delay calculation simulations Type: enum Enum Values: none early_late early_late_corner Default: early_late_corner Edit: Yes Reference: delaycal_combine_mmmc
delaycal_default_net_delay	
	set default net delay. Type: string Default: 1000ps Edit: Yes Reference: delaycal_default_net_delay
delaycal_default_net_load	
	set default net load. Type: string Default: 0.5pf Edit: Yes Reference: delaycal_default_net_load
delaycal_degrade_slew_on_early_nets	
	Controls interconnect slew degradation for early paths. Type: bool Default: true Edit: Yes Reference: delaycal_degrade_slew_on_early_nets

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delaycal_disable_positive_slack_shift	
	disable positive slack shift caused by BEOL variation. Type: bool Default: false Edit: Yes
delaycal_disable_randomcell_in_socv	
	disable randomcell sensitivity in socv Type: bool Default: false Edit: Yes
delaycal_early_irdrop_data_type	
	IRDrop EIV DB EivMethod for early corner Type: enum Enum Values: best_average worst best_average worst_average Default: worst Edit: Yes Reference: delaycal_early_irdrop_data_type
delaycal_enable_accurate_lle_analysis	
	enable accurate lle analysis Type: bool Default: false Edit: Yes
delaycal_enable_corner_based_beol	
	Enable Anchor corner based BEOL analysis Type: bool Default: false Edit: Yes
delaycal_enable_high_fanout	
	Enables the default net delay which will be annotated on high fanout nets Type: bool Default: false Edit: Yes Reference: delaycal_enable_high_fanout
delaycal_enable_input_slew_sensitivity_on_constraint	
	Enable input slew ensitivity on constraint Type: bool Default: false Edit: Yes Reference: delaycal_enable_input_slew_sensitivity_on_constraint
delaycal_enable_quiet_receivers_for_hold	
	Type: bool Default: false Edit: Yes Reference: delaycal_enable_quiet_receivers_for_hold
delaycal_enable_seperate_wire_var_sigma_multiplier	
	Allow seperate reporting sigma multiplier for wire variations. To be used along with option -wire_var_reporting_sigma_multiplier Type: bool Default: false Edit: Yes
delaycal_enable_si	

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	Enables SIAware delay calculation that also includes cross-talk induced delays. Type: enum Enum Values: true false Default: false Edit: Yes Reference: delaycal_enable_si
delaycal_enable_wire_load_model	
	Enables support of Liberty wire-load models and related SDC commands. Type: int Default: 0 Edit: Yes Reference: delaycal_enable_wire_load_model
delaycal_equivalent_waveform_model	
	Controls the equivalent waveform model to be used Type: enum Enum Values: none no_propagation propagation Default: none Edit: Yes Reference: delaycal_equivalent_waveform_model
delaycal_equivalent_waveform_model_for_timing_check	
	Enable EWM for timing check delay. Type: bool Default: false Edit: Yes Reference: delaycal_equivalent_waveform_model_for_timing_check
delaycal_equivalent_waveform_type	
	Equivalent waveform model type to be used Type: enum Enum Values: moments simulation Default: simulation Edit: Yes Reference: delaycal_equivalent_waveform_type
delaycal_honor_slew_propagate_constraint	
	Determines whether to propagate the slew from the disabled timing arcs to the output pin. Type: bool Default: true Edit: Yes Reference: delaycal_honor_slew_propagate_constraint
delaycal_ignore_net_load	
	Uses zero-cap, zero-resistance wire-load model for estimating delays. Type: bool Default: false Edit: Yes Reference: delaycal_ignore_net_load
delaycal_input_transition_delay	
	set default input transition time. Type: string Default: 0ps Edit: Yes Reference: delaycal_input_transition_delay
delaycal_irdrop_data_type	

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	IRDrop EIV DB EivMethod Type: enum Enum Values: best_average worst best average worst_average Default: worst Edit: Yes Reference: delaycal_irdrop_data_type
delaycal_irdrop_window_based	
	Fetch window based EIV values from EIV DB Type: enum Enum Values: none late early both Default: both Edit: Yes Reference: delaycal_irdrop_window_based
delaycal_late_irdrop_data_type	
	IRDrop EIV DB EivMethod for late corner Type: enum Enum Values: best_average worst best average worst_average Default: worst Edit: Yes Reference: delaycal_late_irdrop_data_type
delaycal_library_interpolation_mode	
	library scaling mode for interpolation flow Type: enum Enum Values: linear non_linear non_linear_advanced Default: linear Edit: Yes Reference: delaycal_library_interpolation_mode
delaycal_report_out_bound	
	Generates a report that contains a list of index values (input transition) in the delay tables that are beyond the index range. Type: bool Default: false Edit: Yes Reference: delaycal_report_out_bound
delaycal_separate_params_for_via	
	Create seperate parameters for Via variation Type: bool Default: false Edit: Yes
delaycal_signoff_alignment_settings	
	Turn on/off signoff alignment settings Type: bool Default: false Edit: Yes Reference: delaycal_signoff_alignment_settings
delaycal_skip_slew_merge_from_disabled_path	
	Enable slew merge from disable path Type: bool Default: false Edit: Yes Reference: delaycal_skip_slew_merge_from_disabled_path
delaycal_slew_out_bound_limit_high	

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	Limits the maximum slew used in the analysis to the specified value. Type: double Default: 3.40282e+38 Edit: Yes Reference: delaycal_slew_out_bound_limit_high
delaycal_slew_out_bound_limit_low	
	Limits the minimum slew used in the analysis to the specified value. Type: double Default: 0.5 Edit: Yes Reference: delaycal_slew_out_bound_limit_low
delaycal_socv_accuracy_mode	
	level-based-accuracy-effort in AAE for delays and slews Type: enum Enum Values: low medium high ultra Default: low Edit: Yes Reference: delaycal_socv_accuracy_mode
delaycal_socv_lvf_mode	
	Controls interpretation of LVF data. Type: enum Enum Values: moments early_late Default: early_late Edit: Yes Reference: delaycal_socv_lvf_mode
delaycal_socv_machine_learning_level	
	Enable socv machine learning mode Type: int Default: 0 Edit: Yes Reference: delaycal_socv_machine_learning_level
delaycal_socv_use_lvf_tables	
	Controls which type of variations are considered during analysis. Type: string Default: all Edit: Yes Reference: delaycal_socv_use_lvf_tables
delaycal_support_output_pin_cap	
	useOutputPinCap Type: bool Default: true Edit: Yes Reference: delaycal_support_output_pin_cap
delaycal_support_wire_load_model	
	Enables support of Liberty wire-load models and related SDC commands. Type: int Default: 0 Edit: Yes Reference: delaycal_support_wire_load_model
delaycal_timing_create_clock_use_ideal_slew	

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	When set to true, the software detects created clocks and uses zero input slew. Other slew values, including those set using set_annotated_transition command, will be ignored. When set to false, the software honors set_annotated_transition command settings. By default, this global variable is set to false. Type: <code>bool</code> Default: false Edit: Yes Reference: delaycal_timing_create_clock_use_ideal_slew
delaycal_use_default_delay_limit	
	set threshold to apply the default delay Type: <code>int</code> Default: 1000 Edit: Yes Reference: delaycal_use_default_delay_limit
delaycal_via_corner_sigma_multiplier	
	Provide scaling multiplier for corner scaling Type: <code>string</code> Default: "" Edit: Yes
delaycal_via_correlation_type	
	How to handle inter-layer correlation. accepted values corner Type: <code>enum</code> Enum Values: corner Default: corner Edit: Yes
delaycal_via_reset	
	reset via variation modes to default value Type: <code>bool</code> Default: false Edit: Yes
delaycal_wire_corner_sigma_multiplier	
	Provide scaling multiplier for corner scaling Type: <code>string</code> Default: "" Edit: Yes
delaycal_wire_correlation_type	
	How to handle inter-layer correlation. accepted values statistical binomial corner Type: <code>enum</code> Enum Values: statistical correlated binomial corner Default: statistical Edit: Yes
delaycal_wire_layer_avg_rpt	
	Enable layer averaging reporting(true/false) Type: <code>bool</code> Default: false Edit: Yes
delaycal_wire_rc_correlation_factor	
	same layer R&C correlation. accepted values (1 -1) Type: <code>int</code> Default: 1 Edit: Yes
delaycal_wire_reset	

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	reset wire variation modes to default value Type: bool Default: false Edit: Yes
delaycal_wire_use_nsigma_coupling_capacitance	
	Use mean+nSigma coupling caps(true/false) Type: bool Default: false Edit: Yes
delaycal_wire_use_nsigma_slew_for_timing_check	
	Use Wire Var aware slews for timing check lookup(true/false) Type: bool Default: false Edit: Yes
delaycal_wire_var_reporting_sigma_multiplier	
	Deterministic sigma multiplier for reporting beol variations Type: double Default: 0.0 Edit: Yes
delaycal_wire_variation_correlation_mode	
	Specify wire variatio inter layer correlation Type: enum Enum Values: statistical binomial Default: statistical Edit: Yes
delaycal_wire_variation_rc_correlation	
	Same layer RC correlation for wire variation Type: double Default: 1.0 Edit: Yes
delaycal_wire_wire_cell_variation_correlation	
	Whether to RSS or add the random-cell variation and wire variation for arrivals and slack(true/false) Type: bool Default: false Edit: Yes
delete_flops_on_preserved_net	
	Allows to delete the flops connected to a preserved net. Type: bool Default: true Edit: Yes Reference: delete_flops_on_preserved_net
delete_hier_insts_on_preserved_net	
	Allows to delete the hierarchical instances connected to a preserved net. Type: bool Default: true Edit: Yes Reference: delete_hier_insts_on_preserved_net
delete_unloaded_insts	
	Control deletion of unloaded hierarchical instances. Type: bool Default: true Edit: Yes Reference: delete_unloaded_insts

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delete_unloaded_seqs	
	Control deletion of unloaded sequential instances. Type: bool Default: true Edit: Yes Reference: delete_unloaded_seqs
design_backside_bottom_routing_layer	
	Specifies the highest backside lef layer name for global and detail routing. Type: string Default: "" Edit: Yes Reference: design_backside_bottom_routing_layer
design_backside_top_routing_layer	
	Specifies the lowest backside lef layer name for global and detail routing. Type: string Default: "" Edit: Yes Reference: design_backside_top_routing_layer
design_bottom_routing_layer	
	Specifies the lowest LEF layer name or layer number for global and detail routing. Layer number is from the LEF layer sequence. For example, 2 is equivalent to the second routing layer defined in LEF. Type: string Default: "" Edit: Yes Reference: design_bottom_routing_layer
design_compressed_pg_db	
	Specifies whether the compressed pg feature is enabled. When it's true, read_db will automatically compress the PG data when load the design. You can use "get_db current_design .is_pg_compressed" to check if PG in this design is compressed. Type: bool Default: true Edit: Yes Reference: design_compressed_pg_db
design_cong_effort	
	Specify congestion effort level Type: enum Enum Values: low medium high auto Default: auto Edit: Yes Reference: design_cong_effort
design_dual_rail_via_pitch	
	Min stacked-via pitch for dual std-cell power-rails on first and third routing layers. Space separated min stacked-via pitch for dual std-cell power-rails on first and third routing layers. Type: string Default: "" Edit: Yes Reference: design_dual_rail_via_pitch
design_early_clock_flow	
	Enable early clock flow Type: bool Default: false Edit: Yes Reference: design_early_clock_flow
design_early_pba_mode	

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	Specify early PBA mode Type: enum Enum Values: none high Default: none Edit: Yes Reference: design_early_pba_mode
design_express_route	
	With "-flow_effort express", should routing-stage be forced. Type: bool Default: false Edit: Yes Reference: design_express_route
design_flow_effort	
	Specify flow effort level Type: enum Enum Values: express standard extreme Default: standard Edit: Yes Reference: design_flow_effort
design_high_freq_interposer_flow	
	Specifies the flow is a high frequency interposer flow. When set to true, set_integration_route_constraint will auto turn off the options which are not suitable for interposer routing. Type: bool Default: false Edit: Yes Reference: design_high_freq_interposer_flow
design_ideal_hold_fixing	
	Enable ideal hold fixing flow Type: bool Default: false Edit: Yes Reference: design_ideal_hold_fixing
design_ignore_followpin_vias	
	if true, ignore followpin vias during detailed placement, optimization and routing. Type: bool Default: false Edit: Yes Reference: design_ignore_followpin_vias
design_merge_trim_shapes	
	Specify the setting for trim shapes merge. layers, specify the trim layer list which need merge trim shapes. gap, the trim shapes when end to end gap is less the value. max_trim_num, specify the max trim shapes number are merged. masks, specify the trim shapes color need to be merged. Type: string Default: "" Edit: Yes Reference: design_merge_trim_shapes
design_optimization_density_screen_margin	
	Increase a margin on top of original density screen for optimization Type: double Default: 0.1 Edit: Yes Reference: design_optimization_density_screen_margin
design_pessimistic_mode	

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	Enable pessimistic mode Type: enum Enum Values: true false Default: false Edit: Yes Reference: design_pessimistic_mode
design_power_effort	
	Specify power effort level Type: enum Enum Values: none low high Default: none Edit: Yes Reference: design_power_effort
design_slack_weighting_method	
	Specify slack weighting to be used in implementation flow Type: enum Enum Values: unity_weighting view_based_weighting Default: unity_weighting Edit: Yes Reference: design_slack_weighting_method
design_tech_node	
	set design tech node Type: enum Enum Values: N22 N12 N10 N7 N7Plus N6 N5 N5PlusPlus N4 N3 N3E N2 A16 A14 A12 S11 S10 SF8 SF7 SF5 SF4 SF3 SF2 SF1.4 G7 G5 ICF I7 I5 P1222 P1276 P1277 P1278 P1278x P1280 20A 18A C12 C7 unspecified Default: unspecified Edit: Yes Reference: design_tech_node
design_top_routing_layer	
	Specifies the highest LEF layer name or layer number for global and detail routing. Layer number is from the LEF layer sequence. For example, 2 is equivalent to the second routing layer defined in LEF. Type: string Default: "" Edit: Yes Reference: design_top_routing_layer
design_trim_grid_group	
	Specifies the GROUP name of which set of trim metal grid to be used for placement and routing. See the LEF documentation on the TRIMMETALTRACK keyword for more details. Type: string Default: "" Edit: Yes Reference: design_trim_grid_group
designs	
	All the designs Type: obj(design)* Default: "" Edit: No
dft_1500_hierarchical_parent_child_wrapping	
	Wrap parent in presence of wrapped child hierarchy. Type: bool Default: false Edit: Yes
dft_1500_hierarchical_softcore_wrapping	

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	Ignore wrapped soft-core while wrapping the top level ports. Type: bool Default: false Edit: Yes
dft_add_mux_on_pre_connected_ctl_si	
	Adds a MUX when SI pin of a CTL Segment is pre-connected to a functional pin. Type: bool Default: true Edit: Yes Reference: dft_add_mux_on_pre_connected_ctl_si
dft_allow_dwc_in_top	
	Do not print segment and instance info in report_scan_registers. Type: bool Default: false Edit: Yes Reference: dft_allow_dwc_in_top
dft_apply_sdc_constraints	
	Enables generation and application of SDC constraints for DFT constructs. Type: bool Default: false Edit: Yes
dft_atpg_executable	
	Specifies Encounter Test executable called from Genus. Type: string Default: "" Edit: Yes Reference: dft_atpg_executable
dft_atpg_setup_status	
	This will tell the status of prepare for test point analysis command. Type: string Default: "" Edit: No
dft_auto_create_chains_as_internal	
	Auto-create scan chains as internal. Type: bool Default: false Edit: Yes
dft_auto_identify_shift_register	
	Automatically identifies functional shift register segments. Type: bool Default: false Edit: Yes Reference: dft_auto_identify_shift_register
dft_boundary_cell_module_prefix	
	Specifies the prefix to be added to the boundary scan cell modules. Type: string Default: "" Edit: Yes Reference: dft_boundary_cell_module_prefix
dft_boundary_scan_timing_mode_name	

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	DFT boundary scan timing mode name (default: DFT_BOUNDARY_SCAN_MODE). Type: string Default: DFT_BOUNDARY_SCAN_MODE Edit: Yes Reference: dft_boundary_scan_timing_mode_name
dft_capture_11496_reciever_output	
	Avoids inserting 11496 bsr capture logic on receiver output if founds false. Type: bool Default: true Edit: Yes Reference: dft_capture_11496_reciever_output
dft_capture_timing_mode_name	
	DFT capture timing mode name (default: DFT_CAPTURE_MODE). Type: string Default: DFT_CAPTURE_MODE Edit: Yes Reference: dft_capture_timing_mode_name
dft_check_cfg_mode_aware	
	Allow check_dft_rules to check segments for each configuration mode. Type: bool Default: false Edit: Yes Reference: dft_check_cfg_mode_aware
dft_clock_waveform_divide_fall	
	Default falling edge denominator for test clocks. Type: int Default: 100 Edit: Yes Reference: dft_clock_waveform_divide_fall
dft_clock_waveform_divide_period	
	Default period denominator for test clocks. Type: int Default: 1 Edit: Yes Reference: dft_clock_waveform_divide_period
dft_clock_waveform_divide_rise	
	Default rising edge denominator for test clocks. Type: int Default: 100 Edit: Yes Reference: dft_clock_waveform_divide_rise
dft_clock_waveform_fall	
	Default falling edge numerator (fraction of period) for test clocks. Type: int Default: 90 Edit: Yes Reference: dft_clock_waveform_fall
dft_clock_waveform_period	
	Default period for test clocks (in picoseconds). Type: double Default: 50000.0 Edit: Yes Reference: dft_clock_waveform_period
dft_clock_waveform_rise	

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	Default rising edge numerator (fraction of period) for test clocks. Type: <code>int</code> Default: 50 Edit: Yes
dft_compression_2d_aspect_ratio	
	Specifies the aspect ratio of the 2D compression grid to insert. Type: <code>double</code> Default: no_value Edit: Yes
dft_compression_2d_decomp_pipeline_distance	
	Specifies the number of 2D compression grid cells between two decompressor pipelines. Type: <code>int</code> Default: 0 Edit: Yes
dft_compression_2d_grid_max_x	
	Specifies the x-coordinate to the left of which the 2D compression grid will be inserted. Type: <code>string</code> Default: no_value Edit: Yes
dft_compression_2d_grid_max_y	
	Specifies the y-coordinate below which the 2D compression grid will be inserted. Type: <code>string</code> Default: no_value Edit: Yes
dft_compression_2d_grid_min_x	
	Specifies the x-coordinate to the right of which the 2D compression grid will be inserted. Type: <code>string</code> Default: no_value Edit: Yes
dft_compression_2d_grid_min_y	
	Specifies the y-coordinate above which the 2D compression grid will be inserted. Type: <code>string</code> Default: no_value Edit: Yes
dft_compression_auto_create	
	Specifies to auto-create any required compression test signals that are missing. Type: <code>bool</code> Default: false Edit: Yes
dft_compression_channel_length	
	Specifies the compression channel length to insert. Type: <code>int</code> Default: 0 Edit: Yes
dft_compression_comp_pipeline_max_xor_depth	
	Specifies the maximum XOR depth between two compressor pipelines. Type: <code>int</code> Default: 0 Edit: Yes
dft_compression_compressor_type	

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	Specifies the compressor type to insert (xor misr). Type: string Default: xor Edit: Yes
dft_compression_decomp_pipeline_max_xor_depth	
	Specifies the maximum XOR depth between two decompressor pipelines. Type: int Default: 0 Edit: Yes
dft_compression_decompressor_type	
	Specifies the decompressor type to insert (elastic xor). Type: string Default: elastic Edit: Yes
dft_compression_elastic_parallel_load_support	
	Specify to enable parallel loading of Elastic decompressor. Type: bool Default: false Edit: Yes
dft_compression_elasticity_ratio	
	Specifies the SmartScan deserializer ratio to insert for Elastic decompression. Type: int Default: 1 Edit: Yes
dft_compression_extest_decompressor_type	
	Specifies the extest decompressor type to insert (xor elastic). Type: string Default: xor Edit: Yes
dft_compression_fullscan_support	
	Specifies to insert fullscan muxing to concatenate the compression channels. Type: bool Default: false Edit: Yes
dft_compression_lp_gating_sharing_ratio	
	Specifies how many compression channels are to share a common low power gating register. Type: int Default: 1 Edit: Yes
dft_compression_lp_gating_support	
	Specifies to insert low power compression channel gating. Type: bool Default: false Edit: Yes
dft_compression_mask_and_lp_gating_unload_support	
	Specifies to build unload paths for mask and low power gating registers. Type: bool Default: true Edit: Yes
dft_compression_mask_sharing_ratio	

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	Specifies how many compression channels are to share a common mask register. Type: int Default: 1 Edit: Yes
dft_compression_mask_support	
	Specifies to insert compression (wide1) masking. Type: bool Default: false Edit: Yes
dft_compression_masken_pipeline_depth	
	Specifies the number of pipeline stages to insert on the compression mask enable signal. Type: int Default: 0 Edit: Yes
dft_compression_num_extest_scanin	
	Specifies the number of scan inputs to use for extest compression. Type: int Default: 0 Edit: Yes
dft_compression_num_extest_scanout	
	Specifies the number of scan outputs to use for extest compression. Type: int Default: 0 Edit: Yes
dft_compression_num_scanin	
	Specifies the number of scan inputs to auto-create for compression. Type: int Default: 0 Edit: Yes Reference: dft_compression_num_scanin
dft_compression_num_scanout	
	Specifies the number of scan outputs to auto-create for compression. Type: int Default: 0 Edit: Yes Reference: dft_compression_num_scanout
dft_compression_opcg_unload_support	
	Specifies to build unload paths for OPGC programming registers. Type: bool Default: true Edit: Yes
dft_compression_post_2d_sdc_file	
	Specifies the SDC file to read in after 2D compression is inserted. Type: in_file Default: "" Edit: Yes
dft_compression_post_2d_sdc_mode_name	
	Specifies the name of the timing mode to which the specified SDC file will be applied. Type: string Default: "" Edit: Yes
dft_compression_ratio	

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	Specifies the compression ratio to insert. Type: int Default: 0 Edit: Yes
dft_compression_scanin_pipeline_depth	
	Specifies the number of pipeline stages to insert on every compression scan input. Type: int Default: 0 Edit: Yes
dft_compression_scanout_pipeline_depth	
	Specifies the number of pipeline stages to insert on every compression scan output. Type: int Default: 0 Edit: Yes
dft_compression_serial_load_support	
	Specifies to enable loading/unloading the Elastic decompressor, the MISR, the mask register, or the low power gating register serially. Type: bool Default: false Edit: Yes
dft_configuration_modes	
	List of all dft_configuration_modes Type: obj(dft_configuration_mode)* Default: "" Edit: No
dft_disable_wrapper_inside_non_scan_elem	
	Disable wrapper from being inserted inside a dft_dont_scan hierarchy. Type: bool Default: true Edit: Yes
dft_dont_merge_multibit_lockup	
	Sets dont merge multibit on lockups inserted by connect_scan_chains, add_lockup_element. Type: bool Default: true Edit: Yes Reference: dft_dont_merge_multibit_lockup
dft_dont_wrap_if_shared_threshold_exceed	
	Stop port wrapping if shared threshold exceed. Type: bool Default: false Edit: Yes
dft_enable_flop_placement_in_test_point_insertion	
	Enable physical placement of testpoint flop during testpoint insertion. Requires MOD400 license. Type: bool Default: false Edit: Yes
dft_enable_wir_function_check	
	Only test signals with certain functions are allowed in the WIR. Disabling this check may result in invalid or unsupported configurations. Type: bool Default: true Edit: Yes Reference: dft_enable_wir_function_check
dft_exclude_internal_flops_from_shared_wrapper_threshold	

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	Exclude the internal flops from shared wrapper threshold calculation. Type: bool Default: false Edit: Yes
dft_exclude_tdrcl_fail_seg	
	Exclude abstract segments failing tdrcl checks while connect scan chains. Type: bool Default: false Edit: Yes Reference: dft_exclude_tdrcl_fail_seg
dft_extended_scandef	
	Produce extended scanDEF output to encompass almost all the scan bits in a design. Type: bool Default: true Edit: Yes Reference: dft_extended_scandef
dft_fence_slow_speed_domains	
	Fence slow speed domains. Type: bool Default: false Edit: Yes Reference: dft_fence_slow_speed_domains
dft_generate_atpg_no_testpoint_file	
	Generates a file for Modus that specifies where testpoints cannot be inserted { auto auto_plus_user user_only none }. Type: string Default: auto Edit: Yes Reference: dft_generate_atpg_no_testpoint_file
dft_identify_internal_test_clocks	
	Identify internal test clocks as output pins of multi input gates {false true no_cgic_hier}. Type: enum Enum Values: false true no_cgic_hier Default: false Edit: Yes Reference: dft_identify_internal_test_clocks
dft_identify_non_boundary_shift_registers	
	Determines whether identify_shift_register_scan_segments will exclude boundary registers which can be converted to shared IEEE 1500 wrappers from shift register identification. Type: bool Default: false Edit: Yes
dft_identify_shared_wrapper_cells	
	Process generic design to identify flops to be used as shared wrapper cells. Type: bool Default: false Edit: Yes
dft_identify_test_signals	
	Specifies whether TDRC should auto identify test signals. The default value is true. Type: bool Default: true Edit: Yes Reference: dft_identify_test_signals
dft_identify_top_level_test_clocks	

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	Specifies whether TDRC should auto identify top level test clocks. The default value is true. Type: bool Default: true Edit: Yes Reference: dft_identify_top_level_test_clocks
dft_identify_xsource_violations_from_timing_models	
	Identifies x-source violations from output pins of timing models. Type: bool Default: false Edit: Yes Reference: dft_identify_xsource_violations_from_timing_models
dft_ignore_dont_scan_for_shared_wrapper_processing	
	Exclude non scan flops while marking wrapper cells. Type: bool Default: false Edit: Yes
dft_ignore_non_scan_for_wrapper_processing	
	Ignore non scan logic for wrapper insertion. Type: bool Default: false Edit: Yes
dft_ignore_segment_mode_name	
	Ignore scan segment mode name values during scan chain connection. Type: bool Default: true Edit: Yes
dft_include_controllable_pins_in_abstract_model	
	Enable writing out controllable pins in abstract model {allmodes none test_setup}. Type: enum Enum Values: none test_setup allmodes Default: allmodes Edit: Yes Reference: dft_include_controllable_pins_in_abstract_model
dft_include_test_signal_outputs_in_abstract_model	
	Enable writing out test signal outputs in abstract model. Type: bool Default: true Edit: Yes Reference: dft_include_test_signal_outputs_in_abstract_model
dft_insert_dedicated_inside_sink_hierarchy	
	Insert dedicated wrapper inside sink hierarchy. Type: bool Default: false Edit: Yes
dft_jtag_instance_name	
	Specifies the name of the the JTAG instance. Type: string Default: JTAG_MODULE Edit: Yes Reference: dft_jtag_instance_name
dft_jtag_module_name	

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	Specifies the name of the the JTAG module. Type: string Default: JTAG_MACRO Edit: Yes Reference: dft_jtag_module_name
dft_lbist_capture_timing_mode_name	
	DFT LBIST capture timing mode name (default: DFT_LBIST_CAPTURE_MODE). Type: string Default: DFT_LBIST_CAPTURE_MODE Edit: Yes Reference: dft_lbist_capture_timing_mode_name
dft_lbist_shift_timing_mode_name	
	DFT LBIST shift timing mode name (default: DFT_LBIST_SHIFT_MODE). Type: string Default: DFT_LBIST_SHIFT_MODE Edit: Yes Reference: dft_lbist_shift_timing_mode_name
dft_license_scheme	
	Dft tool selection {Modus Encounter_Test}. Type: enum Enum Values: Modus Encounter_Test Default: Modus Edit: Yes Reference: dft_license_scheme
dft_modedef_internal	
	Use internal modedef file for build testmode when writing out the ATPG scripts. Type: bool Default: false Edit: Yes Reference: dft_modedef_internal
dft_modus_version	
	Set the Modus version. Type: double Default: 22.11 Edit: Yes Reference: dft_modus_version
dft_opcg_block_input_to_flop_paths	
	Propagate DFT clock information from primary inputs. Type: bool Default: false Edit: Yes Reference: dft_opcg_block_input_to_flop_paths
dft_opcg_domain_blocking	
	Enable OPCG domain blocking. Type: bool Default: false Edit: Yes Reference: dft_opcg_domain_blocking
dft_opcg_timing_mode_name	
	DFT OPCG timing mode name (default: DFT_OPCG_MODE). Type: string Default: DFT_OPCG_MODE Edit: Yes Reference: dft_opcg_timing_mode_name

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dft_optimize_chain_wirelength_level	
	Level of optimization needed for physical scan flow, chain wirelength. Appropriate values for this attribute are low , medium or high. Type: string Default: medium Edit: Yes
dft_physical_aware_test_points	
	Enable physically aware test point sharing. Type: bool Default: true Edit: Yes
dft_physical_pd_aware_scan_connection	
	Allow physically aware power domain scan connection. Type: bool Default: false Edit: Yes Reference: dft_physical_pd_aware_scan_connection
dft_pmbist_jtag_timing_mode_name	
	DFT PMBIST jtag timing mode name (default: DFT_PMBIST_JTAG_MODE). Type: string Default: DFT_PMBIST_JTAG_MODE Edit: Yes Reference: dft_pmbist_jtag_timing_mode_name
dft_pmbist_mda_timing_mode_name	
	DFT PMBIST mda timing mode name (default: DFT_PMBIST_MDA_MODE). Type: string Default: DFT_PMBIST_MDA_MODE Edit: Yes Reference: dft_pmbist_mda_timing_mode_name
dft_power_aware_lockup_clock_driver	
	Enable power aware lockup clock driver. Type: bool Default: false Edit: Yes Reference: dft_power_aware_lockup_clock_driver
dft_power_aware_wrapper_insertion	
	Power aware wrapper insertion. Type: bool Default: true Edit: Yes
dft_prefix	
	Prefix to DFT-generated object names. Type: string Default: DFT_ Edit: Yes Reference: dft_prefix
dft_process_multibit_for_shared_wrapper	
	Process multibit cells for shared wrapper cells. Type: bool Default: false Edit: Yes
dft_propagate_test_signals_from_hookup_pins_only	

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	Specifies whether TDRC should propagate test signals values from both the pin and hookup pin or just the hookup pin. The default is false, indicating that test signals values are propagated from both the pin and hookup pin. Type: bool Default: false Edit: Yes Reference: dft_propagate_test_signals_from_hookup_pins_only
dft_report_empty_test_clocks	
	Have dft report commands report information on test clocks that do not drive any registers. Type: bool Default: false Edit: Yes Reference: dft_report_empty_test_clocks
dft_report_scan_register_quiet	
	Do not print segment and instance info in report_scan_registers. Type: bool Default: false Edit: Yes Reference: dft_report_scan_register_quiet
dft_rtl_insertion	
	Check whether DFT inserted RTL generation is requested or not. Type: bool Default: false Edit: Yes Reference: dft_rtl_insertion
dft_run_test_point_analysis_for_compression	
	Run modus directly from genus for testpoint analysis during write_dft_compression_test_points command. Requires MOD400 license. Type: bool Default: false Edit: Yes
dft_run_test_point_analysis_for_lbist	
	Run modus directly from genus for testpoint analysis during write_dft_lbist_test_points command. Requires MOD400 license. Type: bool Default: false Edit: Yes
dft_scan_power_domain_crossing_lockup_latch	
	When attribute is set, connect_scan_chains adds lockup latch at power domain crossing which are not skew safe. Type: bool Default: false Edit: Yes Reference: dft_scan_power_domain_crossing_lockup_latch
dft_scan_style	
	DFT scan style {muxed_scan clocked_issd_scan}. Type: enum Enum Values: muxed_scan clocked_issd_scan Default: muxed_scan Edit: Yes
dft_scanbit_waveform_analysis	
	Applies additional analysis if test clock waveform data is available to determine if a non-scan element in a chain is a scan bit or a lockup. Type: bool Default: false Edit: Yes Reference: dft_scanbit_waveform_analysis
dft_shared_common_logic_threshold	

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	Specifies the threshold for common logic to be included in shared threshold calculation. Type: int Default: 2 Edit: Yes
dft_shared_wrapper_through	
	Trace through the logic to identify the functional flop in the fanin or fanout of pin location for sharing. Type: string Default: "" Edit: Yes
dft_shift_register_identification_mode	
	Identify shift registers under mode setup {logical_only testmode_only test_setup test_setup_shiftenable testmode_shiftenable}. Type: enum Enum Values: logical_only testmode_only test_setup testmode_shiftenable test_setup_shiftenable Default: test_setup_shiftenable Edit: Yes Reference: dft_shift_register_identification_mode
dft_shift_register_max_length	
	Maximum sequential length of the shift register for auto-identification. Type: int Default: inf Edit: Yes
dft_shift_register_min_length	
	Minimum sequential length of the shift register for auto-identification. Type: int Default: 2 Edit: Yes Reference: dft_shift_register_min_length
dft_shift_register_with_mbc1	
	Support shift register with MBC1. Type: bool Default: false Edit: Yes Reference: dft_shift_register_with_mbc1
dft_shift_timing_mode_name	
	DFT shift timing mode name (default: DFT_SHIFT_MODE). Type: string Default: DFT_SHIFT_MODE Edit: Yes Reference: dft_shift_timing_mode_name
dft_tap_lockup_clock_from_adjacent	
	Connect lock up element clock to adjacent element clock pin. Type: bool Default: false Edit: Yes Reference: dft_tap_lockup_clock_from_adjacent
dft_tpi_sharing_scope	
	Controls the scope of sharing for the inserted test points. The valid options are within_hier_only, within_power_domains_only, across_power_domains. Type: string Default: within_power_domains_only Edit: Yes Reference: dft_tpi_sharing_scope
dft_true_time_flow	

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	Enables Encounter Test True Time Flow when writing out the ATPG scripts using 'write_et_atpg'. Type: bool Default: false Edit: Yes Reference: dft_true_time_flow
dft_use_abstract_segment_instance_power_domain_for_scan_connection	
	Use abstract segment instance power domain for scan connection. Type: bool Default: false Edit: Yes Reference: dft_use_abstract_segment_instance_power_domain_for_scan_connection
dft_use_multibit_register_width_for_threshold	
	Specifies whether multibit cell width to be used as threshold. Type: bool Default: false Edit: Yes
dft_use_wck_as_default_wrapper_clock	
	Use fanin/fanout clock if found else use wck clock if present. Type: bool Default: false Edit: Yes Reference: dft_use_wck_as_default_wrapper_clock
disable_when_checks	
	Disable the processing of long when condition having bus operands. Type: bool Default: true Edit: Yes Reference: disable_when_checks
distributed_mmmc_disable_reports_auto_redirection	
	setting 'true' will preserve path specified in report command. User need to specify unique path for each view to avoid all clients writing to same report file. With default setting (false), distributed mmmc run automatically updates report paths to a view unique directory structure. Type: string Default: false Edit: Yes Reference: distributed_mmmc_disable_reports_auto_redirection
double_cell_search_pattern	
	In case LEF is not provided, a search pattern is needed to identify which cells are double cells. Type: string Default: "" Edit: Yes Reference: double_cell_search_pattern
dp_analytical_opt	
	Specifies the effort level for global datapath optimization. Type: enum Enum Values: off standard extreme Default: standard Edit: Yes Reference: dp_analytical_opt
dp_area_mode	
	When true, enables datapath optimizations that focus on improving area results. Type: bool Default: false Edit: Yes Reference: dp_area_mode

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dp_csa	
	Controls CSA operation in 'synthesize -to_generic' (basic/none). Type: enum Enum Values: basic none Default: basic Edit: Yes Reference: dp_csa
dp_csa_factorize	
	Controls datapath csa_factorize in 'syn_generic'. Type: enum Enum Values: basic none Default: basic Edit: Yes Reference: dp_csa_factorize
dp_opt_power_opt	
	Controls datapath power optimization flow. Type: bool Default: false Edit: Yes Reference: dp_opt_power_opt
dp_opt_power_opt_scaling	
	Scaling for power used within datapath power optimization flow. Type: double Default: 1.0 Edit: Yes Reference: dp_opt_power_opt_scaling
dp_opt_toggle_skew_threshold_for_booth_encoding	
	Set an int value for the skew threshold when in the auto toggle rate mode. Type: int Default: 0 Edit: Yes Reference: dp_opt_toggle_skew_threshold_for_booth_encoding
dp_rename	
	Controls the datapath rewriting optimization level in 'synthesize -to_generic' (basic/advanced/none). Type: enum Enum Values: basic advanced none Default: basic Edit: Yes Reference: dp_rename
dp_sharing	
	To control the datapath sharing in 'synthesize -to_generic' (advanced/none). Type: enum Enum Values: advanced none inherited Default: advanced Edit: Yes Reference: dp_sharing
dp_speculation	
	Controls datapath speculation in 'syn_generic'. Type: enum Enum Values: basic none Default: basic Edit: Yes Reference: dp_speculation
dp_ungroup_during_syn_map	

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	Allow ungrouping of datapath and ChipWare hierarchies during syn_map. Type: <code>bool</code> Default: true Edit: Yes Reference: dp_ungroup_during_syn_map
dp_ungroup_separator	
	Separator for instance names of datapath components in ungrouped hierarchies. Type: <code>string</code> Edit: Yes Reference: dp_ungroup_separator
dpopt_balanced_csa	
	Enable additional rebalancing within datapath optimization and balanced CSA tree builder. Type: <code>bool</code> Default: false Edit: Yes
dpopt_combined_costing	
	Enable combined costing within datapath optimization. Type: <code>bool</code> Default: false Edit: Yes
drc_first	
	Use DRC costs first. Type: <code>bool</code> Default: false Edit: Yes Reference: drc_first
drc_max_cap_first	
	Use max_capacitance costs first. Type: <code>bool</code> Default: false Edit: Yes Reference: drc_max_cap_first
drc_max_fanout_first	
	Use max_fanout costs first. Type: <code>bool</code> Default: false Edit: Yes Reference: drc_max_fanout_first
drc_max_trans_first	
	Use max_transition costs first. Type: <code>bool</code> Default: false Edit: Yes Reference: drc_max_trans_first
eco_add_ports	
	Determines whether to add hierarchical module ports during buffering. It only works for manual ECO commands. Type: <code>bool</code> Default: true Edit: Yes Reference: eco_add_ports
eco_batch_mode	

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	Enter/Exit batch mode Type: bool Default: false Edit: Yes Reference: eco_batch_mode
eco_check_logical_equivalence	
	Do logical-equivalence checking Type: bool Default: true Edit: Yes Reference: eco_check_logical_equivalence
eco_delay_calculation_effort	
	Delay calculation effort level { low medium high } used in timing update Type: string Default: high Edit: Yes Reference: eco_delay_calculation_effort
eco_disable_change_net_name_for_flat_netlist	
	Specifies whether nets would be renamed in eco report of eco_compare_netlist for flat netlist. Type: int Default: 0 Edit: Yes Reference: eco_disable_change_net_name_for_flat_netlist
eco_disable_constraints_loading_from_clients	
	Disable Loading Constraints from Clients Type: bool Default: false Edit: Yes
eco_disable_derates_loading_from_clients	
	Disable Loading Derates from Clients Type: bool Default: false Edit: Yes
eco_disable_parasitic_loading_from_clients	
	Disable Loading Parasitics from Clients Type: bool Default: false Edit: Yes
eco_disable_power_format_loading_from_clients	
	Disable Loading Power Format from Clients Type: bool Default: true Edit: Yes
eco_hierarchical_module_list	
	set hierarchical module list Type: string Default: "" Edit: Yes
eco_honor_dont_touch	

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	Check dont_touch on instances, cells and nets Type: bool Default: true Edit: Yes Reference: eco_honor_dont_touch
eco_honor_dont_use	
	Skip library cells marked dont_use Type: bool Default: true Edit: Yes Reference: eco_honor_dont_use
eco_honor_fixed_status	
	Honor fixed (placement) instances Type: bool Default: true Edit: Yes Reference: eco_honor_fixed_status
eco_honor_fixed_wires	
	Honor fixed wires of nets Type: bool Default: true Edit: Yes Reference: eco_honor_fixed_wires
eco_honor_power_intent	
	Perform MSV checks during ECO: Do not allow resize of regular cell with always-on cell (and vice-versa), do not change cells belonging to different power domains, do not allow adding buffer to a power domain it doesn't belong to, do not buffer a cross power domain net, do not allow deletion of an always-on buffer, a level-shifter, or a buffer which will cause redundant isolation cell. Type: bool Default: true Edit: Yes Reference: eco_honor_power_intent
eco_inherit_net_attribute	
	New net inherits attributes of buffered net Type: bool Default: true Edit: Yes Reference: eco_inherit_net_attribute
eco_prefix	
	Prefix name to be used by ECO commands Type: string Default: ECO Edit: Yes Reference: eco_prefix
eco_preserve_hpin_function	
	Determines whether to preserve logical functions at hierarchical module ports. It only works for manual ECO commands. Type: bool Default: false Edit: Yes Reference: eco_preserve_hpin_function
eco_refine_place	
	Refine placement after ECO Type: bool Default: true Edit: Yes Reference: eco_refine_place

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eco_select_partition_list	
	set partition module list Type: string Default: "" Edit: Yes
eco_spread_inverter	
	Spread inverters on buffered net Type: bool Default: true Edit: Yes Reference: eco_spread_inverter
eco_update_only_layers	
	Modify routes only on metal layers in this range (both inclusive) Type: string Default: "" Edit: Yes Reference: eco_update_only_layers
eco_update_timing	
	Update timing results after ECO Type: bool Default: true Edit: Yes Reference: eco_update_timing
edit_wire_align	
	Specifies whether to support align objects. Type: bool Default: true Edit: Yes Reference: edit_wire_align
edit_wire_allow_45_degree	
	Specifies whether to allow creation of a 45-degree wire. Type: bool Default: false Edit: Yes Reference: edit_wire_allow_45_degree
edit_wire_arrow_incremental	
	Specifies the step increments in microns to move wires with the arrow keys. Type: double Default: 1.0 Edit: Yes Reference: edit_wire_arrow_incremental
edit_wire_assign_multi_pattern_color	
	Specifies how to assign the mask color on the DPT layer of the wire segment to be created. Type: enum Enum Values: auto mask1 mask2 mask3 Default: auto Edit: Yes Reference: edit_wire_assign_multi_pattern_color
edit_wire_auto_assign_net	

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	Auto assign the net from touched objects to edit wires/vias. Only enable this behavior for specified edit function. Type: enum Enum Values: add copy move Default: copy Edit: Yes Reference: edit_wire_auto_assign_net
edit_wire_auto_distribute_bus	
	Auto distribute a group of wires, and set same spacing based on option -bus_total_width_vertical or -bus_total_width_horizontal. Type: bool Default: false Edit: Yes Reference: edit_wire_auto_distribute_bus
edit_wire_auto_split_bus	
	Splits buses automatically to avoid DRC violations. Type: bool Default: false Edit: Yes Reference: edit_wire_auto_split_bus
edit_wire_bus_honor_start_parameters	
	Specifies that the wire width and spacing of the start points (pins) should be honored when drawing routes for a bus. Type: bool Default: false Edit: Yes Reference: edit_wire_bus_honor_start_parameters
edit_wire_bus_honor_width_setting	
	Specifies whether the tool should adjust the wire width when snapping bus wires to pin. Type: bool Default: false Edit: Yes Reference: edit_wire_bus_honor_width_setting
edit_wire_bus_total_width_horizontal	
	This option can specify the total width of horizontal bus. So when tool drawing a bus, tool auto set spacing as: wire spacing = ($\frac{\text{total bus width} - \text{total wire width}}{\text{wire number} - 1}$). Type: double Default: 0.0 Edit: Yes Reference: edit_wire_bus_total_width_horizontal
edit_wire_bus_total_width_vertical	
	This option can specify the total width of vertical bus. So when tool drawing a bus, tool auto set spacing as: wire spacing = ($\frac{\text{total bus width} - \text{total wire width}}{\text{wire number} - 1}$). Type: double Default: 0.0 Edit: Yes Reference: edit_wire_bus_total_width_vertical
edit_wire_change_order_at_turn	
	Specifies whether the order of the wires changes or stays the same when the wire makes a 90-degree turn. (Reverse Order Keep Order) Type: string Default: keep_order Edit: Yes Reference: edit_wire_change_order_at_turn
edit_wire_check_design_boundary	

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	Specifies whether stop user to stretch the wire in the forbidden area which defined by edit_wire_pull_back_distance. Type: bool Default: false Edit: Yes Reference: edit_wire_check_design_boundary
edit_wire_close_polygons	
	Specifies whether to close a special route structure toward itself. Type: bool Default: false Edit: Yes Reference: edit_wire_close_polygons
edit_wire_color_align_with_track	
	Specifies whether the wire color is to be changed to match the track color during a move or copy operation. Type: bool Default: false Edit: Yes Reference: edit_wire_color_align_with_track
edit_wire_connect_pin	
	Specifies whether or not generate a via on the pin for specified pin type. Type: enum Enum Values: in out inout nodir Default: in out inout nodir Edit: Yes Reference: edit_wire_connect_pin
edit_wire_connect_with_specified_layer	
	Specifies whether or not the layer of connecting wires should be changed as per the edit_wire_layer_horizontal or edit_wire_layer_vertical setting when moving wires. Type: bool Default: false Edit: Yes Reference: edit_wire_connect_with_specified_layer
edit_wire_create_crossover_vias	
	Specifies whether the software creates a via when you draw a wire that crosses a wire or pin of the same net that is on a different layer. Type: bool Default: true Edit: Yes Reference: edit_wire_create_crossover_vias
edit_wire_create_is_edit_flag	
	Specifies whether to create is_edit flag for wire edit Type: bool Default: false Edit: Yes Reference: edit_wire_create_is_edit_flag
edit_wire_create_via_on_pin	
	Specifies whether the software creates vias at pins. Type: bool Default: true Edit: Yes Reference: edit_wire_create_via_on_pin
edit_wire_cut_class	
	Specifies the cut class name of the generated via when editing wires. Type: string Default: "" Edit: Yes Reference: edit_wire_cut_class

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edit_wire_cut_wire_overlap	
	Specifies whether or not overlaps are created while cutting specified type of wires. Type: enum Enum Values: regular special Default: regular Edit: Yes Reference: edit_wire_cut_wire_overlap
edit_wire_debug_file	
	Specifies the report file name. Type: string Default: "" Edit: Yes Reference: edit_wire_debug_file
edit_wire_delete_pin_with_wire	
	Delete Pin with Wire Type: bool Default: true Edit: Yes Reference: edit_wire_delete_pin_with_wire
edit_wire_delete_tsv	
	Specifies whether to delete the items on TSV layer or not. Type: bool Default: false Edit: Yes Reference: edit_wire_delete_tsv
edit_wire_delete_wire_via_through_layers	
	Delete Wire Via in deep through mode Type: bool Default: false Edit: Yes Reference: edit_wire_delete_wire_via_through_layers
edit_wire_display_wire_length_with_cursor	
	Specifies whether display the total/current wire length with mouse when editing wires. Type: bool Default: true Edit: Yes Reference: edit_wire_display_wire_length_with_cursor
edit_wire_draw_shield	
	Specifies whether draw a shield net only. Type: bool Default: false Edit: Yes Reference: edit_wire_draw_shield
edit_wire_draw_with_group_centerline	
	Turn on this option, tool should draw bus along the center-line of whole wires. Type: bool Default: false Edit: Yes Reference: edit_wire_draw_with_group_centerline
edit_wire_drawing_wire	

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	Specifies which of the nets used with the -nets parameter corresponding to the mouse pointer location when adding an array of wires. Type: int Default: 1 Edit: Yes Reference: edit_wire_drawing_wire
edit_wire_drc_aware_cross_metal	
	Specifies whether to stretch or move wires back to the closest location available before a violation occurs with the crossing metal. This option only work if "setEditMode -stop_at_drc" set to 1. Type: bool Default: true Edit: Yes Reference: edit_wire_drc_aware_cross_metal
edit_wire_drc_on	
	Specifies whether to check DRC rules during wire editing. Type: bool Default: true Edit: Yes Reference: edit_wire_drc_on
edit_wire_drc_use_non_default_spacing	
	Specifies whether non-default spacing defined in the Non-Default Rule of the editing net should be used for spacing check during wire editing. Type: bool Default: false Edit: Yes Reference: edit_wire_drc_use_non_default_spacing
edit_wire_extend_wires	
	Specifies whether, after completing a signal or power route, the specified boundary of wire segment extends and connects to the first logical target Type: enum Enum Values: start end start_cell_boundary end_cell_boundary Default: "" Edit: Yes Reference: edit_wire_extend_wires
edit_wire_final_check_with_verify	
	Specifies whether to do final check with verify Type: bool Default: false Edit: Yes Reference: edit_wire_final_check_with_verify
edit_wire_ignore_drc	
	Specifies which violation type should be ignored during wire editing. Type: enum Enum Values: conn_antenna max_via_stack min_cut min_enclosed_area min_step protrusion Default: "" Edit: Yes Reference: edit_wire_ignore_drc
edit_wire_jog_connect_layer	
	Specifies the number of layers for connecting wires above the layers for moving wires. If set the parameter of 'set_db edit_wire_jog_connect_layer' to 1, disable this option. Type: int Default: 0 Edit: Yes Reference: edit_wire_jog_connect_layer
edit_wire_keep_floating_via	

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	Keep floating vias when user deleting wires. Type: bool Default: false Edit: Yes Reference: edit_wire_keep_floating_via
edit_wire_keep_status	
	Specifies whether to keep object status during editing. Type: bool Default: true Edit: Yes Reference: edit_wire_keep_status
edit_wire_keep_via	
	Specifies whether keep original via in wire move or change width. Type: bool Default: false Edit: Yes Reference: edit_wire_keep_via
edit_wire_lateral_movement_range	
	Specified the max lateral movement distance to avoid DRC when add a wire. Type: double Default: 0.0 Edit: Yes Reference: edit_wire_lateral_movement_range
edit_wire_layer	
	Specifies the layer for patch wires or polygon wires. Type: string Default: M1 Edit: Yes Reference: edit_wire_layer
edit_wire_layer_horizontal	
	Specifies the layer for horizontal wires. Type: string Default: M1 Edit: Yes Reference: edit_wire_layer_horizontal
edit_wire_layer_max	
	Specifies the maximum layer for routing. Type: string Default: UNINITIALIZED Edit: Yes Reference: edit_wire_layer_max
edit_wire_layer_min	
	Specifies the minimum layer for routing. Type: string Default: UNINITIALIZED Edit: Yes Reference: edit_wire_layer_min
edit_wire_layer_vertical	
	Specifies the layer for vertical wires. Type: string Default: M2 Edit: Yes Reference: edit_wire_layer_vertical
edit_wire_look_down_layers	

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	Specifies the number of layers below the current layer that added wires will connect to with a via. Type: int Default: 100 Edit: Yes Reference: edit_wire_look_down_layers
edit_wire_look_up_layers	
	Specifies the number of layers above the current layer that added wires will connect to with a via. Type: int Default: 100 Edit: Yes Reference: edit_wire_look_up_layers
edit_wire_max_pointer_number	
	Specifies the maximum number of pointers to be returned by wire edit commands. Type: int Default: 100 Edit: Yes Reference: edit_wire_max_pointer_number
edit_wire_mirror_bus_route	
	Specifies that whether to reverse the net order for bus route. Type: bool Default: false Edit: Yes Reference: edit_wire_mirror_bus_route
edit_wire_nets	
	Specifies one or more nets for editing. Type: string Default: "" Edit: Yes Reference: edit_wire_nets
edit_wire_no_merge_special_wire	
	Specifies whether automatically merge the added special wires with any existing special wires. Type: bool Default: false Edit: Yes Reference: edit_wire_no_merge_special_wire
edit_wire_only_show_edit_layer	
	Turn on this option, when user editing any wires/vias, only show the layer from edit objects, and dim other objects. Type: bool Default: true Edit: Yes Reference: edit_wire_only_show_edit_layer
edit_wire_orthogonal_connection_only	
	Specifies whether to consider wires and pins that are in the same direction as a target for connection. Only special wire is supported. Type: bool Default: true Edit: Yes Reference: edit_wire_orthogonal_connection_only
edit_wire_outer_shield_spacing	
	Specifies a spacing value in microns to be used by the edit_wire_shield low and edit_wire_shield high parameters. Type: double Default: 0.0 Edit: Yes Reference: edit_wire_outer_shield_spacing
edit_wire_outer_shield_width	

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	Specifies a width value in microns to be used for outer shield(s) depending on the values of edit_wire_shield low or edit_wire_shield high parameters. Type: double Default: 0.0 Edit: Yes Reference: edit_wire_outer_shield_width
edit_wire_override	
	Specifies whether to use the override specification. Type: bool Default: false Edit: Yes Reference: edit_wire_override
edit_wire_partial_overlap_threshold	
	Specifies ratio threshold of special via creation on partial overlap wires. Type: int Default: 0 Edit: Yes Reference: edit_wire_partial_overlap_threshold
edit_wire_pull_back_distance	
	Specify the value for forbidden distance. Default is Auto, it means use the largest min spacing of the layer. Type: string Default: auto Edit: Yes Reference: edit_wire_pull_back_distance
edit_wire_reshape	
	Specifies that when you add new wires, existing redundant wires within the route are automatically removed. Type: bool Default: false Edit: Yes Reference: edit_wire_reshape
edit_wire_return_obj_pointer	
	Specifies whether return pointers after wire edit operations. Type: bool Default: false Edit: Yes Reference: edit_wire_return_obj_pointer
edit_wire_rule	
	Uses the specified LEF rule for regular wires. Type: string Default: "" Edit: Yes Reference: edit_wire_rule
edit_wire_search_route_rule_vias_only	
	Specifies that when the Shift + N/P bindkey is used, the tool should circle through NDR vias only. Type: bool Default: false Edit: Yes Reference: edit_wire_search_route_rule_vias_only
edit_wire_shape	
	Specifies the shape associated with the wire you draw. Type: enum Enum Values: ring stripe followpin iowire corewire blockwire padding blocking fillwire fillwireopc drcfill none Default: stripe Edit: Yes Reference: edit_wire_shape

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edit_wire_shield	
	Specifies whether to add a minimum width shield wire for specified side. Type: enum Enum Values: high low adjacent Default: "" Edit: Yes Reference: edit_wire_shield
edit_wire_shield_look_down_layers	
	Specifies the number of layers below the current layer that an added shield wire can connect using a via. Type: int Default: 100 Edit: Yes Reference: edit_wire_shield_look_down_layers
edit_wire_shield_look_up_layers	
	Specifies the number of layers above the current layer that an added shield wire can connect using a via. Type: int Default: 100 Edit: Yes Reference: edit_wire_shield_look_up_layers
edit_wire_shield_shape	
	Specifies the shape associated with the shield wire you draw. Type: enum Enum Values: ring stripe followpin iowire corewire blockwire padring blockring fillwire drcfill none Default: none Edit: Yes Reference: edit_wire_shield_shape
edit_wire_shielding_nets	
	Specifies net names for shield wires, usually power or ground names. Type: string Default: "" Edit: Yes Reference: edit_wire_shielding_nets
edit_wire_show_drc_info_for_edit_shape	
	Turn on this option, when user editing any wires/vias and generate a violation, tool can display the detail information nearby marker. Type: bool Default: true Edit: Yes Reference: edit_wire_show_drc_info_for_edit_shape
edit_wire_sibling_look_down_layers	
	Specifies the number of layers below the sibling layers that an added shield wire can connect using a via. Type: int Default: 1 Edit: Yes Reference: edit_wire_sibling_look_down_layers
edit_wire_sibling_look_up_layers	
	Specifies the number of layers above the sibling layers that an added shield wire can connect using a via. Type: int Default: 1 Edit: Yes Reference: edit_wire_sibling_look_up_layers
edit_wire_snap	

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	Specifies whether to enable wire and via snap. Type: bool Default: true Edit: Yes Reference: edit_wire_snap
edit_wire_snap_align_to	
	Specifies how to align with a pin when ending a route. Type: enum Enum Values: center low high auto Default: auto Edit: Yes Reference: edit_wire_snap_align_to
edit_wire_snap_bus_to_pin	
	Specifies whether to change wire width, spacing and order according to the connected pin. Type: bool Default: false Edit: Yes Reference: edit_wire_snap_bus_to_pin
edit_wire_snap_end_to	
	Specifies whether to snap the end of wire to specified objects. Type: enum Enum Values: track_regular track_special manufacturing_grid Default: "" Edit: Yes Reference: edit_wire_snap_end_to
edit_wire_snap_objects_to_track	
	Specifies whether to snap added or moved specified wires to the closest routing track in the preferred direction for the layer automatically. Type: enum Enum Values: special regular patch pin Default: regular patch Edit: Yes Reference: edit_wire_snap_objects_to_track
edit_wire_snap_to	
	Specifies whether to snap added or moved wires to specified objects. Type: enum Enum Values: special pg pin row Default: special pg pin Edit: Yes Reference: edit_wire_snap_to
edit_wire_snap_to_track_honor_color	
	Specifies whether to snap wires to a track with the same color automatically. Type: bool Default: true Edit: Yes Reference: edit_wire_snap_to_track_honor_color
edit_wire_snap_trim_metal_to_trim_grid	
	Specify whether snap the trim metal to the closest trim grid during trim metal editing. Type: bool Default: true Edit: Yes Reference: edit_wire_snap_trim_metal_to_trim_grid
edit_wire_spacing	

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	Specifies the distance between wires. Type: string Default: "" Edit: Yes Reference: edit_wire_spacing
edit_wire_spacing_horizontal	
	Specifies the distance between horizontal wires. Type: string Default: "" Edit: Yes Reference: edit_wire_spacing_horizontal
edit_wire_spacing_vertical	
	Specifies the distance between vertical wires. Type: string Default: "" Edit: Yes Reference: edit_wire_spacing_vertical
edit_wire_status	
	Specifies the status associated with the wire you draw. Type: enum Enum Values: cover fixed noshield routed shield auto Default: fixed Edit: Yes Reference: edit_wire_status
edit_wire_stop_at_drc	
	Specifies whether the software can cause a DRC violation by moving or stretching a wire. Type: bool Default: true Edit: Yes Reference: edit_wire_stop_at_drc
edit_wire_stretch_end	
	Specifies the direction in which to stretch or reduce wires. Type: enum Enum Values: high low Default: high Edit: Yes Reference: edit_wire_stretch_end
edit_wire_stretch_with_intersection	
	Enables wires with intersects to be stretched easily. Type: bool Default: false Edit: Yes Reference: edit_wire_stretch_with_intersection
edit_wire_sub_class	
	Specifies the subclass name of wires and vias. Type: string Default: "" Edit: Yes Reference: edit_wire_sub_class
edit_wire_turn_at	

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	Specifies the start location of new wire when do turn. Type: enum Enum Values: center_line wire_edge Default: center_line Edit: Yes Reference: edit_wire_turn_at
edit_wire_type	
	Specifies the type of creating wire. Type: enum Enum Values: regular special patch Default: "" Edit: Yes Reference: edit_wire_type
edit_wire_unrestricted_regular_wire_width	
	Specifies whether the regular wire width is unrestricted in edit. Type: bool Default: false Edit: Yes Reference: edit_wire_unrestricted_regular_wire_width
edit_wire_update_shield_net	
	Specifies whether automatically update shield net attribute on the signal net. Type: bool Default: false Edit: Yes Reference: edit_wire_update_shield_net
edit_wire_use_fix_via	
	Specifies whether to use fix_via for special via with minStep DRC violation. Type: bool Default: false Edit: Yes Reference: edit_wire_use_fix_via
edit_wire_use_interleaving_wire_group	
	Alternates the specified wires in the wire group. Type: bool Default: false Edit: Yes Reference: edit_wire_use_interleaving_wire_group
edit_wire_use_wire_group	
	Groups multiple wires from the same net. Type: bool Default: false Edit: Yes Reference: edit_wire_use_wire_group
edit_wire_use_wire_group_bits	
	Specifies the number of times to replicate a wire in the wire group. Type: int Default: 0 Edit: Yes Reference: edit_wire_use_wire_group_bits
edit_wire_use_wire_group_reinforcement	
	Creates slots in the wire group by creating wires that connect the wires in the wire group, but are orthogonal to them. Type: bool Default: false Edit: Yes Reference: edit_wire_use_wire_group_reinforcement

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edit_wire_use_wire_group_reinforcement_group_via	
	Specifies that the wire group uses group vias. Type: bool Default: false Edit: Yes Reference: edit_wire_use_wire_group_reinforcement_group_via
edit_wire_use_wire_group_reinforcement_spacing	
	Specifies the spacing for the orthogonal wires in the wire groups. Type: double Default: 0.0 Edit: Yes Reference: edit_wire_use_wire_group_reinforcement_spacing
edit_wire_use_wire_group_reinforcement_width	
	Specifies the width of the orthogonal wires in the wire groups. Type: double Default: 0.0 Edit: Yes Reference: edit_wire_use_wire_group_reinforcement_width
edit_wire_verbose	
	Type: bool Default: false Edit: Yes Reference: edit_wire_verbose
edit_wire_via_allow_geometry_drc	
	Check Geometry DRC Type: bool Default: true Edit: Yes Reference: edit_wire_via_allow_geometry_drc
edit_wire_via_auto_snap	
	Controls whether to snap vias to the intersection of wires on the same net or to the manufacturing grid. Type: bool Default: true Edit: Yes Reference: edit_wire_via_auto_snap
edit_wire_via_auto_update	
	Controls whether to replace existing vias. Type: bool Default: true Edit: Yes Reference: edit_wire_via_auto_update
edit_wire_via_cell_name	
	Specifies the name of the via cell from the LEF or DEF file. Type: string Default: "" Edit: Yes Reference: edit_wire_via_cell_name
edit_wire_via_columns	
	Specify the rows of via cut. Type: int Default: 0 Edit: Yes Reference: edit_wire_via_columns

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edit_wire_via_create_by	
	Controls whether to create vias based on the edit_wire_via_cell_name parameter or to create vias based on the other parameters. Type: enum Enum Values: viacell parameters Default: parameters Edit: Yes Reference: edit_wire_via_create_by
edit_wire_via_cut_layer	
	Specifies the via layer name for the via to created or modify. Type: string Default: V12 Edit: Yes Reference: edit_wire_via_cut_layer
edit_wire_via_exclude_spec	
	Specifies the vias to be ignored by wire editor. Type: string Default: "" Edit: Yes Reference: edit_wire_via_exclude_spec
edit_wire_via_override_spec	
	Specifies the default vias to be used by wire editor. Type: string Default: "" Edit: Yes Reference: edit_wire_via_override_spec
edit_wire_via_rows	
	Specify the columns of via cut. Type: int Default: 0 Edit: Yes Reference: edit_wire_via_rows
edit_wire_via_scale_height	
	Specify the via height scale. Type: int Default: 100 Edit: Yes Reference: edit_wire_via_scale_height
edit_wire_via_scale_width	
	Specify the via width scale. Type: int Default: 100 Edit: Yes Reference: edit_wire_via_scale_width
edit_wire_via_snap_honor_color	
	Specifies whether to snap the same mask automatically Type: bool Default: true Edit: Yes Reference: edit_wire_via_snap_honor_color
edit_wire_via_snap_to_intersection	

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	Specifies whether to snap vias to the intersection of wires Type: bool Default: true Edit: Yes Reference: edit_wire_via_snap_to_intersection
edit_wire_via_type	
	Specifies the type of creating via. Type: enum Enum Values: auto regular special Default: auto Edit: Yes Reference: edit_wire_via_type
edit_wire_width	
	Specifies the width of patch wires. Type: double Default: 0.0 Edit: Yes Reference: edit_wire_width
edit_wire_width_horizontal	
	Specifies the width for horizontal wires. Type: double Default: 0.0 Edit: Yes Reference: edit_wire_width_horizontal
edit_wire_width_vertical	
	Specifies the width of vertical wires. Type: double Default: 0.0 Edit: Yes Reference: edit_wire_width_vertical
edit_wire_wire_override_spec	
	Contains the details of the override specification. Type: string Default: "" Edit: Yes Reference: edit_wire_wire_override_spec
eeq_variant_site_start	
	Site variant_id of the first site on rows in core area. This is specified in library LEF58_CELLVARIANTS property by value of STARTVARIANT in 'CELLVARIANTS totalNum [STARTVARIANT num] YFLIPMAP {flippedVariantNum siteVariantNum} ...' Type: int Default: no_value Edit: Yes
eeq_variant_y_flip_table	
	The LEF YFLIPMAP table defined in LEF58_CELLVARIANTS property as 'CELLVARIANTS variantTotalNum YFLIPMAP {flippedVariantNum siteVariantNum} ...' Type: int* Default: "" Edit: No
enable_library_pins_sorting_in_mmmc	
	Sort library pins in MMMC flow. Type: bool Default: true Edit: Yes Reference: enable_library_pins_sorting_in_mmmc

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enable_strict_percent_control	
	Threshold value to be used as hard limit. Type: bool Default: false Edit: Yes Reference: enable_strict_percent_control
enable_ui_precision	
	Enable ui precision reporting. Type: bool Default: false Edit: Yes Reference: enable_ui_precision
exclusive_group_min_gap	
	This is the minimum gap should be maintained between exclusive_groups (all exclusive groups). The value is measured in microns. It can be set by command create_exclusive_groups -min_gap. Type: coord Default: 5 Edit: No
extract_keep_fill_wires	
	Keep metal fill wires for extraction Type: bool Default: false Edit: Yes Reference: extract_keep_fill_wires
extract_rc_assume_metal_fill	
	Assume metal fill Type: double Default: 0.0 Edit: Yes Reference: extract_rc_assume_metal_fill
extract_rc_cap_filter_mode	
	Coupling Capacitance Filtering Mode Type: enum Enum Values: relative_only relative_and_coupling relative_or_coupling Default: relative_only Edit: Yes Reference: extract_rc_cap_filter_mode
extract_rc_cerebrus_license_only	
	Fall back to the regular QTS license(s) when Cerebrus is not available Type: bool Default: false Edit: Yes Reference: extract_rc_cerebrus_license_only
extract_rc_compress_rcdb	
	Create compressed RCDB Type: bool Default: false Edit: Yes Reference: extract_rc_compress_rcdb
extract_rc_coupled	

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	Extract coupling cap Type: bool Default: false Edit: Yes Reference: extract_rc_coupled
extract_rc_coupling_cap_threshold	
	Coupling C threshold value Type: double Default: 3.0 Edit: Yes Reference: extract_rc_coupling_cap_threshold
extract_rc_def_via_cap	
	Consider via caps for extraction Type: bool Default: false Edit: Yes Reference: extract_rc_def_via_cap
extract_rc_effort_level	
	Specifies the effort level of the post-route extraction engine Type: enum Enum Values: low medium high signoff Default: undefined Edit: Yes Reference: extract_rc_effort_level
extract_rc_engine	
	Extraction engine Type: enum Enum Values: pre_route post_route Default: pre_route Edit: Yes Reference: extract_rc_engine
extract_rc_extra_cmd_file	
	Extra command file for Quantus QRC Type: string Default: "" Edit: Yes Reference: extract_rc_extra_cmd_file
extract_rc_hard_block_obs	
	Makes obstructions and power/ground pin shapes visible to the extractor for hard blocks only Type: bool Default: false Edit: Yes Reference: extract_rc_hard_block_obs
extract_rc_incremental	
	Use incremental signoff mode Type: bool Default: false Edit: Yes Reference: extract_rc_incremental
extract_rc_layer_independent	
	Enable layer independent extraction Type: int Default: 1 Edit: Yes Reference: extract_rc_layer_independent

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extract_rc_lef_tech_file_map	
	LEF-QRCTechfile layer map file Type: string Default: "" Edit: Yes Reference: extract_rc_lef_tech_file_map
extract_rc_local_cpu	
	Type: int Default: 0 Edit: Yes Reference: extract_rc_local_cpu
extract_rc_post_route_iquantus_cap_high_accuracy_mode	
	Use for iquantus capacitance high accuracy mode Type: bool Default: false Edit: Yes Reference: extract_rc_post_route_iquantus_cap_high_accuracy_mode
extract_rc_pvs_fill	
	Use to send the PVS fill data attached to the Innovus DB (see set_pvs_fill) to Quantus Type: bool Default: false Edit: Yes Reference: extract_rc_pvs_fill
extract_rc_qrc_cmd_file	
	Quantus QRC user command file name Type: string Default: "" Edit: Yes Reference: extract_rc_qrc_cmd_file
extract_rc_qrc_cmd_type	
	Quantus QRC command input type Type: enum Enum Values: auto partial custom Default: auto Edit: Yes Reference: extract_rc_qrc_cmd_type
extract_rc_qrc_output_mode	
	Quantus QRC output mode Type: enum Enum Values: spef rcdbr Default: spef Edit: Yes Reference: extract_rc_qrc_output_mode
extract_rc_qrc_run_mode	
	Quantus QRC MMC run mode Type: enum Enum Values: concurrent sequential Default: concurrent Edit: Yes Reference: extract_rc_qrc_run_mode
extract_rc_qrc_stream_map_file	

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	Quantus stream/oasis file Type: string Default: "" Edit: Yes Reference: extract_rc_qrc_stream_map_file
extract_rc_quantus_executable	
	QRC Executable file path Type: string Default: "" Edit: Yes Reference: extract_rc_quantus_executable
extract_rc_relative_cap_threshold	
	Relative C threshold value Type: double Default: 0.03 Edit: Yes Reference: extract_rc_relative_cap_threshold
extract_rc_shrink_factor	
	Sets the shrink factor used by the extract_rc command.(Double, default=1) Type: double Default: 1.0 Edit: Yes Reference: extract_rc_shrink_factor
extract_rc_signoff_stream_layer_map	
	QRC stream map file Type: string Default: "" Edit: Yes Reference: extract_rc_signoff_stream_layer_map
extract_rc_signoff_stream_layer_map_ccl	
	QRC stream map file in CCL format Type: string Default: "" Edit: Yes
extract_rc_total_cap_threshold	
	Total C threshold value Type: double Default: 5.0 Edit: Yes Reference: extract_rc_total_cap_threshold
extract_rc_tquantus_model_file	
	TQuantus model file name Type: string Default: "" Edit: Yes Reference: extract_rc_tquantus_model_file
extract_rc_tsv_subckt_file	
	TSV subckt file name Type: string Default: "" Edit: Yes Reference: extract_rc_tsv_subckt_file
extract_rc_turbo_reduce	

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	Reduce XCap (IQuantus) Type: enum Enum Values: true false auto Default: auto Edit: Yes Reference: extract_rc_turbo_reduce
extract_rc_use_qrc_oa_interface	
	Use OA interface for invoking Quantus QRC extraction Type: bool Default: false Edit: Yes Reference: extract_rc_use_qrc_oa_interface
extract_rc_use_shielding_in_detail_mode	
	Considers shielding for a wire segment Type: bool Default: false Edit: Yes Reference: extract_rc_use_shielding_in_detail_mode
extract_rc_via_cap	
	Consider via caps for extraction Type: bool Default: true Edit: Yes Reference: extract_rc_via_cap
extract_rc_write_def_options_for_signoff_extract	
	write_def options Type: string Default: "" Edit: Yes Reference: extract_rc_write_def_options_for_signoff_extract
finish_floorplan_abutted_edge_as_non_row_area	
	Specify whether the abutted edge of power domain should be nonRowArea Type: bool Default: true Edit: Yes Reference: finish_floorplan_abutted_edge_as_non_row_area
finish_floorplan_active_objs	
	active_object list Type: enum Enum Values: macro macro_halo core io_pad io_cell fence hard_blockage soft_blockage partial_blockage route_blockage row die Default: macro macro_halo core Edit: Yes Reference: finish_floorplan_active_objs
finish_floorplan_add_blockage_direction	
	direction Type: enum Enum Values: x y xy Default: xy Edit: Yes Reference: finish_floorplan_add_blockage_direction
finish_floorplan_add_blockage_fill_option	

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	fill option Type: enum Enum Values: channel boundary Default: channel Edit: Yes Reference: finish_floorplan_add_blockage_fill_option
finish_floorplan_drc_region_objs	
	drc_region object list Type: enum Enum Values: macro macro_halo hard_blockage min_gap core_spacing non_row_area row Default: macro macro_halo hard_blockage min_gap core_spacing row Edit: Yes Reference: finish_floorplan_drc_region_objs
finish_floorplan_override	
	override or not Type: bool Default: false Edit: Yes Reference: finish_floorplan_override
fix_min_drcs	
	Fix min_capacitance, min_transition, and min_fanout design rule violations. Type: bool Default: false Edit: Yes Reference: fix_min_drcs
flexible_htrees	
	list of flexible_htree Type: obj(flexible_htree)* Default: "" Edit: No Reference: flexible_htrees
flip_chip_allow_layer_change	
	Specifies if turning on layer change feature Type: bool Default: false Edit: Yes Reference: flip_chip_allow_layer_change
flip_chip_allow_routed_bump_edit	
	By default bumps are allowed to be edited (moved, deleted, reassigned, swapped). If this variable is set to FALSE, any bump edit command for a bump with routing connected to it will give an error message and fail. Type: bool Default: true Edit: Yes Reference: flip_chip_allow_routed_bump_edit
flip_chip_auto_pairing_file	
	Specifies the output file name to dump out route_flip_chip auto pairing information for bump to I/O connection Type: string Default: "" Edit: Yes Reference: flip_chip_auto_pairing_file
flip_chip_bottom_layer	

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	Specifies the bottom-most metal layer by layer number or layer name that the software can use when routing bumps Type: string Default: "" Edit: Yes Reference: flip_chip_bottom_layer
flip_chip_bump_use_octagon_shape	
	route_flip_chip use circumferential octagon to replace the complex pin shape Type: bool Default: false Edit: Yes Reference: flip_chip_bump_use_octagon_shape
flip_chip_check_bump_access_directions	
	Specifies whether check the special bump access directions Type: bool Default: false Edit: Yes Reference: flip_chip_check_bump_access_directions
flip_chip_compaction	
	Specifies if turning on compaction routing Type: bool Default: false Edit: Yes Reference: flip_chip_compaction
flip_chip_connect_power_cell_to_bump	
	Connects all power bumps to the I/O cell pin. It does not connect to a power or ground stripe or ring Type: bool Default: false Edit: Yes Reference: flip_chip_connect_power_cell_to_bump
flip_chip_constraint_file	
	Specifies the file that contains constraints for flip chip routing Type: string Default: "" Edit: Yes Reference: flip_chip_constraint_file
flip_chip_drop_via_on_all_shapes	
	route_flip_chip drop via on all geometries Type: bool Default: false Edit: Yes Reference: flip_chip_drop_via_on_all_shapes
flip_chip_drop_via_on_power_mesh	
	Specifies the metal layer by layer number or layer name that the software can use to drop via Type: string Default: "" Edit: Yes Reference: flip_chip_drop_via_on_power_mesh
flip_chip_extra_config	
	Specifies the name of an extra configuration file for flip chip routing Type: string Default: "" Edit: Yes Reference: flip_chip_extra_config
flip_chip_finger_direction	

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	Specifies main expansion direction for finger routing Type: enum Enum Values: n e s w Default: n Edit: Yes Reference: flip_chip_finger_direction
flip_chip_finger_max_width	
	Specifies the max routing width for one wire segment Type: double Default: 0.0 Edit: Yes Reference: flip_chip_finger_max_width
flip_chip_finger_min_width	
	Specifies the min routing width for one wire segment Type: double Default: 0.0 Edit: Yes Reference: flip_chip_finger_min_width
flip_chip_finger_target_mesh_layer_range	
	Target mesh layer range for finger routing Type: string Default: "" Edit: Yes Reference: flip_chip_finger_target_mesh_layer_range
flip_chip_honor_bump_connect_target_constraint	
	Specifies if enabling port numbering routing Type: bool Default: false Edit: Yes Reference: flip_chip_honor_bump_connect_target_constraint
flip_chip_ignore_pad_type_check	
	Specifies if supporting CLASS PAD cells in flipchip flow Type: bool Default: true Edit: Yes Reference: flip_chip_ignore_pad_type_check
flip_chip_lower_layer_prevent_diagonal_routing	
	Specifies if preventing 45-degree routing for lower layer Type: bool Default: false Edit: Yes Reference: flip_chip_lower_layer_prevent_diagonal_routing
flip_chip_lower_layer_route_width	
	Specifies the value for lower routing layer Type: string Default: "" Edit: Yes Reference: flip_chip_lower_layer_route_width
flip_chip_merge_nearby_pin	
	Merge pins which are close but not touched Type: bool Default: false Edit: Yes Reference: flip_chip_merge_nearby_pin

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flip_chip_multi_pad_routing_style	
	Specifies the routing style for multi-pads routing Type: enum Enum Values: default serial star Default: default Edit: Yes Reference: flip_chip_multi_pad_routing_style
flip_chip_multiple_connection	
	Specifies routing connections between multiple pads and bumps. Default is no multiple connection Type: enum Enum Values: multiple_pads_to_bump multiple_bumps_to_pad multiple_bumps_to_multiple_pads default Default: default Edit: Yes Reference: flip_chip_multiple_connection
flip_chip_pg_mesh_direction	
	Specifies the main routing direction of PG mesh Type: enum Enum Values: default horizontal vertical Default: default Edit: Yes Reference: flip_chip_pg_mesh_direction
flip_chip_pg_mesh_main_width	
	Specifies the main width of PG mesh Type: double Default: 0.0 Edit: Yes Reference: flip_chip_pg_mesh_main_width
flip_chip_pg_mesh_max_width	
	Specifies the max width of PG mesh Type: double Default: 0.0 Edit: Yes Reference: flip_chip_pg_mesh_max_width
flip_chip_pg_mesh_width	
	Specify the exact width of PG mesh. One value for both horizontal and vertical width. And with two values, the first value for horizontal width and the second value for vertical width Type: string Default: "" Edit: Yes Reference: flip_chip_pg_mesh_width
flip_chip_prevent_diagonal_wire_access_bumps	
	Specifies whether use diagonal wire to access bump or not Type: bool Default: false Edit: Yes Reference: flip_chip_prevent_diagonal_wire_access_bumps
flip_chip_prevent_via_under_bump	
	Specifies if generating vias directly under the bump Type: bool Default: false Edit: Yes Reference: flip_chip_prevent_via_under_bump
flip_chip_prevent_via_under_bump_extension	

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	specifies the spacing value around bump to avoid dropping vias Type: string Default: default Edit: Yes Reference: flip_chip_prevent_via_under_bump_extension
flip_chip_route_bump_cluster	
	Route bump cluster after bump grouped Type: bool Default: false Edit: Yes
flip_chip_route_pg_style	
	Specifies the routing pattern for route_flip_chip power routing Type: enum Enum Values: none finger mesh Default: none Edit: Yes Reference: flip_chip_route_pg_style
flip_chip_route_style	
	Specifies routing style Type: enum Enum Values: manhattan diagonal Default: diagonal Edit: Yes Reference: flip_chip_route_style
flip_chip_route_width	
	Specifies routing width Type: double Default: 0.0 Edit: Yes Reference: flip_chip_route_width
flip_chip_top_layer	
	Specifies the top-most metal layer by layer number or layer name that the software can use when routing bumps Type: string Default: "" Edit: Yes Reference: flip_chip_top_layer
flip_chip_via_abut_bump	
	via abut bump only if it is success Type: bool Default: false Edit: Yes Reference: flip_chip_via_abut_bump
flip_chip_wire_to_pad_honor_min_spacing	
	Specifies the min spacing between bump/pad with wire Type: double Default: 0.0 Edit: Yes Reference: flip_chip_wire_to_pad_honor_min_spacing
floorplan_auto_sync_master_clone	
	Specify the objects used to sync-up master to clones. Type: enum Enum Values: bus_guides constraints stdcells macros route_blockages place_blockages Default: "" Edit: Yes Reference: floorplan_auto_sync_master_clone

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floorplan_cells_for_extra_sites	
	Specify cell names whose related sites will be created. Type: string Default: "" Edit: Yes Reference: floorplan_cells_for_extra_sites
floorplan_check_types	
	Specify the check types for check_floorplan. Default is the basic. Type: enum Enum Values: basic fence odd_even_site_row macro_pin color alignment_following_pin alignment_partition_clone bus_guide_connectivity feed_through partition_in_partition multi_layer_pin power_domain partition place same_length_site narrow_channel block_only unified_vdd_on_bottom tsv exclusive_group all Default: basic Edit: Yes Reference: floorplan_check_types
floorplan_cut_off_place_blockage_outside_die	
	When creating placement blockages, will cut the part outside of die and snap it by default. Type: bool Default: true Edit: Yes Reference: floorplan_cut_off_place_blockage_outside_die
floorplan_cut_off_route_blockage_outside_die	
	When creating routing blockages around inst/Hinst/Partition, will cut the part outside of die and snap it by default. If this option is set to false, don't cut and snap it. Type: bool Default: true Edit: Yes Reference: floorplan_cut_off_route_blockage_outside_die
floorplan_default_blockage_name_prefix	
	Name prefix for placement blockage and routing blockage. Type: string Default: "" Edit: Yes Reference: floorplan_default_blockage_name_prefix
floorplan_default_power_domain_site	
	Create rows based on default power domain site instead of the design's default site. Type: bool Default: false Edit: Yes Reference: floorplan_default_power_domain_site
floorplan_default_row_pattern_site	
	Specify rows pattern. Type: string Default: "" Edit: Yes Reference: floorplan_default_row_pattern_site
floorplan_default_tech_site	
	Specify a site as default technical site. After setting a forced one, the auto-calculating default-technical-site functionality will be disabled. Type: string Default: "" Edit: Yes Reference: floorplan_default_tech_site
floorplan_enable_rectilinear_design	

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	Specify whether design boundary can be changed to rectilinear shape. Type: bool Default: false Edit: Yes Reference: floorplan_enable_rectilinear_design
floorplan_extra_row_pattern	
	Specify row pattern site names which will be created. Type: string Default: "" Edit: Yes Reference: floorplan_extra_row_pattern
floorplan_extra_sites	
	Specify site names which will be created. Type: string Default: "" Edit: Yes Reference: floorplan_extra_sites
floorplan_finet_inst_grid	
	The finet_inst grid {x_origin y_origin x_pitch y_pitch}. The x_origin y_origin is lower left corner of core box. (e.g. the core rows). The x_pitch is default technical site's width and separates the vertical grid lines. The y_pitch is FinFET pitch as defined in LEF and separates the horizontal grid lines. Type: rect Default: 0.0 0.0 0.0 0.0 Edit: No Reference: floorplan_finet_inst_grid
floorplan_finet_manufacturing_grid	
	The finet_manufacturing grid {x_origin y_origin x_pitch y_pitch}. The x_origin y_origin is lower left corner of die box (the design .boundary). The x_pitch is Manufacture Grid value and defined in LEF. The y_pitch is FinFET pitch and defined in LEF. Type: rect Default: 0.0 0.0 0.0 0.0 Edit: No Reference: floorplan_finet_manufacturing_grid
floorplan_finet_placement_grid	
	The finet_placement grid {x_origin y_origin x_pitch y_pitch}. The x_origin y_origin is lower left corner of die box (the design .boundary). The x_pitch is the pitch of lowest vertical metal layer and separates the vertical grid lines. The y_pitch is FinFET pitch as defined in LEF and separates the horizontal grid lines. Type: rect Default: 0.0 0.0 0.0 0.0 Edit: No Reference: floorplan_finet_placement_grid
floorplan_first_row_site_index	
	Specify the first row site to be created. Type: int Default: 0 Edit: Yes Reference: floorplan_first_row_site_index
floorplan_include_io_when_init_area	
	Include IO cells when calculating the area of modules. Type: bool Default: false Edit: Yes Reference: floorplan_include_io_when_init_area
floorplan_initial_all_compatible_core_site_rows	

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	If true, initCoreRow will create all SITE with type core. Type: <code>bool</code> Default: <code>false</code> Edit: Yes Reference: floorplan_initial_all_compatible_core_site_rows
floorplan_inst_grid	
	The inst grid {x_origin y_origin x_pitch y_pitch}. The x_origin y_origin is lower left corner of core box (e.g. the core rows). The x_pitch is default technical site's width and separates the vertical grid lines. The y_pitch is default technical site's height and separates the horizontal grid lines. Type: <code>rect</code> Default: <code>0.0 0.0 0.0 0.0</code> Edit: No Reference: floorplan_inst_grid
floorplan_keep_rows_when_moving_power_domain	
	specify whether to keep all existing rows in disjoint power domain when moving, resizing or reshaping a power domain on GUI or by command. Type: <code>bool</code> Default: <code>false</code> Edit: Yes Reference: floorplan_keep_rows_when_moving_power_domain
floorplan_last_row_site_index	
	Specify the last row site to be created. Type: <code>int</code> Default: <code>-1</code> Edit: Yes Reference: floorplan_last_row_site_index
floorplan_layer_track_grid	
	The layer_track grid {x_origin y_origin x_pitch y_pitch}. The x_origin y_origin is lower left corner of die box (the design .boundary). The x_pitch is the pitch of lowest vertical metal layer and separates the vertical grid lines. The y_pitch is the pitch of lowest horizontal metal layer and separates the horizontal grid lines. Type: <code>rect</code> Default: <code>0.0 0.0 0.0 0.0</code> Edit: No Reference: floorplan_layer_track_grid
floorplan_manufacturing_grid	
	The manufacturing grid is in the format {x_origin y_origin x_pitch y_pitch} from the LEF global MANUFACTURINGGRID statement. The x_origin y_origin is lower left corner of die box(the design .boundary). See the layer .mfg_grid values if the mfg_grid varies per layer. Type: <code>rect</code> Default: <code>0.0 0.0 0.0 0.0</code> Edit: No Reference: floorplan_manufacturing_grid
floorplan_max_io_height	
	Use maximum IO height to calculate a die box with IO placement. By default, minimum IO height is used. Type: <code>bool</code> Default: <code>false</code> Edit: Yes Reference: floorplan_max_io_height
floorplan_minimum_sites	
	Specify the number of minimal sites (N) that check_floorplan will check for the same length site rule. When same_length_site is specified in floorplan_check_types, this option is required. Type: <code>int</code> Default: <code>0</code> Edit: Yes Reference: floorplan_minimum_sites
floorplan_move_child_constraint_with_constraint	

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	Specify child constraints move with parent constraint modules at the same time. Type: bool Default: false Edit: Yes Reference: floorplan_move_child_constraint_with_constraint
floorplan_move_macros_with_constraint	
	Specify macros move with constraint modules at the same time. Type: enum Enum Values: guide region fence none all Default: fence Edit: Yes Reference: floorplan_move_macros_with_constraint
floorplan_move_preplaced_std_cell_only	
	Specify only pre-placed std cells move with constraint modules at the same time. Type: bool Default: false Edit: Yes Reference: floorplan_move_preplaced_std_cell_only
floorplan_move_std_cell_with_constraint	
	Specify std cells move with constraint modules at the same time. Type: enum Enum Values: guide region fence none all Default: none Edit: Yes Reference: floorplan_move_std_cell_with_constraint
floorplan_narrow_channel_threshold	
	Reports narrow channels whose width (in micros) is smaller than the specified value. When narrow_channel is specified in floorplan_check_types, this option is required. Type: double Default: 0.0 Edit: Yes Reference: floorplan_narrow_channel_threshold
floorplan_no_cut_row	
	Specify no need to do cut row before other operations. Type: bool Default: false Edit: Yes Reference: floorplan_no_cut_row
floorplan_placement_grid	
	The placement grid {x_origin y_origin x_pitch y_pitch}. The x_origin y_origin is lower left corner of die box (the design .boundary). The x_pitch is the pitch of lowest vertical metal layer and separates the vertical grid lines. The y_pitch is pitch of lowest horizontal metal layer and separates the horizontal grid lines. Type: rect Default: 0.0 0.0 0.0 0.0 Edit: No Reference: floorplan_placement_grid
floorplan_power_rail_layer	
	Specify the layers for calculating power/ground on bottom attribute of cell or techsite. By default, the lowest metal layer is used. Type: string Default: "" Edit: Yes Reference: floorplan_power_rail_layer
floorplan_row_height_increment_corner_to_corner	

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	Specify the offset value for user defined row height constraint with corner to corner edge. Type: int Default: 0 Edit: Yes Reference: floorplan_row_height_increment_corner_to_corner
floorplan_row_height_increment_in_corner_to_corner	
	Specify the offset value for user defined row height constraint with incorner to corner edge. Type: int Default: 0 Edit: Yes Reference: floorplan_row_height_increment_in_corner_to_corner
floorplan_row_height_increment_in_corner_to_in_corner	
	Specify the offset value for user defined row height constraint with incorner to incorner edge. Type: int Default: 0 Edit: Yes Reference: floorplan_row_height_increment_in_corner_to_in_corner
floorplan_row_height_multiple	
	Specify the pitch value for user defined row height constraint. Type: int Default: 0 Edit: Yes Reference: floorplan_row_height_multiple
floorplan_row_site_height	
	Specify the odd even row height constraints. Type: enum Enum Values: any odd even computed Default: any Edit: Yes Reference: floorplan_row_site_height
floorplan_row_site_width	
	Specify the odd even row width constraints. Type: enum Enum Values: any odd even Default: any Edit: Yes Reference: floorplan_row_site_width
floorplan_skip_violations	
	Specify message ID then the message will be not reported and shown in violation browser. (list, optional) Type: string Default: "" Edit: Yes Reference: floorplan_skip_violations
floorplan_snap_all_corners_to_grid	
	Specify to snap all corners of object to grid. Type: bool Default: false Edit: Yes Reference: floorplan_snap_all_corners_to_grid
floorplan_snap_block_grid	

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	Specify the block snap rules. It can be set to snap to manufacture grid, instance grid, placement grid, user-define grid, layer_track grid, finfet manufacture grid, finfet instance grid, finfet placement grid. The default is manufacturing_grid. For Finfet design, the default is finfet_manufacturing_grid. Type: enum Enum Values: manufacturing inst placement user_define layer_track finfet_inst finfet_manufacturing finfet_placement Default: manufacturing Edit: Yes Reference: floorplan_snap_block_grid
floorplan_snap_constraint_grid	
	Specify the constraint snap rules. It can be set to snap to manufacture grid, instance grid, placement grid, user-define grid, layer_track grid, finfet manufacture grid, finfet instance grid, finfet placement grid. The default is inst_grid. For Finfet design, the default is finfet_inst_grid. Type: enum Enum Values: manufacturing inst placement user_define layer_track finfet_inst finfet_manufacturing finfet_placement Default: inst Edit: Yes Reference: floorplan_snap_constraint_grid
floorplan_snap_core_grid	
	Specify the core snap rules. It can be set to snap to manufacture grid, instance grid, placement grid, user-define grid, layer_track grid, finfet manufacture grid, finfet instance grid, finfet placement grid. The default is placement_grid. For Finfet design, the default is finfet_placement_grid. Type: enum Enum Values: manufacturing inst placement user_define layer_track finfet_inst finfet_manufacturing finfet_placement Default: placement Edit: Yes Reference: floorplan_snap_core_grid
floorplan_snap_die_grid	
	Specify the die snap rules. It can be set to snap to manufacture grid, instance grid, placement grid, user-define grid, layer_track grid, finfet manufacture grid, finfet instance grid, finfet placement grid. The default is placement_grid. For Finfet design, the default is finfet_placement_grid. Type: enum Enum Values: manufacturing inst placement user_define layer_track finfet_inst finfet_manufacturing finfet_placement Default: placement Edit: Yes Reference: floorplan_snap_die_grid
floorplan_snap_io_grid	
	Specify the IO snap rules. It can be set to snap to manufacture grid, instance grid, placement grid, user-define grid, layer_track grid, finfet manufacture grid, finfet instance grid, finfet placement grid. The default is manufacturing_grid. For Finfet design, the default is finfet_manufacturing_grid. Type: enum Enum Values: manufacturing inst placement user_define layer_track finfet_inst finfet_manufacturing finfet_placement Default: manufacturing Edit: Yes Reference: floorplan_snap_io_grid
floorplan_snap_place_blockage_grid	
	Specify the placement blockage snap rules. The default is inst. Type: enum Enum Values: manufacturing inst user_define Default: inst Edit: Yes Reference: floorplan_snap_place_blockage_grid
floorplan_snap_place_blockage_type	
	Specify the snapping type. Type: enum Enum Values: closest upsize sequential Default: closest Edit: Yes Reference: floorplan_snap_place_blockage_type
floorplan_user_define_grid	

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	Specify the user-define grid. Format is {x_offset y_offset x_pitch y_pitch}. The x_offset y_offset is the offset of user-define grid. The x_pitch separates the vertical grid lines. The y_pitch separates the horizontal grid lines. The values are in microns. Type: rect Default: 0.0 0.0 0.0 0.0 Edit: Yes Reference: floorplan_user_define_grid
floorplan_vertical_row	
	Specifies that the rows in the floorplan are vertical (1) or horizontal (0). Type: int Default: 0 Edit: Yes Reference: floorplan_vertical_row
flow_branch	
	Branch being run for a hierarchical flow. Type: string Default: "" Edit: Yes Reference: flow_branch
flow_caller_data	
	Data used by the caller of the tool to identify this flow. Type: string Default: "" Edit: Yes Reference: flow_caller_data
flow_db_directory	
	Flow directory where results databases are stored. Type: string Default: dbs Edit: Yes Reference: flow_db_directory
flow_error_errorinfo	
	Tcl error stack in error database. Type: string Default: "" Edit: Yes Reference: flow_error_errorinfo
flow_error_message	
	Tcl error message in error database. Type: string Default: "" Edit: Yes Reference: flow_error_message
flow_error_write_db	
	Write a database when a flow step results in an error condition. Type: bool Default: true Edit: Yes Reference: flow_error_write_db
flow_exclude_time_for_init_flow	
	Marks all snapshots created during the init_flow command as 'exclude_time_metric'. This includes user steps added to the init_flow internal plugin locations. Type: bool Default: false Edit: Yes Reference: flow_exclude_time_for_init_flow

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flow_exit_when_done	
	Exit after running final step. Type: bool Default: false Edit: Yes Reference: flow_exit_when_done
flow_feature_values	
	Global feature settings. Type: string Default: "" Edit: Yes Reference: flow_feature_values
flow_features	
	Global feature definitions. Type: string Default: "" Edit: Yes Reference: flow_features
flow_footer_tcl	
	TCL script to run when a flow ends. Type: string Default: "" Edit: Yes Reference: flow_footer_tcl
flow_header_tcl	
	TCL script to run when a flow starts. Type: string Default: "" Edit: Yes Reference: flow_header_tcl
flow_hier_path	
	Path of current flow within the flow hierarchy. Type: string Default: "" Edit: Yes Reference: flow_hier_path
flow_history	
	Complete flow run history. Type: string Default: "" Edit: Yes Reference: flow_history
flow_log_directory	
	Flow directory where log files are stored. Type: string Default: logs Edit: Yes Reference: flow_log_directory
flow_log_prefix_generator	

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	TCL script to create log filenames in Flow tool. Type: string Default: "" Edit: Yes Reference: flow_log_prefix_generator
flow_mail_on_error	
	Email to flow_mail_to if an error is detected. Type: bool Default: false Edit: Yes Reference: flow_mail_on_error
flow_mail_to	
	Email address where results are sent. Type: string Default: "" Edit: Yes Reference: flow_mail_to
flow_metrics_file	
	Flow metrics file for reporting results. Type: string Default: "" Edit: Yes Reference: flow_metrics_file
flow_metrics_snapshot_parent_uuid	
	The snapshot uuid the results from this flow will be appended to. Type: string Default: "" Edit: Yes Reference: flow_metrics_snapshot_parent_uuid
flow_metrics_snapshot_uuid	
	The snapshot uuid of the most recent flow step executed. Type: string Default: "" Edit: Yes Reference: flow_metrics_snapshot_uuid
flow_overwrite_db	
	Enables overwriting databases when saving. Type: bool Default: false Edit: Yes Reference: flow_overwrite_db
flow_plugin_names	
	Names of all internal plug-in points. Type: string Default: "" Edit: Yes Reference: flow_plugin_names
flow_plugin_steps	
	List of plug-in steps that have been added through edit_flow. Type: string Default: "" Edit: Yes Reference: flow_plugin_steps

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flow_post_db_overwrite	
	If set within a skip_db flow_step, it allows the user to identify the name and type of database to present to future flows as the flow_starting_db. This can be set to no_db to force subsequent flows to begin without a starting db. Type: string Default: "" Edit: Yes Reference: flow_post_db_overwrite
flow_remark	
	Remark from the last flow yaml file loaded. Type: string Default: "" Edit: Yes Reference: flow_remark
flow_report_directory	
	Flow directory where reports are written. Type: string Default: reports Edit: Yes Reference: flow_report_directory
flow_run_tag	
	Tags for this particular flow run. Type: string Default: "" Edit: Yes Reference: flow_run_tag
flow_schedule	
	Set of flows to run after the current flow completes. Type: string Default: "" Edit: Yes Reference: flow_schedule
flow_starting_db	
	Starting database for the flow run. Type: string Default: "" Edit: Yes Reference: flow_starting_db
flow_startup_directory	
	Directory where the tool was started. Type: string Default: /home/manishk/dbschema/novus/newscript Edit: Yes Reference: flow_startup_directory
flow_status_file	
	File for flow status. Type: string Default: "" Edit: Yes Reference: flow_status_file
flow_step_begin_tcl	
	Defines the Tcl to run before running the body of each step. Type: string Default: "" Edit: Yes

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flow_step_canonical_current	
	Currently running canonical flow step. Type: string Default: "" Edit: Yes Reference: flow_step_canonical_current
flow_step_check_tcl	
	Procedure to check the steps. Type: string Default: "" Edit: Yes Reference: flow_step_check_tcl
flow_step_current	
	Currently running flow step. Type: string Default: "" Edit: Yes Reference: flow_step_current
flow_step_end_tcl	
	Defines the Tcl to run after running the body of each step. Type: string Default: "" Edit: Yes
flow_step_last	
	Last flow step to have been completed. Type: string Default: "" Edit: Yes Reference: flow_step_last
flow_step_last_msg	
	Message provided for the last step that was run. Type: string Default: "" Edit: Yes Reference: flow_step_last_msg
flow_step_last_status	
	Status for the last flow step run. Type: string Default: not_run Edit: Yes Reference: flow_step_last_status
flow_step_next	
	Next step to run in the flow. Type: string Default: "" Edit: Yes Reference: flow_step_next
flow_steps	
	List of all created flow_steps Type: obj(flow_step)* Default: "" Edit: No
flow_summary_tcl	

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	TCL script to run during run_flow summary. Type: string Default: "" Edit: Yes Reference: flow_summary_tcl
flow_template_feature_definition	
	Features list and status for the current template. Type: string Default: "" Edit: Yes Reference: flow_template_feature_definition
flow_template_tools	
	List of tools included in flow templates being run. Type: string Default: "" Edit: Yes Reference: flow_template_tools
flow_template_type	
	Type of template being run. Type: string Default: "" Edit: Yes Reference: flow_template_type
flow_template_version	
	Version of the template being run. Type: string Default: "" Edit: Yes Reference: flow_template_version
flow_top	
	Flow to run by default. Type: string Default: "" Edit: Yes Reference: flow_top
flow_user_templates	
	Flow user template definitions. Type: string Default: "" Edit: Yes Reference: flow_user_templates
flow_verbose	
	Enables printing run information in the logfile. Type: bool Default: true Edit: Yes Reference: flow_verbose
flow_working_directory	
	Flow directory where the flow is being run. Type: string Default: . Edit: Yes Reference: flow_working_directory
flow_write_db_snapshot	

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	Create a snapshot for write_db on flow steps which save a database. Type: <code>bool</code> Default: true Edit: Yes Reference: flow_write_db_snapshot
flow_write_db_snapshot_exclude_time	
	Marks all snapshots created for write_db on flow steps which save a database as 'exclude_time_metric'. Type: <code>bool</code> Default: true Edit: Yes Reference: flow_write_db_snapshot_exclude_time
flow_yamllint_exec	
	Yamllint executable command and arguments for checking yaml files. Type: <code>string</code> Default: yamllint Edit: Yes Reference: flow_yamllint_exec
flows	
	List of all created flows Type: <code>obj(flow)*</code> Default: "" Edit: No
flowtool_exit_timeout	
	Maximum amount of time in seconds Flow tool will wait after a tool exits for a completed status. Type: <code>int</code> Default: 30 Edit: Yes Reference: flowtool_exit_timeout
flowtool_extra_arguments	
	Extra arguments passed to Flowtool when running in tool mode. Type: <code>string</code> Default: "" Edit: Yes Reference: flowtool_extra_arguments
flowtool_metrics_qor_excel	
	Metrics file to write after tools exit. Type: <code>string</code> Default: "" Edit: Yes Reference: flowtool_metrics_qor_excel
flowtool_metrics_qor_html	
	Metrics file to write after tools exit. Type: <code>string</code> Default: "" Edit: Yes Reference: flowtool_metrics_qor_html
flowtool_metrics_qor_text	
	Metrics file to write after tools exit. Type: <code>string</code> Default: "" Edit: Yes Reference: flowtool_metrics_qor_text
flowtool_metrics_qor_vivid	

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	Metrics file to write after tools exit. Type: string Default: "" Edit: Yes Reference: flowtool_metrics_qor_vivid
flowtool_predict_full_names	
	Show skip_metric flow name in prediction. Type: bool Default: false Edit: Yes Reference: flowtool_predict_full_names
flowtool_schedule_flow_immediate	
	Start schedule_flow processes immediately instead of waiting for current tool exit. Type: bool Default: false Edit: Yes
flowtool_summary_tcl	
	Tcl script to run at the end of flowtool. Type: string Default: puts [report_metric -format text] Edit: Yes Reference: flowtool_summary_tcl
foreign_cells	
	Short-cut for [get_db base_cells .foreign_cells] Type: obj(foreign_cell)* Default: "" Edit: No
formal_verification_constraints	
	List of all formal_verification_constraints Type: obj(formal_verification_constraint)* Default: "" Edit: No
gcell_grids	
	Short-cut to all the gcell_grid objects in the design. Type: obj(gcell_grid)* Default: "" Edit: No
gcells	
	The gcells for this design. The gcells are only created after global route has occurred. Type: obj(gcell)* Default: "" Edit: No
generate_special_via_accuracy_effort	
	Specifies viagen effort to generate vias. Type: enum Enum Values: low medium high Default: high Edit: Yes Reference: generate_special_via_accuracy_effort
generate_special_via_add_pin_to_pin_vias	

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	Inserts vias to connect pins between cover macros Type: bool Default: false Edit: Yes Reference: generate_special_via_add_pin_to_pin_vias
generate_special_via_align_merged_stack_via_metal	
	Aligns merged stack vias Type: bool Default: false Edit: Yes Reference: generate_special_via_align_merged_stack_via_metal
generate_special_via_allow_via_expand	
	Allows via expansion to meet LEF rules Type: bool Default: true Edit: Yes Reference: generate_special_via_allow_via_expand
generate_special_via_allow_wire_shape_change	
	Allows via to change the metal shape of existing wire Type: bool Default: true Edit: Yes Reference: generate_special_via_allow_wire_shape_change
generate_special_via_area_only	
	adds via only within the specified area Type: bool Default: false Edit: Yes Reference: generate_special_via_area_only
generate_special_via_bot_enclosure	
	Customize bot enclosure Type: string Default: "" Edit: Yes
generate_special_via_check_signal_routes	
	Honors pre-existing signal routes Type: enum Enum Values: 0 1 2 Default: 0 Edit: Yes Reference: generate_special_via_check_signal_routes
generate_special_via_create_double_row_cut_via	
	Expands vias to have an additional cut than original for one-row or one-column vias if possible Type: enum Enum Values: no_extra_cut add_extra_cut add_extra_cut_and_metal Default: no_extra_cut Edit: Yes Reference: generate_special_via_create_double_row_cut_via
generate_special_via_create_max_row_cut_via	
	Creates max rows of via cuts Type: bool Default: false Edit: Yes Reference: generate_special_via_create_max_row_cut_via

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generate_special_via_cut_class_preference	
	Specifies via cut class preference Type: string Default: default Edit: Yes Reference: generate_special_via_cut_class_preference
generate_special_via_cut_spacing	
	Customize cut spacing Type: string Default: "" Edit: Yes
generate_special_via_disable_via_merge	
	Disables via merging Type: bool Default: false Edit: Yes Reference: generate_special_via_disable_via_merge
generate_special_via_enable_check_drc	
	Invokes check_drc Type: bool Default: false Edit: Yes Reference: generate_special_via_enable_check_drc
generate_special_via_extend_out_wire_end	
	Extends via outside of existing wires end Type: bool Default: true Edit: Yes Reference: generate_special_via_extend_out_wire_end
generate_special_via_full_cut_via_only	
	allows to create full cut via only Type: bool Default: false Edit: Yes Reference: generate_special_via_full_cut_via_only
generate_special_via_hookup_auto_set_viacell	
	Hookup auto set viaCell Type: bool Default: false Edit: Yes
generate_special_via_hookup_contact_max	
	Sets the max distance to create hookup via from contact, and the max distance between two hookup via on a continuous M0 PG pin. Type: string Default: 0, 0 Edit: Yes Reference: generate_special_via_hookup_contact_max
generate_special_via_hookup_contact_pg_track	
	Sets start point and the pitch of PG track used by hookup contact. Type: string Default: 0, 0 Edit: Yes Reference: generate_special_via_hookup_contact_pg_track
generate_special_via_hookup_fixed_grid	

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	Specifies to add hookup via over the fixed Grid. Type: bool Default: false Edit: Yes Reference: generate_special_via_hookup_fixed_grid
generate_special_via_hookup_min_distance	
	min distance between two pg vias Type: string Default: min Edit: Yes Reference: generate_special_via_hookup_min_distance
generate_special_via_hookup_preplace_fix	
	Enhance hookup via generation runtime Type: int Default: 0 Edit: Yes Reference: generate_special_via_hookup_preplace_fix
generate_special_via_hookup_rail_pair	
	Create hookup vias over pairing followpins Type: bool Default: false Edit: Yes Reference: generate_special_via_hookup_rail_pair
generate_special_via_hookup_via_distance	
	Sets via hookup distance and boundary spacing Type: string Default: 4pitch 4pitch 0 Edit: Yes Reference: generate_special_via_hookup_via_distance
generate_special_via_hookup_via_rule	
	Specifies the viarule used in specified area Type: string Default: "" Edit: Yes Reference: generate_special_via_hookup_via_rule
generate_special_via_hookup_via_style	
	Sets via hookup style Type: enum Enum Values: none loose compact moderate Default: none Edit: Yes Reference: generate_special_via_hookup_via_style
generate_special_via_hookup_virtual_trim_grid	
	Specifies the virtual trim grid for pg hookup Type: string Default: "" Edit: Yes Reference: generate_special_via_hookup_virtual_trim_grid
generate_special_via_ignore_design_boundary	
	Allow vias to be on design boundary Type: bool Default: false Edit: Yes Reference: generate_special_via_ignore_design_boundary

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generate_special_via_ignore_drc	
	Ignores DRC Type: bool Default: false Edit: Yes Reference: generate_special_via_ignore_drc
generate_special_via_ignore_rule_enclosure	
	Ignores enclosures defined in VIARULE GENERATE Type: bool Default: true Edit: Yes Reference: generate_special_via_ignore_rule_enclosure
generate_special_via_inherit_wire_status	
	Inherits wire status Type: bool Default: false Edit: Yes Reference: generate_special_via_inherit_wire_status
generate_special_via_keep_existing_via	
	Keeps existing vias Type: enum Enum Values: remove keep_with_new_via keep_with_no_new_via Default: remove Edit: Yes Reference: generate_special_via_keep_existing_via
generate_special_via_keep_fixed_via	
	Keeps fixed vias not modifiable Type: bool Default: true Edit: Yes Reference: generate_special_via_keep_fixed_via
generate_special_via_mincut_preference	
	Specifies min cut preference Type: int Default: 1 Edit: Yes Reference: generate_special_via_mincut_preference
generate_special_via_opt_cross_via	
	Controls the creation of additional via cuts Type: bool Default: false Edit: Yes Reference: generate_special_via_opt_cross_via
generate_special_via_opt_via_on_routing_track	
	Frees up via tracks Type: bool Default: false Edit: Yes Reference: generate_special_via_opt_via_on_routing_track
generate_special_via_parameterized_via_only	

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	generate parameterized vias only, auto sets true for OA flow, false for non-OA flow. Type: enum Enum Values: auto true false Default: auto Edit: Yes Reference: generate_special_via_parameterized_via_only
generate_special_via_partial_overlap_threshold	
	Specifies ratio threshold of via creation on partial overlap wires Type: double Default: 0.0 Edit: Yes Reference: generate_special_via_partial_overlap_threshold
generate_special_via_preferred_vias_only	
	Generate only vias set in -viarule_preference Type: enum Enum Values: use_lef open keep Default: use_lef Edit: Yes Reference: generate_special_via_preferred_vias_only
generate_special_via_prefix	
	Specifies prefix name for all vias Type: string Default: "" Edit: Yes Reference: generate_special_via_prefix
generate_special_via_reference_boundary	
	Sets reference boundary Type: enum Enum Values: design core Default: design Edit: Yes Reference: generate_special_via_reference_boundary
generate_special_via_respect_stdcell_geometry	
	Respects all standard cell patterns Type: bool Default: false Edit: Yes Reference: generate_special_via_respect_stdcell_geometry
generate_special_via_respect_wire_ndr	
	Control whether viagen support ndr rules Type: bool Default: false Edit: Yes Reference: generate_special_via_respect_wire_ndr
generate_special_via_rule_preference	
	Specifies via rule preference Type: string Default: default Edit: Yes Reference: generate_special_via_rule_preference
generate_special_via_set_via_expand_dir	

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	Sets via expansion direction Type: enum Enum Values: auto horizontal vertical Default: auto Edit: Yes Reference: generate_special_via_set_via_expand_dir
generate_special_via_snap_via_center_to_grid	
	Specifies the via center alignment style per routing layer Type: string Default: "" Edit: Yes Reference: generate_special_via_snap_via_center_to_grid
generate_special_via_split_long_via_global_grid	
	Controls global grids so vias on them are in line Type: string Default: 0 0 0 0 0 0 0 Edit: Yes Reference: generate_special_via_split_long_via_global_grid
generate_special_via_symmetrical_via_only	
	adds symmetrical via only, auto sets true for OA flow, false for non-OA flow. Type: enum Enum Values: auto true false Default: auto Edit: Yes Reference: generate_special_via_symmetrical_via_only
generate_special_via_top_enclosure	
	Customize top enclosure Type: string Default: "" Edit: Yes
generate_special_via_use_track_offset	
	Uses track offset Type: bool Default: false Edit: Yes Reference: generate_special_via_use_track_offset
generate_special_via_use_trim_metal_enclosure	
	Force tool to use enclosures with trim layer Type: bool Default: false Edit: Yes Reference: generate_special_via_use_trim_metal_enclosure
get_db_display_limit	
	Controls the number of objects that will be displayed when using get_db with an attribute pattern like "*" rather than single attribute name. If the value is 10, 'get_db insts.*' will only display 10 inst objects with all their attributes, and will only display the first 10 objects for any attribute that is an object list (e.g. the first 10 pins for each inst.pins value). Type: int Default: 10 Edit: Yes Reference: get_db_display_limit
group_instance_suffix	

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	base name of group hinst is concatenation of group module name and suffix defined by this tcl global Type: string Default: i Edit: Yes Reference: group_instance_suffix
groups	
	Short-cut for [get_db current_design .groups] Type: obj(group)* Default: "" Edit: No
gui_lines	
	Short-cut to all the gui_line objects in the design. Type: obj(gui_line)* Default: "" Edit: No
gui_polygons	
	Short-cut to all the gui_polygon objects in the design. Type: obj(gui_polygon)* Default: "" Edit: No
gui_rects	
	Short-cut to all the gui_rect objects in the design. Type: obj(gui_rect)* Default: "" Edit: No
gui_shapes	
	Short-cut to all the gui_shape objects in the design. Type: obj(gui_rect)* obj(gui_line)* obj(gui_polygon)* Default: "" Edit: No
gui_texts	
	Short-cut to all the gui_text objects in the design. Type: obj(gui_text)* Default: "" Edit: No
gui_verbose	
	Controls which GUI actions are logged. The choices are: - all: all GUI actions; - db: GUI actions that affect the DB like gui_select, set_layer_preference, but not display-only actions like gui_zoom, gui_pan; - none: no gui_* commands are logged, so the .log and .cmd file are incomplete Type: enum Enum Values: all db none Default: db Edit: Yes Reference: gui_verbose
hdl_allow_inout_constant_port_connect	
	When set to true, allows constants to be connected to inout ports. Type: bool Default: false Edit: Yes Reference: hdl_allow_inout_constant_port_connect
hdl_allow_instance_name_conflict	

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	Allow a Verilog instance and wire to have the same name. Type: bool Default: false Edit: Yes Reference: hdl_allow_instance_name_conflict
hdl_allow_positional_connections_for_pg_inst	
	Allows positional port association for instantiation of cell with power and ground pins.. Type: bool Default: false Edit: Yes Reference: hdl_allow_positional_connections_for_pg_inst
hdl_append_generic_ports	
	When set to false, does not append generic port information to the module name. Type: bool Default: true Edit: Yes Reference: hdl_append_generic_ports
hdl_array_naming_style	
	Format for naming individual bits of array variables. Type: string Default: %s[%d] Edit: Yes Reference: hdl_array_naming_style
hdl_async_set_reset	
	Specifies that the listed signals be used to drive latch set/reset pins when possible. Type: string Default: "" Edit: Yes Reference: hdl_async_set_reset
hdl_auto_async_set_reset	
	Forces use of asynchronous set and reset pins, rather than data input pins, to implement all asynchronous set and reset operations for latches. Type: bool Default: false Edit: Yes Reference: hdl_auto_async_set_reset
hdl_auto_sync_set_reset	
	When set to true, the set/reset signal will be preserved with the flip-flop during generic optimization. Type: bool Default: true Edit: Yes Reference: hdl_auto_sync_set_reset
hdl_bus_wire_naming_style	
	Format for naming individual bits of bus wires. Type: string Default: %s[%d] Edit: Yes Reference: hdl_bus_wire_naming_style
hdl_case_mux_threshold	
	Applies to muxes generated for case statements. When the the number of data inputs is less than the threshold, AND/OR logic is generated rather than a mux. Type: int Default: 2 Edit: Yes Reference: hdl_case_mux_threshold

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hdl_case_sensitive_instances	
	When set to false, allows case insensitive name matching for instances in Verilog whose definition exists in VHDL or Verilog files. Type: string Default: none Edit: Yes Reference: hdl_case_sensitive_instances
hdl_convert_one_bit_vector_to_scalar	
	When set to true converts one bit vector to scalar. Type: bool Default: false Edit: Yes Reference: hdl_convert_one_bit_vector_to_scalar
hdl_convert_onebit_vector_wire_to_scalar	
	Converts a one-bit vector declaration into a scalar. Type: bool Default: false Edit: Yes Reference: hdl_convert_onebit_vector_wire_to_scalar
hdl_create_label_for_unlabeled_generate	
	Specifies whether a name is assigned to unnamed generate blocks. Type: bool Default: true Edit: Yes Reference: hdl_create_label_for_unlabeled_generate
hdl_decimal_parameter_name	
	When set to true it prints out parameters in decimal format. Type: bool Default: false Edit: Yes Reference: hdl_decimal_parameter_name
hdl_delete_transparent_latch	
	Deletes latches that are always enabled. Type: bool Default: true Edit: Yes Reference: hdl_delete_transparent_latch
hdl_early_redundant_logic_removal	
	Do an early cleanup of redundant steering logic nodes. Type: bool Default: true Edit: Yes Reference: hdl_early_redundant_logic_removal
hdl_enable_proc_name	
	Enable generation of instance hdl_proc_name attributes. Type: bool Default: false Edit: Yes Reference: hdl_enable_proc_name
hdl_enable_real_support	
	Enable elaboration of constant expressions of real type. Type: bool Default: false Edit: Yes Reference: hdl_enable_real_support

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hdl_error_on_blackbox	
	An error is issued if a blackbox is found in a design. Type: bool Default: false Edit: Yes Reference: hdl_error_on_blackbox
hdl_error_on_latch	
	An error is issued if a latch is inferred for a design. Type: bool Default: false Edit: Yes Reference: hdl_error_on_latch
hdl_error_on_logic_abstract	
	An error is issued if a logic abstract is found in a design. Type: bool Default: false Edit: Yes Reference: hdl_error_on_logic_abstract
hdl_error_on_negedge	
	Specifies that an error be issued when a falling edge-triggered flip-flop is inferred. Type: bool Default: false Edit: Yes Reference: hdl_error_on_negedge
hdl_exclude_parameter_in_cell_search	
	This attribute applies to situation when a parameterized module instantiation is resolved with a Liberty complex cell as opposed to an hdl module. This attribute will contain the list of names of the cells for which the search is done with the name of the module but without the parameter information. Type: string Default: "" Edit: Yes Reference: hdl_exclude_parameter_in_cell_search
hdl_ff_keep_explicit_feedback	
	Controls how flip-flop stable states are implemented for feedback assignments that are explicitly specified in the RTL, e.g., D = Q.Explicit feedback is always kept when hdl_ff_keep_feedback is true. Type: bool Default: true Edit: Yes Reference: hdl_ff_keep_explicit_feedback
hdl_ff_keep_feedback	
	Controls how flip-flop stable states are implemented: with a feedback path from the Q output to the D input (true), or with a synchronous enable signal (false). Type: bool Default: false Edit: Yes Reference: hdl_ff_keep_feedback
hdl_file_row_column_consider_modules_as_leaf_insts	
	Specifies list of primitive modules. Type: string Default: "" Edit: Yes Reference: hdl_file_row_column_consider_modules_as_leaf_insts
hdl_flatten_complex_port	

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	To convert complex ports to vector ports. Type: bool Default: false Edit: Yes Reference: hdl_flatten_complex_port
hdl_flatten_complex_port_in_bottom_up_flow	
	Used for creating one dimensional packed version of complex ports in bottom up flow. Type: bool Default: true Edit: Yes Reference: hdl_flatten_complex_port_in_bottom_up_flow
hdl_generate_index_style	
	Format of 'for generate' block labels. Type: string Default: %s[%d] Edit: Yes Reference: hdl_generate_index_style
hdl_generate_separator	
	String used to separate generate block labels. Type: string Default: . Edit: Yes Reference: hdl_generate_separator
hdl_ignore_pragma_names	
	Specifies pragmas to be ignored. Type: string Default: coverage Edit: Yes Reference: hdl_ignore_pragma_names
hdl_index_mux_threshold	
	Applies to muxes generated for variable index array references, such as 'y = x[i]'. When the the number of data inputs is less than the threshold, AND/OR logic is generated rather than a mux. Type: int Default: 0 Edit: Yes Reference: hdl_index_mux_threshold
hdl_inout_assign	
	When set to true, interprets Verilog continuous assign statements as being bidirectional. Type: bool Default: false Edit: Yes Reference: hdl_inout_assign
hdl_inout_wand_wor_assign	
	When set to true treats assignment to wand/wor net type as bidirectional. Type: bool Default: false Edit: Yes Reference: hdl_inout_wand_wor_assign
hdl_instance_array_naming_style	
	Format of instance array labels. Type: string Default: %s[%d] Edit: Yes Reference: hdl_instance_array_naming_style

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hdl_interface_separator	
	String used to separate interface elements. Type: string Default: _ Edit: Yes Reference: hdl_interface_separator
hdl_keep_first_module_definition	
	Keep first seen definition of a module. Type: bool Default: false Edit: Yes Reference: hdl_keep_first_module_definition
hdl_language	
	Specifies the default HDL language mode. Type: string Default: v2001 Edit: Yes Reference: hdl_language
hdl_latch_keep_feedback	
	Controls how explicitly-specified latch stable states (e.g, q <= q) are implemented: with a feedback path from the Q output to the D input (true), or with a latch enable signal (false). Type: bool Default: false Edit: Yes Reference: hdl_latch_keep_feedback
hdl_link_from_any_lib	
	Resolve an instantiation with a 'unique' binding found across all present libs. Type: bool Default: true Edit: Yes Reference: hdl_link_from_any_lib
hdl_max_loop_limit	
	Determines the maximum number of iterations for unfolding a loop construct of any type. Type: int Default: 4096 Edit: Yes Reference: hdl_max_loop_limit
hdl_max_map_to_mux_control_width	
	Specifies the maximum multiplexer size, in terms of the width of the control input, that can be generated for HDL 'case' statements marked with the 'map_to_mux' pragma. Type: int Default: 10 Edit: Yes Reference: hdl_max_map_to_mux_control_width
hdl_max_memory_address_range	
	Sets the maximum addressable memory range to prevent synthesis of large memories. Type: int Default: 4194304 Edit: Yes Reference: hdl_max_memory_address_range
hdl_max_recursion_limit	

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	Sets the maximum number of elaborations for recursive instantiations to prevent possible infinite recursions. Type: int Default: 1024 Edit: Yes Reference: hdl_max_recursion_limit
hdl_module_definition_resolution	
	Resolves the definition of module in case of clash to first seen module or to last seen module or to module according to the directory rules. It can take four values viz last, first, first_dir_netlist and first_dir_rtl. Type: string Default: last Edit: Yes Reference: hdl_module_definition_resolution
hdl_nc_compatible_module_link	
	When true, implements NC compatible binding rules for linking module/entity instantiations with corresponding definitions. Type: bool Default: true Edit: Yes Reference: hdl_nc_compatible_module_link
hdl_new_inout_assign	
	When set to true, interprets Verilog continuous assign statements as being bidirectional. Type: bool Default: true Edit: Yes Reference: hdl_new_inout_assign
hdl_overwrite_command_line_macros	
	Enables overwriting of command line macros. Type: bool Default: true Edit: Yes Reference: hdl_overwrite_command_line_macros
hdl_parameter_module_name	
	When set to false, skips all parameter values in parameterized module name for top-level design or design specified with 'elaborate' command. Type: bool Default: true Edit: Yes Reference: hdl_parameter_module_name
hdl_parameter_naming_style	
	Format of the suffix added to original module name per each parameter overwrite. Type: string Default: <code>_%s%d</code> Edit: Yes Reference: hdl_parameter_naming_style
hdl_preserve_async_set_reset_priority_logic	
	When true, prevents RC from generating redundant logic when the mapper selects a flip-flop library cell that has the same priority for the asynchronous set and reset operations as is specified in the input HDL. Type: bool Default: false Edit: Yes Reference: hdl_preserve_async_set_reset_priority_logic
hdl_preserve_dangling_output_nets	
	When true, preserves dangling nets connected to instance output ports. Type: bool Default: true Edit: Yes Reference: hdl_preserve_dangling_output_nets

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hdl_preserve_supply_nets	
	Preserve supply nets in user input. Type: bool Default: true Edit: Yes Reference: hdl_preserve_supply_nets
hdl_preserve_sync_control_logic	
	Instructs Genus to preserve the sync control logic structure. Setting this attribute can impact QoR. Type: bool Default: false Edit: Yes Reference: hdl_preserve_sync_control_logic
hdl_preserve_sync_set_reset	
	Instructs Genus to preserve the sync set/reset control logic structure. Setting this attribute can impact QoR. Type: bool Default: false Edit: Yes Reference: hdl_preserve_sync_set_reset
hdl_preserve_unused_flop	
	When set to true, preserves flip-flops that do not affect any outputs. Type: bool Default: false Edit: Yes Reference: hdl_preserve_unused_flop
hdl_preserve_unused_latch	
	When set to true, preserves latches that do not affect any outputs. Type: bool Default: false Edit: Yes Reference: hdl_preserve_unused_latch
hdl_preserve_unused_registers	
	When set to true, preserves flip-flops and latches that do not affect any outputs. Type: bool Default: false Edit: Yes Reference: hdl_preserve_unused_registers
hdl_primitive_input_multibit	
	Allow a Verilog primitive gate instantiated in RTL to have multi bit input. Type: bool Default: false Edit: Yes Reference: hdl_primitive_input_multibit
hdl_record_naming_style	
	Format for naming individual bits of record variables. Type: string Default: %s[%s] Edit: Yes Reference: hdl_record_naming_style
hdl_register_array_naming_style	
	Format for naming array type registers. Type: string Default: "" Edit: Yes Reference: hdl_register_array_naming_style

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hdl_register_naming_style	
	Format for naming register instances. Type: string Default: %s_reg%s Edit: Yes Reference: hdl_register_naming_style
hdl_register_record_naming_style	
	Format for naming record type registers. Type: string Default: "" Edit: Yes Reference: hdl_register_record_naming_style
hdl_rename_cdn_flop_pins	
	Provides the control to change the pin names of internal flops. Type: string Default: "" Edit: Yes Reference: hdl_rename_cdn_flop_pins
hdl_rename_cdn_latch_pins	
	Provides the control to change the pin names of internal latches. Type: string Default: "" Edit: Yes Reference: hdl_rename_cdn_latch_pins
hdl_report_case_info	
	When set to true the case pragma information will be reported. Type: bool Default: false Edit: Yes Reference: hdl_report_case_info
hdl_resolve_instance_with_lib_cell	
	Use a technology element rather than a Verilog module or VHDL architecture for any instantiation. Type: bool Default: false Edit: Yes Reference: hdl_resolve_instance_with_lib_cell
hdl_resolve_parameterized_instance_with_structural_module	
	Uses a module named with a parameterized name for any instance of parameterized Verilog modules or VHDL entities. Type: bool Default: false Edit: Yes Reference: hdl_resolve_parameterized_instance_with_structural_module
hdl_sv_module_wrapper	
	Enables generation of System Verilog module wrapper. Type: bool Default: false Edit: Yes Reference: hdl_sv_module_wrapper
hdl_sync_set_reset	
	Specifies that the listed signals be used to drive flip-flop set/reset pins when possible. Type: string Default: "" Edit: Yes Reference: hdl_sync_set_reset

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hdl_track_filename_row_col	
	Enable/disable HDL file:row:col info tracking on instances. Type: bool Default: false Edit: Yes Reference: hdl_track_filename_row_col
hdl_track_module_elab_memory_and_runtime	
	Track elaboration memory and runtime for all modules. Type: bool Default: false Edit: Yes Reference: hdl_track_module_elab_memory_and_runtime
hdl_unconnected_marking_done	
	Specifies whether the undriven signals have been connected based on the value of the 'hdl_unconnected_value' attribute. Type: bool Default: false Edit: Yes
hdl_unconnected_value	
	Connects any undriven signal (undriven input in module or cell instantiation/ undriven output/ undriven signal in a module to the specified value unless the 'none' value is specified. Type: string Default: 0 Edit: Yes Reference: hdl_unconnected_value
hdl_use_block_prefix	
	Whether block statement labels should be used to prefix instances. Type: bool Default: true Edit: Yes Reference: hdl_use_block_prefix
hdl_use_case_generate_prefix	
	Specifies whether 'case' generate labels should be used to prefix instances. Type: bool Default: true Edit: Yes Reference: hdl_use_case_generate_prefix
hdl_use_current_dir_before_hdl_search_path	
	Uses current directory before hdl search path. Type: bool Default: false Edit: Yes Reference: hdl_use_current_dir_before_hdl_search_path
hdl_use_cw_first	
	Uses a ChipWare component rather than a Verilog module or VHDL architecture for any instantiation. Type: bool Default: false Edit: Yes Reference: hdl_use_cw_first
hdl_use_default_parameter_values_in_name	
	When set to true, uses all parameter values in parameterized module name. Type: bool Default: false Edit: Yes Reference: hdl_use_default_parameter_values_in_name

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hdl_use_for_generate_prefix	
	Specifies whether 'for' generate labels should be used to create instance names. Type: bool Default: true Edit: Yes Reference: hdl_use_for_generate_prefix
hdl_use_if_generate_prefix	
	Specifies whether 'if' generate labels should be used to prefix instances. Type: bool Default: true Edit: Yes Reference: hdl_use_if_generate_prefix
hdl_use_port_default_value	
	When true, uses default initial values of VHDL entity input ports. Type: bool Default: true Edit: Yes Reference: hdl_use_port_default_value
hdl_verilog_defines	
	A list of Verilog macro definitions that are passed to read_hdl and read_netlist commands. Type: string Default: SYNTHESIS Edit: Yes Reference: hdl_verilog_defines
hdl_vhdl_assign_width_mismatch	
	When set to true, allows arrays with size mismatch to be assigned in VHDL. Type: bool Default: false Edit: Yes Reference: hdl_vhdl_assign_width_mismatch
hdl_vhdl_case	
	Causes the VHDL analyzer to store VHDL identifiers and operators in lower case, upper case, or the case given in the source file. Type: string Default: original Edit: Yes Reference: hdl_vhdl_case
hdl_vhdl_environment	
	Specifies the selection of the predefined arithmetic libraries. Type: string Default: common Edit: Yes Reference: hdl_vhdl_environment
hdl_vhdl_preferred_architecture	
	Sets the name of preferred architecture to use with an entity when there are multiple architectures. Type: string Default: "" Edit: Yes Reference: hdl_vhdl_preferred_architecture
hdl_vhdl_range_opt	
	When set to false disables range optimization in VHDL. Type: bool Default: false Edit: Yes Reference: hdl_vhdl_range_opt

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hdl_vhdl_read_version	
	Specifies the VHDL version to be used when reading VHDL designs. Type: string Default: 1993 Edit: Yes Reference: hdl_vhdl_read_version
hdl_vhdl_strict_compliance	
	Enforces a more strict interpretation of the VHDL LRM. Type: bool Default: false Edit: Yes Reference: hdl_vhdl_strict_compliance
hier_flow_budgets	
	This will set budgeting mode options via setBudgetingMode command for accurate or preliminary budgeting whichever is specified. It has lower precedence than other user specified budgeting mode settings. Type: enum Enum Values: accurate preliminary none Default: none Edit: Yes
hinsts	
	Short-cut for [get_db current_design .hinsts] Type: obj(hinst)* Default: "" Edit: No
hnets	
	Short-cut for [get_db current_design .hnets] Type: obj(hnet)* Default: "" Edit: No
honor_valid_location	
	Controls whether to honor the valid location defined on the library cell. Type: bool Default: false Edit: Yes Reference: honor_valid_location
hpin_busses	
	Short-cut for [get_db current_design .hpin_busses] Type: obj(hpin_bus)* Default: "" Edit: No
hpins	
	Short-cut for [get_db current_design .hpins] Type: obj(hpin)* Default: "" Edit: No
hport_busses	
	Short-cut for [get_db current_design .hport_busses] Type: obj(hport_bus)* Default: "" Edit: No
hports	

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	Short-cut for [get_db hinsts .hports] Type: obj(hport)* Default: "" Edit: No
hybrid_hier_feedthru	
	Specifies to enable or disable feed-through insertion Type: bool Default: true Edit: Yes
hybrid_hier_feedthru_plugin	
	Specifies the file name having customized feed-through flow from user Type: string Default: "" Edit: Yes
hybrid_hier_out_dir	
	Specifies directory name for feedthru, budget and pin assignment reports Type: string Default: . Edit: Yes
hybrid_hier_pre_timing_budget_plugin	
	User file to specify additional budgeting mode settings. Type: string Default: "" Edit: Yes
hybrid_hier_timing_budget	
	Specifies to enable or disable timing budget Type: bool Default: true Edit: Yes
hybrid_hier_write_db	
	Specifies to save innovus design after intermediate steps Type: bool Default: true Edit: Yes
ignore_max_via_stack_rule	
	If true, the MaxVIASTACK rule in tlef file is ignored. Type: bool Default: false Edit: Yes Reference: ignore_max_via_stack_rule
ignore_pin_error_in_test_cell_function	
	Ignore pin error in the test cell function. Type: bool Default: false Edit: Yes Reference: ignore_pin_error_in_test_cell_function
ignore_scan_combinational_arcs	
	Should we ignore combinational arcs involving scan pins. Type: bool Default: true Edit: Yes Reference: ignore_scan_combinational_arcs
ignore_sigma_arc_mismatch	

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	Ignores sigma arc inconsistency across views during mmmc consistency check. Type: bool Default: true Edit: Yes Reference: ignore_sigma_arc_mismatch
ilm_filter_internal_path	
	filterInternalPath Type: bool Default: false Edit: Yes Reference: ilm_filter_internal_path
ilm_keep_async	
	Keep paths from asynchronous input ports Type: bool Default: false Edit: Yes Reference: ilm_keep_async
ilm_keep_flatten	
	Instruct Innovus to keep in flattened state Type: bool Default: true Edit: Yes Reference: ilm_keep_flatten
ilm_keep_high_fanout_ports	
	Keep high fanout input port paths Type: bool Default: true Edit: Yes Reference: ilm_keep_high_fanout_ports
ilm_keep_inst_in_sdc	
	Keep instances which have constraints on it Type: bool Default: false Edit: Yes Reference: ilm_keep_inst_in_sdc
ilm_keep_loopback	
	Keep loop back paths Type: bool Default: false Edit: Yes Reference: ilm_keep_loopback
ilm_max_num_insts	
	Specify the maximum number of instances per port (used to define high_fanout_ports) Type: int Default: 1000 Edit: Yes Reference: ilm_max_num_insts
ilm_max_num_registers	
	Specify the maximum number of registers per port (used to define high_fanout_ports) Type: int Default: 40 Edit: Yes Reference: ilm_max_num_registers
ilm_slack_driven	

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	slackDriven Type: bool Default: false Edit: Yes Reference: ilm_slack_driven
incremental_joules_instance_threshold_count_multi_thread	
	Incremental joules instance threshold count in MT. Type: int Default: 1000 Edit: Yes
init_blackbox_for_undefined	
	When set to true, empty modules are treated as unresolved references. Type: bool Default: true Edit: Yes
init_check_netlist	
	After reading in the netlist, will do the same checks as check_design -netlist Type: bool Default: false Edit: Yes Reference: init_check_netlist
init_check_output_pin_constant	
	Stops the Verilog reader if any output (or inout) signal pins have any 1'b0 or 1'b1 constants that would cause shorts to power/ground. Type: bool Default: true Edit: Yes Reference: init_check_output_pin_constant
init_delete_floating_hnets	
	If true, an hnet (Verilog wire statement) that has nothing connected to it, is deleted from the netlist. The default value is false in Innovus to allow for netlist repair, but true in Tempus to delete non-timed nets and reduce memory. Type: bool Default: false Edit: Yes Reference: init_delete_floating_hnets
init_design_mmmc_skip_inactive	
	Enables runtime improvement during loading libraries. Type: bool Default: false Edit: Yes
init_design_netlist_type	
	Specifies the source of the design netlist. You can specify either "Verilog" or "OA". Type: string Default: Verilog Edit: Yes Reference: init_design_netlist_type
init_design_uniquify	
	Specifies whether the top is uniquified. When set to 1 (true) the design will be uniquified during the read and flatten process. Optimization is not allowed on a non-unique design. For use in a master/clone partitioning flow, the value should be 0 (false). Default = 0 (false). Type: bool Default: false Edit: Yes Reference: init_design_uniquify
init_enable_drc_region_layer	

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	List of drc_region_layer to restore active DRC region layer. Type: string Default: "" Edit: Yes
init_ground_nets	
	List of global Ground nets. Type: string Default: "" Edit: Yes Reference: init_ground_nets
init_hdl_search_path	
	Specifies the search path for HDL files. Type: string Default: . Edit: Yes Reference: init_hdl_search_path
init_ignore_pg_pin_polarity_check	
	List of leaf cell pin names to ignore power/ground pin polarity check. By default connect_global_net and CPF check for matching polarity between the net and pin. Type: string Default: "" Edit: Yes Reference: init_ignore_pg_pin_polarity_check
init_keep_empty_modules	
	By default, usage of an empty Verilog module that has no matching library cell name becomes an empty hinst. If this value is false, a dummy library cell is created, and usage of it becomes an inst. You should then use other commands to 'fix' the dummy library cell before proceeding in the flow. Type: bool Default: true Edit: Yes Reference: init_keep_empty_modules
init_lef_files	
	List of LEF files to be read. Mutually exclusive with init_oa_ref_lib. Type: string Default: "" Edit: Yes Reference: init_lef_files
init_min_dbu_per_micron	
	Minimum DBU per micron Type: int Default: 0 Edit: Yes Reference: init_min_dbu_per_micron
init_mmmc_files	
	Path to MMMC View Definition file. Type: string Default: "" Edit: Yes Reference: init_mmmc_files
init_no_new_assigns	

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	Turns on an assign-free flow. By default, applications like optimization can add Verilog assigns to remove unnecessary buffers and reduce power. If you have external tools that do not support Verilog assigns, you can set this value to true to prevent applications from adding any new assign statements to the Verilog netlist. If there are existing assigns in the Verilog netlist you can remove them with the delete_assigns command. Type: bool Default: false Edit: Yes Reference: init_no_new_assigns
init_oa_abstract_views	
	List of OpenAccess view names to be processed as abstracts (LEF MACRO equivalent). The read_netlist command uses init_oa_abstract_views list for on-demand loading of missing cells. Used by the read_physical and read_netlist commands. Type: string Default: "" Edit: Yes Reference: init_oa_abstract_views
init_oa_default_rule	
	The default rule to be used in OpenAccess. Typically used to specify a metal stack LEFDefaultRouteSpec type constraint group to read from the technology library if the library contains multiple metal stacks. Used by the read_physical command. Type: string Default: "" Edit: Yes Reference: init_oa_default_rule
init_oa_design_cell	
	OpenAccess design cell name that was read. Set by the read_netlist -oa_cell_view command. Type: string Default: "" Edit: Yes Reference: init_oa_design_cell
init_oa_design_lib	
	OpenAccess design lib name that was read. Set by the read_netlist -oa_cell_view command. Type: string Default: "" Edit: Yes Reference: init_oa_design_lib
init_oa_design_view	
	OpenAccess design view name that was read. Set by the read_netlist -oa_cell_view command. Type: string Default: "" Edit: Yes Reference: init_oa_design_view
init_oa_foundry_rule	
	The foundry constraint group to be used in OpenAccess. Typically used to specify a foundry_innovus_XXXX constraint group to read from the technology library if the library contains multiple foundry_innovus_XXXX constraint groups. Used by the read_physical command. Type: string Default: "" Edit: Yes Reference: init_oa_foundry_rule
init_oa_layout_views	
	List of OpenAccess view names to be processed for cell layout viewing (GDSII equivalent). Also used in the case where a Verilog netlist is read and views from the init_oa_layout_views list are used in the Quick Abstract Inference flow to derive abstract information from layout views for cells that were not processed from the libraries in the read_physical -oa_ref_libs list. The read_netlist command uses init_oa_layout_view list for on-demand loading of missing cells. Used by the read_physical and read_netlist commands. Type: string Default: "" Edit: Yes Reference: init_oa_layout_views

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init_oa_ref_libs	
	List of OpenAccess reference libraries. The first library in the list indicates the library that was used for technology information unless read_physical -oa_tech_lib is specified. All cells from the specified libraries are read. It is important to include the standard cell library in the list to make sure that cells that are not in the current netlist are read and available for optimization. The init_oa_abstract_views and init_oa_layout_views lists specify which views of the cells are read. Set by the read_physical -oa_ref_libs command. Type: string Default: "" Edit: Yes Reference: init_oa_ref_libs
init_oa_search_libs	
	List of OpenAccess reference libraries to be searched when cells are used in the Verilog netlist that are not found in the list of cells read from the libraries specified by read_physical -oa_ref_libs command, by default all libraries in the cds.lib are searched. The init_oa_abstract_views and init_oa_layout_views lists are used to indicate which view names are looked for during the search. Applies only when read_netlist command is used to process a Verilog netlist specify the connectivity to be read. Set by the read_physical -oa_search_libs command and used by the read_netlist command. Type: string Default: * Edit: Yes Reference: init_oa_search_libs
init_oa_special_rule	
	The special rule to be used in OpenAccess to define the via search order for power routing. Typically used to specify a metal stack LEFSpecialRouteRouteSpec type constraint group to read from the technology library if the library contains multiple metal stacks. Used by the read_physical command. Type: string Default: "" Edit: Yes Reference: init_oa_special_rule
init_oa_tech_lib	
	OpenAccess library to use as source of technology information. If not specified, the first library in the read_physical -oa_ref_libs list will be used. Applies only when read_netlist is used to process a Verilog netlist. Set by the read_physical -oa_tech_lib command. Type: string Default: "" Edit: Yes Reference: init_oa_tech_lib
init_power_intent_files	
	Power intent files(s) read by last 'read_power_intent' command. Type: string Default: "" Edit: No Reference: init_power_intent_files
init_power_nets	
	List of global Power nets. Type: string Default: "" Edit: Yes Reference: init_power_nets
init_read_netlist_allow_port_mismatch	
	If false, a Verilog instance with ports that do not match the Verilog module or library cell definition will cause an error and exit. If true, an error message occurs, but the extra ports are added to the module or library cell definition to allow the inconsistent netlist to be read in and debugged further. The default value is false i.e. 'dirty netlists' are not tolerated. Type: bool Default: false Edit: Yes Reference: init_read_netlist_allow_port_mismatch
init_read_netlist_allow_undefined_cells	

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	If false, a Verilog reference to an undefined cell (e.g. a Verilog module name that has no matching library cell and no Verilog module definition) causes an error and an exit after parsing the netlist. If true, an empty Verilog module is created with a warning. The default value is true in Innovus to allow repair of 'dirty netlists' while it is false in Tempus to stop on 'dirty netlists'. An empty module will become an hinst or inst depending on the value of init_keep_empty_modules. Type: bool Default: true Edit: Yes Reference: init_read_netlist_allow_undefined_cells
init_read_netlist_files	
	A list of Verilog files read in during initialization. read_netlist will set this root attribute to the list of input files given to it. The original input file names are retained thru the flow for reference, and used by write_do_lec for the default golden netlist comparison file names. Type: string Default: "" Edit: Yes Reference: init_read_netlist_files
init_sync_relative_path	
	Synchronize Relative paths Type: bool Default: false Edit: Yes Reference: init_sync_relative_path
init_timing_enabled	
	If read_physical is issued before read_mmmc or read_libs in RC and Tempus, an error will be issued. In EDI, doing this will result in init_timing_enabled being set to false by the tool. EDI is now in physical only mode, and any attempt to set timing data will result in an error. By default, init_timing_enabled is set to true. Type: bool Default: true Edit: No Reference: init_timing_enabled
input_assert_one_cold_pragma	
	Specifies the list of alias names of the pragma 'one_cold'. Type: string Default: assert_one_cold one_cold Edit: Yes Reference: input_assert_one_cold_pragma
input_assert_one_hot_pragma	
	Specifies the list of alias names of the pragma 'one_hot'. Type: string Default: assert_one_hot one_hot Edit: Yes Reference: input_assert_one_hot_pragma
input_asynchro_reset_blk_pragma	
	Specifies the list of alias names of the pragma 'asynchro_reset_blk'. Type: string Default: async_set_reset_local asynchro_reset_blk Edit: Yes Reference: input_asynchro_reset_blk_pragma
input_asynchro_reset_pragma	
	Specifies the list of alias names of the pragma 'asynchro_reset'. Type: string Default: async_set_reset asynchro_reset Edit: Yes Reference: input_asynchro_reset_pragma
input_case_cover_pragma	

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	Specifies the list of alias names of the pragma 'full_case'. Type: string Default: full_case Edit: Yes Reference: input_case_cover_pragma
input_case_decode_pragma	
	Specifies the list of alias names of the pragma 'parallel_case'. Type: string Default: parallel_case Edit: Yes Reference: input_case_decode_pragma
input_map_to_mux_pragma	
	Specifies the list of alias names of the pragma 'map_to_mux'. Type: string Default: map_to_mux infer_mux_override Edit: Yes Reference: input_map_to_mux_pragma
input_pragma_keyword	
	Specifies pragma keywords. Type: string Default: cadence synopsys ambit pragma synthesis Edit: Yes Reference: input_pragma_keyword
input_synchro_enable_blk_pragma	
	Specifies the list of alias names of the pragma 'synchro_enable_blk'. Type: string Default: synchro_enable_blk Edit: Yes Reference: input_synchro_enable_blk_pragma
input_synchro_enable_pragma	
	Specifies the list of alias names of the pragma 'synchro_enable'. Type: string Default: synchro_enable Edit: Yes Reference: input_synchro_enable_pragma
input_synchro_reset_blk_pragma	
	Specifies the list of alias names of the pragma 'synchro_reset_blk'. Type: string Default: sync_set_reset_local synchro_reset_blk Edit: Yes Reference: input_synchro_reset_blk_pragma
input_synchro_reset_pragma	
	Specifies the list of alias names of the pragma 'synchro_reset'. Type: string Default: sync_set_reset synchro_reset Edit: Yes Reference: input_synchro_reset_pragma
insert_clock_gating	
	Enable/disable RTL clock-gating. Type: bool Default: false Edit: Yes Reference: insert_clock_gating
insts	

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	insts of current design Type: <code>obj(inst)*</code> Default: "" Edit: No
integrity_dynamic_prompt_name	
	If turned on, the prompt title would update dynamically current die name switched by switchTopCellAndLefSet. Type: <code>bool</code> Default: false Edit: Yes
io_constraints	
	Short-cut for [get_db current_design .io_constraints] Type: <code>obj(io_constraint)*</code> Default: "" Edit: No
is_ilm_flattened	
	Is true if you have ILMs, and the flatten_ilm command has flattened them so the internal ILM insts are visible. unflatten_ilm will set it back to false. Type: <code>bool</code> Default: false Edit: No
is_ilm_read	
	Is true if you have read in any ILMs with read_ilm. It is not affected by flatten_ilm/unflatten_ilm. Type: <code>bool</code> Default: false Edit: No
jedai_catalog_api_path	
	Type: <code>string</code> Default: "" Edit: Yes
jedai_catalog_url	
	Type: <code>string</code> Default: "" Edit: Yes
joules_incremental_quiet	
	To silence the incremental joules info/warning messages. Type: <code>bool</code> Default: true Edit: Yes
joules_quiet	
	To silence the joules info/warning messages. Type: <code>bool</code> Default: false Edit: Yes
large_cell_arc_threshold	
	Arc threshold count to create separate bdd's. Type: <code>int</code> Default: 200 Edit: Yes Reference: <code>large_cell_arc_threshold</code>
layer_name_no_abbreviation	

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	By default this value is true, and most command -layer <layer> options require a LEF/OA layer name, or a routing-layer index value (e.g. 1 for the first routing layer, 2 for the second, etc.) for input. If this global is false, then an older style abbreviation is also allowed (e.g. M1 for the first routing layer, M2 for the second routing layer, etc.). This layer abbreviation style is not recommended, and is only allowed for backward compatibility to migrate older scripts. Eventually this global will be removed, so you should modify your scripts to use LEF/OA layer names, or routing-layer index values for -layer options to avoid problems in the future. Type: <code>bool</code> Default: <code>true</code> Edit: Yes Reference: <code>layer_name_no_abbreviation</code>
layers	
	All the layers defined in the LEF or OA techfile. Type: <code>obj(layer)*</code> Default: <code>""</code> Edit: No
lib_arcs	
	lib_arcs Type: <code>obj(lib_arc)*</code> Default: <code>""</code> Edit: No
lib_cells	
	lib_cells Type: <code>obj(lib_cell)*</code> Default: <code>""</code> Edit: No
lib_pins	
	lib_pins Type: <code>obj(lib_pin)*</code> Default: <code>""</code> Edit: No
libraries	
	Returns the information about the libraries in the design. Type: <code>obj(library)*</code> Default: <code>""</code> Edit: No
library_sets	
	Returns the information about the library sets in the design. Type: <code>obj(library_set)*</code> Default: <code>""</code> Edit: No
link_library	
	Link library for instantiated cells in the design. Type: <code>string</code> Default: <code>""</code> Edit: Yes Reference: <code>link_library</code>
log_file	
	The log file name of the program. It can be changed in the session to direct log file output to a different file. Type 'man log_file' for more details. Type: <code>string</code> Default: <code>""</code> Edit: Yes Reference: <code>log_file</code>
log_verbose_style	

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	When verbose logging is enabled (see <code>source_verbose</code>, <code>flow_verbose</code>, or <code>set_proc_verbose</code>), the logging style for each command is controllable. It is sometimes helpful to include line numbers and comments for script debug and other times this is too much information. You can switch between settings in the middle of a script. 'debug' will output the file name, line number, commands, comments, and empty lines. 'echo' will output only the commands and line number. 'echo_comments' will output commands, comments, empty lines, and line number. Type: enum Enum Values: debug echo echo_comments Default: echo Edit: Yes Reference: log_verbose_style
log_verbose_type	
	When verbose logging is enabled (see <code>source_verbose</code>, <code>flow_verbose</code>, or <code>set_proc_verbose</code>), the default logging of each commands is pre-Tcl expansion. In some cases it is helpful to see post-Tcl expansion, or even both pre-Tcl and post-Tcl expansion when doing detailed debugging. You can switch between settings in the middle of a script. Type 'man log_file' for more details. Type: enum Enum Values: pre post both Default: pre Edit: Yes Reference: log_verbose_type
logv_file	
	The verbose log file name of the program. It can be changed in the session to direct logv file output to a different file. Type 'man log_file' for more details. Type: string Default: "" Edit: Yes Reference: logv_file
map_clock_tree	
	Enables mapping of all clock logic to a specified set of library cells. Type: bool Default: false Edit: Yes Reference: map_clock_tree
map_drc_first	
	Give preference to 'drc' cost during mapping. Type: bool Default: false Edit: Yes Reference: map_drc_first
map_latch_allow_async_decomp	
	Allow implementing the asynchronous control logic externally to the latch. Type: bool Default: false Edit: Yes Reference: map_latch_allow_async_decomp
map_prefer_non_inverted_clock_line	
	The tool will try to avoid inserting inverters in the clock line, but if that is not possible still can add them. If set to false, then the tool can chose the solution which is best from QoR point of view, i.e. has no preference for solutions without inverter. Type: bool Default: true Edit: Yes Reference: map_prefer_non_inverted_clock_line
map_respect_rtl_clk_phase	
	Respect the clock phase specification of registers from the RTL. Type: bool Default: false Edit: Yes Reference: map_respect_rtl_clk_phase

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mark_async_pin_using_timing_arcs	
	Mark library pin having recovery or removal timing arcs as asynchronous. Type: bool Default: true Edit: Yes Reference: mark_async_pin_using_timing_arcs
mark_macro_as_power_switch_cell	
	Power switch cell defined as macro is also marked as power switch cell. Type: bool Default: false Edit: Yes Reference: mark_macro_as_power_switch_cell
mark_valid_lp_cell_as_usable	
	Mark valid low power cells as usable. Type: bool Default: false Edit: Yes Reference: mark_valid_lp_cell_as_usable
markers	
	Short-cut to all the marker objects in the design. Type: obj(marker) Default: "" Edit: No
merge_combinational_hier_instances	
	Allow merging of combinational hierarchical instances. Type: bool Default: true Edit: Yes Reference: merge_combinational_hier_instances
merge_non_scan_to_scan_flops	
	Controls whether non scan single bit flops can be merged to multibit scan flops. Type: bool Default: false Edit: Yes Reference: merge_non_scan_to_scan_flops
mesg_severity_downgrade	
	Allow message severity to be downgraded. Type: bool Default: false Edit: Yes Reference: mesg_severity_downgrade
messages	
	A list of all the message objects. Type: obj(message)* Default: "" Edit: No
metric_advanced_url_endpoint	
	Base URL for the advanced metric server. Type: string Default: "" Edit: Yes Reference: metric_advanced_url_endpoint
metric_capture_3d_hotspots	

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	Capture 3d hotspot metrics. Type: bool Default: true Edit: Yes Reference: metric_capture_3d_hotspots
metric_capture_depth	
	Depth for capturing hinst design and power metrics. Type: int Default: 1 Edit: Yes Reference: metric_capture_depth
metric_capture_design_image	
	Capture design image. Type: bool Default: true Edit: Yes Reference: metric_capture_design_image
metric_capture_design_image_blockages	
	Capture blockages in design image. Type: bool Default: true Edit: Yes Reference: metric_capture_design_image_blockages
metric_capture_design_image_blockages_exclude_layers	
	Layers to exclude when capturing blockages in metric image. Type: string Default: "" Edit: Yes
metric_capture_design_image_blockages_threshold	
	Threshold for capturing blockages in metric image. Type: int Default: 100 Edit: Yes Reference: metric_capture_design_image_blockages_threshold
metric_capture_design_image_power_intent	
	Capture power intent objects in design image. Type: bool Default: true Edit: Yes Reference: metric_capture_design_image_power_intent
metric_capture_design_image_route_drc	
	Capture route drc markers in design image. Type: bool Default: true Edit: Yes Reference: metric_capture_design_image_route_drc
metric_capture_max_drc_markers	
	Maximum number of DRC markers to include in metric image. Type: int Default: 1000 Edit: Yes Reference: metric_capture_max_drc_markers
metric_capture_min_count	

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	Minimum instance count for capturing hinst design and power metrics. Type: <code>int</code> Default: 1000 Edit: Yes Reference: metric_capture_min_count
metric_capture_overwrite	
	Overwrite pending metrics during create_snapshot category capture. Type: <code>bool</code> Default: false Edit: Yes Reference: metric_capture_overwrite
metric_capture_pba_tns_histogram	
	Capture path based analysis histogram data for TNS. Type: <code>bool</code> Default: false Edit: Yes Reference: metric_capture_pba_tns_histogram
metric_capture_per_view	
	Capture timing metrics per analysis_view. Type: <code>bool</code> Default: true Edit: Yes Reference: metric_capture_per_view
metric_capture_reg2reg_metrics	
	Force the capture of reg2reg metrics regardless of path groups. Type: <code>bool</code> Default: false Edit: Yes Reference: metric_capture_reg2reg_metrics
metric_capture_timing_analysis_mode	
	Timing analysis mode for metric capture during create_snapshot categories setup and hold; gba, pba, epba, ipba. Type: <code>enum</code> Enum Values: gba pba epba ipba Default: gba Edit: Yes Reference: metric_capture_timing_analysis_mode
metric_capture_timing_path_groups	
	Defines user path groups to capture for detailed display. Type: <code>string</code> Default: "" Edit: Yes Reference: metric_capture_timing_path_groups
metric_capture_timing_paths	
	Number of paths to capture for detailed display. Type: <code>int</code> Default: 10 Edit: Yes Reference: metric_capture_timing_paths
metric_capture_tns_histogram	
	Capture histogram data for TNS. Type: <code>bool</code> Default: true Edit: Yes Reference: metric_capture_tns_histogram

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metric_capture_tns_histogram_buckets	
	Number of buckets for the TNS histogram. Type: int Default: 50 Edit: Yes Reference: metric_capture_tns_histogram_buckets
metric_capture_tns_histogram_max_slack	
	Maximum slack for the TNS histogram. Type: int Default: 0 Edit: Yes Reference: metric_capture_tns_histogram_max_slack
metric_capture_tns_histogram_paths	
	Number of paths to capture for the TNS histogram. Type: int Default: 10000 Edit: Yes Reference: metric_capture_tns_histogram_paths
metric_capture_vth_metrics	
	Capture vth metrics. Type: bool Default: true Edit: Yes Reference: metric_capture_vth_metrics
metric_capture_vth_per_power_domain	
	Capture vth metrics per power domain. Type: bool Default: false Edit: Yes Reference: metric_capture_vth_per_power_domain
metric_capture_wns_average_paths	
	Defines average top-N paths for WNS. Type: string Default: "" Edit: Yes
metric_category_default	
	Default metric categories to capture if none are provided. Type: string Default: design Edit: Yes Reference: metric_category_default
metric_current_run_id	
	Current unique run ID returned by the advanced metric server. Type: string Default: "" Edit: Yes Reference: metric_current_run_id
metric_enable	
	enable metric snapshot capture with create_snapshot Type: bool Default: false Edit: Yes Reference: metric_enable

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metric_summary_metrics	
	Summary metrics to show when creating snapshots. Type: string Default: flow.cputime flow.realtime timing.setup.tns timing.setup.wns Edit: Yes Reference: metric_summary_metrics
minimize_uniquify	
	Try to avoid uniquification of subdesigns. Type: bool Default: false Edit: Yes Reference: minimize_uniquify
modules	
	Short-cut for [get_db current_design .modules] Type: obj(module)* Default: "" Edit: No
multibit_allow_async_phase_map	
	Allow interchanging async pins for multibit mapping. Type: bool Default: true Edit: Yes Reference: multibit_allow_async_phase_map
multibit_allow_unused_bits	
	Allows unused bits in multibit instance. Type: bool Default: true Edit: Yes Reference: multibit_allow_unused_bits
multibit_area_power_scoring	
	Controls whether area and power are to be scored in merging. Type: enum Enum Values: none area_only area_power Default: none Edit: Yes Reference: multibit_area_power_scoring
multibit_auto_exclude_registers_with_exceptions	
	Enable this to automatically apply dont_infer_multibit on all sequential instances with exceptions.By default exceptional instances are merged only when all exceptions are transferable to multibit formed. Type: bool Default: false Edit: Yes Reference: multibit_auto_exclude_registers_with_exceptions
multibit_aware_seq_mapping	
	Mapper prefers flops which have corresponding multibit variants. Type: enum Enum Values: auto true false Default: auto Edit: Yes Reference: multibit_aware_seq_mapping
multibit_aware_seq_mapping_higher_priority	

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	Multibit-aware mapping has higher priority than other constraints like <code>avoid_tied_inputs</code>, <code>map_prefer_non_inverted_clock_line</code>, <code>exact_match_seq_sync_ctrls</code>, <code>exact_match_seq_async_ctrls</code>, etc. Nonetheless, it remains lower priority than <code>map_to_register</code>, <code>map_to_ser_register</code>, <code>scan</code> and <code>state retention</code> constraints. Type: <code>bool</code> Default: <code>false</code> Edit: Yes Reference: multibit_aware_seq_mapping_higher_priority
multibit_cells_from_different_busses	
	Allows to merge 1-bit sequential, tristate and combinational instances from different busses into single multibit cell. Type: <code>bool</code> Default: <code>true</code> Edit: Yes Reference: multibit_cells_from_different_busses
multibit_combinational_name_concatenate_string	
	Separator string to be used for concatenation when naming multibit combinational and special low power cells. Recommended values for Genus-LEC flow are <code>'_MB_'</code> and <code>'_mb_'</code>. Type: <code>string</code> Default: <code>_MB_</code> Edit: Yes
multibit_debug	
	Logs Multibit Inferencing debug messages. Type: <code>bool</code> Default: <code>false</code> Edit: Yes Reference: multibit_debug
multibit_invert_clock_phase	
	Used for enabling merging of cells with inverted clock phase. Type: <code>bool</code> Default: <code>false</code> Edit: Yes Reference: multibit_invert_clock_phase
multibit_mapping_effort_level	
	Effort level(high, auto, low) for multibit mapping. Type: <code>enum</code> Enum Values: high auto low Default: high Edit: Yes Reference: multibit_mapping_effort_level
multibit_predefined_allow_unused_bits	
	Allows unused bits in predefined multibit. Type: <code>bool</code> Default: <code>false</code> Edit: Yes Reference: multibit_predefined_allow_unused_bits
multibit_prefix_string	
	User to specify multi bit flop name prefix Type: <code>string</code> Default: <code>CDN_MBIT_</code> Edit: Yes Reference: multibit_prefix_string
multibit_preserve_inferred_instances	

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	Preserve size_delete_ok inferred multibit instances. Type: bool Default: false Edit: Yes Reference: multibit_preserve_inferred_instances
multibit_preserved_net_check	
	Merge instances into multibit only if no preserve on connected nets. Type: bool Default: false Edit: Yes Reference: multibit_preserved_net_check
multibit_seqs_instance_naming_style	
	Naming style to be used for newly created sequential multibit instance. Type: string Default: concat Edit: Yes Reference: multibit_seqs_instance_naming_style
multibit_seqs_members_naming_style	
	Naming style for sequential instances to be merged into multibit cell. Type: string Default: %s%d %s%d_reg %s_%d_%s_%d_reg {%s[%d]} {%s[%d]_reg} Edit: Yes Reference: multibit_seqs_members_naming_style
multibit_seqs_name_concat_string	
	User to specify flop merge and split name separator Type: string Default: _MB_ Edit: Yes Reference: multibit_seqs_name_concat_string
multibit_short_prefix_string	
	Prefix string to be used while using short naming style for multibit instances. Recommended values for Genus-LEC flow are '_CDN_CPX_'. Type: string Default: _CDN_CPX_ Edit: Yes Reference: multibit_short_prefix_string
multibit_split_string	
	Naming style string to be used for naming single bit instances when source netlist already has the multibit cells. Default value is '_split_'. Type: string Default: _split_ Edit: Yes Reference: multibit_split_string
multibit_unused_input_value	
	Connects each unconnected input pin in Multibit cell to specified value unless 'none' is supplied. Type: string Default: 0 Edit: Yes Reference: multibit_unused_input_value
net_groups	
	Short-cut for [get_db current_design .net_groups] Type: obj(net_group)* Default: "" Edit: No
nets	

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	Short-cut for [get_db current_design .nets]. All the nets inside this design, including logical Verilog nets, physical-only nets, Verilog supply0/supply1, and Verilog 1'b0/1'b1 nets. Type: obj(net)* Default: "" Edit: No
non_dft_timing_mode_name	
	Non-DFT timing mode name (default: NON_DFT_MODE). Type: string Default: NON_DFT_MODE Edit: Yes Reference: non_dft_timing_mode_name
num_eeq_variants	
	Maximum number of EEQ variants of any cell in library, from LIBRARY LEF58_CELLVARIANTS property 'CELLVARIANTS variantTotalNum YFLIPMAP {flippedVariantNum siteVariantNum} ...' Type: int Default: "" Edit: No
oa_allow_analysis_only	
	Allow analysis only flow Type: bool Default: false Edit: Yes Reference: oa_allow_analysis_only
oa_allow_bit_connection	
	Whether to create instance terminal creations bitwise, if bus not found in OA. Type: bool Default: false Edit: Yes Reference: oa_allow_bit_connection
oa_allow_tech_update	
	Whether to allow the tech.db to be updated or not. Type: bool Default: true Edit: Yes Reference: oa_allow_tech_update
oa_bindkey_file	
	Virtuoso bind key file Type: string Default: "" Edit: Yes Reference: oa_bindkey_file
oa_cell_view_dir	
	System defined transient global referring to origin of restored design. Type: string Default: "" Edit: Yes Reference: oa_cell_view_dir
oa_convert_diagonal_path_to	
	Whether to convert non-conforming pathsegs/paths to polygon or convert the endStyle to truncate on orthogonal and to custom on diagonal pathsegs Type: enum Enum Values: polygon default_style Default: polygon Edit: Yes Reference: oa_convert_diagonal_path_to

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oa_cut_rows	
	Whether to use oa_cut_rows or not Type: bool Default: false Edit: Yes Reference: oa_cut_rows
oa_display_resource_file	
	Display resource file Type: string Default: "" Edit: Yes Reference: oa_display_resource_file
oa_display_resource_file_in_library	
	Whether to use display resource file in the tech library Type: bool Default: false Edit: Yes Reference: oa_display_resource_file_in_library
oa_drc_fill_purpose	
	Whether to save drcFill shapes to gap_fill or drawing purpose. By default DEF routes with + SHAPE DRCFILL are mapped to the OA 'gapFill' purpose. Some OA techfiles do not have this purpose defined, and require these shapes be mapped to 'drawing' purpose. In that case, they get an additional internal attribute so they can round-trip back to DRCFILL shapes properly. Type: enum Enum Values: gap_fill drawing Default: gap_fill Edit: Yes Reference: oa_drc_fill_purpose
oa_enclose_quick_abstract_pins	
	Include shapes that enclose layout pin shapes Type: bool Default: true Edit: Yes Reference: oa_enclose_quick_abstract_pins
oa_full_layer_list	
	Whether to support full list of layer import Type: bool Default: true Edit: Yes Reference: oa_full_layer_list
oa_full_path	
	Whether to use full path or not Type: bool Default: false Edit: Yes Reference: oa_full_path
oa_hybrid_rows	
	Read hybrid rows from PDK Type: bool Default: false Edit: Yes Reference: oa_hybrid_rows
oa_inst_placed_if_none	

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	Whether to map OA Placement status oacNonePlacementStatus of an instance to Placed(true)/ Unplaced(false). Type: <code>bool</code> Default: true Edit: Yes Reference: oa_inst_placed_if_none
oa_lib_create_mode	
	Library creation method during write_db for newly created libraries Type: <code>enum</code> Enum Values: none attach copy reference Default: reference Edit: Yes Reference: oa_lib_create_mode
oa_lock	
	Whether to lock the OA database during read_db Type: <code>enum</code> Enum Values: false true error_if_locked Default: false Edit: Yes Reference: oa_lock
oa_logic_only_import	
	Whether to run read_oa after init_design or not Type: <code>bool</code> Default: false Edit: Yes Reference: oa_logic_only_import
oa_merge_trim	
	Merge the trim metal shapes during save when the trim metal gap less than or equal to the max_gap in micron. Type: <code>bool</code> Default: true Edit: Yes Reference: oa_merge_trim
oa_new_lib_compress_level	
	OA Database compression level Type: <code>int</code> Default: 1 Edit: Yes Reference: oa_new_lib_compress_level
oa_non_rectilinear_shapes	
	Translates non-rectilinear shapes to rectilinear shapes. Type: <code>enum</code> Enum Values: bbox approximate Default: bbox Edit: Yes Reference: oa_non_rectilinear_shapes
oa_pin_purpose	
	Whether to set Pin Purpose as Pin or not Type: <code>bool</code> Default: false Edit: Yes Reference: oa_pin_purpose
oa_push_pin_constraint	

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	whether write_db to push interoperable pin constraint into its corresponding blackbox abstract cellview. Note: Interoperable pin constraint means pin constraints that are understood by Innovus, Virtuoso and PVS. Type: bool Default: false Edit: Yes Reference: oa_push_pin_constraint
oa_quick_abstract_for_custom_cells	
	Trim design shapes of Custom Cells Type: enum Enum Values: pcells stdcells all off Default: all Edit: Yes Reference: oa_quick_abstract_for_custom_cells
oa_read_system_reserved_layers	
	Read system reserved layers from PDK Type: enum Enum Values: all text off Default: all Edit: Yes Reference: oa_read_system_reserved_layers
oa_silently_ignore_unsupported_vias	
	Whether read_db should issue messages about vias that are not supported in Innovus. Type: bool Default: false Edit: Yes Reference: oa_silently_ignore_unsupported_vias
oa_text_purpose	
	Specify the purpose name that will be used when text labels (add_text) are written. Type: string Default: drawing Edit: Yes Reference: oa_text_purpose
oa_tie_net	
	tieHigh tieLow net names Type: string Default: "" Edit: Yes Reference: oa_tie_net
oa_update_mode	
	Whether to use (save/restore)OaDesign in update mode or not Type: enum Enum Values: false true auto Default: false Edit: Yes Reference: oa_update_mode
oa_use_virtuoso_bindkey	
	Whether to use Virtuoso bind keys Type: bool Default: false Edit: Yes Reference: oa_use_virtuoso_bindkey
oa_use_virtuoso_color	

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	Whether to use Virtuoso color and stipple Type: bool Default: false Edit: Yes Reference: oa_use_virtuoso_color
oa_view_sub_type	
	Subtype of view of the OA design Type: enum Enum Values: vce vxl none Default: vxl Edit: Yes Reference: oa_view_sub_type
oa_write_cds_fixed_vias	
	Save fixed via Type: enum Enum Values: new_vias_only off Default: new_vias_only Edit: Yes Reference: oa_write_cds_fixed_vias
oa_write_mask_data_locked	
	Whether to save colored shapes with locked attribute Type: bool Default: false Edit: Yes Reference: oa_write_mask_data_locked
oa_write_net_voltage	
	Whether to allow write_db to write voltage information to the oaBitNet's voltage attribute Type: bool Default: false Edit: Yes Reference: oa_write_net_voltage
oa_write_relative_path	
	Whether to save relative path in config file. Type: bool Default: true Edit: Yes Reference: oa_write_relative_path
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (root) Default: "" Edit: No
obj_types	
	object definitions Type: obj(obj_type)* Default: "" Edit: No
opconds	
	Returns the information about the opconds in the design. Type: obj(opcond)* Default: "" Edit: No
operating_conditions	

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	Operating condition for timing (from technology library). Type: string Default: "" Edit: Yes Reference: operating_conditions
opt_activity_refresh_args	
	Specify the list of setStim options to be passed for Joules Xreplay. Call "setOptMode -activity_refresh_args help" to get detailed usage. Type: string Default: "" Edit: Yes Reference: opt_activity_refresh_args
opt_activity_refresh_stages	
	Control for what points in synthesize_design should xreplay be performed. Valid values are one or both of: pre_gen post_gen. Requires opt_activity_refresh_args to be set. Type: string Default: "" Edit: Yes
opt_add_always_on_feed_through_buffers	
	Activates adding Always On Feed Thru buffers Type: bool Default: false Edit: Yes Reference: opt_add_always_on_feed_through_buffers
opt_add_insts	
	Enables adding of new instances Type: bool Default: true Edit: Yes Reference: opt_add_insts
opt_add_ports	
	Enables adding ports for optimal buffering Type: bool Default: true Edit: Yes Reference: opt_add_ports
opt_add_repeater_report_failure_reason	
	To enable printing detailed failure reason report file for addRepeaterByRule Type: bool Default: false Edit: Yes Reference: opt_add_repeater_report_failure_reason
opt_all_end_points	
	Enables the all end points optimization mode Type: bool Default: false Edit: Yes Reference: opt_all_end_points
opt_allow_floating_outputs	
	Allow floating outputs for non-sequential cells during optimization. Type: bool Default: false Edit: Yes Reference: opt_allow_floating_outputs
opt_allow_multi_bit_on_flop_with_sdc	

Stylus Common UI Database Object Information

Database Objects--root

	allow merge or split works on flops which have SDC Type: enum Enum Values: true false merge_only split_only Default: true Edit: Yes Reference: opt_allow_multi_bit_on_flop_with_sdc
opt_allow_only_cell_swapping	
	Forces the post_route optimization to perform only same-size Vth swapping for timing improvements Type: bool Default: false Edit: Yes Reference: opt_allow_only_cell_swapping
opt_area_recovery	
	Enables area reclamation step Type: enum Enum Values: true false default Default: default Edit: Yes Reference: opt_area_recovery
opt_area_recovery_setup_target_slack	
	Specifies target slack value (in nano secs) for area and power reclaim during setup timing optimization Type: double Default: 0.0 Edit: Yes Reference: opt_area_recovery_setup_target_slack
opt_boundary_optimize_invert_hpins_rename_extension	
	Extension to be appended on pin or net name, when boundary pin is inverted. Type: string Default: _BAR Edit: Yes Reference: opt_boundary_optimize_invert_hpins_rename_extension
opt_clock_traverse_by_level	
	Traverse Instances in clock path 1 level at a time. Type: bool Default: false Edit: Yes Reference: opt_clock_traverse_by_level
opt_clone_insts_list	
	list of instance list Type: string Default: "" Edit: Yes Reference: opt_clone_insts_list
opt_concatenate_default_and_user_prefixes	
	Enables combining default prefix for instances(nets) with the prefix specified by opt_new_inst_prefix(opt_new_net_prefix) Type: bool Default: true Edit: Yes Reference: opt_concatenate_default_and_user_prefixes
opt_consider_routing_congestion	

Stylus Common UI Database Object Information

Database Objects--root

	Enables routing congestion check at the post_route stage during optimization of setup, hold, drv, & glitch. It also impacts post_cts hold fixing. Type: enum Enum Values: auto false true Default: auto Edit: Yes Reference: opt_consider_routing_congestion
opt_constant_inputs	
	Force optimization of instances with constant inputs, even if it worsens objective gain Type: bool Default: false Edit: Yes Reference: opt_constant_inputs
opt_constant_nets	
	Enables optimization of constant nets Type: bool Default: true Edit: Yes Reference: opt_constant_nets
opt_delete_insts	
	Enables instance deletion on all paths Type: bool Default: true Edit: Yes Reference: opt_delete_insts
opt_detail_drv_failure_reason	
	To enable printing detailed drv failure reason report file Type: bool Default: false Edit: Yes Reference: opt_detail_drv_failure_reason
opt_detail_drv_failure_reason_max_num_nets	
	Specify how many violating nets printed Type: int Default: 50 Edit: Yes Reference: opt_detail_drv_failure_reason_max_num_nets
opt_dont_use_qn_sequential_pins	
	Do not use inverted outputs of sequential cells. Type: bool Default: false Edit: Yes Reference: opt_dont_use_qn_sequential_pins
opt_down_size_insts	
	Enables instance downsizing Type: bool Default: true Edit: Yes Reference: opt_down_size_insts
opt_drv	
	Enables fixing max cap/max tran violations on constrained nets Type: bool Default: true Edit: Yes Reference: opt_drv

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opt_drv_margin	
	Used for scaling max cap/max tran constraints Type: double Default: 0.0 Edit: Yes Reference: opt_drv_margin
opt_drv_with_miller_cap	
	Type: bool Default: false Edit: Yes Reference: opt_drv_with_miller_cap
opt_duplicate_cte_constrained_hport	
	allow buffering to duplicate hterms/hports with CTE constraints. Type: bool Default: true Edit: Yes Reference: opt_duplicate_cte_constrained_hport
opt_early_hold_fixing	
	set the ideal or propagaged clock mode for hold fixing in preCTS Type: enum Enum Values: none ideal propagated Default: none Edit: Yes Reference: opt_early_hold_fixing
opt_enable_always_on_type_in_remove_assign	
	Enable always on type in remove assign. Type: bool Default: false Edit: Yes Reference: opt_enable_always_on_type_in_remove_assign
opt_enable_clock_pulse_width_checks	
	Fix and honor pulse width check violations Type: bool Default: false Edit: Yes Reference: opt_enable_clock_pulse_width_checks
opt_enable_data_to_data_checks	
	Type: enum Enum Values: true false Default: false Edit: Yes Reference: opt_enable_data_to_data_checks
opt_enable_podv2_clock_opt_flow	
	enable PodV2 and clock_opt_design flow Type: bool Default: true Edit: Yes Reference: opt_enable_podv2_clock_opt_flow
opt_enable_restructure	

Stylus Common UI Database Object Information

Database Objects--root

	Enables netlist restructuring Type: bool Default: true Edit: Yes Reference: opt_enable_restructure
opt_enable_targeted_synthesis	
	Enable targeted synthesis mode Type: bool Default: false Edit: Yes Reference: opt_enable_targeted_synthesis
opt_fix_fanout_load	
	Enables fixing fanOut load violations Type: bool Default: false Edit: Yes Reference: opt_fix_fanout_load
opt_flop_pins_report	
	print full hierarchical flop pins names for flops deleted in simplifyNetlist Type: enum Enum Values: false true Default: false Edit: Yes Reference: opt_flop_pins_report
opt_flops_report	
	print full hierarchical flop names for flops deleted in simplifyNetlist Type: enum Enum Values: false true Default: false Edit: Yes Reference: opt_flops_report
opt_hier_add_antenna_cell	
	add antenna diode to interface load Type: bool Default: false Edit: Yes Reference: opt_hier_add_antenna_cell
opt_hier_opt_stage	
	Turn on default settings for hierarchical flows at different optimization stage. Type: enum Enum Values: pre_cts post_cts unset Default: unset Edit: Yes Reference: opt_hier_opt_stage
opt_high_effort_cells	
	Provide a list of cells to be used for high effort optimization. Dont-use and Dont-touch constraints for these cells will be ignored for high effort optimization. Type: string Default: "" Edit: Yes Reference: opt_high_effort_cells
opt_hold_allow_overlap	

Stylus Common UI Database Object Information
Database Objects--root

	Enables hold optimization to insert cells with overlaps when inadequate free space is available. The placement will then be legalized before ECO routing. Type: enum Enum Values: auto false true Default: auto Edit: Yes Reference: opt_hold_allow_overlap
opt_hold_allow_resize	
	Enables the resizing capability inside hold optimization. Type: enum Enum Values: false true Default: true Edit: Yes Reference: opt_hold_allow_resize
opt_hold_allow_setup_tns_degradation	
	Allows setup total negative slack to degrade during hold optimization. Type: bool Default: true Edit: Yes Reference: opt_hold_allow_setup_tns_degradation
opt_hold_cells	
	Provide a list of buffer & delay cells to be used for hold buffering. The dont_use constraints for these cells will be ignored. Type: string Default: "" Edit: Yes Reference: opt_hold_cells
opt_hold_ignore_path_groups	
	Specifies path groups to be excluded in hold timing optimization Type: string Default: "" Edit: Yes Reference: opt_hold_ignore_path_groups
opt_hold_on_excluded_clock_nets	
	Do hold fixing on the excluded clock nets Type: bool Default: false Edit: Yes Reference: opt_hold_on_excluded_clock_nets
opt_hold_slack_threshold	
	Specifies slack threshold value beyond which the hold timing violations will not be fixed Type: double Default: -1000.0 Edit: Yes Reference: opt_hold_slack_threshold
opt_hold_target_slack	
	Specifies target slack value (in nano secs) for hold timing optimization Type: double Default: 0.0 Edit: Yes Reference: opt_hold_target_slack
opt_honor_density_screen	

Stylus Common UI Database Object Information
Database Objects--root

	Control density screen Type: bool Default: false Edit: Yes Reference: opt_honor_density_screen
opt_honor_fences	
	Specifies that the timing optimization takes fences constraints into account. Placement of cells will be legalized within each fence. Type: bool Default: false Edit: Yes Reference: opt_honor_fences
opt_icg_enable_pin_rebuffering	
	Rebuffer nets to enable pin of ICG instances after clock tree insertion Type: enum Enum Values: true auto false Default: false Edit: Yes Reference: opt_icg_enable_pin_rebuffering
opt_leakage_to_dynamic_ratio	
	relative effort for leakage and dynamic power optimization in the range [0.0 - 1.0] Type: double Default: 1.0 Edit: Yes Reference: opt_leakage_to_dynamic_ratio
opt_max_density	
	Specifies maximum area utilization Type: double Default: 0.95 Edit: Yes Reference: opt_max_density
opt_max_length	
	Enables max length constraints in opt_design Type: double Default: -1.0 Edit: Yes Reference: opt_max_length
opt_move_insts	
	Enables moving instances while fixing setup violations Type: bool Default: true Edit: Yes Reference: opt_move_insts
opt_multi_bit_combinational_merge_timing_effort	
	Specifies the timing effort level for multibit combinational merging Type: enum Enum Values: low medium high Default: medium Edit: Yes Reference: opt_multi_bit_combinational_merge_timing_effort
opt_multi_bit_combinational_mode	

Stylus Common UI Database Object Information

Database Objects--root

	Enables the user specific multi bit combination cell merge and split based on area and power Type: enum Enum Values: auto power area Default: auto Edit: Yes Reference: opt_multi_bit_combinational_mode
opt_multi_bit_combinational_opt	
	enables or disables the multi-bit combinational optimization flow Type: enum Enum Values: false true mergeOnly splitOnly Default: splitOnly Edit: Yes Reference: opt_multi_bit_combinational_opt
opt_multi_bit_combinational_split_timing_effort	
	Specifies the timing effort level for multibit combinational splitting Type: enum Enum Values: low medium high Default: low Edit: Yes Reference: opt_multi_bit_combinational_split_timing_effort
opt_multi_bit_flop_merge_bank_label_inference	
	Enables soft merging of flops with the same bank label Type: bool Default: false Edit: Yes Reference: opt_multi_bit_flop_merge_bank_label_inference
opt_multi_bit_flop_merge_timing_effort	
	Timing effort level for MBFF merging Type: enum Enum Values: low medium high Default: medium Edit: Yes Reference: opt_multi_bit_flop_merge_timing_effort
opt_multi_bit_flop_name_prefix	
	User to specify multi bit flop name prefix Type: string Default: CDN_MBIT_ Edit: Yes Reference: opt_multi_bit_flop_name_prefix
opt_multi_bit_flop_name_separator	
	User to specify flop merge and split name separator Type: string Default: _MB_ Edit: Yes Reference: opt_multi_bit_flop_name_separator
opt_multi_bit_flop_name_suffix	
	User to specify multi bit flop name suffix, For example: _FF Type: string Default: "" Edit: Yes Reference: opt_multi_bit_flop_name_suffix
opt_multi_bit_flop_opt	

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	specifies multi bit flop merge/split opt Type: enum Enum Values: false true mergeOnly splitOnly Default: splitOnly Edit: Yes Reference: opt_multi_bit_flop_opt
opt_multi_bit_flop_reorder_bits	
	Enable Swap MBFF bits to take advantage of mixed-driving strength cells. false: Disable; true: enable for timing and power optimization; timing: enable for timing optimization; power: enable for power optimization. Type: enum Enum Values: false true timing power Default: false Edit: Yes Reference: opt_multi_bit_flop_reorder_bits
opt_multi_bit_flop_split_report_failure_reason	
	To enable printing split failure reasons Type: bool Default: true Edit: Yes Reference: opt_multi_bit_flop_split_report_failure_reason
opt_multi_bit_flop_split_timing_effort	
	Timing effort level for MBFF splitting Type: enum Enum Values: low medium high Default: low Edit: Yes Reference: opt_multi_bit_flop_split_timing_effort
opt_multi_bit_merge_scan_head	
	merge of the scan chain head/tail Type: bool Default: false Edit: Yes
opt_multi_bit_split_scan_head	
	split of the scan chain head/tail Type: bool Default: false Edit: Yes
opt_multi_bit_unused_bit_count	
	specifies how many maximum unused bits to be allowed in each multibit cell Type: int Default: 1 Edit: Yes Reference: opt_multi_bit_unused_bit_count
opt_multi_bit_unused_bits	
	specifies whether a mapped multibit instance can have one or more unused bits Type: enum Enum Values: false true pre_place pre_cts Default: pre_place Edit: Yes Reference: opt_multi_bit_unused_bits
opt_multibit_allow_shift_register_head_flop_merge	

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	Controls whether head flops of shift register can be merged to multibit. Type: bool Default: false Edit: Yes Reference: opt_multibit_allow_shift_register_head_flop_merge
opt_multibit_combinational_name_concatenate_string	
	Separator string to be used for concatenation when naming multibit combinational and special low power cells. Recommended values for Genus-LEC flow are '_MB_' and '_mb_'. Type: string Default: _MB_ Edit: Yes Reference: opt_multibit_combinational_name_concatenate_string
opt_multibit_seqs_name_concatenate_string	
	Separator string to be used for concatenation when naming multibit sequential instances. Recommended values for Genus-LEC flow are '_MB_' and '_mb_'. Type: string Default: _MB_ Edit: Yes Reference: opt_multibit_seqs_name_concatenate_string
opt_new_inst_prefix	
	Prefix string for name of new instances Type: string Default: "" Edit: Yes Reference: opt_new_inst_prefix
opt_new_net_prefix	
	Prefix string for name of new nets Type: string Default: "" Edit: Yes Reference: opt_new_net_prefix
opt_one_pass_lec	
	Set options one pass lec opt support Type: bool Default: false Edit: Yes Reference: opt_one_pass_lec
opt_pin_swapping	
	Enables pin swapping Type: bool Default: true Edit: Yes Reference: opt_pin_swapping
opt_podv2_flow_effort	
	Effort level for PodV2 flow Type: enum Enum Values: auto standard extreme Default: auto Edit: Yes Reference: opt_podv2_flow_effort
opt_post_route_allow_overlap	

Stylus Common UI Database Object Information
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	Enables post_route optimization to insert cells with overlaps when inadequate free space is available. The placement will then be legalized before ECO routing. Type: bool Default: true Edit: Yes Reference: opt_post_route_allow_overlap
opt_post_route_area_reclaim	
	To enable area reclaim in post_route optimization Type: enum Enum Values: none setup_aware hold_and_setup_aware Default: none Edit: Yes Reference: opt_post_route_area_reclaim
opt_post_route_art_flow	
	Enables optimizing the top-level interface logic for a hierarchical floorplan Type: bool Default: false Edit: Yes Reference: opt_post_route_art_flow
opt_post_route_check_antenna_rules	
	Allows optimization to check/ignore antenna rules in lefsafe swaps Type: bool Default: true Edit: Yes Reference: opt_post_route_check_antenna_rules
opt_post_route_drv_recovery	
	Enables post_route drv recovery in gigaopt. Type: enum Enum Values: auto true false Default: auto Edit: Yes Reference: opt_post_route_drv_recovery
opt_post_route_fix_clock_drv	
	enables DRV fixing on clock net Type: bool Default: true Edit: Yes Reference: opt_post_route_fix_clock_drv
opt_post_route_fix_glitch	
	Enables glitch fixing in AAE flow Type: bool Default: true Edit: Yes Reference: opt_post_route_fix_glitch
opt_post_route_fix_si_transitions	
	Enables sislew fixing in AAE flow Type: bool Default: false Edit: Yes Reference: opt_post_route_fix_si_transitions
opt_post_route_hold_recovery	

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	Type: enum Enum Values: auto true false Default: auto Edit: Yes Reference: opt_post_route_hold_recovery
opt_post_route_report_si_transitions	
	report sislew violations Type: bool Default: false Edit: Yes Reference: opt_post_route_report_si_transitions
opt_post_route_setup_recovery	
	Enables post_route SI setup timing recovery in gigaopt. Type: enum Enum Values: auto true false Default: auto Edit: Yes Reference: opt_post_route_setup_recovery
opt_power_effort	
	Effort level for power aware mode in optimization Type: enum Enum Values: none low high Default: none Edit: Yes Reference: opt_power_effort
opt_pre_route_ndr_aware	
	Provide a list of NDR names that will be used NDR-based optimization Type: string Default: "" Edit: Yes Reference: opt_pre_route_ndr_aware
opt_preserve_all_sequential	
	Preserve sequential element during the simplify netlist optimization step Type: bool Default: false Edit: Yes Reference: opt_preserve_all_sequential
opt_preserve_hpin_function	
	Enables preserving logical functions at hierarchical ports Type: bool Default: false Edit: Yes Reference: opt_preserve_hpin_function
opt_print_ports_nets_preserved_for_callback	
	Print nets and pins preserved for verification. Type: bool Default: false Edit: Yes Reference: opt_print_ports_nets_preserved_for_callback
opt_remove_redundant_insts	

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	Enables simplifying the netlist during optimization Type: bool Default: true Edit: Yes Reference: opt_remove_redundant_insts
opt_report_multi_bit_unmerged_reasons	
	When enabled, the reasons for unmerged multibit flops/latches will be printed in the MBFF summary Type: bool Default: false Edit: Yes Reference: opt_report_multi_bit_unmerged_reasons
opt_resize_flip_flops	
	Enables FF resizing Type: bool Default: true Edit: Yes Reference: opt_resize_flip_flops
opt_resize_level_shifter_and_iso_insts	
	Enables resizing of shifters and isolation instances Type: bool Default: false Edit: Yes Reference: opt_resize_level_shifter_and_iso_insts
opt_resize_power_switch_insts	
	Enables resizing of power switch instances during optimization Type: bool Default: false Edit: Yes Reference: opt_resize_power_switch_insts
opt_retime_move_mux_loop_with_register	
	Move the feedback mux loop along with the register during retiming. Type: bool Default: true Edit: Yes Reference: opt_retime_move_mux_loop_with_register
opt_retime_register_naming_suffix	
	Suffix that gets appended to the name of the retimed flops. Type: string Default: _reg Edit: Yes Reference: opt_retime_register_naming_suffix
opt_route_opt_recovery	
	Type: enum Enum Values: auto true false Default: auto Edit: Yes Reference: opt_route_opt_recovery
opt_sequential_genus_restructure_report_failure_reason	
	To enable printing SeqCRR failure reasons Type: bool Default: false Edit: Yes Reference: opt_sequential_genus_restructure_report_failure_reason

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opt_setup_target_slack	
	Specifies target slack value (in nano secs) for setup timing optimization Type: double Default: 0.0 Edit: Yes Reference: opt_setup_target_slack
opt_signoff_add_inst	
	Specifies whether timing optimizer is allowed to add buffer/delay/inverter cell instances. Type: bool Default: true Edit: Yes Reference: opt_signoff_add_inst
opt_signoff_add_load	
	Specifies whether Hold optimizer is allowed to add dummy cell instances Type: bool Default: false Edit: Yes Reference: opt_signoff_add_load
opt_signoff_allow_skewing	
	useful skew based setup opt Type: bool Default: false Edit: Yes Reference: opt_signoff_allow_skewing
opt_signoff_along_route_buffering	
	Allow buffer/inverter to be inserted along the route and not only at driver or sink terms. Type: bool Default: true Edit: Yes Reference: opt_signoff_along_route_buffering
opt_signoff_buffer_cell_list	
	Specifies a list of buffer/delay cells that can be used for DRV and Hold optimizers, regardless of whether those cells have a "dont_use" attribute. Type: string Default: "" Edit: Yes Reference: opt_signoff_buffer_cell_list
opt_signoff_check_type	
	Check type for path based analysis. Type: enum Enum Values: early late both Default: both Edit: Yes Reference: opt_signoff_check_type
opt_signoff_clock_cell_list	
	Restricts the list of buffers/inverter/gating cells that are allowed to be used by ECO on the clock tree. Type: string Default: "" Edit: Yes Reference: opt_signoff_clock_cell_list
opt_signoff_clock_max_level	

Stylus Common UI Database Object Information
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	max level allowed for clock tree Type: int Default: 0 Edit: Yes Reference: opt_signoff_clock_max_level
opt_signoff_delete_inst	
	delete buffer for optimization Type: bool Default: true Edit: Yes Reference: opt_signoff_delete_inst
opt_signoff_disable_geometry_checks	
	Performs Vth swapping without checking pins geometry. Type: bool Default: false Edit: Yes Reference: opt_signoff_disable_geometry_checks
opt_signoff_drv_margin	
	Specifies a margin value by which ratio the drv check would be relaxed or tightened Type: double Default: 0.0 Edit: Yes Reference: opt_signoff_drv_margin
opt_signoff_eco_file_prefix	
	Specifies the prefix name for the ECO files generated. Type: string Default: "" Edit: Yes Reference: opt_signoff_eco_file_prefix
opt_signoff_em_type	
	Specifies the current waveform calculation method when doing EM fixing. Type: enum Enum Values: all avg rms peak Default: all Edit: Yes Reference: opt_signoff_em_type
opt_signoff_fix_cell_em	
	Enables Cell EM violations fixing during DRV optimization. Type: bool Default: false Edit: Yes Reference: opt_signoff_fix_cell_em
opt_signoff_fix_clock_drv	
	fix max_cap/max_tran violations on clock nets Type: bool Default: false Edit: Yes Reference: opt_signoff_fix_clock_drv
opt_signoff_fix_data_drv	
	fix max_cap/max_tran violations on data nets Type: bool Default: true Edit: Yes Reference: opt_signoff_fix_data_drv

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opt_signoff_fix_glitch	
	fix nets with glitch Type: bool Default: false Edit: Yes Reference: opt_signoff_fix_glitch
opt_signoff_fix_hold_allow_setup_degrade	
	Allow setup slack fixing in hold fixing. Type: enum Enum Values: none low medium high Default: none Edit: Yes Reference: opt_signoff_fix_hold_allow_setup_degrade
opt_signoff_fix_hold_allow_setup_optimization	
	simultaneous hold and setup fixing Type: bool Default: false Edit: Yes Reference: opt_signoff_fix_hold_allow_setup_optimization
opt_signoff_fix_hold_allow_setup_tns_degrade	
	When active, the Setup Total Negative Slack can be degraded during Hold time violations fixing while still maintaining the Setup Worst Negative Slack in every Setup active views. Type: bool Default: false Edit: Yes Reference: opt_signoff_fix_hold_allow_setup_tns_degrade
opt_signoff_fix_hold_variation_aware	
	Variation aware hold fixing. Type: bool Default: false Edit: Yes Reference: opt_signoff_fix_hold_variation_aware
opt_signoff_fix_hold_with_margin	
	Specifies a slack margin value in nanoseconds for Hold time violations fixing. The purpose is to overfix any Hold violations by the amount of the margin value. This margin is not applied on nets that are already meeting the Hold target slack. Type: double Default: 0.0 Edit: Yes Reference: opt_signoff_fix_hold_with_margin
opt_signoff_fix_ir_drop	
	fix ir-drop hotspots Type: bool Default: false Edit: Yes Reference: opt_signoff_fix_ir_drop
opt_signoff_fix_max_cap	
	Enables max_cap violations fixing during DRV optimization. Type: bool Default: true Edit: Yes Reference: opt_signoff_fix_max_cap
opt_signoff_fix_max_transition	

Stylus Common UI Database Object Information

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	Enables max_tran violations fixing during DRV optimization. Type: bool Default: true Edit: Yes Reference: opt_signoff_fix_max_transition
opt_signoff_fix_osus_glitch	
	fix nets with osus glitch Type: bool Default: false Edit: Yes Reference: opt_signoff_fix_osus_glitch
opt_signoff_fix_si_slew	
	Performs SI Slew fixing during DRV fixing using resizing and buffering techniques. Type: bool Default: false Edit: Yes Reference: opt_signoff_fix_si_slew
opt_signoff_fix_slack_in_range_lower_limit	
	Sets the lower slack limit for a violated path to be considered during timing closure. Type: double Default: -1e+30 Edit: Yes Reference: opt_signoff_fix_slack_in_range_lower_limit
opt_signoff_fix_slack_in_range_upper_limit	
	Sets the upper slack limit for a violated path to be considered during timing closure. Type: double Default: 1e+30 Edit: Yes Reference: opt_signoff_fix_slack_in_range_upper_limit
opt_signoff_fix_xtalk	
	Reduces the crosstalk on the net that violates the thresholds set by the user. Type: bool Default: false Edit: Yes Reference: opt_signoff_fix_xtalk
opt_signoff_fix_xtalk_preserve_hold	
	preserve hold slack in xtalk fixing Type: bool Default: false Edit: Yes Reference: opt_signoff_fix_xtalk_preserve_hold
opt_signoff_fix_xtalk_preserve_setup	
	preserve setup slack in xtalk fixing Type: bool Default: false Edit: Yes Reference: opt_signoff_fix_xtalk_preserve_setup
opt_signoff_hold_target_slack	
	Specifies a target slack value in nanoseconds for hold timing optimize Type: double Default: 0.0 Edit: Yes Reference: opt_signoff_hold_target_slack
opt_signoff_hold_target_slack_per_view	

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	Specifies view-wise target slack values in nanoseconds for hold timing optimization Type: string Default: "" Edit: Yes Reference: opt_signoff_hold_target_slack_per_view
opt_signoff_hold_xtalk_delta_threshold	
	Crosstalk delta delay threshold for selecting nets in hold timing views when running Xtalk fixing. Type: double Default: 0.3 Edit: Yes Reference: opt_signoff_hold_xtalk_delta_threshold
opt_signoff_hold_xtalk_slack_threshold	
	Slack threshold for selecting nets in hold timing views when running Xtalk fixing. Type: double Default: 1000.0 Edit: Yes Reference: opt_signoff_hold_xtalk_slack_threshold
opt_signoff_ignore_drv_checks	
	Ignores DRV checks during timing/area/power optimization. Type: bool Default: false Edit: Yes Reference: opt_signoff_ignore_drv_checks
opt_signoff_keep_tmp_files	
	To keep all the temporary files generated by D-MMMC and Tempus clients for debugging purposes. Type: bool Default: false Edit: Yes Reference: opt_signoff_keep_tmp_files
opt_signoff_legal_only	
	Enables/disables legal location checks during resizing and buffering Type: bool Default: true Edit: Yes Reference: opt_signoff_legal_only
opt_signoff_load_cell_list	
	Specifies the list of load cells Type: string Default: "" Edit: Yes Reference: opt_signoff_load_cell_list
opt_signoff_max_cap_margin	
	Specifies the ratio by which the max_cap check would be relaxed or tightened Type: double Default: 0.0 Edit: Yes Reference: opt_signoff_max_cap_margin
opt_signoff_max_local_density	
	threshold value for local density check. To enable give a value more than 0 Type: double Default: -1.0 Edit: Yes Reference: opt_signoff_max_local_density
opt_signoff_max_opt_runtime	

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	If set to a positive value, would restrict the optimization runtime by that value, in minutes. Type: int Default: 0 Edit: Yes Reference: opt_signoff_max_opt_runtime
opt_signoff_max_paths	
	Specifies how many paths are getting retimed for ECO DB generation in PBA mode. Type: int Default: 5000000 Edit: Yes Reference: opt_signoff_max_paths
opt_signoff_max_runtime	
	If set to a positive value, would restrict the command runtime by that value, in minutes. Type: int Default: 0 Edit: Yes Reference: opt_signoff_max_runtime
opt_signoff_max_slack	
	Specifies the maximum GBA slack value to be used when selecting paths to be retimed for ECO DB generation in PBA mode. Type: double Default: 0.0 Edit: Yes Reference: opt_signoff_max_slack
opt_signoff_max_transition_margin	
	Specifies the ratio by which the max_tran check would be relaxed or tightened Type: double Default: 0.0 Edit: Yes Reference: opt_signoff_max_transition_margin
opt_signoff_num_report_paths	
	Number of paths printed after signoff timing analysis per view Type: int Default: 50 Edit: Yes Reference: opt_signoff_num_report_paths
opt_signoff_nworst	
	Specifies the number of paths to be retimed per endpoints for ECO DB generation in PBA mode. Type: int Default: 50 Edit: Yes Reference: opt_signoff_nworst
opt_signoff_optimize_core_only	
	Enables the tool to optimize timing or recover area/power only on register to register paths. Type: bool Default: false Edit: Yes Reference: opt_signoff_optimize_core_only
opt_signoff_optimize_replicated_modules	
	Allows the tool to perform timing optimization inside replicated hierarchies while generating only one ECO file per sets of clones. The cell name of the replicated hierarchies have to be listed in a file pointed by the opt_signoff_partition_list_file option. Type: bool Default: false Edit: Yes Reference: opt_signoff_optimize_replicated_modules

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opt_signoff_optimize_sequential_cells	
	Specifies whether the timing/area/power optimizer is allowed to optimize sequential cells. Type: bool Default: true Edit: Yes Reference: opt_signoff_optimize_sequential_cells
opt_signoff_partition_list_file	
	File containing list of partitions in the design Type: string Default: "" Edit: Yes Reference: opt_signoff_partition_list_file
opt_signoff_pba_eco_route	
	Enables PBA timing to be passed to ecoRoute for higher timing driven routing accuracy Type: bool Default: false Edit: Yes Reference: opt_signoff_pba_eco_route
opt_signoff_pba_effort	
	When enabling "high" effort mode, all the optimization engines will perform more aggressive timing closure or area/power recovery with respect to PBA timing and the expectation is to get better quality of results. Type: enum Enum Values: medium high Default: high Edit: Yes Reference: opt_signoff_pba_effort
opt_signoff_post_mask	
	enable post mask ECO mode Type: bool Default: false Edit: Yes Reference: opt_signoff_post_mask
opt_signoff_post_mask_decap_cells	
	Specify list of PF decap cells that are allowed to be removed in post mask mode Type: string Default: "" Edit: Yes Reference: opt_signoff_post_mask_decap_cells
opt_signoff_post_mask_filler_cells	
	Specify list of PF filler cells that are allowed to be removed/reinserted in post mask mode Type: string Default: "" Edit: Yes Reference: opt_signoff_post_mask_filler_cells
opt_signoff_post_mask_gate_array_filler_cells	
	original of enable post mask mode to use cells from the specified list at GA filler sites Type: string Default: "" Edit: Yes Reference: opt_signoff_post_mask_gate_array_filler_cells
opt_signoff_post_mask_logic_cells	

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	Specify list of PF filler logic/ECO cells that are allowed to be used during optimization in post mask mode Type: string Default: "" Edit: Yes Reference: opt_signoff_post_mask_logic_cells
opt_signoff_post_sta_tcl	
	File containing the TCL code that will be executed immediately after signoff timing analysis per view Type: string Default: "" Edit: Yes Reference: opt_signoff_post_sta_tcl
opt_signoff_power_aware	
	power awareness in all optimizers and transforms Type: bool Default: false Edit: Yes Reference: opt_signoff_power_aware
opt_signoff_power_opt_focus	
	focus for power optimization Type: enum Enum Values: total leakage dynamic leakageOnly dynamicOnly Default: total Edit: Yes Reference: opt_signoff_power_opt_focus
opt_signoff_pre_sta_tcl	
	File containing the TCL code that will be executed immediately prior to signoff timing analysis per view Type: string Default: "" Edit: Yes Reference: opt_signoff_pre_sta_tcl
opt_signoff_prefix	
	Specifies the prefix to be assigned to the new instances inserted. Type: string Default: ESO Edit: Yes Reference: opt_signoff_prefix
opt_signoff_preserve_filler	
	To enable preserve and restore of fillers Type: bool Default: true Edit: Yes Reference: opt_signoff_preserve_filler
opt_signoff_read_eco_opt_db	
	Path of directory where ECO Timing DB files must be loaded from. Type: string Default: "" Edit: Yes Reference: opt_signoff_read_eco_opt_db
opt_signoff_read_irdrop_db	
	Path of directory where command debug_irdrop generated IR drop DB files are kept Type: string Default: "" Edit: Yes Reference: opt_signoff_read_irdrop_db

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opt_signoff_resize_inst	
	enable/disable resizing transform Type: bool Default: true Edit: Yes Reference: opt_signoff_resize_inst
opt_signoff_retime	
	To enable PBA-based timing report and optimization for of all optimizers. Type: enum Enum Values: none aocv path_slew_propagation aocv_path_slew_propagation Default: path_slew_propagation Edit: Yes Reference: opt_signoff_retime
opt_signoff_retime_depth	
	Retime depth for exhaustive path based analysis. Type: enum Enum Values: default infinite Default: default Edit: Yes Reference: opt_signoff_retime_depth
opt_signoff_retime_depth_early	
	Retime depth for exhaustive path based early analysis. Type: enum Enum Values: default infinite Default: default Edit: Yes Reference: opt_signoff_retime_depth_early
opt_signoff_retime_mode	
	Retime mode for path based analysis. Type: enum Enum Values: none path exhaustive Default: path Edit: Yes Reference: opt_signoff_retime_mode
opt_signoff_route_congestion_threshold	
	threshold value for route congestion buffering, value between 0 and 1 Type: double Default: 0.7 Edit: Yes Reference: opt_signoff_route_congestion_threshold
opt_signoff_routing_congestion_aware	
	Check routing congestion before adding buffers Type: bool Default: false Edit: Yes Reference: opt_signoff_routing_congestion_aware
opt_signoff_select_drv_net_file	
	list of nets to be optimized during DRV fixing Type: string Default: "" Edit: Yes Reference: opt_signoff_select_drv_net_file
opt_signoff_select_drv_pin_file	

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	list of pins to be optimized during DRV fixing Type: string Default: "" Edit: Yes Reference: opt_signoff_select_drv_pin_file
opt_signoff_select_hold_endpoints	
	select end points for hold fixing Type: string Default: "" Edit: Yes Reference: opt_signoff_select_hold_endpoints
opt_signoff_select_setup_endpoints	
	select end points for setup fixing Type: string Default: "" Edit: Yes Reference: opt_signoff_select_setup_endpoints
opt_signoff_set_hold_endpoints_margin	
	specify end point specific margin for hold fixing Type: string Default: "" Edit: Yes Reference: opt_signoff_set_hold_endpoints_margin
opt_signoff_set_setup_endpoints_margin	
	specify end point specific margin for setup fixing Type: string Default: "" Edit: Yes Reference: opt_signoff_set_setup_endpoints_margin
opt_signoff_setup_recovery	
	Forces Setup fixing to perform Setup timing recovery by using Vth swapping only. Useful feature to recover timing degradation after large ECO change during Power optimization. Type: bool Default: false Edit: Yes Reference: opt_signoff_setup_recovery
opt_signoff_setup_target_slack	
	Specifies a target slack value in nanoseconds for setup timing optimize Type: double Default: 0.0 Edit: Yes Reference: opt_signoff_setup_target_slack
opt_signoff_setup_target_slack_per_view	
	Specifies view-wise target slack values in nanoseconds for setup timing optimization Type: string Default: "" Edit: Yes Reference: opt_signoff_setup_target_slack_per_view
opt_signoff_setup_xtalk_delta_threshold	
	Crosstalk delta delay threshold for selecting nets in setup timing views when running Xtalk fixing. Type: double Default: 0.3 Edit: Yes Reference: opt_signoff_setup_xtalk_delta_threshold

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opt_signoff_setup_xtalk_slack_threshold	
	Slack threshold for selecting nets in setup timing views when running Xtalk fixing. Type: double Default: 1000.0 Edit: Yes Reference: opt_signoff_setup_xtalk_slack_threshold
opt_signoff_skew_core_only	
	dont skew flops which eventually launches/captures any IO path Type: bool Default: false Edit: Yes Reference: opt_signoff_skew_core_only
opt_signoff_skip_drv_net_file	
	list of nets to be excluded during DRV fixing Type: string Default: "" Edit: Yes Reference: opt_signoff_skip_drv_net_file
opt_signoff_swap_inst	
	Enables/disables VT swap resizing. Setup/leakage optimizers do not honor this option and always perform VT-swapping. Type: bool Default: true Edit: Yes Reference: opt_signoff_swap_inst
opt_signoff_upsize_in_power_opt	
	Allow upsizeing of cells for total power Type: bool Default: false Edit: Yes Reference: opt_signoff_upsize_in_power_opt
opt_signoff_verbose	
	Specifies whether the command should output log file reporting with verbosity or not. Type: bool Default: false Edit: Yes Reference: opt_signoff_verbose
opt_signoff_write_eco_opt_db	
	Path of directory where ECO Timing DB files are saved. Type: string Default: ecoTimingDB Edit: Yes Reference: opt_signoff_write_eco_opt_db
opt_skew	
	Enables triggering the skew_clock command from within opt_design Type: bool Default: true Edit: Yes Reference: opt_skew
opt_skew_apply_delay_limits_to_full_flow	
	delay limits apply to whole flow Type: bool Default: false Edit: Yes Reference: opt_skew_apply_delay_limits_to_full_flow

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opt_skew_ccopt	
	Effort for useful skew optimization within ccopt_design Type: enum Enum Values: none standard extreme Default: standard Edit: Yes Reference: opt_skew_ccopt
opt_skew_delay_pre_cts	
	delaying mode in pre_cts Type: bool Default: true Edit: Yes Reference: opt_skew_delay_pre_cts
opt_skew_macro_only	
	skew hard blocks Type: bool Default: false Edit: Yes Reference: opt_skew_macro_only
opt_skew_max_allowed_delay	
	max allowed skewing delay, unit is 'ns' Type: double Default: 1.0 Edit: Yes Reference: opt_skew_max_allowed_delay
opt_skew_min_allowed_delay	
	min allowed skewing delay, unit is 'ns' Type: double Default: 0.0 Edit: Yes Reference: opt_skew_min_allowed_delay
opt_skew_no_boundary	
	consider boundary sequential elements Type: bool Default: false Edit: Yes Reference: opt_skew_no_boundary
opt_skew_post_route	
	Enables useful skew optimization during post_route timing optimization. Type: bool Default: true Edit: Yes Reference: opt_skew_post_route
opt_skew_pre_cts	
	Enables useful skew optimization during pre_cts timing optimization. Type: bool Default: true Edit: Yes Reference: opt_skew_pre_cts
opt_target_based_opt_file	

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	Target based opt input file Type: string Default: "" Edit: Yes Reference: opt_target_based_opt_file
opt_target_based_opt_file_only	
	Use data from target based opt file for optimization Type: enum Enum Values: false true Default: true Edit: Yes Reference: opt_target_based_opt_file_only
opt_target_based_opt_hold_file	
	Target based opt input file for hold Type: string Default: "" Edit: Yes Reference: opt_target_based_opt_hold_file
opt_tied_inputs	
	Force optimization of instances with inputs tied together, even if it worsens objective gain Type: bool Default: false Edit: Yes Reference: opt_tied_inputs
opt_time_design_compress_reports	
	save reports in compressed format Type: bool Default: true Edit: Yes Reference: opt_time_design_compress_reports
opt_time_design_expanded_view	
	To report expanded views in time_design Type: bool Default: false Edit: Yes Reference: opt_time_design_expanded_view
opt_time_design_num_paths	
	Report specified number of paths in time_design Type: int Default: 50 Edit: Yes Reference: opt_time_design_num_paths
opt_time_design_report_net	
	To report net delays in time_design Type: bool Default: true Edit: Yes Reference: opt_time_design_report_net
opt_time_design_report_path_type	
	Type: enum Enum Values: end summary full full_clock end_slack_only summary_slack_only Default: full_clock Edit: Yes

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opt_time_design_vertical_timing_summary	
	Prints vertical timing summary after time_design and final summary report Type: bool Default: false Edit: Yes Reference: opt_time_design_vertical_timing_summary
opt_tns	
	Optimize all negative slack endpoints. Type: bool Default: true Edit: Yes Reference: opt_tns
opt_unfix_clock_insts	
	Enables unfixing clock instances flow Type: bool Default: true Edit: Yes Reference: opt_unfix_clock_insts
opt_use_default_related_pg_pin_for_always_on	
	Use default power or ground pin for the input or output pin's related_power_pin or related_ground_pin of always on buffer or inverter cell respectively. Type: bool Default: true Edit: Yes Reference: opt_use_default_related_pg_pin_for_always_on
opt_use_next_state_type_only_to_assign_sync_control	
	Forces to use only nextstate_type for sync control assignment. Type: bool Default: false Edit: Yes Reference: opt_use_next_state_type_only_to_assign_sync_control
opt_verbose	
	Enables verbose information logging Type: bool Default: false Edit: Yes Reference: opt_verbose
opt_xor_gating_exclude_flops	
	The list of flops to be excluded from XorGating Type: string Default: "" Edit: Yes
optimize_constant_0_flops	
	Allow constant 0 propagation through flip-flops. Type: bool Default: true Edit: Yes Reference: optimize_constant_0_flops
optimize_constant_1_flops	
	Allow constant 1 propagation through flip-flops. Type: bool Default: true Edit: Yes Reference: optimize_constant_1_flops

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optimize_constant_across_preserved	
	Enable constant optimization across preserved instances. Type: bool Default: true Edit: Yes Reference: optimize_constant_across_preserved
optimize_constant_feedback_seqs	
	Allow optimization of uninitialized feedback flip-flops. Type: bool Default: true Edit: Yes Reference: optimize_constant_feedback_seqs
optimize_constant_latches	
	Allow constant propagation through latches. Type: bool Default: true Edit: Yes Reference: optimize_constant_latches
optimize_merge_flops	
	Allow merging of equivalent flip-flops. Type: bool Default: true Edit: Yes Reference: optimize_merge_flops
optimize_merge_latches	
	Allow merging of equivalent latches. Type: bool Default: true Edit: Yes Reference: optimize_merge_latches
optimize_seq_x_to	
	Value that should be used to optimize constant X sequential instances. Type: enum Enum Values: 0 1 x Default: 0 Edit: Yes Reference: optimize_seq_x_to
optimize_yield	
	Optimize design for yield. Type: bool Default: false Edit: Yes Reference: optimize_yield
override_library_max_drc	
	Allow user to override design rule constraints set on library. Type: bool Default: false Edit: Yes Reference: override_library_max_drc
package_objects	
	Short-cut for [get_db current_design .package_objects] Type: obj(package_object)* Default: "" Edit: No

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parse_lib_moments_table	
	Parse library moments table while loading library file. Type: <code>bool</code> Default: <code>false</code> Edit: Yes Reference: parse_lib_moments_table
partition_floorplan_export	
	Export setting Type: <code>enum</code> Enum Values: macro special_route pin_constraint Default: special_route Edit: Yes Reference: partition_floorplan_export
partition_floorplan_import	
	Import setting Type: <code>enum</code> Enum Values: macro special_route pin_constraint Default: "" Edit: Yes Reference: partition_floorplan_import
partitions	
	Short-cut for [get_db current_design .partitions] Type: <code>obj(partition)*</code> Default: "" Edit: No
pbs_db_directory	
	Specifies the directory where partition based synthesis can write temporary files. Type: <code>string</code> Default: . Edit: Yes Reference: pbs_db_directory
pbs_mim_flow	
	Use folded hierarchies for partition based synthesis flow. Type: <code>bool</code> Default: false Edit: Yes Reference: pbs_mim_flow
percent_control_tolerance_for_map	
	Tolerance to add for tall and short limits for mapping (does not apply to doubles). Type: <code>double</code> Default: 10.0 Edit: Yes Reference: percent_control_tolerance_for_map
pg_base_pins	
	All the base_pins in all the base_cells. Type: <code>obj(pg_base_pin)*</code> Default: "" Edit: No
pg_nets	
	Short-cut for [get_db current_design .pg_nets]. List of power/ground nets in the design. This include physical-only PG nets, and Verilog supply0/supply1 nets. Note that Verilog nets assigned to 1'b0/1'b1 are returned by the nets attribute, not the pg_nets attribute. Type: <code>obj(net)*</code> Default: "" Edit: No

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pg_pins	
	Short-cut for [get_db current_design .pg_pins]. Type: obj(pg_pin)* Default: "" Edit: No
pg_ports	
	Short-cut for [get_db current_design .ports] Type: obj(port)* Default: "" Edit: No
phys_early_skew_flow	
	Enable early clock flow based early skewing. Type: bool Default: false Edit: Yes
phys_insts	
	Short-cut for [get_db current_design .phys_insts] Type: obj(inst)* Default: "" Edit: No
pin_blockages	
	Short-cut for [get_db current_design .pin_blockages] Type: obj(pin_blockage)* Default: "" Edit: No
pin_groups	
	Short-cut for [get_db current_design .pin_groups] Type: obj(pin_group)* Default: "" Edit: No
pin_guides	
	Short-cut for [get_db current_design .pin_guides] Type: obj(pin_guide)* Default: "" Edit: No
pins	
	Short-cut for [get_db current_design .pins] Type: obj(pin)* Default: "" Edit: No
place_3d_use_existing_bump_tsv	
	reuse pre-placed bumps or TSVs Type: bool Default: false Edit: Yes
place_blockages	
	Short-cut for [get_db current_design .place_blockages] Type: obj(place_blockage)* Default: "" Edit: No
place_cell_edge_spacing	

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	List of cell edge-type spacing values from LEF CELLEDGESPACINGTABLE or OA techfile that affects the placer. It specifies the min-spacing between different edge-types of std-cells, in a list format: {cell_edge_type1 cell_edge_type2 spacing}. The edge-type names match the base_cell's right_edge_type and left_edge_type attribute values. Type: string Default: "" Edit: No Reference: place_cell_edge_spacing
place_design_enable_3d	
	enable 3d placement Type: bool Default: false Edit: Yes Reference: place_design_enable_3d
place_design_floorplan_mode	
	run placement in floorplan mode Type: bool Default: false Edit: Yes Reference: place_design_floorplan_mode
place_design_integrity_ir_fix_effort	
	set power integrity placement IR fixing effort Type: enum Enum Values: low medium high Default: medium Edit: Yes Reference: place_design_integrity_ir_fix_effort
place_design_refine_macro	
	specify whether to run refine_macro_place after place_design -concurrent_macro Type: bool Default: false Edit: Yes Reference: place_design_refine_macro
place_design_refine_place	
	if set to false, calls to refinePlace from other apps (such as place_opt_design, opt_design) will not run refinePlace Type: bool Default: true Edit: Yes Reference: place_design_refine_place
place_detail_activity_power_driven	
	Type: bool Default: false Edit: Yes Reference: place_detail_activity_power_driven
place_detail_allow_border_pin_abut	
	allow instance abut with its neighbors when it has pins near by the cell boundary Type: bool Default: false Edit: Yes Reference: place_detail_allow_border_pin_abut
place_detail_allow_single_height_row_symmetry_x	

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	allow single height row to be symmetryX Type: string Default: "" Edit: Yes Reference: place_detail_allow_single_height_row_symmetry_x
place_detail_check_cut_spacing	
	Check via cut spacing during DRC checking Type: bool Default: false Edit: Yes Reference: place_detail_check_cut_spacing
place_detail_check_inst_space_group	
	check inst space group Type: bool Default: false Edit: Yes Reference: place_detail_check_inst_space_group
place_detail_check_route	
	do DRV checks against FIXED wires during legalization, along with preroutes Type: bool Default: false Edit: Yes Reference: place_detail_check_route
place_detail_color_aware_legal	
	enable adjacent cell color conflict check in legalization Type: bool Default: false Edit: Yes Reference: place_detail_color_aware_legal
place_detail_context_aware_legal	
	specify the types of cell context rules the placer needs to honor. The argument values [optional, required, user ignore_soft] can be used separately or combined, but they are exclusive to all none (default=all) Type: enum Enum Values: none all optional required user ignore_soft Default: all Edit: Yes Reference: place_detail_context_aware_legal
place_detail_eco_max_distance	
	specify max distance (in micron) for refinePlace ECO mode Type: double Default: 0.0 Edit: Yes Reference: place_detail_eco_max_distance
place_detail_eco_priority_insts	
	select instance priority for refinePlace ECO mode Type: enum Enum Values: placed fixed eco Default: placed Edit: Yes Reference: place_detail_eco_priority_insts
place_detail_fixed_shifter	

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	mark shifters FIXED once placed Type: bool Default: false Edit: Yes Reference: place_detail_fixed_shifter
place_detail_honor_inst_pad	
	honor padding from set_inst_padding in detail placement Type: bool Default: false Edit: Yes Reference: place_detail_honor_inst_pad
place_detail_io_pin_blockage	
	ioPins from top-level are treated as pre-routes during DRV checks in legalization Type: bool Default: false Edit: Yes Reference: place_detail_io_pin_blockage
place_detail_ir_aware_max_drive_strength	
	Type: double Default: 0.0 Edit: Yes Reference: place_detail_ir_aware_max_drive_strength
place_detail_irdrop_aware_effort	
	selects which IR drop optimization effort level to use during refinePlace Type: enum Enum Values: none low medium high Default: none Edit: Yes Reference: place_detail_irdrop_aware_effort
place_detail_irdrop_aware_timing_effort	
	selects which timing effort level to use during refinePlace Type: enum Enum Values: none standard high Default: high Edit: Yes Reference: place_detail_irdrop_aware_timing_effort
place_detail_irdrop_region_number	
	value for debug_irdrop -nregion in native API Type: int Default: 100 Edit: Yes Reference: place_detail_irdrop_region_number
place_detail_legalization_inst_gap	
	minimum gap between instances (unit sites) Type: int Default: 0 Edit: Yes Reference: place_detail_legalization_inst_gap
place_detail_m3_stripe_push_down	
	Width threshold in site-units for 'virtual' push-down of M3 stripe to M2 for M1 pin-access. Value of 0 means all M3 stripes will be pushed-down. Type: int Default: -1 Edit: Yes Reference: place_detail_m3_stripe_push_down

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place_detail_m3_stripe_shrink	
	Shrink the virtual M2 stripe (see spgM3StripePushDown) by given total site units, splitting value to each side, starting on left. e.g. value of 1 shrinks by 1 site on left, 2 shrinks by 1 site on each side, value of 3 shrinks 2 sites on left and 1 on right. Type: int Default: 0 Edit: Yes Reference: place_detail_m3_stripe_shrink
place_detail_max_shifter_column_depth	
	maximum distance from vertical power domain boundary that a shifter can be placed Type: double Default: 9999.0 Edit: Yes Reference: place_detail_max_shifter_column_depth
place_detail_max_shifter_depth	
	maximum distance from a horizontal or vertical power domain boundary that a shifter can be placed Type: double Default: 9999.0 Edit: Yes Reference: place_detail_max_shifter_depth
place_detail_max_shifter_row_depth	
	maximum distance from a horizontal power domain boundary that a shifter can be placed Type: double Default: 9999.0 Edit: Yes Reference: place_detail_max_shifter_row_depth
place_detail_no_filler_without_implant	
	there is no cell which has no implant obs Type: bool Default: false Edit: Yes Reference: place_detail_no_filler_without_implant
place_detail_pad_fixed_insts	
	honor cell-padding for FIXED instances in refinePlace Type: bool Default: false Edit: Yes Reference: place_detail_pad_fixed_insts
place_detail_pad_physical_cells	
	padding overlap with physical cells would be honored Type: bool Default: false Edit: Yes Reference: place_detail_pad_physical_cells
place_detail_pgftv_insertion_cell_list	
	Specify list of cells as PG FTV cells Type: string Default: "" Edit: Yes
place_detail_preroute_as_obs	
	specify layers on which preroute are taken as OBS Type: string Default: "" Edit: Yes Reference: place_detail_preroute_as_obs

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place_detail_preserve_routing	
	do not delete routed wires during refinePlace Type: bool Default: true Edit: Yes Reference: place_detail_preserve_routing
place_detail_remove_affected_routing	
	delete only wires whose nets were touched due to moved cells Type: bool Default: false Edit: Yes Reference: place_detail_remove_affected_routing
place_detail_swap_eeq_cells	
	replace master cells by its EEQ cell during legalization, to keep max-dist move low Type: bool Default: false Edit: Yes Reference: place_detail_swap_eeq_cells
place_detail_use_check_drc	
	Use FGC based DRC engine in legalization Type: bool Default: false Edit: Yes Reference: place_detail_use_check_drc
place_detail_use_diffusion_transition_fill	
	insert OD transition filler in 1X gap Type: bool Default: false Edit: Yes Reference: place_detail_use_diffusion_transition_fill
place_detail_use_gate_array_filler_groups	
	filler cell insts will only be replaced by logic insts that belong to same GA groups. Type: bool Default: false Edit: Yes Reference: place_detail_use_gate_array_filler_groups
place_detail_use_no_diffusion_one_site_filler	
	override -fillerGapMinGap to 0 and all source-drain spacing rule to 1 site during legalizing Type: bool Default: false Edit: Yes Reference: place_detail_use_no_diffusion_one_site_filler
place_detail_wire_length_opt_effort	
	selects which wire-length optimization effort level to use during refinePlace Type: enum Enum Values: none medium high Default: medium Edit: Yes Reference: place_detail_wire_length_opt_effort
place_global_activity_power_driven	

Stylus Common UI Database Object Information
Database Objects--root

	identifies and constrains power-critical nets to reduce switching power Type: enum Enum Values: false true Default: false Edit: Yes Reference: place_global_activity_power_driven
place_global_activity_power_driven_effort	
	-place_global_activity_power_driven_effort mode Type: enum Enum Values: standard high Default: standard Edit: Yes Reference: place_global_activity_power_driven_effort
place_global_align_macro	
	enable mixed-placer to align nearby same-size macros during global placement Type: bool Default: true Edit: Yes Reference: place_global_align_macro
place_global_allow_3d_stack	
	Type: bool Default: false Edit: Yes Reference: place_global_allow_3d_stack
place_global_auto_blockage_in_channel	
	Placement will (temporarily) block channels between areas with limited routing capacity Type: enum Enum Values: none soft partial Default: partial Edit: Yes Reference: place_global_auto_blockage_in_channel
place_global_clock_gate_aware	
	find better placement for clock gating elements towards the center of gravity for fanout Type: bool Default: true Edit: Yes Reference: place_global_clock_gate_aware
place_global_clock_power_driven	
	clock power driven Type: bool Default: true Edit: Yes Reference: place_global_clock_power_driven
place_global_clock_power_driven_effort	
	-place_global_clock_power_driven_effort mode Type: enum Enum Values: low standard high Default: low Edit: Yes Reference: place_global_clock_power_driven_effort
place_global_cong_effort	

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Database Objects--root

	level of effort for congestion driven global placer Type: enum Enum Values: low medium high extreme auto Default: auto Edit: Yes Reference: place_global_cong_effort
place_global_cpg_effort	
	level of effort for CPG Type: enum Enum Values: low medium high extreme Default: low Edit: Yes Reference: place_global_cpg_effort
place_global_cpg_file	
	Type: string Default: "" Edit: Yes Reference: place_global_cpg_file
place_global_enable_advanced_pipeline	
	Type: bool Default: false Edit: Yes Reference: place_global_enable_advanced_pipeline
place_global_enable_distributed_place	
	enable distributed placement platform Type: bool Default: false Edit: Yes Reference: place_global_enable_distributed_place
place_global_ignore_scan	
	ignore scan net during placement Type: enum Enum Values: true 1 false 0 auto Default: true Edit: Yes Reference: place_global_ignore_scan
place_global_ignore_spare	
	discard spare cell connections during global placement Type: bool Default: false Edit: Yes Reference: place_global_ignore_spare
place_global_max_density	
	placement strives to not let density exceed given value, in any part of design Type: double Default: -1.0 Edit: Yes Reference: place_global_max_density
place_global_module_aware_spare	

Stylus Common UI Database Object Information

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	Spare insts are placed randomly along with the containing module's insts Type: bool Default: false Edit: Yes Reference: place_global_module_aware_spare
place_global_module_padding	
	define the padding factor for the given module to reduce the local density and congestion Type: string Default: "" Edit: Yes Reference: place_global_module_padding
place_global_place_io_pins	
	place IO Pins concurrently with std.cell placement, and do layer assignment Type: bool Default: false Edit: Yes Reference: place_global_place_io_pins
place_global_reorder_scan	
	turn on reorder scan during placement Type: bool Default: true Edit: Yes Reference: place_global_reorder_scan
place_global_soft_guide_strength	
	level of effort for user defined softGuide for global placer Type: enum Enum Values: low medium high Default: low Edit: Yes Reference: place_global_soft_guide_strength
place_global_timing_effort	
	level of effort for timing driven global placer Type: enum Enum Values: medium high Default: medium Edit: Yes Reference: place_global_timing_effort
place_global_uniform_density	
	enable even cell distribution for designs with less than 70% utilization Type: enum Enum Values: false true Default: false Edit: Yes Reference: place_global_uniform_density
place_hard_fence	
	honor fence and region constraints in refinePlace Type: bool Default: true Edit: Yes Reference: place_hard_fence
place_hierarchical_flow	

Stylus Common UI Database Object Information
Database Objects--root

	enable hierarchical pin assignment flow Type: bool Default: false Edit: Yes Reference: place_hierarchical_flow
place_opt_post_place_tcl	
	a tcl script to be sourced after initial placement and before preCTS optimization in place_opt_design Type: string Default: "" Edit: Yes Reference: place_opt_post_place_tcl
place_opt_run_global_place	
	place opt run global place Type: enum Enum Values: none seed full Default: full Edit: Yes Reference: place_opt_run_global_place
place_sdp_clock_location	
	ICG cell location Type: string Default: center Edit: Yes Reference: place_sdp_clock_location
place_sdp_disable_extend_core	
	disable the extended core flag Type: bool Default: true Edit: Yes Reference: place_sdp_disable_extend_core
place_sdp_honor_dont_use	
	honor dont_use on base_cells during EEQ swapping in SDP legalization Type: bool Default: false Edit: Yes Reference: place_sdp_honor_dont_use
place_sdp_honor_orient	
	honor user specified orient during legalization Type: bool Default: false Edit: Yes Reference: place_sdp_honor_orient
place_sdp_legalization	
	Specify fixed corner of sdp box during legalizing sdp instance Type: enum Enum Values: NONE SW SE NW NE AUTO Default: NONE Edit: Yes Reference: place_sdp_legalization
place_sdp_legalization_effort	

Stylus Common UI Database Object Information
Database Objects--root

	Specifies the legalization effort in placing SDP groups Type: enum Enum Values: low medium high Default: low Edit: Yes Reference: place_sdp_legalization_effort
place_sdp_max_move_distance	
	max sdp group move distance, default is half perimeter of the core box Type: double Default: 0.0 Edit: Yes Reference: place_sdp_max_move_distance
place_sdp_num_column	
	number of column for sdp group Type: int Default: 2 Edit: Yes Reference: place_sdp_num_column
place_sdp_place_report	
	Output a detailed sdp placement data. Use empty string to reset to no output Type: string Default: "" Edit: Yes Reference: place_sdp_place_report
place_sdp_pre_fixed_cells_blockage_direction	
	Specify the direction when covering preplaced std cells using placement blockages Type: enum Enum Values: NONE X Y Default: NONE Edit: Yes Reference: place_sdp_pre_fixed_cells_blockage_direction
place_spare_update_timing_graph	
	update timing graph during place_spare_modules Type: bool Default: true Edit: Yes Reference: place_spare_update_timing_graph
plan_3dic_power_map_unit	
	power unit for power map file Type: string Default: mw Edit: Yes
plan_design_domain	
	Format: -domain {{PD1 <value1> net1} {PD2 <value2> net2}..} Type: string Default: "" Edit: Yes Reference: plan_design_domain
plan_design_rail_model	
	Select rail model: virtual/existing. Type: enum Enum Values: virtual existing Default: existing Edit: Yes Reference: plan_design_rail_model

Stylus Common UI Database Object Information

Database Objects--root

plan_design_total_power	
	total power defined by user. Type: double Default: 0.0 Edit: Yes Reference: plan_design_total_power
planner_keep_one_level_logics	
	Specifies the option to keep one level logics Type: bool Default: false Edit: Yes
pmbist_enable_multiple_views	
	Check whether support for multiple views of the same memory is requested or not. Type: bool Default: false Edit: Yes Reference: pmbist_enable_multiple_views
pmbist_full_async_reset	
	Allow all PMBIST logic to have full asynchronous reset. Type: bool Default: true Edit: Yes Reference: pmbist_full_async_reset
port_busses	
	Short-cut for [get_db current_design .port_busses] Type: obj(port_bus)* Default: "" Edit: No
port_shapes	
	Short-cut for [get_db current_design .port_shapes] Type: obj(port_shape)* Default: "" Edit: No
ports	
	Short-cut for [get_db current_design .ports] Type: obj(port)* Default: "" Edit: No
power_add_simulation	
	Add simulation for design coverage Type: bool Default: false Edit: Yes Reference: power_add_simulation
power_adjust_input_activity_in_iterations	
	Adjust input activity in state propagation. Type: bool Default: true Edit: Yes Reference: power_adjust_input_activity_in_iterations
power_adjust_macro_activity_in_iterations	

Stylus Common UI Database Object Information
Database Objects--root

	Adjust macro activity in state propagation. Type: bool Default: true Edit: Yes Reference: power_adjust_macro_activity_in_iterations
power_analysis_temperature	
	Sets the temperature for static power calculation. If not set, then uses the current opcond temperature Type: double Default: -274.000000 Edit: Yes Reference: power_analysis_temperature
power_annotation_detail_report	
	Generate detailed annotation report showing objects in their specific type blocks. Type: bool Default: false Edit: Yes Reference: power_annotation_detail_report
power_auto_twf_delay_annotation	
	Auto detect zero delay vectors Type: bool Default: false Edit: Yes Reference: power_auto_twf_delay_annotation
power_average_rise_fall_cap	
	When set to true, the software uses the average of rise and fall capacitance from the liberty file. Type: bool Default: false Edit: Yes Reference: power_average_rise_fall_cap
power_block_independent_peakpower	
	Power profiling done independently for each vector, design block Type: bool Default: false Edit: Yes Reference: power_block_independent_peakpower
power_bulk_pins	
	Defines power and ground bulk LEF pins for the design. Type: string Default: "" Edit: Yes Reference: power_bulk_pins
power_capacity	
	Specifies the segment count for profiler Type: enum Enum Values: low medium high Default: low Edit: Yes Reference: power_capacity
power_case_insensitive_mapping	
	Case insensitive auto-mapping, ignoring the case(upper or lower) of a character while matching rtl with design names. Type: bool Default: false Edit: Yes Reference: power_case_insensitive_mapping

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Database Objects--root

power_clock_source_as_clock	
	Specifies to use the correct clock frequency for activity calculation. Type: bool Default: false Edit: Yes Reference: power_clock_source_as_clock
power_comprehensive_automapping	
	Comprehensively perform instance name mapping between the RTL netlist and GATE level netlist. Type: bool Default: false Edit: Yes Reference: power_comprehensive_automapping
power_compress_reports	
	choose whether to compress power reports and set compression ratio Type: string Default: false Edit: Yes
power_constant_override	
	Specifies to give higher precedence to propagated constants defined using the set_case_analysis command. Type: bool Default: false Edit: Yes Reference: power_constant_override
power_corner	
	Defines the library corner (for non-MMMC setup). Type: string Default: "" Edit: Yes Reference: power_corner
power_create_driver_db	
	Specifies to determine all the drivers/receivers on the ground net. Type: bool Default: false Edit: Yes Reference: power_create_driver_db
power_current_generation_method	
	choose avg or peak current generation method Type: string Default: avg Edit: Yes Reference: power_current_generation_method
power_db_name	
	Specifies the name of the binary power database file (filename.db). Type: string Default: "" Edit: Yes Reference: power_db_name
power_debug_instance	
	print debug messages for instances Type: string Default: "" Edit: Yes
power_decap_cell_list	

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Database Objects--root

	Specifies the physical only cells, such as decap cells, to include in the power report. Type: string Default: "" Edit: Yes Reference: power_decap_cell_list
power_default_frequency	
	Specifies to set the default frequency in the MHz unit for net not annotated by TWF. Type: double Default: -1.0 Edit: Yes Reference: power_default_frequency
power_default_slew	
	Specifies to set the default slew in the ns unit for net not annotated by TWF. Type: string Default: "" Edit: Yes Reference: power_default_slew
power_default_supply_voltage	
	Specifies to set a default voltage for power nets in a scenario where the power engine cannot determine the voltage of these nets. Type: string Default: "" Edit: Yes Reference: power_default_supply_voltage
power_detailed_view_current_only	
	Ignore switching current for instances with detailed view Type: bool Default: true Edit: Yes Reference: power_detailed_view_current_only
power_disable_clock_gate_clipping	
	Specifies to enable clock gate output clipping so that the output transition density of an ICG cell does not exceed the input clock pin transition density. Type: bool Default: true Edit: Yes Reference: power_disable_clock_gate_clipping
power_disable_leakage_scaling	
	Specifies to exclude leakage power during scaling factor computation when target power is specified using the set_power command. Type: bool Default: false Edit: Yes Reference: power_disable_leakage_scaling
power_disable_static	
	When set to true in the dynamic vector-based or dynamic vectorless flows, the command performs only dynamic analysis and turns-off static power calculation. Type: bool Default: false Edit: Yes Reference: power_disable_static
power_display_negative_internal_power	
	For displaying negative Internal power results. Type: bool Default: true Edit: Yes Reference: power_display_negative_internal_power

Stylus Common UI Database Object Information
Database Objects--root

power_distribute_switching_power	
	distribute switching power to all connected I/O instances if the net does not have a driver instance Type: bool Default: false Edit: Yes Reference: power_distribute_switching_power
power_distributed_combine_report_format	
	control which power report files need to be combined Type: string Default: reduced Edit: Yes Reference: power_distributed_combine_report_format
power_distributed_setup	
	Specifies a file containing the customized setup details for running power analysis in the distributed mode. Type: string Default: "" Edit: Yes Reference: power_distributed_setup
power_domain_based_clipping	
	Determines the frequency to be used to clip propagated activity that is too high. Type: bool Default: false Edit: Yes Reference: power_domain_based_clipping
power_domains	
	Short-cut for [get_db current_design .power_domains] Type: obj(power_domain)* Default: "" Edit: No
power_dynamic_glitch_filter	
	set absolute time threshold value (in ns) for dynamic glitch filter Type: double Default: -1.0 Edit: Yes Reference: power_dynamic_glitch_filter
power_dynamic_scale_clock_by_frequency	
	set clock frequency scaling ratio by frequency value Type: string Default: "" Edit: Yes Reference: power_dynamic_scale_clock_by_frequency
power_dynamic_scale_clock_by_name	
	set clock frequency scaling ratio by clock name Type: string Default: "" Edit: Yes Reference: power_dynamic_scale_clock_by_name
power_dynamic_vectorless_ranking_methods	
	choose ranking options from: load, slack, clock and vector_activity Type: string Default: "" Edit: Yes Reference: power_dynamic_vectorless_ranking_methods

Stylus Common UI Database Object Information
Database Objects--root

power_enable_auto_mapping	
	Specifies to automatically perform instance name mapping between the RTL netlist and GATE level netlist. Type: bool Default: false Edit: Yes Reference: power_enable_auto_mapping
power_enable_auto_queue	
	Automatically assign power job to proper queue Type: bool Default: false Edit: Yes Reference: power_enable_auto_queue
power_enable_disk_mapping	
	Enable disk caching of toggles. Type: bool Default: false Edit: Yes Reference: power_enable_disk_mapping
power_enable_duty_propagation_with_global_activity	
	Specifies to propagate duty cycle with the global switching activity setting for all data nets in the dynamic vectorless analysis flow. Type: bool Default: false Edit: Yes Reference: power_enable_duty_propagation_with_global_activity
power_enable_dynamic_current_slew_load_interpolation	
	enable dynamic current slew load interpolation Type: bool Default: false Edit: Yes Reference: power_enable_dynamic_current_slew_load_interpolation
power_enable_dynamic_scaling	
	Enables set_power scaling for dynamic power analysis. Type: bool Default: false Edit: Yes Reference: power_enable_dynamic_scaling
power_enable_flop_state_propagation	
	Enable flop state propagation Type: bool Default: false Edit: Yes Reference: power_enable_flop_state_propagation
power_enable_force_propagation	
	Enable force propagation Type: bool Default: false Edit: Yes
power_enable_generated_clock	
	Specifies to get generated clock frequency during activity propagation for static power analysis. Type: bool Default: true Edit: Yes Reference: power_enable_generated_clock

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Database Objects--root

power_enable_input_net_power	
	Specifies to calculate the switching power of input nets. Type: bool Default: false Edit: Yes Reference: power_enable_input_net_power
power_enable_interactive_reports	
	Enable voltus interactive reports Type: bool Default: true Edit: Yes Reference: power_enable_interactive_reports
power_enable_mt_state_propagation	
	Enable Multithreading in state propagation Type: bool Default: true Edit: Yes Reference: power_enable_mt_state_propagation
power_enable_multithread_reports	
	to enable multithreaded power report generation Type: bool Default: false Edit: Yes Reference: power_enable_multithread_reports
power_enable_pba_for_tempus_pi	
	ensure path based timing analysis in tempus-pi flow. Type: bool Default: true Edit: Yes Reference: power_enable_pba_for_tempus_pi
power_enable_power_target_flow	
	enable vectorless state propagation with power target Type: bool Default: false Edit: Yes Reference: power_enable_power_target_flow
power_enable_rtl_dynamic_vector_based	
	Specifies to take an RTL or partial VCD/FSDB file as input and use that for dynamic vector-based flow. Type: bool Default: false Edit: Yes Reference: power_enable_rtl_dynamic_vector_based
power_enable_scan_report	
	enable scan report automatical generation Type: bool Default: false Edit: Yes Reference: power_enable_scan_report
power_enable_single_cycle_scheduling	
	Stage-propagation based vectorless flow schedules new flops/macros every cycle to achieve faster toggle coverage. Type: bool Default: false Edit: Yes Reference: power_enable_single_cycle_scheduling

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Database Objects--root

power_enable_slew_based_ccs_pin_cap	
	honor ccs pin cap Type: bool Default: false Edit: Yes Reference: power_enable_slew_based_ccs_pin_cap
power_enable_stable_clock_gating	
	Use stable clock gate in vless flow Type: bool Default: false Edit: Yes Reference: power_enable_stable_clock_gating
power_enable_stable_flop_scheduling	
	Enable stable flop scheduling Type: bool Default: false Edit: Yes Reference: power_enable_stable_flop_scheduling
power_enable_state_propagation	
	enable vectorless state propagation dynamic analysis Type: bool Default: false Edit: Yes Reference: power_enable_state_propagation
power_enable_tempus_pi	
	enable timing critical path analysis in state propagation based vectorless flow. Type: bool Default: false Edit: Yes Reference: power_enable_tempus_pi
power_enable_xp	
	standalone power analysis in voltus-XP Type: bool Default: false Edit: Yes Reference: power_enable_xp
power_engine	
	For selecting legacy, joules power engine. Type: enum Enum Values: legacy joules Default: joules Edit: Yes Reference: power_engine
power_enhanced_blackbox_avg	
	Specifies to use the average toggle rate of related inputs for enhanced blackbox propagation. Type: bool Default: false Edit: Yes Reference: power_enhanced_blackbox_avg
power_enhanced_blackbox_max	

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Database Objects--root

	Specifies to use the maximum toggle rate of related inputs for enhanced blackbox propagation. Type: bool Default: false Edit: Yes Reference: power_enhanced_blackbox_max
power_equivalent_annotation	
	Enable Qbar annotation Type: bool Default: false Edit: Yes Reference: power_equivalent_annotation
power_event_based_leakage_power	
	Event based leakage power Type: bool Default: false Edit: Yes Reference: power_event_based_leakage_power
power_external_load_config_file	
	Allow user to provide external load file to set external load manually. Type: string Default: "" Edit: Yes Reference: power_external_load_config_file
power_extraction_tech_file	
	technology file for exaction Type: string Default: "" Edit: Yes Reference: power_extraction_tech_file
power_extractor_include	
	extraction include file Type: string Default: "" Edit: Yes Reference: power_extractor_include
power_fanout_limit	
	Specifies the fanout limit for nets. Type: int Default: -1 Edit: Yes Reference: power_fanout_limit
power_force_library_merging	
	force library merging even if there is some error during merging Type: bool Default: false Edit: Yes Reference: power_force_library_merging
power_from_x_transition_factor	
	activity count for x to 0/1 transition Type: double Default: 0.5 Edit: Yes Reference: power_from_x_transition_factor
power_from_z_transition_factor	

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Database Objects--root

	activity count for z to 0/1 transition Type: double Default: 0.25 Edit: Yes Reference: power_from_z_transition_factor
power_generate_activity_mapping_report	
	turn on/off generation of activity_mapping.rpt Type: bool Default: false Edit: Yes Reference: power_generate_activity_mapping_report
power_generate_current_for_rail	
	Generates current files for the specified rail. Type: string Default: "" Edit: Yes Reference: power_generate_current_for_rail
power_generate_flop_ranking_data	
	Specifies the name of the dir containing the ranking data. Type: string Default: "" Edit: Yes Reference: power_generate_flop_ranking_data
power_generate_leakage_power_map_based_on_calculated_leakage	
	Print leakage power table based on calculated leakage power. Type: bool Default: false Edit: Yes Reference: power_generate_leakage_power_map_based_on_calculated_leakage
power_generate_static_report_from_state_propagation	
	generate static report from state propagation flow Type: bool Default: false Edit: Yes Reference: power_generate_static_report_from_state_propagation
power_get_state_dependent_leakage_power	
	To get state independent cell leakage power. Type: bool Default: true Edit: Yes
power_grid_libraries	
	Specifies the name of the Cadence power cell libraries (.cl). Type: string Default: "" Edit: Yes Reference: power_grid_libraries
power_handle_glitch	
	A transition is defined as a glitch when the time difference between two sequential toggles are less than half of the rise transition time + fall transition time in the vector-based static power calculation. Type: bool Default: false Edit: Yes Reference: power_handle_glitch
power_handle_transport_glitch	

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	Enable transport glitch support Type: bool Default: false Edit: Yes Reference: power_handle_transport_glitch
power_handle_tristate	
	When set to true , all tristate device enable pin values will be taken into account when determining the propagation of the activity through tristate gates. Type: bool Default: false Edit: Yes Reference: power_handle_tristate
power_hier_delimiter	
	Specifies the hierarchical delimiter in the DEF file. The default hierarchical delimiter is a forward slash (/) but can be changed by setting the - hier_delimiter parameter. Type: string Default: "" Edit: Yes Reference: power_hier_delimiter
power_honor_combinational_logic_on_clock_net	
	Specifies the behavior of combination logic on clock net. Type: bool Default: true Edit: Yes Reference: power_honor_combinational_logic_on_clock_net
power_honor_negative_energy	
	A value of true specifies that Voltus will keep negative internal energy numbers from the .lib internal power table. Type: bool Default: true Edit: Yes Reference: power_honor_negative_energy
power_honor_net_activity	
	honor net activity from activity file Type: bool Default: true Edit: Yes Reference: power_honor_net_activity
power_honor_non_mission_leakage	
	Option to activate Non-Mission leakage flow Type: bool Default: false Edit: Yes Reference: power_honor_non_mission_leakage
power_hybrid_analysis	
	Vbased + Vless runs done separately and results combined Type: bool Default: false Edit: Yes Reference: power_hybrid_analysis
power_ignore_control_signals	
	A value of true specifies that control signals will be ignored when propagating activity. Type: bool Default: true Edit: Yes Reference: power_ignore_control_signals

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power_ignore_data_phase_for_clock	
	Type: bool Default: false Edit: Yes Reference: power_ignore_data_phase_for_clock
power_ignore_end_toggles_in_profile	
	ignore toggles on end boundary in profiler analysis Type: bool Default: false Edit: Yes Reference: power_ignore_end_toggles_in_profile
power_ignore_glitches_at_same_time_stamp	
	Specifies to ignore glitches at the same transient time in the vector-based static and dynamic power analysis flow. Type: bool Default: true Edit: Yes Reference: power_ignore_glitches_at_same_time_stamp
power_ignore_inout_pin_cap	
	When set to true , ignores the bidirectional pin capacitances (direction : inout) defined for I/O cells in the .lib file, when calculating switching power and internal power. Type: bool Default: false Edit: Yes Reference: power_ignore_inout_pin_cap
power_ignore_macro_leakage_scale_for_temperature	
	Option for ignoring leakage power scaling on macro cell Type: bool Default: false Edit: Yes Reference: power_ignore_macro_leakage_scale_for_temperature
power_include_file	
	Used for adding additional setting for power analysis Type: string Default: "" Edit: Yes Reference: power_include_file
power_include_initial_x_transitions	
	Controls how the initial X state toggle should be counted. Type: bool Default: true Edit: Yes Reference: power_include_initial_x_transitions
power_include_sequential_clock_pin_power	
	Reports the clock pin power of flip-flops as part of the clock network power. Type: bool Default: false Edit: Yes Reference: power_include_sequential_clock_pin_power
power_include_timing_in_current_file	

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	enable timing bit writing in current file Type: bool Default: false Edit: Yes Reference: power_include_timing_in_current_file
power_intent_add_pin_name_to_power_instance	
	Add interface pin name to the low power instance that commit_power_intent inserts for 1801 flow. Type: bool Default: false Edit: Yes Reference: power_intent_add_pin_name_to_power_instance
power_intent_allow_back_to_back_isolation	
	Supports back-to-back isolation insertion along cross-domain net based on power intent ISO rule Type: bool Default: false Edit: Yes Reference: power_intent_allow_back_to_back_isolation
power_intent_allow_dont_touch_cells	
	Allow use of dont touch cells for commit_power_intent. Type: bool Default: false Edit: Yes Reference: power_intent_allow_dont_touch_cells
power_intent_allow_dont_use_cells	
	Allow use of avoided cells for commit_power_intent. Type: bool Default: false Edit: Yes Reference: power_intent_allow_dont_use_cells
power_intent_allow_inverted_level_shifter	
	Allow use of level shifters with inverter functionality. Type: bool Default: true Edit: Yes Reference: power_intent_allow_inverted_level_shifter
power_intent_allow_nested_default_domain	
	Allows a non-default power domain member to be logically nested in a default power domain Type: bool Default: true Edit: Yes Reference: power_intent_allow_nested_default_domain
power_intent_allow_power_domain_min_gap_zero	
	Type: bool Default: true Edit: Yes Reference: power_intent_allow_power_domain_min_gap_zero
power_intent_allow_shifter_voltage_mismatch	
	Type: bool Default: true Edit: Yes Reference: power_intent_allow_shifter_voltage_mismatch
power_intent_assume_iso_enable_pin_is_always_on	

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	To mark all Isolation cell's enable pin as always on Type: bool Default: true Edit: Yes Reference: power_intent_assume_iso_enable_pin_is_always_on
power_intent_check_all_nets_for_domain_crossing	
	Checks all the nets for domain crossings Type: bool Default: true Edit: Yes Reference: power_intent_check_all_nets_for_domain_crossing
power_intent_disable_always_on_buffering	
	Disable always on buffering for some hierarchies where always on cell may not be available. Type: bool Default: false Edit: Yes Reference: power_intent_disable_always_on_buffering
power_intent_do_not_use_top_domain_for_port_voltage	
	Controls voltage using top fterm domain for IO pin voltage Type: bool Default: false Edit: Yes Reference: power_intent_do_not_use_top_domain_for_port_voltage
power_intent_enable_third_domain_buffering	
	Enable buffering in third domain on data to output path of isolation or level shifter cells. Type: bool Default: false Edit: Yes Reference: power_intent_enable_third_domain_buffering
power_intent_honor_power_domain_for_domain_crossing_route	
	to handle nets belong to multiple power domains Type: bool Default: true Edit: Yes Reference: power_intent_honor_power_domain_for_domain_crossing_route
power_intent_honor_power_domain_for_intra_domain_route	
	To route intra-power domain nets only within their power domains Type: bool Default: true Edit: Yes Reference: power_intent_honor_power_domain_for_intra_domain_route
power_intent_include_dot_lib_related_pg_pin	
	Honors timing library related PG pins Type: bool Default: true Edit: Yes Reference: power_intent_include_dot_lib_related_pg_pin
power_intent_insert_auto_ls	

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	Enable capability to infer LS and to merge ISO into ELS depending on shifting requirements. Possible values are : combo_ls : insert LS and merge existing ISOs to ELS if required ls_only : insert only LS none : do not insert auto-ls Type: enum Enum Values: none combo_ls ls_only Default: none Edit: Yes
power_intent_insert_on_switch_network	
	Enable or disable insertion of isolation and level shifter cells on switch network. Type: bool Default: false Edit: Yes Reference: power_intent_insert_on_switch_network
power_intent_invert_preserved_net	
	Enable inversion on preserved nets for isolation insertion. Type: bool Default: false Edit: Yes Reference: power_intent_invert_preserved_net
power_intent_iso_level_shifter_skip_constant_propagation_loads	
	Type of loads of isolation or level shifters to skip constant propagation. Type: string Default: none Edit: Yes Reference: power_intent_iso_level_shifter_skip_constant_propagation_loads
power_intent_iso_nor	
	Prefer isonor for cell selection. Type: bool Default: false Edit: Yes Reference: power_intent_iso_nor
power_intent_parse_error_on_missing_objects	
	Treat missing objects as error. Type: string Default: "" Edit: Yes Reference: power_intent_parse_error_on_missing_objects
power_intent_parse_honor_dont_use_cells	
	Power intent parser to honor avoid attribute on library cell(s). Type: bool Default: false Edit: Yes Reference: power_intent_parse_honor_dont_use_cells
power_intent_read_enable_exhaustive_search	
	Set this attribute to control the searches done in power intent parser. Type: string Default: true Edit: Yes Reference: power_intent_read_enable_exhaustive_search
power_intent_share_well_always_on_buffering_support	

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	Type: bool Default: true Edit: Yes Reference: power_intent_share_well_always_on_buffering_support
power_intent_upf_insert_on_floating_pins	
	Type: bool Default: true Edit: Yes Reference: power_intent_upf_insert_on_floating_pins
power_intent_use_cpf_global_connect_for_always_on_buffer	
	Honors connection specs for AO buffer based on the power intent rules Type: bool Default: false Edit: Yes Reference: power_intent_use_cpf_global_connect_for_always_on_buffer
power_intent_use_cpf_global_connect_for_shifter	
	Honors connection specs for shifters based on the CPF rules Type: bool Default: false Edit: Yes Reference: power_intent_use_cpf_global_connect_for_shifter
power_intent_use_effective_domain_for_iso_shifter_insertion	
	Honors effective domains for ISO/LS insertions Type: bool Default: true Edit: Yes Reference: power_intent_use_effective_domain_for_iso_shifter_insertion
power_ir_derated_timing_view	
	set timing view in tempus-pi flow. Type: string Default: "" Edit: Yes Reference: power_ir_derated_timing_view
power_keep_clock_gate_ratio_in_iterations	
	Keep clock gate ratio in state propagation. Type: bool Default: false Edit: Yes Reference: power_keep_clock_gate_ratio_in_iterations
power_leakage_scale_factor_for_temperature	
	Performs a linear scaling of the leakage power for temperature in all libraries. Type: double Default: 1.0 Edit: Yes Reference: power_leakage_scale_factor_for_temperature
power_lib_files	
	Specify the lib files to be used for power Analysis. Libs specified through this option are not read in during design loading Type: string Default: "" Edit: Yes Reference: power_lib_files
power_library_preference	

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	option to choose instance binding to closest voltage or ecdm/ccsp library. Type: string Default: voltage Edit: Yes Reference: power_library_preference
power_macro_pgv_leakage	
	Use PG pin based leakage current from PGV instead of from liberty. Type: bool Default: true Edit: Yes Reference: power_macro_pgv_leakage
power_macro_toggle_pin_percentage	
	Set percentage of the number of Macro bits to schedule Type: double Default: 0.0 Edit: Yes Reference: power_macro_toggle_pin_percentage
power_match_state_for_logic_x	
	Controls how the logic X will be evaluated in the boolean function of the 'when' state of a power table. Type: string Default: x Edit: Yes Reference: power_match_state_for_logic_x
power_memory_current_scaling_method	
	NLPM Current Scaling Method for Memories [1/2/3] Type: int Default: 0 Edit: Yes Reference: power_memory_current_scaling_method
power_merge_switched_net_currents	
	Specifies to merge switched net currents into always-on currents. Type: bool Default: false Edit: Yes Reference: power_merge_switched_net_currents
power_method	
	Specifies the type of analysis to be performed. Type: enum Enum Values: static dynamic dynamic_vectorless dynamic_vectorbased dynamic_mixed_mode vector_profile event_based Default: static Edit: Yes Reference: power_method
power_min_leaf_count	
	min number of leaf instances count Type: int Default: 0 Edit: Yes Reference: power_min_leaf_count
power_modes	
	List of the power mode names in the design. Type: string Default: "" Edit: No
power_multi_scenario_simulation	

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	Enable multi-scenario simulation Type: bool Default: false Edit: Yes Reference: power_multi_scenario_simulation
power_multibit_flop_toggle_behavior	
	select mbff toggle: simultaneous independent(default) sbff pin_percentage during vectorless state propagation dynamic analysis Type: enum Enum Values: simultaneous independent sbff pin_percentage Default: independent Edit: Yes Reference: power_multibit_flop_toggle_behavior
power_multibit_flop_toggle_pin_percentage	
	The percentage of MBFF bits to schedule Type: double Default: 0.0 Edit: Yes Reference: power_multibit_flop_toggle_pin_percentage
power_off_pg_bulk_leakage	
	Contribute the bulk pin leakage to the connected always on rail and zero leakage to the off rail Type: bool Default: false Edit: Yes
power_off_pg_nets	
	The name of the power rails that will be turned off. Type: string Default: "" Edit: Yes Reference: power_off_pg_nets
power_output_current_data_prefix	
	Specifies a prefix to the static or dynamic current files generated by the software. Type: string Default: "" Edit: Yes Reference: power_output_current_data_prefix
power_output_dir	
	Used to save power engine output to the defined directory. If not set, will use the current working dir. Type: string Default: "" Edit: Yes Reference: power_output_dir
power_partition_count	
	Number of physical partitions for distributed run. Type: int Default: 0 Edit: Yes Reference: power_partition_count
power_partition_twf	
	turn on/off twf partitioning in distributed flow Type: bool Default: false Edit: Yes Reference: power_partition_twf
power_pin_based_twf	

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	turn on/off generation of pin based TWF in dynamic method Type: bool Default: false Edit: Yes Reference: power_pin_based_twf
power_pre_simulation_empty_period	
	Set pre-simulation empty period for power analysis Type: string Default: "" Edit: Yes Reference: power_pre_simulation_empty_period
power_pre_simulation_period	
	Set pre-simulation period for power analysis Type: string Default: "" Edit: Yes Reference: power_pre_simulation_period
power_pre_simulation_power_exclude_period	
	To specify pre simulation power exclude duration Type: string Default: "" Edit: Yes Reference: power_pre_simulation_power_exclude_period
power_precision	
	Specifies precision of decimal range [1-8] to ensure consistent decimal places are displayed for each power component. Type: int Default: 8 Edit: Yes Reference: power_precision
power_quit_on_activity_coverage_threshold	
	Specifies the activity (VCD/FSDB/TCF) coverage threshold value. Type: double Default: 0.0 Edit: Yes Reference: power_quit_on_activity_coverage_threshold
power_read_rcdb	
	Specifies to write a SPEF file containing only the total C for signal nets and pass it to the Dynamic Power engine. Type: bool Default: false Edit: Yes Reference: power_read_rcdb
power_relax_arc_match	
	Fallback arc lookup when no exact match Type: bool Default: false Edit: Yes Reference: power_relax_arc_match
power_report_black_boxes	
	Specify this parameter to report cells that are used as black boxes. Type: bool Default: false Edit: Yes Reference: power_report_black_boxes
power_report_clock_instance_switching_info	

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	Generate a report for clock instance switched during state propagation Type: string Default: None Edit: Yes Reference: power_report_clock_instance_switching_info
power_report_idle_instances	
	report non-switching instances in vectorless state propagation dynamic analysis Type: bool Default: false Edit: Yes Reference: power_report_idle_instances
power_report_instance_switching_info	
	all or output_logic or none Type: enum Enum Values: all output_logic none Default: none Edit: Yes Reference: power_report_instance_switching_info
power_report_instance_switching_list	
	specify file which is having list of instances Type: string Default: "" Edit: Yes Reference: power_report_instance_switching_list
power_report_library_usage	
	generate report for lib usage Type: bool Default: false Edit: Yes Reference: power_report_library_usage
power_report_missing_bulk_connectivity	
	Specifies to report missing bulk pins in the missing input report (-report_missing_input). Type: bool Default: false Edit: Yes Reference: power_report_missing_bulk_connectivity
power_report_missing_input	
	Specifies to report missing netlist information in input files, such as library, LEF/DEF, PGV, SPEF, TWF, and missing logical connectivity for cell instance (PGNET). Type: bool Default: false Edit: Yes Reference: power_report_missing_input
power_report_missing_nets	
	Controls reporting of missing nets in both static and dynamic power analysis. Type: bool Default: false Edit: Yes Reference: power_report_missing_nets
power_report_scan_chain_statistics	
	report scan chain analysis statistics Type: bool Default: false Edit: Yes Reference: power_report_scan_chain_statistics

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power_report_statistics	
	Reports a set of statistics on the instance power, instance power density, clock power, or transition density. Type: bool Default: false Edit: Yes Reference: power_report_statistics
power_report_time_display_fraction_digits	
	specify display fraction digits of reporting time Type: int Default: -1 Edit: Yes Reference: power_report_time_display_fraction_digits
power_report_twf_attributes	
	TWF attributes report format Type: string Default: "" Edit: Yes Reference: power_report_twf_attributes
power_reuse_flop_ranking_data	
	Specifies the name of the dir containing the ranking data. Type: string Default: "" Edit: Yes Reference: power_reuse_flop_ranking_data
power_reuse_flop_ranking_data_hier	
	A hierarchy name from the user that will be used to search the ranking data for the block level being analyzed. Type: string Default: "" Edit: Yes Reference: power_reuse_flop_ranking_data_hier
power_scale_to_sdc_clock_frequency	
	Specifies to enable the VCD clock frequency to be scaled up to the SDC clock frequency. Type: bool Default: false Edit: Yes Reference: power_scale_to_sdc_clock_frequency
power_scan_chain_activity	
	define scan chain name and activity Type: string Default: "" Edit: Yes Reference: power_scan_chain_activity
power_scan_chain_name_pattern	
	define scan chain name and pattern Type: string Default: "" Edit: Yes Reference: power_scan_chain_name_pattern
power_scan_control_file	
	Specifies the scan control file name required to run the Scan Mode Analysis flow. Type: string Default: "" Edit: Yes Reference: power_scan_control_file

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power_scan_multi_bit_flop_chain_type	
	Specifies the type of multi-bit scan flip-flop. Type: string Default: liberty Edit: Yes Reference: power_scan_multi_bit_flop_chain_type
power_set_power_event_based	
	Constrained power analysis in event based power Type: bool Default: false Edit: Yes Reference: power_set_power_event_based
power_settling_buffer	
	Specifies the settling buffer time (in ps unit) between multiple windows of a VCD file. Type: string Default: "" Edit: Yes Reference: power_settling_buffer
power_show_cell_usage_stats	
	show cell usage stats in power report Type: bool Default: false Edit: Yes
power_shutoff_aware_estimate	
	Enable power calculation considering power domains which are shutoff. Type: bool Default: true Edit: Yes Reference: power_shutoff_aware_estimate
power_smart_window	
	Max power metric window extraction Type: bool Default: false Edit: Yes Reference: power_smart_window
power_split_bus_power	
	A value of false applies the internal power number to each individual bit. Type: bool Default: false Edit: Yes Reference: power_split_bus_power
power_start_time_alignment	
	Specifies to align all the signals of the activity file at time zero. Type: bool Default: true Edit: Yes Reference: power_start_time_alignment
power_state_dependent_leakage	
	When set to false , power analysis performs state independent leakage power calculation. Type: bool Default: true Edit: Yes Reference: power_state_dependent_leakage

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power_stateprop_ignore_unannot_pins	
	Enable ignore unannotated pins in state propagation Type: bool Default: false Edit: Yes Reference: power_stateprop_ignore_unannot_pins
power_static_multi_mode_scenario_file	
	set static multi-mode scenario file name Type: string Default: "" Edit: Yes Reference: power_static_multi_mode_scenario_file
power_static_netlist	
	Specifies to perform static power analysis using a Verilog or DEF only netlist. Type: string Default: verilog Edit: Yes Reference: power_static_netlist
power_switching_power_on_rise_only	
	Switching power analysis done only for rise toggles Type: bool Default: false Edit: Yes Reference: power_switching_power_on_rise_only
power_thermal_input_file	
	Specifies to read a power map file to perform thermal analysis. Type: string Default: "" Edit: Yes Reference: power_thermal_input_file
power_thermal_leakage_temperature_scale_table_file	
	Defines library/cell/instance based leakage power scaling factors on each temperature point. Type: string Default: "" Edit: Yes Reference: power_thermal_leakage_temperature_scale_table_file
power_to_x_transition_factor	
	activity count for 0/1 to x transition Type: double Default: 0.5 Edit: Yes Reference: power_to_x_transition_factor
power_to_z_transition_factor	
	activity count for 0/1 to z transition Type: double Default: 0.25 Edit: Yes Reference: power_to_z_transition_factor
power_toggle_rate_unit	
	Toggle rate time unit. Type: string Default: /ns Edit: Yes Reference: power_toggle_rate_unit

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power_transition_factor_based_duty	
	Duty calculation using transition factor of toggle Type: bool Default: false Edit: Yes Reference: power_transition_factor_based_duty
power_transition_time_method	
	Specifies the minimum, maximum, or average transition time method that will be used with the integrated timer or the external TWF. Type: string Default: max Edit: Yes Reference: power_transition_time_method
power_twf_delay_annotation	
	Allows you to choose between min, max, or avg arrival times from the timing window file (TWF). Type: string Default: avg Edit: Yes Reference: power_twf_delay_annotation
power_twf_load_cap	
	Allows you to ignore or to select minimum, maximum, or average value of TWF external load or capacitance for power calculation. Type: string Default: max Edit: Yes Reference: power_twf_load_cap
power_unified_power_switch_flow	
	if enable ups flow Type: bool Default: false Edit: Yes Reference: power_unified_power_switch_flow
power_unit	
	Power unit. Type: string Default: nW Edit: Yes Reference: power_unit
power_use_cell_leakage_power_density	
	Specifies to use library leakage density times area (default_leakage_power_density). Type: bool Default: true Edit: Yes Reference: power_use_cell_leakage_power_density
power_use_fastest_clock_for_dynamic_scheduling	
	Specifies to use only the respective fastest clock associated with each net or pin to schedule events for a given simulation period. Type: bool Default: false Edit: Yes Reference: power_use_fastest_clock_for_dynamic_scheduling
power_use_lef_for_missing_cells	
	Allows a combination of LEF and PGV in the dynamic analysis flow. Type: bool Default: false Edit: Yes Reference: power_use_lef_for_missing_cells

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power_use_only_ddv_current	
	use only DDV current from PGV Type: bool Default: true Edit: Yes Reference: power_use_only_ddv_current
power_use_physical_partition	
	use geometric partition in DP Type: bool Default: false Edit: Yes Reference: power_use_physical_partition
power_use_twf_stable_randomized_arrival_delay	
	Allows tool to choose among all, data_only and none for stable randomized delay annotation from timing window file. Type: string Default: none Edit: Yes Reference: power_use_twf_stable_randomized_arrival_delay
power_use_zero_delay_vector_file	
	Enables zero delay mode in vector-based dynamic analysis to avoid pessimism in current estimation. Type: bool Default: false Edit: Yes Reference: power_use_zero_delay_vector_file
power_vector_based_multithread	
	Specifies to enable multi-threading for the VCD/FSDB based dynamic vector-based flows. Type: bool Default: true Edit: Yes Reference: power_vector_based_multithread
power_vector_profile_mode	
	activity or density Type: enum Enum Values: activity event_based power_density transient Default: event_based Edit: Yes Reference: power_vector_profile_mode
power_worst_case_vector_activity	
	Specifies to use the worst activity value when multiple vectors are specified in the static power calculation flow. Type: bool Default: false Edit: Yes Reference: power_worst_case_vector_activity
power_worst_step_size	
	Set worst step size for power analysis Type: string Default: "" Edit: Yes Reference: power_worst_step_size
power_worst_window_count	

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	Set worst window count for power analysis Type: int Default: 1 Edit: Yes Reference: power_worst_window_count
power_worst_window_coverage	
	Set worst window coverage type Type: string Default: "" Edit: Yes Reference: power_worst_window_coverage
power_worst_window_reports	
	generate Power reports for full simulation and worst power window Type: string Default: full Edit: Yes Reference: power_worst_window_reports
power_worst_window_size	
	Set worst window size for power analysis Type: string Default: "" Edit: Yes Reference: power_worst_window_size
power_worst_window_type	
	Set worst window type for power analysis Type: string Default: "" Edit: Yes Reference: power_worst_window_type
power_write_bbox	
	save the instance bbox to a text file when output power db Type: bool Default: false Edit: Yes Reference: power_write_bbox
power_write_db	
	A value of true creates a binary database of power results. Type: bool Default: false Edit: Yes Reference: power_write_db
power_write_default_pti_files	
	For instances that are not hooked to any power/ground rail, the current for these instances are reflected in the default static current files. Type: bool Default: true Edit: Yes Reference: power_write_default_pti_files
power_write_dynamic_currents	
	Current write with method event_based Type: bool Default: false Edit: Yes Reference: power_write_dynamic_currents
power_write_gui_db	

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	A value of true creates a binary database of power results for gui. Type: bool Default: true Edit: Yes Reference: power_write_gui_db
power_write_profiling_db	
	turn on/off profiling database writing Type: bool Default: false Edit: Yes Reference: power_write_profiling_db
power_write_simulation_db	
	Save functional waveform foreach instance when parsing FSDB/VCD Type: bool Default: false Edit: Yes Reference: power_write_simulation_db
power_write_static_currents	
	A value of true tells Voltus to generate the current data files per net. Type: bool Default: false Edit: Yes Reference: power_write_static_currents
power_x_count_transition_using_3_states	
	Toggle count using 3 states in VCD Type: bool Default: false Edit: Yes Reference: power_x_count_transition_using_3_states
power_x_transition_factor	
	When using VCD/FSDB one can specify how transitions to and from X are counted. Type: double Default: 0.5 Edit: Yes Reference: power_x_transition_factor
power_z_transition_factor	
	When using VCD/FSDB one can specify how transitions to and from Z are counted. Type: double Default: 0.25 Edit: Yes Reference: power_z_transition_factor
power_zero_delay_vector_toggle_shift	
	Specifies to shift the current waveform in the zero-delay VCD flow. Type: string Default: "" Edit: Yes Reference: power_zero_delay_vector_toggle_shift
preserve_combinational_loop_ports_nets	
	Preserve nets and pins involved in combinational loops as a reference for verification. Type: bool Default: true Edit: Yes Reference: preserve_combinational_loop_ports_nets
print_full_message_summary	

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	Used to control the format of message summary. If true, print full message, otherwith, print truncated message. Type: bool Default: false Edit: Yes Reference: print_full_message_summary
program_major_version	
	The major version plus one digit. So if program_version = 19.11-e062_1 it would return '19.1'. Type: double Default: "" Edit: No Reference: program_major_version
program_name	
	name of the program Type: string Default: "" Edit: No Reference: program_name
program_short_name	
	short name of the program Type: string Default: "" Edit: No Reference: program_short_name
program_version	
	version of the program Type: string Default: "" Edit: No Reference: program_version
propagate_constant_from_timing_model	
	Allow constant propagation from timing model instances. Type: bool Default: true Edit: Yes Reference: propagate_constant_from_timing_model
proto_allow_model_with_io	
	Supports model with I/Os. Type: bool Default: false Edit: Yes Reference: proto_allow_model_with_io
proto_allow_power_domain_in_flexmodel	
	Allow powerdomain in flexmodel Type: bool Default: false Edit: Yes Reference: proto_allow_power_domain_in_flexmodel
proto_create_dir	
	Specifies the directory to save created models. Type: string Default: proto_model Edit: Yes Reference: proto_create_dir
proto_create_high_fanout_ps_per_micron	

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	Create high fanout psPM models. Type: bool Default: true Edit: Yes Reference: proto_create_high_fanout_ps_per_micron
proto_create_lib	
	Specifies the OA library to save created models. Type: string Default: proto_model_lib Edit: Yes Reference: proto_create_lib
proto_create_metal_fill_ndr	
	Will be used by create_proto_net_delay_model for creating psPM model for nets with non default rules. Type: double Default: 0.0 Edit: Yes Reference: proto_create_metal_fill_ndr
proto_create_metal_fill_nominal	
	Will be used by create_proto_net_delay_model for creating psPM model for normal nets. Type: double Default: 0.5 Edit: Yes Reference: proto_create_metal_fill_nominal
proto_create_multi_corner_ps_per_micron	
	Create multi corner psPM models. Type: bool Default: false Edit: Yes Reference: proto_create_multi_corner_ps_per_micron
proto_create_ndr_ps_per_micron	
	Control whether create_proto_net_delay_model should generate psPM model for non default rule or not Type: enum Enum Values: auto on off Default: auto Edit: Yes Reference: proto_create_ndr_ps_per_micron
proto_create_no_flex_filler	
	Create proto model without adding flex filler. Type: bool Default: false Edit: Yes Reference: proto_create_no_flex_filler
proto_create_partition_as_flexmodel	
	Create partition specified as flexmodel automatically. Type: bool Default: false Edit: Yes Reference: proto_create_partition_as_flexmodel
proto_create_pipeline_flop	
	Specifies a list of pipeline flop instance patterns that are used in the design. Type: string Default: "" Edit: Yes Reference: proto_create_pipeline_flop

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proto_create_power_domain_ps_per_micron	
	Create power domain psPM models. Type: bool Default: false Edit: Yes Reference: proto_create_power_domain_ps_per_micron
proto_create_timing_budget	
	Keep consistent setting of create_timing_budget Type: bool Default: false Edit: Yes Reference: proto_create_timing_budget
proto_design_cover_fixed_macros	
	force flexModel cover fixed macros belong to it when using new plan_design algorithm Type: bool Default: false Edit: Yes Reference: proto_design_cover_fixed_macros
proto_design_effort	
	Specifies proto_design effort Type: enum Enum Values: low medium high Default: low Edit: Yes Reference: proto_design_effort
proto_design_flexmodel_constraint_type	
	Convert flexModel to the specified constraint type in proto_design. Type: enum Enum Values: guide region fence Default: region Edit: Yes Reference: proto_design_flexmodel_constraint_type
proto_design_keep_guide	
	-keep_guide {true false} #default=true Type: bool Default: true Edit: Yes Reference: proto_design_keep_guide
proto_design_place_macro	
	control if do macro placement in plan_design Type: bool Default: true Edit: Yes Reference: proto_design_place_macro
proto_design_remove_overlap	
	control if do remove overlap in plan_design Type: bool Default: true Edit: Yes Reference: proto_design_remove_overlap
proto_feasible_target	

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	Controls the feasible target feature to ignore huge negative slack endpoints from target computation and optimization. Type: bool Default: false Edit: Yes Reference: proto_feasible_target
proto_feasible_target_adjust_slack_pct	
	Controls percentage of slack value to be used as positive adjust value in path_adjust exception. Type: int Default: 100 Edit: Yes Reference: proto_feasible_target_adjust_slack_pct
proto_feasible_target_threshold	
	Controls minimum threshold delay for which path_adjust should be applied. Type: string Default: no_value Edit: Yes Reference: proto_feasible_target_threshold
proto_feasible_target_threshold_clock_pct	
	Controls slack to clock period percentage for which path_adjust should be applied. Type: int Default: 75 Edit: Yes Reference: proto_feasible_target_threshold_clock_pct
proto_hdl	
	Continue elaboration, in presence of certain types of inconsistencies(e.g. module name case mismatch between module instance and module definition). Type: string Default: "" Edit: Yes Reference: proto_hdl
proto_identify_engine	
	identify_proto algorithm <default=auto> Type: enum Enum Values: auto module_based instgroup_based Default: auto Edit: Yes Reference: proto_identify_engine
proto_identify_estimated_flexmodel_number	
	Specified the estimated flexModel number to identify for the design when using instance group based identify_proto_model. Type: int Default: 20 Edit: Yes Reference: proto_identify_estimated_flexmodel_number
proto_identify_exclude_module	
	Excludes specified module(s) but not its sub-modules for being marked as models. Type: string Default: "" Edit: Yes Reference: proto_identify_exclude_module
proto_identify_exclude_module_and_parent	
	Excludes specified module(s) and their parents for being marked as models. Type: string Default: "" Edit: Yes Reference: proto_identify_exclude_module_and_parent

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proto_identify_exclude_module_tree	
	Excludes specified module(s) and its sub-modules for being marked as models. Type: string Default: "" Edit: Yes Reference: proto_identify_exclude_module_tree
proto_identify_honor_objects_hierarchy	
	Specified the list of objects which instance group based identify_proto_model will honor its hierarchy. Type: string Default: "" Edit: Yes Reference: proto_identify_honor_objects_hierarchy
proto_identify_max_inst	
	Specifies the maximum instance count threshold of a module to be marked as model.(default=8% of total instance count and no lower than 300K instances) Type: int Default: -1 Edit: Yes Reference: proto_identify_max_inst
proto_identify_min_inst	
	Specifies the minimum instance count threshold of a module to be marked as model. Type: int Default: 1000 Edit: Yes Reference: proto_identify_min_inst
proto_keep_inst_file_only	
	FlexModel only contains inst in keep_inst_file. Type: bool Default: false Edit: Yes Reference: proto_keep_inst_file_only
proto_keep_instance_defined_in_sdc	
	FlexModel will keep inst defined in sdc file Type: bool Default: false Edit: Yes Reference: proto_keep_instance_defined_in_sdc
proto_keep_slack_improve_ndr	
	Will be used by report_NDR_WNS_gain to keep top critical nets as non default rule nets . Type: bool Default: false Edit: Yes Reference: proto_keep_slack_improve_ndr
proto_max_report_ndr_net	
	Will be used by create_proto_net_delay_model for creating psPM model for nets with non default rules. Type: double Default: 0.05 Edit: Yes Reference: proto_max_report_ndr_net
proto_route_net_ndr	

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	Will be used by create_proto_net_delay_model for creating psPM model for nets with non default rules. Type: string Default: "" Edit: Yes Reference: proto_route_net_ndr
proto_timing_net_delay	
	Specifies the estimated delay per micron for each routing layer. Layer range is from 1 to maximum routing layers. Type: string Default: "" Edit: Yes Reference: proto_timing_net_delay
proto_use_timing_net_delay_model	
	set mode of calculating wire delay Type: enum Enum Values: best_layer_no_detour best_layer_blockage_aware use_actual_wire Default: best_layer_no_detour Edit: Yes Reference: proto_use_timing_net_delay_model
proto_verbose	
	Output detailed information into .log file. Type: bool Default: false Edit: Yes Reference: proto_verbose
rc_corners	
	Returns the information about the rc corners in the design. Type: obj(rc_corner)* Default: "" Edit: No
read_db_directory	
	Specifies the directory of current loaded DB. Type: string Default: "" Edit: Yes Reference: read_db_directory
read_db_file_check	
	By default, read_db will check all files inside the saved DB directory are unchanged. If any files have been edited, renamed or deleted, then an error will occur unless this value is set to false. Type: bool Default: true Edit: Yes Reference: read_db_file_check
read_db_keep_updated_macro_lefs	
	If set true, the lef files of updated macros that has been saved in DB would be honored, even if there are new lef files for these macros are specified in read_db -lef_files {xxx}. If set false, the updated macros' lef files in db would be ignored. Only the lef files that are specified in read_db -lef_files {xxx} would be updated. Type: bool Default: false Edit: Yes
read_db_stop_at_design_in_memory	
	By default, tool will error and stop if call read_db more than once in the same Innovus session unless this value is set to false. Type: bool Default: true Edit: Yes Reference: read_db_stop_at_design_in_memory

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read_db_tool_name	
	Indicates the tool name of the tool that created the restored db on the disk database Type: string Default: "" Edit: Yes Reference: read_db_tool_name
read_db_version	
	Indicates the version of the tool that generates an on disk database Type: string Default: "" Edit: Yes Reference: read_db_version
read_def_check_mask_shifts	
	If on/bypass, will check for conflicts between DEF COMPONENT MASKSHIFT and MACRO MASKSHIFT values with the LEF FIXEDMASK and LAYERMASKSHIFT values in the technology and MACRO sections, error out if on and error without out if bypass Type: enum Enum Values: off on bypass Default: off Edit: Yes Reference: read_def_check_mask_shifts
read_physical_check_antenna	
	If true, LEF (or OA cells) will have process antenna consistency checks done for LEF MACRO PIN ANTENNA* data (or equivalent OA data) attached to their pins. Input pins with 0 gate-area, output pins with 0 diff-area, and inout pins with 0 gate-area or 0 diff-area values will generate warnings. If false, no consistency checking is done. Type: bool Default: true Edit: Yes Reference: read_physical_check_antenna
read_physical_check_colored_shape	
	Specify how to check and ignore colored shapes on colorless layers during reading lef or def data. If true, error out if a colored shape is on a colorless layer; if false, give error message and ignored the color. Type: bool Default: false Edit: Yes Reference: read_physical_check_colored_shape
read_physical_check_mask_shifts	
	If on/bypass, will check for conflicts between LEF FIXEDMASK and LAYERMASKSHIFT values in the technology section and MACRO section when reading in any LEF files, error out if on and error without out if bypass Type: enum Enum Values: off on bypass Default: off Edit: Yes Reference: read_physical_check_mask_shifts
read_physical_extend_cell_obs_shapes_under_trim	
	A list of routing layer. LEF OBS shapes on each <layer> that are overlapped partially by a shape on TRIMMETAL layer that trims the routing layer, will be extended in the preferred routing direction to be fully covered by the trim shape. This is a workaround for some advanced node libraries to avoid trim/OBS short violations. Type: string Default: "" Edit: Yes Reference: read_physical_extend_cell_obs_shapes_under_trim
read_physical_extend_polygon_cell_shapes	

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	A list of {{<layer> <value>} ...} pairs. LEF PIN/OBS shapes on each <layer> that are polygons (not simple rects), will be extended with OBS SPACING 0 rects by <value> (in microns) in the preferred routing direction. This is a workaround for some advanced node libraries to avoid spacing violations during routing. Type: string Default: "" Edit: Yes Reference: read_physical_extend_polygon_cell_shapes
read_physical_ignore_layer_names	
	Specify the layer list that needs to be ignored. Type: string Default: "" Edit: Yes Reference: read_physical_ignore_layer_names
read_physical_merge_multi_port_to_single_port	
	If true, multiple ports in a LEF PIN will be merged into single port. Type: bool Default: false Edit: Yes Reference: read_physical_merge_multi_port_to_single_port
read_physical_must_join_all_ports	
	If false, the LEF_MUSTJOINALLPORTS property on a LEF PIN auto-creates MUSTJOIN pins for each PORT. If true, the MUSTJOIN pins are not created, but the router must connect to every port with a signal wire or signal via that physically overlaps all the ports. Type: bool Default: true Edit: Yes Reference: read_physical_must_join_all_ports
read_write_lef_check_uncolored_shapes	
	Specifies whether to check for uncolored LEF PIN and OBS shapes on multi-mask routing layers when reading or writing LEF files. If the check is on, LEF read or write commands will error out if there are any uncolored routing layer shapes (cut layers are not checked). If set to pin_only, only check LEF PIN shapes and ignore LEF OBS shapes. Type: enum Enum Values: on off pin_only Default: off Edit: Yes Reference: read_write_lef_check_uncolored_shapes
reload_when_for_macro_cell	
	Reload when condition processing for macro library cell. Type: bool Default: true Edit: Yes Reference: reload_when_for_macro_cell
reorder_scan_add_scan_port_prefix	
	prefix of scan hport Type: string Default: "" Edit: Yes Reference: reorder_scan_add_scan_port_prefix
reorder_scan_clock_aware	
	Clock tree aware reorder Type: bool Default: false Edit: Yes Reference: reorder_scan_clock_aware
reorder_scan_comp_logic	

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	trace considering multiple input logic Type: bool Default: false Edit: Yes Reference: reorder_scan_comp_logic
reorder_scan_effort	
	Specifies effort level for reorder_scan Type: enum Enum Values: low medium high auto Default: auto Edit: Yes Reference: reorder_scan_effort
reorder_scan_enable_for_partition	
	enable scanReorder for design with partition Type: bool Default: false Edit: Yes Reference: reorder_scan_enable_for_partition
reorder_scan_keep_hinst_port_name	
	Specifies filename which contains HInsts Type: string Default: "" Edit: Yes Reference: reorder_scan_keep_hinst_port_name
reorder_scan_keep_hport	
	Maintains hierarchical ports without adding or deleting shifters Type: bool Default: true Edit: Yes Reference: reorder_scan_keep_hport
reorder_scan_prefer_horizontal	
	Makes horizontal scan chain connections if possible Type: bool Default: false Edit: Yes Reference: reorder_scan_prefer_horizontal
reorder_scan_prefer_vertical	
	Makes vertical scan chain connections if possible Type: bool Default: false Edit: Yes Reference: reorder_scan_prefer_vertical
reorder_scan_skip_mode	
	Specifies method for handling buffers and inverters Type: enum Enum Values: skip_none skip_buffer skip_floating_buffer Default: skip_none Edit: Yes Reference: reorder_scan_skip_mode
reorder_scan_swap_effort	

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	Specifies effort level for reorder_scan different chains swapping Type: enum Enum Values: low medium high Default: high Edit: Yes Reference: reorder_scan_swap_effort
report_library_message_summary	
	Enables library message summary report. Type: bool Default: true Edit: Yes Reference: report_library_message_summary
report_obj_display_limit	
	Controls the number of objects that will be displayed when using report_obj. If the value is 10, 'report_obj insts' will only display 10 inst objects with all their non-default attributes. Type: int Default: 1000 Edit: Yes Reference: report_obj_display_limit
report_pin_density_map_display_step	
	range step for displaying the pin density map Type: double Default: 0.05 Edit: Yes Reference: report_pin_density_map_display_step
report_pin_density_map_grid_size	
	specify the value (in row-height) of side of square grid for pin density calculations. Type: int Default: 10 Edit: Yes Reference: report_pin_density_map_grid_size
report_pin_density_map_grid_unit	
	specify the value (in micron) of side of square grid for pin density calculations. Type: double Default: 50.0 Edit: Yes Reference: report_pin_density_map_grid_unit
report_pin_density_map_threshold	
	threshold value over which the densities must be reported. Type: double Default: 0.5 Edit: Yes Reference: report_pin_density_map_threshold
report_place_density_map_grid_size	
	specify the value (in row-height) of side of square grid for density calculations. Type: int Default: 10 Edit: Yes Reference: report_place_density_map_grid_size
report_place_density_map_grid_unit	
	specify the value (in micron) of side of square grid for density calculations. Type: double Default: 50.0 Edit: Yes Reference: report_place_density_map_grid_unit

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report_place_density_map_ignore_block	
	Type: bool Default: false Edit: Yes Reference: report_place_density_map_ignore_block
report_place_density_map_ignore_filler	
	Type: bool Default: false Edit: Yes Reference: report_place_density_map_ignore_filler
report_place_density_map_threshold	
	threshold value over which the densities must be reported. Type: double Default: 0.75 Edit: Yes Reference: report_place_density_map_threshold
resize_blockages	
	Short-cut for [get_db current_design .resize_blockages] Type: obj(resize_blockage)* Default: "" Edit: No
resize_floorplan_cong_aware	
	resizes and shifts the floorplan objects by estimating the congestion for the floorplan and automatically deciding where to draw a resize line to avoid the congested area. Type: bool Default: false Edit: Yes Reference: resize_floorplan_cong_aware
resize_floorplan_honor_halo	
	honors placement halo and preserves the space between macros held by halo after the floorplan resize. Type: bool Default: true Edit: Yes Reference: resize_floorplan_honor_halo
resize_floorplan_io_fix	
	fixes I/Os at current location. Type: bool Default: false Edit: Yes Reference: resize_floorplan_io_fix
resize_floorplan_io_move_with_edge	
	moves pins on the moved edge orthogonally to edge direction so pins stay on edge after movement. Type: bool Default: false Edit: Yes Reference: resize_floorplan_io_move_with_edge
resize_floorplan_io_relative	
	resize the space among I/Os proportionally. Type: bool Default: true Edit: Yes Reference: resize_floorplan_io_relative

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resize_floorplan_maintain_resource_ratio	
	Enable to maintain the resource ratio after resize_floorplan in shift-based mode. Type: bool Default: false Edit: Yes Reference: resize_floorplan_maintain_resource_ratio
resize_floorplan_relative	
	resizes the space among floorplan objects proportionally. Type: bool Default: false Edit: Yes Reference: resize_floorplan_relative
resize_floorplan_shift_based	
	shifts floorplan objects at appropriate location(s) without changing the proportional spacing. Type: bool Default: true Edit: Yes Reference: resize_floorplan_shift_based
resize_floorplan_shrink_fence	
	enables resized lines to go through fences and regions in floorplan. Type: bool Default: false Edit: Yes Reference: resize_floorplan_shrink_fence
resize_floorplan_snap_to_track	
	snaps resize values (shrink/expand) of the floorplan to a multiple integer of the metal layer pitch. Type: bool Default: true Edit: Yes Reference: resize_floorplan_snap_to_track
retime_async_reset	
	Should flops with asynchronous reset be retimed. Type: bool Default: true Edit: Yes Reference: retime_async_reset
retime_effort_level	
	Set effort level during retiming (high, medium(default), or low). Type: enum Enum Values: low medium high Default: medium Edit: Yes Reference: retime_effort_level
retime_optimize_reset	
	Determines whether the reset should be dropped if the computation results in a dont_care value. Type: bool Default: false Edit: Yes Reference: retime_optimize_reset
retime_verification_flow	

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	Control the annotation in the netlist for retiming verification. Type: bool Default: true Edit: Yes Reference: retime_verification_flow
route_add_antenna_inst_prefix	
	prefix for the diode insts added to fix antenna Type: string Default: "" Edit: Yes Reference: route_add_antenna_inst_prefix
route_adjust_auto_via_weight	
	adjust auto via weight for via swapping Type: bool Default: true Edit: Yes Reference: route_adjust_auto_via_weight
route_allow_inst_overlaps	
	don't do instance overlap check Type: bool Default: true Edit: Yes Reference: route_allow_inst_overlaps
route_allow_pin_as_feedthru	
	allow pin as feedthrough Type: string Default: true Edit: Yes Reference: route_allow_pin_as_feedthru
route_antenna_cell_name	
	specify the antenna diode cell names for antenna fixing Type: string Default: "" Edit: Yes Reference: route_antenna_cell_name
route_antenna_diode_insertion	
	insert antenna diode to fix antenna violation Type: bool Default: false Edit: Yes Reference: route_antenna_diode_insertion
route_blockages	
	Short-cut for [get_db current_design .route_blockages] Type: obj(route_blockage)* Default: "" Edit: No
route_concurrent_minimize_via_count_effort	
	specify the effort of concurrent via minimization Type: string Default: medium Edit: Yes Reference: route_concurrent_minimize_via_count_effort
route_connect_to_bumps	

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	connect to bumps in native NanoRoute in Innovus Type: <code>bool</code> Default: false Edit: Yes Reference: route_connect_to_bumps
route_design_high_freq_bus_mixed_layer	
	bus mixed layer {false true auto only} Type: <code>string</code> Default: false Edit: Yes Reference: route_design_high_freq_bus_mixed_layer
route_detail_add_passive_fill_only_on_layers	
	specify layers where passive fill will be added Type: <code>string</code> Default: "" Edit: Yes Reference: route_detail_add_passive_fill_only_on_layers
route_detail_allow_passive_fill_only_in_layers	
	specify layers where do passive fill only Type: <code>string</code> Default: false Edit: Yes Reference: route_detail_allow_passive_fill_only_in_layers
route_detail_antenna_eco_list_file	
	set file name for antenna eco list Type: <code>string</code> Default: "" Edit: Yes Reference: route_detail_antenna_eco_list_file
route_detail_auto_stop	
	control whether Nanoroute continues routing if there are many violations Type: <code>string</code> Default: true Edit: Yes Reference: route_detail_auto_stop
route_detail_check_mar_on_cell_pin	
	Enable mar checking on used cell pins Type: <code>bool</code> Default: false Edit: Yes Reference: route_detail_check_mar_on_cell_pin
route_detail_end_iteration	
	specify the last iteration pass in a detailed routing step Type: <code>string</code> Default: auto Edit: Yes Reference: route_detail_end_iteration
route_detail_fix_antenna	
	fix antenna violation by jumping metal layers Type: <code>enum</code> Enum Values: auto true false Default: auto Edit: Yes Reference: route_detail_fix_antenna

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route_detail_fix_antenna_on_secondary_pg_nets	
	fix antenna violation on sencondary pg nets Type: bool Default: false Edit: Yes Reference: route_detail_fix_antenna_on_secondary_pg_nets
route_detail_fix_antenna_with_gate_array_filler_mode	
	detect GA filler flow while doing antenna violation fixing Type: bool Default: false Edit: Yes Reference: route_detail_fix_antenna_with_gate_array_filler_mode
route_detail_merge_abutting_cut	
	merge two abutting cell cut shapes into a new one Type: bool Default: true Edit: Yes Reference: route_detail_merge_abutting_cut
route_detail_min_length_for_spread_wire	
	specify the min. length of a spreaded wire Type: string Default: 0 Edit: Yes Reference: route_detail_min_length_for_spread_wire
route_detail_min_length_for_widen_wire	
	specify the min. length of a widened wire Type: double Default: 1.0 Edit: Yes Reference: route_detail_min_length_for_widen_wire
route_detail_min_slack_for_opt_wire	
	specify the min slack for a net wire optimization Type: double Default: 0.0 Edit: Yes Reference: route_detail_min_slack_for_opt_wire
route_detail_no_taper_in_layers	
	Specifies the range of the layer where taper is not allowed Type: string Default: "" Edit: Yes Reference: route_detail_no_taper_in_layers
route_detail_no_taper_on_output_pin	
	forbid wire tapering at output pin Type: enum Enum Values: false true auto Default: false Edit: Yes Reference: route_detail_no_taper_on_output_pin
route_detail_on_grid_only	

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	route on grid only Type: string Default: none Edit: Yes Reference: route_detail_on_grid_only
route_detail_post_route_litho_repair	
	litho hotspot repair in post route stage Type: bool Default: false Edit: Yes Reference: route_detail_post_route_litho_repair
route_detail_post_route_spread_wire	
	spread wire in post route stage Type: string Default: auto Edit: Yes Reference: route_detail_post_route_spread_wire
route_detail_post_route_swap_via	
	to enforce post route via swapping mode in detail route command. Type: string Default: false Edit: Yes Reference: route_detail_post_route_swap_via
route_detail_post_route_via_pillar_effort	
	Set opt effort/range for opportunistic via pillar Type: string Default: high Edit: Yes Reference: route_detail_post_route_via_pillar_effort
route_detail_post_route_wire_widen	
	widen wire in post route stage Type: string Default: none Edit: Yes Reference: route_detail_post_route_wire_widen
route_detail_post_route_wire_widen_rule	
	specify the rule used to widen wire Type: string Default: "" Edit: Yes Reference: route_detail_post_route_wire_widen_rule
route_detail_postroute_via_priority	
	Set net priority for via swapping based on timing Type: string Default: auto Edit: Yes Reference: route_detail_postroute_via_priority
route_detail_search_and_repair	
	run search-and-repair step after the initial detailed routing Type: bool Default: true Edit: Yes Reference: route_detail_search_and_repair
route_detail_signoff_effort	

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	specify when to give up on irresolvable violations Type: string Default: high Edit: Yes Reference: route_detail_signoff_effort
route_detail_stub_routing_in_first_layer	
	to use Minimum routing in the first routing layer defined in the LEF Type: string Default: false Edit: Yes Reference: route_detail_stub_routing_in_first_layer
route_detail_use_multi_cut_via_effort	
	specify the effort to use multi-cut via during routing Type: string Default: low Edit: Yes Reference: route_detail_use_multi_cut_via_effort
route_diode_insertion_for_clock_nets	
	allow diode insertion on clock nets Type: bool Default: false Edit: Yes Reference: route_diode_insertion_for_clock_nets
route_disable_route_rule_on_via_pillar_to_special_net_wire	
	Disable non-default rules on via pillar to special net wire Type: string Default: false Edit: Yes
route_early_global_effort_level	
	Specifies the congestion effort level. Default is "standard", where congestion is more accurate and runtime is larger. If set to "medium", runtime will be much smaller and congestion report will have less accuracy. Setting to "low", it will be faster than "medium" however less accurate. Type: enum Enum Values: low medium standard Default: standard Edit: Yes Reference: route_early_global_effort_level
route_early_global_honor_partition_allow_feedthru	
	Honor partition fences with feedthrough for <list_of_ptn_cell_names>. Type: string Default: "" Edit: Yes Reference: route_early_global_honor_partition_allow_feedthru
route_early_global_honor_partition_fence	
	Honor partition fences for <list_of_ptn_cell_names>. Type: string Default: "" Edit: Yes Reference: route_early_global_honor_partition_fence
route_early_global_honor_partition_pin	
	Honor partition fences and single-entry constraint for <list_of_ptn_cell_names>. Type: string Default: "" Edit: Yes Reference: route_early_global_honor_partition_pin

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route_early_global_honor_partition_pin_guide	
	Honor partition pin guides. Default is true Type: bool Default: true Edit: Yes Reference: route_early_global_honor_partition_pin_guide
route_early_global_honor_power_domain	
	consider MSV constraints while routing Type: bool Default: false Edit: Yes Reference: route_early_global_honor_power_domain
route_early_global_num_tracks_per_clock_wire	
	Specify special number of tracks per clock wires. Type: int Default: 0 Edit: Yes Reference: route_early_global_num_tracks_per_clock_wire
route_early_global_reverse_direction_regions	
	Reverse routing direction in the given region on the specified layer-range. Example: "(x1 y1 x2 y2) M1:M2 (x3 y3 x4 y4) M3:M4 ..." Type: string Default: "" Edit: Yes Reference: route_early_global_reverse_direction_regions
route_early_global_route_bump_nets	
	Do not ignore special nets if they are connecting bumps Type: bool Default: true Edit: Yes
route_early_global_route_selected_net_only	
	Only route the nets which are selected in the DB. Type: bool Default: false Edit: Yes Reference: route_early_global_route_selected_net_only
route_early_global_secondary_pg	
	Specifies whether secondary PG pins needs to be routed or not Type: bool Default: false Edit: Yes Reference: route_early_global_secondary_pg
route_early_global_secondary_pg_max_fanout	
	Specifies the max fanout limit for each secondary PG subnet Type: int Default: no_value Edit: Yes Reference: route_early_global_secondary_pg_max_fanout
route_early_global_stripe_layer_range	
	Route secondary PG pins to access to the PG stripes in the given layer range. Type: string Default: "" Edit: Yes Reference: route_early_global_stripe_layer_range

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route_eco_ignore_existing_route	
	Ignore existing route in eco, mode: {all signal} Type: string Default: "" Edit: Yes Reference: route_eco_ignore_existing_route
route_enable_mixed_size_tsv_via	
	Enable mixed size tsv vias in routing Type: bool Default: false Edit: Yes
route_enable_route_rule_si_limit_length	
	use MAR wire length as NDR PRL requirement for metal spacing check Type: string Default: false Edit: Yes Reference: route_enable_route_rule_si_limit_length
route_enforce_route_rule_on_special_net_wire	
	enforce ndr rule on all special wire segments of specified nets Type: string Default: false Edit: Yes Reference: route_enforce_route_rule_on_special_net_wire
route_extra_via_enclosure	
	specify an extra via enclosure to use when connecting to block pins and special net wires Type: double Default: 0.0 Edit: Yes Reference: route_extra_via_enclosure
route_fix_clock_nets	
	set clock nets routing status to fixed or routed Type: bool Default: false Edit: Yes Reference: route_fix_clock_nets
route_high_freq_constraint_groups	
	only route specified constraint groups, legal values are {net match bus pair shield}, add "order" in front to control route order Type: string Default: "" Edit: Yes Reference: route_high_freq_constraint_groups
route_high_freq_match_report_file	
	specify match report file name Type: string Default: "" Edit: Yes Reference: route_high_freq_match_report_file
route_high_freq_num_reserved_layers	
	number of layers reserved for standard cell pin access Type: int Default: 1 Edit: Yes Reference: route_high_freq_num_reserved_layers

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route_high_freq_remove_floating_shield	
	remove floating shield segments for high frequency nets Type: bool Default: false Edit: Yes Reference: route_high_freq_remove_floating_shield
route_high_freq_search_repair	
	run search and repair to remove violations, legal value is one of {auto false true only} Type: string Default: auto Edit: Yes Reference: route_high_freq_search_repair
route_high_freq_shield_trim_length	
	specify minimum length of shielding wire to be kept Type: double Default: 0.0 Edit: Yes Reference: route_high_freq_shield_trim_length
route_honor_exclusive_region	
	honor exclusive region routing Type: bool Default: true Edit: Yes Reference: route_honor_exclusive_region
route_honor_power_domain	
	honor power domain routing Type: bool Default: false Edit: Yes Reference: route_honor_power_domain
route_ignore_antenna_top_cell_pin	
	ignore antenna check on block I/O pins Type: bool Default: true Edit: Yes Reference: route_ignore_antenna_top_cell_pin
route_ignore_follow_pin_shapes	
	Ignore follow pin via shapes Type: bool Default: false Edit: Yes Reference: route_ignore_follow_pin_shapes
route_interposer_allow_diagonal_trunk	
	allow_diagonal_trunk Type: string Default: "" Edit: Yes Reference: route_interposer_allow_diagonal_trunk
route_interposer_control_options	
	control options Type: string Default: "" Edit: Yes Reference: route_interposer_control_options

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route_interposer_interlayer_shielding_layers	
	interlayer shielding layers Type: string Default: "" Edit: Yes Reference: route_interposer_interlayer_shielding_layers
route_interposer_interlayer_shielding_nets	
	interlayer shielding nets Type: string Default: "" Edit: Yes Reference: route_interposer_interlayer_shielding_nets
route_interposer_interlayer_shielding_offsets	
	interlayer shielding offsets Type: string Default: "" Edit: Yes Reference: route_interposer_interlayer_shielding_offsets
route_interposer_interlayer_shielding_widths	
	interlayer shielding widths Type: string Default: "" Edit: Yes Reference: route_interposer_interlayer_shielding_widths
route_interposer_same_layer_shielding_net	
	same_layer_shielding_net Type: string Default: "" Edit: Yes Reference: route_interposer_same_layer_shielding_net
route_interposer_same_layer_shielding_width_spacing	
	same_layer_shielding_width_spacing Type: string Default: "" Edit: Yes Reference: route_interposer_same_layer_shielding_width_spacing
route_interposer_trunk_routing_layers	
	trunk_routing_layers Type: string Default: "" Edit: Yes Reference: route_interposer_trunk_routing_layers
route_interposer_trunk_routing_width_spacing	
	trunk_routing_width_spacing Type: string Default: "" Edit: Yes Reference: route_interposer_trunk_routing_width_spacing
route_number_fail_limit	
	set limit for number of fails Type: int Default: 0 Edit: Yes Reference: route_number_fail_limit

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route_number_thread	
	set the number of processors to be used in one workstation for multi-threading Type: int Default: 1 Edit: Yes Reference: route_number_thread
route_number_warning_limit	
	set limit for number of warnings Type: int Default: 0 Edit: Yes Reference: route_number_warning_limit
route_process_node	
	Specify the process node Type: string Default: "" Edit: Yes Reference: route_process_node
route_rc_extraction_corner	
	specific which RC extraction corner to use in routing Type: string Default: "" Edit: Yes Reference: route_rc_extraction_corner
route_relaxed_route_rule_spacing_to_power_ground_nets	
	relax the spacing requirement from NDR spacing for the layers Type: string Default: none Edit: Yes Reference: route_relaxed_route_rule_spacing_to_power_ground_nets
route_reserve_space_for_multi_cut	
	Reserves space to insert multicut vias in postroute stage. This option has to be set before routing. After routing with this parameter specified, you can add double-cut vias or larger overhang vias by using the "route_design -via_opt" command. For examples, see route_design. Type: bool Default: false Edit: Yes Reference: route_reserve_space_for_multi_cut
route_reverse_direction	
	reverse routing direction in area: (lx ly ux uy [bot_lyr : top_lyr]) ... , top_lyr and bot_lyr could be either layer name or layer routing id Type: string Default: "" Edit: Yes Reference: route_reverse_direction
route_route_clock_nets_first	
	route clock nets first Type: bool Default: true Edit: Yes Reference: route_route_clock_nets_first
route_rules	

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	All the route_rules. It is initialized from LEF NONDEFAULTRULES section or from OA equivalents. It includes the reserved name "default" for the implicit LEF default rule or the OA LEFDefaultRouteSpec. It also includes design specific route_rules from create_route_rule or read in from DEF or an OA cellview. Type: obj(route_rule)* Default: "" Edit: No
route_selected_net_only	
	route selected net only Type: bool Default: false Edit: Yes Reference: route_selected_net_only
route_shield_crosstie_offset	
	Specifies the offset in terms of number of tracks for adding crossties. The default is 0 for all layers. The syntax is 'layer_name:numTrack1 layer_name2:numTrack2... '. Type: string Default: "" Edit: Yes Reference: route_shield_crosstie_offset
route_shield_length_threshold	
	min route segment length below which there is no shielding needed Type: double Default: -1.0 Edit: Yes Reference: route_shield_length_threshold
route_shield_report_skip_status	
	generate report file on if a net is skipped for shielding Type: bool Default: false Edit: Yes Reference: route_shield_report_skip_status
route_shield_stripe_layer_range	
	specify the target layer range of stripes for tie net connection Type: string Default: "" Edit: Yes Reference: route_shield_stripe_layer_range
route_shield_tap_cell_insertion	
	insert tap cell for shielding Type: bool Default: false Edit: Yes Reference: route_shield_tap_cell_insertion
route_shield_tap_cell_name	
	specify the tap cell names for shielding Type: string Default: "" Edit: Yes Reference: route_shield_tap_cell_name
route_skip_analog	
	skip routing nets or pins marked + USE ANALOG in the DEF file Type: bool Default: false Edit: Yes Reference: route_skip_analog

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route_special_allow_non_preferred_direction_route	
	allow wrong way route Type: bool Default: false Edit: Yes Reference: route_special_allow_non_preferred_direction_route
route_special_avoid_over_core_row_layer	
	specify number of layers to avoid routing over core row area Type: string Default: max layer of standard cell pin Edit: Yes Reference: route_special_avoid_over_core_row_layer
route_special_block_pin_connect_ring_pin_corners	
	Extend the ring pins at the corners Type: bool Default: false Edit: Yes Reference: route_special_block_pin_connect_ring_pin_corners
route_special_block_pin_route_with_pin_width	
	Connect using pin width only Type: bool Default: false Edit: Yes Reference: route_special_block_pin_route_with_pin_width
route_special_connect_broken_core_pin	
	Break followpin for standard cell pin or obs Type: bool Default: false Edit: Yes Reference: route_special_connect_broken_core_pin
route_special_core_pin_ignore_obs	
	ignore specified blockage when connecting follow pin Type: enum Enum Values: none placement_blockage overlap_obs block_halo Default: none Edit: Yes Reference: route_special_core_pin_ignore_obs
route_special_core_pin_length	
	connect core pins of at least specified length Type: double Default: 0.0 Edit: Yes Reference: route_special_core_pin_length
route_special_core_pin_length_as_inst	
	connect core pins if their lengths are the same as the instances Type: bool Default: false Edit: Yes Reference: route_special_core_pin_length_as_inst
route_special_core_pin_max_via_scale	

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	Maximum via width and height at stripe crossover for standcell pins (%) Type: string Default: "" Edit: Yes Reference: route_special_core_pin_max_via_scale
route_special_core_pin_merge_limit	
	gaps smaller than this variable cause followpin rails to be joined Type: double Default: 0.0 Edit: Yes Reference: route_special_core_pin_merge_limit
route_special_core_pin_refer_to_follow_pin	
	M2 followpin generation uses follow pin wire Type: bool Default: false Edit: Yes Reference: route_special_core_pin_refer_to_follow_pin
route_special_core_pin_reference_macro	
	use specified macro as reference for followpin creation Type: string Default: "" Edit: Yes Reference: route_special_core_pin_reference_macro
route_special_core_pin_site_rail_width	
	follow pin width at layer for specified site Type: string Default: "" Edit: Yes Reference: route_special_core_pin_site_rail_width
route_special_core_pin_snap_to	
	snap M2 followpin referred to M1 pin to position Type: enum Enum Values: m1_pin opt_routing_track grid half_grid Default: m1_pin Edit: Yes Reference: route_special_core_pin_snap_to
route_special_core_pin_stop_route	
	Type: enum Enum Values: RowEnd CellPinEnd Default: RowEnd Edit: Yes Reference: route_special_core_pin_stop_route
route_special_endcap_as_core	
	route_special treat Endcap macro as Core macro Type: bool Default: false Edit: Yes Reference: route_special_endcap_as_core
route_special_extend_nearest_target	
	extend nearest target so that it can be connected Type: bool Default: false Edit: Yes Reference: route_special_extend_nearest_target

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route_special_jog_threshold_ratio	
	distance ratio between a jogging target and a straight target Type: double Default: 10.0 Edit: Yes Reference: route_special_jog_threshold_ratio
route_special_layer_non_preferred_direction_cost	
	layer cost of non-preferred direction Type: string Default: "" Edit: Yes Reference: route_special_layer_non_preferred_direction_cost
route_special_layer_preferred_direction_cost	
	layer cost of preferred direction Type: string Default: "" Edit: Yes Reference: route_special_layer_preferred_direction_cost
route_special_pad_pin_min_via_size	
	Minimum via width percent between pad and ring. Type: double Default: 20.0 Edit: Yes Reference: route_special_pad_pin_min_via_size
route_special_pad_pin_split	
	Split width and spacing for wide pad pins Type: string Default: "" Edit: Yes Reference: route_special_pad_pin_split
route_special_pad_ring_use_lef	
	Keep user LEF input for pad rings Type: bool Default: true Edit: Yes Reference: route_special_pad_ring_use_lef
route_special_pg_pin_as_signal	
	List of cell/pin information NOT extract from CPF Type: string Default: "" Edit: Yes Reference: route_special_pg_pin_as_signal
route_special_secondary_pin_max_gap	
	gaps bigger than this variable cause level shifter rail to break for them Type: double Default: 0.0 Edit: Yes Reference: route_special_secondary_pin_max_gap
route_special_secondary_pin_rail_width	
	width of followpin rail Type: double Default: 0.0 Edit: Yes Reference: route_special_secondary_pin_rail_width

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route_special_signal_pin_as_pg	
	signal pins as PG pins Type: bool Default: false Edit: Yes Reference: route_special_signal_pin_as_pg
route_special_split_long_via	
	Split vias longer than <threshold> into smaller vias with specified <step> and bottom/left end <offset> and vertical/horizontal <height> Type: string Default: 0 0 -1 -1 Edit: Yes Reference: route_special_split_long_via
route_special_target_number	
	number of target to connect Type: int Default: 0 Edit: Yes Reference: route_special_target_number
route_special_target_search_distance	
	target search distance Type: double Default: 0.0 Edit: Yes Reference: route_special_target_search_distance
route_special_time_limit	
	time limit (in seconds) that the router is allowed in routing each port Type: double Default: 0.0 Edit: Yes Reference: route_special_time_limit
route_special_via_connect_to_shape	
	Make via connection to specified shapes Type: enum Enum Values: padding ring stripe blockring blockpin coverpin noshape blockwire corewire followpin iowire Default: padding ring stripe blockring blockpin coverpin noshape blockwire corewire followpin iowire Edit: Yes Reference: route_special_via_connect_to_shape
route_special_via_through_to_closest_ring	
	A via should connect only to closest ring layer when multiple rings overlap Type: bool Default: false Edit: Yes Reference: route_special_via_through_to_closest_ring
route_special_welltap_as_endcap	
	route_special treat Welltap macro as Endcap macro Type: string Default: "" Edit: Yes Reference: route_special_welltap_as_endcap
route_strict_honor_route_rule	

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	Strictly enforce non-default rules Type: string Default: false Edit: Yes Reference: route_strict_honor_route_rule
route_strictly_honor_1d_routing	
	Strictly honor 1d routing Type: string Default: false Edit: Yes Reference: route_strictly_honor_1d_routing
route_stripe_layer_range	
	specify the target layer range of stripes for tie net connection Type: string Default: "" Edit: Yes Reference: route_stripe_layer_range
route_third_party_data	
	Allow third party data as input Type: bool Default: false Edit: Yes Reference: route_third_party_data
route_tieoff_to_shapes	
	specify the target special wire shapes or target instance pin for tie net connection: [auto stripe ring powergroundpin] Type: string Default: auto Edit: Yes Reference: route_tieoff_to_shapes
route_trial_max_print_ignored_pad_nets	
	Number of nets to be printed connecting a pad term to a fterm without geometry Type: int Default: 0 Edit: Yes Reference: route_trial_max_print_ignored_pad_nets
route_trim_pull_back_distance_from_boundary	
	route_trim_pull_back_distance_from_boundary {<layer>:<value> ...} Type: string Default: "" Edit: Yes Reference: route_trim_pull_back_distance_from_boundary
route_trunk_with_cluster_target_size	
	global control of trunk routing pattern for nets with TRUNK pattern attribute. 0: nets with TRUNK pattern will be routed as Steiner tree; 1: default, nets will be routed as traditional trunk pattern, i.e., each pin directly connects to SNET trunk separately; >1: define a proximate max cluster size (in number of pins), and nets will be routed as fishbone style, i.e., several pins in a column are clustered together, and then connects to SNET trunk. Type: int Default: 1 Edit: Yes Reference: route_trunk_with_cluster_target_size
route_types	
	All the route_types from create_route_type command. Type: obj(route_type)* Default: "" Edit: No

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route_use_auto_via	
	allow to use internal generated vias Type: string Default: auto Edit: Yes Reference: route_use_auto_via
route_via_weight	
	set weight for vias Type: string Default: "" Edit: Yes Reference: route_via_weight
route_with_eco	
	enable eco routing Type: bool Default: false Edit: Yes Reference: route_with_eco
route_with_litho_driven	
	enable the litho-driven routing Type: bool Default: false Edit: Yes Reference: route_with_litho_driven
route_with_si_driven	
	enable si driven routing Type: bool Default: false Edit: Yes Reference: route_with_si_driven
route_with_timing_driven	
	enable timing driven routing Type: bool Default: false Edit: Yes Reference: route_with_timing_driven
route_with_trim_metal	
	set cut metal dgrid for short pin extension generated Type: string Default: "" Edit: Yes Reference: route_with_trim_metal
route_with_via_in_pin	
	enclose via in pin shape Type: string Default: false Edit: Yes Reference: route_with_via_in_pin
route_with_via_only_for_block_cell_pin	
	enclose via in pin shape for macro cell pins Type: string Default: false Edit: Yes Reference: route_with_via_only_for_block_cell_pin

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route_with_via_only_for_stdcell_pin	
	disable planar access to standard cell pins Type: string Default: false Edit: Yes Reference: route_with_via_only_for_stdcell_pin
rows	
	Short-cut for [get_db current_design .rows] Type: obj(row)* Default: "" Edit: No
run_abstract_abstract_blockage_cut_around_pin	
	For std cell, we do not want cover obs on pin. So: If -input_cell_type {std}, setting the value to all metal and via layer name list. If -input_cell_type {block} or -input_cell_type {io}, setting the value to all via layer name list. Type: string Default: "" Edit: Yes Reference: run_abstract_abstract_blockage_cut_around_pin
run_abstract_antenna_connectivity	
	Specifies the layers for which connectivity is to be extracted. The syntax is {<layer1> <connect_layer> <layer2>}. Type: string Default: "" Edit: Yes Reference: run_abstract_antenna_connectivity
run_abstract_antenna_diffusion_geometry	
	Specifies diffusion layer, followed by expression to remove gate channel area from diffusion region. Type: string Default: "" Edit: Yes Reference: run_abstract_antenna_diffusion_geometry
run_abstract_antenna_gate_geometry	
	Specifies poly layer that forms a gate, and then the layer expression to derive the gate area. This option also triggers the creation of antenna models. Type: string Default: "" Edit: Yes Reference: run_abstract_antenna_gate_geometry
run_abstract_blockage_detailed_layers	
	Specifies the layers for which a detailed blockage model is to be created. The detailed blockage model generates blockages only where there are real shapes in the cell on the layer. This only applies to IO cells and block cells, it does not apply to standard cells. Type: string Default: "" Edit: Yes Reference: run_abstract_blockage_detailed_layers
run_abstract_boundary_layers	
	By default this list is empty, and the prBoundary is used. The prBoundary comes from an explicit GDS layer that is used to compute the LEF SIZE (and OVERLAP OBS shapes for rectilinear boundaries). If there is no explicit GDS prBoundary layer, the boundary can be computed from the bounding box of geometries on this list of layers. Note, this option is almost never correct for core standard cells. Type: string Default: "" Edit: Yes Reference: run_abstract_boundary_layers
run_abstract_cell_symmetry	

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	Specifies a property value for cell symmetry. The valid values for the property symmetry include: {R0 X Y R90 X Y X R90 Y R90 X Y R90}. Default {X Y} Type: string Default: "" Edit: Yes Reference: run_abstract_cell_symmetry
run_abstract_export_lef_version	
	Version of LEF to export. The default version is the same as the version of technology LEF file being imported. Type: enum Enum Values: 5.3 5.4 5.5 5.6 5.7 5.8 Default: 5.7 Edit: Yes Reference: run_abstract_export_lef_version
run_abstract_extract_layers_power	
	Specifies the layers through which the extractor extracts each power net. This is used to reduce the search space if abstract is taking too much run time for each cell. Usually not applicable to standard cells. For blocks it is common to only extract the power nets on the top one or two metal layers. Type: string Default: "" Edit: Yes Reference: run_abstract_extract_layers_power
run_abstract_extract_layers_signal	
	Specifies the layers through which the extractor extracts each net. This is used to reduce the search space if abstract is taking too much run time for each cell. Usually not applicable to standard cells. Type: string Default: "" Edit: Yes Reference: run_abstract_extract_layers_signal
run_abstract_extract_pin_layers_power	
	Determines which of the shapes found by the extractor should be turned into pins. This argument is valid only if the associated layer, geometry specification and connectivity have been provided in the technology LEF. Usually not applicable to standard cells. Type: string Default: "" Edit: Yes Reference: run_abstract_extract_pin_layers_power
run_abstract_extract_pin_layers_signal	
	Determines which of the shapes found by the extractor should be turned into pins. This argument is valid only if the associated layer, geometry specification and connectivity have been provided in the technology LEF. Usually not applicable to standard cells. Type: string Default: "" Edit: Yes Reference: run_abstract_extract_pin_layers_signal
run_abstract_input_cell_type	
	Choose the cell type being translated with abstract. The tool adjusts its settings based on this selection. Optional. Type: enum Enum Values: std io block Default: std Edit: Yes Reference: run_abstract_input_cell_type
run_abstract_input_lef_tech_file	
	Input technology LEF file. This is used to get all the layer names, their types (routing, cut, etc.) and the layer order for extracting connectivity. Required. Type: string Default: "" Edit: Yes Reference: run_abstract_input_lef_tech_file

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run_abstract_input_stream_layer_map_file

Specify the file that maps GDS layer number / data type to LEF layer names in the Virtuoso format. Required.
For example, the gds file is not colorized, you could use the common virtuoso format gds layer map to stream gds file.
#layer name #purpose name #stream layer #stream data type

V0 drawing 35 0

V0 drawing 35 235

V0 drawing 35 236

M1 drawing 15 0

M1 drawing 15 235

M1 drawing 15 236

M1 drawing 15 239

For example, the gds file is colorized, you could specify the mask color in gds layer map to stream gds file.

#layer name #purpose name #stream layer #stream data type #photo mask color #color state

V0 drawing 35 0

V0 drawing 35 235 mask1Color unlocked

V0 drawing 35 236 mask2Color unlocked

M1 drawing 15 0

M1 drawing 15 235 mask1Color unlocked

M1 drawing 15 236 mask2Color unlocked

M1 drawing 15 239 mask3Color unlocked

Type: [string](#)

Default: ""

Edit: Yes

Reference: [run_abstract_input_stream_layer_map_file](#)

run_abstract_keep_output_files

Do not clean up temporary files/libraries that are created in the process after command finishes.

Type: [bool](#)

Default: false

Edit: Yes

Reference: [run_abstract_keep_output_files](#)

run_abstract_pins_analog_names

Identifies analog pin names based on label strings in the Abstract to set the LEF "USE ANALOG" property. Enter a list of possible names for the analog pins. This can be a list of regular expressions, each separated by a space.

Type: [string](#)

Default: ""

Edit: Yes

Reference: [run_abstract_pins_analog_names](#)

run_abstract_pins_clock_names

Identifies clock pin names based on label strings in the Abstract to set the LEF "USE CLOCK" property. Enter a list of possible names for the clock pins. This can be a list of regular expressions, each separated by a space.

Type: [string](#)

Default: ""

Edit: Yes

Reference: [run_abstract_pins_clock_names](#)

run_abstract_pins_ground_names

Identifies ground pins based on label strings in the Abstract to set the LEF "USE GROUND" property. Enter a list of possible names for the ground pins. This can be a list of regular expressions, each separated by a space.

Type: [string](#)

Default: ""

Edit: Yes

Reference: [run_abstract_pins_ground_names](#)

run_abstract_pins_output_names

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	Identifies output pins based on label strings in the GDS to set the LEF direction to OUTPUT. Enter a list of regular expressions separated by spaces. Type: string Default: "" Edit: Yes Reference: run_abstract_pins_output_names
run_abstract_pins_power_names	
	Identifies power pin names based on label strings in the Abstract to set the LEF "USE POWER" property. Enter a list of possible names for the power pins. This can be a list of regular expressions, each separated by a space. Type: string Default: "" Edit: Yes Reference: run_abstract_pins_power_names
run_abstract_pins_text_pin_map	
	Maps text labels to pins on nets. Specify the layer purpose pairs based on which you want the abstract generator to search for geometry of any given layer-purpose pair with a text label. Type: string Default: "" Edit: Yes Reference: run_abstract_pins_text_pin_map
run_abstract_pre_abstract_script	
	a skill file that will be executed in abstract before the auto-generated script is executed. Optional. Type: string Default: "" Edit: Yes Reference: run_abstract_pre_abstract_script
run_abstract_selected_cells	
	Select cells to create abstract for using regex style expressions. Optional. Type: string Default: "" Edit: Yes Reference: run_abstract_selected_cells
run_abstract_site_name	
	Site to be used for macros. This site will be added to all the cells. Type: string Default: "" Edit: Yes Reference: run_abstract_site_name
run_abstract_verbose	
	Verbose mode with extra print. Type: bool Default: false Edit: Yes Reference: run_abstract_verbose
safety_dcls_isolate_halo_type	
	The type of isolation halo required for dcls (dual core lock step) groups. Supported values: Type: enum Enum Values: soft hard Default: soft Edit: No Reference: safety_dcls_isolate_halo_type
safety_dcls_isolate_type	

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	The type of isolation required for dcls (dual core lock step) groups. Supported values: Type: enum Enum Values: soft hard Default: soft Edit: No Reference: safety_dcls_isolate_type
safety_dcls_net2net_spacing	
	Specifies the net to net spacing threshold of dcls groups. Type: float Default: no_value Edit: Yes Reference: safety_dcls_net2net_spacing
safety_dcls_route_types	
	The type of nets on which to enforce routing rules for dcls (dual core lock step) groups. Supported values are: - internal: this net's pins are all logically inside a single dcls group. This net cannot route outside of the group boundary. - interface: this net has pins logically inside a dcls group, and pins logically outside of that group AND logically outside of it's exclusive_groups. This net cannot route over the boundary of it's exclusive_groups. - common: this net has pins logically inside two or more dcls exclusive_groups. This net can route inside the boundary of those groups to complete the connections, but a violation should be reported. - top: this net's pins are all logically outside of dcls groups. This net can route over the boundary of any dcls group, but cannot route over the boundary of two or more exclusive_groups. Type: enum Enum Values: internal interface common top Default: "" Edit: No Reference: safety_dcls_route_types
safety_dmr_isolate_clock	
	The type of isolation required for the clock signals driving dmr (dual modular redundancy) flops. Supported values: Type: enum Enum Values: none split_network split_net Default: none Edit: No
safety_dmr_isolate_reset	
	The type of isolation required for the reset signals driving dmr (dual modular redundancy) flops. Supported values: Type: enum Enum Values: none split_network split_net Default: none Edit: No
safety_dmr_isolate_scan_enable	
	The type of isolation required for the scan_enable signals driving dmr (dual modular redundancy) flops. Supported values: Type: enum Enum Values: none split_network split_net Default: none Edit: No
safety_dmr_spacing	
	This is the radial spacing in any direction that should be maintained between the 2 flops in a dmr (dual modular redundancy) group. The value is measured in microns. Attributes "safety_dmr_spacing_x safety_dmr_spacing_y" and "safety_dmr_spacing" are mutually exclusive. Type: coord Default: 0.0 Edit: No
safety_dmr_spacing_x	

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	This is the horizontal spacing that should be maintained between the 2 flops in a dmr (dual modular redundancy) group. The value is measured in microns. Only one of either "safety_dmr_spacing_x" or "safety_dmr_spacing_y" needs to be met. Attributes "safety_dmr_spacing_x safety_dmr_spacing_y" and "safety_dmr_spacing" are mutually exclusive. Type: coord Default: 0.0 Edit: No
safety_dmr_spacing_y	
	This is the vertical spacing that should be maintained between the 2 flops in a dmr (dual modular redundancy) group. The value is measured in microns. Only one of either "safety_dmr_spacing_x" or "safety_dmr_spacing_y" needs to be met. Attributes "safety_dmr_spacing_x safety_dmr_spacing_y" and "safety_dmr_spacing" are mutually exclusive. Type: coord Default: 0.0 Edit: No
safety_flow_enable	
	Master switch to enable the implementation and analysis of the safety-mechanisms defined in the USF file.. Type: bool Default: false Edit: No Reference: safety_flow_enable
safety_midas_enable	
	Enables USF (Unified Safety Format) technology from Midas. Supported values: Type: enum Enum Values: none fmeda physical Default: none Edit: Yes Reference: safety_midas_enable
safety_ser_cells	
	The list of base_cells allowed for those insts with an ser (soft error resilience) safety mechanism. This root attribute is for designs without power_domains. Type: obj(base_cell)* Default: "" Edit: No Reference: safety_ser_cells
safety_ser_type	
	The type of ser (soft error resilience) safety mechanism. Supported values: Type: enum Enum Values: hard soft Default: hard Edit: No Reference: safety_ser_type
safety_tmr_isolate_clock	
	The type of isolation required for the clock signals driving tmr (triple modular redundancy) flops. Supported values: Type: enum Enum Values: none split_net split_network Default: none Edit: No Reference: safety_tmr_isolate_clock
safety_tmr_isolate_reset	

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	The type of isolation required for the reset signals driving tmr (triple modular redundancy) flops. Supported values: Type: enum Enum Values: none split_network split_net Default: none Edit: No Reference: safety_tmr_isolate_reset
safety_tmr_isolate_scan_enable	
	The type of isolation required for the scan_enable signals driving tmr (triple modular redundancy) flops. Supported values: Type: enum Enum Values: none split_network split_net Default: none Edit: No Reference: safety_tmr_isolate_scan_enable
safety_tmr_spacing	
	This is the radial spacing in any direction that should be maintained between the 3 flops in a tmr (triple modular redundancy) group. The value is measured in microns. Attributes "safety_tmr_spacing_x safety_tmr_spacing_y" and "safety_tmr_spacing" are mutually exclusive. Type: coord Default: 0.0 Edit: No Reference: safety_tmr_spacing
safety_tmr_spacing_x	
	This is the horizontal spacing that should be maintained between the 3 flops in a tmr (triple modular redundancy) group. The value is measured in microns. Only one of either "safety_tmr_spacing_x" or "safety_tmr_spacing_y" needs to be met. Attributes "safety_tmr_spacing_x safety_tmr_spacing_y" and "safety_tmr_spacing" are mutually exclusive. Type: coord Default: 0.0 Edit: No Reference: safety_tmr_spacing_x
safety_tmr_spacing_y	
	This is the vertical spacing that should be maintained between the 3 flops in a tmr (triple modular redundancy) group. The value is measured in microns. Only one of either "safety_tmr_spacing_x" or "safety_tmr_spacing_y" needs to be met. Attributes "safety_tmr_spacing_x safety_tmr_spacing_y" and "safety_tmr_spacing" are mutually exclusive. Type: coord Default: 0.0 Edit: No Reference: safety_tmr_spacing_y
safety_tmr_well_tap_cells	
	The well-tap cells to be placed next to the left and right edges of tmr (triple modular redundancy) flops. This root attribute is for designs without power_domains. Type: obj(base_cell)* Default: "" Edit: No Reference: safety_tmr_well_tap_cells
scan_chains	
	List of all scan_chains Type: obj(scan_chain)* Default: "" Edit: No
scan_segments	
	List of all scan_segments Type: obj(scan_segment)* Default: "" Edit: No
script_begin	

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	Specifies the list of alias names of the pragma 'script_script'. Type: string Default: dc_script_begin dc_tcl_script_begin script_begin Edit: Yes Reference: script_begin
script_end	
	Specifies the list of alias names of the pragma 'script_end'. Type: string Default: dc_script_end dc_tcl_script_end script_end Edit: Yes Reference: script_end
script_search_path	
	The variable provides a set of directories for the software to search for files that sourced using the Tcl 'source' command. The software will search in each search path directory for the specified file. Type: string Default: . Edit: Yes Reference: script_search_path
selected	
	The list of currently selected objects. It can be set to a list of objects as long as they are selectable. Type: string Default: "" Edit: Yes Reference: selected
set_db_verbose	
	Specifies additional messages are issued describing what attribute is being set on what object. Type: bool Default: false Edit: Yes Reference: set_db_verbose
setup_views	
	Returns the information about the setup analysis views in the design. Type: obj(analysis_view)* Default: "" Edit: No
si_accumulated_small_aggressor_factor	
	Specifies the multiplication factor that controls the effect of the cap mode accumulated small attacker. Type: double Default: 1.0 Edit: Yes Reference: si_accumulated_small_aggressor_factor
si_accumulated_small_aggressor_mode	
	Specifies the multiplication factor that controls the effect of the cap mode accumulated small attacker. Type: enum Enum Values: cap current zero_mean numerical Default: "" Edit: Yes Reference: si_accumulated_small_aggressor_mode
si_accumulated_small_aggressor_threshold	
	Specifies whether to use the current-matching based method for accumulated small attackers. Type: double Default: 10.01 Edit: Yes Reference: si_accumulated_small_aggressor_threshold

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si_aggressor_alignment	
	Specifies the attacker alignment. Type: enum Enum Values: path path_overlap timing_aware_edge input_arrival Default: path Edit: Yes Reference: si_aggressor_alignment
si_clock_synchronicity	
	Specifies the value above which the clock delta delay should be calculated. Type: enum Enum Values: asynchronous synchronous Default: synchronous Edit: Yes Reference: si_clock_synchronicity
si_delay_clock_delta_threshold	
	Specifies the value above which the clock delta delay should be calculated. Type: double Default: -1.0 Edit: Yes Reference: si_delay_clock_delta_threshold
si_delay_delta_annotation_mode	
	Calculates the delay for each arc (cell arc and net arc) in a path. Type: enum Enum Values: lumpedOnNet arc Default: arc Edit: Yes Reference: si_delay_delta_annotation_mode
si_delay_delta_threshold	
	delta delay analysis threshold Type: double Default: -1.0 Edit: Yes Reference: si_delay_delta_threshold
si_delay_double_switching	
	Specifies type for which double switching analysis will happen Type: enum Enum Values: none clock data_sequential all Default: none Edit: Yes Reference: si_delay_double_switching
si_delay_enable_double_clocking_check	
	Enables double clocking check in AAE. Type: enum Enum Values: true false Default: false Edit: Yes Reference: si_delay_enable_double_clocking_check
si_delay_enable_logical_correlation	
	Enables the AAE-SI logical correlation. Type: enum Enum Values: true false Default: false Edit: Yes Reference: si_delay_enable_logical_correlation
si_delay_enable_report	

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	Enables SI delay reports. Type: bool Default: false Edit: Yes Reference: si_delay_enable_report
si_delay_separate_on_data	
	Separates the delta delay on data (delta delay is always separated for clock). Type: bool Default: false Edit: Yes Reference: si_delay_separate_on_data
si_enable_bus_attacker_correlation	
	Type: bool Default: false Edit: Yes Reference: si_enable_bus_attacker_correlation
si_enable_drv_with_delta_slew	
	Enables SI slews reporting. Type: bool Default: false Edit: Yes Reference: si_enable_drv_with_delta_slew
si_enable_glitch_overshoot_undershoot	
	Enables overshoot/undershoot analysis and reporting. Type: bool Default: false Edit: Yes Reference: si_enable_glitch_overshoot_undershoot
si_enable_glitch_propagation	
	Enables glitch propagation and driver weakening through single stage cells. Type: bool Default: false Edit: Yes Reference: si_enable_glitch_propagation
si_enable_glitch_propagation_spice_deck	
	Enables glitch propagation in spice deck. Type: bool Default: false Edit: Yes Reference: si_enable_glitch_propagation_spice_deck
si_enable_logical_correlation_for_bus	
	Type: bool Default: false Edit: Yes
si_enable_two_stage_driver_weakening	
	Type: bool Default: true Edit: Yes Reference: si_enable_two_stage_driver_weakening
si_enable_user_defined_bus_merging	

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	Enables bus merging feature - if a net is in two groups, then the groups are merged Type: bool Default: false Edit: Yes
si_enable_virtual_attacker_constituent_report	
	Type: bool Default: false Edit: Yes Reference: si_enable_virtual_attacker_constituent_report
si_glitch_accuracy_level	
	Specifies the accuracy levels for glitch analysis Type: int Default: 0 Edit: Yes Reference: si_glitch_accuracy_level
si_glitch_analysis_type	
	Type: enum Enum Values: source full_propagation Default: source Edit: Yes Reference: si_glitch_analysis_type
si_glitch_constrained_input_threshold_failure_point	
	Type: enum Enum Values: input both Default: input Edit: Yes Reference: si_glitch_constrained_input_threshold_failure_point
si_glitch_enable_dynamic_receiver_peak_limits	
	Type: bool Default: false Edit: Yes Reference: si_glitch_enable_dynamic_receiver_peak_limits
si_glitch_enable_report	
	Enables AAE SI glitch analysis. Type: bool Default: false Edit: Yes Reference: si_glitch_enable_report
si_glitch_input_full_propagation_threshold	
	Type: double Default: 0.75 Edit: Yes Reference: si_glitch_input_full_propagation_threshold
si_glitch_input_full_propagation_voltage_high_threshold	
	Type: double Default: 0.75 Edit: Yes Reference: si_glitch_input_full_propagation_voltage_high_threshold

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si_glitch_input_full_propagation_voltage_low_threshold	
	Type: double Default: 0.75 Edit: Yes Reference: si_glitch_input_full_propagation_voltage_low_threshold
si_glitch_input_threshold	
	Specifies the input glitch failure threshold value during AAE analysis. Type: double Default: 0.4 Edit: Yes Reference: si_glitch_input_threshold
si_glitch_input_voltage_high_threshold	
	Specifies the input glitch vh failure threshold value during AAE analysis. Type: double Default: 0.4 Edit: Yes Reference: si_glitch_input_voltage_high_threshold
si_glitch_input_voltage_low_threshold	
	Specifies the input glitch vl failure threshold value during AAE analysis. Type: double Default: 0.4 Edit: Yes Reference: si_glitch_input_voltage_low_threshold
si_glitch_receiver_clock_peak_limit	
	Defines the glitch check limit for clock inputs in receiver peak mode. This parameter is specified as a ratio of vdd. Type: double Default: 0.05 Edit: Yes Reference: si_glitch_receiver_clock_peak_limit
si_glitch_receiver_latch_peak_limit	
	Defines the glitch check limit for latch inputs in receiver peak mode. This parameter is specified as a ratio of vdd. Type: double Default: 0.1 Edit: Yes Reference: si_glitch_receiver_latch_peak_limit
si_glitch_receiver_peak_limit	
	Defines the receiver peak/glitch peak limit at all nodes. The value specified for this parameter is a ratio of vdd. Type: double Default: 0.15 Edit: Yes Reference: si_glitch_receiver_peak_limit
si_individual_aggressor_clock_threshold	
	Sets the glitch tolerance for individual attacker nets on victim clock nets. Type: double Default: 0.015 Edit: Yes Reference: si_individual_aggressor_clock_threshold
si_individual_aggressor_simulation_filter	
	Enables electrical filtering based on attackers' simulated glitch peaks. Type: bool Default: true Edit: Yes Reference: si_individual_aggressor_simulation_filter

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si_individual_aggressor_threshold	
	AAE virtual attacker threshold Type: double Default: 0.015 Edit: Yes Reference: si_individual_aggressor_threshold
si_max_bus_attacker_count	
	Max number of bus attacker per victim net Type: int Default: 999 Edit: Yes
si_max_virtual_attacker_constituents	
	Type: int Default: 5 Edit: Yes Reference: si_max_virtual_attacker_constituents
si_nonlinear_aggressor_slew	
	Uses a non-linear attacker. If set to false , it uses a linear attacker. Type: enum Enum Values: true false Default: true Edit: Yes Reference: si_nonlinear_aggressor_slew
si_num_iteration	
	Specifies the maximum number of timing window iterations that should be performed during SI delay analysis. Type: int Default: 2 Edit: Yes Reference: si_num_iteration
si_pessimistic_mode	
	Pessimistic Mode Settings Type: enum Enum Values: advanced advanced_node reduced_pessimism increased_pessimism high_pessimism Default: advanced Edit: Yes Reference: si_pessimistic_mode
si_reselection	
	Specifies the SI reselection criteria during timing window iterations. Type: enum Enum Values: delta_delay slack xcap_ratio Default: slack Edit: Yes Reference: si_reselection
si_reselection_delay_threshold	
	Allows to change the delay reselection criteria. Type: double Default: 1e-11 Edit: Yes Reference: si_reselection_delay_threshold
si_reselection_hold_slack	

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	Specifies the slack reselection threshold for hold. Type: double Default: -1e-12 Edit: Yes Reference: si_reselection_hold_slack
si_reselection_setup_slack	
	Specifies the slack reselection threshold for setup. Type: double Default: -1e-12 Edit: Yes Reference: si_reselection_setup_slack
si_secondary_attacker_decoupling_factor	
	Specifies decoupling factor for secondary aggressors. Type: double Default: 1.0 Edit: Yes Reference: si_secondary_attacker_decoupling_factor
si_skip_noise_model_check	
	Specifies a list of pins for which noise model check will not take place. Type: string Default: "" Edit: Yes Reference: si_skip_noise_model_check
si_skip_timing_window	
	Specifies the list of nets for which timing window is to be skipped. Type: string Default: "" Edit: Yes Reference: si_skip_timing_window
si_switch_probability	
	Sets the attacker switching probability. Type: double Default: 0.3 Edit: Yes Reference: si_switch_probability
si_unconstrained_net_use_infinite_timing_window	
	Specifies that unconstrained nets will have an infinite timing window. Type: enum Enum Values: true false Default: true Edit: Yes
si_use_infinite_timing_window	
	Type: enum Enum Values: true false Default: false Edit: Yes Reference: si_use_infinite_timing_window
si_view_specific_settings	
	Specifies a global variable value or macro settings for particular view using {viewName:eng_global macro:value} Type: string Default: "" Edit: Yes
si_virtual_inter_die_attacker_slew	

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	This setting specifies the virtual inter die attacker slew value, default is 25ps Type: double Default: 2.5e-11 Edit: Yes
sites	
	All the sites defined in LEF or OA. Type: obj(site)* Default: "" Edit: No
skew_groups	
	list of skew_group Type: obj(skew_group)* Default: "" Edit: No Reference: skew_groups
socv_library	
	To specify SOCV library. Type: string Default: "" Edit: Yes Reference: socv_library
soft_stack_size_limit	
	Soft stacksize limit in mbytes. When the tool is launched, auto-enlarge the soft stacksize limit to 0.2%RAM. If 0.2%RAM is larger than hard limit, set to the hard limit. Type: string Default: 8 Edit: Yes Reference: soft_stack_size_limit
source_continue_on_error	
	Enable/Disable 'source' to continue the script on TCL_ERROR. If true, the source command will ignore Tcl errors, and continue processing the script files. If false, a Tcl error will halt the script. Type: bool Default: false Edit: Yes Reference: source_continue_on_error
source_echo_filename	
	If this is true, and verbose 'source' command logging is off (e.g. source_verbose is false), any 'source' command will be logged with 'Sourcing <filename>'. This can be useful to find which source files are calling other source files, or to log all the startup Tcl files that are loaded even though verbose logging is off. Type 'man log_file' for more details. Type: bool Default: false Edit: Yes Reference: source_echo_filename
source_verbose	
	Writes out each command inside the source script to the log and cmd files. Type 'man log_file' for more details. Type: bool Default: true Edit: Yes Reference: source_verbose
source_verbose_line_length_limit	

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	Limits the command printed to stdout and the .log file to the specified number of characters for verbose script source. This is normally only needed if you log commands "post Tcl expansion" when long Tcl lists may occur in the command args. Type 'man log_file' for more details. Type: int Default: -1 Edit: Yes Reference: source_verbose_line_length_limit
st_launch_wait_time	
	Maximum waiting time in minute for super thread launching. Type: int Default: 10 Edit: Yes Reference: st_launch_wait_time
stack_via_rules	
	The stack_via_rules available in this design from the LEF or create_stack_via_rules. Type: obj(stack_via_rule)* Default: {} Edit: No
support_internal_pg_pins	
	Support internal power and ground pins. Type: bool Default: true Edit: Yes Reference: support_internal_pg_pins
support_master_slave_flop	
	Support master slave flop library cell. Type: bool Default: false Edit: Yes Reference: support_master_slave_flop
support_multi_seq_elements	
	Support library cell having multiple sequential elements. Type: bool Default: true Edit: Yes Reference: support_multi_seq_elements
support_multi_seq_scan_latch	
	Support scan latch library cell defined using 2 latches. Type: bool Default: false Edit: Yes Reference: support_multi_seq_scan_latch
support_ui_units	
	Support for ui_units_timing and ui_units_capacitance attributes. Type: bool Default: false Edit: Yes Reference: support_ui_units
syn_exact_match_sequential_async_control	
	Do not tie off (degenerate) asynchronous controls of sequential cells. Type: bool Default: false Edit: Yes Reference: syn_exact_match_sequential_async_control
syn_exact_match_sequential_sync_control	

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	Forces the synchronous inputs of flop-flops to be mapped to corresponding library pins where possible. Type: bool Default: false Edit: Yes Reference: syn_exact_match_sequential_sync_control
synthesis_off_command	
	Specifies the list of alias names of the 'synthesis_off' pragma. Type: string Default: translate_off synthesis_off Edit: Yes Reference: synthesis_off_command
synthesis_on_command	
	Specifies the list of alias names of the 'synthesis_on' pragma. Type: string Default: translate_on synthesis_on Edit: Yes Reference: synthesis_on_command
target_library	
	Target library for technology mapping. Type: string Default: "" Edit: Yes Reference: target_library
tcl_error_info	
	If user set the value to true, application can print following error info if meet a tcl error. Type: bool Default: false Edit: Yes Reference: tcl_error_info
tcl_partial_cmd_argument_matching	
	Use this switch to control command argument partial matching as well as enum value partial matching. When it's turned on, argument '-opt' will match '-option' if no other argument name begins with -opt. Default is 'quiet', which means to silently allow partial-matching when there's no ambiguity. 'warn' means allow partial-matching with a warning message. 'error' means to disable partial-matching, and error out. Type: enum Enum Values: quiet warn error Default: quiet Edit: Yes Reference: tcl_partial_cmd_argument_matching
tcl_report_header_with_value	
	If user set the value to true, print the Tcl variable value in the report header. Type: bool Default: false Edit: Yes Reference: tcl_report_header_with_value
tcl_return_display_length_limit	
	Limits the number of characters displayed to stdout and the .log file. The Tcl return value itself is not affected--only the display is truncated. A value of -1 indicates that there is no display limit. This is useful to limit the output to the display of any command that can return long lists of objects or values. Type: int Default: 10000 Edit: Yes Reference: tcl_return_display_length_limit
tech_db_units	

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	Database units per micron from LEF or OA techfile, unless init_min_dbu_per_micron is larger. Type: int Default: "" Edit: No Reference: tech_db_units
tech_finfet_grid_direction	
	An enum value indicating the direction of the finfet grid from the LEF FINFET statement or OA. Type: enum Enum Values: vertical horizontal Default: vertical Edit: Yes Reference: tech_finfet_grid_direction
tech_finfet_grid_offset	
	The finfet grid offset from 0 in microns from the LEF FINFET statement or OA. Type: coord Default: no_value Edit: Yes Reference: tech_finfet_grid_offset
tech_finfet_grid_pitch	
	Returns a single floating point number indicating the pitch of the FINFET grid as described in LEF. Type: coord Default: no_value Edit: Yes Reference: tech_finfet_grid_pitch
tech_inst_mask_shift_layers	
	Ordered list of layer pointers to layers that allow instance mask shifting. This is the list of layers that will show up in the DEF COMPONENTMASK statement for layers that can have mask values shifted in standard cells. Type: obj(layer)* Default: "" Edit: No Reference: tech_inst_mask_shift_layers
tech_mfg_grid	
	Manufacturing grid from LEF or OA techfile. Note, this is the default value for each layer, but it can be overridden for specific layers. In some technologies the mfg_grid for higher metal layers is coarser than for the lower layers. Type: coord Default: "" Edit: No Reference: tech_mfg_grid
techs	
	All of the tech objects in the DB.. Type: obj(tech)* Default: "" Edit: No
test_clock_domains	
	List of all test_clock_domains Type: obj(test_clock_domain)* Default: "" Edit: No
test_clocks	
	List of all test_clocks Type: obj(test_clock)* Default: "" Edit: No
test_signals	

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	List of all test_signals Type: obj(test_signal)* Default: "" Edit: No
texts	
	Short-cut to all the text objects in the current_design. Type: obj(text) Default: "" Edit: No
timing_all_registers_filter_clock_pins_by_clock	
	When enabled, and -clocks/-clocks_rise/-clocks_fall is specified along with -clock_pins to all_registers, command will filter and return only those clock_pins where atleast 1 clock from specified list arrives Type: bool Default: false Edit: Yes Reference: timing_all_registers_filter_clock_pins_by_clock
timing_all_registers_identify_macros_include_is_macro_cell	
	Enable to include cells as macro cells if is_macro_cell is true in library Type: bool Default: true Edit: Yes Reference: timing_all_registers_identify_macros_include_is_macro_cell
timing_all_registers_include_icg_cells	
	Controls whether to include ICG cells as register in all_registers -edge_triggered Type: bool Default: true Edit: Yes Reference: timing_all_registers_include_icg_cells
timing_allow_input_delay_on_clock_source	
	When set to true allows you to apply input delay constraints on a pin where a clock was previously asserted Type: bool Default: false Edit: Yes Reference: timing_allow_input_delay_on_clock_source
timing_analysis_aging	
	Aging Analysis Type: bool Default: false Edit: Yes
timing_analysis_aocv	
	AOCV Analysis Type: bool Default: false Edit: Yes Reference: timing_analysis_aocv
timing_analysis_async_checks	
	Async checks Type: enum Enum Values: async no_async async_only auto Default: auto Edit: Yes Reference: timing_analysis_async_checks
timing_analysis_case_analysis	

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	caseAnalysis Type: bool Default: true Edit: Yes Reference: timing_analysis_case_analysis
timing_analysis_check_type	
	Report Violations for hold, setup Type: enum Enum Values: setup hold Default: setup Edit: Yes Reference: timing_analysis_check_type
timing_analysis_clock_gating	
	clockGatingCheck Type: bool Default: true Edit: Yes Reference: timing_analysis_clock_gating
timing_analysis_clock_net_marking_mode	
	clkNetsMarking Type: enum Enum Values: before_constant_propagation after_constant_propagation Default: before_constant_propagation Edit: Yes Reference: timing_analysis_clock_net_marking_mode
timing_analysis_clock_propagation_mode	
	Timing arc attributes effect on timing analysis Type: enum Enum Values: sdc_control forced_ideal Default: sdc_control Edit: Yes Reference: timing_analysis_clock_propagation_mode
timing_analysis_clock_source_paths	
	clkSrcPath Type: bool Default: true Edit: Yes Reference: timing_analysis_clock_source_paths
timing_analysis_cpvr	
	Removes pessimism from clock paths that have a portion of the clock network in common between the clock source and clock destination paths Type: enum Enum Values: both none setup hold Default: none Edit: Yes Reference: timing_analysis_cpvr
timing_analysis_enable_inter_power_domain	
	Inter Power Domain Analysis Type: bool Default: false Edit: Yes
timing_analysis_enable_robustness_mode	
	Timing Robustness Analysis Type: bool Default: false Edit: Yes

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timing_analysis_engine	
	Statistical or Static or Rql Type: enum Enum Values: statistical static rql Default: static Edit: Yes Reference: timing_analysis_engine
timing_analysis_honor_active_logic_view	
	Honor Active Logic View Type: bool Default: false Edit: Yes Reference: timing_analysis_honor_active_logic_view
timing_analysis_multi_input_switching_mode	
	Specify MIS mode: 0,Simultaneous input switching is not considered 1,MIS analysis consider cells for which multiple inputs of a cell tied together (tied-inputs) 2,MIS analysis consider both tied-inputs and inputs of a cells switching independent to each other 3,derate based MIS analysis Type: int Default: 0 Edit: Yes Reference: timing_analysis_multi_input_switching_mode
timing_analysis_precision_ps	
	Control delay precision 0.1ps or 0.01ps Type: double Default: 0.01 Edit: Yes Reference: timing_analysis_precision_ps
timing_analysis_self_loops_paths_no_skew	
	Eliminates clock skew due to clock uncertainty for a path starting and ending at the same register. If the clock skew is not eliminated, the timing for such paths is pessimistic Type: bool Default: false Edit: Yes Reference: timing_analysis_self_loops_paths_no_skew
timing_analysis_socv	
	SOCV Analysis Type: bool Default: false Edit: Yes Reference: timing_analysis_socv
timing_analysis_type	
	Single or BCWC or OCV TimingAnalysis type Type: enum Enum Values: single best_case_worst_case ocv Default: best_case_worst_case Edit: Yes Reference: timing_analysis_type
timing_aocv_analysis_mode	
	Controls handling of AOCV analysis modes Type: enum Enum Values: launch_capture clock_only separate_data_clock combine_launch_capture Default: launch_capture Edit: Yes Reference: timing_aocv_analysis_mode
timing_aocv_chip_size	

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	Specifies the diagonal length of the chip, in microns Type: double Default: 1e+30 Edit: Yes Reference: timing_aocv_chip_size
timing_aocv_core_size	
	Specifies the diagonal length of the core area, in microns Type: double Default: 1e+30 Edit: Yes Reference: timing_aocv_core_size
timing_aocv_derate_mode	
	Controls the AOCV derating mode Type: enum Enum Values: aocv_multiplicative aocv_additive Default: aocv_multiplicative Edit: Yes Reference: timing_aocv_derate_mode
timing_aocv_slack_threshold	
	Analyze aocv slack based at specified threshold Type: double Default: 0.0 Edit: Yes
timing_aocv_stage_count_update_on_timing_reset	
	When set to true, the aocv stage counts are recalculated when timing is reset. Type: bool Default: false Edit: Yes Reference: timing_aocv_stage_count_update_on_timing_reset
timing_apply_check_derate_to_external_output_delay	
	Applies check derate to output external delay Type: bool Default: false Edit: Yes Reference: timing_apply_check_derate_to_external_output_delay
timing_apply_default_primary_input_assertion	
	When set to true, primary input and bidirectional ports that do not have an explicit arrival time specified are provided a default arrival time Type: bool Default: true Edit: Yes Reference: timing_apply_default_primary_input_assertion
timing_apply_exceptions_to_data_check_related_pin	
	When set to true, any false path assertion which blocks the data path to the related (-from) pin of the data-to-data check also causes the check to be disabled Type: bool Default: true Edit: Yes Reference: timing_apply_exceptions_to_data_check_related_pin
timing_apply_setup_hold_exceptions_to_data_check_related_pin	
	When set to false, application of path exception with only -setup/hold on reference path doesn't affect signal path Type: bool Default: true Edit: Yes Reference: timing_apply_setup_hold_exceptions_to_data_check_related_pin

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timing_cap_unit	
	If set, this value is passed to set_library_unit -cap to set the capacitance units used in timing library and .sdc files and timing reports. Legal values are 1pf and 1ff. Type: string Default: "" Edit: Yes Reference: timing_cap_unit
timing_case_analysis_for_icg_propagation	
	Determines whether constant propagation continues through integrated clock gating (ICG) cells Type: enum Enum Values: false require_seq_prop always case_only constant_only Default: false Edit: Yes Reference: timing_case_analysis_for_icg_propagation
timing_case_analysis_for_sequential_propagation	
	When set to true, calculates constants on the outputs of sequential elements Type: enum Enum Values: false true 0 1 case_only constant_only Default: false Edit: Yes Reference: timing_case_analysis_for_sequential_propagation
timing_case_analysis_propagation	
	Overrides sequential and ICG propagation behaviors. Type: bool Default: true Edit: Yes Reference: timing_case_analysis_propagation
timing_check_timing_report_all_checks	
	Enables reporting of all checks performed by check_timing Type: bool Default: false Edit: Yes Reference: timing_check_timing_report_all_checks
timing_check_timing_signal_level_high_to_low_threshold	
	Threshold for warnig about difference in signal level between hight_drv and low_rcv Type: double Default: 0.0 Edit: Yes Reference: timing_check_timing_signal_level_high_to_low_threshold
timing_check_timing_signal_level_low_to_high_threshold	
	Threshold for warnig about difference in signal level between low_drv and high_rcv Type: double Default: 0.0 Edit: Yes Reference: timing_check_timing_signal_level_low_to_high_threshold
timing_clock_phase_propagation	
	Lets you select appropriate clock phases at register clock pins when both positive and negative phases of the same clock signal are seen in the clock network Type: enum Enum Values: positive negative both Default: both Edit: Yes Reference: timing_clock_phase_propagation
timing_clock_pulse_width_sensitivity_checks_threshold	

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	Type: double Default: 1.0 Edit: Yes
timing_clock_source_paths_unconstrained_mark_clock_used_as_data	
	global is set to true, is_clock_used_as_data returns true for unconstrained clock source network Type: bool Default: true Edit: Yes Reference: timing_clock_source_paths_unconstrained_mark_clock_used_as_data
timing_clock_source_use_driving_cell	
	When set to false, the create_clock command for output pins of cells uses default slew instead of the propagated slew from the primary input ports Type: bool Default: true Edit: Yes Reference: timing_clock_source_use_driving_cell
timing_clock_uncertainty_from_to_precedence	
	Gives higher precedence to clock-to-clock uncertainty when set to true Type: bool Default: false Edit: Yes Reference: timing_clock_uncertainty_from_to_precedence
timing_collection_all_fanin_fanout_traversal_mode	
	Controls the way software processes netlist for all_fanin/all_fanout command processing Type: enum Enum Values: hierarchical flat Default: flat Edit: Yes Reference: timing_collection_all_fanin_fanout_traversal_mode
timing_collection_result_display_limit	
	Limits the number of objects of a collection to be displayed in the output report Type: int Default: 100 Edit: Yes Reference: timing_collection_result_display_limit
timing_collection_variable_assignment_compatibility	
	Enables printing names of output of get objects commands when set to false Type: bool Default: true Edit: Yes Reference: timing_collection_variable_assignment_compatibility
timing_conditions	
	Returns the information about the timing conditions in the design. Type: obj(timing_condition)* Default: "" Edit: No
timing_constraint_disable_min_max_input_delay_worst_casing	
	Used for disabling the override of -max value from -min value in set_input_delay. Type: bool Default: false Edit: Yes Reference: timing_constraint_disable_min_max_input_delay_worst_casing
timing_constraint_enable_detailed_report_invalid_begin_end_points	

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	Enables detailed reporting of invalid begin points/end points specified with set_false_path/set_multicycle_path constraints else only checks structural end points Type: bool Default: false Edit: Yes Reference: timing_constraint_enable_detailed_report_invalid_begin_end_points
timing_constraint_enable_drv_limit_override	
	Enables support of drv override flow Type: bool Default: true Edit: Yes Reference: timing_constraint_enable_drv_limit_override
timing_constraint_enable_logging	
	Enables dumping of constraint command in log file Type: bool Default: false Edit: Yes Reference: timing_constraint_enable_logging
timing_constraint_enable_report_invalid_begin_end_points	
	Enables reporting of invalid begin points/end points specified with set_false_path/set_multicycle_path constraints Type: bool Default: false Edit: Yes Reference: timing_constraint_enable_report_invalid_begin_end_points
timing_constraint_enable_search_path	
	Enable searching of constraint files in search_path directory variable Type: bool Default: false Edit: Yes Reference: timing_constraint_enable_search_path
timing_constraint_enable_separate_multicycle_data_checks	
	When set to true, disables multicycle path constraint application on data checks Type: bool Default: false Edit: Yes Reference: timing_constraint_enable_separate_multicycle_data_checks
timing_constraint_load_minimal_set_for_automated_eco	
	Enable/disable constraints while loading for opt signoff Type: bool Default: false Edit: Yes Reference: timing_constraint_load_minimal_set_for_automated_eco
timing_constraint_path_delay_exclude_check_delay_from_ignore_clock_latency	
	Excludes check delays from set_max/min_delay -ignore_clock_latency slack computation Type: bool Default: false Edit: Yes Reference: timing_constraint_path_delay_exclude_check_delay_from_ignore_clock_latency
timing_constraint_path_delay_exclude_io_delay_from_ignore_clock_latency	
	Excludes input/output delays from set_max/min_delay -ignore_clock_latency slack computation Type: bool Default: false Edit: Yes Reference: timing_constraint_path_delay_exclude_io_delay_from_ignore_clock_latency

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timing_constraint_path_delay_exclude_unconstrained_endpoints	
	When set to true, does not constrain unconstrained path with path delay. Type: bool Default: false Edit: Yes Reference: timing_constraint_path_delay_exclude_unconstrained_endpoints
timing_constraint_path_delay_include_clock_pin_endpoints	
	When set to true, considers register clock pins as endpoint's for path delay constraints applied upstream Type: bool Default: false Edit: Yes Reference: timing_constraint_path_delay_include_clock_pin_endpoints
timing_constraint_update_io_latency_averaging_mode	
	Controls the analysis to update io latency for rise and fall both or for rise only Type: enum Enum Values: rise_fall rise_only Default: rise_fall Edit: Yes Reference: timing_constraint_update_io_latency_averaging_mode
timing_constraint_warn_for_timing_derate_exceeding_max_limit	
	Sets the limit on derate value for giving the warning in case the larger derate is applied Type: double Default: 100.0 Edit: Yes Reference: timing_constraint_warn_for_timing_derate_exceeding_max_limit
timing_constraints_enable_path_exception_efficient_file_line_debug_mode	
	Reports the honored and ignored exceptions that were applicable on the path with the file name and line number with compression On Type: bool Default: false Edit: Yes
timing_context_apply_default_primary_input_assertion	
	propagates @(D) phase at block level for unmapped clocks Type: bool Default: false Edit: Yes Reference: timing_context_apply_default_primary_input_assertion
timing_context_apply_port_sdc_exceptions	
	honor path exceptions on block constraints in HC flow Type: bool Default: false Edit: Yes Reference: timing_context_apply_port_sdc_exceptions
timing_context_clock_mapping_check_waveform	
	Controls the waveform checking Type: bool Default: true Edit: Yes Reference: timing_context_clock_mapping_check_waveform
timing_context_clock_mapping_percentage_tolerance	
	Controls the tolerance value in clock mapping Type: double Default: 0.0 Edit: Yes Reference: timing_context_clock_mapping_percentage_tolerance

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timing_context_clock_phase_based_clock_mapping	
	Enable clock mapping on block data ports based on clock mapped on block clock ports Type: bool Default: false Edit: Yes Reference: timing_context_clock_phase_based_clock_mapping
timing_context_continue_on_invalid_module_list	
	When all specified modules/instances are invalid, continue with timing analysis without generating timing context Type: bool Default: false Edit: Yes Reference: timing_context_continue_on_invalid_module_list
timing_context_data_phase_based_clock_mapping	
	Enable clock mapping on block data ports based on clock mapped on other block data ports Type: bool Default: false Edit: Yes Reference: timing_context_data_phase_based_clock_mapping
timing_context_port_based_clock_mapping	
	Enable clock mapping based on block ports. Type: bool Default: false Edit: Yes Reference: timing_context_port_based_clock_mapping
timing_continue_on_error	
	When set to true, directs software to skip the error and continue processing when an error occurs during timing analysis Type: bool Default: false Edit: Yes Reference: timing_continue_on_error
timing_cpnr_enable_mismatch_transition_mode	
	Enables CPNR compatibility mode when transitions mismatch at the common pin Type: bool Default: true Edit: Yes Reference: timing_cpnr_enable_mismatch_transition_mode
timing_cpnr_opposite_edge_mean_scale_factor	
	Controls mean delay component for CPNR credit for opposite transition edges at common pin Type: double Default: 1.0 Edit: Yes Reference: timing_cpnr_opposite_edge_mean_scale_factor
timing_cpnr_opposite_edge_sigma_scale_factor	
	Controls sigma delay component for CPNR credit for opposite transition edges at common pin Type: double Default: 1.0 Edit: Yes Reference: timing_cpnr_opposite_edge_sigma_scale_factor
timing_cpnr_opposite_edge_sigma_scale_factor_cell	
	Controls cell sigma delay component for CPNR credit for opposite transition edges at common pin Type: double Default: 1.0 Edit: Yes Reference: timing_cpnr_opposite_edge_sigma_scale_factor_cell

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timing_cppr_opposite_edge_sigma_scale_factor_net	
	Controls net sigma delay component for CPPR credit for opposite transition edges at common pin Type: double Default: 1.0 Edit: Yes Reference: timing_cppr_opposite_edge_sigma_scale_factor_net
timing_cppr_propagate_thru_latches	
	When set to true, the cppr of a timing path crossing a latch (or latches) is calculated using the common pin of its origin flop and the capture device at the end of the path. Otherwise the usual segment-based cppr calculation is performed. This cppr setting is considered only when the latch thru analysis mode is enabled. Type: bool Default: false Edit: Yes Reference: timing_cppr_propagate_thru_latches
timing_cppr_remove_clock_to_data_pessimism	
	When set to true, removes clock reconvergence pessimism (CRP) for clock source paths Type: bool Default: false Edit: Yes Reference: timing_cppr_remove_clock_to_data_pessimism
timing_cppr_self_loop_mode	
	When set to true in case of self-loop paths, computes CPPR adjustment by taking the difference between early and late clock arrival time of the common point Type: bool Default: true Edit: Yes Reference: timing_cppr_self_loop_mode
timing_cppr_skip_clock_reconvergence	
	Specifies the branch point to use for computing clock path pessimism removal (CPPR) adjustment when there is reconvergence in the clock tree Type: bool Default: false Edit: Yes Reference: timing_cppr_skip_clock_reconvergence
timing_cppr_skip_clock_reconvergence_for_unmatched_clocks	
	When set to true, this enables CPPR branch point search in reconverging clock tree only when launching and capturing clocks are different. Type: bool Default: false Edit: Yes Reference: timing_cppr_skip_clock_reconvergence_for_unmatched_clocks
timing_cppr_threshold_ps	
	Specifies the maximum amount of pessimism that clock path pessimism removal (CPPR) analysis is allowed to leave in the path Type: double Default: 20.0 Edit: Yes Reference: timing_cppr_threshold_ps
timing_cppr_transition_sense	
	Specifies the transition sense of the launching and capturing clocks at the common node, to calculate clock path pessimism removal (CPPR) Type: enum Enum Values: normal same_transition same_transition_expanded Default: normal Edit: Yes Reference: timing_cppr_transition_sense
timing_create_clock_default_propagated	

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	Enables clocks to be created in propagated mode Type: bool Default: false Edit: Yes Reference: timing_create_clock_default_propagated
timing_default_opcond_per_lib	
	When set to true, use default operating conditions for each lib Type: bool Default: true Edit: Yes Reference: timing_default_opcond_per_lib
timing_defer_mmmc_obj_updates	
	You can set the timing_defer_mmmc_object_updates global to true, so that the software allows a sequence of MMMC updates to be accumulated before new data is loaded and analyzed Type: bool Default: false Edit: Yes Reference: timing_defer_mmmc_obj_updates
timing_derate_aocv_dynamic_delays	
	This global controls whether or not AOCV derating factors are applied to the dynamic, SI-induced delay component or not. With a value of '1', the AOCV derating factor will be applied to both the static and dynamic components of the delay arc. When set to '0', the AOCV derate will only apply to the static component of the delay. Type: bool Default: true Edit: Yes Reference: timing_derate_aocv_dynamic_delays
timing_derate_aocv_reference_point	
	Allows specification of reference point for AOCV factors Type: int Default: 1 Edit: Yes Reference: timing_derate_aocv_reference_point
timing_derate_dynamic_compatibility	
	This global controls how the set_timing_derate factors that are not specified using either -static or -dynamic options are applied to the delay. When this global is set to '1' the behavior will be compatible with previous releases. In this mode, the static and dynamic components will be summed before applying derating. With a setting of '0', the static and dynamic components will be derated separately and then combined. A setting of '0' is equivalent to using separate derate assertions with the -dynamic and -static options explicitly specified. Type: bool Default: true Edit: Yes Reference: timing_derate_dynamic_compatibility
timing_derate_incremental_adjust_additive_mode	
	By default or with a setting of 'false', successive application of incremental derating using the -increment option (or deprecated -incremental_adjust option) will overwrite any existing derate value with the new value. When this global is set to 'true', successive application of the derates will be accumulated into a final derate value. Type: bool Default: false Edit: Yes Reference: timing_derate_incremental_adjust_additive_mode
timing_derate_incremental_multiply_accumulative_mode	
	Type: bool Default: false Edit: Yes Reference: timing_derate_incremental_multiply_accumulative_mode
timing_derate_negative_delay_backward_compatibility	

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	Type: bool Default: true Edit: Yes Reference: timing_derate_negative_delay_backward_compatibility
timing_derate_ocv_reference_point	
	Allows specification of reference point for OCV factors Type: int Default: 1 Edit: Yes Reference: timing_derate_ocv_reference_point
timing_derate_spatial_distance_unit	
	Set the distance unit. Type: enum Enum Values: default 1um 1nm Default: default Edit: Yes Reference: timing_derate_spatial_distance_unit
timing_derate_voltage_scaling_mode	
	Type: enum Enum Values: first_library snap_to_nearest interpolate Default: first_library Edit: Yes Reference: timing_derate_voltage_scaling_mode
timing_disable_black_box_endpoints	
	Ignore any boundary model endpoints on black box instances Type: bool Default: false Edit: Yes
timing_disable_bus_contention_check	
	Checks for setup and hold violations in three-state bus designs Type: bool Default: false Edit: Yes Reference: timing_disable_bus_contention_check
timing_disable_clock_period_checks	
	When set to true, disables timing model clock period checks during timing analysis Type: bool Default: false Edit: Yes Reference: timing_disable_clock_period_checks
timing_disable_constant_propagation_for_sequential_cells	
	When set to true, disables constant prop. across sequential cells if sequential_prop global is false Type: bool Default: false Edit: Yes Reference: timing_disable_constant_propagation_for_sequential_cells
timing_disable_drv_report_on_constant_nets	
	Disables DRV checks for nets having disabled constant propagation: Type: bool Default: false Edit: Yes Reference: timing_disable_drv_report_on_constant_nets

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timing_disable_floating_bus_check	
	When set to true, disables propagation of minimum delay through three state disable timing arcs and maximum delay through three state enable arcs. These checks are only valid during floating bus conditions Type: bool Default: false Edit: Yes Reference: timing_disable_floating_bus_check
timing_disable_genclk_combinational_blocking	
	Disables blocking of generated clocks with '-combinational' option which are downstream of another generated clock Type: bool Default: false Edit: Yes Reference: timing_disable_genclk_combinational_blocking
timing_disable_inferred_clock_gating_checks	
	When set to true, disables clock gating checks that are inferred on combinational elements in the clock path. Explicit clock gating checks that are described in the timing library are not affected by this global variable Type: bool Default: false Edit: Yes Reference: timing_disable_inferred_clock_gating_checks
timing_disable_inout_output_side_timing_checks	
	When set to false, timing checks on both the input and output sides of the bidirectional pin are analyzed Type: bool Default: false Edit: Yes Reference: timing_disable_inout_output_side_timing_checks
timing_disable_internal_inout_cell_paths	
	When set to false, enables internal bidirectional feedback paths that are completely contained in one instance Type: bool Default: true Edit: Yes Reference: timing_disable_internal_inout_cell_paths
timing_disable_internal_inout_net_arcs	
	When set to true, this global disables internal bidirectional feedback paths that span multiple instances Type: bool Default: true Edit: Yes Reference: timing_disable_internal_inout_net_arcs
timing_disable_lib_pulse_width_checks	
	When set to true, disables timing model pulse width checks during timing analysis Type: bool Default: false Edit: Yes Reference: timing_disable_lib_pulse_width_checks
timing_disable_library_data_to_data_checks	
	When set to true, this global disables data-to-data checks that are coded in the library as non-sequential timing checks Type: bool Default: false Edit: Yes Reference: timing_disable_library_data_to_data_checks
timing_disable_library_tieoffs	

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	Disables constant functions in a library Type: bool Default: false Edit: Yes Reference: timing_disable_library_tieoffs
timing_disable_netlist_constants	
	When set to true, ignores constants defined in the Verilog netlist Type: bool Default: false Edit: Yes Reference: timing_disable_netlist_constants
timing_disable_nochange_checks	
	When set to true, disables no change timing model checks during timing analysis Type: bool Default: false Edit: Yes Reference: timing_disable_nochange_checks
timing_disable_non_sequential_checks	
	When set to true, disables the timing arcs between any data-to-clock or clock-to-clock checks Type: bool Default: false Edit: Yes Reference: timing_disable_non_sequential_checks
timing_disable_output_as_clock_port	
	Controls clock to output port to be treated as data or clock irrespective of constraints set on the port Type: bool Default: false Edit: Yes Reference: timing_disable_output_as_clock_port
timing_disable_parallel_arcs	
	to control enabling/disabling both CTE's parallel arc reduction, and the related task compression on the AAE side. Setting this variable to 'true' should yield AAE task compression mode '2'. Setting it to 'false' will result in task compression mode '0' Type: bool Default: true Edit: Yes Reference: timing_disable_parallel_arcs
timing_disable_pulse_width_same_edge_si_cprr_mode	
	Enables SI CPPr credit up to the last divergent pin for pulse width checks for zero width pulse clocks Type: bool Default: true Edit: Yes Reference: timing_disable_pulse_width_same_edge_si_cprr_mode
timing_disable_report_header_info	
	Controls whether timing reports are generated using a common report header. Type: bool Default: false Edit: Yes Reference: timing_disable_report_header_info
timing_disable_retime_clock_path_slew_propagation	
	Controls whether analysis is performed on data or clock paths Type: bool Default: true Edit: Yes Reference: timing_disable_retime_clock_path_slew_propagation

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timing_disable_sdf_retain_arc_merging	
	Controls handling of retain arcs in SDF2.1 Type: bool Default: false Edit: Yes Reference: timing_disable_sdf_retain_arc_merging
timing_disable_skew_checks	
	When set to true, disables library skew checks for timing analysis Type: bool Default: false Edit: Yes Reference: timing_disable_skew_checks
timing_disable_test_signal_arc	
	When set to true, timing analysis will not analyze the signal arcs coming from or going to the test pin Type: bool Default: false Edit: Yes Reference: timing_disable_test_signal_arc
timing_disable_timing_model_latch_inferencing	
	Controls whether latch behavior is inferred for cell descriptions tagged with the Liberty timing_model_type attribute, including all values: abstracted, extracted, or qtm Type: bool Default: true Edit: Yes Reference: timing_disable_timing_model_latch_inferencing
timing_disable_tristate_disable_arcs	
	When set to true, disables all 0/1->Z transitions of tristate arcs during timing analysis Type: bool Default: false Edit: Yes Reference: timing_disable_tristate_disable_arcs
timing_disable_user_data_to_data_checks	
	When set to true, disables data-to-data checks that are created by the set_data_check command Type: bool Default: false Edit: Yes Reference: timing_disable_user_data_to_data_checks
timing_driving_cell_override_library	
	Controls the selection of library cells specified using the set_driving_cell command Type: bool Default: false Edit: Yes Reference: timing_driving_cell_override_library
timing_drv_analysis_ignore_latency_phases	
	Enables ignoring latency phases in drv analysis Type: bool Default: false Edit: Yes
timing_enable_aocv_slack_based	
	Enable aocv analysis using slack based method Type: bool Default: false Edit: Yes Reference: timing_enable_aocv_slack_based

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timing_enable_cap_fanout_in_copy_drv_assertions	
	Enables copying of max/min capacitance and max/min fanout assertions Type: bool Default: true Edit: Yes
timing_enable_case_analysis_conflict_warning	
	Controls to report case analysis conflict warnings to CTE_constant_mismatch.rpt file. Type: bool Default: true Edit: Yes Reference: timing_enable_case_analysis_conflict_warning
timing_enable_clock_phase_based_rise_fall_derating	
	Determines the interpretation of edge-specific derating factors for clock paths, specified using the set_timing_derate -rise/-fall parameters Type: bool Default: false Edit: Yes Reference: timing_enable_clock_phase_based_rise_fall_derating
timing_enable_clock_to_clock_clock_gating_check	
	When set to true, performs clock to clock gating checks at gating elements Type: bool Default: false Edit: Yes Reference: timing_enable_clock_to_clock_clock_gating_check
timing_enable_data_through_clock_gating	
	When set to false, blocks signals arriving on the enable of the clock-gating check Type: bool Default: true Edit: Yes Reference: timing_enable_data_through_clock_gating
timing_enable_db_attribute_query	
	Enables querying DB attribute through CTE commands Type: bool Default: true Edit: Yes
timing_enable_derating_for_pulse_width_checks	
	Applies check derating for pulse width checks Type: bool Default: false Edit: Yes Reference: timing_enable_derating_for_pulse_width_checks
timing_enable_early_late_data_slews_for_setuphold_mode_checks	
	When set to true, the global variable enables propagation of early and late slews on data paths. Delay calculation then uses these early and late slews to calculate timing checks. You can use this global in simultaneous setup and hold mode only. Type: bool Default: false Edit: Yes Reference: timing_enable_early_late_data_slews_for_setuphold_mode_checks
timing_enable_edge_based_path_adjust_group_mode	
	Enables transition-based set_path_adjust_groups. Type: bool Default: true Edit: Yes
timing_enable_genclock_divide_by_inherit_parent_duty_cycle	

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	Controls inheritance of duty cycle from master clock for generated clocks with divide by option Type: <code>bool</code> Default: false Edit: Yes Reference: timing_enable_genclk_divide_by_inherit_parent_duty_cycle
timing_enable_genclk_source_path_register_limit	
	Limits generated clock source latency path to traverse across one register only Type: <code>bool</code> Default: false Edit: Yes Reference: timing_enable_genclk_source_path_register_limit
timing_enable_generated_clock_edge_based_source_latency	
	Controls how the software chooses generated clock source latency paths Type: <code>bool</code> Default: true Edit: Yes Reference: timing_enable_generated_clock_edge_based_source_latency
timing_enable_get_obj_escaped_name_backward_compatible	
	Enables printing of escape characters in the get_object_name command output. Type: <code>bool</code> Default: false Edit: Yes Reference: timing_enable_get_obj_escaped_name_backward_compatible
timing_enable_get_objects_regexp_compatibility	
	Enables the -regexp option compatibility for get_* commands Type: <code>bool</code> Default: false Edit: Yes Reference: timing_enable_get_objects_regexp_compatibility
timing_enable_get_pins_of_lib_cell_pins	
	Enable get_pins -of lib cell pins Type: <code>bool</code> Default: true Edit: Yes
timing_enable_hier_context_unmapped_clock_analysis	
	For unmapped block clocks, enable timing analysis based on blok SDC constraints set on block ports. Type: <code>bool</code> Default: false Edit: Yes
timing_enable_hierarchical_get_nets_support	
	Enable the hierarchical segment support for get_nets. Type: <code>bool</code> Default: false Edit: Yes Reference: timing_enable_hierarchical_get_nets_support
timing_enable_individual_distance_inter_die_analysis	
	Enables individual inter die calculation in 3DIC analysis Type: <code>bool</code> Default: false Edit: Yes
timing_enable_latch_thru_mode	

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	Enable the latch thru analysis mode in which timing paths can propagate across latches, depending on their arrival time with respect to the latch transparency window. Type: bool Default: false Edit: Yes Reference: timing_enable_latch_thru_mode
timing_enable_latency_through_clock_gating	
	Controls propagation of latency phases across a clock gating element. Type: bool Default: true Edit: Yes Reference: timing_enable_latency_through_clock_gating
timing_enable_minimal_constraints_loading_for_opt_signoff	
	Enable/disable constraints while loading for opt signoff Type: bool Default: false Edit: Yes
timing_enable_mmmc_loop_breaking	
	When set to true, loop breaking is handled independently per analysis view Type: bool Default: true Edit: Yes Reference: timing_enable_mmmc_loop_breaking
timing_enable_multi_drive_net_reduction_with_assertions	
	Controls whether multi-drive net reduction (if enabled) will attempt to reduce multi-drive nets that also have only either/both set_annotated_delay or set_annotated_transitions assertions present. By default, any assertions present on different drivers of multi-drive nets will prevent reduction of the multi-drive net. Type: enum Enum Values: none all delay Default: none Edit: Yes Reference: timing_enable_multi_drive_net_reduction_with_assertions
timing_enable_multi_frequency_latch_analysis	
	Enables multi-frequency latch timing analysis of the latch time borrowing for when a data signal coming to a latch is controlled by a clock with a frequency different to a clock of the latch enabling signal Type: bool Default: false Edit: Yes Reference: timing_enable_multi_frequency_latch_analysis
timing_enable_multicycle_data_check_compatibility	
	Enables checking for SDC compatible data checks when set_multicycle_path -start parameter is specified Type: bool Default: false Edit: Yes Reference: timing_enable_multicycle_data_check_compatibility
timing_enable_path_delay_to_unconstrained_endpoints_compatibility	
	When set to true, does not constrain path with path delay if path ends at unconnected input pin of combo cell. Type: bool Default: false Edit: Yes Reference: timing_enable_path_delay_to_unconstrained_endpoints_compatibility
timing_enable_pessimistic_cpnr_for_reconvergent_clock_paths	

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	When set to true, enables pessimistic cpr adjustment for re-convergent clock paths Type: bool Default: false Edit: Yes Reference: timing_enable_pessimistic_cpr_for_reconvergent_clock_paths
timing_enable_power_ground_constants	
	Controls whether case analysis is inferred from power and ground rail connections. Type: bool Default: false Edit: Yes Reference: timing_enable_power_ground_constants
timing_enable_preset_clear_arcs	
	Determines whether timing arcs are created to model the transition to active state (assertion) of the preset or clear pin, and the subsequent transition of the output to controlled state Type: bool Default: false Edit: Yes Reference: timing_enable_preset_clear_arcs
timing_enable_pulse_latch	
	Enables pulse-latch analysis Type: bool Default: true Edit: Yes Reference: timing_enable_pulse_latch
timing_enable_set_reset_latch_preset_clear_arcs	
	Enable timing through clear/preset arcs of SR latches. Type: bool Default: false Edit: Yes Reference: timing_enable_set_reset_latch_preset_clear_arcs
timing_enable_si_cpr	
	Enables more accurate CPPR analysis when incremental delays are present and the timing_remove_clock_reconvergence_pessimism global variable is set to true Type: bool Default: true Edit: Yes Reference: timing_enable_si_cpr
timing_enable_simultaneous_setup_hold_mode	
	Controls whether setup and hold checks are analyzed separately, or together on the same timing graph Type: bool Default: false Edit: Yes Reference: timing_enable_simultaneous_setup_hold_mode
timing_enable_spatial_derate_mode	
	When set to true, enables spatial derate feature. Type: bool Default: false Edit: Yes Reference: timing_enable_spatial_derate_mode
timing_enable_timing_window_pessimism_removal	
	When set to true, removes common clock path pessimism between aggressor and victim pins in the design Type: bool Default: false Edit: Yes Reference: timing_enable_timing_window_pessimism_removal

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timing_enable_tristate_clock_gating	
	When this global is set to true, inferred gated-clock checks are added when clock and data signals converge through the tristate enable and data input of tristate buffers Type: bool Default: false Edit: Yes Reference: timing_enable_tristate_clock_gating
timing_enable_uncertainty_for_clock_checks	
	Specifies clock uncertainty check type true all false none pulse_width period skew Type: string Default: true Edit: Yes Reference: timing_enable_uncertainty_for_clock_checks
timing_enable_uncertainty_for_pulse_width_checks	
	When set to true, considers clock uncertainty when performing minimum pulse width checks Type: bool Default: false Edit: Yes Reference: timing_enable_uncertainty_for_pulse_width_checks
timing_enable_unique_vt_mode_for_gba	
	when true Enables Unique VtMode for GBA analysis Type: bool Default: false Edit: Yes
timing_enable_vtskew_derate_mode	
	Type: bool Default: false Edit: Yes Reference: timing_enable_vtskew_derate_mode
timing_enable_zero_delay_analysis_mode	
	When set to true, uses zero delays in timing analysis. Type: bool Default: false Edit: Yes Reference: timing_enable_zero_delay_analysis_mode
timing_extract_model_aocv_mode	
	This global sets the AOCV mode to be used during model extraction Type: enum Enum Values: none graph_based path_based Default: none Edit: Yes Reference: timing_extract_model_aocv_mode
timing_extract_model_case_analysis_in_library	
	When set to false, specifies that port propagated constants are written to the generated constraints file Type: bool Default: true Edit: Yes Reference: timing_extract_model_case_analysis_in_library
timing_extract_model_check_arcs_as_lvf	

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	With this global turned on, check arcs will be modeled as LVF arcs Type: <code>bool</code> Default: true Edit: Yes Reference: timing_extract_model_check_arcs_as_lvf
timing_extract_model_consider_design_level_drv	
	When set to false, specifies that user asserted design level DRVs should not be considered while writing to the extracted timing model Type: <code>bool</code> Default: true Edit: Yes Reference: timing_extract_model_consider_design_level_drv
timing_extract_model_disable_cycle_adjustment	
	When set to true, the cycle adjustment for the multicycle-paths that could not be pushed out, is disabled Type: <code>bool</code> Default: false Edit: Yes
timing_extract_model_enable_combinational_arc_to_clock_source_on_output_ports	
	When ON, models the combinational arc ending on an end point with clock asserted on it. Type: <code>bool</code> Default: false Edit: Yes
timing_extract_model_enable_unconstrained_port_modelling	
	When ON, extracts unconstrained path to the output ports. Type: <code>bool</code> Default: false Edit: Yes
timing_extract_model_exhaustive_validation_dir	
	Specifies the output directory where the validation reports will be written Type: <code>string</code> Default: "" Edit: Yes Reference: timing_extract_model_exhaustive_validation_dir
timing_extract_model_exhaustive_validation_mode	
	Enables ETM validation at the minimum and maximum indices of the slew / load indices Type: <code>bool</code> Default: false Edit: Yes Reference: timing_extract_model_exhaustive_validation_mode
timing_extract_model_gating_as_nochange_arc	
	When set to false, disables conversion of clock gating checks to nochange arcs in the extracted timing model Type: <code>bool</code> Default: true Edit: Yes Reference: timing_extract_model_gating_as_nochange_arc
timing_extract_model_half_cycle_non_interface_paths	
	When ON, models half cycle non interface paths as MPW checks in ETM Type: <code>bool</code> Default: false Edit: Yes
timing_extract_model_ideal_clock_latency_arc	

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	Specifies if latency arc from ideal master clock need to be extracted in timing model Type: bool Default: true Edit: Yes Reference: timing_extract_model_ideal_clock_latency_arc
timing_extract_model_include_applied_load_in_characterization_range	
	When set to true, specifies that actual load visible to timer, will be included in the load characterization range for path endpoints Type: bool Default: true Edit: Yes Reference: timing_extract_model_include_applied_load_in_characterization_range
timing_extract_model_include_applied_slew_in_characterization_range	
	When set to true, specifies that actual timer slew must be included in the characterization range for timing model. Type: bool Default: true Edit: Yes Reference: timing_extract_model_include_applied_slew_in_characterization_range
timing_extract_model_max_feedthru_characterization_load	
	Load specified will be used by model extractor for feed through path characterization. Type: double Default: 0.0 Edit: Yes
timing_extract_model_non_borrowing_latch_path_as_setup	
	If turned ON, models the non borrowing interface latch paths as setup check. Type: bool Default: false Edit: Yes Reference: timing_extract_model_non_borrowing_latch_path_as_setup
timing_extract_model_slew_propagation_mode	
	Specifies the type of slew propagation to use for generating extracted timing model Type: enum Enum Values: worst_slew path_based_slew Default: worst_slew Edit: Yes Reference: timing_extract_model_slew_propagation_mode
timing_extract_model_write_clock_checks_as_arc	
	Allows timing model to write min pulse width or min period checks as arcs. Type: bool Default: true Edit: Yes Reference: timing_extract_model_write_clock_checks_as_arc
timing_extract_model_write_clock_checks_as_scalar_tables	
	Allows timing model to write min pulse width or min period checks as scalar arcs. Type: bool Default: false Edit: Yes Reference: timing_extract_model_write_clock_checks_as_scalar_tables
timing_extract_model_write_lvf	
	When set to true, generated ETM contains LVF model. Type: bool Default: false Edit: Yes Reference: timing_extract_model_write_lvf
timing_extract_model_write_min_max_clock_tree_path	

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	When set to true, writes the min max worst latency for each clock source. Type: bool Default: true Edit: Yes Reference: timing_extract_model_write_min_max_clock_tree_path
timing_generate_normalized_driver_waveform	
	To generate ndw if it is not defined in the library. Type: bool Default: true Edit: Yes Reference: timing_generate_normalized_driver_waveform
timing_generated_clocks_allow_nested_assertions	
	Enables clock latency handling for nested generated clocks created on the same timing pin as their master clocks Type: bool Default: false Edit: Yes Reference: timing_generated_clocks_allow_nested_assertions
timing_generated_clocks_inherit_ideal_latency	
	Controls generated clock's inheritance of parent's ideal network latency. Type: enum Enum Values: false true actual_partial_latency_path ideal_partial_latency_path Default: false Edit: Yes Reference: timing_generated_clocks_inherit_ideal_latency
timing_get_of_objects_hier_compatibility	
	When set to true, the get_pins -of_objects parameter will return hierarchical leaf pins connected to the specified objects Type: bool Default: false Edit: Yes Reference: timing_get_of_objects_hier_compatibility
timing_hier_obj_name_compatibility	
	Controls how hierarchical delimiters are interpreted in the search pattern, when the -hier parameter is used with the get_* collection command Type: bool Default: true Edit: Yes Reference: timing_hier_obj_name_compatibility
timing_ignore_lumped_rc_assertions	
	When set to false, the set_load and set_resistance values override the actual extracted representation Type: bool Default: false Edit: Yes Reference: timing_ignore_lumped_rc_assertions
timing_io_use_clock_network_latency	
	Controls whether network latency of a reference clock is added or not to the data arrival time on the port Type: enum Enum Values: always ideal Default: ideal Edit: Yes Reference: timing_io_use_clock_network_latency
timing_ipd_allow_paths_for_asynchronous_clocks	
	When enabled, ensure the paths between the asynchronous clocks are allowed to be treated as IPD paths. Type: bool Default: true Edit: Yes

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timing_library_allow_invalid_primary_power_pins_lib_cell	
	Library cell having invalid primary_power pins is treated as unusable. Type: bool Default: false Edit: Yes Reference: timing_library_allow_invalid_primary_power_pins_lib_cell
timing_library_allow_multiple_sync_control	
	Allow multiple synchronous clear and preset pins. Type: bool Default: false Edit: Yes Reference: timing_library_allow_multiple_sync_control
timing_library_build_async_deassert_arc	
	Controls whether input to output arcs from the preset or clear pins transitioning to inactivestate are included when the timing system is initialized Type: bool Default: true Edit: Yes Reference: timing_library_build_async_deassert_arc
timing_library_convert_async_setuphold_to_recrem	
	Controls whether single-edged setup and hold checks on asynchronous pins in the Liberty library are inferred as recovery and removal checks. This is mainly to support older modeling styles in legacy libraries. Type: int Default: 1 Edit: Yes Reference: timing_library_convert_async_setuphold_to_recrem
timing_library_generated_clock_use_group_name	
	When set to true, the software uses the generated_clock group name when creating a generated clock from a library-generated clock group Type: bool Default: false Edit: Yes Reference: timing_library_generated_clock_use_group_name
timing_library_hold_constraint_corner_sigma_multiplier	
	User control multiplier to generate constraint table applied for hold arc. Type: double Default: 0.0 Edit: Yes Reference: timing_library_hold_constraint_corner_sigma_multiplier
timing_library_hold_sigma_multiplier	
	User control multiplier to generate constraint table applied for hold arc. Type: double Default: 0.0 Edit: Yes Reference: timing_library_hold_sigma_multiplier
timing_library_ignore_attribute_check_during_pin_convert	
	Ignore liberty attributes check while converting library's logical pin defined as power or ground in LEF. Type: bool Default: true Edit: Yes Reference: timing_library_ignore_attribute_check_during_pin_convert
timing_library_ignore_combinational_arc_mismatch_tables	
	Type: bool Default: false Edit: Yes

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timing_library_infer_async_pins_from_timing_arcs	
	This global marks the pins as async pins based on the timing arcs associated with the pin. Type: bool Default: false Edit: Yes Reference: timing_library_infer_async_pins_from_timing_arcs
timing_library_infer_cap_range_from_c1cn_dynamic_pin_cap_model	
	Enable Syntech support to infer cap range from n-piece dynamic pincap. Type: bool Default: false Edit: Yes Reference: timing_library_infer_cap_range_from_c1cn_dynamic_pin_cap_model
timing_library_infer_cap_range_from_ccs_receiver_model	
	When true infer the rise/fall capacitance range values from ccs receiver capacitance model Type: bool Default: false Edit: Yes Reference: timing_library_infer_cap_range_from_ccs_receiver_model
timing_library_infer_cap_range_from_dynamic_pin_cap_model	
	Enable Syntech support to infer cap range from dynamic pincap. Type: bool Default: false Edit: Yes Reference: timing_library_infer_cap_range_from_dynamic_pin_cap_model
timing_library_infer_cap_range_from_ecsm_receiver_model	
	When this global is enabled would infer cap range from ecsm capacitance group of library. Type: bool Default: false Edit: Yes Reference: timing_library_infer_cap_range_from_ecsm_receiver_model
timing_library_infer_socv_from_aocv	
	Infer sensitivity data from AOCV libraries Type: bool Default: false Edit: Yes Reference: timing_library_infer_socv_from_aocv
timing_library_interpolate_drv_values	
	Allows the software to use a range of trilib DRV values for performing delay calculations Type: bool Default: false Edit: Yes Reference: timing_library_interpolate_drv_values
timing_library_limit_messages	
	Controls the printing of LBR messages. Type: bool Default: true Edit: Yes
timing_library_look_up_drv_per_frequency	
	Interpolation method for 'max_cap' 1D table (linear worst). Type: enum Enum Values: worst linear Default: worst Edit: Yes Reference: timing_library_look_up_drv_per_frequency

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timing_library_multibit_force_merge_combinational_cells	
	Merge combinational instances into multibit ignoring qor change. Type: bool Default: false Edit: Yes Reference: timing_library_multibit_force_merge_combinational_cells
timing_library_multibit_force_merge_iso_cells	
	Merge combinational instances into multibit ignoring qor change. Type: bool Default: false Edit: Yes Reference: timing_library_multibit_force_merge_iso_cells
timing_library_multibit_force_merge_sequential_cells	
	Merge sequential instances into multibit ignoring qor change. Type: bool Default: false Edit: Yes Reference: timing_library_multibit_force_merge_sequential_cells
timing_library_multibit_use_combinational_cells	
	Allows to merge 1-bit combinational cells into multibit cells. Type: bool Default: false Edit: Yes Reference: timing_library_multibit_use_combinational_cells
timing_library_read_ccs_noise_data	
	can read ccs noise construct from library when enabled. Type: bool Default: true Edit: Yes Reference: timing_library_read_ccs_noise_data
timing_library_read_without_ecsm	
	Switch off loading of ECSM data Type: bool Default: false Edit: Yes Reference: timing_library_read_without_ecsm
timing_library_read_without_power	
	can read library without power by setting this global to true Type: bool Default: false Edit: Yes Reference: timing_library_read_without_power
timing_library_read_without_sensitivity	
	Switch off loading of ECSM timing sensitivity data Type: bool Default: false Edit: Yes Reference: timing_library_read_without_sensitivity
timing_library_respect_async_control_priority	
	Consider the liberty description of asynchronous inputs priority for sequential lib_cells. Type: bool Default: true Edit: Yes Reference: timing_library_respect_async_control_priority

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timing_library_scale_aocv_to_socv_to_n_sigma	
	AOCV derates are expected to be derived based on 3-sigma variation Type: double Default: 3.0 Edit: Yes Reference: timing_library_scale_aocv_to_socv_to_n_sigma
timing_library_sequential_in_out_phase_opt	
	Allows to propagate inverters through sequential instances. Type: bool Default: false Edit: Yes Reference: timing_library_sequential_in_out_phase_opt
timing_library_setup_constraint_corner_sigma_multiplier	
	User control multiplier to generate constraint table using sigma values applied for setup arc. Type: double Default: 0.0 Edit: Yes Reference: timing_library_setup_constraint_corner_sigma_multiplier
timing_library_setup_sigma_multiplier	
	User control multiplier to generate constraint table using sigma values applied for setup arc. Type: double Default: 0.0 Edit: Yes Reference: timing_library_setup_sigma_multiplier
timing_library_support_3d_table_power_arc	
	Support three dimensional power arc. Type: bool Default: false Edit: Yes Reference: timing_library_support_3d_table_power_arc
timing_library_support_combinational_clock	
	Support data-pins marked as clock on combinational cells. Type: bool Default: false Edit: Yes Reference: timing_library_support_combinational_clock
timing_library_support_merge_libs	
	Enables appending libraries with same names & PVT conditions in different .lib files. Type: bool Default: false Edit: Yes Reference: timing_library_support_merge_libs
timing_library_support_serial_scan_in_multibit_cell	
	Support the serial scan input multibit cell in which serial scan input is specified using internal node instead of scan input pin for all the outputs expect the first. Type: bool Default: true Edit: Yes Reference: timing_library_support_serial_scan_in_multibit_cell
timing_library_support_transparent_latch_group	

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	Support library cell having tlatch group. Type: bool Default: true Edit: Yes Reference: timing_library_support_transparent_latch_group
timing_library_suppress_library_messages	
	Controls the printing of library parsing messages. Type: bool Default: false Edit: Yes
timing_library_term_voltage_from_lib_pin	
	Use Library pin voltage for reporting the term voltage. Type: int Default: 0 Edit: Yes Reference: timing_library_term_voltage_from_lib_pin
timing_library_ui_respect_preserve	
	Cause netlist editing commands to fail if they would modify preserved logic. Type: bool Default: true Edit: Yes Reference: timing_library_ui_respect_preserve
timing_library_use_max_cap_look_up_table	
	Honor frequency based max_cap lut values from library for design rule violations. Type: bool Default: true Edit: Yes Reference: timing_library_use_max_cap_look_up_table
timing_library_use_sequential_test_cell	
	Use test cell function if main cell function is unusable. Type: bool Default: false Edit: Yes Reference: timing_library_use_sequential_test_cell
timing_library_use_two_piece_receiver_cap	
	When libraries have both 2 piece and multi piece pin caps, give priority to two piece models. Type: bool Default: false Edit: Yes Reference: timing_library_use_two_piece_receiver_cap
timing_library_zero_negative_timing_check_arcs	
	When true check for negative values in timing arcs Type: bool Default: false Edit: Yes Reference: timing_library_zero_negative_timing_check_arcs
timing_max_transition_use_si_transition	
	When set to true, enables noisy slew reporting in DRV flows Type: bool Default: false Edit: Yes Reference: timing_max_transition_use_si_transition
timing_multi_frequency_clock_rounding_factor	

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	Rounds off irrational numbers so that they can be read appropriately Type: double Default: 1e-05 Edit: Yes Reference: timing_multi_frequency_clock_rounding_factor
timing_normalized_driver_waveform_clip_linear_part	
	To control the clipping of the generated ndw which is generated by the global timing_generate_normalized_driver_waveform. Type: bool Default: false Edit: Yes Reference: timing_normalized_driver_waveform_clip_linear_part
timing_normalized_driver_waveform_weight_factor	
	Decides the weight factor for the exponential and the linear part in the waveform generated by the global timing_generate_normalized_driver_waveform. Type: double Default: 0.5 Edit: Yes Reference: timing_normalized_driver_waveform_weight_factor
timing_nsigma_multiplier	
	Controls sigma multiplier to be used in SOCV mode Type: double Default: 3.0 Edit: Yes Reference: timing_nsigma_multiplier
timing_null_collection_return_compatibility	
	Allows you to migrate previous release scripts to the new use model Type: bool Default: false Edit: Yes Reference: timing_null_collection_return_compatibility
timing_path_based_enable_bounding_box_for_io_paths	
	when true, enables bounding box mode for i/o paths Type: bool Default: false Edit: Yes Reference: timing_path_based_enable_bounding_box_for_io_paths
timing_path_based_enable_closest_common_pin_driver_for_bounding_box	
	when TRUE, enables bounding box calculation using closest common pin driver from clock mesh Type: bool Default: false Edit: Yes Reference: timing_path_based_enable_closest_common_pin_driver_for_bounding_box
timing_path_based_enable_exhaustive_depth_bounded_by_gba	
	Enable the tool to bound the PBA exhaustive analysis by next worst GBA slack incase the depth is exhausted for any given endpoint Type: bool Default: true Edit: Yes Reference: timing_path_based_enable_exhaustive_depth_bounded_by_gba
timing_path_based_enable_report_launch_clock_path	
	set it to false to skip reporting launch clock path in pba path_type full report Type: bool Default: true Edit: Yes Reference: timing_path_based_enable_report_launch_clock_path

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timing_path_based_enable_verbose_mode	
	controls the verbosity of path based analysis messages printed during and at the end of the analysis Type: enum Enum Values: 0 1 2 true false Default: 1 Edit: Yes Reference: timing_path_based_enable_verbose_mode
timing_path_based_epba_bounded_mode	
	Turn on EPBA Bounded mode Type: bool Default: false Edit: Yes Reference: timing_path_based_epba_bounded_mode
timing_path_based_exhaustive_enable_design_coverage	
	Enables the exhaustive path based analysis to evaluate all violating endpoints Type: bool Default: false Edit: Yes Reference: timing_path_based_exhaustive_enable_design_coverage
timing_path_based_exhaustive_max_paths_limit	
	Set the maximum number of paths which can be retimed during exhaustive path based analysis Type: int Default: 2000000 Edit: Yes Reference: timing_path_based_exhaustive_max_paths_limit
timing_path_based_ipba_endpoints_max_limit	
	Set endpoints max limit, for infinite PBA analysis of top max GBA endpoints Type: int Default: 100000 Edit: Yes Reference: timing_path_based_ipba_endpoints_max_limit
timing_path_based_low_memory_mode	
	A lower value for this variable will reduce the peak memory footprint of path based analysis (PBA) at the cost of some addition runtime. Type: double Default: 10.0 Edit: Yes
timing_path_based_report_analysis_summary_max_paths_limit	
	specifies the limit of maximum PBA FEPs reported Type: int Default: 2000000 Edit: Yes Reference: timing_path_based_report_analysis_summary_max_paths_limit
timing_pba_exhaustive_path_nworst_limit	
	Specify the nworst exhaustive limit to be honored by tool during PBA exhaustive limit Type: int Default: 10000 Edit: Yes Reference: timing_pba_exhaustive_path_nworst_limit
timing_pll_clock_use_driving_cell	

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	When set to false, the software considers the PLL clock output to be electrically isolated from the input-side reference clock, and creates the output clock with an idealized transition. You can use <code>set_annotated_transition</code> to customize the output clock transition. When set to true, delay calculation of all enabled arcs to the PLL clock output will be used to compute the transition. If no driving arcs are available in the model or if they are disabled, the same idealized transition will be used. Type: <code>bool</code> Default: <code>false</code> Edit: Yes Reference: timing_pll_clock_use_driving_cell
timing_prefix_module_name_with_library_generated_clock	
	When set to true, the software appends the instance name to the clock pin name when creating a generated clock Type: <code>bool</code> Default: <code>true</code> Edit: Yes Reference: timing_prefix_module_name_with_library_generated_clock
timing_propagate_latch_data_uncertainty	
	When set to true, uses the clock phase associated with a flush latch's data pin as the from clock phase for downstream uncertainty timing calculations Type: <code>bool</code> Default: <code>false</code> Edit: Yes Reference: timing_propagate_latch_data_uncertainty
timing_property_arrival_clocks_consider_clock_source_data_phase	
	When set to true, the <code>arrival_clocks</code> property will consider clock source data phases Type: <code>bool</code> Default: <code>true</code> Edit: Yes Reference: timing_property_arrival_clocks_consider_clock_source_data_phase
timing_property_arrival_clocks_consider_data_phase	
	When set to true, the <code>arrival_clocks</code> property will return pure data phase Type: <code>bool</code> Default: <code>true</code> Edit: Yes Reference: timing_property_arrival_clocks_consider_data_phase
timing_property_arrival_clocks_consider_latency_phase	
	When set to true, the <code>arrival_clocks</code> property will return latency phase Type: <code>bool</code> Default: <code>true</code> Edit: Yes Reference: timing_property_arrival_clocks_consider_latency_phase
timing_property_arrival_window_enable_tcl_dict_format	
	When set to TRUE, new format for the <code>arrival_windows</code> property will be used to print. Type: <code>bool</code> Default: <code>false</code> Edit: Yes Reference: timing_property_arrival_window_enable_tcl_dict_format
timing_property_return_null_collection_with_quiet	
	Enables returning properties values for all objects when <code>-quiet</code> option is used in <code>get_property</code> Type: <code>bool</code> Default: <code>false</code> Edit: Yes Reference: timing_property_return_null_collection_with_quiet
timing_rail_swing_checks_high_voltage_threshold	

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	user to specify voltage thresholds for high rail swing checks Type: double Default: 0.95 Edit: Yes Reference: timing_rail_swing_checks_high_voltage_threshold
timing_rail_swing_checks_low_voltage_threshold	
	user to specify voltage thresholds for low rail swing checks Type: double Default: 0.05 Edit: Yes Reference: timing_rail_swing_checks_low_voltage_threshold
timing_rcdb_allow_mismatch_option	
	Add the -allow_mismatch option to read_parasitics for BM rcdb loading. Type: bool Default: true Edit: Yes Reference: timing_rcdb_allow_mismatch_option
timing_recompute_sdf_in_setuphold_mode	
	Controls the recomputing of SDF delays when the software is in simultaneous setup and hold analysis mode Type: bool Default: false Edit: Yes Reference: timing_recompute_sdf_in_setuphold_mode
timing_reduce_multi_drive_net_arcs	
	Controls the reduction of the number of net arcs created for timing analysis for nets driven by parallel buffers Type: bool Default: true Edit: Yes Reference: timing_reduce_multi_drive_net_arcs
timing_reduce_multi_drive_net_arcs_threshold	
	Sets a threshold number used by the tool to trigger the reduction of timing arcs of nets driven by parallel buffers Type: int Default: 10000 Edit: Yes Reference: timing_reduce_multi_drive_net_arcs_threshold
timing_report_analysis_summary_csv_extended_new_format	
	This global keep the report_analysis_summary csv_extended report in sync with report_analysis_summary report. Type: bool Default: false Edit: Yes Reference: timing_report_analysis_summary_csv_extended_new_format
timing_report_arrival_property_worstcase_mode	
	To be set to 0, 1, 2 or 3 to differentiate clock and data phase on properties arrival_* Type: int Default: 0 Edit: Yes
timing_report_begin_end_pair_max_path_limit	
	Controls the number of max_paths to be reported by report_timing in begin_end_pair option. Higher max_paths limit may lead to high memory and runtime. Type: int Default: 2000000 Edit: Yes Reference: timing_report_begin_end_pair_max_path_limit
timing_report_check_timing_unconstrained_endpoints_due_to_constants	

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	Will split unconstrained endpoint warning into another warning unconstrained endpoints due to case analysis. Type: bool Default: false Edit: Yes Reference: timing_report_check_timing_unconstrained_endpoints_due_to_constants
timing_report_clock_pin_as_begin_point	
	Enable whether to show CLK as start point Type: bool Default: false Edit: Yes Reference: timing_report_clock_pin_as_begin_point
timing_report_constraint_enable_extended_drv_format	
	Enables reporting of report_constraint as per clock/data phases and rise/fall Type: bool Default: false Edit: Yes Reference: timing_report_constraint_enable_extended_drv_format
timing_report_constraint_extended_cell_em_flow	
	Enables extended Cell EM flow Type: bool Default: false Edit: Yes Reference: timing_report_constraint_extended_cell_em_flow
timing_report_constraint_format	
	Enables custom drv reporting in specified format Type: string Default: "" Edit: Yes Reference: timing_report_constraint_format
timing_report_constraint_max_paths_limit	
	Set the maximum number of violations (Data/Clock checks) to be reported. Type: int Default: 2000000 Edit: Yes
timing_report_constraint_rise_fall_clock_period_check	
	Enables rise-fall check for clock_period check type in report_constraint Type: bool Default: false Edit: Yes Reference: timing_report_constraint_rise_fall_clock_period_check
timing_report_constraint_use_infinity_slack_for_unconstrained	
	Enables reporting of timing slack as INFINITY for unconstrained paths for configurable DRV report. The default behavior is to report timing slacks using star "*" character Type: bool Default: false Edit: Yes Reference: timing_report_constraint_use_infinity_slack_for_unconstrained
timing_report_default_frequency_for_unconstrained_nets	
	Defines frequency for unconstrained nets to be used in max_cap drv reporting Type: double Default: 1e+09 Edit: Yes Reference: timing_report_default_frequency_for_unconstrained_nets
timing_report_disable_max_paths_per_group	

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	This global controls whether or not group-based mode semantics are used when -group and -max_paths are used with report_timing. Type: bool Default: false Edit: Yes Reference: timing_report_disable_max_paths_per_group
timing_report_drv_enable_clock_source_as_clock	
	Enables clock drv for pure clock source path in place of data drv Type: bool Default: false Edit: Yes Reference: timing_report_drv_enable_clock_source_as_clock
timing_report_drv_enable_frequency_per_view	
	Enables view based frequency drv flow Type: bool Default: true Edit: Yes Reference: timing_report_drv_enable_frequency_per_view
timing_report_drv_enable_ghz_notation_for_drv_fields	
	Enables reporting of clock frequency in Ghz notation for custom drv reporting Type: bool Default: false Edit: Yes Reference: timing_report_drv_enable_ghz_notation_for_drv_fields
timing_report_drv_enable_slew_threshold_scaling	
	Enables transition constraints scaling as per slew threshold Type: bool Default: false Edit: Yes Reference: timing_report_drv_enable_slew_threshold_scaling
timing_report_drv_per_frequency	
	Enables worst case reporting of max_transition/max_capacitance and considers set_max_cap(tran)_per_freq, set_max_transition(capacitance) and library defined values in comparison Type: bool Default: true Edit: Yes Reference: timing_report_drv_per_frequency
timing_report_drv_per_frequency_per_input_slew	
	Enables frequency dependent drv lookup from library Type: bool Default: false Edit: Yes Reference: timing_report_drv_per_frequency_per_input_slew
timing_report_drv_use_worst_timing_slack	
	Enables reporting of the worst DRV slack across all views for configurable DRV report Type: bool Default: false Edit: Yes Reference: timing_report_drv_use_worst_timing_slack
timing_report_enable_clock_to_for_unconstrained_paths	
	Enables Unconstrained Path reporting for -to_clock option Type: bool Default: false Edit: Yes
timing_report_enable_cpnr_point	

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	Enable Cppr Point reporting in report timing command Type: bool Default: false Edit: Yes Reference: timing_report_enable_cppr_point
timing_report_enable_em_index_clipping_report	
	Enables index clipping report in em drv reports Type: bool Default: false Edit: Yes Reference: timing_report_enable_em_index_clipping_report
timing_report_enable_flag_field_symbols	
	Enables reporting of Instance/Net related bits in report_timing Flags column Type: bool Default: false Edit: Yes Reference: timing_report_enable_flag_field_symbols
timing_report_enable_lead_trail_to_rise_fall_map	
	Enable reporting of Rise/Fall edge times instead of Lead/Trail edge times of clock in report_clocks command Type: bool Default: false Edit: Yes Reference: timing_report_enable_lead_trail_to_rise_fall_map
timing_report_enable_markers	
	Enables marker for report_timing for Pin/timing points columns Type: bool Default: true Edit: Yes Reference: timing_report_enable_markers
timing_report_enable_max_capacitance_drv_for_constant_nets	
	Enables max capacitance DRV checks for nets having disabled constant propagation Type: bool Default: false Edit: Yes Reference: timing_report_enable_max_capacitance_drv_for_constant_nets
timing_report_enable_max_path_limit_warning	
	warning message if max_paths option specified is not enough for coverage Type: bool Default: false Edit: Yes Reference: timing_report_enable_max_path_limit_warning
timing_report_enable_qpin_as_start_for_begin_end_pair	
	Enables reporting of multiple paths with different q-pins but same clock pin Type: bool Default: false Edit: Yes
timing_report_enable_report_clock_timing_across_clock_pin	
	Enables fanout tracing across clock pins if any combinational arc is present Type: bool Default: false Edit: Yes Reference: timing_report_enable_report_clock_timing_across_clock_pin
timing_report_enable_si_debug	

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	For decision regarding SI attribute debugging through path collection objects Type: bool Default: false Edit: Yes Reference: timing_report_enable_si_debug
timing_report_enable_unique_pins_multiple_capture_clock_paths	
	enables reporting different paths w.r.t ref clocks in unique_pins Type: bool Default: false Edit: Yes Reference: timing_report_enable_unique_pins_multiple_capture_clock_paths
timing_report_enable_verbose_ssta_mode	
	Enables new format for header in report_timing SOCV mode Type: bool Default: false Edit: Yes Reference: timing_report_enable_verbose_ssta_mode
timing_report_enable_write_twf_third_party_for_input_pins	
	Enable writing of timing information in third party twf for input pins Type: bool Default: false Edit: Yes
timing_report_fields	
	Specifies a report_timing format Type: string Default: instance arc cell delay arrival required Edit: Yes Reference: timing_report_fields
timing_report_generated_clock_info	
	When set to true (the default), generated clock information is automatically added to the report if generated clocks are encountered in either the launching or latching clock paths Type: bool Default: true Edit: Yes Reference: timing_report_generated_clock_info
timing_report_group_based_mode	
	Groups paths by clock domain Type: bool Default: false Edit: Yes Reference: timing_report_group_based_mode
timing_report_limit_max_path_count	
	Set the maximum number of violations (Data/Clock checks) to be reported. Type: int Default: 2000000 Edit: Yes
timing_report_max_transition_check_using_nsigma_slew	
	Enables report_constraint drv checking to use mean+nsigma slew values in socv flow Type: bool Default: false Edit: Yes Reference: timing_report_max_transition_check_using_nsigma_slew
timing_report_property_fastest_clock_consider_data_phase	

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	When on, it will consider data phase also while finding the fastest clock. Type: bool Default: false Edit: Yes Reference: timing_report_property_fastest_clock_consider_data_phase
timing_report_pulse_width_matching_launch_capture_paths	
	Reports only those pulse_width paths that have same pins on launch and capture path Type: bool Default: false Edit: Yes Reference: timing_report_pulse_width_matching_launch_capture_paths
timing_report_redirect_message_types	
	Redirect reporting messages to report file Type: enum Enum Values: info warning error all none Default: none Edit: Yes Reference: timing_report_redirect_message_types
timing_report_retime_formatting_mode	
	This variable can be used to manage the retime fields in reporting format. Based upon given setting it would automatically replaces (or add) the default columns with respective retime columns. For example in case of delay column it will add(or replace) the 'Retime Delay' automatically. Type: enum Enum Values: manual retime_compare retime_replace Default: manual Edit: Yes Reference: timing_report_retime_formatting_mode
timing_report_skip_constraint_loop_check	
	skip reporting of loop in case timing graph inside loop is broken due to create_clock, set_input_delay, set_output_delay defined inside the loop Type: bool Default: false Edit: Yes Reference: timing_report_skip_constraint_loop_check
timing_report_skip_icg_from_all_fanin_fanout_collection	
	Skip ICG clock pins from all_fanin/fanout collection when queried with -startpoint/endpoint Type: bool Default: false Edit: Yes
timing_report_socv_summary_mean_sigma	
	Enables new format for header in report_timing SOCV mode Type: bool Default: false Edit: Yes Reference: timing_report_socv_summary_mean_sigma
timing_report_split_other_end_arrival	
	This global splits the printing of other end arrival time in timing reports into other end path delay and clock edge time. Type: bool Default: false Edit: Yes
timing_report_timing_header_detail_info	

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	Controls whether the report_timing reports are generated using the default orextended report header Type: enum Enum Values: default extended Default: default Edit: Yes Reference: timing_report_timing_header_detail_info
timing_report_unconstrained_path_early_late_header	
	When set to true, prints late and early type for unconstrained path Type: bool Default: false Edit: Yes Reference: timing_report_unconstrained_path_early_late_header
timing_report_unconstrained_paths	
	When set to true, the report_timing command reports unconstrained paths if it cannot find a constrained path to report Type: bool Default: false Edit: Yes Reference: timing_report_unconstrained_paths
timing_report_use_receiver_model_capacitance	
	Uses ECSM pin capacitance values for load or DRV calculation Type: bool Default: false Edit: Yes Reference: timing_report_use_receiver_model_capacitance
timing_report_use_worst_parallel_cell_arc	
	When set to false, setting the -nworst parameter reports several paths by selecting parallel arcs in the library cell between two pins Type: bool Default: false Edit: Yes Reference: timing_report_use_worst_parallel_cell_arc
timing_resolve_driver_conflicts	
	Resolves values of nets having multiple drivers Type: enum Enum Values: conservative aggressive Default: aggressive Edit: Yes Reference: timing_resolve_driver_conflicts
timing_scaling_for_negative_checks	
	Modifies the scaling of the negative timing analysis check value Type: enum Enum Values: default divider multiplier Default: default Edit: Yes Reference: timing_scaling_for_negative_checks
timing_scaling_for_negative_delays	
	Modifies the scaling of the negative delay value to be used during timing analysis Type: enum Enum Values: default divider multiplier Default: default Edit: Yes Reference: timing_scaling_for_negative_delays
timing_sdf_adjust_negative_setuphold	

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	Controls how Setup and Hold check values are adjusted when both are initially negative. Type: bool Default: false Edit: Yes Reference: timing_sdf_adjust_negative_setuphold
timing_sdf_enable_setuphold_scond_ccond	
	To enable proper generation of SDF (Standard Delay Format) with scond and ccond qualifiers on SETUPHOLD and RECREM timing checks Type: bool Default: false Edit: Yes Reference: timing_sdf_enable_setuphold_scond_ccond
timing_self_loop_paths_no_skew_max_depth	
	Identifies self-loop paths at the specified depth. Type: int Default: 10 Edit: Yes Reference: timing_self_loop_paths_no_skew_max_depth
timing_self_loop_paths_no_skew_max_slack	
	Identifies self-loop paths at the specified threshold Type: double Default: 0.0 Edit: Yes Reference: timing_self_loop_paths_no_skew_max_slack
timing_set_clock_source_to_output_as_data	
	When set to true, causes a clock source path leading to an output or bidi port to be treated as a data path if there is a set_output_delay or set_data_check assertion on the port Type: bool Default: false Edit: Yes Reference: timing_set_clock_source_to_output_as_data
timing_socv_preserve_variation_with_annotations	
	Type: bool Default: false Edit: Yes Reference: timing_socv_preserve_variation_with_annotations
timing_socv_rc_variation_mode	
	Enable interconnect variation mode in SOCV mode Type: bool Default: true Edit: Yes Reference: timing_socv_rc_variation_mode
timing_socv_sigma_spatial_multiplier	
	The value specified for this global provides a fractional percentage multiplier which allows the user to select that all, none, or some of the spatial derate be applied Type: double Default: 1.0 Edit: Yes
timing_socv_view_based_nsigma_multiplier_mode	
	Enables setting of view-based separate setup and hold sigma multipliers Type: bool Default: true Edit: Yes Reference: timing_socv_view_based_nsigma_multiplier_mode

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timing_spatial_derate_chip_size	
	Type: double Default: 1e+30 Edit: Yes Reference: timing_spatial_derate_chip_size
timing_spatial_derate_distance_mode	
	Type: enum Enum Values: bounding_box chip_size arc_bounding_box Default: bounding_box Edit: Yes Reference: timing_spatial_derate_distance_mode
timing_suppress_escape_characters	
	Suppresses the reporting of escape characters in timing object names Type: bool Default: true Edit: Yes Reference: timing_suppress_escape_characters
timing_suppress_ilm_constraint_mismatches	
	When set to true, the software suppresses all error and warning messages related to objects not found when loading SDC constraint files for the ILM flow Type: bool Default: false Edit: Yes Reference: timing_suppress_ilm_constraint_mismatches
timing_time_unit	
	This value is passed to set_library_unit -time to set the time units used in timing library and .sdc files and timing reports. Legal values are none, 1ns, 1ps, 10ps, 100ps. Type: string Default: none Edit: Yes Reference: timing_time_unit
timing_use_clock_pin_attribute_for_clock_net_marking	
	To allow the propagation of clocks to pins with clock attributes regardless of the presence of check arcs or trigger arcs at the pins Type: bool Default: false Edit: Yes Reference: timing_use_clock_pin_attribute_for_clock_net_marking
timing_use_incremental_si_transition	
	When set to true, enables usage of incremental slew during DRV violation reporting Type: bool Default: false Edit: Yes Reference: timing_use_incremental_si_transition
timing_use_latch_early_launch_edge	
	Enables the latch use early launch edge feature. Type: bool Default: true Edit: Yes Reference: timing_use_latch_early_launch_edge
timing_use_latch_time_borrow	

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	When set to false, does not consider time borrowing during timing analysis. Time borrowing is the amount of time borrowed by a previous logic Type: bool Default: true Edit: Yes Reference: timing_use_latch_time_borrow
timing_use_verilog_for_model_netlist	
	Whether to write scope and boundary model netlists in verilog format rather than binary Type: int Default: 0 Edit: Yes Reference: timing_use_verilog_for_model_netlist
timing_waveform_aware_pulse_width_checks_high_voltage_level	
	User to specify high voltage level to be used for waveform aware pulse-width checks Type: double Default: 0.95 Edit: Yes Reference: timing_waveform_aware_pulse_width_checks_high_voltage_level
timing_waveform_aware_pulse_width_checks_low_voltage_level	
	User to specify low voltage level to be used for waveform aware pulse-width checks Type: double Default: 0.05 Edit: Yes Reference: timing_waveform_aware_pulse_width_checks_low_voltage_level
timing_write_sdf_no_escape_backslash	
	Supports various patterns of the escape character. Type: bool Default: false Edit: Yes Reference: timing_write_sdf_no_escape_backslash
tns_critical_range	
	Slack-delta to consider critical (relative to the worst endpoint slack). Type: string Default: 0.0 Edit: Yes Reference: tns_critical_range
top_sdps	
	Short-cut to all top level structured data path (sdp) objects. Type: obj(sdp)* Default: "" Edit: No
trace_obj_macro_pins	
	specify macro pin name patterns in regular expression Type: string Default: "" Edit: Yes Reference: trace_obj_macro_pins
trace_obj_max_fanin_fanout	
	Specify the maximum number of a pin's fanin/fanout,a pin will be ignored in tracing connectivity if its fanin/fanout larger than the specified number Type: int Default: 1000 Edit: Yes Reference: trace_obj_max_fanin_fanout
trace_obj_register_inputs	

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	Specify register input pin names Type: string Default: "" Edit: Yes Reference: trace_obj_register_inputs
trace_obj_register_outputs	
	Specify register output pin names Type: string Default: "" Edit: Yes Reference: trace_obj_register_outputs
track_patterns	
	Short-cut to all the track_pattern objects in the design. Type: obj(track_pattern)* Default: "" Edit: No
treat_non_seq_arc_cell_as_unusable	
	Libcell having non sequential setup arc is treated as unusable if 'timing_disable_non_sequential_checks' is set to false. Type: bool Default: true Edit: Yes Reference: treat_non_seq_arc_cell_as_unusable
trim_grids	
	Short-cut to all the trim_grid objects in the design from the LEF TRIMMETALTRACK statement. Type: obj(trim_grid)* Default: "" Edit: No
turbo_lib_loading	
	Populate timing arcs, power arcs and library establishment data only on need basis for runtime speedup. Type: int Default: 2 Edit: Yes Reference: turbo_lib_loading
ui_precision	
	Specifies the number of significant digits to be displayed in timing reports for data with no specific type. Type: int Default: 3 Edit: Yes Reference: ui_precision
ui_precision_capacitance	
	Specifies the number of significant digits to be displayed in timing reports for data of type capacitance. Type: int Default: 3 Edit: Yes Reference: ui_precision_capacitance
ui_precision_derating	
	Specifies the number of significant digits to be displayed in timing reports for derating factors such as those specified via set_timing_derate constraints or AOCV derating libraries. Type: int Default: 3 Edit: Yes Reference: ui_precision_derating
ui_precision_power	

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	Specifies the number of significant digits to be displayed in timing reports for data of type power. Type: int Default: 3 Edit: Yes Reference: ui_precision_power
ui_precision_sensitivities	
	Specifies the number of significant digits to be displayed in timing reports for statistical sensitivity values and SOCV sigma values. Type: int Default: 3 Edit: Yes Reference: ui_precision_sensitivities
ui_precision_timing	
	Specifies the number of significant digits to be displayed for timing reports for delay type values including cell and net delays, transitions, arrival and required times, and slacks. Type: int Default: 3 Edit: Yes Reference: ui_precision_timing
ui_units_capacitance	
	capacitance unit for CUI Type: enum Enum Values: 1pf 1ff 10ff 100ff Default: "" Edit: Yes
ui_units_capacitance_reporting	
	Capacitance reporting unit for CUI Type: enum Enum Values: 1pf 1ff 10ff 100ff Default: "" Edit: No
ui_units_power_reporting	
	Power reporting unit for CUI Type: string Default: "" Edit: No
ui_units_timing	
	timing unit for CUI Type: enum Enum Values: 1ns 1ps 10ps 100ps Default: "" Edit: Yes
ui_units_timing_reporting	
	Time reporting unit for CUI Type: enum Enum Values: 1ns 1ps 10ps 100ps Default: "" Edit: No
uniquify_naming_style	
	the default uniquify naming style as defined by this tcl global Type: string Default: %s_%d Edit: Yes
unmap_scan_flops	

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	Whether it is OK to unmap scan flops {false 0 not_mapped_for_dft true 1}. Type: enum Enum Values: false not_mapped_for_dft true Default: false Edit: Yes Reference: unmap_scan_flops
use_area_from_lef	
	Controls whether the cell area from the LEF files should be used. Type: string Default: false Edit: Yes Reference: use_area_from_lef
use_compatibility_based_grouping	
	Allows to merge 1-bit sequential, combinational and tristate cells into multibit cells based on compatible groups. Type: bool Default: true Edit: Yes Reference: use_compatibility_based_grouping
use_main_cell_output_function_for_test_cell	
	Use main cell's output function for test cell. Type: bool Default: true Edit: Yes Reference: use_main_cell_output_function_for_test_cell
use_multibit_cells	
	Allows to merge 1-bit sequential, combinational, tristate and iso/ls cells into multibit cells. Type: bool Default: false Edit: Yes Reference: use_multibit_cells
use_multibit_iso_cells	
	Allows to merge 1-bit iso cells into multibit cells. Type: bool Default: false Edit: Yes Reference: use_multibit_iso_cells
use_multibit_seq_and_tristate_cells	
	Allows to merge 1-bit sequential and tristate cells into multibit cells. Type: bool Default: false Edit: Yes Reference: use_multibit_seq_and_tristate_cells
use_scan_seqs_for_non_dft	
	Allow the use of scan sequential cells for non-DFT purposes (true/false/degenerated_only). Type: enum Enum Values: true false degenerated_only Default: true Edit: Yes Reference: use_scan_seqs_for_non_dft
verification_directory_naming_style	

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	Format of directory in which verification files are written when the 'write_verification_files' attribute is 'true'. The directory will be created if it does not already exist. Type: string Default: fv/%s Edit: Yes Reference: verification_directory_naming_style
via_def_rules	
	All the via_def_rules defined in the LEF or OpenAccess techfile. Type: obj(via_def_rule)* Default: "" Edit: No
via_defs	
	All the via_defs (via master definitions). Type: obj(via_def)* Default: "" Edit: No
violations	
	List of all violations Type: obj(violation)* Default: "" Edit: No
wcdc_clock_dom_comb_propagation	
	To control clock domain propagation through combinational elements in CDC. Valid values are exact_phase or either_phase or wire or logic_phase or logic_nophase. Type: string Default: logic_nophase Edit: Yes Reference: wcdc_clock_dom_comb_propagation
write_clp_lib_statetable	
	Whether 'read library' has the '-statetable' switch. Type: bool Default: true Edit: Yes Reference: write_clp_lib_statetable
write_db_auto_save_user_globals	
	If true, write_db automatically saves user created Tcl global variables. User globals are defined as any "info globals" names that do not exist at startup (e.g. created by user executed code). See the define_variables command to add or delete specific Tcl variables from the automatic save list. Type: bool Default: false Edit: Yes Reference: write_db_auto_save_user_globals
write_db_binary_timing_constraints	
	If true, write_db will save timing constraints in binary format. The value of the global should be persistent in subsequent write_db and read_db operations. Type: bool Default: true Edit: Yes Reference: write_db_binary_timing_constraints
write_db_cmd_file_limit	

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	write_db saves the complete .cmd history across multiple sessions in the inn.cmd.gz file inside the save directory up to this file length limit in units of Mb. If it is set to 0, no history is saved at all Type: int Default: 10 Edit: Yes Reference: write_db_cmd_file_limit
write_db_copy_timing_constraints_always	
	Specifies whether SDC files are always copied to the DB directory rather than links to external SDC files. By default they are only copied if the constraints were modified. Type: bool Default: false Edit: Yes Reference: write_db_copy_timing_constraints_always
write_db_create_read_file	
	If true, 'write_db my_db' will also create a 'my_db.read' file with a 'read_db my_db' command inside it. This is a Tcl file that can be passed to the Linux executable to make it easy to load the DB at startup like this: 'innovus -files my_db.read'. For OA usage, it would have the appropriate 'read_db -oa_lib_cell_view {<lib> <cell> <view>}' command inside it. Type: bool Default: false Edit: Yes Reference: write_db_create_read_file
write_db_include_metal_fill_rules	
	specify whether to save metalfill settings Type: int Default: 0 Edit: Yes Reference: write_db_include_metal_fill_rules
write_db_restrict_under_batch_mode	
	By default, a new db can be saved under batch mode, tool errors out when this attribute is set to true. Type: bool Default: false Edit: Yes Reference: write_db_restrict_under_batch_mode
write_db_restrict_under_eco_batch_mode	
	Controls whether to restrict db saving in eco batch mode. Type: bool Default: false Edit: Yes
write_db_save_unused_lcf_block_names	
	Specifies whether to remember a list of unused macros during saving Type: bool Default: false Edit: Yes Reference: write_db_save_unused_lcf_block_names
write_def_compress_vias	
	Compress the via statement in SPECIALNETS section. This variable controls write_def to sort vias with two dimension array, so it can use 'DO numX BY numY STEP stepX stepY' statement. Type: int Default: 0 Edit: Yes Reference: write_def_compress_vias
write_def_hierarchy_delimiter	

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	Specify the write_def hierarchy delimiter. Type: string Default: "" Edit: Yes Reference: write_def_hierarchy_delimiter
write_def_include_lef_ndr	
	Output LEF NONDEFAULTRULES in DEF. This variable controls write_def to output LEF nondefault rule information. By default, write_def does not write out nondefault rules defined in LEF. Set this variable to true to output nondefault rules defined in LEF . Type: bool Default: false Edit: Yes Reference: write_def_include_lef_ndr
write_def_include_lef_vias	
	Output LEF vias in DEF. This variable controls write_def to output LEF via information. By default, write_def does not write out via defined in LEF. Set this variable to true to output LEF via. Type: bool Default: false Edit: Yes Reference: write_def_include_lef_vias
write_def_lef_out_version	
	Specify LEF/DEF output version. Possible string value are 5.5, 5.6, 5.7, 5.8 and 6.0. The default is 5.8. Type: enum Enum Values: 5.5 5.6 5.7 5.8 6.0 Default: 5.8 Edit: Yes Reference: write_def_lef_out_version
write_def_polygon_die_area	
	By default write_def writes out polygon DIEAREA if the design is rectilinear. To write out rectangular DIEAERA along with blockages in cut-out area for a rectilinear design, set this variable to false. Type: bool Default: true Edit: Yes Reference: write_def_polygon_die_area
write_def_stream_check_uncolored	
	Specify how to check and ignore uncolored shapes on DPT layers during DEF out and stream out. If true or 1, checks uncolored shapes on all DPT layers; if false or 0, doesn't do check; if metal_only, only checks uncolored shapes on metal layers, uncolored shapes on cut layers are ignored. It's automatically set to metal_only for TSMC N5 and below nodes. Type: enum Enum Values: true 1 false 0 metal_only Default: false Edit: Yes Reference: write_def_stream_check_uncolored
write_do_lec_add_no_black_box_retime_subdesign	
	Whether to write 'add noblack box -submodules' for the retime modules in the dofile. Type: bool Default: true Edit: Yes Reference: write_do_lec_add_no_black_box_retime_subdesign
write_do_lec_analyze_abort	
	Whether to enable 'analyze_abort' flow in LEC. Type: bool Default: false Edit: Yes Reference: write_do_lec_analyze_abort
write_do_lec_analyze_setup	

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	Whether to enable 'analyze setup' flow in LEC. Type: bool Default: false Edit: Yes Reference: write_do_lec_analyze_setup
write_do_lec_auto_analyze	
	Whether to enable 'auto analyze' in LEC. Type: bool Default: true Edit: Yes Reference: write_do_lec_auto_analyze
write_do_lec_black_box_ilm_modules	
	Black box the ILM modules. Type: bool Default: false Edit: Yes Reference: write_do_lec_black_box_ilm_modules
write_do_lec_compare_threads	
	Specifies number of threads for 'set compare option -threads'. Type: int Default: 4 Edit: Yes Reference: write_do_lec_compare_threads
write_do_lec_composite_compare	
	Specifies two-step comparison: RTL-to-fv_map and fv_map-to-revised. This attribute has no effect when root attribute 'one_pass_formal_verification' is 'true'. Type: bool Default: true Edit: Yes Reference: write_do_lec_composite_compare
write_do_lec_dft_constraint_file	
	DFT constraints file. This file must exist when verification is performed. Type: string Default: "" Edit: Yes Reference: write_do_lec_dft_constraint_file
write_do_lec_gzip_lec_json	
	Compress the implementation information files. Conformal 19.10-s300 or newer is required to execute the dofile. Type: bool Default: false Edit: Yes Reference: write_do_lec_gzip_lec_json
write_do_lec_hier_append_string	
	When not empty, the value is used as the '-append_string' argument to the Conformal LEC 'write_hier_compare_dofile' command written to the dofile by 'write_do_lec'. Type: string Default: "" Edit: Yes Reference: write_do_lec_hier_append_string
write_do_lec_hier_comp_threshold	
	The threshold limit for minimum number of instances in a module in order to perform hierarchical comparison. Type: int Default: 50 Edit: Yes Reference: write_do_lec_hier_comp_threshold

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write_do_lec_hier_compare_string	
	When not empty, the value is used as the '-compare_string' argument to the Conformal LEC 'write_hier_compare_dofile' command written to the dofile by 'write_do_lec'. Type: string Default: "" Edit: Yes Reference: write_do_lec_hier_compare_string
write_do_lec_hier_prepend_string	
	When not empty, the value is used as the '-prepend_string' argument to the Conformal LEC 'write_hier_compare_dofile' command written to the dofile by 'write_do_lec'. Type: string Default: "" Edit: Yes Reference: write_do_lec_hier_prepend_string
write_do_lec_lib_statetable	
	Whether 'read library' has the '-statetable' switch. Type: bool Default: true Edit: Yes Reference: write_do_lec_lib_statetable
write_do_lec_low_power_analysis	
	Adds the Conformal Low Power commands to compare the power domains between the golden and revised designs. Power intent must be present. Type: bool Default: false Edit: Yes Reference: write_do_lec_low_power_analysis
write_do_lec_multithread_license_list	
	Specifies the license, or a list of licenses, to use for multi-threaded processing. If an empty string, the LEC default list will be used. Type: string Default: "" Edit: Yes Reference: write_do_lec_multithread_license_list
write_do_lec_no_dft_constraints	
	Do not include automatically generated DFT constraints in the dofile. Type: bool Default: false Edit: Yes Reference: write_do_lec_no_dft_constraints
write_do_lec_parallel_threads	
	Specifies number of threads for 'set parallel option -threads'. Type: int Default: 4 Edit: Yes Reference: write_do_lec_parallel_threads
write_do_lec_post_add_no_translate_modules	
	An extra dofile to be source'd from dofile right after any 'add_notranslate_modules' commands but before reading the library. Type: string Default: "" Edit: Yes Reference: write_do_lec_post_add_no_translate_modules
write_do_lec_run_hier_check_non_equivalent	

Stylus Common UI Database Object Information

Database Objects--root

	Adds '-check_noneq' option to run_hier_compare / go_hier_compare. Type: bool Default: false Edit: Yes Reference: write_do_lec_run_hier_check_non_equivalent
write_do_lec_set_cdn_syn_root	
	Whether to write out the setting for CDN_SYNTH_ROOT in the dofile. Type: bool Default: false Edit: Yes Reference: write_do_lec_set_cdn_syn_root
write_do_lec_uniquify	
	Whether to have a 'uniquify -all' command. Type: bool Default: true Edit: Yes Reference: write_do_lec_uniquify
write_do_lec_use_lec_model	
	Whether to use LEC built-in CW/DW models. Type: bool Default: true Edit: Yes Reference: write_do_lec_use_lec_model
write_do_lec_use_smart_lec	
	Adds adaptive-proof commands and information on massively parallel hierarchical compare to the dofile. Conformal Smart LEC license required. Type: bool Default: false Edit: Yes Reference: write_do_lec_use_smart_lec
write_do_lec_verbose	
	Add more verbose reports to the dofile. Type: bool Default: false Edit: Yes Reference: write_do_lec_verbose
write_lec_dft_constraints	
	This attribute is a TCL dict and holds the LEC dft pin constraints from the last LEC run by Genus. It is passed forward through write_design. The write_do_lec command will automatically include this constraints in the do file. Type: string Default: "" Edit: Yes Reference: write_lec_dft_constraints
write_lec_directory_naming_style	
	The directory name where 'write_do_lec' will write verification files when the 'write_lec_files' attribute is 'true'. The directory will be created if it does not already exist, and will overwrite an existing directory of the same name. A %s in the string is replaced with the design name (e.g. it will overwrite the directory for the same design but not for a different design). A %d in the string is replaced with a unique integer to avoid overwriting any existing directory. Type: string Default: fv/invs/%s Edit: Yes Reference: write_lec_directory_naming_style
write_lec_files	

Stylus Common UI Database Object Information

Database Objects--root

	Specifies whether 'write_do_lec' should write intermediate files to the verification directory specified by the 'write_lec_directory_naming_style' attribute. If false, no files are written. Type: bool Default: true Edit: Yes Reference: write_lec_files
write_lef_abstract_customer_header	
	Write lef abstract files with customer specified header information. Type: string Default: "" Edit: Yes Reference: write_lef_abstract_customer_header
write_netlist_full_pin_out	
	Output all pins. Type: bool Default: false Edit: Yes Reference: write_netlist_full_pin_out
write_netlist_port_association_style	
	Do implicit port mapping. Type: bool Default: false Edit: Yes Reference: write_netlist_port_association_style
write_stream_allow_path_type4	
	If false, converts PATHTYPE4 to PATHTYPE0. Type: bool Default: true Edit: Yes Reference: write_stream_allow_path_type4
write_stream_cell_instance_color	
	stream out the instance based color. The instance mask shift is honored in write_stream. When the option is on and -merge option is used, the merged gds files with mask shift suffix must be provided through option -merge. Type: bool Default: false Edit: Yes Reference: write_stream_cell_instance_color
write_stream_cell_master_color	
	stream out the cell based color from innovus color engine. Type: bool Default: true Edit: Yes Reference: write_stream_cell_master_color
write_stream_cell_name_prefix	
	Add a prefix to ALL cell names being written out, use user specified prefix can avoid potential conflict when integrating different macros, which potentially 2 different cells (in 2 different libraries) with the same name. Type: string Default: "" Edit: Yes Reference: write_stream_cell_name_prefix
write_stream_cell_name_suffix	

Stylus Common UI Database Object Information

Database Objects--root

	Instead of appending color pattern to normal cell names, use user specified suffix. This applies to fixed color shift (flip) pattern only. i.e., only single shift (flip) pattern applied to the design. Type: string Default: "" Edit: Yes Reference: write_stream_cell_name_suffix
write_stream_check_map_file	
	Check the specified map file. If the objects existed in innovus, but the objects have no objects mapping in map file, print the missed objects mapping and missed objects in log file, only the main objects mapping are checked by the option. Type: bool Default: false Edit: Yes Reference: write_stream_check_map_file
write_stream_compatible	
	Output instance only is FOREIGN is specified. Type: bool Default: false Edit: Yes Reference: write_stream_compatible
write_stream_define_via_name	
	Specifies the naming convention that the write_stream command uses for via cells declared in the DEF file. Type: string Default: default Edit: Yes Reference: write_stream_define_via_name
write_stream_ignore_fixed_mask	
	Ignore macro FIXEDMASK keywords, enable color flip function. Type: bool Default: false Edit: Yes Reference: write_stream_ignore_fixed_mask
write_stream_label_all_pin_shape	
	If true, label pin name to its all shapes. Type: bool Default: false Edit: Yes Reference: write_stream_label_all_pin_shape
write_stream_merge_append	
	Append cell color information in current design to cell geometries that are from merging files. This option is for TSMC 10nm fixed-mask design. It is exclusive with write_stream_ignore_inst_color attribute and write_stream_ignore_inst_cce_color attribute, it will set those two attributes to false automatically when it is set to true. Type: bool Default: false Edit: Yes Reference: write_stream_merge_append
write_stream_no_pin_label_on_nets	
	Do not label pin text on specified nets. Type: string Default: "" Edit: Yes Reference: write_stream_no_pin_label_on_nets
write_stream_oasis_cell_offset	

Stylus Common UI Database Object Information
Database Objects--root

	Include cell offset table to declare the byte offset from the beginning of the file (byte 0) to where the corresponding CELL record appears in the file. Type: bool Default: false Edit: Yes Reference: write_stream_oasis_cell_offset
write_stream_oasis_compression	
	CBLOCK compression is partial-selective encryption technology uses OASIS CBLOCK records to encrypt arbitrarily encapsulated sets of data records within an OASIS layout file. REPETITION compression supports an 'array' of cell placements, geometries, or text elements, instead of write each the element separate only different in position, this can save duplicate information for an OASIS layout file size. Type: bool Default: true Edit: Yes Reference: write_stream_oasis_compression
write_stream_oasis_layer_name	
	Support LAYERNAME in OASIS file. Type: bool Default: false Edit: Yes Reference: write_stream_oasis_layer_name
write_stream_pin_text_orientation	
	Output pin name on the boundary of design to improve readability. Type: string Default: "" Edit: Yes Reference: write_stream_pin_text_orientation
write_stream_remove_nets	
	Don't output nets in the list. Type: string Default: "" Edit: Yes Reference: write_stream_remove_nets
write_stream_remove_short_metal_within_cell_boundary	
	The trim metal short inside FTV_cell_list should be cut off. Type: string Default: "" Edit: Yes Reference: write_stream_remove_short_metal_within_cell_boundary
write_stream_snap_to_mfg	
	Snap net's wire to manufacture grid if it is not. Type: bool Default: false Edit: Yes Reference: write_stream_snap_to_mfg
write_stream_stream_convert_rect_to_path	
	When enabled, DEF SPECIALNETS path/rect and FILLS shapes are converted to GDSII/stream PATH data instead of BOUNDARY data. Conversion to path data can result in significantly smaller output database when there are many rectangle shapes (example: after metal fill is included). Type: bool Default: false Edit: Yes Reference: write_stream_stream_convert_rect_to_path
write_stream_stream_version	

Stylus Common UI Database Object Information

Database Objects--route_blockage

	Specify GDSII version to output. Type: int Default: 3 Edit: Yes Reference: write_stream_stream_version
write_stream_text_size	
	Changes the size of the text used in the text labels. Type: double Default: 1.0 Edit: Yes Reference: write_stream_text_size
write_stream_uniquify_cell_names_prefix	
	Adds a prefix, instead of a suffix, to uniquified cell names. Type: bool Default: false Edit: Yes Reference: write_stream_uniquify_cell_names_prefix
write_stream_via_names	
	Determines the naming convention used for vias declared in the LEF file. Type: bool Default: false Edit: Yes Reference: write_stream_via_names
write_stream_virtual_connection	
	Specifies whether a colon (:) label is appended to pin names for DEF pins with the .extraN syntax and to LEF pins with multiple ports (if - output_macros is specified for LEF pin ports that have disjoint shapes). Type: bool Default: true Edit: Yes Reference: write_stream_virtual_connection
write_verification_files	
	Specifies whether to write intermediate files to the verification directory, which is specified by the 'verification_directory_naming_style' attribute. Type: bool Default: true Edit: Yes Reference: write_verification_files

route_blockage

Parent Objects

[design](#), [root](#)

Definition

Routing blockage

Attribute	Description
base_name	
	name of blockage Type: string Default: "" Edit: Yes
density	

Stylus Common UI Database Object Information

Database Objects--route_blockage

	The density percentage allowed for a partial routing blockage. It causes the global router to only use up to this percent of the routing resource on the layer in the blockage area, so the global router will see higher congestion and put fewer routes on that layer in the area. Type: <code>int</code> Default: <code>""</code> Edit: Yes
design_rule_width	
	Specifies that the blockage has an effective width for the purposes of spacing calculations to other shapes on the same layer. A value of 0 indicates that there is no specified design_rule_width value. The design_rule_width and spacing attributes are not allowed to have non-default values at the same time, so the design_rule_width value cannot be changed when the spacing attribute's value is not the default. Type: <code>coord</code> Default: <code>no_value</code> Edit: Yes
inst	
	The instance that the routing blockage is associated with (equivalent to DEF BLOCKAGES + COMPONENT) Type: <code>obj(inst)</code> Default: <code>""</code> Edit: No
is_except_pg_net	
	Indicates that Power/Ground routing is ignored when checking for DRC violations (including shorts) involving the current shape (equivalent to DEF BLOCKAGES + EXCEPTPGNET) Type: <code>bool</code> Default: <code>false</code> Edit: Yes
is_fixed	
	Disable set_db to change any attribute for route blockages with .is_fixed=1. Type: <code>bool</code> Default: <code>false</code> Edit: Yes
is_no_wrong_way	
	Block wrong way routing. Only real routing wires are honored to not have jogs in this region. Type: <code>bool</code> Default: <code>""</code> Edit: No
is_only_pg_net	
	Indicate that the routing blockage applies only to power or ground SPECIALNETS routing. (equivalent to DEF BLOCKAGES + ONLYPGNET) Type: <code>bool</code> Default: <code>false</code> Edit: Yes
is_pushdown	
	Indicates that routing blockage has been pushed down from a higher level in the design hierarchy. The idea is that the routing blockage is owned by a higher level(equivalent to DEF BLOCKAGES + PUSHDOWN) Type: <code>bool</code> Default: <code>false</code> Edit: Yes
layer	
	layer of blockage Type: <code>obj(layer)</code> Default: <code>""</code> Edit: Yes
name	

Stylus Common UI Database Object Information

Database Objects--route_rule

	name of blockage Type: string Default: "" Edit: Yes
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (route_blockage) Default: "" Edit: No
rects	
	List of non-overlapping rectangles that defines the shape of the route_blockage. If the blockage was defined with a polygon, it is broken up into non-overlapping rectangles Type: rect* Default: "" Edit: No
shapes	
	List of shapes that define the blockage area Type: obj(shape)* Default: "" Edit: No
spacing	
	Specifies the minimum spacing allowed between the blockage and any other shape on the same layer (e.g. DEF BLOCKAGES with +SPACING). If there is no spacing value defined, the default is -1 in DB units, which will appear like a small value < 0 (e.g.-0.0005 if the get_db db_units is 2000). The design_rule_width and spacing attributes are not allowed to have non-default values at the same time, so the spacing value cannot be changed when the design_rule_width attribute's value is also set. To reset the spacing value use [set_db -dbu route_blockages .spacing -1]. Type: coord Default: no_value Edit: Yes
type	
	The routing blockage type. default: means the blockage is completely blocked and check_drc will treat it like min-width routing for spacing checks. partial: means a percentage of the routing resource is available on the layer (see density attribute). fills: means do not add any metal fill in the area. except_fills: means do not add objects except metal fills in the area.slots: means don't add slots to the area. slots is not used in Innovus but is allowed to match DEF. The type corresponds to the DEF BLOCKAGES + FILLS or + SLOTS value. There is no DEF syntax for 'partial', so partial blockages are currently lost when written to DEF. core_eol is an end-of-line blockage added by finish_floorplan - add_core_end_of_line_blockage using LEF COREEOLBLOCKAGE statements. block_eol: An end-of-line blockage added by finish_floorplan - add_block_eol_blockage using LEF BLOCKEOLBLOCKAGE statements. Type: enum Enum Values: default slots fills except_fills partial core_eol block_eol boundary_eol Default: "" Edit: Yes

route_rule

Parent Objects

[patch_wire](#), [track_pattern](#), [wire](#), [pg_base_pin](#), [net](#), [tech](#), [via](#), [route_type](#), [root](#), [base_pin](#)

Definition

Rule information

Attribute	Description
base_name	

Stylus Common UI Database Object Information

Database Objects--route_type

	Name of non-default rule. Type: string Default: "" Edit: No
is_from_lib	
	Indicates whether the rule came from the library technology (true: from LEF or OA tech) or from the design (false: from DEF, OA database, or create_route_rule). Type: bool Default: "" Edit: No
is_hard_spacing	
	Indicates that any spacing values that exceed the LEF LAYER spacing requirements are 'hard' rules instead of 'soft' rules. Type: bool Default: "" Edit: No
layer_rules	
	List of layers rules Type: obj(layer_rule)* Default: "" Edit: No
min_cuts	
	List of cut layer and minimum number of cuts allowed for any via using the specified cut layer Type: string* Default: "" Edit: No
name	
	Name of non-default rule. Type: string Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (route_rule) Default: "" Edit: No
tech	
	The tech for this route_rule. Type: obj(tech) Default: "" Edit: No
via_defs	
	List of via, default or USEVIA or derived from mincut. Type: obj(via_def)* Default: "" Edit: No

route_type

Parent Objects

[root](#)

Stylus Common UI Database Object Information

Database Objects--route_type

Definition

route_type

Attribute	Description
base_name	
	Name of route_type object. Type: <code>string</code> Default: "" Edit: No
bottom_mask_layer_num	
	Specify the bottom layer number that the mask constraint should be applied. Type: <code>int</code> Default: 0 Edit: Yes
bottom_one_side_layer_num	
	Specify the bottom layer number that one side spacing constrain should be applied on. By default, NDR spacing is applied to both sides of an NDR net or wire. Use this to specify the layer range for wires that have only one neighboring wire with minimum spacing, which means only one side needs to follow larger NDR spacing. Type: <code>int</code> Default: 0 Edit: Yes
bottom_preferred_layer	
	The preferred lowest routing layer. This attribute is a soft limit; that is, NanoRoute might use a layer below the specified layer if necessary to complete routing. Type: <code>obj(layer)</code> Default: "" Edit: No
driver_use_multi_cut_via	
	If true, then driver pins will use multi-cut vias. Type: <code>bool</code> Default: false Edit: Yes
em_route_rule	
	Specifies the EM route_rule to associate with this route type. When routing within the distance specified in route_em_rule_distance from the output pin, the router will use this route_rule to route. By default no route_em_rule is used. Type: <code>string</code> Default: "" Edit: Yes
em_route_rule_distance	
	Specifies the distance from the output pin, when the route_em_rule is applied to the net. This allows a larger width route_rule for a short distance to avoid EM violations near the output pin. When routing outside this distance from the output pin, the router uses the normal net route_rule. Type: <code>double</code> Default: no_value Edit: Yes
is_table	
	Indicates that whether the route type is table based. Type: <code>bool</code> Default: "" Edit: No
mask	

Stylus Common UI Database Object Information

Database Objects--route_type

	Indicates mask number for multiple mask layer usage. Refer to layers .numMask attribute for valid range, 0 indicates unconstrained. Layers that do not support the specified value will be treated as unconstrained. (Legal range: 0-3). Type: int Default: 0 Edit: Yes
min_stack_layer	
	The net should use a stacked via from output pins up to the given layer before starting normal routing. This is normally used to force the routing to higher layers with wider widths to reduce wiring resistance or avoid EM current limits for high-drive outputs. If output_stack_via_rule or input_stack_via_rule is also specified, the specified stack via rule is used for the input or output pins accordingly. If both output_stack_via_rule/input_stack_via_rule are not specified, a single-cut stacked-via will be used for the output pins only. Type: obj(layer) Default: "" Edit: Yes
name	
	Name of route_type object. Type: string Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (route_type) Default: "" Edit: No
route_effort	
	route effort Type: enum Enum Values: low medium high Default: "" Edit: Yes
route_rule	
	The non-default rule corresponding to the net, nets with the default routing rule will return NULL (0x0). Type: obj(route_rule) Default: "" Edit: No
shield_net	
	shield net Type: obj(net) Default: "" Edit: No
shield_side	
	Specifies whether to perform one sided or two sided shielding for the route type specified. Type: enum Enum Values: one_side both_side Default: "" Edit: Yes
shield_tap_instance_insertion_effort	

Stylus Common UI Database Object Information

Database Objects--row

	different route type for clock shielding, including high frequency(high) and low frequency(standard) that main difference is what ground tie cell instance/via used to achieve shield segments electrical connection. None: default Standard: use existing standard cell vss m0 pins as much as possible, create a new instance of a ground tie cell under the shield route if cannot find an existing vss m0 pin within a user controlled search distance of a required connection, the new cell vss pins can have a "max fanout" of 2 shield nets. High: must create ground tie cell instances, and user can specify the physical distance of these new ground tie cells along the clock segments. new instance m0 vss pin to the clock shield segments will be a via stack of single vias on ever layer to the shield nets. It is acceptable for the router to create a n-1 metal shape to connect adjacent vss shield segments together and then route this single vss shape down to m0 ground tie pin. Shield segments may be electrically connected between adjacent route layers through vias. If this is done, there shall be no Manhattan distance between ground tie cells that is larger than the user provided value. Newly inserted ground tie cells cannot be shared between clocks. Type: enum Enum Values: none standard high Default: none Edit: Yes
stack_distance	
	Specifies that the cut distance of cuts on adjacent layers in the stacked vias are defined in min_stack_layer. Type: double Default: no_value Edit: Yes
top_mask_layer_num	
	Specify the top layer number that the mask constraint should be applied. Type: int Default: 0 Edit: Yes
top_one_side_layer_num	
	Specify the top layer number that one side spacing constrain should be applied on. By default, NDR spacing is applied to both sides of an NDR net or wire. Use this to specify the layer range for wires that have only one neighboring wire with minimum spacing, which means only one side needs to follow larger NDR spacing. Type: int Default: 0 Edit: Yes
top_preferred_layer	
	The preferred highest routing layer. This attribute is a soft limit; that is, NanoRoute might use a layer above the specified layer if necessary to complete routing. Type: obj(layer) Default: "" Edit: No

row

Parent Objects

[design](#), [root](#)

Definition

Row (core), constructed from sites (equivalent to DEF ROWS)

Attribute	Description
base_name	
	Name of row (generated name) Type: string Default: "" Edit: No

Stylus Common UI Database Object Information

Database Objects--sdp

name	
	Name of row (generated name) Type: string Default: "" Edit: No
num_x	
	Number of sites in X direction (refer to DEF ROW syntax) Type: int Default: "" Edit: Yes
num_y	
	Number of sites in Y direction (refer to DEF ROW syntax) Type: int Default: "" Edit: Yes
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (row) Default: "" Edit: No
orient	
	Orientation of the sites in the row Type: enum Enum Values: r0 r90 r180 r270 mx mx90 my my90 Default: "" Edit: Yes
rect	
	Rectangle that defines the row shape Type: rect Default: "" Edit: No
site	
	The site used in the row Type: obj(site) Default: "" Edit: Yes
step_x	
	Step in the X direction (refer to DEF ROW syntax) Type: coord Default: no_value Edit: Yes
step_y	
	Step in the Y direction (refer to DEF ROW syntax) Type: coord Default: no_value Edit: Yes

sdp

Parent Objects

[inst](#), [design](#), [root](#)

Stylus Common UI Database Object Information

Database Objects--sdp

Definition

A structured datapath object. Each sdp is formed hierarchically from a list of sdps below it. Each sdp can be a row, column, space or inst (see .type). At the leaf-level, an sdp can only be a space or inst. See the create_sdp_group command for more help.

Attribute	Description
bbox	
	Bounding box of the data path. Type: <code>rect</code> Default: "" Edit: No
flip	
	Specifies if an sdp group or an sdp element is flipped in vertical, horizontal, or both directions. If an sdp group is flipped, its members will also be flipped. Type: <code>enum</code> Enum Values: none flip_x flip_y flip_xy Default: "" Edit: Yes
hier_name	
	Specifies the hierarchical path name prepended to inst names inside the sdp file. This attribute is only available when .is_top = 1. Type: <code>string</code> Default: "" Edit: Yes
insts	
	All the insts from a hierarchical descent below this sdp to the bottom of the tree. Type: <code>obj(inst)*</code> Default: "" Edit: No
is_placed	
	Specifies if the data path is placed. This attribute is only available when .is_top = 1. Type: <code>bool</code> Default: "" Edit: No
is_top	
	Indicates that the data path is a top data path group. Type: <code>bool</code> Default: "" Edit: No
justify_by	
	Specifies the anchor point that will be used for aligning an sdp group or an sdp element. If the justify_by constraint is not specified at current level, it will be inherited from its parent level. Type: <code>enum</code> Enum Values: sw se nw ne mid Default: "" Edit: Yes
local_list	
	An ordered list of sdp objects and inst objects contained by this sdp (e.g. just one-level of the sdp hierarchy). Type: <code>obj(sdp)* obj(inst)*</code> Default: "" Edit: No
location	

Stylus Common UI Database Object Information

Database Objects--sdp

	Lower left location of the data path. Type: point Default: "" Edit: Yes
name	
	Name of Data Path. Type: string Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (sdp) Default: "" Edit: No
orient	
	The orientation of an sdp group or an sdp element. If the orientation is specified at sdp group level, it will be applied to instances that belong to this sdp group. Type: enum Enum Values: r0 r180 mx my none Default: "" Edit: Yes
padding	
	Edge extend value of top SDP bounding box. Type: point Default: "" Edit: Yes
parent	
	Return parent sdp group that the sdp belongs to. Type: obj(sdp) Default: "" Edit: No
pin_alignment	
	The sdp edge to use for pin alignment. The w, e, mid values are only available for sdp objects in a column sdp group (.type = column), that have .pin_names set. All others will have a value of unknown. Type: enum Enum Values: unknown w e mid Default: "" Edit: Yes
pin_names	
	The pin names by which sdps are aligned using the .pin_alignment setting. Type: string* Default: "" Edit: Yes
sdps	
	List of all the sdps from a hierarchical descent inside and below this sdp. Type: obj(sdp)* Default: "" Edit: No
snap_row_site_index	
	Specifies the single row site that SDP bbox should snap to. Type: int Default: -1 Edit: Yes

Stylus Common UI Database Object Information

Database Objects--shape

space	
	Specifies a space value to be skipped. If the space value is defined in a column, then this value is for row skipping and represents the number of skipped rows. If the space value is defined in a row, then this value is for column skipping and represents the number of M2 tracks (pitch of first vertical routing layer). This attribute is only valid when .type = space. Type: string Default: "" Edit: Yes
type	
	Type of data path object. A row or column sdp has a .local_list with an ordered list of sdp and inst objects, while a space type only has a .space value. Type: enum Enum Values: row column space Default: "" Edit: No

shape

Parent Objects

[route_blockage](#), [place_blockage](#), [layer_shape](#)

Definition

A LEF PIN or OBS single-layer shape (RECT, POLYGON or PATH statement). Note that PATH statements are converted to one or more equivalent rect shapes. See [shape_vias](#) for a LEF VIA used in a PIN or OBS statement.

Attribute	Description
mask	
	Indicates mask number for multiple mask layer usage. Refer to layer's .num_masks attribute for legal range, 0 indicates uncolored. Type: int Default: 0 Edit: Yes
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (shape) Default: "" Edit: No
polygon	
	Points of the polygon for either rect or polygon shapes. Type: polygon Default: "" Edit: No
rect	
	Bounding box for either rect or polygon shapes. Type: rect Default: "" Edit: No
type	
	The shape type (rect or polygon). A LEF PATH statement is converted to one or more rect shapes. Type: enum Enum Values: rect polygon Default: "" Edit: No

shape_via

Parent Objects

[base_cell](#), [physical_pin](#)

Definition

layer shape via

Attribute	Description
bottom_mask	
	Indicates mask number for bottom layer for multiple mask layer usage. Refer to layer's .numMask attribute for legal range, 0 indicates uncolored Type: int Default: 0 Edit: Yes
bottom_rects	
	List of rectangles (typically only one) on bottom routing layer in terms of design coordinates (equivalent attribute on the via master is in coordinates local to the via master) Type: rect Default: "" Edit: No
cut_mask	
	Indicates mask number for cut layer for multiple mask layer usage. Applies to lower left cut of the via, other cuts are rotated from the reference cut in the lower left corner. Refer to layer's .numMask attribute for legal range, 0 indicates uncolored},cutMask,, Type: int Default: 0 Edit: Yes
cut_rects	
	List of rectangles on cut layer in terms of design coordinates (equivalent attribute on the via master is in coordinates local to the via master). Type: rect* Default: "" Edit: No
location	
	Via location Type: point Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (shape_via) Default: "" Edit: No
top_mask	
	Indicates mask number for top layer for multiple mask layer usage. Refer to layer's .numMask attribute for legal range, 0 indicates uncolored Type: int Default: 0 Edit: Yes
top_rects	

Stylus Common UI Database Object Information

Database Objects--site

	List of rectangles (typically only one) on top routing layer in terms of design coordinates (equivalent attribute on the via master is in coordinates local to the via master) Type: rect Default: "" Edit: No
via_def	
	The via master of this via Type: obj(via_def) Default: "" Edit: No

site

Parent Objects

[base_cell](#), [design](#), [tech](#), [row](#), [power_domain](#), [root](#)

Definition

LEF SITE

Attribute	Description
base_name	
	Name of site (equivalent to LEF SITE NAME) Type: string Default: "" Edit: No
class	
	Site class (equivalent to LEF SITE CLASS) Type: enum Enum Values: pad core Default: "" Edit: No
density_in_design	
	Returns the density of the specified tech site in the design. User should query density info through <code>get_db current_design .core_site.density_in_design</code> . If user queries this attribute through <code>get_db sites .density_in_design</code> when no design is loaded, the result would be meaningless. Type: double Default: "" Edit: No
name	
	Name of site (equivalent to LEF SITE NAME) Type: string Default: "" Edit: No
obj_type	
	The <code>obj_type</code> name for this object. To see all <code>obj_type</code> names, go to Contents Type: enum (site) Default: "" Edit: No
row_pattern	

Stylus Common UI Database Object Information
Database Objects--skew_group

	The row pattern for the site. (equivalent to LEF SITE ROWPATTERN) Type: string Default: "" Edit: No
size	
	Size of the site (equivalent to LEF SITE SIZE) Type: point Default: "" Edit: No
symmetry	
	site symmetry. Type: enum Enum Values: none x y xy any Default: "" Edit: Yes
tech	
	The tech for this site. Type: obj(tech) Default: "" Edit: No

skew_group

Parent Objects

[pin](#), [root](#), [port](#)

Definition

skew_group

Attribute	Description
cts_skew_group_constrains	
	Specifies how this skew group constrains the balancing of sinks during CCOpt. If set to "default", this skew group will constrain sinks during "clock_design" and during the initial global balancing step of a regular "ccopt_design" run. If set to "all", this skew group will constrain both "clock_design" and the whole of "ccopt_design" - not just the initial solution. If set to "none" this specifies that the skew group will only be used for reporting purposes. Valid values: none default all Type: string Default: default Edit: Yes
cts_skew_group_created_from_clock	
	This contains the name of the SDC clock that this skew group has been created to represent the balancing constraints in CCOpt. Valid values: string Type: string Default: "" Edit: Yes
cts_skew_group_created_from_constraint_modes	

Stylus Common UI Database Object Information

Database Objects--skew_group

	This contains the names of the constraint mode that this skew_group has been created to represent the balancing constraints in CCOpt. Valid values: string Type: string Default: "" Edit: Yes
cts_skew_group_created_from_delay_corners	
	This contains the delay corners associated with this skew_group that has been created to represent the balancing constraints in CTS. Valid values: string Type: string Default: "" Edit: Yes
cts_skew_group_include_source_latency	
	Specifies whether clock tree source latency should be included when timing the skew group. Valid values: true false Type: bool Default: false Edit: Yes
cts_skew_group_target_insertion_delay	
	The target insertion delay used for clock tree synthesis. This may be set to the following values: auto - Allow the minimum clustered insertion delay to be pushed up a little (around 5%) to facilitate clock tree power reduction. A numeric value - Attempt to balance the clock tree to the specified insertion delay (specified in library units). CTS will attempt to have a longest clock path delay of no more than this value plus half of the skew target, and a shortest path delay of no less than this value minus half the skew target. The value must be positive and real valued. Valid values: auto double Type: string Default: auto Edit: Yes
cts_target_skew	
	This specifies the target skew for clock tree balancing. This may be set to a numeric value, or one of 'auto', 'ignore' or 'default'. If set to 'auto' this indicates that an appropriate skew target should be computed. If set to 'ignore' this indicates that skew should not be balanced for this corner/path combination. If unspecified then the value of this attribute is 'default'. If the value of the attribute is 'default' the target skew for late delays in the primary delay corner is interpreted as 'auto' and as 'ignore' otherwise. Valid values: default auto ignore double Type: string Allowed -index values: delay_corner Default: default Edit: Yes
ignore_pins	
	ignored pins within the skew group Type: obj(pin)* obj(port)* Default: "" Edit: No
name	

Stylus Common UI Database Object Information

Database Objects--special_via

	name of skew_group Type: string Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (skew_group) Default: "" Edit: No
sinks	
	sinks (pins or ports) of the skew group Type: obj(pin)* obj(port)* Default: "" Edit: No
sinks_active	
	active sinks (pins or ports) of the skew group Type: obj(pin)* obj(port)* Default: "" Edit: No
sources	
	source(s) (pins or ports) of the skew group Type: obj(pin)* obj(port)* Default: "" Edit: No

special_via

Parent Objects

[net](#)

Definition

A special_via is normally used for power-vias, flip-chip routing, and sometimes pre-routed nets. special_vias (along with special_wires) are not changed during signal routing. They appear in the DEF SPECIALNETS section.

Attribute	Description
bottom_mask	
	Is the mask number for the lower, left shape on the bottom layer of the via. Normally there is only one shape on the bottom layer of a via, but if there are two or more bottom layer shapes, then the mask for the other shapes on the bottom layer are derived from the corresponding via_def mask values by "shifting" the via_def's mask values to match. See the DEF manual section on 'Multi-Mask Layers with Special Wiring' for figures and examples. A value of 0 indicates the bottom layer is uncolored, or the layer is not a multi-mask layer. Type: int Default: 0 Edit: Yes
bottom_rects	
	List of rectangles (typically only one) on bottom routing layer in terms of design coordinates (equivalent attribute on the via master is in coordinates local to the via master) Type: rect* Default: "" Edit: No
bottom_rects_mask	

Stylus Common UI Database Object Information

Database Objects--special_via

	List of mask values for each rect in bottom_rects in the same order as bottom_rects. A value of 0 means it is uncolored, or this layer is not a multi-mask layer. Type: int* Default: "" Edit: No
cut_mask	
	Is the mask number for the lower, left cut of the via. The mask for the other cuts of the special_via are derived from the via_def by "shifting" the via_def's cut masks to match. So, if the via_def lower, left cut is mask 1, and the special_via cut_mask is set to 3, then all the via_def cuts on mask 1 become mask 3 for this special_via, and similarly cuts on 2 shift to 1, and cuts on 3 shift to 2. See the layer .num_masks attribute for the max mask value allowed. See the DEF manual section on 'Multi-Mask Layers with Special Wiring' for figures and examples. A value of 0 indicates the cut layer is uncolored, or the layer is not a multi-mask layer. Type: int Default: 0 Edit: Yes
cut_rects	
	List of rectangles on cut layer in terms of design coordinates (equivalent attribute on the via master is in coordinates local to the via master) Type: rect* Default: "" Edit: No
cut_rects_mask	
	List of mask values for each rect in cut_rects in the same order as cut_rects. A value of 0 means it is uncolored, or this layer is not a multi-mask layer. Type: int* Default: "" Edit: No
net	
	The net that the special_via belongs to Type: obj(net) Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (special_via) Default: "" Edit: No
point	
	Location of Via Type: point Default: "" Edit: No
shape	
	DEF SPECIALNETS + SHAPE equivalents (ring, stripe, etc.) Type: enum Enum Values: notype ring stripe followpin iowire corewire blockwire fillwire blockagewire padding blockring drcfill fillwireopc Default: "" Edit: Yes
shield_net	
	The net that is shielded if status is shield Type: obj(net) Default: "" Edit: No
status	

Stylus Common UI Database Object Information

Database Objects--special_wire

	Wiring status (equivalent to DEF SPECIALNETS special wiring status) Type: enum Enum Values: routed fixed cover shield unknown Default: "" Edit: Yes
top_mask	
	Is the mask number for the lower, left shape on the top layer of the via. Normally there is only one shape on the top layer of a via, but if there are two or more top layer shapes, then the mask for the other shapes on the top layer are derived from the corresponding via_def mask values by "shifting" the via_def's mask values to match. See the DEF manual section on 'Multi-Mask Layers with Special Wiring' for figures and examples. A value of 0 indicates the top layer is uncolored, or the layer is not a multi-mask layer. Type: int Default: 0 Edit: Yes
top_rects	
	List of rectangles (typically only one) on top routing layer in terms of design coordinates (equivalent attribute on the via master is in coordinates local to the via master) Type: rect* Default: "" Edit: No
top_rects_mask	
	List of mask values for each rect in top_rects in the same order as top_rects. A value of 0 means it is uncolored, or this layer is not a multi-mask layer. Type: int* Default: "" Edit: No
user_class	
	User_class value (empty string is returned if no user_class is specified) Type: string Default: "" Edit: Yes
via_def	
	The via master Type: obj(via_def) Default: "" Edit: No

special_wire

Parent Objects

[net](#)

Definition

Special wire (equivalent to DEF SPECIALNETS wiring)

Attribute	Description
area	
	Area of the special wire as defined by the LEF MACRO SIZE or OVERLAP information Type: area Default: "" Edit: No
begin_extension	

Stylus Common UI Database Object Information
Database Objects--special_wire

	Extension of path at the first point (only on path type) Type: coord Default: "" Edit: No
end_extension	
	Extension of path at the last point (only on path type) Type: coord Default: "" Edit: No
has_trim_metal	
	Indicate the special wire has trim metal or not Type: bool Default: "" Edit: No
is_patch	
	Indicate this special wire is a patch wire Type: bool Default: "" Edit: No
layer	
	The layer of special_wire Type: obj(layer) Default: "" Edit: Yes
mask	
	Indicates mask number for multiple mask layer usage. Refer to layer's .numMask attribute for legal range, 0 indicates uncolored. Type: int Default: 0 Edit: Yes
net	
	The net that the special_wire belongs to Type: obj(net) Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (special_wire) Default: "" Edit: No
path	
	2 points for pathSeg center-line, n points for path center-line, n points for polygon, null for rect Type: point* Default: "" Edit: No
polygon	
	Polygon boundary for the object, the first point is not repeated as the last point in the list Type: polygon Default: "" Edit: No
rect	

Stylus Common UI Database Object Information

Database Objects--special_wire

	Rectangle that defines the special_wire shape if type = rect, or if type = path_seg and it is orthogonal. If type = polygon, or 45-degree path_seg, this is the bounding box of the shape Type: rect Default: "" Edit: Yes
shape	
	DEF SPECIALNETS + SHAPE equivalents (ring, stripe, etc.) Type: enum Enum Values: notype ring stripe followpin iowire corewire blockwire fillwire blockagewire padding blockring drofill fillwireopc Default: "" Edit: Yes
shield_net	
	The net that is shielded if status is shieldNet Type: obj(net) Default: "" Edit: No
status	
	Wiring status (equivalent to DEF SPECIALNETS special wiring status) Type: enum Enum Values: routed fixed cover shield unknown Default: "" Edit: Yes
trim_metal_color	
	Color of this special wire trim metal Type: int Default: "" Edit: No
trim_metal_rect	
	The trim_metal rect if the special_wire has a trim_metal shape attached. This only occurs for some advanced node layers that use self-aligned patterning. {0 0 0 0} is returned if there is no trim_metal attached. Type: rect Default: "" Edit: No
type	
	Type of shape (rect, polygon, etc.) Type: enum Enum Values: rect polygon path_seg path Default: "" Edit: No
user_class	
	User_class value (empty string is returned if no user_class is specified) Type: string Default: "" Edit: Yes
width	
	Width of path/pathSeg type Type: coord Default: no_value Edit: Yes

stack_via_rule

Parent Objects

[pin](#), [pg_base_pin](#), [tech](#), [root](#), [pg_pin](#), [base_pin](#)

Definition

Stack via rule information

Attribute	Description
name	
	Stack Via Rule name. Type: string Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (stack_via_rule) Default: "" Edit: No

tech

Parent Objects

[via_def](#), [route_rule](#), [base_cell](#), [design](#), [pg_base_pin](#), [root](#), [site](#), [via_def_rule](#), [layer](#), [base_pin](#)

Definition

The LEF or OA technology and library data used by a given design. It has all the layers, base_cells, route_rules, via_defs, etc. for a given process technology node. It may be used by one or more design objects.

Attribute	Description
base_cells	
	All of the base_cells for this tech. Type: obj (base_cell)* Default: "" Edit: No
base_pins	
	All of the base_pins for this tech. Type: obj (base_pin)* Default: "" Edit: No
layers	
	All of the layers for this tech. Type: obj (layer)* Default: "" Edit: No
lef_files	

Stylus Common UI Database Object Information
Database Objects--stack_via_rule

	All of the LEF files loaded to create this tech object. Type: string Default: "" Edit: No
name	
	The name of the tech. Type: string Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (tech) Default: "" Edit: No
pg_base_pins	
	All of the pg_base_pins for this tech. Type: obj (pg_base_pin)* Default: "" Edit: No
route_rule	
	All of the route_rules for this tech. Type: obj (route_rule) Default: "" Edit: No
sites	
	All of the sites for this tech. Type: obj (site)* Default: "" Edit: No
stack_via_rules	
	All of the stack_via_rules for this tech. Type: obj (stack_via_rule)* Default: "" Edit: No
tap_distance_rules	
	Returns the LEF TAPDISTANCERULE info. Type: obj (tap_distance_rule)* Default: "" Edit: No
via_def_rules	
	All of the via_def_rules for this tech. Type: obj (via_def)* Default: "" Edit: No
via_defs	
	All of the via_defs for this tech. Type: obj (via_def)* Default: "" Edit: No

text

Parent Objects

[design](#), [root](#)

Definition

Interoperable text labels

Attribute	Description
alignment	
	Horizontal and vertical alignment of the text string Type: enum Enum Values: upper_left center_left lower_left upper_center center_center lower_center upper_right center_right lower_right Default: "" Edit: No
drafting	
	Indicates if the text is always displayed left-to-right or top-to-bottom. Text will remain readable even if rotated and mirrored if this value is true. Type: bool Default: "" Edit: No
font_name	
	Font name Type: enum Enum Values: euro_style gothic math roman script stick fixed swedish mil_spec Default: "" Edit: No
font_number	
	Font Number Type: enum Enum Values: 0 1 2 3 4 5 6 7 8 Default: "" Edit: No
height	
	Text height Type: coord Default: no_value Edit: Yes
label	
	Text string value Type: string Default: "" Edit: Yes
layer	
	The layer of text Type: obj(layer) Default: "" Edit: Yes
oa_purpose	

Stylus Common UI Database Object Information

Database Objects--timing_condition

	User specified purpose name for OA text layer purpose pair support. Only values available that exist in the library's tech graph are allowed. The default value is 'drawing'. Type: string Default: "" Edit: Yes
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (text) Default: "" Edit: No
orient	
	Text Orientation Type: enum Enum Values: r0 r90 r180 r270 mx mx90 my my90 Default: "" Edit: No
point	
	Text location Type: point Default: "" Edit: No

timing_condition

Parent Objects

[delay_corner](#), [root](#)

Definition

A timing_condition represents a set of libraries at a specific operating condition - effectively defining a device corner. When assigned to a power domain - the domain uses the library and PVT bindings defined by the timing_condition. Timing conditions may be assigned to one or several power domains in the design via the delay_corner object's timing condition attributes. Use the create_timing_condition and update_timing_condition commands to create and modify timing_conditions.

Attribute	Description
library_sets	
	Specifies the associated library_set object(s). A list of library_sets (a bundle) can be used for voltage interpolation. Type: obj(library_set)* Default: "" Edit: No
name	
	Provides the name of this timing_condition object as specified by create_timing_condition. Type: string Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (timing_condition) Default: "" Edit: No
opcond	

Stylus Common UI Database Object Information

Database Objects--timing_path

	Specifies the associated opcond object. If this is not set, the opcond is derived from timing library itself. The root attribute timing_default_opcond_per_lib can also affect which opcond is used if it is not set here. Type: obj(opcond) Default: "" Edit: No
opcond_library	
	The library of the opcond for this timing condition. Can be empty if this is a virtual opcond. Type: obj(library) Default: "" Edit: No

timing_path

Parent Objects

Definition

cte timing path

Attribute	Description
actual_clock_period	
	Returns the actual_clock_period property for a timing_path Type: double Default: "" Edit: No
actual_pulse_width	
	Returns the actual_pulse_width property for a timing_path Type: double Default: "" Edit: No
applied_exception	
	Returns the applied exception on timing path. Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
arrival	
	arrival Type: double Default: "" Edit: No
arrival_mean	
	Returns the arrival_mean property for a timing_path Type: double Default: "" Edit: No
arrival_sigma	
	Returns the arrival_sigma property for a timing_path Type: double Default: "" Edit: No
asserted_network_insertion_delay	

Stylus Common UI Database Object Information

Database Objects--timing_path

	Returns the network insertion delay asserted on this timing path. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
asserted_source_insertion_delay	
	Returns the source insertion delay asserted on this timing path. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
borrowing_path_q_pin_and_transition	
	Returns the borrowing_path_q_pin_and_transition property for a timing_path Type: string Default: "" Edit: No
borrowing_path_q_pin_transition	
	Returns the borrowing_path_q_pin_transition property for a timing_path Type: string Default: "" Edit: No
capture_clock_path	
	Returns the capture_clock_path property for a timing_path Type: obj(timing_path)* Default: "" Edit: No
capture_parasitics_capacitance	
	Returns the parasitics info of launch and data nets on timing path post cppr. Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
capture_parasitics_resistance	
	Returns the parasitics info of launch and data nets on timing path post cppr. Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
capturing_clock	
	Returns the capturing_clock property for a timing_path Type: obj(clock)* Default: "" Edit: No
capturing_clock_close_edge_time	
	Returns the capturing_clock_close_edge_time property for a timing_path Type: double Default: "" Edit: No
capturing_clock_close_edge_type	
	Returns the capturing_clock_close_edge_type property for a timing_path Type: string Default: "" Edit: No

Stylus Common UI Database Object Information
Database Objects--timing_path

capturing_clock_is_inverted	
	Returns the capturing_clock_is_inverted property for a timing_path Type: <code>bool</code> Default: "" Edit: No
capturing_clock_is_propagated	
	Returns the capturing_clock_is_propagated property for a timing_path Type: <code>bool</code> Default: "" Edit: No
capturing_clock_latency	
	Returns the capturing_clock_latency property for a timing_path Type: <code>double</code> Default: "" Edit: No
capturing_clock_latency_mean	
	Returns the capturing_clock_latency_mean property for a timing_path Type: <code>double</code> Default: "" Edit: No
capturing_clock_latency_sigma	
	Returns the capturing_clock_latency_sigma property for a timing_path Type: <code>double</code> Default: "" Edit: No
capturing_clock_latency_skewness	
	Returns the capturing_clock_latency_skewness property for a timing_path Type: <code>double</code> Default: "" Edit: No
capturing_clock_latency_stddev	
	Returns the capturing_clock_latency_stddev property for a timing_path Type: <code>double</code> Default: "" Edit: No
capturing_clock_network_insertion_delay	
	Returns the network insertion delay of the capturing clock path of this timing path. Type: <code>double</code> Allowed -index values: analysis_view clock Default: "" Edit: No
capturing_clock_open_edge_time	
	capturing_clock_open_edge_time Type: <code>double</code> Default: "" Edit: No
capturing_clock_open_edge_type	
	Returns the capturing_clock_open_edge_type property for a timing_path Type: <code>string</code> Default: "" Edit: No
capturing_clock_pin	

Stylus Common UI Database Object Information
Database Objects--timing_path

	Returns the capturing_clock_pin property for a timing_path Type: obj(pin)* Default: "" Edit: No
capturing_clock_reference_edge_type	
	capturing_clock_reference_edge_type Type: string Default: "" Edit: No
capturing_clock_root_open_edge_time	
	capturing_clock_root_open_edge_time Type: double Default: "" Edit: No
capturing_clock_root_open_edge_type	
	capturing_clock_root_open_edge_type Type: string Default: "" Edit: No
capturing_clock_source_arrival_time	
	Returns the capturing_clock_source_arrival_time property for a timing_path Type: double Default: "" Edit: No
capturing_clock_source_arrival_time_mean	
	Returns the capturing_clock_source_arrival_time_mean property for a timing_path Type: double Default: "" Edit: No
capturing_clock_source_arrival_time_sigma	
	Returns the capturing_clock_source_arrival_time_sigma property for a timing_path Type: double Default: "" Edit: No
capturing_clock_source_insertion_delay	
	Returns the source insertion delay of the capturing clock path of this timing path. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
capturing_point	
	Returns a pointer to the end point of the timing path. Type: obj(port)* obj(pin)* Default: "" Edit: No
capturing_point_is_level_sensitive	
	Returns a value of true if the end point of the data path is level sensitive. Type: bool Default: "" Edit: No
check_delay	

Stylus Common UI Database Object Information
Database Objects--timing_path

	Returns the check_delay property for a timing_path Type: double Default: "" Edit: No
check_delay_mean	
	Returns the check_delay_mean property for a timing_path Type: double Default: "" Edit: No
check_delay_sigma	
	Returns the check_delay_sigma property for a timing_path Type: double Default: "" Edit: No
check_type	
	Returns the check_type property for a timing_path Type: string Default: "" Edit: No
clock_path_end_point	
	clock_path_end_point Type: obj(timing_point)* Default: "" Edit: No
clock_source_jitter	
	clock_source_jitter Type: double Default: "" Edit: No
clock_uncertainty	
	clock_uncertainty Type: double Default: "" Edit: No
clock_uncertainty_offset	
	Returns the clock uncertainty offset of timing path. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
clock_uncertainty_offset_pin	
	Returns the pin name of the clock uncertainty offset of the timing path. Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
cpr_adjustment	
	Returns the CPR adjustment of the timing path. Type: double Default: "" Edit: No
cpr_adjustment_mean	

Stylus Common UI Database Object Information
Database Objects--timing_path

	Returns the cpr_adjustment_mean property for a timing_path Type: double Default: "" Edit: No
cpr_adjustment_sigma	
	Returns the cpr_adjustment_sigma property for a timing_path Type: double Default: "" Edit: No
cpr_branch_point	
	Returns the CPPR branch point of the given timing path. Type: obj(port)* obj(pin)* Default: "" Edit: No
cumulative_manhattan_length	
	Returns the cumulative_manhattan_length property for a timing_path. Type: double Default: "" Edit: No
cumulative_manhattan_length_x	
	Returns the cumulative_manhattan_length_x property for a timing_path. Type: double Default: "" Edit: No
cumulative_manhattan_length_y	
	Returns the cumulative_manhattan_length_y property for a timing_path. Type: double Default: "" Edit: No
cycle_adjustment	
	Returns the cycle_adjustment property for a timing_path Type: double Default: "" Edit: No
data_path	
	Returns the data path arrival. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
data_path_mean	
	Returns the data path arrival mean. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
data_path_sigma	
	Returns the data path arrival sigma. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
dominant_vt_group	

Stylus Common UI Database Object Information
Database Objects--timing_path

	Returns the name of the dominant vt group of the timing path. Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
drive_adjustment	
	Returns the drive_adjustment property for a timing_path Type: double Default: "" Edit: No
drive_adjustment_mean	
	Returns the drive_adjustment_mean property for a timing_path Type: double Default: "" Edit: No
drive_adjustment_sigma	
	Returns the drive_adjustment_sigma property for a timing_path Type: double Default: "" Edit: No
external_delay	
	external_delay Type: double Default: "" Edit: No
gba_slack	
	Returns the gba_slack property for a timing_path Type: double Default: "" Edit: No
gba_slack_mean	
	Returns the gba_slack_mean property for a timing_path Type: double Default: "" Edit: No
gba_slack_sigma	
	Returns the gba_slack_sigma property for a timing_path Type: double Default: "" Edit: No
hold	
	hold Type: double Default: "" Edit: No
inter_die_skew_adjust	
	Returns the inter die skew adjustment value of the timing path. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
ipba_slack_adj	

Stylus Common UI Database Object Information

Database Objects--timing_path

	Returns the IPBA slack adjustment value. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
is_borrow_edge_adjust	
	Returns true if time borrow adjustment is done for the timing path. Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
is_clock_gating_hold	
	Returns a value of true if the path is a clock gating hold check. Type: bool Default: "" Edit: No
is_clock_gating_setup	
	Returns a value of true if the path is a clock gating setup check. Type: bool Default: "" Edit: No
is_epba_depth_exhausted	
	Returns true if epba depth is exhausted for the timing path. Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
is_invalid	
	Returns the is_invalid property for a timing_path Type: bool Default: "" Edit: No
is_ipba_bounded_path	
	Returns true if the path is IPBA bounded path Type: bool Allowed -index values: analysis_view clock Default: "" Edit: No
is_path_borrowing	
	Returns the is_path_borrowing property for a timing_path Type: bool Default: "" Edit: No
is_retimed	
	Returns the is_retimed property for a timing_path Type: bool Default: "" Edit: No
is_time_given	
	Returns the is_time_given property for a timing_path Type: bool Default: "" Edit: No

Stylus Common UI Database Object Information
Database Objects--timing_path

is_transparent_latch	
	Returns a value of true if the path is a transparent latch path. Type: bool Default: "" Edit: No
launch_clock_path	
	launch_clock_path Type: obj(timing_path)* Default: "" Edit: No
launch_parasitics_capacitance	
	Returns the parasitics info of launch and data nets on timing path post cpr. Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
launch_parasitics_resistance	
	Returns the parasitics info of launch and data nets on timing path post cpr. Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
launching_clock	
	launching_clock Type: obj(clock)* Default: "" Edit: No
launching_clock_is_inverted	
	launching_clock_is_inverted Type: bool Default: "" Edit: No
launching_clock_is_propagated	
	launching_clock_is_propagated Type: bool Default: "" Edit: No
launching_clock_latency	
	launching_clock_latency Type: double Default: "" Edit: No
launching_clock_network_insertion_delay	
	Returns the network insertion delay of the launching clock path of this timing path. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
launching_clock_open_edge_time	
	launching_clock_open_edge_time Type: double Default: "" Edit: No

Stylus Common UI Database Object Information

Database Objects--timing_path

launching_clock_open_edge_type	
	launching_clock_open_edge_type Type: <i>string</i> Default: "" Edit: No
launching_clock_source_arrival_time	
	launching_clock_source_arrival_time Type: <i>double</i> Default: "" Edit: No
launching_clock_source_arrival_time_mean	
	Returns the launching_clock_source_arrival_time_mean property for a timing_path Type: <i>double</i> Default: "" Edit: No
launching_clock_source_arrival_time_sigma	
	Returns the launching_clock_source_arrival_time_sigma property for a timing_path Type: <i>double</i> Default: "" Edit: No
launching_clock_source_insertion_delay	
	Returns the source insertion delay of the capturing clock path of this timing path. Type: <i>double</i> Allowed -index values: analysis_view clock Default: "" Edit: No
launching_input_delay	
	launching_input_delay Type: <i>double</i> Default: "" Edit: No
launching_point	
	Returns a pointer to the start point of the timing path. Type: <i>obj(port)* obj(pin)*</i> Default: "" Edit: No
launching_point_is_level_sensitive	
	Returns a value of true if the begin point of the data path is level sensitive. Type: <i>bool</i> Default: "" Edit: No
lending_path_d_pin	
	Returns the lending_path_d_pin property for a timing_path Type: <i>string</i> Default: "" Edit: No
lending_path_d_pin_transition	
	Returns the lending_path_d_pin_transition property for a timing_path Type: <i>string</i> Default: "" Edit: No
nets	

Stylus Common UI Database Object Information
Database Objects--timing_path

	Returns a collection of nets of the timing path. Type: obj(net)* Default: "" Edit: No
num_cell_arcs	
	Returns the num_cell_arcs property for a timing_path Type: double Default: "" Edit: No
num_net_arcs	
	Returns the num_net_arcs property for a timing_path Type: double Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (timing_path) Default: "" Edit: No
other_end_arrival	
	Returns the other_end_arrival property for a timing_path Type: double Default: "" Edit: No
other_end_arrival_mean	
	Returns the other_end_arrival_mean property for a timing_path Type: double Default: "" Edit: No
other_end_arrival_sigma	
	Returns the other_end_arrival_sigma property for a timing_path Type: double Default: "" Edit: No
other_end_arrival_skewness	
	Returns the other_end_arrival_skewness property for a timing_path Type: double Default: "" Edit: No
other_end_arrival_stddev	
	Returns the other_end_arrival_stddev property for a timing_path Type: double Default: "" Edit: No
path_adjust_value	
	Returns the path_adjust_value property for a timing_path Type: double Default: "" Edit: No
path_cell_delay	

Stylus Common UI Database Object Information
Database Objects--timing_path

	Returns the path_cell_delay property for a timing_path Type: double Default: "" Edit: No
path_cell_delay_mean	
	Returns the path_cell_delay_mean property for a timing_path Type: double Default: "" Edit: No
path_cell_delay_sigma	
	Returns the path_cell_delay_sigma property for a timing_path Type: double Default: "" Edit: No
path_delay	
	Returns the path_delay property for a timing_path Type: double Default: "" Edit: No
path_delay_adjustment	
	Returns the path_delay_adjustment property for a timing_path Type: double Default: "" Edit: No
path_delay_mean	
	Returns the path_delay_mean property for a timing_path Type: double Default: "" Edit: No
path_delay_sigma	
	Returns the path_delay_sigma property for a timing_path Type: double Default: "" Edit: No
path_group	
	path_group Type: obj(path_group)* Default: "" Edit: No
path_group_name	
	Returns the name of the path group for the timing path. Type: string Default: "" Edit: No
path_net_delay	
	Returns the path_net_delay property for a timing_path Type: double Default: "" Edit: No
path_net_delay_mean	

Stylus Common UI Database Object Information
Database Objects--timing_path

	Returns the path_net_delay_mean property for a timing_path Type: double Default: "" Edit: No
path_net_delay_sigma	
	Returns the path_net_delay_sigma property for a timing_path Type: double Default: "" Edit: No
path_ptr_loss	
	Returns the path_ptr_loss property for a timing_path Type: double Default: "" Edit: No
path_type	
	Reports max type for late path and min type for early path Type: string Default: "" Edit: No
period	
	Returns the period property for a timing_path Type: double Default: "" Edit: No
phase_shift	
	Returns the phase_shift property for a timing_path Type: double Default: "" Edit: No
ptr_mean	
	Returns the ptr_mean property for a timing_path Type: double Default: "" Edit: No
ptr_sigma	
	Returns the ptr_sigma property for a timing_path Type: double Default: "" Edit: No
ptr_skewness	
	Returns the ptr_skewness property for a timing_path Type: double Default: "" Edit: No
ptr_slack	
	Returns the PTR Slack Time property for a timing_path Type: double Default: "" Edit: No
real_gba_arrival	

Stylus Common UI Database Object Information
Database Objects--timing_path

	Returns the real_gba_arrival property for a timing_path Type: double Default: "" Edit: No
real_gba_arrival_mean	
	Returns the real_gba_arrival_mean property for a timing_path Type: double Default: "" Edit: No
real_gba_arrival_sigma	
	Returns the real_gba_arrival_sigma property for a timing_path Type: double Default: "" Edit: No
real_gba_arrival_skewness	
	Returns the real_gba_arrival_skewness property for a timing_path Type: double Default: "" Edit: No
real_gba_arrival_stddev	
	Returns the real_gba_arrival_stddev property for a timing_path Type: double Default: "" Edit: No
real_gba_slack	
	Returns the real_gba_slack property for a timing_path Type: double Default: "" Edit: No
real_gba_slack_mean	
	Returns the real_gba_slack_mean property for a timing_path Type: double Default: "" Edit: No
real_gba_slack_sigma	
	Returns the real_gba_slack_sigma property for a timing_path Type: double Default: "" Edit: No
real_gba_slack_skewness	
	Returns the real_gba_slack_skewness property for a timing_path Type: double Default: "" Edit: No
real_gba_slack_stddev	
	Returns the real_gba_slack_stddev property for a timing_path Type: double Default: "" Edit: No
recovery	

Stylus Common UI Database Object Information

Database Objects--timing_path

	Returns the recovery time at the end point of the path. Type: double Default: "" Edit: No
removal	
	removal Type: double Default: "" Edit: No
required_time	
	Returns the required time for the timing path. Type: double Default: "" Edit: No
required_time_mean	
	Returns the required_time_mean property for a timing_path Type: double Default: "" Edit: No
required_time_sigma	
	Returns the required_time_sigma property for a timing_path Type: double Default: "" Edit: No
setup	
	The setup time at the end point of the path. Type: double Default: "" Edit: No
skew	
	Returns the skew property for a timing_path Type: double Default: "" Edit: No
slack	
	Returns the slack of the timing path. Type: double Default: "" Edit: No
slack_adjusted	
	Returns the adjusted slack value. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
slack_mean	
	Returns the slack_mean property for a timing_path Type: double Default: "" Edit: No
slack_sigma	

Stylus Common UI Database Object Information
Database Objects--timing_path

	Returns the slack_sigma property for a timing_path Type: double Default: "" Edit: No
spatial_distance	
	Returns the spatial_distance property for a timing_path Type: double Default: "" Edit: No
time_borrowed	
	Returns the amount of time borrowed from the timing end point. Type: double Default: "" Edit: No
time_lent	
	Returns the amount of time lent to the timing start point. Type: double Default: "" Edit: No
timing_points	
	timing_points Type: obj(timing_point)* Default: "" Edit: No
total_cell_delta_delay	
	Returns the total_cell_delta_delay property for a timing_path Type: double Default: "" Edit: No
total_cell_delta_delay_mean	
	Returns the total_cell_delta_delay_mean property for a timing_path Type: double Default: "" Edit: No
total_cell_delta_delay_sigma	
	Returns the total_cell_delta_delay_sigma property for a timing_path Type: double Default: "" Edit: No
total_delta_delay	
	Returns the total_delta_delay property for a timing_path Type: double Default: "" Edit: No
total_delta_delay_mean	
	Returns the total_delta_delay_mean property for a timing_path Type: double Default: "" Edit: No
total_delta_delay_sigma	

Stylus Common UI Database Object Information
Database Objects--timing_path

	Returns the total_delta_delay_sigma property for a timing_path Type: double Default: "" Edit: No
total_net_delta_delay	
	Returns the total_net_delta_delay property for a timing_path Type: double Default: "" Edit: No
total_net_delta_delay_mean	
	Returns the total_net_delta_delay_mean property for a timing_path Type: double Default: "" Edit: No
total_net_delta_delay_sigma	
	Returns the total_net_delta_delay_sigma property for a timing_path Type: double Default: "" Edit: No
underated_slack	
	Returns the underated slack time for the timing path. The underated slack value is available with timing path object only if it is generated using the -derate parameter. Type: double Default: "" Edit: No
view_name	
	The name of the analysis view for the timing_path Type: string Default: "" Edit: No
worst_cell_delay	
	Returns the worst_cell_delay property for a timing_path Type: double Default: "" Edit: No
worst_cell_delay_mean	
	Returns the worst_cell_delay_mean property for a timing_path Type: double Default: "" Edit: No
worst_cell_delay_sigma	
	Returns the worst_cell_delay_sigma property for a timing_path Type: double Default: "" Edit: No
worst_cell_delta_delay	
	Returns the worst_cell_delta_delay property for a timing_path Type: double Default: "" Edit: No
worst_cell_delta_delay_mean	

Stylus Common UI Database Object Information
Database Objects--timing_path

	Returns the worst_cell_delta_delay_mean property for a timing_path Type: double Default: "" Edit: No
worst_cell_delta_delay_sigma	
	Returns the worst_cell_delta_delay_sigma property for a timing_path Type: double Default: "" Edit: No
worst_delay	
	Returns the worst_delay property for a timing_path Type: double Default: "" Edit: No
worst_delay_mean	
	Returns the worst_delay_mean property for a timing_path Type: double Default: "" Edit: No
worst_delay_sigma	
	Returns the worst_delay_sigma property for a timing_path Type: double Default: "" Edit: No
worst_delta_delay	
	Returns the worst_delta_delay property for a timing_path Type: double Default: "" Edit: No
worst_delta_delay_mean	
	Returns the worst_delta_delay_mean property for a timing_path Type: double Default: "" Edit: No
worst_delta_delay_sigma	
	Returns the worst_delta_delay_sigma property for a timing_path Type: double Default: "" Edit: No
worst_manhattan_length	
	Returns the worst_manhattan_length property for a timing_path Type: double Default: "" Edit: No
worst_manhattan_length_net_name	
	Returns the worst_manhattan_length_net_name property for a timing_path. Type: string Default: "" Edit: No
worst_net_delay	

Stylus Common UI Database Object Information

Database Objects--timing_point

	Returns the worst_net_delay property for a timing_path Type: double Default: "" Edit: No
worst_net_delay_mean	
	Returns the worst_net_delay_mean property for a timing_path Type: double Default: "" Edit: No
worst_net_delay_sigma	
	Returns the worst_net_delay_sigma property for a timing_path Type: double Default: "" Edit: No
worst_net_delta_delay	
	Returns the worst_net_delta_delay property for a timing_path Type: double Default: "" Edit: No
worst_net_delta_delay_mean	
	Returns the worst_net_delta_delay_mean property for a timing_path Type: double Default: "" Edit: No
worst_net_delta_delay_sigma	
	Returns the worst_net_delta_delay_mean property for a timing_path Type: double Default: "" Edit: No

timing_point

Parent Objects

[timing_path](#)

Definition

cte timing point

Attribute	Description
arrival	
	Returns the arrival time at the timing point. Type: double Default: "" Edit: No
arrival_mean	
	Returns the arrival_mean property for a timing_point Type: double Default: "" Edit: No
arrival_sigma	

Stylus Common UI Database Object Information

Database Objects--timing_point

	Returns the arrival_sigma property for a timing_point Type: double Default: "" Edit: No
arrival_skewness	
	Returns the skewness in arrival at timing point. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
clock_unroll_adjustment	
	Returns the clock unrolling adjustment at the timing point. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
delay	
	Returns the delay property for a timing_point. Type: double Default: "" Edit: No
delay_effective	
	Returns the effective delay at the timing point. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
delay_mean	
	Returns the delay_mean property of a timing_point Type: double Default: "" Edit: No
delay_sigma	
	Returns the delay_sigma property for a timing_point. Type: double Default: "" Edit: No
delay_skewness	
	Returns the skewness in delay at the timing point. Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
delta_delay	
	Returns the delta_delay property for a timing_point Type: double Default: "" Edit: No
direction	
	Returns the direction property for a timing_point Type: string Default: "" Edit: No

Stylus Common UI Database Object Information
Database Objects--timing_point

hierarchical_name	
	Returns the hierarchical_name property for a timing_point Type: string Default: "" Edit: No
is_fanin_arc_cell_arc	
	Returns cell arc in the fanin of the timing point Type: bool Default: "" Edit: No
is_hierarchical	
	Returns the is_hierarchical property for a timing_point Type: bool Default: "" Edit: No
load	
	Returns the total load seen at the timing point Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
local_ref_clock	
	Returns the local reference clock name of the timing point Type: string Allowed -index values: analysis_view clock Default: "" Edit: No
local_slack	
	Returns the local slack at the timing point. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
local_slack_adjusted	
	Returns the local adjusted slack at the timing point. Type: double Allowed -index values: analysis_view clock Default: "" Edit: No
net	
	Reports the nets connected to the timing point Type: obj(net)* Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (timing_point) Default: "" Edit: No
pin	

Stylus Common UI Database Object Information
Database Objects--timing_point

	Returns the pin property for a timing_point Type: <code>obj(port)*obj(pin)*</code> Default: "" Edit: No
si_object	
	Returns the si_object property for a timing_point Type: <code>obj(delay_noise_obj)</code> Default: "" Edit: No
slack	
	Returns the slack property for a timing_point Type: <code>double</code> Default: "" Edit: No
slack_mean	
	Returns the slack_mean property for a timing_point Type: <code>double</code> Default: "" Edit: No
slack_sigma	
	Returns the slack_sigma property for a timing_point Type: <code>double</code> Default: "" Edit: No
slack_skewness	
	Returns the skewness in slack at the timing point. Type: <code>double</code> Allowed -index values: analysis_view clock Default: "" Edit: No
slew	
	Returns the path-specific slew of the timing point. Type: <code>double</code> Default: "" Edit: No
slew_mean	
	Returns the slew_mean property for a timing_point. Type: <code>double</code> Default: "" Edit: No
slew_sigma	
	Returns the slew_sigma property for a timing_point Type: <code>double</code> Default: "" Edit: No
slew_skewness	
	Returns the skewness in slew at the timing point Type: <code>double</code> Allowed -index values: analysis_view clock Default: "" Edit: No
spatial_derate	

Stylus Common UI Database Object Information

Database Objects--track_pattern

	Returns the spatial_derate property for a timing_point Type: double Default: "" Edit: No
spatial_derate_sigma	
	Returns the spatial_derate_sigma property for a timing_point Type: double Default: "" Edit: No
ssi_derate	
	Returns the path-specific ssi_derate of the timing point. Type: double Default: "" Edit: No
total_derate	
	Returns the total_derate property for a timing_point Type: double Default: "" Edit: No
total_derate_sigma	
	Returns the total_derate_sigma property for a timing_point Type: double Default: "" Edit: No
transition_type	
	Returns the transition type of the timing point: rise or fall . Type: string Default: "" Edit: No
user_derate	
	Returns the user_derate property for a timing_point Type: double Default: "" Edit: No
user_derate_sigma	
	Returns the user_derate_sigma property for a timing_point Type: double Default: "" Edit: No
voltage	
	Returns the voltage property for a timing_point Type: double Default: "" Edit: No

track_pattern

Parent Objects

design, root

Stylus Common UI Database Object Information

Database Objects--track_pattern

Definition

Floorplan track information (DEF TRACKS equivalent)

Attribute	Description
direction	
	Specifies the location and direction of the first track defined. x indicates vertical lines; y indicates horizontal lines. Type: enum Enum Values: y x Default: "" Edit: No
layers	
	List of layers Type: obj(layer)* Default: "" Edit: No
mask	
	Specifies the mask number for the first track. Type: int Default: "" Edit: No
num_tracks	
	Specifies the number of tracks to create for the grid Type: int Default: "" Edit: Yes
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (track_pattern) Default: "" Edit: No
route_rule	
	Pointer to route_rule object (DEF NONDEFAULTRULE) to be used as a constraint for wiring that can be created on the track. An empty value indicates that there is no constraint on the rules of the wires on the track. This value can be set with add_tracks and is intended for advanced nodes that do not allow different route_rules on the same track for lower routing layers. Type: obj(route_rule) Default: "" Edit: No
start	
	Specifies the coordinate of the first line Type: coord Default: no_value Edit: Yes
step	
	Specifies the spacing between the tracks Type: coord Default: no_value Edit: Yes
width	

	Width constraint for wiring that can be created on the track. A value of 0 indicates that there is no constraint on the width of wires on the track. This value can be set with <code>add_tracks</code> and is intended for advanced nodes that do not allow different widths on the same track for lower routing layers. Type: <code>coord</code> Default: "" Edit: No
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trim_grid

Parent Objects

[design](#), [root](#)

Definition

Trim grid pattern. Equivalent to LEF TRIMMETALTRACK statement that defines the grids that trim shapes must align to. See the LEF manual for more descriptions, figures and examples of how these attributes are defined.

Attribute	Description
group	
	Specifies the group names of the trim_grid from the LEF TRIMMETALSTACK GROUP value. Type: <code>string</code> Default: {} Edit: No
layer	
	The trim layer of this trim_grid from the LEF TRIMMETALSTACK trimLayer value Type: <code>obj(layer)</code> Default: {} Edit: No
mask	
	Specifies the mask number for the trim_grid, from the LEF TRIMMETALSTACK MASK value. Type: <code>int</code> Default: {} Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: <code>enum (trim_grid)</code> Default: "" Edit: No
offset	
	Specifies the offset from the lower left point of the core box to the first trim_grid from the LEF TRIMMETALSTACK COREOFFSET value. Type: <code>coord</code> Default: {} Edit: No
on_track	
	True if the LEF TRIMMETALSTACK ONTRACK keyword is given for this trim_grid. Type: <code>bool</code> Default: false Edit: No
pitch	

Stylus Common UI Database Object Information

Database Objects--via

	Specifies the spacing between each trim_grid from the LEF TRIMMETALSTACK PITCH value. Type: coord Default: {} Edit: No
track_count	
	The number of times to repeat the pitch for this track_index from the LEF TRIMMETALSTACK METALTRACKPITCH value. See the LEF manual for figures and examples of how this is used. Type: int Default: 0 Edit: No
track_index	
	Specifies the index value of the trim_grid from the LEF TRIMMETALSTACK METALTRACKOFFSET value. The offset and pitch values apply to this track_index for track_count times. See the LEF manual for figures and examples of how this is used to overlay different track_index settings on one layer on top of each other to create non-uniform grid patterns. Type: int Default: 0 Edit: No

via

Parent Objects

[net](#)

Definition

DEF NETS via instance

Attribute	Description
bottom_mask	
	Is the mask number for the lower, left shape on the bottom layer of the via. Normally there is only one shape on the bottom layer of a via, but if there are two or more bottom layer shapes, then the mask for the other shapes on the bottom layer are derived from the corresponding via_def mask values by "shifting" the via_def's mask values to match. See the DEF manual section on 'Multi-Mask Patterns for Routing Points' for figures and examples. A value of 0 indicates the bottom layer is uncolored, or the layer is not a multi-mask layer. Type: int Default: 0 Edit: Yes
bottom_rects	
	List of rectangles (typically only one) on bottom routing layer in terms of design coordinates (equivalent attribute on the via master is in coordinates local to the via master) Type: rect* Default: "" Edit: No
bottom_rects_mask	
	List of mask values for each rect in bottom_rects in the same order as bottom_rects. A value of 0 means it is uncolored, or this layer is not a multi-mask layer. Type: int* Default: "" Edit: No
cut_mask	

Stylus Common UI Database Object Information

Database Objects--via

	Is the mask number for the lower, left cut of the via. The mask for the other cuts of the via are derived from the via_def by "shifting" the via_def's cut masks to match. So, if the via_def lower, left cut is mask 1, and the via cut_mask is set to 3, then all the via_def cuts on mask 1 become mask 3 for this via, and similarly cuts on 2 shift to 1, and cuts on 3 shift to 2. See the layer .num_masks attribute for the max mask value allowed. See the DEF manual section on 'Multi-Mask Patterns for Routing Points' for figures and examples. A value of 0 indicates the cut layer is uncolored, or the layer is not a multi-mask layer. Type: int Default: no_value Edit: Yes
cut_rects	
	List of rectangles on cut layer in terms of design coordinates (equivalent attribute on the via master is in coordinates local to the via master) Type: rect Default: "" Edit: No
cut_rects_mask	
	List of mask values for each rect in cut_rects in the same order as cut_rects. A value of 0 means it is uncolored, or this layer is not a multi-mask layer. Type: int* Default: "" Edit: No
location	
	Location of Via Type: point Default: "" Edit: No
net	
	The net that the via belongs to Type: obj(net) Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (via) Default: "" Edit: No
route_rule	
	The non-default rule corresponding to the via, vias with the default routing rule will return an empty string. Type: obj(route_rule) Default: "" Edit: No
status	
	Wiring status (equivalent to DEF NETS regular wiring status) Type: enum Enum Values: unknown routed fixed cover noshield Default: "" Edit: Yes
top_mask	
	Indicates mask number for top layer for multiple mask layer usage. Refer to layer's .numMask attribute for legal range, 0 indicates uncolored Type: int Default: 0 Edit: Yes
top_rects	

Stylus Common UI Database Object Information

Database Objects--via_def

	List of rectangles (typically only one) on top routing layer in terms of design coordinates (equivalent attribute on the via master is in coordinates local to the via master) Type: rect Default: "" Edit: No
top_rects_mask	
	List of mask values for each rect in top_rects in the same order as top_rects. A value of 0 means it is uncolored, or this layer is not a multi-mask layer. Type: int* Default: "" Edit: No
via_def	
	The via cell Type: obj(via_def) Default: "" Edit: No

via_def

Parent Objects

[route_rule](#), [tech](#), [shape_via](#), [via](#), [special_via](#), [root](#)

Definition

A via definition. This is equivalent to the LEF/DEF VIA section statements. A via_def can be either a parameterized via (VIA with VIARULE parameters), or a fixed via (VIA with just RECT statements). See the LEF/DEF manual VIA section for more details.

Attribute	Description
bottom_layer	
	The bottom routing layer Type: obj(layer) Default: "" Edit: No
bottom_rects	
	List of rectangles (typically only one) on bottom routing layer. Type: rect* Default: "" Edit: No
bottom_rects_mask	
	List of mask values for each rect in bottom_rects in the same order as bottom_rects. A value of 0 means it is uncolored, or this layer is not a multi-mask layer. Type: int* Default: no_value Edit: Yes
cut_class	
	Returns the name of the CUTCLASS definition (from LEF or OA tech) for the cut_rects in this via_def. It returns an empty string if no CUTCLASS exists for the cut layer, or no CUTCLASS matches the size of the cut_rects. Type: string Default: "" Edit: No
cut_columns	

Stylus Common UI Database Object Information

Database Objects--via_def

	The number of cut columns. It is only set for generated vias created from a via_def_rule. See the LEF/DEF manual VIA definition with VIARULE and ROWCOL values for more details. It is 0 for fixed vias (e.g. a LEF/DEF VIA definition with only RECT values). Type: int Default: 0 Edit: No
cut_layer	
	The cut layer of via Type: obj(layer) Default: "" Edit: No
cut_rects	
	List of rectangles on cut layer Type: rect* Default: "" Edit: No
cut_rects_mask	
	List of mask values for each rect in cut_rects in the same order as cut_rects. A value of 0 means it is uncolored, or this layer is not a multi-mask layer. Type: int* Default: no_value Edit: Yes
cut_rows	
	The number of cut rows. It is only set for generated vias created from a via_def_rule. See the LEF/DEF manual VIA definition with VIARULE and ROWCOL values for more details. It is 0 for fixed vias (e.g. a LEF/DEF VIA definition with only RECT values). Type: int Default: 0 Edit: No
cut_size	
	The {width height} of the first rect in cut_rects in microns. It is only set for generated vias created from a via_def_rule. See the LEF/DEF manual VIA definition with VIARULE and CUTSIZE values for more details. It is {0 0} for fixed vias (e.g. a LEF/DEF VIA definition with only RECT values). Type: point Default: "" Edit: No
from_design	
	Indicates that the via is from design from_design = true && from_lib = false: The via is generated from DEF or create_via_definition. from_design = false && from_lib = true: The via is generated from LEF. from_design = true && from_lib = true: The via is generated by NanoRoute. from_design = false && from_lib = false: The via is generated by VIAGEN. Type: bool Default: "" Edit: No
from_lib	
	Indicates that the via is from library. See the from_design attribute for more details. Type: bool Default: "" Edit: No
is_default	
	Indicates that the via is a default via (LEF VIA DEFAULT) Type: bool Default: "" Edit: No
is_non_default	

Stylus Common UI Database Object Information

Database Objects--via_def

	Indicates that the via is declared in a LEF NONDEFAULTRULE Type: <code>bool</code> Default: "" Edit: No
name	
	Via name Type: <code>string</code> Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: <code>enum</code> (via_def) Default: "" Edit: No
resistance	
	Via resistance in ohms that is derived from LEF or OA data. This may not match the resistance of RC extraction results derived from extraction coefficient data. If the via is a fixed via with a resistance value defined in the LEF VIA definition statement or OA via_def, that value is returned. For vias without a resistance value, the resistance is computed from the cut-layer resistance_per_cut value and the number of cuts in the via (or equivalent cuts for a LEF CUTCLASS or OA cut_class via with different cut sizes). If both the via definition, and the cut-layer has no resistance value, then 0.0 is returned Type: <code>double</code> Default: "" Edit: No
tech	
	The tech for this via_def. Type: <code>obj(tech)</code> Default: "" Edit: No
top_layer	
	The top routing layer of via Type: <code>obj(layer)</code> Default: "" Edit: No
top_rects	
	List of rectangles (typically only one) on top routing layer. Type: <code>rect*</code> Default: "" Edit: No
top_rects_mask	
	List of mask values for each rect in top_rects in the same order as top_rects. A value of 0 means it is uncolored, or this layer is not a multi-mask layer. Type: <code>int*</code> Default: no_value Edit: Yes
via_def_rule	
	The via_def_rule for this via_def. It is only set for generated vias created from via_def_rule parameters. See the LEF/DEF manual VIA statement with VIARULE for more details. It returns {} for fixed vias (e.g. a LEF/DEF VIA with only RECT values). Type: <code>obj(via_def_rule)</code> Default: "" Edit: No

via_def_rule

Parent Objects

via_def, root

Definition

Equivalent of one LEF VIARULE GENERATE statement that has the rules to create new generated vias (e.g. LEF/DEF VIA statements with VIARULE parameters). See the LEF manual for more details.

Attribute	Description
bottom_enclosure	
	Two minimum enclosure values for the cuts in the via. The order of the two values does not matter. The bottom layer shape must enclose all the cuts by one of the enclosure values in one direction (e.g. either X or Y), and by the other enclosure value in the other direction. If it is {0 0}, which is recommended for newer technologies, then only the DRC rules are used to compute the minimum enclosure.. Type: point Default: "" Edit: No
bottom_layer	
	The bottom routing layer for the via. Type: obj(layer) Default: "" Edit: No
bottom_width	
	Optional min and max width. If given, this rule should only be used if the bottom wire width is greater than or equal to the first value (min-width), and less than or equal to the second value (max-width). For example, {1 2} means min-width is ≥ 1.0 , and max_width is ≤ 2.0 . If not given, the default is {0 0}, which means this rule can be used for any width. Type: point Default: "" Edit: No
cut_layer	
	The cut layer for the via. Type: obj(layer) Default: "" Edit: No
cut_rect	
	The size of one cut rectangle. Type: rect Default: "" Edit: No
cut_spacing	
	Minimum center-to-center spacing in the X and Y directions.. Type: point Default: "" Edit: No
name	
	Via rule name. Type: string Default: "" Edit: No
obj_type	

Stylus Common UI Database Object Information

Database Objects--virtual_wire

	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (via_def_rule) Default: "" Edit: No
resistance_per_cut	
	Optional via resistance per cut in ohms that is defined in LEF or OA data. This value is useful for estimation, but will not match the resistance extracted by RC extraction commands that use more accurate coefficient files. It is 0.0 if not given in the library data. Type: double Default: "" Edit: No
tech	
	The tech for this via_def_rule. Type: obj(tech) Default: "" Edit: No
top_enclosure	
	Two minimum enclosure values for the cuts in the via. The order of the two values does not matter. The top layer shape must enclose all the cuts by one of the enclosure value in one direction (e.g. either X or Y), and by the other enclosure value in the other direction. If it is {0 0}, which is recommended, then only the DRC rules are used to compute the minimum enclosure. Type: point Default: "" Edit: No
top_layer	
	The top routing layer for the via. Type: obj(layer) Default: "" Edit: No
top_width	
	Optional min and max width. If given, this rule should only be used if the top wire width is greater than or equal to the first value (min-width), and less than or equal to the second value (max-width). For example, {1 2} means min-width is >= 1.0, and max_width is <= 2.0). If not given, the default is {0 0}, which means this rule can be used for any width. Type: point Default: "" Edit: No

virtual_wire

Parent Objects

[net](#)

Definition

DEF NETS VIRTUAL wire

Attribute	Description
begin_layer	
	The begin layer for the virtual connection Type: obj(layer) Default: "" Edit: No
begin_point	

Stylus Common UI Database Object Information

Database Objects--what_if_via

	Reference point to the begin symbolic location Type: point Default: "" Edit: No
end_layer	
	The end layer for the virtual connection Type: obj(layer) Default: "" Edit: No
end_point	
	Reference point to the end symbolic location Type: point Default: "" Edit: No
net	
	The net that the virtual connection belongs to Type: obj(net) Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (virtual_wire) Default: "" Edit: No
status	
	Wiring status (equivalent to DEF NET regular wiring status) Type: enum Enum Values: unknown routed fixed cover noshield Default: "" Edit: No

what_if_via

Parent Objects

[net](#)

Definition

After loading a Voltus IR-drop analysis result, you can use add_what_if_shapes to add 'what if' power vias to see how they would improve the IR-drop results without modifying the real power-mesh.

Attribute	Description
bottom_layer	
	The bottom routing layer for this what_if_via. Type: obj(layer) Default: "" Edit: No
cut_layer	
	The cut_layer for this what_if_via. Type: obj(layer) Default: "" Edit: No

Stylus Common UI Database Object Information

Database Objects--what_if_wire

cut_rect	
	The rect on the cut_layer for this what_if_via. Type: rect Default: "" Edit: No
net	
	The net for this what_if_via. Type: obj(net) Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (what_if_via) Default: "" Edit: No
top_layer	
	The top routing layer for this what_if_via. Type: obj(layer) Default: "" Edit: No

what_if_wire

Parent Objects

[net](#)

Definition

After loading a Voltus IR-drop analysis result, you can use add_what_if_shapes to add 'what if' power wires to see how they would improve the IR-drop results without modifying the real power-mesh.

Attribute	Description
layer	
	The layer for the wire. Type: obj(layer) Default: "" Edit: No
net	
	The net of the wire. Type: obj(net) Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (what_if_wire) Default: "" Edit: No
rect	

Stylus Common UI Database Object Information

Database Objects--wire

	The rect for the wire. Type: <code>rect</code> Default: "" Edit: No
width	
	The width of the wire. Type: <code>coord</code> Default: "" Edit: No

wire

Parent Objects

`net`

Definition

A 2-point wire segment intended for symbolic routing where the end-points normally align with other wire or via end-points, or end on a pin shape or special_wire. The wire, via, and patch_wire objects correspond to the DEF NETS symbolic routing data.

Attribute	Description
begin_extension	
	Extension of wire at the first point Type: <code>coord</code> Default: no_value Edit: Yes
direction	
	Direction of wire, consistent with layer direction from LEF/OpenAccess Type: <code>enum</code> Enum Values: vertical other horizontal Default: "" Edit: No
end_extension	
	Extension of wire at the second point Type: <code>coord</code> Default: no_value Edit: Yes
layer	
	The layer of wire Type: <code>obj(layer)</code> Default: "" Edit: No
length	
	The center-line length of the wire between the two end points. It does not include the extension values Type: <code>coord</code> Default: "" Edit: No
mask	
	Indicates mask number for multiple mask layer usage. Refer to layer's .numMask attribute for legal range, 0 indicates uncolored. Type: <code>int</code> Default: 0 Edit: Yes

Stylus Common UI Database Object Information

Database Objects--port_bus

net	
	The net that the wire belongs to Type: obj(net) Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (wire) Default: "" Edit: No
points	
	2 points (center-line) for the wire Type: point* Default: "" Edit: No
rect	
	Rectangle that defines the wire shape, including any begin extension or end extension. Type: rect Default: "" Edit: No
route_rule	
	The non-default rule corresponding to the wire, wires with the default routing rule will return empty string. Type: obj(route_rule) Default: "" Edit: Yes
status	
	Wiring status (equivalent to DEF NETS regular wiring status) Type: enum Enum Values: unknown routed fixed cover noshield Default: "" Edit: Yes
width	
	Width of wire Type: coord Default: "" Edit: No

port_bus

Parent Objects

[design](#), [root](#), [port](#)

Definition

The bus on port

Attribute	Description
base_name	

Stylus Common UI Database Object Information
Database Objects--hport_bus

	Non-vector portion of the bus name. (e.g. A for bus A[7:0]). Type: string Default: "" Edit: No
bits	
	A list of individual ports that constitute the bus. Type: obj(port)* Default: "" Edit: No
design	
	The design of the port bus. Type: obj(design) Default: "" Edit: No
direction	
	Returns the direction of the port bus. Type: enum Enum Values: internal in out inout Default: "" Edit: No
escaped_name	
	The escaped name of the port bus. Type: string Default: "" Edit: No
lsb	
	Least significant bit. Returns the bottom bit index of the bus. Type: int Default: "" Edit: No
msb	
	Most significant bit. Returns the top bit index of the bus. Type: int Default: "" Edit: No
name	
	The name of the port bus Type: string Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (port_bus) Default: "" Edit: No

hport_bus

Parent Objects

[hinst](#), [design](#), [root](#), [hport](#)

Stylus Common UI Database Object Information

Database Objects--hport_bus

Definition

The bus on hport

Attribute	Description
base_name	
	Non-vector portion of the bus name (eg. A for bus A[7:0]) Type: string Default: "" Edit: No
bits	
	The ports of the bus. All the bits of one bus will have the same obj_type. Type: obj(hport)* Default: "" Edit: No
direction	
	Returns the direction of the hport_bus. Type: enum Enum Values: internal in out inout Default: "" Edit: No
escaped_name	
	The escaped name of the hport_bus Type: string Default: "" Edit: No
hinst	
	Returns the hierarchical instance to which the hport_bus belongs. Type: obj(hinst) Default: "" Edit: No
lsb	
	Bottom bit index of hport_bus. Type: int Default: "" Edit: No
msb	
	Top bit index of hport_bus. Type: int Default: "" Edit: No
name	
	The name of hport_bus. Type: string Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (hport_bus) Default: "" Edit: No

hpin_bus

Parent Objects

[hinst](#), [design](#), [hpin](#), [root](#)

Definition

The bus on hpin

Attribute	Description
base_name	
	Non-vector portion of the bus name (eg. A for bus A[7:0]) Type: string Default: "" Edit: No
bits	
	The ports of the bus. All the bits of one bus will have the same obj_type. Type: obj(hpin)* Default: "" Edit: No
direction	
	Returns the direction of the hpin_bus. Type: enum Enum Values: internal in out inout Default: "" Edit: No
escaped_name	
	The escaped name of the hpin_bus Type: string Default: "" Edit: No
hinst	
	Returns the hierarchical instance to which the hpin_bus belongs. Type: obj(hinst) Default: "" Edit: No
lsb	
	Bottom bit index of bus. Type: int Default: "" Edit: No
msb	
	Top bit index of bus. Type: int Default: "" Edit: No
name	
	The name of hpin_bus. Type: string Default: "" Edit: No

obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (hpin_bus) Default: "" Edit: No

test_signal

Parent Objects

[inst](#), [design](#), [test_clock](#), [root](#), [dft_configuration_mode](#)

Definition

test signal

Attribute	Description
active	
	Value of test signal during test mode. Type: enum Enum Values: high low Default: "" Edit: Yes
atpg_use	
	ATPG usage of test signal. Type: enum Enum Values: none opcg_enable opcg_trigger opcg_edge_mode opcg_custom_se opcg_blocking_se master_test_enable misr_reset_clock Default: none Edit: Yes
base_name	
	Name of test_signal Type: string Default: "" Edit: No
dedicated_pin	
	Whether a dedicated test pin is used for the test signal. Type: bool Default: false Edit: Yes
default_shift_enable	
	When 'true', indicates this test signal is the default shift enable. Type: bool Default: false Edit: Yes
dft_compression_signal	
	Type of Compression test signal. Type: enum Enum Values: none compression_enable spreader mask_enable mask_load Default: none Edit: Yes
dft_hookup_pin	

Stylus Common UI Database Object Information

Database Objects--test_signal

	Port/pin where the test signal actually hooks up inside the core. Type: <code>obj(pin) obj(hpin) obj(port) obj(hport)</code> Default: "" Edit: No
dft_hookup_polarity	
	Value of test signal during test mode at the hookup pin. Type: <code>string</code> Default: "" Edit: No
divide_fall	
	Falling edge denominator. Type: <code>int</code> Default: no_value Edit: No
divide_period	
	Clock period denominator. Type: <code>int</code> Default: no_value Edit: No
divide_rise	
	Rising edge denominator. Type: <code>int</code> Default: no_value Edit: No
escaped_name	
	Name of test_signal including escape characters Type: <code>string</code> Default: "" Edit: No
fall	
	Falling edge numerator (fraction of period). Type: <code>int</code> Default: no_value Edit: No
function	
	Function of the test signal object. Type: <code>enum</code> Enum Values: parallel_sdi parallel_sdo compress_sdi compress_sdo serial_sdi serial_sdo select_wir select_serial select_bypass shift_enable test_mode scan_clock lssd_scan_clock_a lssd_scan_clock_b aux_scan_clock async_set_reset test_clock compression_clock compression_enable spread_enable pipe_sen mask_load mask_enable misr_reset_enable in_pipe_bypass out_pipe_bypass decomp_pipe_bypass comp_pipe_bypass codec_pipe_bypass cpi_over_npi first_core wint wext wrapper_reset wrapper_clock wrapper_and_compression_clock shift_wr capture_wr update_wr wir_test custom broadcast_enable serial_wrck elastic_reset smartscan_enable smartscan_broadcast opcg_enable opcg_trigger opcg_load opcg_load_clock opcg_pllclk_capture opcg_los_enable lbist_test_enable lbist_direct_reset lbist_direct_done lbist_direct_fail lbist_direct_pass lbist_direct_enable lbist_setbist lbist_runbist lbist_run misr_bypass mbist_retention_test_out lp_staggered_shift_mode misr_reset pipe_reset lp_gating_enable lp_gating_load swizzle_enable async_hold segment_bypass Default: "" Edit: No
has_fanout	
	Whether a dedicated test pin is used for the test signal by fanout analysis. Type: <code>enum</code> Enum Values: false true test_only Default: false Edit: Yes

Stylus Common UI Database Object Information

Database Objects--test_signal

ideal	
	Whether the test signal network should be treated as ideal. Type: bool Default: false Edit: Yes
index	
	Index of the test signal object. Type: int Default: 0 Edit: No
lec_value	
	Whether a lec value is attached to the test signal. Type: string Default: "" Edit: Yes
master_signal	
	Master test signal. Type: bool Default: true Edit: Yes
name	
	Name of test_signal Type: string Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (test_signal) Default: "" Edit: No
period	
	Clock period numerator. Type: double Default: 0.0 Edit: No
pin	
	Port/pin associated to test signal. Type: obj(pin) obj(hpin) obj(port) obj(hport) Default: "" Edit: Yes
pmbist_use	
	Test signal used during ATPG to control programmable MBIST logic (possible values are: {none test_block_async_reset test_clock_select test_rambypass test_ramsequential retention_pause_continue retention_pause_active}). Type: string Default: "" Edit: Yes
rise	
	Rising edge numerator (fraction of period). Type: int Default: no_value Edit: No
scan_shift	

Stylus Common UI Database Object Information

Database Objects--formal_verification_constraint

	Whether the test signal will be captured as a clock in the ATPG file. Type: <code>bool</code> Default: false Edit: Yes
type	
	Type of test signal. Type: <code>enum</code> Enum Values: test_mode shift_enable scan_clock scan_clock_a scan_clock_b aux_scan_clock hier_test_signal Default: "" Edit: Yes
user_defined_signal	
	Whether the test signal was defined by the user or automatically generated by the tool. Type: <code>enum</code> Enum Values: true false simulated Default: true Edit: Yes
wir_reset_value	
	Value of the WIR output pin when the wir reset signal is asserted. Type: <code>enum</code> Enum Values: high low Default: "" Edit: Yes
wir_signal	
	Whether test signal is a WIR signal. Type: <code>bool</code> Default: false Edit: Yes
wir_tm_value	
	Value of the WIR output pin when the wir is being scan tested. Type: <code>enum</code> Enum Values: high low Default: "" Edit: Yes

formal_verification_constraint

Parent Objects

`design`, `root`

Definition

formal verification constraint

Attribute	Description
base_name	
	Name of formal_verification_constraint Type: <code>string</code> Default: "" Edit: No
escaped_name	

Stylus Common UI Database Object Information

Database Objects--test_clock

	Name of formal_verification_constraint including escape characters Type: string Default: "" Edit: No
name	
	Name of formal_verification_constraint Type: string Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (formal_verification_constraint) Default: "" Edit: No

test_clock

Parent Objects

[inst](#), [design](#), [root](#)

Definition

test clock

Attribute	Description
at_speed	
	Whether the test clock is an at-speed test clock. Type: bool Default: false Edit: Yes
base_name	
	Name of test_clock Type: string Default: "" Edit: No
blocking_se	
	The blocking shift enable corresponding to the test clock. Type: obj (test_signal) Default: "" Edit: Yes
controllable	
	Whether the test clock is controllable in test mode. Type: bool Default: false Edit: Yes
controllable_from	
	Source the test clock is controllable from. Type: obj (pin) obj (hpin) obj (port) obj (hport) Default: "" Edit: No
dft_hookup_pin	

Stylus Common UI Database Object Information
Database Objects--test_clock

	Port/pin where the test clock actually hooks up inside the core. Type: obj(pin) obj(hpin) obj(port) obj(hport) Default: "" Edit: No
dft_hookup_polarity	
	Value of test clock during test mode at the hookup pin. Type: string Default: "" Edit: No
divide_fall	
	Falling edge denominator. Type: int Default: "" Edit: No
divide_period	
	Clock period denominator. Type: int Default: 1 Edit: No
divide_rise	
	Rising edge denominator. Type: int Default: "" Edit: No
domain_se	
	The domain shift enable corresponding to the test clock. Type: obj(test_signal) Default: "" Edit: Yes
escaped_name	
	Name of test_clock including escape characters Type: string Default: "" Edit: No
fall	
	Falling edge numerator (fraction of period). Type: int Default: "" Edit: No
name	
	Name of test_clock Type: string Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (test_clock) Default: "" Edit: No
off_state	

Stylus Common UI Database Object Information

Database Objects--scan_chain

	Off state during shift mode. Type: <code>int</code> Default: "" Edit: No
period	
	Clock period numerator. Type: <code>double</code> Default: "" Edit: No
rise	
	Rising edge numerator (fraction of period). Type: <code>int</code> Default: "" Edit: No
root_source_pins	
	Root source pin(s) of the test clock. Type: <code>obj(pin)* obj(hpin)* obj(port)* obj(hport)*</code> Default: "" Edit: No
root_source_polarity	
	Polarity between root pin and clock pin of the test clock. Type: <code>string</code> Default: "" Edit: No
sources	
	Source pin(s) of the test clock. Type: <code>obj(pin)* obj(hpin)* obj(port)* obj(hport)*</code> Default: "" Edit: No
user_defined_signal	
	Whether the test clock was defined by the user or automatically generated by the tool. Type: <code>bool</code> Default: true Edit: Yes

scan_chain

Parent Objects

[hinst](#), [inst](#), [design](#), [hpin](#), [pin](#), [actual_scan_segment](#), [root](#), [violation](#), [hport](#), [port](#)

Definition

scan chain

Attribute	Description
base_name	
	Name of scan_chain Type: <code>string</code> Default: "" Edit: No
body	

Stylus Common UI Database Object Information
Database Objects--scan_chain

	Body (segment) of scan chain. Type: <code>object*</code> Default: "" Edit: No
complete	
	Scan chain was completely specified by user or analysis. Type: <code>bool</code> Default: false Edit: No
compressed	
	Identifies if a chain has undergone compression. Type: <code>bool</code> Default: false Edit: Yes
dft_hookup_pin_sdi	
	Core side hookup pin for SDI. Type: <code>obj(pin)</code> <code>obj(hpin)</code> <code>obj(port)</code> <code>obj(hport)</code> Default: "" Edit: No
dft_hookup_pin_sdo	
	Core side hookup pin for SDO. Type: <code>obj(pin)</code> <code>obj(hpin)</code> <code>obj(port)</code> <code>obj(hport)</code> Default: "" Edit: No
domain	
	Clock domain of scan chain. Type: <code>string</code> Default: "" Edit: No
edge	
	Edge of clock domain in scan chain. Type: <code>string</code> Default: any Edit: No
escaped_name	
	Name of scan_chain including escape characters Type: <code>string</code> Default: "" Edit: No
head	
	Head (segment) of scan chain. Type: <code>string</code> Default: "" Edit: No
max_length	
	User specified length for this scan chain (might be slightly different than the actual length). Type: <code>int</code> Default: no_value Edit: No
name	

Stylus Common UI Database Object Information

Database Objects--scan_chain

	Name of scan_chain Type: string Default: "" Edit: No
non_shared_scan_out	
	Non shared scan out of scan chain is on functional data path. Type: bool Default: false Edit: Yes
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (scan_chain) Default: "" Edit: No
scan_clock_a	
	Scan clock a (signal) of scan chain. Type: string Default: "" Edit: No
scan_clock_b	
	Scan clock b (signal) of scan chain. Type: string Default: "" Edit: No
scan_in	
	Scan in pin of scan chain. Type: string Default: "" Edit: No
scan_out	
	Scan out pin of scan chain. Type: string Default: "" Edit: No
sdi_compression_signal	
	Whether the scan in of the chain is shared with mask enable. Type: string Default: "" Edit: Yes
shared_input	
	Scan chain shares scan input with another functional pin. Type: bool Default: false Edit: No
shared_output	
	Scan chain shares scan output with another functional pin. Type: bool Default: false Edit: No
shared_select	

Stylus Common UI Database Object Information

Database Objects--actual_scan_chain

	Control signal to the multiplexer added because the scan out pin is shared. Type: string Default: "" Edit: No
shift_enable	
	Scan shift enable (signal) of scan chain. Type: string Default: "" Edit: No
tail	
	Tail (segment) of scan chain. Type: string Default: "" Edit: No
terminal_lockup	
	Type of the terminal lockup element in the scan chain. Type: string Default: none Edit: No

actual_scan_chain

Parent Objects

[hinst](#), [inst](#), [design](#), [hpin](#), [pin](#), [actual_scan_segment](#), [root](#), [violation](#), [hport](#), [port](#)

Definition

actual scan chain

Attribute	Description
analyzed	
	Whether this is an analyzed chain. Type: bool Default: false Edit: No
base_name	
	Name of actual_scan_chain Type: string Default: "" Edit: No
compressed	
	Identifies if a chain has undergone compression. Type: bool Default: false Edit: Yes
connected_shift_enable	
	Connected shift enable attribute of an actual scan chain. Type: bool Default: false Edit: No
dft_hookup_pin_sdi	

Stylus Common UI Database Object Information
Database Objects--actual_scan_chain

	Core side hookup pin for SDI. Type: <code>obj(pin) obj(hpin) obj(port) obj(hport)</code> Default: "" Edit: No
dft_hookup_pin_sdo	
	Core side hookup pin for SDO. Type: <code>obj(pin) obj(hpin) obj(port) obj(hport)</code> Default: "" Edit: No
domain	
	Clock domain of scan chain. Type: <code>string</code> Default: "" Edit: No
edge	
	Edge of clock domain in scan chain. Type: <code>string</code> Default: any Edit: No
elements	
	Elements of scan chain (a Tcl list). Type: <code>object*</code> Default: "" Edit: No
escaped_name	
	Name of actual_scan_chain including escape characters Type: <code>string</code> Default: "" Edit: No
head_lockup	
	Instance of the head lockup element in the scan chain. Type: <code>string</code> Default: none Edit: No
mode_name	
	Mode to which the scan chain belongs to. Type: <code>string</code> Default: "" Edit: No
name	
	Name of actual_scan_chain Type: <code>string</code> Default: "" Edit: No
non_shared_scan_out	
	Non shared scan out of scan chain is on functional data path. Type: <code>bool</code> Default: false Edit: Yes
obj_type	

Stylus Common UI Database Object Information
Database Objects--actual_scan_chain

	The obj_type name for this object. To see all obj_type names, go to Contents Type: <code>enum</code> (<code>actual_scan_chain</code>) Default: "" Edit: No
other_clocks	
	Other clock pins of the segment and their active values as a Tcl list. Type: <code>string</code> Default: "" Edit: No
power_domain	
	All the power domains of this scan chain. Type: <code>string</code> Default: "" Edit: No
reg_count	
	Number of flip-flops in scan chain. Type: <code>int</code> Default: "" Edit: No
scan_clock_a	
	Scan clock a (signal) of scan chain. Type: <code>string</code> Default: "" Edit: No
scan_clock_b	
	Scan clock b (signal) of scan chain. Type: <code>string</code> Default: "" Edit: No
scan_in	
	Scan in pin of scan chain. Type: <code>string</code> Default: "" Edit: No
scan_out	
	Scan out pin of scan chain. Type: <code>string</code> Default: "" Edit: No
sdi_compression_signal	
	Whether the scan in of the chain is shared with mask enable. Type: <code>string</code> Default: "" Edit: Yes
shared_input	
	Scan chain shares scan input with another functional pin. Type: <code>bool</code> Default: <code>false</code> Edit: No
shared_output	

Stylus Common UI Database Object Information

Database Objects--violation

	Scan chain shares scan output with another functional pin. Type: <code>bool</code> Default: <code>false</code> Edit: No
shared_select	
	Control signal to the multiplexer added because the scan out pin is shared. Type: <code>string</code> Default: <code>""</code> Edit: No
shift_enable	
	Scan shift enable (signal) of scan chain. Type: <code>string</code> Default: <code>""</code> Edit: No
terminal_lockup	
	Instance of the terminal lockup element in the scan chain. Type: <code>string</code> Default: <code>none</code> Edit: No

violation

Parent Objects

`design`, `root`

Definition

violation

Attribute	Description
base_name	
	Name of violation Type: <code>string</code> Default: <code>""</code> Edit: No
description	
	Violation description. Type: <code>string</code> Default: <code>""</code> Edit: No
endpoints	
	End points. Type: <code>obj(pin)* obj(hpin)* obj(port)* obj(hport)*</code> Default: <code>""</code> Edit: No
escaped_name	
	Name of violation including escape characters Type: <code>string</code> Default: <code>""</code> Edit: No
fixed	

Stylus Common UI Database Object Information

Database Objects--violation

	Whether fixed by fix_dft_violations. Type: <code>bool</code> Default: <code>false</code> Edit: No
id	
	Violation ID. Type: <code>int</code> Default: <code>""</code> Edit: No
name	
	Name of violation Type: <code>string</code> Default: <code>""</code> Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: <code>enum</code> (violation) Default: <code>""</code> Edit: No
reg_count	
	Register count. Type: <code>int</code> Default: <code>""</code> Edit: No
registers	
	Affected registers. Type: <code>obj(inst)* obj(hinst)*</code> Default: <code>""</code> Edit: No
root_node	
	Root port/pin. Type: <code>obj(pin) obj(hpin) obj(port) obj(hport)</code> Default: <code>""</code> Edit: No
segments	
	Affected abstract segments. Type: <code>obj(scan_chain)* obj(scan_segment)* obj(actual_scan_segment)* obj(actual_scan_chain)*</code> Default: <code>""</code> Edit: No
tristate_net_drivers	
	Driver pins for tristate net. Type: <code>obj(pin)* obj(hpin)* obj(port)* obj(hport)*</code> Default: <code>""</code> Edit: No
tristate_net_load	
	Load pin for tristate net. Type: <code>obj(pin) obj(hpin) obj(port) obj(hport)</code> Default: <code>""</code> Edit: No
type	

	Violation type. Type: string Default: "" Edit: No
--	---

dft_configuration_mode

Parent Objects

[design](#), [root](#)

Definition

dft configuration mode

Attribute	Description
base_name	
	Name of dft_configuration_mode Type: string Default: "" Edit: No
current_mode	
	Current scan mode. Type: bool Default: false Edit: No
decoded_pin	
	Configuration mode decoded pin. Type: obj(pin) obj(hpin) obj(port) obj(hport) Default: "" Edit: Yes
escaped_name	
	Name of dft_configuration_mode including escape characters Type: string Default: "" Edit: No
mode_enable_high	
	Test Signals that are held to active high value for the mode. Type: obj(test_signal)* Default: "" Edit: No
mode_enable_low	
	Test Signals that are held to active low value for the mode. Type: obj(test_signal)* Default: "" Edit: No
name	
	Name of dft_configuration_mode Type: string Default: "" Edit: No
obj_type	

	The obj_type name for this object. To see all obj_type names, go to Contents Type: <code>enum</code> (<code>dft_configuration_mode</code>) Default: "" Edit: No
type	
	Configuration mode type. Type: <code>string</code> Default: "" Edit: No
usage	
	Configuration mode usage. Type: <code>string</code> Default: "" Edit: No
user_defined	
	Is user defined configuration mode. Type: <code>bool</code> Default: false Edit: No

tap_distance_rule

Parent Objects

[tech](#)

Definition

Returns the LEF TAPDISTANCERULE info.

Attribute	Description
default	
	The effective default tap distance of the rule. Type: <code>coord</code> Default: "" Edit: No
name	
	The name of the tap_distance_rule defined in LEF. Type: <code>string</code> Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: <code>enum</code> (<code>tap_distance_rule</code>) Default: "" Edit: No
tap_layer_distances	
	Returns the layer and its tap distance. Type: <code>obj(tap_layer_distance)*</code> Default: "" Edit: No

tap_layer_distance

Parent Objects

[tap_distance_rule](#)

Definition

Returns layer's distance info in LEF TAPDISTANCERULE.

Attribute	Description
distance	
	Returns the tap distance of the specified layer. Type: coord Default: "" Edit: No
layer	
	Returns the layer object. Type: obj(layer) Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (tap_layer_distance) Default: "" Edit: No

interposer_route_rule

Parent Objects

[net](#)

Definition

Return net's interposer route rule

Attribute	Description
layer	
	The layer rule. Type: obj(layer) Default: "" Edit: No
net	
	The net rule. Type: obj(net) Default: "" Edit: No
obj_type	

Stylus Common UI Database Object Information

Database Objects--seq_function

	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (interposer_route_rule) Default: "" Edit: No
offset	
	The offset rule. Type: coord Default: "" Edit: No
spacing	
	The offset rule. Type: coord Default: "" Edit: No
width	
	The width rule. Type: coord Default: "" Edit: No

seq_function

Parent Objects

[lib_cell](#)

Definition

cte seq_function

Attribute	Description
base_name	
	Returns base name of the seq function object. Type: string Allowed -index values: analysis_view Default: "" Edit: No
d_function	
	Returns d function of the seq function object. Type: string Allowed -index values: analysis_view Default: "" Edit: No
escaped_name	
	Returns escaped name of the seq function object. Type: string Allowed -index values: analysis_view Default: "" Edit: No
lib_cell	

Stylus Common UI Database Object Information

Database Objects--test_clock_domain

	Returns lib cell associated with that seq function object. Type: <code>obj(lib_cell)*</code> Allowed -index values: analysis_view Default: "" Edit: No
name	
	Returns name of the seq function object. Type: <code>string</code> Allowed -index values: analysis_view Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: <code>enum (seq_function)</code> Default: "" Edit: No

test_clock_domain

Parent Objects

[design](#), [root](#)

Definition

test clock domain

Attribute	Description
base_name	
	Name of test_clock_domain Type: <code>string</code> Default: "" Edit: No
escaped_name	
	Name of test_clock_domain including escape characters Type: <code>string</code> Default: "" Edit: No
name	
	Name of test_clock_domain Type: <code>string</code> Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: <code>enum (test_clock_domain)</code> Default: "" Edit: No

scan_segment

Parent Objects

[hinst](#), [inst](#), [design](#), [hpin](#), [pin](#), [actual_scan_segment](#), [root](#), [violation](#), [hport](#), [port](#)

Definition

scan segment

Attribute	Description
active	
	Shift enable active value for an abstract segment. Type: string Default: "" Edit: No
base_name	
	Name of scan_segment Type: string Default: "" Edit: No
clock	
	Head clock pin for an abstract segment. Type: string Default: "" Edit: No
clock_edge	
	Active (head) clock edge at the local clock pin for an abstract segment. Type: string Default: "" Edit: No
clock_gating_shift_enable	
	Clock gating shift enable pin for an abstract, preserved or shift_reg segment. Type: string Default: "" Edit: No
connected_scan_clock_a	
	Whether an abstract or preserved segment has its common scan clock a pin already globally connected. Type: bool Default: false Edit: No
connected_scan_clock_b	
	Whether an abstract or preserved segment has its common scan clock b pin already globally connected. Type: bool Default: false Edit: No
connected_shift_enable	
	Whether an abstract or preserved segment has its common shift enable pin already globally connected. Type: bool Default: false Edit: No
core_wrapper	

Stylus Common UI Database Object Information
Database Objects--scan_segment

	Whether the segment is for a core wrapper cell. Type: <code>bool</code> Default: <code>false</code> Edit: Yes
core_wrapper_ports	
	Specifies the ports associated with the wrapper. Type: <code>obj(pin)* obj(hpin)* obj(port)* obj(hport)*</code> Default: <code>""</code> Edit: Yes
core_wrapper_type	
	Specifies the wrapper type. Type: <code>string</code> Default: <code>""</code> Edit: Yes
core_wrapper_usage	
	Specifies the wrapper usage. Type: <code>string</code> Default: <code>""</code> Edit: Yes
ctl_defined	
	Scan segment defined in ctl abstract model. Type: <code>bool</code> Default: <code>false</code> Edit: Yes
dft_dont_scan	
	Exclude this abstract segment from scan-synthesis. Type: <code>bool</code> Default: <code>false</code> Edit: Yes
dft_exclude_abstract_segment_from_los_pipeline	
	Avoid insertion of launch-off-shift pipelined logic to shift enable of abstract segment. Type: <code>bool</code> Default: <code>false</code> Edit: Yes
dft_hookup_pin_sdi	
	Core side hookup pin for SDI. Type: <code>obj(pin) obj(hpin) obj(port) obj(hport)</code> Default: <code>""</code> Edit: No
dft_hookup_pin_sdo	
	Core side hookup pin for SDO. Type: <code>obj(pin) obj(hpin) obj(port) obj(hport)</code> Default: <code>""</code> Edit: No
dft_part_of_segment	
	The type of segment this segment belongs to {abstract fixed floating preserved shift_register}. Type: <code>enum</code> Enum Values: fixed floating preserve abstract shift_register Default: <code>""</code> Edit: No
dft_status	

Stylus Common UI Database Object Information
Database Objects--scan_segment

	DFT rule check status as analyzed by check_dft_rules (Passes DFT rules Fails DFT rules Abstract Segment Dont scan Dont scan Misc. non scan). Type: string Default: "" Edit: No
dft_tail_test_clock	
	Top level tail test_clock for an abstract segment. Type: string Default: "" Edit: No
dft_tail_test_clock_edge	
	Active top level tail test_clock edge for an abstract segment. Type: string Default: "" Edit: No
dft_tail_test_clock_waveform_edge	
	Active top level tail test_clock waveform leading or trailing edge for an abstract segment. Type: string Default: "" Edit: No
dft_test_clock	
	Top level (head) test_clock for an abstract segment. Type: string Default: "" Edit: No
dft_test_clock_edge	
	Active top level (head) test_clock edge for an abstract segment. Type: string Default: "" Edit: No
dft_test_clock_waveform_edge	
	Active top level (head) test_clock waveform leading or trailing edge for an abstract segment. Type: string Default: "" Edit: No
dft_violation	
	DFT rule violation as analyzed by check_dft_rules. Type: string Default: "" Edit: No
elements	
	Elements of scan segment (a Tcl list). Type: object* Default: "" Edit: No
escaped_name	
	Name of scan_segment including escape characters Type: string Default: "" Edit: No
head_skew_safe	

Stylus Common UI Database Object Information

Database Objects--scan_segment

	Whether an abstract segment has a lockup element at the head. Type: <code>bool</code> Default: <code>false</code> Edit: No
instance	
	Instance for an abstract segment. Type: <code>obj(inst) obj(hinst)</code> Default: <code>""</code> Edit: No
model_defined	
	Scan segment defined in abstract model. Type: <code>bool</code> Default: <code>false</code> Edit: Yes
name	
	Name of scan_segment Type: <code>string</code> Default: <code>""</code> Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: <code>enum (scan_segment)</code> Default: <code>""</code> Edit: No
other_clocks	
	Other clock pins of the segment and their active values as a Tcl list. Type: <code>string</code> Default: <code>""</code> Edit: No
power_domain	
	All the power domains of this scan chain. Type: <code>string</code> Default: <code>""</code> Edit: No
reg_count	
	Number of flip-flops in scan segment. Type: <code>int</code> Default: <code>""</code> Edit: No
reorderable	
	Whether the preserved segment is reorderable for scanDEF purposes. Type: <code>bool</code> Default: <code>false</code> Edit: No
scan_in	
	SDI pin for an abstract, preserved or shift_reg segment. Type: <code>string</code> Default: <code>""</code> Edit: No
scan_out	

Stylus Common UI Database Object Information
Database Objects--actual_scan_segment

	SDO pin for an abstract, preserved or shift_reg segment. Type: <i>string</i> Default: "" Edit: No
shift_enable	
	Shift enable pin for an abstract, preserved or shift_reg segment. Type: <i>string</i> Default: "" Edit: No
skew_safe	
	Whether an abstract segment has a lockup element at the end. Type: <i>bool</i> Default: false Edit: No
tail_clock	
	Tail clock pin for an abstract segment. Type: <i>string</i> Default: "" Edit: No
tail_clock_edge	
	Active tail clock edge at the local clock pin for an abstract segment. Type: <i>string</i> Default: "" Edit: No
test_modes	
	Test mode pins of the segment and their active values as a Tcl list. Type: <i>string</i> Default: "" Edit: No
type	
	Type of scan segment. Type: <i>string</i> Default: "" Edit: No
user_defined_segment	
	Whether the segment was defined by the user or automatically identified by the tool. Type: <i>bool</i> Default: true Edit: No

actual_scan_segment

Parent Objects

[hinst](#), [inst](#), [design](#), [hpin](#), [pin](#), [root](#), [violation](#), [hport](#), [port](#)

Definition

scan segment

Attribute	Description
active	

Stylus Common UI Database Object Information
Database Objects--actual_scan_segment

	Shift enable active value for an abstract segment. Type: <i>string</i> Default: "" Edit: No
base_name	
	Name of actual_scan_segment Type: <i>string</i> Default: "" Edit: No
clock	
	Head clock pin for an abstract segment. Type: <i>string</i> Default: "" Edit: No
clock_edge	
	Active (head) clock edge at the local clock pin for an abstract segment. Type: <i>string</i> Default: "" Edit: No
clock_gating_shift_enable	
	Clock gating shift enable pin for an abstract, preserved or shift_reg segment. Type: <i>string</i> Default: "" Edit: No
connected_scan_clock_a	
	Whether an abstract or preserved segment has its common scan clock a pin already globally connected. Type: <i>bool</i> Default: false Edit: No
connected_scan_clock_b	
	Whether an abstract or preserved segment has its common scan clock b pin already globally connected. Type: <i>bool</i> Default: false Edit: No
connected_shift_enable	
	Whether an abstract or preserved segment has its common shift enable pin already globally connected. Type: <i>bool</i> Default: false Edit: No
core_wrapper	
	Whether the segment is for a core wrapper cell. Type: <i>bool</i> Default: false Edit: Yes
core_wrapper_type	
	Specifies the wrapper type. Type: <i>string</i> Default: "" Edit: Yes
core_wrapper_usage	

Stylus Common UI Database Object Information

Database Objects--actual_scan_segment

	Specifies the wrapper usage. Type: <code>string</code> Default: "" Edit: Yes
ctl_defined	
	Scan segment defined in ctl abstract model. Type: <code>bool</code> Default: false Edit: Yes
dft_exclude_abstract_segment_from_los_pipeline	
	Avoid insertion of launch-off-shift pipelined logic to shift enable of abstract segment. Type: <code>bool</code> Default: false Edit: Yes
dft_hookup_pin_sdi	
	Core side hookup pin for SDI. Type: <code>obj(pin)</code> <code>obj(hpin)</code> <code>obj(port)</code> <code>obj(hport)</code> Default: "" Edit: No
dft_hookup_pin_sdo	
	Core side hookup pin for SDO. Type: <code>obj(pin)</code> <code>obj(hpin)</code> <code>obj(port)</code> <code>obj(hport)</code> Default: "" Edit: No
dft_part_of_segment	
	The type of segment this segment belongs to {abstract fixed floating preserved shift_register}. Type: <code>enum</code> Enum Values: fixed floating preserve abstract shift_register Default: "" Edit: No
dft_scan_chain	
	The actual scan chain this segment belongs to. Type: <code>obj(scan_chain)</code> <code>obj(scan_segment)</code> <code>obj(actual_scan_segment)</code> <code>obj(actual_scan_chain)</code> Default: "" Edit: No
dft_tail_test_clock	
	Top level tail test_clock for an abstract segment. Type: <code>string</code> Default: "" Edit: No
dft_tail_test_clock_edge	
	Active top level tail test_clock edge for an abstract segment. Type: <code>string</code> Default: "" Edit: No
dft_tail_test_clock_waveform_edge	
	Active top level tail test_clock waveform leading or trailing edge for an abstract segment. Type: <code>string</code> Default: "" Edit: No
dft_test_clock	

Stylus Common UI Database Object Information
Database Objects--actual_scan_segment

	Top level (head) test_clock for an abstract segment. Type: string Default: "" Edit: No
dft_test_clock_edge	
	Active top level (head) test_clock edge for an abstract segment. Type: string Default: "" Edit: No
dft_test_clock_waveform_edge	
	Active top level (head) test_clock waveform leading or trailing edge for an abstract segment. Type: string Default: "" Edit: No
elements	
	Elements of scan segment (a Tcl list). Type: object* Default: "" Edit: No
escaped_name	
	Name of actual_scan_segment including escape characters Type: string Default: "" Edit: No
head_skew_safe	
	Whether an abstract segment has a lockup element at the head. Type: bool Default: false Edit: No
instance	
	Instance for an abstract segment. Type: obj(inst) obj(hinst) Default: "" Edit: No
name	
	Name of actual_scan_segment Type: string Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (actual_scan_segment) Default: "" Edit: No
other_clocks	
	Other clock pins of the segment and their active values as a Tcl list. Type: string Default: "" Edit: No
power_domain	

Stylus Common UI Database Object Information
Database Objects--actual_scan_segment

	All the power domains of this scan chain. Type: <i>string</i> Default: "" Edit: No
reg_count	
	Number of flip-flops in scan segment. Type: <i>int</i> Default: "" Edit: No
reorderable	
	Whether the preserved segment is reorderable for scanDEF purposes. Type: <i>bool</i> Default: false Edit: No
scan_in	
	SDI pin for an abstract, preserved or shift_reg segment. Type: <i>string</i> Default: "" Edit: No
scan_out	
	SDO pin for an abstract, preserved or shift_reg segment. Type: <i>string</i> Default: "" Edit: No
shift_enable	
	Shift enable pin for an abstract, preserved or shift_reg segment. Type: <i>string</i> Default: "" Edit: No
skew_safe	
	Whether an abstract segment has a lockup element at the end. Type: <i>bool</i> Default: false Edit: No
tail_clock	
	Tail clock pin for an abstract segment. Type: <i>string</i> Default: "" Edit: No
tail_clock_edge	
	Active tail clock edge at the local clock pin for an abstract segment. Type: <i>string</i> Default: "" Edit: No
type	
	Type of scan segment. Type: <i>string</i> Default: "" Edit: No

isolation_rule

Parent Objects

[power_domain](#)

Definition

power_intent isolation rule

Attribute	Description
applies_to	
	isolation_rule's applies_to Type: string Default: "" Edit: No
applies_to_boundary	
	isolation_rule's applies_to_boundary Type: string Default: "" Edit: No
clamp_value	
	isolation_rule's clamp_value Type: string Default: "" Edit: No
diff_supply_only	
	isolation_rule's diff_supply_only Type: bool Default: "" Edit: No
domain	
	isolation_rule's domain Type: string Default: "" Edit: No
elements	
	isolation_rule's elements Type: string* Default: "" Edit: No
exclude_elements	
	isolation_rule's exclude_elements Type: string* Default: "" Edit: No
isolation_sense	
	isolation_rule's isolation_sense Type: string Default: "" Edit: No
isolation_signal	

Stylus Common UI Database Object Information

Database Objects--level_shifter_rule

	isolation_rule's isolation_signal Type: string Default: "" Edit: No
isolation_supply	
	isolation_rule's isolation_supply Type: string Default: "" Edit: No
location	
	isolation_rule's location Type: string Default: "" Edit: No
name	
	isolation_rule's name Type: string Default: "" Edit: No
no_isolation	
	isolation_rule's no_isolation Type: bool Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (isolation_rule) Default: "" Edit: No
sink	
	isolation_rule's sink Type: string Default: "" Edit: No
source	
	isolation_rule's source Type: string Default: "" Edit: No

level_shifter_rule

Parent Objects

[power_domain](#)

Definition

power_intent level_shifter rule

Attribute	Description
applies_to	

Stylus Common UI Database Object Information
Database Objects--level_shifter_rule

	level_shifter_rule's applies_to Type: string Default: "" Edit: No
applies_to_boundary	
	level_shifter_rule's applies_to_boundary Type: string Default: "" Edit: No
domain	
	level_shifter_rule's domain Type: string Default: "" Edit: No
elements	
	level_shifter_rule's elements Type: string* Default: "" Edit: No
exclude_elements	
	level_shifter_rule's exclude_elements Type: string* Default: "" Edit: No
input_supply	
	level_shifter_rule's input_supply Type: string Default: "" Edit: No
location	
	level_shifter_rule's location Type: string Default: "" Edit: No
name	
	level_shifter_rule's name Type: string Default: "" Edit: No
no_shift	
	level_shifter_rule's no_shift Type: bool Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (level_shifter_rule) Default: "" Edit: No
output_supply	

Stylus Common UI Database Object Information

Database Objects--retention_rule

	level_shifter_rule's output_supply Type: string Default: "" Edit: No
rule	
	level_shifter_rule's rule Type: string Default: "" Edit: No
sink	
	level_shifter_rule's sink Type: string Default: "" Edit: No
source	
	level_shifter_rule's source Type: string Default: "" Edit: No

retention_rule

Parent Objects

[power_domain](#)

Definition

power_intent retention rule

Attribute	Description
domain	
	retention_rule's domain Type: string Default: "" Edit: No
elements	
	retention's elements Type: string* Default: "" Edit: No
exclude_elements	
	retention's exclude_elements Type: string* Default: "" Edit: No
name	
	retention_rule's name Type: string Default: "" Edit: No
no_retention	

Stylus Common UI Database Object Information
Database Objects--retention_rule

	retention_rule's no_retention Type: <code>bool</code> Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: <code>enum</code> (<code>retention_rule</code>) Default: "" Edit: No
restore_condition	
	retention_rule's restore_condition Type: <code>string</code> Default: "" Edit: No
restore_signal	
	retention_rule's restore_signal Type: <code>string</code> Default: "" Edit: No
retention_supply	
	retention_rule's retention_supply Type: <code>string</code> Default: "" Edit: No
save_condition	
	retention_rule's save_condition Type: <code>string</code> Default: "" Edit: No
save_signal	
	retention_rule's save_signal Type: <code>string</code> Default: "" Edit: No
zero_pin_live_slave_clock_isolation_cell	
	retention_rule's zero_pin_live_slave_clock_isolation_cell Type: <code>string</code> Default: "" Edit: Yes
zero_pin_live_slave_reset_isolation_cell	
	retention_rule's zero_pin_live_slave_reset_isolation_cell Type: <code>string</code> Default: "" Edit: Yes
zero_pin_live_slave_set_isolation_cell	
	retention_rule's zero_pin_live_slave_set_isolation_cell Type: <code>string</code> Default: "" Edit: Yes

power_switch_rule

Parent Objects

[power_domain](#)

Definition

power_intent power_switch rule

Attribute	Description
ack_port	
	power_switch_rule's ack_port Type: string* Default: "" Edit: No
control_port	
	power_switch_rule's control_port Type: string* Default: "" Edit: No
domain	
	power_switch_rule's domain Type: string Default: "" Edit: No
input_supply_port	
	power_switch_rule's input_supply_port Type: string* Default: "" Edit: No
name	
	power_switch_rule's name Type: string Default: "" Edit: No
obj_type	
	The obj_type name for this object. To see all obj_type names, go to Contents Type: enum (power_switch_rule) Default: "" Edit: No
off_state	
	power_switch_rule's off_state Type: string* Default: "" Edit: No
on_state	
	power_switch_rule's on_state Type: string* Default: "" Edit: No
output_supply_port	

Stylus Common UI Database Object Information

Database Objects--power_switch_rule

	power_switch_rule's output_supply_port Type: string Default: "" Edit: No
supply_set	
	power_switch_rule's supply_set Type: string Default: "" Edit: No